System Design Patterns — Vol 1: Fundamentals and Algorithms

Every basic concept from zero (Module 0) and the algorithms behind real systems, step by step (Module 19)

2 modules · 60+ patterns · every lesson explained, with course diagrams and links to the originals

what it is · problem · how it works · trade-offs · real world · interview · self-check · TL;DR

Personal study notes summarised from the DesignGurus course (Arslan Ahmad). Not for redistribution.

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Module 0 — System Design Fundamentals

Module Introduction

What this module is. Every other module assumes the basic vocabulary of system design: what *latency* means, what a *load balancer* does, why a *cache* helps, what *sharding* or *eventual consistency* are. This module teaches all of it from zero. If you have only written programs that run on one computer, start here.

How it is organised. Six parts, each building on the one before:

Part	What you learn
1 — Foundations and performance	Clients and servers, latency, throughput, percentiles, estimation, availability and the nines, SLAs, scaling up vs out, stateless services, bottlenecks, how a design interview runs, non-functional requirements and their design levers
2 — Networking and communication	IP, ports, DNS, TCP/UDP/QUIC, HTTP versions, TLS, REST/gRPC/GraphQL and RPC internals, WebSockets and friends, proxies, API gateways, load balancers (every type and algorithm), CDNs
3 — Data storage	Relational databases, ACID, isolation levels, NoSQL families, indexes, replication, partitioning and sharding, consistent hashing, block/file/object storage, OLTP vs OLAP, durability
4 — Caching, async and consistency	Cache strategies and eviction, queues and streams, delivery guarantees, idempotency, consistency models, CAP/PACELC, quorums, consensus, clocks, 2PC and sagas, Bloom filters, ID generation
5 — Architecture, operations and security	Monolith vs microservices vs serverless, resilience patterns, rate limiting, autoscaling, containers and Kubernetes, CI/CD and deployments, observability, disaster recovery, auth, encryption, web security
6 — Putting it together	A request's journey end to end, a worked design, a decision cheat sheet, 60 interview questions and a master A–Z glossary

How to read it. Each concept follows the same rhythm: *What it is → Why it matters → How it works → Types → Trade-offs → Common mistakes → Real-world examples → In the interview → TL;DR*. Read Parts 1–5 in order the first time; afterwards use the Part 6 cheat sheet and the master A–Z glossary (which holds every part's terms) as a reference.

> These notes are written in my own words from official documentation, papers and engineering blogs; specific numbers were checked against their sources.

Part 1 — Foundations and performance

The ground floor: what "system design" means, the basic pieces every web application is built from, and the vocabulary for speed (latency, throughput, percentiles), size (estimation, units), staying up (availability, reliability, durability) and growing (scalability). It ends with a step-by-step method for a system design interview. It assumes only that a website runs on a computer called a **server**, which you reach over the internet.

What system design is

What it is

System design is deciding which parts a piece of software needs, what each part does, how the parts talk to each other, and where data lives, so that the whole meets its goals: serving a certain number of

users, responding within a certain time, staying available when machines break, and costing an acceptable amount.

Writing code answers "how does this function work?". System design answers "how do these twenty programs, on two hundred machines in three countries, work together without losing anyone's data?" Like an architect who decides floors, lifts and fire exits but lays no bricks, the designer shapes the system and programmers fill in each room.

Why it matters

A program that works for 10 users can collapse at 10 million because the *shape* of the system is wrong: one database every request queues for, one machine whose failure takes the site down, data so far from users that every page takes seconds. These are fixed by design decisions, not better loops. Design also drives cost: two working designs can differ tenfold in their cloud bill.

How it works

Most system design follows the same loop:

1. **Understand the goal.** What must the system do (features), and how well (speed, size, uptime)?
2. **Estimate the load.** Users, requests per second, gigabytes per day.
3. **Sketch the parts.** Clients, servers, databases, caches, queues and how requests flow between them.
4. **Find the weak spots.** Which part runs out of capacity first? Which part, if it dies, takes everything with it?
5. **Fix and justify.** Add or change parts, and state what each change costs.

You repeat the loop as the system grows; a design is right for a given load and set of goals, never "finished".

Abstraction is the main tool. An **abstraction** hides how a part works behind a simple contract, so you can reason about the whole system without holding every detail in your head. Distributed systems lean on a few standard ones: a remote call that looks like a local function (*RPC internals*, Part 2), a database that looks like one reliable table although it is many replicated machines (Part 3), a **consistency model** that states what readers may see (Part 4), and a **failure model** that states how parts may break (*Faults, failures and fault tolerance*, this part). Abstractions **leak**: the network is slow and fails partially, replicas lag, clocks drift. Good design uses the contract and names the leaks it must handle.

Trade-offs

Almost every choice gives something up: faster reads may mean slower writes; more copies of data mean better safety but more cost and more chance copies disagree; simpler systems are easier to run but may not scale. Name the trade-off out loud instead of pretending a choice is free.

Common mistakes

- Designing for Google-scale traffic when the product has 1,000 users. Extra parts add cost and failure points.
- Jumping to product names ("use Kafka") before stating the problem the product solves.
- Ignoring failure. Every machine, disk and network link eventually breaks.

In the interview

Q: What is the interviewer testing in a system design round?

A: Turning a vague product idea into a reasoned architecture: asking the right questions, estimating scale, choosing components, explaining trade-offs. There is rarely one correct answer; the reasoning is graded.

Q: How is system design different from coding?

A: Coding is correctness inside one program. System design is how many programs and machines cooperate, how they behave under load and failure, and what that costs.

TL;DR

System design is choosing the parts of a software system and how they fit together so it is fast, big, reliable and cheap enough. Every choice is a trade-off you should be able to explain.

Clients, servers and the request/response cycle

What it is

A **client** is any program that asks for something: a browser, a phone app, or another server. A **server** is a program (and, loosely, the machine it runs on) that waits for requests and answers them.

They follow the **request/response** pattern: the client sends a **request** ("give me the home page", "save this photo"), the server does some work and sends back a **response** (the page, or "saved, here is its ID"). On the web these messages usually use **HTTP** (HyperText Transfer Protocol), a standard format for requests and responses (Part 2). Like a diner ordering from a waiter, the client only needs to know how to order, not how the kitchen works.

How it works

Opening `https://example.com/profile` in a browser, at the simplest level:

1. The browser finds the server's address (using **DNS**, the internet's phone book, Part 2).
2. It opens a network connection to that server.
3. It sends an HTTP request such as:

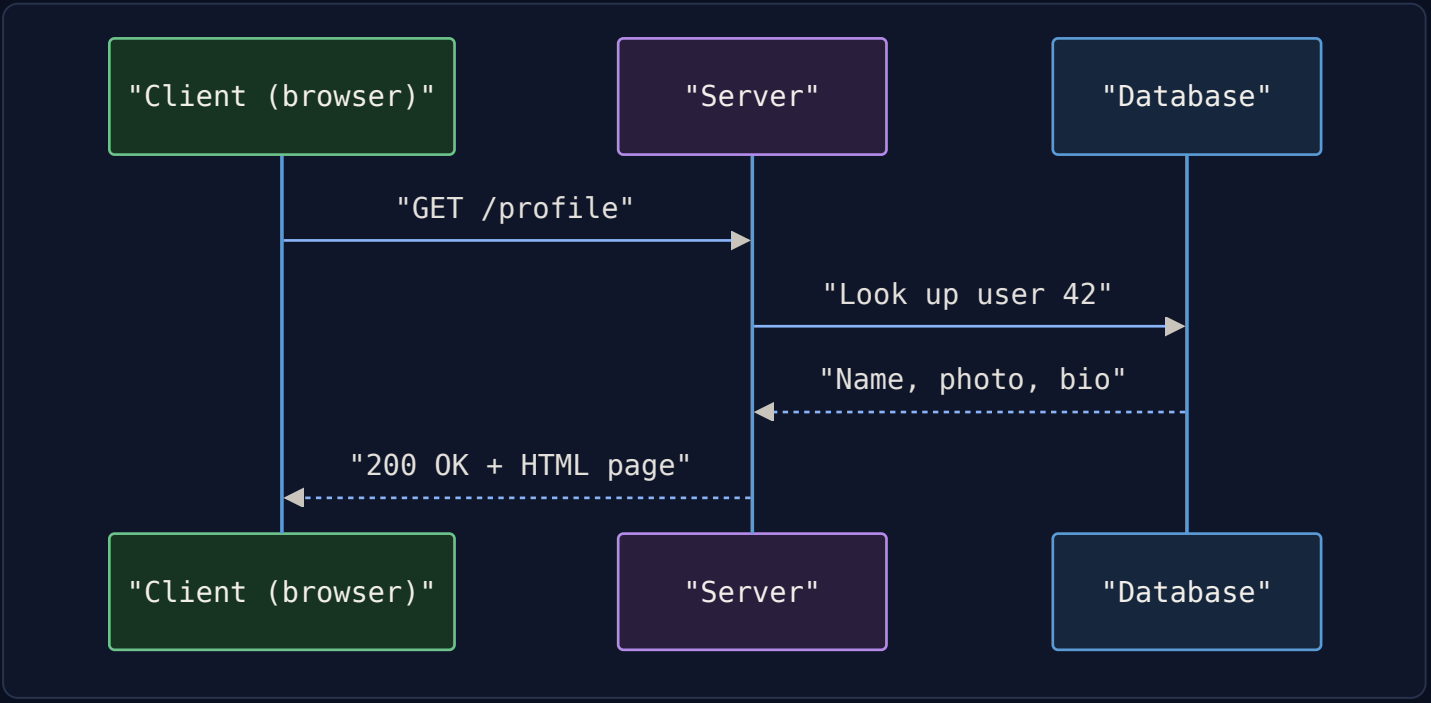
```
GET /profile HTTP/1.1
Host: example.com
```

4. The server reads the request, looks up your profile (perhaps in a database), and builds a response.
5. It sends back something like:

```
HTTP/1.1 200 OK
Content-Type: text/html

<html> ... your profile ... </html>
```

6. The browser draws the page.



Types

- **Thin vs thick client.** A thin client (a basic web page) relies on the server; a thick client (a mobile game, a desktop app) does a lot locally and talks to the server only for shared data.
- **Server-to-server.** Inside a large system servers are constantly clients of other servers (the "profile service" calls the "photo service"). The same request/response idea applies.
- **Push instead of pull.** Normally the client asks (pulls). Some features need the server to send data unasked (a chat message arriving); techniques for this (WebSockets, Server-Sent Events) are in Part 2.

Common mistakes

- Thinking "server" means one physical box. A server is a role: one machine can run many server programs, and one logical server (`api.example.com`) is often hundreds of machines behind a load balancer.
- Forgetting that the network between client and server is slow and unreliable compared with work inside one machine.

In the interview

Q: Can a server also be a client?

A: Yes. Any program sending a request is a client for that request; in a microservice system almost every service is both.

TL;DR

A client asks, a server answers. Almost everything on the web is a chain of request/response conversations, often with servers acting as clients of other servers.

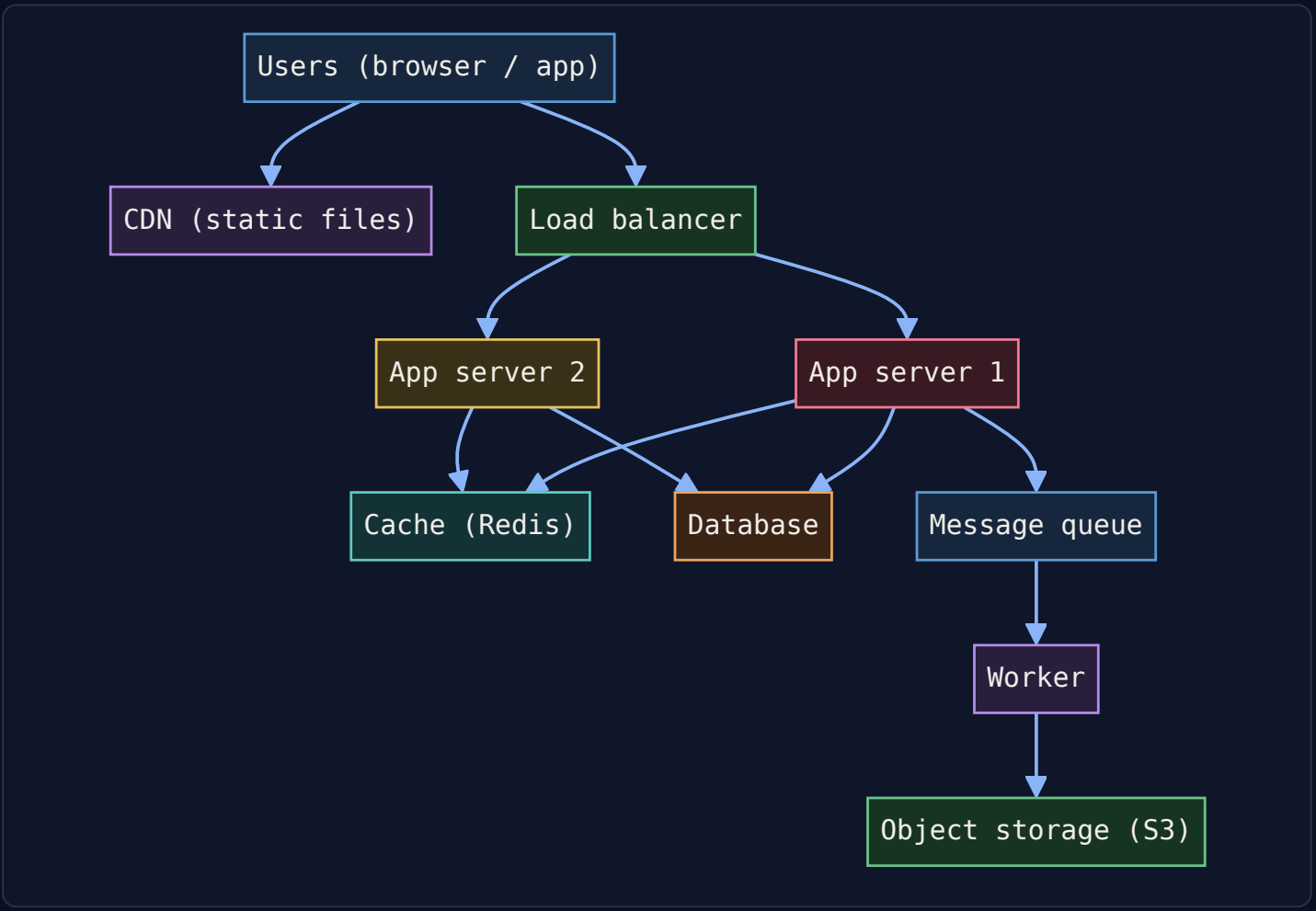
The building blocks of a web application

What it is

Nearly every web application, from a blog to a bank, is built from the same handful of parts:

Part	What it does	Examples (covered in)
Client (browser/app)	Renders the interface and calls the backend over HTTP	Browser, mobile app
Load balancer	Sits in front of the app servers and forwards each request to one of them, so none is overloaded and dead ones are skipped	NGINX, AWS ALB (Part 2)
App server (application server)	Receives requests, applies business rules ("only friends may see this post"), reads and writes data, returns a response; run several identical copies so one crash doesn't stop the site	Java, Python, Go, Node.js programs
Database	Stores data durably (survives power loss) and lets you search it; usually the hardest part to scale because it holds **state** (data that must be remembered)	PostgreSQL, MySQL, MongoDB, DynamoDB (Part 3)
Cache	Small, fast store (usually RAM) holding copies of frequently read data so the app skips the slower database: a RAM read takes well under a millisecond, a database query several ms or more	Redis, Memcached (Part 4)
Message queue	Buffer where one part drops a job ("send a welcome email to user 42") and another picks it up later, so the web request returns at once while slow work runs in the background	RabbitMQ, Amazon SQS, Apache Kafka (Part 4)
Workers	Background programs that take jobs from the queue and do them	—
Object storage	Stores big files ("objects") cheaply by key, e.g. `photos/42/avatar.jpg`; databases are bad at large files	Amazon S3 (Part 3)
CDN (Content Delivery Network)	Servers around the world keeping copies of static files (images, JavaScript, video) near users, so Sydney doesn't fetch every image from Virginia	Cloudflare, CloudFront (Part 2)

How it works



Why it matters

In an interview you draw some version of this diagram within fifteen minutes. Knowing what each box is for lets you justify it ("we add a cache because reads outnumber writes 100 to 1") instead of adding boxes because "big systems have them".

Common mistakes

- Adding every component by default. A small internal tool may need one app server and one database.
- Storing images inside the database. Put files in object storage and keep only their key in the database.
- Doing slow work (emails, video encoding) inside the web request. Put it on a queue.

Real-world examples

- Instagram's early stack (2011–2012) was Django app servers, PostgreSQL, Redis and Memcached, with photos in Amazon S3 — almost exactly the diagram above.
- Cloud tutorials (AWS, Google Cloud, Azure) describe a "three-tier" app: presentation (client), logic (app servers), data (database), with caches and queues added as it grows.

In the interview

Q: Why not keep everything on one server?

A: One server is a single point of failure and has a hard capacity limit. Stateless app servers behind a load balancer, plus a separate database, let you add servers and survive crashes.

Q: When do you add a queue?

A: When work is slow, can be done later, or arrives in bursts. The user gets a fast response and the queue smooths spikes for the workers.

TL;DR

Client, load balancer, app servers, database, cache, queue with workers, object storage and CDN are the standard building blocks. Each solves one problem; add it only when you can name that problem.

Latency, throughput and bandwidth

What it is

Three different kinds of speed:

- **Latency** is how long one thing takes: the delay between asking and getting the answer, in milliseconds (ms, thousandths of a second) or microseconds (µs, millionths; written "us" in plain text).
- **Throughput** is how many things get done per unit of time: requests, messages or megabytes per second.
- **Bandwidth** is the maximum data a network link (or disk, or memory bus) can carry per second — a capacity, like the width of a pipe, in bits per second (Mbps, Gbps).

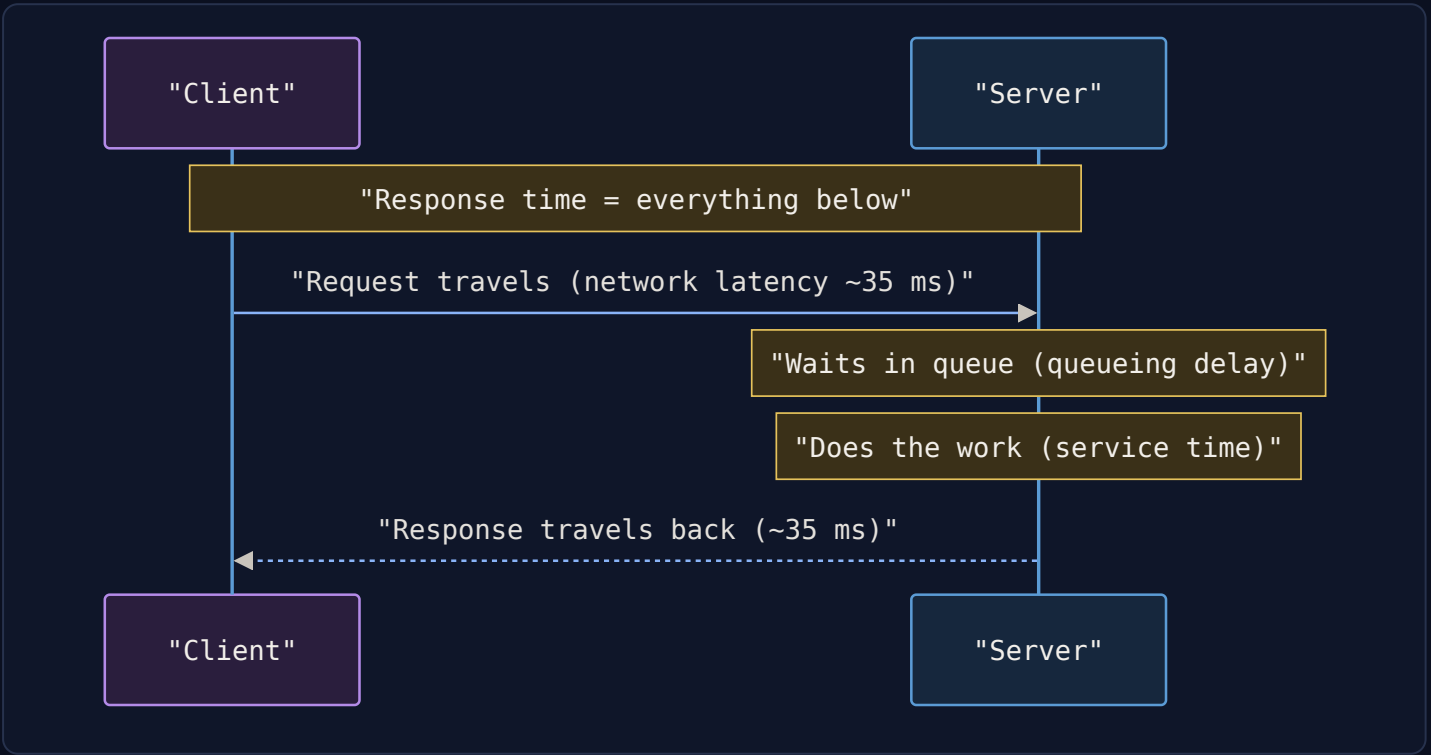
On a motorway, latency is how long one car takes from London to Birmingham, bandwidth is the number of lanes, and throughput is how many cars actually arrive per hour (lower than the lanes allow if there is a crash or toll booth).

Response time vs latency

Often used as synonyms, but strictly:

- **Response time** is what the client experiences: the full time from sending a request to receiving the complete answer, including network travel both ways, queueing and processing.
- **Latency**, used strictly, covers only delay: the stretch before a server starts on the request (sitting in a queue, travelling a network hop), not the work itself.
- **Service time** is the time the server spends actually working on it.

So: response time = network delay + queueing delay + service time. In interviews "latency" almost always means response time; be clear which you mean when it matters.



Why it matters

- Users feel latency. Under about 100 ms feels instant; over a second breaks concentration. Google and Amazon have both reported that small added delays measurably reduce searches and sales.

- Throughput decides how many machines you need: at 1,000 requests per second per server and 50,000 expected, you need at least 50 (more for headroom).
- Bandwidth decides whether large transfers (video, backups) are possible in time.

How it works

Where latency comes from. A request picks up delay at every step:

1. **Propagation delay:** signals in fibre travel at about two thirds the speed of light, ~200 km per millisecond. London–New York is ~5,600 km, so one way takes at least 28 ms; a real **round trip** (RTT) is typically ~70 ms because cables don't run straight and routers add delay.
2. **Transmission delay:** time to push the bits onto the wire. 1 MB (8 megabits) takes 80 ms at 100 Mbps, 8 ms at 1 Gbps. This is where bandwidth matters.
3. **Queueing delay:** waiting behind other requests at routers, load balancers, thread pools, database locks.
4. **Processing delay:** the actual work: parsing, running code, reading disk.

Latency and bandwidth are independent. A geostationary satellite link may have huge bandwidth and ~600 ms round trips. A small request is limited by latency, a large download by bandwidth; more bandwidth does not make a small API call faster.

Throughput and latency are linked through load. At low load throughput rises with offered load and latency stays flat. Near full use, requests wait for each other: latency shoots up while throughput levels off. A server with its CPU (central processing unit) 95% busy usually has terrible latency.



Low-latency checklist

Every latency fix attacks one of the four delays above. Start with a **latency budget**: split the target (say p99 < 200 ms for an API call) across the hops, for example 60 ms of network, 20 ms at the gateway, 80 ms in the service and its database, and 40 ms of slack. Any hop that overspends its share is where you work first. Then go down the list, roughly in order of payoff:

#	Lever	What to do	Which delay it cuts	Explained in
1	Remove round trips	Batch many small calls into one; run independent calls in parallel, not one after another; avoid N+1 query loops	Propagation × number of trips	*Connection pooling, ORMs and N+1 queries* (Part 3)
2	Reuse connections	Connection pools and keep-alive; HTTP/2 or HTTP/3 multiplexing; TLS 1.3 and session resumption, so a request does not pay 2–3 handshake RTTs	Propagation	*TCP, UDP and QUIC*, *HTTPS and TLS* (Part 2)
3	Move bytes closer	CDN for static and cacheable content; edge termination of TLS; regional deployments or read replicas near users	Propagation	*CDNs and anycast* (Part 2), *Replication* (Part 3)
4	Serve from memory	In-process or distributed cache for hot reads; precompute results (feeds, counts, materialised views) at write time	Processing	*Caching basics* (Part 4)
5	Do less while the user waits	Move anything the user doesn't need in the response (emails, analytics, thumbnails) onto a queue	Processing	*Message queues* (Part 4)
6	Shrink payloads	Compress (gzip, Brotli); compact binary formats (Protobuf); paginate; return only the fields the screen needs	Transmission	*API styles* (Part 2)
7	Make each query cheap	Index the access path; denormalise the hot read; keep data in a store whose layout matches the query	Processing	*Indexes*, *Normalization and denormalization* (Part 3)
8	Keep headroom	Run at 50–70% peak utilisation; autoscale before saturation; shed load rather than queue forever	Queueing	*Autoscaling*, *Load shedding* (Part 5)
9	Cut the tail	Deadlines on every call; hedged requests to a second replica; limit fan-out width; avoid long garbage-collection pauses and shared locks	Queueing and processing at p99	*Percentiles, tail latency* (this part)
10	Measure the right thing	Track p50/p95/p99 per hop with histograms and distributed tracing; load-test with open-loop tools that don't hide slow responses	All of them (you can't fix what you can't see)	*Observability* (Part 5)

Two rules keep the list honest. **Fix the biggest term first:** a trans-Atlantic round trip (70 ms) dwarfs a 2 ms JSON parse, so moving the data closer beats micro-optimising code. **Every lever has a price:** caches can serve stale data, replicas cost money and add consistency questions, hedging adds load, and async work means the user doesn't see the result right away. Say which price you're paying.

Trade-offs

- **Batching** improves throughput but hurts latency: collecting 100 writes to send together is efficient, but the first write waits for the other 99.
- **Spare capacity** keeps latency low but costs money; many teams target 50–70% peak utilisation so bursts don't create queues.

Common mistakes

- Mixing bits and bytes. Network speeds are in bits (Mbps), file sizes in bytes (MB); 1 byte = 8 bits, so 100 Mbps moves at most 12.5 MB per second.
- Assuming more bandwidth fixes slowness. If the problem is round trips (a page making 40 sequential API calls), only fewer trips or shorter distances help.
- Measuring throughput without latency (or the reverse). "10,000 requests per second" means nothing if each takes 5 seconds.

Real-world examples

- Video streaming (Netflix, YouTube) is bandwidth-bound: a 4K stream needs roughly 15–25 Mbps.
- High-frequency trading firms pay for microwave links between Chicago and New York to shave a few milliseconds of propagation delay compared with fibre.
- Online games are latency-bound: players feel the difference between 20 ms and 120 ms "ping" (round-trip time).

In the interview

Q: What is the difference between latency and throughput?

A: Latency is how long one request takes; throughput is how many complete per second. You can have high throughput with high latency (a cargo ship) or low latency with low throughput (a motorbike courier).

Q: A user in Australia complains the site is slow; servers are in Virginia. What helps?

A: The round trip is ~200 ms, so cut round trips: CDN for static files near them, fewer sequential calls, reused connections, perhaps a regional deployment. More bandwidth won't help much.

TL;DR

Latency is time per request, throughput is requests per time, bandwidth is the pipe's maximum capacity. Push a system near its limit and latency rises sharply.

Percentiles, tail latency and why averages lie

What it is

A **percentile** is the value below which a given percentage of measurements fall. If **p95** (95th percentile) latency is 300 ms, 95% of requests finished in 300 ms or less and 5% took longer.

Name	Meaning	1 in N requests slower
p50 (median)	Half are faster, half slower	1 in 2
p90	90% are faster	1 in 10
p95	95% are faster	1 in 20
p99	99% are faster	1 in 100
p999 (p99.9)	99.9% are faster	1 in 1,000

Tail latency is the slow end of the distribution: p99, p999 and the maximum — on a latency graph the slow requests form a long thin tail to the right. (If one bus trip in ten takes 90 minutes, a twice-a-day rider is stuck several times a month; "the average trip is 20 minutes" hides that.)

Why averages lie

The **mean** (average) mixes fast and slow requests into one number that may match no real request. Ten requests:

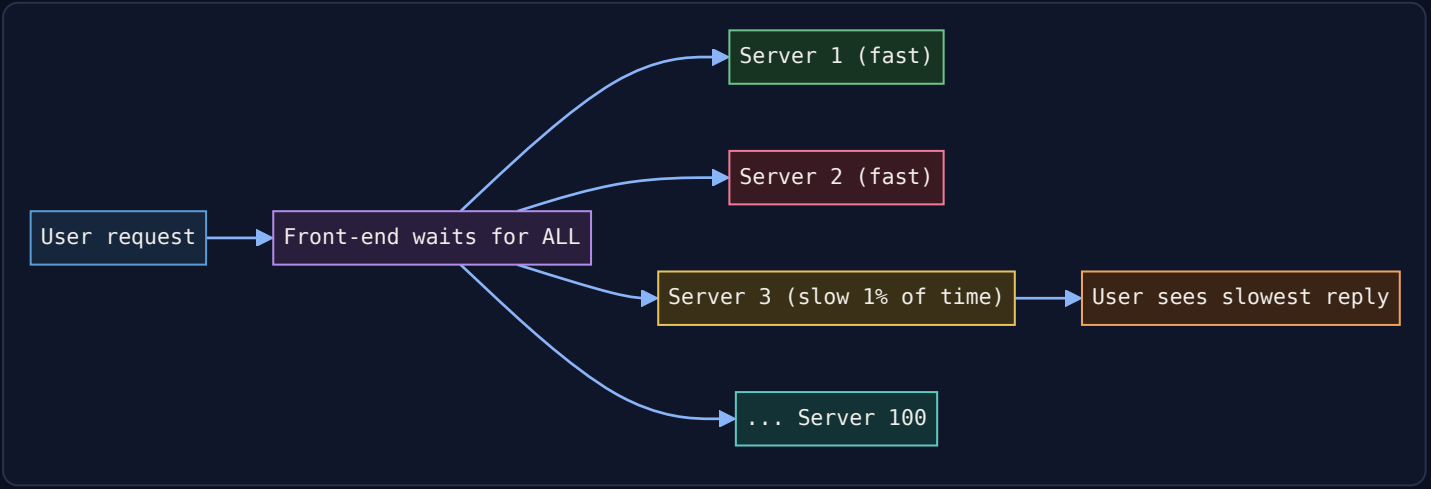
```
10, 10, 10, 10, 10, 10, 10, 10, 10, 1000   (milliseconds)
```

- Mean = 1090 / 10 = 109 ms. No request took 109 ms.
- Median (p50) = 10 ms. The typical user is happy.
- p95 and p99 = 1000 ms (they land on the slowest request). One user in ten waits a full second.

Real latency is almost never a bell curve; it is skewed, with a long tail from garbage collection pauses, cache misses, slow disks, retries and queueing. Percentiles show the shape; averages flatten it.

Why the tail matters more than you think

1. **Heavy users hit the tail most.** A user loading 100 pages a day hits your p99 about once a day — often your most valuable customers (most data, biggest carts).
2. **Fan-out amplifies the tail.** If one user request calls 100 backend servers in parallel and waits for all, it is as slow as the slowest. If each server is slow (1 second) just 1% of the time, the chance that *at least one* of 100 is slow is 1 − 0.99^100 ≈ 63%: a 1-in-100 problem per server becomes a nearly 2-in-3 problem for the user (Google's "The Tail at Scale", Dean and Barroso, 2013).



How it works

Measuring percentiles. Record each latency, sort, and pick the value at the required position (the 990th of 1,000 sorted values ≈ p99). Monitoring systems can't store every value, so they use **histograms** (counts per latency bucket, e.g. 0–10 ms, 10–25 ms…) or sketches such as HDRHistogram or t-digest, which estimate percentiles with small error.

Do not average percentiles. The mean of server A's p99 and server B's p99 is not the p99 of the combined traffic. Merge the histograms and compute from the merged data.

Reducing tail latency:

- **Hedged requests:** if no reply within (say) the p95 time, send the same request to a second replica and use whichever answers first.
- **Timeouts and fewer fan-out calls.**
- Removing sources of pauses: tune garbage collection, avoid shared locks, keep spare capacity so queues stay short.

Common mistakes

- Reporting only the average in dashboards or SLOs.
- Measuring latency only on the server. The client also sees network time, DNS lookups and connection setup.
- **Coordinated omission:** a load-test tool that waits for each slow response before sending the next sends fewer requests during slow periods, so it under-reports the tail.

Real-world examples

- Amazon has long described setting internal service targets at p99.9 rather than averages, so even its heaviest customers get good performance.
- Prometheus, Datadog and Grafana show latency as p50/p95/p99 lines by default.

In the interview

Q: Why use p99 instead of the average for an SLO?

A: The average hides slow outliers and describes no real user. p99 says "99% of requests are at least this fast", capturing the unlucky but common slow cases.

Q: Your service's p99 is fine but users still complain. Why?

A: Each page may call many services or make many requests, so the chance of at least one slow call per page is much higher than 1%. Look at end-to-end page latency and fan-out, not each service in isolation.

TL;DR

Describe latency with percentiles (p50, p95, p99, p999), not averages. Heavy users and fan-out make rare slowness common.

Little's law

What it is

Little's law holds for any stable system where items arrive, wait or get served, and leave:

$$L = \lambda \times W$$

- **L** = average number of items inside the system at once (requests in flight, people in a shop).
- **λ** (lambda) = average arrival rate (requests per second); in a stable system this equals throughput.
- **W** = average time each item spends inside (latency).

If 2 customers arrive per minute and each stays 10 minutes, on average 20 are in the shop — whatever the barista does.

Why it matters

It gives any one of the three numbers from the other two, which is handy for sizing:

- **Concurrent connections or threads needed:** 1,000 requests/s × 0.2 s each = **200 requests in flight**, so at least 200 worker threads or connections (plus headroom), or requests queue.
- **Maximum throughput:** a pool of 20 database connections with 5 ms queries serves at most 20 / 0.005 = **4,000 queries per second**.
- **Why latency hurts throughput:** with fixed concurrency limits, if latency doubles (a slow dependency), maximum throughput halves.

How it works

1. Pick the boundary of your "system": one server, one thread pool, one queue, the whole service.
2. Measure or assume two of the numbers over a period long enough for the system to be stable (arrivals ≈ departures).
3. Compute the third: W = L / λ (latency from in-flight count and rate); λ = L / W (throughput from concurrency limit and latency).

Common mistakes

- Applying it to an unstable system where the queue keeps growing (arrival rate above capacity): there is no steady state and L grows without limit.
- Forgetting units. Convert milliseconds to seconds before multiplying by "per second".

In the interview

Q: A service handles 5,000 requests/s with 40 ms average latency. How many requests are in progress at any moment?

A: L = 5,000 × 0.04 = 200 in flight on average; with 50 instances, about 4 per instance.

Q: A thread pool has 100 threads and each request holds a thread for 250 ms. Max throughput?

A: λ = 100 / 0.25 = 400 requests/s. Beyond that requests queue and latency rises.

TL;DR

Items in the system = arrival rate × time in system. Use it to size thread pools, connection pools and fleets, and to explain why slower dependencies cut throughput.

Latency numbers every engineer should know

What it is

Rough costs of common operations, popularised by Google's Jeff Dean and Peter Norvig. Exact numbers change with hardware (the table uses rounded figures for current machines; the original list gave, for example, 150 us for an SSD random read); what matters is the **order of magnitude** (nanoseconds, microseconds or milliseconds) and the ratios.

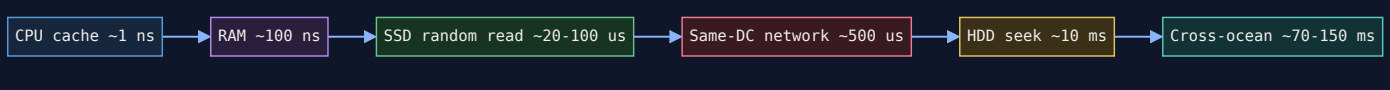
Units: 1 ns (nanosecond) = one billionth of a second; 1 µs (microsecond) = 1,000 ns; 1 ms (millisecond) = 1,000 µs.

Operation	Approximate time	Scaled: if 1 ns were 1 second
CPU L1 cache read	1 ns	1 second
CPU L2 cache read	4 ns	4 seconds
Main memory (RAM) read	100 ns	about 2 minutes
Compress 1 KB (fast algorithm)	2 us	about half an hour
Send 1 KB over 1 Gbps network	10 us	about 3 hours
Read 4 KB at random from NVMe SSD	20-100 us	6 hours to a day
Read 1 MB sequentially from RAM	50-250 us	14 hours to 3 days
Round trip inside one data centre	0.5 ms (500 us)	about 6 days
Read 1 MB sequentially from SSD	0.2-1 ms	2 to 12 days
Hard disk (HDD) seek	10 ms	about 4 months
Read 1 MB sequentially from HDD	5-20 ms	2 to 8 months
Round trip London to New York	70 ms	about 2 years
Round trip California to Europe	150 ms	about 5 years

(A **CPU cache** is tiny, very fast memory inside the processor. **RAM** is the main working memory, lost on power-off. An **SSD** is a solid-state drive with no moving parts; an **HDD** is a spinning magnetic disk with a moving arm, which is why "seeking" to a new position is slow.)

Why it matters

- **RAM is about 100–1,000× faster than SSD for random reads**, and SSD about 100× faster than spinning disk. That's why caches live in RAM.
- **A data-centre round trip (0.5 ms) costs as much as thousands of memory reads.** A loop making 1,000 small database calls spends at least half a second on round trips. Batch them.
- **Crossing an ocean costs 70–150 ms per round trip**, and light is not getting faster. Only moving data closer to users (CDNs, regional servers) fixes it.
- **Sequential reads are far faster than random reads**, especially on HDD, so many storage engines (append-only logs, the log-structured merge trees or LSM-trees of Part 3) write sequentially.



Common mistakes

- Memorising exact numbers. Interviewers want orders of magnitude: "RAM is nanoseconds, SSD microseconds, network and disk milliseconds".
- Treating a remote cache (Redis over the network) as "as fast as memory". The RAM read is 100 ns but the round trip is ~0.5 ms — still far faster than a typical database query, but not free.

In the interview

Q: Why is a cache faster than a database?

A: A cache serves from RAM (nanoseconds plus one network hop); a database may read disk, parse a query, take locks and do more work. Cache hits take well under 1 ms; database queries often several ms or more.

Q: Why does a service that calls another 50 times per request feel slow though each call is "fast"?

A: Each call costs at least a round trip (~0.5 ms in one data centre, more across zones), and sequential calls add up: 50 × 1 ms = 50 ms before real work. Batch or parallelise.

TL;DR

Memory is nanoseconds, SSD tens of microseconds, data-centre network half a millisecond, spinning disks and cross-continent trips tens to hundreds of milliseconds. Keep the hot path at the fast end.

Units, powers of two and data sizes

What it is

Data is measured in **bits** (a single 0 or 1) and **bytes** (8 bits, enough for one English letter).

Unit	Bytes (decimal, used in estimates)	Power of two (exact)	Roughly holds
KB (kilobyte)	1,000	2^10 = 1,024	A short text message or tweet with metadata
MB (megabyte)	1,000,000 (10^6)	2^20 ≈ 1.05 million	A high-quality photo, a minute of MP3
GB (gigabyte)	10^9	2^30 ≈ 1.07 billion	An hour of streamed HD video (a few GB)
TB (terabyte)	10^12	2^40 ≈ 1.1 trillion	A typical laptop drive
PB (petabyte)	10^15	2^50	A large company's data warehouse

Above PB come EB (exabyte, 10^18) and ZB (zettabyte, 10^21). Strictly the power-of-two units have their own names (KiB, MiB, GiB), but estimates round 2^10 to 1,000; the 2–10% error doesn't matter.

How it works

Powers of two to remember:

- 2^10 ≈ 1 thousand (10^3); 2^20 ≈ 1 million (10^6); 2^30 ≈ 1 billion (10^9); 2^40 ≈ 1 trillion (10^12)
- 2^32 ≈ 4.3 billion (the number of IPv4 addresses and the limit of an unsigned 32-bit integer; a 32-bit ID column runs out at about 2 billion signed or 4 billion unsigned)
- 2^64 ≈ 1.8 × 10^19 (plenty for any ID)

Typical sizes:

Item	Rough size
A number (integer)	4 or 8 bytes
A timestamp	8 bytes
A UUID (random unique ID)	16 bytes (36 characters as text)
A short text post with metadata	0.5-1 KB
A web page's HTML	50-100 KB
A compressed photo	200 KB - 2 MB
One minute of HD video	roughly 25-60 MB

Time conversions: 1 day = 86,400 s ≈ **10^5 seconds** (round it); 1 month ≈ 2.6 × 10^6 s; 1 year ≈ 3 × 10^7 s (31.5 million).

Shortcut: 1 million requests per day ≈ 12 per second (10^6 / 86,400 ≈ 11.6), so "X million per day" ≈ "12 × X per second".

Common mistakes

- Mixing bits and bytes: a "1 Gbps" network card moves 125 MB per second at best.
- Carrying too many digits. Round to one or two significant figures and say so.

TL;DR

KB, MB, GB, TB, PB each step up by about 1,000 (2^10). A day is about 10^5 seconds, so 1 million per day is about 12 per second.

Back-of-the-envelope estimation

What it is

A **back-of-the-envelope estimate** is a quick, rough calculation that sizes a problem — requests per second, storage, bandwidth, servers. The goal is the right **order of magnitude** (10, 1,000 or 1,000,000?), not precision.

Why it matters

The estimate decides the architecture. 100 requests per second fits on one server with one database; 100,000 needs many servers, caching, and probably sharded data (split across many databases). 10 GB fits in memory on one machine; 10 PB needs distributed storage.

How it works

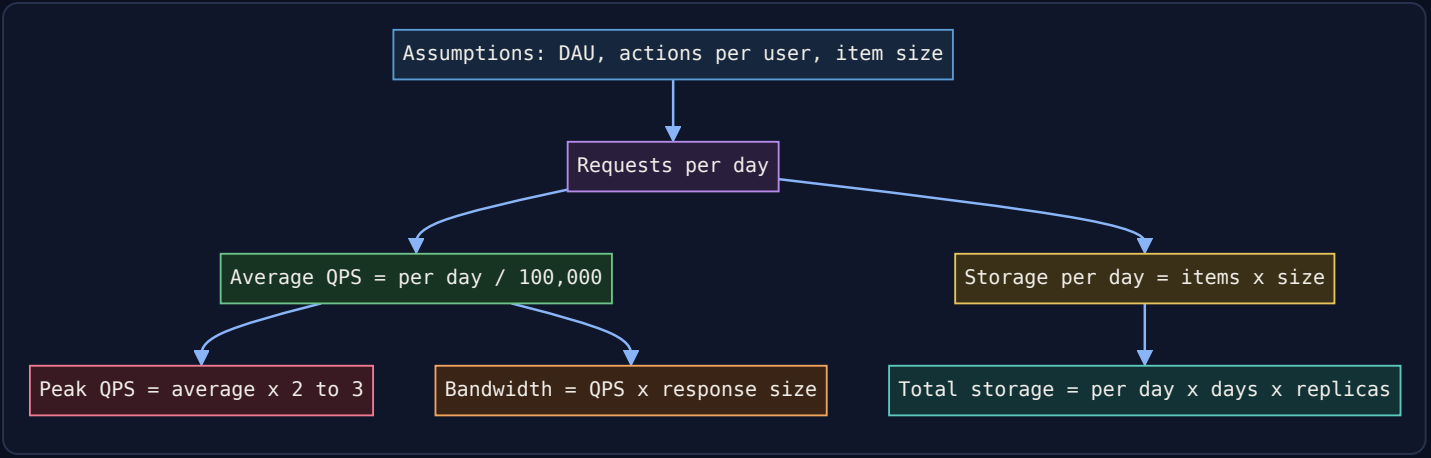
The key terms:

- **DAU (Daily Active Users):** distinct users on a given day; **MAU** the same per month. DAU is usually 20–50% of MAU for social apps.
- **QPS (Queries Per Second)** or **RPS (Requests Per Second):** requests reaching the system each second. **Average QPS** = requests per day / 86,400.
- **Peak factor:** traffic is not flat (evenings, weekends, a sale, a football final). Peak QPS is commonly **2–3×** average, sometimes 5–10× for spiky products.
- **Read/write ratio:** reads per write. Social feeds might be 100:1; logging is almost all writes.
- **Storage per day** = new items per day × item size. **Bandwidth** = QPS × response (or request) size.

Worked example — a photo-sharing app. Assumptions (state them out loud): 10 million DAU; each user views 50 photos and uploads 0.2 per day; average photo 500 KB, metadata 1 KB; keep data 5 years.

1. **Read QPS:** 10M × 50 = 500M views/day; 500M / 10^5 ≈ **5,000 QPS** average, ≈ **15,000** at peak (×3).

2. **Write QPS:** 10M × 0.2 = 2M uploads/day ≈ **20/s** average, ~60 at peak. Read:write ≈ 250:1 — read-heavy, so caching and a CDN matter a lot.
3. **Storage per day:** 2M × 500 KB = **1 TB/day** of photos, plus 2M × 1 KB = 2 GB/day of metadata.
4. **5 years:** 1 TB × 365 × 5 ≈ **1.8 PB** of photos (typically × 3 with replicas); metadata ≈ 3.6 TB. Photos go in object storage; metadata fits in a (sharded) database.
5. **Outgoing bandwidth:** 5,000 views/s × 500 KB = 2.5 GB/s ≈ **20 Gbps** average — a strong argument for a CDN.
6. **Servers:** 15,000 peak QPS at ~1,000 simple requests/s per server is ~15 servers before headroom and redundancy; the worked example below does this step properly.



Rules of thumb for capacity (very rough, simple workloads on modern hardware):

Component	Rough capacity per machine
Web/app server (simple API)	1,000-10,000 requests/s
Relational database (simple indexed queries)	1,000-10,000 queries/s, more for cached reads
Redis/Memcached cache	100,000+ operations/s
Kafka broker	hundreds of MB/s

Worked example: how many servers do we need?

"Divide peak QPS by a rule-of-thumb capacity" is a start, but interviewers like to see the four corrections that turn it into a real number: **measured cost per request, a utilisation ceiling, a concurrency check, and failure headroom**.

Assumptions (say them out loud): an API tier with **30,000 requests/s at peak**. Load testing shows each request uses **4 ms of CPU time** (time the processor actually spends on it) and takes **50 ms end to end** (mostly waiting on the cache and database). Machines have **16 cores**. Responses average **20 KB**. We run in **3 availability zones** and must survive losing one.

Step	Calculation	Result
1. Capacity of one core	1,000 ms ÷ 4 ms CPU per request	250 requests/s per core
2. Capacity of one server at 100%	250 × 16 cores	4,000 requests/s
3. Apply a utilisation ceiling	Keep CPU ≤ 60% so queues stay short (see *Latency, throughput and bandwidth*): 4,000 × 0.6	**2,400 requests/s per server**
4. Servers for peak	30,000 ÷ 2,400 = 12.5, round up	**13 servers**
5. Concurrency check (*Little's law*)	In flight = arrival rate × time in system = 30,000 × 0.05 s = 1,500 requests; ÷ 13 servers	~115 in flight per server: fine for an async server; with one thread per request, that means ~115 threads and a database pool sized to match
6. Survive a zone loss	13 servers must remain after one of 3 zones dies, so each of the 2 survivors needs [13 ÷ 2] = 7	7 per zone × 3 zones = **21 servers**
7. Deployment surge	Rolling deploys take ~10% out of service at a time: 21 × 1.1	**~23 servers**
8. Network check	30,000 × 20 KB = 600 MB/s ≈ 4.8 Gbps in total; ÷ 21 servers	~0.23 Gbps each: far below a 10 Gbps network card, so CPU, not network, is the limit

Reading the answer. The naive "30,000 ÷ 4,000 = 8 servers" was off by about 3×; the utilisation ceiling and the zone-loss rule, both deliberate and defensible, explain nearly all of it. CPU per request decided the number: halving it halves the fleet, which is why profiling is a cost lever.

Repeat per tier. Do the same for each layer with its own bottleneck resource: the cache tier is usually limited by **memory** (working set ÷ usable RAM per node, times replicas), the database by **IOPS or write throughput** per primary (which is what forces sharding), and a streaming tier by **network and disk bandwidth** per broker. The tier whose count grows fastest as traffic grows is the one to design around.

In one sentence for the interview: "At 4 ms of CPU per request a 16-core box does 4,000 requests/s, I keep it at 60%, so 2,400; 30,000 at peak needs 13, and to survive a zone loss across three zones I deploy 7 per zone, 21 total, plus a little for rolling deploys."

Trade-offs

More precise estimates take longer and rarely change the design. Spend 3–5 minutes in an interview, write numbers down, and move on.

Common mistakes

- Forgetting the peak. Systems fall over at peak, not at average.
- Forgetting replication (3 copies triple storage) and metadata, indexes and logs (often +20–50%).
- False precision ("11,574.07 QPS"). Say "about 12,000".
- Doing maths silently. Say each assumption so the interviewer can correct it.

Real-world examples

- WhatsApp said in late 2020 it delivers about 100 billion messages per day: 10^11 / 10^5 ≈ 1 million messages per second on average, an anchor for "very large" chat scale.
- YouTube said in 2019 that more than 500 hours of video are uploaded every minute — about 720,000 hours a day.

In the interview

Q: 100 million DAU, each making 20 requests per day. What QPS should you plan for?

A: 2 billion requests/day ÷ 10^5 = 20,000 QPS average; with a peak factor of 3, plan for about 60,000.

Q: Why bother estimating if the numbers are guesses?

A: The order of magnitude decides the architecture: whether one database is enough, whether data fits in memory, whether you need a CDN. Rough numbers still rule designs in or out.

TL;DR

From DAU and per-user actions, divide per-day counts by about 10^5 for per second, multiply by 2–3 for peak, and compute storage and bandwidth from item sizes. Round aggressively and state every assumption.

Availability and the nines

What it is

Availability is the fraction of time a system is up and doing its job (answering requests correctly), usually written with many 9s: "three nines" (99.9%), "four nines" (99.99%).

Availability = uptime / (uptime + downtime), or, measured by requests, successful requests / total valid requests. A 24-hour shop closed by surprise for 9 hours in a year was available about 99.9% of the time.

The nines table

(Year = 365 days, month = 30 days.)

Availability	Name	Downtime per year	Downtime per month	Downtime per day
99%	two nines	3.65 days	7.2 hours	14.4 minutes
99.9%	three nines	8.76 hours	43.2 minutes	1.44 minutes
99.95%	three and a half	4.38 hours	21.6 minutes	43 seconds
99.99%	four nines	52.6 minutes	4.32 minutes	8.6 seconds
99.999%	five nines	5.26 minutes	26 seconds	0.86 seconds

Each extra nine cuts allowed downtime by 10× and usually costs much more than 10× the engineering effort.

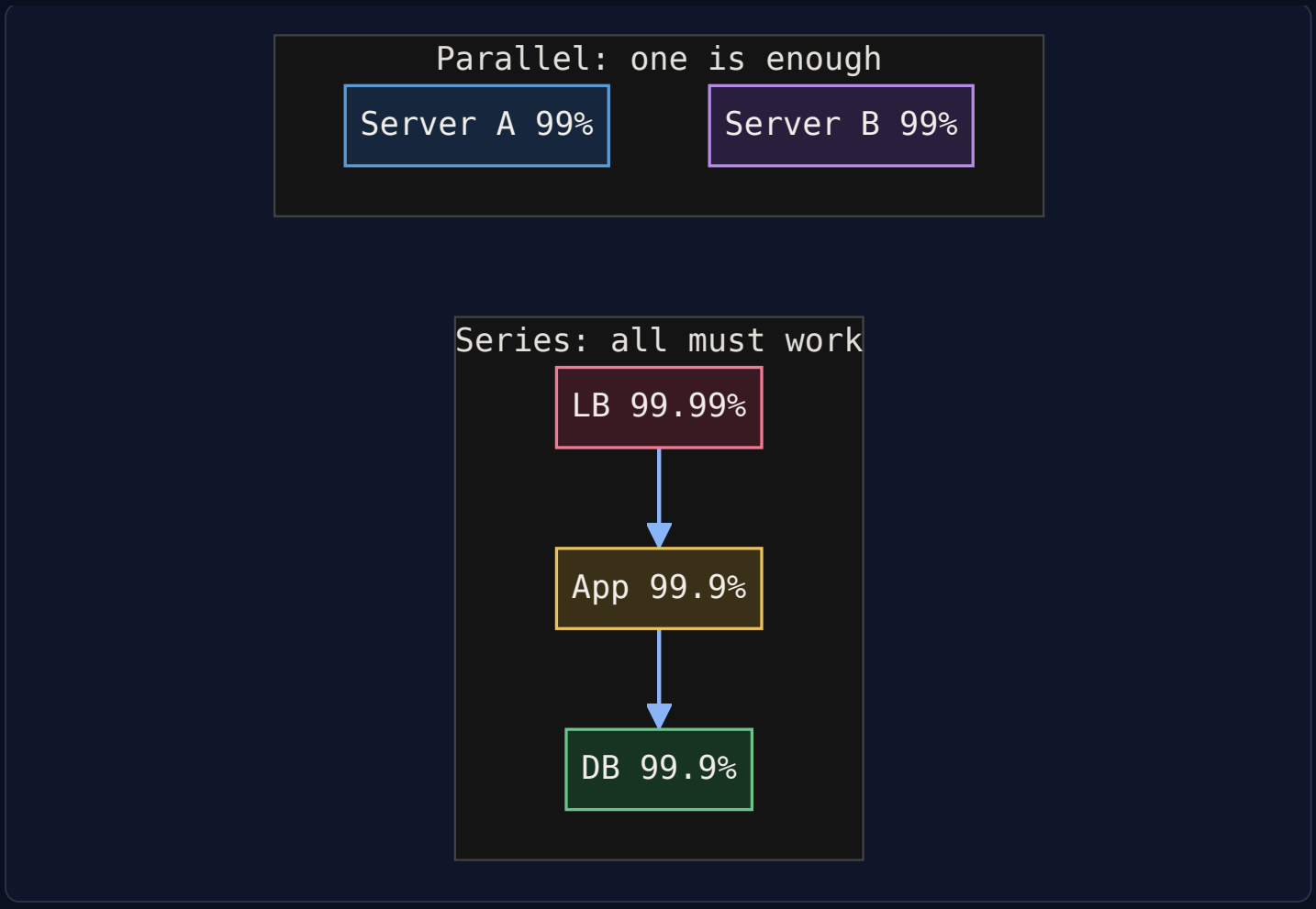
Why it matters

- Downtime costs money and trust: a shop down for an hour on Black Friday loses that hour's sales permanently.
- The target decides the design. 99% survives manual recovery ("someone restarts the server in the morning"). 99.99% leaves 4 minutes a month — less than a human needs to wake up and log in — so recovery must be **automatic**.

How it works

Parts in series (all must work). A request through a load balancer (99.99%), an app server (99.9%) and a database (99.9%): 0.9999 × 0.999 × 0.999 ≈ **99.79%**. The chain is weaker than its weakest link; every dependency added in series lowers availability.

Parts in parallel (any one is enough). Two independent servers, each 99%, are down only when both are: 1 − (0.01 × 0.01) = **99.99%**. Redundancy turns two "two nines" machines into a "four nines" pair — if failures are independent (often not fully: both in one rack lose power together).



Trade-offs

Higher availability needs redundancy (more machines, more data centres), automation, testing and on-call staff. 99.9% → 99.99% might mean multiple data centres with automatic failover, roughly doubling infrastructure and adding complexity. Many internal tools are fine at 99.5%.

Common mistakes

- Promising five nines without the budget. Even big cloud providers' SLAs for single services are usually 99.9%–99.99%.
- Forgetting dependencies. You can't be more available than what you need in series (unless you design fallbacks).
- Counting planned maintenance as "not downtime". Users don't care why it's down.

Real-world examples

- Amazon S3 Standard is designed for 99.99% availability; its SLA promises 99.9% per month (service credits below that).
- Telephone networks were historically engineered for five nines.

In the interview

Q: How much downtime per month does 99.9% allow?

A: About 43 minutes; 99.99% about 4.3 minutes.

Q: How do you get higher availability than any single machine provides?

A: Redundant copies in parallel with automatic failover, no single points of failure, copies spread across separate failure domains (racks, zones, regions).

TL;DR

Availability is the share of time a system works. Each extra nine is ten times less downtime: 99.9% ≈ 43 minutes a month, 99.99% ≈ 4. Series dependencies multiply down; parallel redundancy multiplies up.

SLA, SLO, SLI and error budgets

What it is

From most technical to most legal:

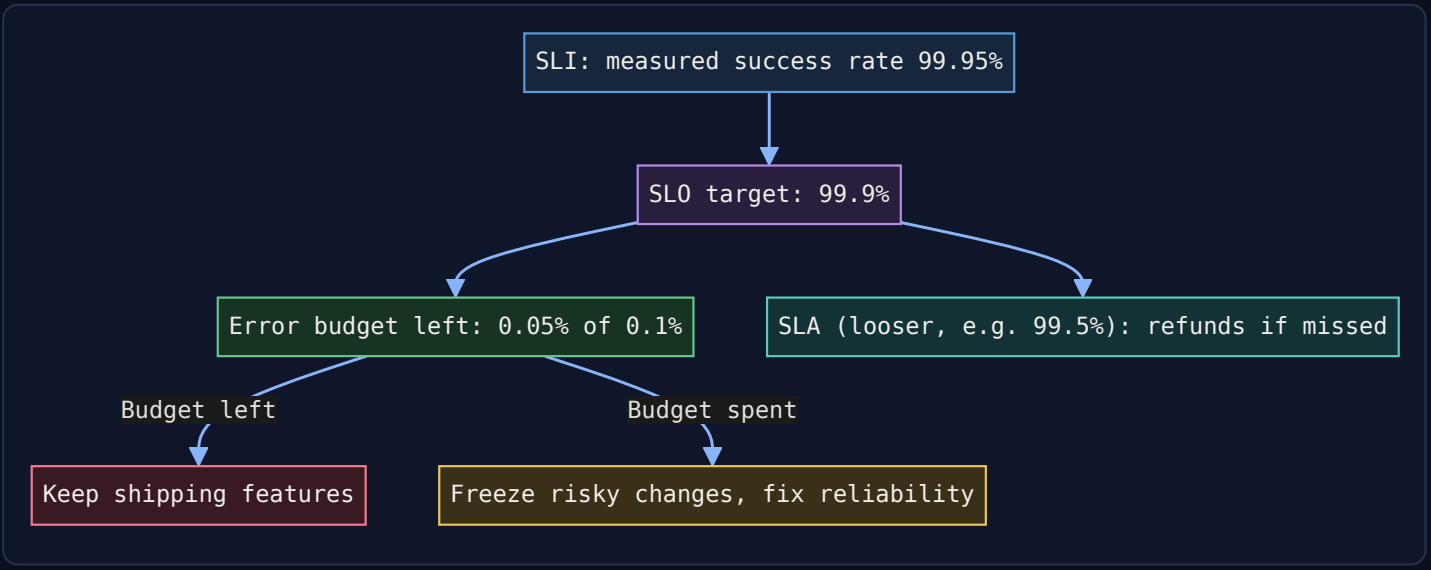
- **SLI (Service Level Indicator):** a *measurement* of how the service is doing, e.g. "percentage of HTTP requests that succeeded in under 300 ms over the last 30 days."
- **SLO (Service Level Objective):** an internal *target* for an SLI, e.g. "99.9% of requests succeed in under 300 ms, measured over 30 days."
- **SLA (Service Level Agreement):** a *contract* with customers saying what happens if a target is missed, usually refunds or service credits, e.g. "below 99.9% monthly availability, customers get 10% of their bill back."

(Pizza shop: the SLI is how long deliveries actually take, the SLO the manager's "95% within 30 minutes", the SLA the printed "over 45 minutes and it's free".) The SLA is normally looser than the SLO, so the team gets warning (missing the SLO) before the company has to pay (missing the SLA).

Error budgets

An **error budget** is the unreliability the SLO allows: with a 99.9% SLO it is 0.1% — in a month with 100 million requests up to 100,000 may fail (or about 43 minutes of full outage).

The budget turns reliability into a shared decision: while budget remains, the team ships fast and takes risks; when it is spent, releases slow and reliability work takes priority until it recovers. The idea comes from Google's **Site Reliability Engineering (SRE)**, the discipline of running production systems with software engineering methods.



How it works

1. Choose SLIs users feel: availability (success rate), latency (p99 under X ms), freshness (data no older than Y), durability.
2. Set the SLO slightly below what users actually need, never "100%" (impossible, and leaves no room to change anything).
3. Measure over a window (often 28 or 30 days, rolling).
4. Alert when the budget burns too fast (burn-rate alerts, Part 5).

Common mistakes

- Setting the SLO to 100%: any deployment risk then "violates" it.
- Choosing SLIs users don't feel (CPU usage) instead of ones they do (failed checkouts).
- Making the SLA as tight as the SLO, leaving no warning margin.

In the interview

Q: What's the difference between an SLO and an SLA?

A: An SLO is an internal target for a measured indicator; an SLA is an external contract with penalties, usually looser.

Q: What is an error budget used for?

A: It quantifies acceptable failure (1 − SLO). Teams spend it on releases and experiments; when it's gone, they prioritise reliability.

TL;DR

SLI is the measurement, SLO the target, SLA the contract with penalties. The error budget (1 − SLO) is how much failure you can "spend" before you must slow down and fix reliability.

MTBF, MTTR and measuring failure

What it is

- **MTBF (Mean Time Between Failures):** average time a system runs before it fails. Higher is better.
- **MTTR (Mean Time To Repair / Recover):** average time from a failure until service is restored. Lower is better.
- **MTTD (Mean Time To Detect):** how long before anyone notices. **MTTF (Mean Time To Failure):** used for parts that are replaced rather than repaired (like disks).

How it works

```
Availability = MTBF / (MTBF + MTTR)
```

A server failing every 1,000 hours on average and taking 1 hour to recover: 1,000 / 1,001 ≈ 99.9%.

Two levers: **increase MTBF** (better hardware, testing, careful deployments) or **decrease MTTR** (fast detection by monitoring and alerts, automatic failover, quick rollback, runbooks). In large distributed systems failures are constant, so cutting MTTR is usually cheaper: automatic 10-second recovery instead of 1 hour gives the same failure rate about 99.9997% availability.

Common mistakes

- Only trying to prevent failures. You can't prevent them all; plan to recover fast.
- Ignoring detection time. An outage nobody notices for 30 minutes has at least 30 minutes of MTTR.

In the interview

Q: Which is easier to improve at scale, MTBF or MTTR?

A: Usually MTTR. With thousands of machines something always fails, so automatic detection and failover (seconds) buy more availability than making each machine fail less often.

TL;DR

MTBF is time between failures, MTTR time to recover. Availability ≈ MTBF / (MTBF + MTTR); in practice, shrink MTTR with monitoring and automation.

Reliability, availability and durability

What it is

- **Reliability:** the system keeps doing the *right thing* over time, even when things go wrong — correct results, no corrupted data.
- **Availability:** the system is *up and responding* when you ask. It can be available but unreliable (answers quickly, with wrong data).
- **Durability:** saved data *stays saved*, even if machines crash or disks die. A system can be unavailable but durable (down for an hour, no data lost).

(At a bank: the cash machine works when you walk up; it gives the right amount and debits the right account; your balance survives a flood.)

Property	Question it answers	Typical measure
Availability	Can I use it right now?	% uptime, % successful requests
Reliability	Does it work correctly over time?	Failure rate, MTBF, correctness
Durability	Will my saved data still exist later?	Probability of losing an object per year

How it works

- Availability: redundancy and failover (below).
- Durability: multiple copies on separate disks, machines and locations; writing to disk before confirming (write-ahead logs, Part 3); backups.
- Reliability is the umbrella: testing, validation, handling faults gracefully, and both of the above.

Real-world examples

- Amazon S3 Standard is designed for 99.999999999% ("eleven nines") yearly durability per object — store ten million objects and on average one goes missing roughly once in ten thousand years — but only 99.99% availability (SLA credits below 99.9%). Durability and availability are separate promises.

Common mistakes

- Saying "highly available" when you mean "doesn't lose data". Say durable.
- Assuming replication equals backup. A bug that deletes data is faithfully replicated to every copy; backups protect against that.

In the interview

Q: Can a system be durable but not available?

A: Yes. A database cluster down for maintenance can't be read (unavailable), but all data is safely on disk (durable).

TL;DR

Availability = can I use it now; reliability = does it work correctly over time; durability = will saved data survive. Different techniques improve each, and they are promised separately.

Faults, failures and fault tolerance

What it is

- A **fault** is one part going wrong: a disk dies, a process crashes, a cable is cut, a bug returns a wrong value.
- A **failure** is the *whole system* no longer providing its service to users.

Faults are inevitable; failures are what we prevent. **Fault tolerance** (resilience) is continuing to work, perhaps in a reduced way, when some parts have faults — like a twin-engine plane landing on one engine.

Types

Fault type	Example	Typical defence
Hardware	Disk dies, power supply fails	Redundant machines, replicated data
Software	Bug, memory leak, crash on bad input	Testing, gradual rollouts, restarts
Network	Packets lost, link cut, partition	Timeouts, retries, replicas in other zones
Human	Wrong config pushed, table deleted	Reviews, automation, backups, rollback
Dependency	Third-party API is slow or down	Timeouts, fallbacks, circuit breakers

Failure models. Faults also differ by *how the broken part behaves*. Distributed-systems research names a spectrum of **failure models** (formal assumptions about what a faulty part may do); roughly, each row below permits more misbehaviour than the one above, so it is harder to tolerate:

Model	What the faulty part does	Can the others tell?	Typical defence and cost	Example
Fail-stop	Halts, and every other part reliably learns that it halted	Yes, by definition	Simplest: with guaranteed detection, f+1 copies can tolerate f failures	An idealisation; approximated by fencing or "shoot the other node" power switches
Crash (crash-stop)	Halts and sends nothing more, but nobody is told	No: on a network, a crashed node looks exactly like a slow one	Timeouts plus majority quorums: consensus needs 2f+1 nodes to tolerate f crashes	Process killed, kernel panic
Crash-recovery	Crashes, later restarts with only what it saved to disk	Not reliably	Write state durably before acknowledging (write-ahead log), rejoin and catch up	Server reboot, container restart
Omission	Keeps running but drops some messages it should send or receive	Only through missing acknowledgements	Acknowledgements, retries, idempotent handlers	Full network buffer, lossy link
Timing (performance)	Gives correct answers but outside the expected time bound, or its clock drifts	Indistinguishable from omission until the answer arrives	Timeouts, deadlines, leases with clock-error margins, hedged requests	Long garbage-collection pause, overloaded disk
Byzantine (arbitrary)	Anything at all: wrong, contradictory or deliberately malicious answers	No	3f+1 replicas to agree despite f liars; signatures and checksums	Memory corruption, a compromised node, a buggy replica

Most systems inside one company's data centres assume **crash-recovery plus omission plus timing faults**, and add checksums to catch the one Byzantine-like fault that really happens: corrupted data. Blockchains and some aerospace systems design for full Byzantine faults. The model you assume decides how many replicas you need and which algorithms are correct, so state it out loud.

How it works

Detect the fault (health checks, timeouts, heartbeats), **isolate** it so it doesn't spread (timeouts, bulkheads), **recover** (route around it, restart, fail over to a copy). Parts 4 and 5 cover the tools: retries, circuit breakers, replication, consensus.

Common mistakes

- Treating "slow" as fine. A dependency taking 30 seconds to fail can tie up all your threads; a fast error is often better.
- Assuming faults are independent. A bad deployment or shared power supply can break many "redundant" copies at once (**correlated failure**).

Real-world examples

- Netflix's Chaos Monkey randomly shuts down production servers so engineers must build services that tolerate instance faults.
- Large cloud outages are often a configuration change pushed everywhere at once: human and software faults matter as much as hardware.

In the interview

Q: What is the difference between a fault and a failure?

A: A fault is a component misbehaving; a failure is the system as a whole not serving users. Fault tolerance stops faults becoming failures.

Q: Why does Raft need 3 nodes to survive 1 failure, while a Byzantine-tolerant protocol needs 4?

A: Raft assumes crash faults: a live majority (2 of 3) always overlaps any other majority, so 2f+1 suffices. With Byzantine faults a node can lie, so honest nodes must outnumber liars inside every quorum, which takes 3f+1 nodes (Lamport, Shostak and Pease, 1982).

TL;DR

Faults happen to parts; failures happen to the system. Name your failure model (crash, omission, timing, Byzantine), because it sets how many replicas you need; then detect, isolate and recover so faults don't become user-visible failures.

Sources: R. Schlichting and F. Schneider, "Fail-stop processors", *ACM TOCS* 1(3), 1983; V. Hadzilacos and S. Toueg, "A modular approach to fault-tolerant broadcasts and related problems", Cornell TR 94-1425, 1994; L. Lamport, R. Shostak, M. Pease, "The Byzantine Generals Problem", *ACM TOPLAS* 4(3), 1982; D. Ongaro and J. Ousterhout, "In Search of an Understandable Consensus Algorithm (Raft)", USENIX ATC 2014.

Redundancy, single points of failure and failover

What it is

- **Redundancy:** more than one of something (servers, data copies, network links, power supplies), so a spare can take over.
- A **single point of failure (SPOF)** is any part whose failure alone brings down the whole system: one database, one load balancer, one DNS provider, one data centre, even one engineer who knows the password.
- **Failover** is switching work from a failed component to a standby (as a hospital switches to its backup generators within seconds of a blackout).

How it works

Finding SPOFs: ask of every box and arrow, "if this disappears, what happens?" If "everything stops", it's a SPOF.

Removing them:

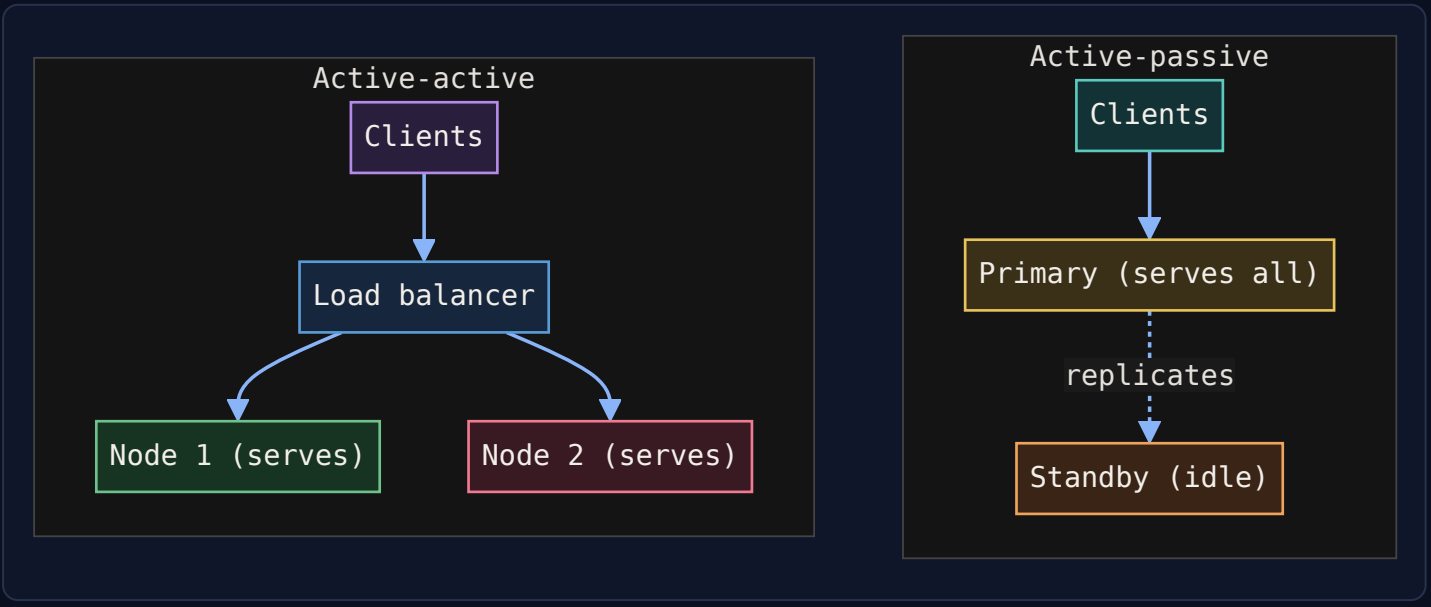
1. Run at least two of every critical component in separate **failure domains**: different racks, **availability zones** (separate data centres in one region, with their own power and networking), or regions.
2. Put something in front that routes around dead copies (a load balancer, DNS, a client library).
3. Add **health checks** — periodic "are you alive?" tests — so dead copies are detected.
4. Automate failover so it happens in seconds, not when a human notices.

Types

Active-passive (primary-standby): one copy handles all traffic; an up-to-date standby takes over when the primary fails. Simple, but the standby is idle capacity and failover takes seconds to minutes.

Active-active: all copies serve traffic at once; if one fails the others absorb its share. Better hardware use and faster recovery, but harder for stateful parts like databases (two copies accepting writes can conflict).

Cold, warm and hot standby: **cold** — exists but must be started and loaded with data (minutes to hours); **warm** — running and partially up to date (minutes); **hot** — running, fully in sync, ready immediately (seconds).



N+1 and N+2: if N servers handle peak load, run N+1 (survive one failure) or N+2 (survive one failure during maintenance of another).

Trade-offs

- Redundancy costs money: standby machines, extra copies of data.
- Failover is risky. If the system wrongly decides a slow or cut-off primary is dead, two primaries may both accept writes: **split brain**. Parts 3 and 4 cover prevention (quorums, fencing).
- Failover must be tested regularly; untested failover often fails when needed.

Common mistakes

- Making app servers redundant but leaving one database or one load balancer.
- Putting all "redundant" copies in the same rack or zone.
- Forgetting the standby needs capacity for the full load.

Real-world examples

- AWS recommends spreading instances across at least two availability zones; Amazon RDS Multi-AZ keeps a synchronous standby in another zone and fails over automatically, typically within a minute or two.
- Load balancers are deployed in pairs (or as a managed, internally redundant service) so they are not the SPOF.

In the interview

Q: How do you find single points of failure in a design?

A: Go component by component (including DNS, load balancers, databases, config stores, third-party APIs) and ask what happens if it disappears. Anything that stops the whole service needs a redundant copy in a separate failure domain and automatic failover.

Q: Active-passive or active-active?

A: Active-passive is simpler and common for single-writer databases. Active-active uses capacity better, recovers faster and suits stateless services, but needs conflict handling for stateful data.

TL;DR

Redundancy means spares; a SPOF is any part with no spare; failover moves work to the spare. Duplicate critical parts across failure domains and automate failover, while guarding against split brain.

Scalability

What it is

Scalability is handling more load (users, requests, data) by adding resources, without a redesign and without performance collapsing: cost and effort grow roughly in proportion to load instead of hitting a wall.

Load is described by **load parameters**: requests per second, concurrent users, read/write ratio, data size, followers per user, and so on. "Scalable" only makes sense relative to one of them: "we can scale to 10× the requests per second by adding servers". (A restaurant that serves more diners by adding tables and cooks scales; one whose tiny kitchen tops out at 50 meals an hour does not.)

Why it matters

Successful products grow, sometimes overnight. A system needing a rewrite at each 10× step slows the business at exactly the wrong moment.

How it works

Dimensions to scale along:

- **Load scalability:** more requests per second.
- **Data scalability:** more stored data.
- **Geographic scalability:** users worldwide needing low latency everywhere.
- **Organisational scalability:** more engineers working without stepping on each other (one reason teams split into microservices, Part 5).

Core techniques, explained later:

1. Scale out stateless app servers behind a load balancer (next section).
2. Cache hot data (Part 4).
3. Copy data to read replicas for read-heavy loads (Part 3).
4. Split data across shards for write-heavy or huge data (Part 3).
5. Move slow work to queues and workers (Part 4).
6. Put static content on a CDN (Part 2).

Common mistakes

- Calling a system "scalable" without saying which load parameter and by how much.
- Scaling the wrong part: more app servers do nothing if the database is the bottleneck.
- Building for 1,000× growth on day one. Make the next 10× easy, then re-evaluate.

In the interview

Q: What does it mean for a system to be scalable?

A: It handles growing load, stated in concrete numbers, by adding resources roughly in proportion, while staying within performance targets.

TL;DR

Scalability is handling more load by adding resources without a redesign. Always say what load (QPS, data size, users) and by how much.

Vertical and horizontal scaling

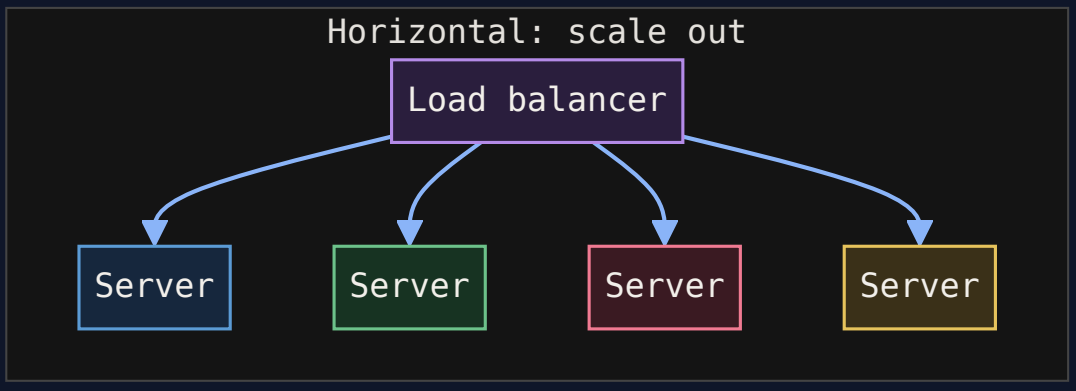
What it is

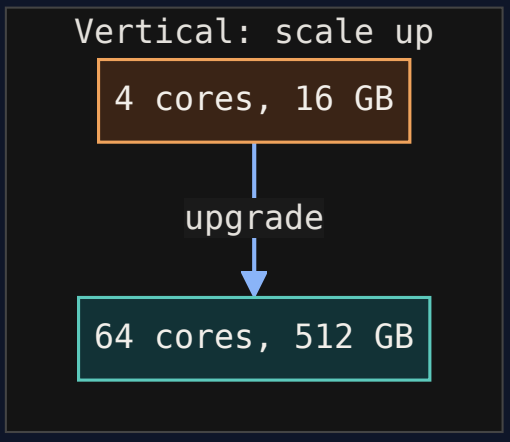
- **Vertical scaling (scale up):** make one machine bigger: more CPU cores, RAM, faster disks (a bigger bus).
- **Horizontal scaling (scale out):** add more machines and spread the work across them (more buses).

How it works

Vertical scaling is usually a configuration change in the cloud: stop the machine, pick a larger size, start it; no code changes. Cloud machines go very large: AWS offers instances with hundreds of cores and multiple terabytes of RAM.

Horizontal scaling needs divisible work: stateless app servers are easy (add copies behind a load balancer); databases are harder — reads can go to replicas, but spreading writes needs sharding (each machine holds part of the data).





Trade-offs

	Vertical (up)	Horizontal (out)
Simplicity	Very simple, no code change	Needs load balancing, stateless design or sharding
Limit	Hard ceiling: the biggest machine	Practically unlimited
Fault tolerance	One machine is still a SPOF	Losing one node loses only a fraction
Cost curve	Big machines cost disproportionately more	Many cheap commodity machines
Downtime to resize	Often needs a restart	Add or remove nodes live
Consistency	Easy (one copy of data)	Harder (data spread across machines)

Common mistakes

- Dismissing vertical scaling. One large database server handles a lot; scaling up first is often the right, cheap answer.
- Scaling out a stateful service without dealing with its state (next sections).

Real-world examples

- Stack Overflow famously served a very large site from a few powerful servers (in 2016: 11 web servers and 4 SQL Server machines, plus a few Redis, search and tag-engine boxes) — vertical scaling done well.
- Google, Meta and Amazon are each estimated to run millions of commodity servers: horizontal scaling at the extreme.

In the interview

Q: When would you choose vertical scaling?

A: Early on, or for components hard to distribute (a relational database's primary). It's simple and buys time. Go horizontal when nearing the largest machine, needing fault tolerance, or when costs climb steeply.

Q: What makes horizontal scaling hard?

A: State. Stateless servers scale out trivially; anything storing data must be replicated or partitioned, bringing consistency and coordination problems.

TL;DR

Scale up = bigger machine (simple, limited, still a SPOF). Scale out = more machines (unlimited and fault tolerant, but needs stateless design or data partitioning). Most real systems do both.

Elasticity

What it is

Elasticity is adding and removing resources *automatically and quickly* as load changes, so you pay for roughly what you use. Scalability says "we *can* grow"; elasticity says "we grow and shrink by ourselves, in minutes" — like a supermarket opening checkouts when queues lengthen.

How it works

Watch a metric (CPU, requests per second, queue length); past a threshold (CPU above 70% for 5 minutes) launch servers, below another (30%) remove some, while the load balancer starts and stops sending them traffic. This is **autoscaling** (*Autoscaling*, Part 5: reactive, scheduled, predictive), provided by cloud

platforms (AWS Auto Scaling groups, Kubernetes Horizontal Pod Autoscaler) and **serverless** platforms (AWS Lambda, which runs your code per request and scales to zero).

Trade-offs

- New servers take time to start (seconds for containers, minutes for virtual machines); a sudden spike can outrun capacity.
- Only stateless parts are easily elastic; databases scale much more slowly.
- Scaling down too eagerly causes "flapping" (adding and removing repeatedly).

In the interview

Q: What's the difference between scalability and elasticity?

A: Scalability is the capability to handle more load by adding resources; elasticity is doing it automatically, in both directions, as demand changes.

TL;DR

Elasticity is automatic, fast scaling up and down with demand, mostly for stateless parts, so you pay only for what you need.

Stateless and stateful services

What it is

State is data a program remembers between requests: who is logged in, a shopping cart, the contents of a database.

- A **stateless service** keeps no client-specific data in its own memory between requests. Every request carries (or points to) everything needed, so any copy can handle any request — like a call centre where any agent can pull up your file.
- A **stateful service** remembers data locally between requests, so a client or piece of data is tied to a particular server — like a family doctor who keeps your history in their head: great until they're on holiday.

Why it matters

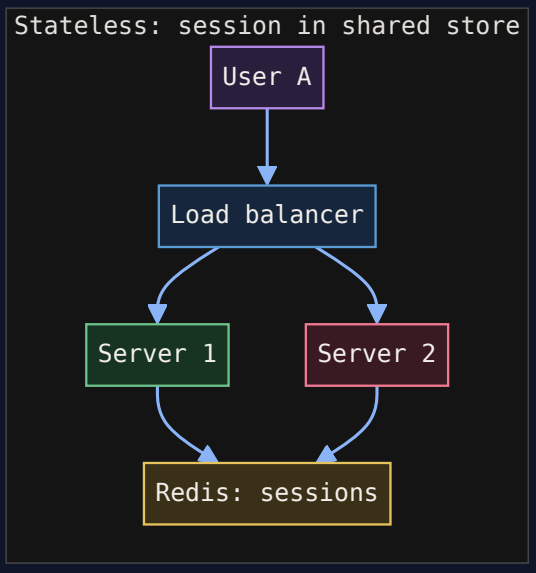
Stateless services are easy to scale out, load balance, restart and replace: kill any copy and nothing is lost. Hence the standard advice: "**make app servers stateless and push state into dedicated stores**" (databases, caches, object storage) designed for replication.

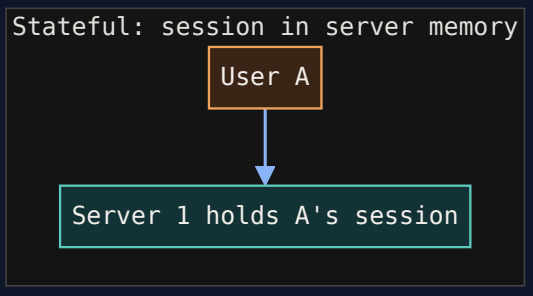
How it works

To make a web app stateless:

1. Store login sessions in a shared store (Redis) or a signed token the client sends with each request (a JWT, JSON Web Token, Part 5), not in server memory.
2. Store uploaded files in object storage, not the server's local disk.
3. Keep shopping carts and similar data in a database or cache.

Then the load balancer can send each request to any server.





If a service stays stateful (a game server holding a live match, a chat server holding open connections), the load balancer must send each client to the same server — **sticky sessions** (Part 2) — and you must plan for that server dying.

Trade-offs

- Stateless servers need an extra network hop to fetch state (usually under a millisecond to a cache).
- Stateful designs can be faster (data already in memory) and are sometimes necessary (databases, real-time games, WebSocket servers), but are harder to scale and recover.

Common mistakes

- Keeping sessions in local memory, then adding a second server and finding users randomly logged out.
- Writing uploaded files to one server's local disk.
- Calling a system stateless when state just moved elsewhere; the state store now needs scaling and redundancy too.

In the interview

Q: Why should application servers be stateless?

A: So any instance can handle any request: add or remove instances freely, balance evenly, and lose one without losing user data. State lives in stores built for durability and replication.

TL;DR

Stateless services remember nothing between requests and scale easily; stateful ones hold data and are harder to scale and recover. Keep app servers stateless and put state in databases and caches.

Performance vs scalability

What it is

- **Performance** is how fast the system is for *one* request or user at the current load: low latency, efficient resource use.
- **Scalability** is how well performance holds up as load *grows*.

Diagnostic: send one request to an otherwise idle system. If even that is slow, it's **performance**; if it's quick but things drag as traffic piles up, it's **scalability**. (A sports car is fast for one person but seats two; a train is slower off the mark but carries a thousand.)

Trade-offs

Some changes help one and hurt the other: a network hop to a shared cache slightly hurts single-request performance but lets you scale out; keeping everything in one process's memory is fast but caps you at one machine.

In the interview

Q: The site is slow even at 3 am with almost no traffic. Performance or scalability?

A: Performance: something is slow per request (an unindexed query, a slow external call, a big page). Adding servers won't help; profile the request.

TL;DR

Performance = fast for one; scalability = stays fast for many. Diagnose which before fixing.

Concurrency vs parallelism

What it is

- **Concurrency:** several tasks in progress during the same period, the program switching between them whenever one waits. It describes how work is *organised*.
- **Parallelism:** several tasks running at the same instant on multiple CPU cores or machines. It describes how work is *executed*.

Rob Pike (Go co-creator): concurrency is *dealing with* lots of things at once, parallelism is *doing* lots of things at once. One barista interleaving orders while the milk steams is concurrent; three baristas each making a drink at once is parallel. Concurrency works on a single core (by switching); parallelism needs multiple cores.

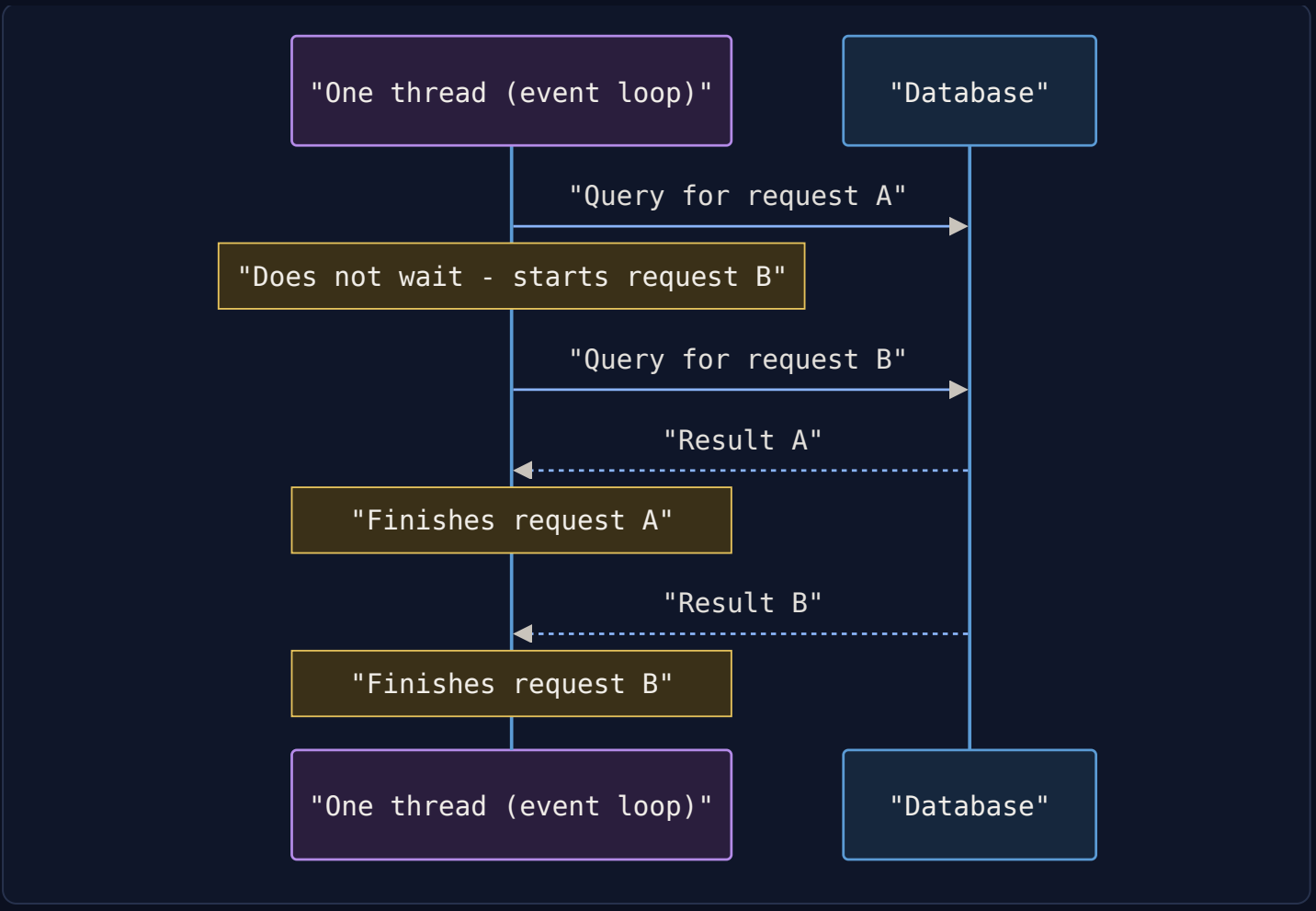
Why it matters

Most web servers spend their time **waiting** (network, database, disk), not computing. Concurrency lets one server keep thousands of requests in progress while each waits. Parallelism speeds up **CPU-heavy** work (image processing, encryption, analytics) by splitting it across cores.

How it works

Ways to achieve concurrency:

- **Processes:** separate programs with their own memory. Heavy but isolated.
- **Threads:** multiple flows of execution inside one process sharing memory. Lighter, but shared memory needs locks.
- **Event loop / async I/O:** one thread handles many requests by never blocking: start an I/O operation, handle something else, return when the result is ready (Node.js, NGINX, Python's asyncio).
- **Lightweight threads / coroutines:** cheap tasks scheduled by the language runtime (Go's goroutines, Java's virtual threads): simple blocking-style code with event-loop efficiency.



Concurrency hazards when tasks share data: a **race condition** (the result depends on timing — two requests both read a balance of 100 and both write 90 after a withdrawal of 10, losing one withdrawal), **deadlock** (two tasks each wait forever for a lock the other holds) and contention (tasks queue for the same lock). Locks, atomic operations and database transactions (Part 3) address them.

Amdahl's law: the speed-up from parallelism is limited by the part that must run serially. If 10% of a job is serial, even infinite cores give at most 10×.

Common mistakes

- Adding threads to a CPU-bound task beyond the number of cores: just switching overhead.
- Assuming parallel code is automatically faster; coordination and locking can make it slower.

In the interview

Q: Is a single-threaded Node.js server concurrent?

A: Yes: an event loop and non-blocking I/O handle many requests concurrently. It isn't parallel for JavaScript on one core; run several processes to use multiple cores.

TL;DR

Concurrency is juggling many tasks (great for waiting on I/O); parallelism is doing several at the same instant on multiple cores (great for CPU-heavy work). Shared data under concurrency needs care to avoid races and deadlocks.

Bottlenecks: CPU, memory, disk and network

What it is

A **bottleneck** is the part that limits overall capacity, like a bottle's narrow neck; improving anything else won't help until it is fixed. Every machine has four basic resources that can become one:

Resource	What it limits	Signs it's the bottleneck	Usual fixes
CPU	Computation (encryption, parsing, compression)	CPU near 100%, latency rises with load	Optimise code, cache results, add cores or servers
Memory (RAM)	How much data fits in fast storage	Swapping to disk, out-of-memory crashes, cache misses	More RAM, smaller data, evict wisely
Disk	Reading and writing stored data	High disk wait time, IOPS or MB/s at limit	SSDs, caching, sequential writes, sharding
Network	Moving data between machines	Link saturated, packet loss, high RTT	Compression, CDNs, fewer calls, closer servers

How it works

Finding the bottleneck:

1. Measure each resource under load (dashboards, `top`, `iostat`, cloud metrics).
2. Look for the one at or near its limit, or queues building in front of something (full thread pools, exhausted database connections, lock waits).
3. Fix it and measure again. The bottleneck moves: fix the database and the network may be next.

Non-hardware bottlenecks are just as common: a single lock everybody waits for, a too-small connection pool, one "hot" row every request updates, an external API's rate limit.

A task is **CPU-bound** when limited by computation and **I/O-bound** when limited by waiting for disk or network. Most web services are I/O-bound.

IOPS and disk performance

IOPS (Input/Output Operations Per Second) is how many separate read or write operations a storage device performs per second; it matters most for **random** access (many small reads at different locations), such as database lookups. Disks have two separate limits: **IOPS** (small random reads/writes) and **throughput** in MB/s (large sequential transfers such as streaming a video or a backup). A disk can hit one limit without the other.

Device	Rough random IOPS	Rough sequential throughput
Spinning HDD (7,200 rpm)	about 75-150	100-250 MB/s
SATA SSD	tens of thousands	about 500 MB/s
NVMe SSD	hundreds of thousands to about 1 million	several GB/s
Cloud block disk (e.g. AWS gp3 baseline)	3,000 (can buy more)	125 MB/s (can buy more)

An HDD's IOPS is low because each random read must physically move the arm and wait for the platter (roughly 5–10 ms); hence databases moved to SSDs, and HDD systems favour sequential writes. Example: 20,000 random cache-missing reads per second would need over 100 HDDs (~150 IOPS each); one NVMe SSD handles it comfortably.

Common mistakes

- Guessing the bottleneck instead of measuring.
- Looking only at averages: a resource at 50% average can be at 100% for the busy minute that matters.
- Forgetting cloud disks have provisioned IOPS limits: a database can be throttled by its disk while CPU looks idle.

In the interview

Q: Latency rises under load but CPU is only at 30%. Where would you look?

A: Another bottleneck: disk IOPS, network, a full connection pool, a lock, or a slow dependency. Look for where requests queue.

Q: What's the difference between IOPS and disk throughput?

A: IOPS counts operations per second (key for small random reads like database lookups); throughput counts bytes per second (key for large sequential transfers).

TL;DR

The bottleneck is whichever resource (CPU, memory, disk, network, or a lock or pool) runs out first. Measure to find it; disks are limited separately by IOPS and by MB/s.

How to approach a system design interview

What it is

A 45–60 minute, deliberately open-ended conversation: "design a URL shortener / a chat app / a news feed". A repeatable structure helps you cover everything and shows how you think.

How it works

A widely used sequence, with rough timings for a 45-minute interview:



1. **Clarify requirements.** Never start drawing straight away. Ask questions and write two lists:
   - **Functional requirements:** *what* the system does, the features. Example for a URL shortener: "create a short link from a long URL", "redirect a short link to the original", "optional custom aliases", "links expire after a year".

- **Non-functional requirements:** *how well* it does it, the qualities. Scale (100M new links per month), latency (redirect in under 50 ms at p99), availability (99.99% for redirects), durability (never lose a link), consistency (is a newly created link visible everywhere immediately?), security, maintainability, cost. The table after step 7 lists each one with the design levers that meet it.
- Also agree what's **out of scope** (analytics dashboards, user accounts).

2. Estimate scale. Back-of-the-envelope: QPS (average and peak), read/write ratio, storage over time, bandwidth. Keep it quick and say how the numbers affect the design ("100:1 reads, so caching matters").

3. Define the API. The main endpoints with inputs and outputs; this pins down exactly what the system must do:

```
POST /links        body: {long_url, custom_alias?}  -> {short_code}
GET  /{short_code}                                  -> 301 redirect to long_url
```

4. Sketch the data model. Entities, key fields and access patterns, e.g. a `links` table: `short_code` (primary key), `long_url`, `created_at`, `expires_at`. Mention which kind of database fits the access pattern (Part 3).

5. Draw the high-level design. Boxes and arrows (clients, load balancer, app servers, cache, database, queue…). Walk one write and one read end to end. Keep it simple; it must meet the functional requirements.

6. Deep dives. Go deeper on the hardest parts, often steered by the interviewer: how to generate unique short codes without collisions, how to shard the database, how to cache hot links, what happens when a data centre fails, how to handle a celebrity's viral link. Use your estimates to justify choices.

7. Discuss trade-offs and wrap up. Name weaknesses and next steps: bottlenecks, single points of failure, monitoring, what changes at 10× scale.

Non-functional requirements and the levers that meet them

A non-functional requirement only helps if it is **measurable** ("p99 < 200 ms", not "fast") and if you can name the **design lever** that delivers it. Use this table twice: in step 1 to make sure you asked about each row, and in step 7 to check that every number you agreed has a mechanism behind it.

Requirement	Make it measurable as	Main design levers	What the lever costs	Explained in
Availability	% of successful requests per month (SLO); allowed downtime	Redundancy in several zones, health checks and failover, no single point of failure, graceful degradation	More machines; consistency questions between copies	*Availability and the nines*, *Redundancy* (Part 1)
Latency	p50/p95/p99 per endpoint	Caching, CDNs, fewer round trips, async work, headroom	Staleness, money, complexity (see *Low-latency checklist*)	*Latency, throughput and bandwidth* (Part 1)
Scalability / throughput	Peak QPS, data size and growth per year	Stateless services behind load balancers, sharding, queues to absorb bursts	Cross-shard queries and transactions get hard	*Scalability*, *Partitioning and sharding* (Parts 1, 3)
Consistency	"After a write, who must see it, and how soon?"	Single leader with synchronous replicas, quorums, consensus; or eventual consistency with conflict handling	Higher write latency and lower availability during partitions (CAP/PACELC)	*Consistency models*, *CAP theorem* (Part 4)
Durability	"Never lose an acknowledged write"; RPO (how much data may be lost)	Write-ahead log, replication before acknowledging, backups, cross-region copies	Write latency, storage cost	*Durability* (Part 3), *Disaster recovery* (Part 5)
Reliability / fault tolerance	Error rate, MTTR	Timeouts, retries with backoff, idempotency, circuit breakers, bulkheads	Extra load from retries; code complexity	Part 5 resilience sections
Security and privacy	Who may do what; which data is sensitive; compliance rules	Authentication and authorisation at the edge, encryption in transit and at rest, least privilege, audit logs	Latency of checks, operational overhead	Part 5 security sections
Maintainability	Time to diagnose an incident; time to ship a typical change; deploys per day	See the three parts below	Up-front engineering time	This section
Cost	Cost per request, per user or per GB stored	Right-size machines, autoscale, tier storage (hot/cold), cache to avoid expensive reads	Less headroom; slower cold data	*Cost awareness* (Part 5)

Maintainability is the requirement candidates forget, yet most of a system's lifetime cost is spent operating and changing it after launch. Break it into three parts:

Part	Meaning	What it looks like in a design
Operability	How easy it is for the people running the system to keep it healthy	Metrics, logs and traces on every component; dashboards and alerts tied to SLOs; runbooks; safe, automated deploys and rollbacks; sensible defaults; no manual steps that only one person knows
Simplicity	How easy it is for a new engineer to understand the system	Few component types; clear boundaries with one owner each; well-named APIs that hide internals; avoiding accidental complexity (complexity that comes from the implementation, not from the problem itself)
Evolvability	How easy it is to change the system as requirements change	Versioned APIs and backward-compatible schemas; loose coupling through queues or well-defined interfaces; feature flags; data migrations that can run online; tests that make refactoring safe

In an interview, maintainability is usually one or two sentences at the wrap-up: "each service has one owning team and its own SLO dashboard, schemas are versioned so producers and consumers deploy independently, and every deployment is a canary with automatic rollback". That sentence shows you have run systems, not just drawn them.

Sources: ISO/IEC 25010:2023 product quality model (maintainability and its sub-characteristics); M. Kleppmann, *Designing Data-Intensive Applications*, ch. 1 (O'Reilly, 2017), for the operability/simplicity/evolvability split; B. Beyer et al., *Site Reliability Engineering* (Google, 2016), chapters on SLOs and monitoring.

Trade-offs

Breadth vs depth: covering every box shallowly looks junior; going deep on one box and never finishing the design is worse. Aim for a complete simple design first, then depth.

Common mistakes

- Jumping into detailed design before clarifying requirements, or staying silent — think out loud.
- Naming technologies without reasons ("use Cassandra"). Always say why ("write-heavy, simple key lookups, needs multi-region").
- Ignoring the non-functional requirements, which usually decide the design.
- Over-engineering: microservices, Kafka and five databases for a system with 100 QPS.
- Not checking the design against the original requirements and numbers at the end.

In the interview

Q: What is the difference between functional and non-functional requirements?

A: Functional = features, what it does (post a photo, follow a user). Non-functional = qualities, how well (latency, availability, scale, consistency, durability, cost); these usually drive the architecture.

Q: The interviewer says "design Twitter". What do you do first?

A: Narrow scope with clarifying questions: which features (post, follow, home timeline, search?), how many users, read/write mix, latency and consistency expectations. Write the agreed requirements down before estimating or drawing.

Q: How do you choose what to deep-dive into?

A: The hardest parts given the requirements and estimates (usually the biggest scale or consistency challenge), or wherever the interviewer steers. Ask if unsure.

TL;DR

Requirements (functional and non-functional), estimates, API, data model, high-level design, deep dives, trade-offs. Think out loud, justify every component with a requirement or a number, and finish with weaknesses and next steps.

Part 1 glossary

Every Part 1 term is defined in one line in the Master A–Z glossary, which merges the glossaries of all six parts into a single alphabetical list.

Part 2 — Networking and communication

Part 1 was about the boxes of a design diagram; this part is about the arrows: how one computer finds another, how bytes travel, which protocols they speak, and which helpers sit in the middle (proxies, load balancers, CDNs). You need not be a network engineer, but you do need to know what happens when a browser opens `www.example.com`, why HTTP/2 beats HTTP/1.1, and what an L7 load balancer can do that an L4 one cannot.

A **protocol** is an agreed set of rules for how two programs talk ("say hello, wait for a reply, then speak"). Networking is a stack of protocols, each solving one problem and handing the rest to the layer above or below.

IP addresses, NAT, ports and sockets

What it is

An **IP address** (Internet Protocol address) identifies a machine's network connection, like a street address. **Routers** forward traffic across the internet reading only this address on each **packet** (a small chunk of data, usually up to about 1,500 bytes on Ethernet).

- **IPv4**: 32 bits, four numbers 0–255 (`93.184.215.14`); about 4.3 billion addresses (2^32), which ran out in the 2010s.
- **IPv6**: 128 bits, eight groups of hex digits (`2606:2800:21f:cb07:6820:80da:af6b:8b2c`, runs of zeros shortened with `::`); about 3.4 × 10^38 addresses — effectively unlimited.

A **port** (0–65535) picks which program on the machine gets the data — the apartment number in the building. Well-known: 22 (SSH), 53 (DNS), 80 (HTTP), 443 (HTTPS), 5432 (PostgreSQL), 6379 (Redis).

A **socket** is one end of a conversation: IP address plus port. A TCP connection is uniquely identified by a **5-tuple**: protocol, source IP, source port, destination IP, destination port — which is why one server on port 443 can hold hundreds of thousands of connections (each client brings a different source IP or port).

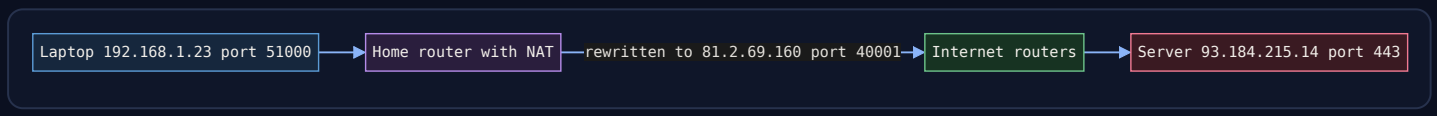
Public vs private IP. A **public IP** is unique worldwide and reachable from anywhere. A **private IP** is meaningful only inside one local network and reused by millions of networks; the reserved IPv4 ranges are `10.0.0.0/8`, `172.16.0.0/12` and `192.168.0.0/16`. (`/8` is **CIDR notation**: the number of fixed leading bits; `10.0.0.0/8` is every address starting with 10, about 16.7 million.) Your home laptop is probably something like `192.168.1.23`.

NAT (Network Address Translation) lets many private machines share one public IP: the home router rewrites each outgoing packet's source from `192.168.1.23:51000` to its public IP and a free port (`81.2.69.160:40001`), records the mapping, and reverses it on replies. In the cloud a **NAT gateway** does this so servers in a private subnet can download updates without being reachable from outside.

Why it matters

- Your servers live mostly on private IPs inside a **VPC** (Virtual Private Cloud — your own isolated network in a cloud provider); only the load balancer or gateway gets a public address. This is the first layer of security.
- NAT is why a server cannot simply "call" a phone or browser: the client has no stable public address, so push systems keep a long-lived connection open *from* the client (WebSockets, below).
- The 5-tuple sets connection limits: a proxy opening connections from one IP to one backend IP and port has at most about 64,000 source ports (about 28,000 with Linux's default range) — **ephemeral port exhaustion** on busy proxies.

How it works



1. The browser asks the OS for a socket to `93.184.215.14:443`; the OS picks a random **ephemeral port** (temporary source port, typically 32768–60999 on Linux) such as 51000.
2. The router's NAT swaps the private source for its public one and records the mapping.
3. Internet routers forward the packet hop by hop on the destination IP alone; a typical path crosses 10–20 routers.
4. The server replies to `81.2.69.160:40001`; the router looks up its table and forwards to `192.168.1.23:51000`.

Common mistakes

- Thinking an IP identifies a person or device. Behind **carrier-grade NAT** (NAT run by a mobile operator or ISP) thousands of users share one public IP, so rate limiting by IP alone can block a whole office or mobile network.
- Forgetting that IPs change. Cloud instances get new private IPs on restart, so systems use DNS names and service discovery, not hard-coded addresses.

TL;DR

An IP address finds the machine, a port finds the program, the 5-tuple identifies one connection. Private IPs hide behind NAT; only a few edge components should have public IPs.

DNS — the internet's phone book

What it is

DNS (Domain Name System) turns names like `www.example.com` into IP addresses, like a phone's contacts list. Without it you would type `93.184.215.14`, and a company could never move servers without telling every user.

A **domain name** reads right to left as a hierarchy: `.` (the root, usually invisible) → `com` (the **top-level domain**, TLD) → `example` (the registered domain) → `www` (a **subdomain**). Each level is managed by a different organisation, which lets DNS scale to hundreds of millions of domains.

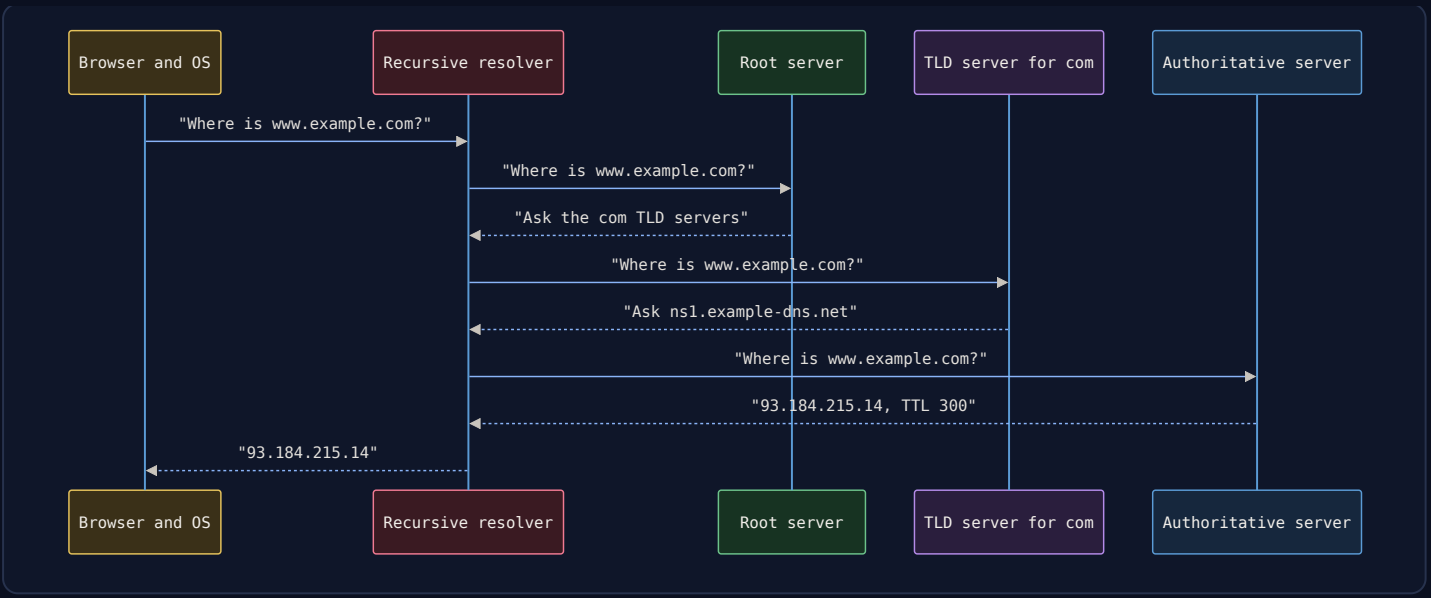
Why it matters

- DNS is the first step of almost every request, so its speed adds directly to page load (a cold lookup can cost 20–120 ms).
- It is a cheap, global traffic-steering tool: answering with different IPs spreads users across data centres or sends them to the closest.
- When DNS breaks, everything looks down though servers are healthy — the 2016 Dyn attack, and Facebook's 2021 outage, when its DNS servers became unreachable after a routing change.

How it works

The players:

- **Stub resolver** — the tiny DNS client in your OS; it just asks someone else.
- **Recursive resolver** — does the legwork for you and caches answers; run by your ISP (internet service provider), your company, or public services like Google `8.8.8.8` or Cloudflare `1.1.1.1`.
- **Root name servers** — the top of the tree: 13 named identities (`a.root-servers.net` to `m.root-servers.net`), each copied worldwide using anycast (CDN section), so close to 2,000 physical instances.
- **TLD name servers** — know who is in charge of each domain under `.com`, `.org`, `.uk` …
- **Authoritative name server** — holds the real records for a domain (Route 53, Cloudflare DNS, or your own); its answer is the final truth.



1. The browser checks its own cache, then the OS cache (and the `hosts` file).
2. On a miss, the stub resolver sends a **recursive query** ("give me the final answer") to the recursive resolver.
3. With nothing cached, the resolver asks a root server, which replies with a **referral** ("here are the `.com` servers"). These follow-ups are **iterative queries** — each server just points to the next.
4. The `.com` TLD server refers it to the domain's authoritative servers (its **NS records**).
5. The authoritative server returns the answer.
6. The resolver caches every step (root and TLD referrals for days), so the next `.com` lookup skips steps 3–4.

In practice a busy resolver has root and TLD answers cached, so most lookups are one hop to the authoritative server or a pure cache hit (under 1 ms on the same machine, ~10–30 ms to a public resolver).

Types

DNS record types:

Record	Maps	Example
A	name → IPv4 address	`www → 93.184.215.14`
AAAA	name → IPv6 address	`www → 2606:2800::1`
CNAME	name → another name (alias)	`shop → shops.myplatform.com`
MX	domain → mail servers, with priority	`10 mail1.example.com`
NS	domain → its authoritative name servers	`ns1.example-dns.net`
TXT	name → free text	domain-ownership proofs, SPF rules listing which servers may send a domain's email

Others: **SOA** (start of authority — admin data for a zone), **SRV** (service location with port), **PTR** (reverse lookup, IP → name), **CAA** (which certificate authorities may issue certificates for the domain). A **zone** is the set of records one authoritative server is responsible for.

A CNAME cannot sit at the bare domain (`example.com` , the **apex**) because the apex must also hold NS and SOA records; providers work around this with non-standard "ALIAS"/"ANAME" records or Cloudflare's "CNAME flattening".

```
$ dig www.example.com A +noall +answer
www.example.com.    300    IN    A    93.184.215.14
```

`300` is the **TTL**, `IN` means internet class, `A` is the record type.

TTL and DNS caching. Every record carries a **TTL (time to live)** in seconds: how long any resolver or browser may cache it (300 = reuse for 5 minutes). Long TTLs (hours, a day) mean fewer lookups and faster pages but slow changes — "propagation" is not a push; resolvers keep the old answer until their copy times out. Short TTLs (30–60 s) allow quick failover but cause more queries. Common trick: lower the TTL a day before a planned migration, switch the record, raise it again. Some clients and resolvers ignore TTLs and cache longer, so never rely on DNS alone for instant failover.

DNS-based load balancing. The authoritative server returns several A records and rotates their order per query — **round-robin DNS**. Clients usually try the first, so load spreads roughly evenly. Free and simple but blind: DNS doesn't know if a server is busy or dead unless the provider runs health checks and removes bad IPs.

GeoDNS answers by where the query comes from: Paris gets the Frankfurt data centre, Tokyo the Tokyo one. The location is guessed from the resolver's IP or from the **EDNS Client Subnet** extension, which passes part of the user's own IP. **Latency-based routing** (AWS Route 53) picks the region with the lowest measured latency rather than the nearest on a map.

Trade-offs

Choice	Benefit	Cost
Long TTL	Fewer lookups, resilient if DNS provider has a blip	Slow to change or fail over
Short TTL	Fast failover and traffic shifts	More DNS traffic and lookup latency
DNS load balancing	Global, cheap, no extra hop	Coarse, cached, no knowledge of server load

Common mistakes

- Expecting a DNS change to take effect instantly; cached answers live until their TTL expires, sometimes longer.
- One DNS provider with no redundancy. Many large sites now use two.
- Treating DNS round robin as a real load balancer; it cannot react to a slow server.

Real-world examples

Amazon Route 53, Cloudflare DNS, Google Cloud DNS and NS1 are authoritative DNS services with health checks, GeoDNS and latency routing. Netflix and many others use DNS to steer users between AWS regions during regional failures.

In the interview

Q: What happens when you type a URL in the browser, up to the IP address?

A: Browser cache, OS cache, then a recursive resolver, which (if nothing is cached) walks root → TLD → authoritative server, caches each answer for its TTL, and returns the IP. The browser then opens a TCP (or QUIC) connection to it.

Q: How would you move traffic away from a failed region using DNS?

A: Health-checked records with a short TTL (say 60 s) so the provider stops returning the failed region's IP. Some clients will still hit the old IP for a while, so the failed region's load balancer should also redirect or fail fast.

TL;DR

DNS maps names to IPs through a cached hierarchy of root, TLD and authoritative servers. TTL trades speed against how quickly changes spread; DNS can do coarse global load balancing (round robin, GeoDNS, latency-based).

Network layers — OSI and TCP/IP models

What it is

Networking is split into **layers**, each solving one job and relying on the layer below — like a parcel: letter (application), apartment number (transport), street address (network), the van (link and physical). Each layer only reads its own label.

The **OSI model** has seven layers and is mainly a teaching vocabulary; the **TCP/IP model** has four and describes what the internet runs. "L4" or "L7" always means OSI layer numbers.

OSI layer	Name	What it handles	Examples
7	Application	Meaning of messages	HTTP, DNS, gRPC, SMTP
6	Presentation	Encoding, encryption	TLS (roughly), JSON (a text data format)
5	Session	Conversations	(rarely discussed)
4	Transport	Process-to-process delivery, ports	TCP, UDP, QUIC
3	Network	Machine-to-machine routing	IP, ICMP
2	Data link	One hop on a local network	Ethernet, Wi-Fi, MAC addresses
1	Physical	Bits on wire, light or radio	Copper, fibre, radio

TCP/IP merges 5–7 into "application" and 1–2 into "link": Application, Transport, Internet, Link.

Why it matters

A component's layer decides what it can see and do:

- **L3** (router, firewall rule on IPs): only source and destination IPs.
- **L4** (L4 load balancer): also ports and TCP/UDP state, not content — "send port 443 traffic to these servers", but not "send `/api/` to these".
- **L7** (L7 load balancer, API gateway): reads the HTTP request — path, headers, cookies — so it can route by path, rewrite headers and block bad requests, at the cost of more CPU and decrypting TLS (the encryption behind HTTPS, below).

How it works

Each layer wraps the data with its own header — **encapsulation**: HTTP request → TCP segment (ports, sequence numbers) → IP packet (IPs) → Ethernet frame (**MAC addresses**, the hardware address of a network card). The receiver unwraps in reverse.

TL;DR

L3 is IP addresses, L4 ports and connections (TCP/UDP), L7 application content (HTTP). "L4 vs L7" means "can it read the request or only the connection?"

TCP, UDP and QUIC

What it is

The **transport-layer** protocols that carry data between two programs:

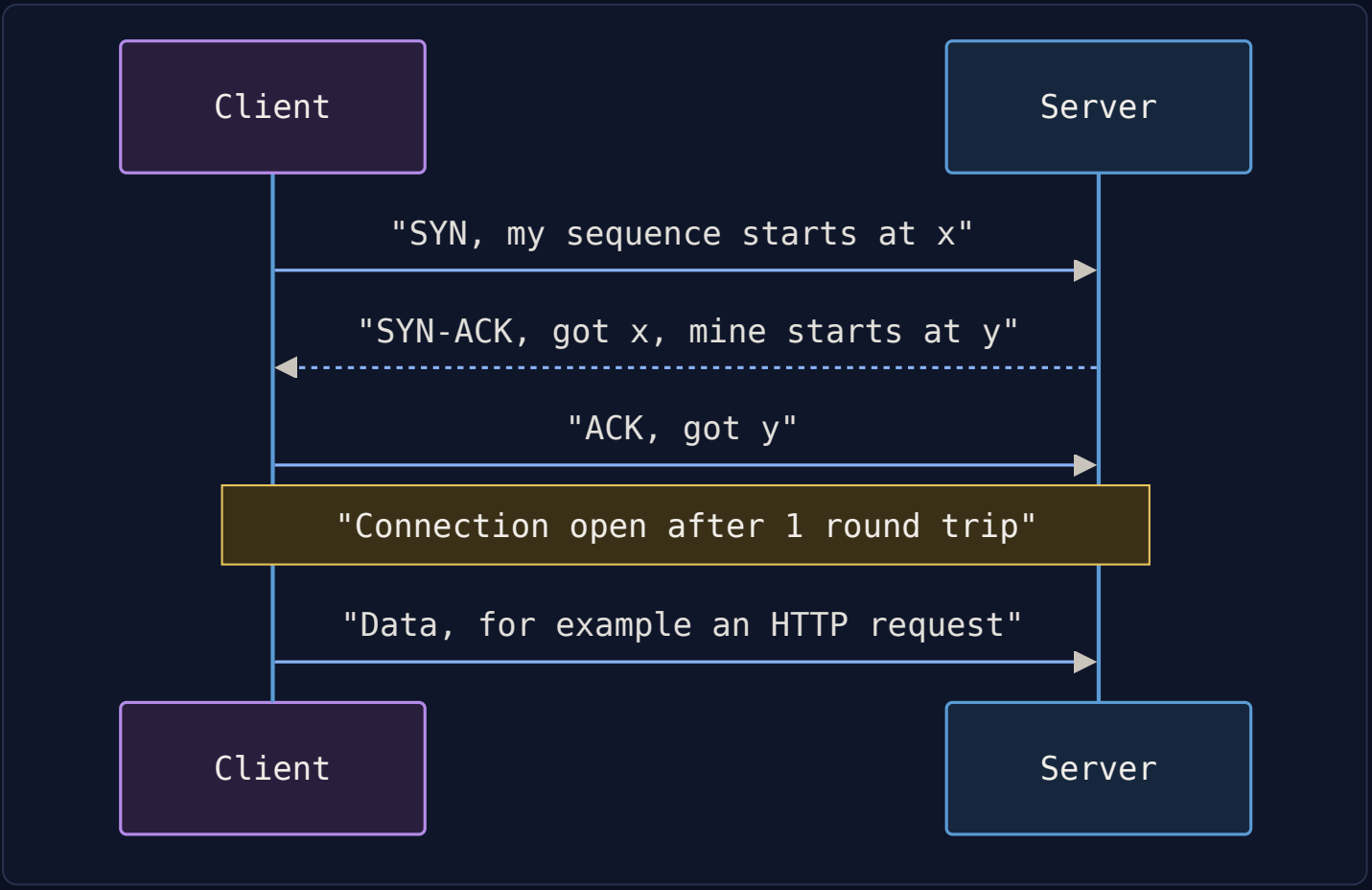
- **TCP (Transmission Control Protocol)**: a phone call with a careful note-taker — connect first; everything arrives, in order; lost words are repeated.
- **UDP (User Datagram Protocol)**: postcards shouted across a room — no setup, independent messages, some lost or out of order, nothing repeated.
- **QUIC**: a newer protocol over UDP with TCP-like reliability plus built-in encryption, minus some of TCP's weaknesses. HTTP/3 runs on it.

Why it matters

The choice shapes latency and behaviour under packet loss. Web pages, APIs and databases need every byte correct, so they use TCP (or QUIC). Voice, video calls, games and DNS lookups prefer UDP: a late packet is useless, better skipped than waited for.

How it works

TCP three-way handshake. Before any data, the two sides agree starting sequence numbers:



This costs one **round trip (RTT)** before the first request byte — 70 ms London–New York — and TLS adds more (HTTPS). Hence reusing connections matters.

Reliability and ordering. TCP numbers every byte (**sequence numbers**); the receiver sends **acknowledgements (ACKs)**; if an ACK is late the sender **retransmits**. The receiver reorders segments and hands the application a clean, ordered byte stream.

Flow control. The receiver advertises a **receive window** (its free buffer space) so a fast sender cannot overwhelm a slow receiver.

Congestion control. To avoid overwhelming the network in between, TCP starts slowly (**slow start**: a few packets, roughly doubling each round trip while ACKs return) and backs off on loss. Algorithms: Reno, CUBIC (Linux default) and Google's BBR, which models bandwidth and RTT instead of waiting for loss. A new connection cannot use full bandwidth immediately — another reason to keep connections warm.

Closing. Either side sends **FIN**; the other acknowledges. The side closing first keeps the socket in **TIME_WAIT** briefly (60 s on Linux) to catch stray packets — a source of port exhaustion on clients that open and close many connections.

Head-of-line (HOL) blocking. TCP delivers bytes strictly in order, so one lost packet holds up everything behind it until retransmitted, even data already arrived. If several independent requests share one TCP connection (HTTP/2), one loss stalls them all.

UDP has no handshake, ACKs, ordering or congestion control; each **datagram** is sent and forgotten. Its header is 8 bytes versus TCP's 20+. Apps needing some reliability add it themselves.

QUIC (standardised by the IETF, the Internet Engineering Task Force, in 2021; originally developed at Google) runs over UDP and merges what took TCP + TLS:

1. **Faster setup** — transport and TLS 1.3 handshakes combined: 1 RTT for a new connection; a returning client can send data in its first packet (**0-RTT**).
2. **Independent streams** — a lost packet stalls only its own stream; no transport-level HOL blocking between streams.
3. **Connection migration** — a connection is identified by a **connection ID**, not the 5-tuple, so a phone moving from Wi-Fi to 4G keeps it.
4. **Always encrypted** — even most transport headers.
5. **Lives in user space** — evolves without OS upgrades. Downsides: more CPU per byte than kernel TCP, and some corporate firewalls block UDP, so clients fall back to TCP.

Types

	TCP	UDP	QUIC
Connection setup	1 RTT (plus TLS)	none	1 RTT incl. TLS, 0-RTT on resume
Reliable, ordered	Yes, one stream	No	Yes, per stream
HOL blocking	Yes	No	Only within a stream
Encryption	Separate (TLS)	None built in	Built in
Typical use	Web, APIs, databases	DNS, video calls, games	HTTP/3

Trade-offs

TCP: correctness, at the cost of setup latency and stalls under loss. UDP: speed and control, but you build any reliability yourself. QUIC fixes TCP's main web pains but costs more CPU and is sometimes blocked.

Common mistakes

- "UDP is faster" without context. On a clean network a long-lived TCP connection is as fast for bulk data; UDP wins on setup and on not waiting for retransmissions.
- Forgetting that every new TCP + TLS connection costs 2–3 RTTs before the first byte. Use connection pooling and keep-alive.

Real-world examples

Zoom, WebRTC (browser video calls) and most online games use UDP. DNS uses UDP for normal queries and TCP for large responses. Google, YouTube, Meta and Cloudflare serve a large share of traffic over QUIC / HTTP/3.

In the interview

Q: Why would a video-calling app use UDP instead of TCP?

A: A frame 300 ms late is worthless. TCP would pause all later frames while retransmitting the lost one (HOL blocking); with UDP the app skips it and stays real-time.

Q: What problem does QUIC solve?

A: TCP's head-of-line blocking across multiplexed streams and the slow combined TCP + TLS handshake. QUIC gives independent streams, a 1-RTT (or 0-RTT) encrypted handshake, and survives network changes.

TL;DR

TCP = connected, reliable, ordered, congestion-controlled, with handshake cost and head-of-line blocking. UDP = fire-and-forget datagrams. QUIC = reliable, encrypted, multi-stream transport over UDP that powers HTTP/3.

HTTP — the language of the web

What it is

HTTP (HyperText Transfer Protocol) is the request–response protocol of browsers and APIs. The client sends a **request** (method, path, headers, optional body); the server sends a **response** (status code, headers, optional body).

```
GET /users/42 HTTP/1.1
Host: api.example.com
Accept: application/json
Authorization: Bearer eyJhbGciOi...

HTTP/1.1 200 OK
Content-Type: application/json
Cache-Control: max-age=60

{"id": 42, "name": "Asha"}
```

Why it matters

Almost every API you design in an interview is HTTP. The right method and status code make APIs predictable, cache-friendly and safe to retry; the HTTP version decides how many connections you need and how much latency you pay.

How it works

Methods (verbs).

Method	Meaning	Safe?	Idempotent?
GET	Read a resource	Yes	Yes
HEAD	Like GET, headers only	Yes	Yes
POST	Create, or run an action	No	No
PUT	Replace a resource fully	No	Yes
PATCH	Change part of a resource	No	Not guaranteed
DELETE	Remove a resource	No	Yes
OPTIONS	Ask what is allowed (used by CORS, the browser's cross-origin permission check)	Yes	Yes

Safe = changes nothing on the server. **Idempotent** = doing it once or ten times leaves the same server state, so a client can retry after a timeout: `DELETE /orders/7` twice still leaves order 7 deleted (the second may return 404). `POST /orders` twice creates two orders, which is why payment APIs add an **idempotency key** header so the server recognises a retry.

Status codes — the first digit is the category:

Range	Meaning	Common codes
1xx	Informational	101 Switching Protocols (WebSocket upgrade)
2xx	Success	200 OK, 201 Created, 202 Accepted, 204 No Content
3xx	Redirect	301 Moved Permanently, 302 Found, 304 Not Modified
4xx	Client's fault	400 Bad Request, 401 Unauthorized, 403 Forbidden, 404 Not Found, 409 Conflict, 429 Too Many Requests
5xx	Server's fault	500 Internal Server Error, 502 Bad Gateway, 503 Service Unavailable, 504 Gateway Timeout

401 = "we don't know who you are" (not logged in); 403 = "we know who you are and you may not do this". 502 and 504 usually come from a proxy or load balancer: the backend gave a bad answer (502) or none in

time (504). 202 Accepted = "queued, will be processed later".

Headers are key–value metadata: `Host` (which site — many share one IP), `Content-Type` (body format), `Authorization` (credentials), `Cache-Control` (who may cache and how long), `ETag` and `If-None-Match` (server replies 304 Not Modified if the client's copy is current), `User-Agent`, `Accept-Encoding: gzip, br` (compression), `X-Forwarded-For` (original client IP, added by proxies).

Statelessness and cookies. HTTP is **stateless**: each request carries everything needed and the server doesn't remember the previous one, so many identical servers can sit behind a load balancer. Logins and carts use **cookies**: the server sends `Set-Cookie: session=abc123; HttpOnly; Secure; SameSite=Lax` and the browser returns `Cookie: session=abc123` on every later request; the session lives in a shared store (Redis) or a signed token (flags and tokens: *Sessions, cookies, tokens and JWT*, Part 5).

Keep-alive (persistent connections). HTTP/1.0 opened a TCP connection per request; HTTP/1.1 keeps it open by default for reuse, saving the handshake each time.

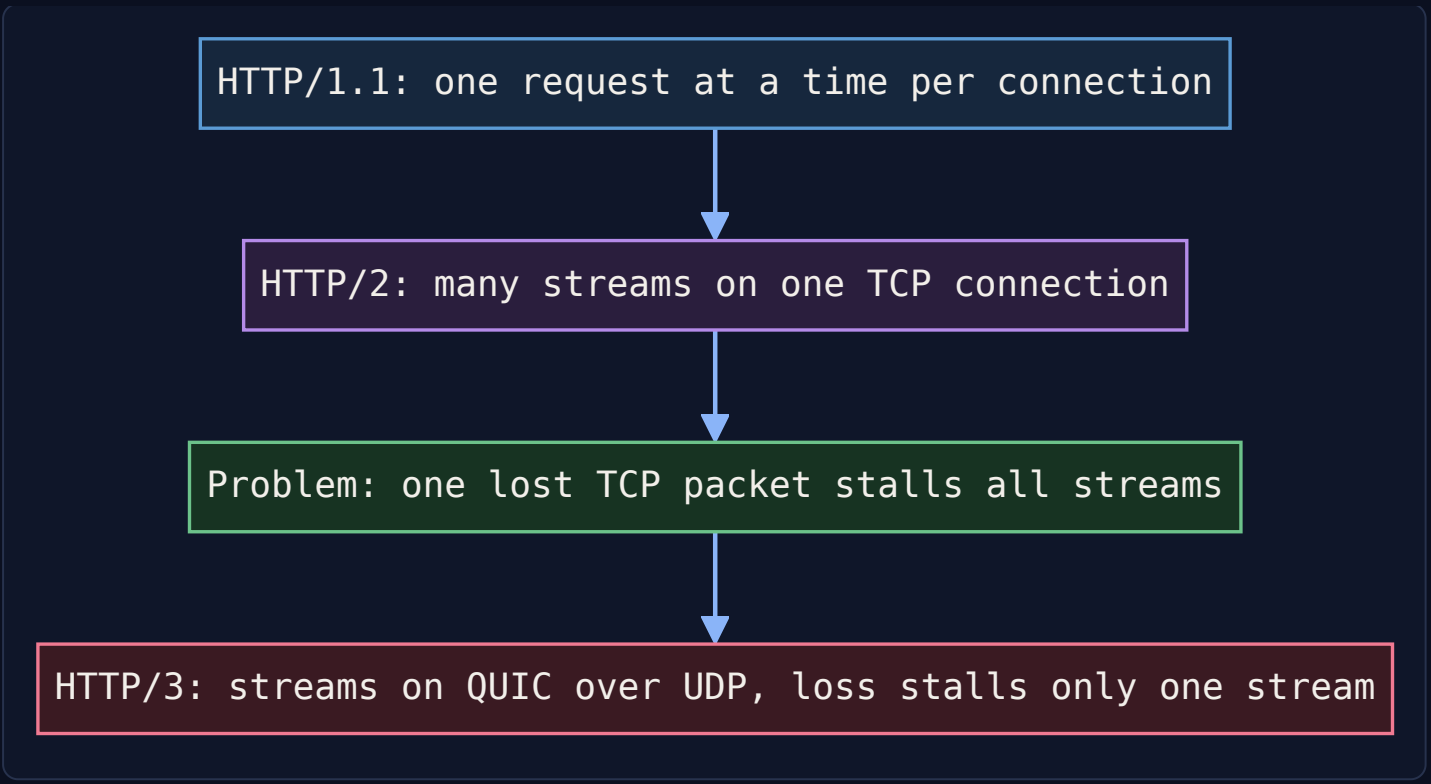
Types

HTTP/1.1 (1997). Text-based; one request at a time per connection, the next waiting for the previous response (**HTTP-level HOL blocking**). "Pipelining" was specified but never worked reliably, so browsers open about 6 connections per host and sites used tricks like domain sharding and sprite images.

HTTP/2 (2015). Binary framing. **Multiplexing**: many requests and responses interleave as **streams** over one TCP connection, so one slow response no longer blocks others at the HTTP level. **Header compression** (HPACK) shrinks repetitive headers. Server push was little used and has been removed from Chrome. Weakness: one TCP connection, so one lost packet stalls every stream (TCP-level HOL blocking) — on lossy mobile networks sometimes worse than six HTTP/1.1 connections.

HTTP/3 (2022). HTTP/2's semantics on QUIC over UDP: independent streams remove TCP HOL blocking, setup is faster, connections survive network switches. Header compression is QPACK. Browsers discover HTTP/3 through an `Alt-Svc` header or a DNS **HTTPS record**, falling back to HTTP/2 if UDP is blocked.

	HTTP/1.1	HTTP/2	HTTP/3
Transport	TCP	TCP	QUIC (UDP)
Format	Text	Binary frames	Binary frames
Parallel requests	~6 connections	Many streams, 1 connection	Many streams, 1 connection
HOL blocking	HTTP and TCP level	TCP level	Per stream only
Encryption	Optional	Required by browsers	Always



Common mistakes

- GET for actions that change data (a crawler or prefetcher may trigger them).
- Returning 200 with `{"error": ...}` in the body; proxies, caches, monitoring and retries rely on real status codes.
- Retrying non-idempotent POSTs blindly. Use idempotency keys.
- Session state in one server's memory, which breaks with two servers.

In the interview

Q: Which HTTP methods are idempotent and why does it matter?

A: GET, HEAD, PUT, DELETE, OPTIONS. Networks time out; if the client doesn't know whether a request succeeded, it can safely retry an idempotent one. POST needs an idempotency key.

Q: What did HTTP/2 improve and what did HTTP/3 fix on top?

A: HTTP/2 multiplexed many requests over one connection with header compression, removing HTTP-level head-of-line blocking. HTTP/3 moved to QUIC to remove the remaining TCP-level blocking and speed up connection setup.

TL;DR

HTTP is a stateless request–response protocol of methods, status codes and headers. Cookies add sessions, keep-alive reuses connections, HTTP/2 multiplexes streams over TCP, HTTP/3 over QUIC.

HTTPS and TLS

What it is

HTTPS is HTTP inside a **TLS (Transport Layer Security)** encrypted tunnel. TLS gives **confidentiality** (eavesdroppers see gibberish), **integrity** (bytes can't be changed undetected) and **authentication** (you really are talking to `bank.com`) — a sealed, tamper-evident envelope plus a passport check. Its older name is **SSL**, hence "SSL certificate" and "SSL termination". Current version TLS 1.3 (2018); TLS 1.2 still common; anything older deprecated.

Why it matters

Browsers mark plain HTTP "Not secure", HTTP/2 in browsers and HTTP/3 always require TLS, and passwords, tokens and personal data on the wire must be encrypted. This is **encryption in transit** (vs **encryption at rest**, protecting stored data on disk). TLS costs latency and CPU, which affects where you decrypt (TLS termination, load balancer section).

How it works

Certificates. A server proves its identity with a **certificate**: domain name, the server's **public key**, an expiry date, and a digital signature from a **Certificate Authority (CA)** trusted by browsers and operating systems (Let's Encrypt, DigiCert). The browser follows the **chain of trust**: server certificate → intermediate CA → **root CA** pre-installed in the browser; if the chain is valid, the name matches and it hasn't expired, the server is trusted. Let's Encrypt issues free certificates renewed automatically with the ACME protocol. They have been valid for 90 days, and lifetimes are shrinking (Let's Encrypt plans 64 days in 2027 and 45 in 2028), so automated renewal is mandatory.

Public-key vs symmetric encryption. Asymmetric (public-key) cryptography uses a key pair: anything signed with the private key can be verified with the public key. It is slow, so TLS uses it only in the handshake to authenticate the server and agree a shared secret; data is then encrypted with fast **symmetric** encryption (one shared key, e.g. AES-GCM or ChaCha20).

TLS 1.3 handshake (simplified):



1. The client sends supported versions, cipher suites and a **key share** (its half of a **Diffie–Hellman** exchange, where two sides each mix in a secret and arrive at the same key without ever sending it).
2. The server answers with its key share, certificate, and a signature proving it owns the certificate's private key.
3. Both derive the same symmetric keys; the client sends Finished and immediately the encrypted request.

TLS 1.3 needs **1 RTT** (TLS 1.2 needed 2). Plus TCP's 1 RTT, a fresh HTTPS connection costs about 2 RTTs before the request goes out — 140 ms London–New York. **Session resumption** and **0-RTT** let returning clients skip work (0-RTT data can be replayed by an attacker, so use it only for idempotent requests). Fresh Diffie–Hellman keys per session give **forward secrecy**: stealing the server's private key later doesn't decrypt old recorded traffic.

SNI (Server Name Indication). The ClientHello carries the hostname so one IP can serve certificates for many domains — essential for load balancers and CDNs.

mTLS (mutual TLS) — both sides present certificates; used between internal services (service meshes) so each proves its identity.

Common mistakes

- Letting certificates expire — it has caused outages at large companies. Automate renewal and alert on expiry.
- Terminating TLS at the edge and sending plaintext across untrusted networks. Inside one VPC it may be acceptable; across data centres, re-encrypt.

In the interview

Q: Why does HTTPS use both asymmetric and symmetric encryption?

A: Asymmetric crypto handles authentication and key agreement without a shared secret but is slow; once a key is shared, symmetric encryption is fast enough for bulk data.

Q: How many round trips does a new HTTPS request cost?

A: 1 RTT for TCP, 1 for TLS 1.3, 1 for request and response — 3 in total. HTTP/3 over QUIC merges the first two, and resumed connections can send data immediately.

TL;DR

TLS encrypts and authenticates connections with certificates signed by trusted CAs; the handshake uses public-key crypto to agree a symmetric key in 1 RTT (TLS 1.3). HTTPS is HTTP inside TLS.

API styles — REST, RPC and gRPC, GraphQL

What it is

An **API (Application Programming Interface)** is the menu a service offers other programs: which operations exist, their inputs and outputs. Three styles dominate:

- **REST (Representational State Transfer)** models everything as **resources** (nouns) with URLs, manipulated with standard HTTP methods: `GET /users/42`, `POST /users/42/orders`.
- **RPC (Remote Procedure Call)** models the API as functions called on a remote machine as if local: `getUser(42)`, `chargeCard(order, amount)`.
- **GraphQL** gives clients one endpoint and a query language to ask for exactly the fields they want, across related objects, in one round trip.

Why it matters

The API is the contract between teams and with users. A poor style causes chatty clients, **over-fetching** (downloading data you don't need) or **under-fetching** (many calls to build one screen).

How it works

REST in practice.

1. Name resources with plural nouns: `/users`, `/users/42`, `/users/42/orders`.
2. Use methods for actions: GET reads, POST creates, PUT replaces, PATCH updates part, DELETE removes.
3. Return proper status codes (201 Created with a `Location` header, 404, 409 Conflict).
4. Keep requests stateless, so any server can answer and GET responses can be cached by browsers and CDNs.
5. Usually JSON over HTTP/1.1 or HTTP/2.

Idempotent methods (GET, PUT, DELETE) are safe to retry — a big reason to prefer `PUT /users/42/avatar` over `POST /changeAvatar`.

gRPC. Google's open-source RPC framework on HTTP/2. Services and messages are defined in a `.proto` file using **Protocol Buffers (Protobuf)**, a compact binary format with a schema:

```
service UserService {
  rpc GetUser (GetUserRequest) returns (User) {}
}
message GetUserRequest { int64 id = 1; }
message User { int64 id = 1; string name = 2; }
```

The numbers (`= 1` , `= 2`) are field tags identifying each field on the wire. A code generator produces client and server code in many languages, so calling another service looks like a local, compile-time type-checked function call. Protobuf messages are typically several times smaller and faster to parse than JSON, and numbered fields allow **backward-compatible** schema changes (add new numbers, never reuse old ones). Four call shapes: unary (one request, one response), server streaming, client streaming, bidirectional streaming. Browsers can't speak native gRPC, so it is mostly **service-to-service**, with gRPC-Web or a REST gateway for browsers.

GraphQL. Created at Facebook (open-sourced 2015). The server publishes a typed **schema**; the client sends a query:

```
query {
  user(id: 42) {
    name
    orders(last: 3) { id total }
  }
}
```

and gets back exactly that shape. Writes are **mutations**, live updates **subscriptions**; each field is filled by a **resolver** function. Ideal when many clients (web, iOS, Android) need different slices of the same data. Downsides: usually one POST endpoint, so ordinary HTTP caching doesn't work; a careless query can be very expensive, so servers limit query depth and cost; naive resolvers cause the **N+1 query problem** (one query for the list, then one per item), fixed with batching (the DataLoader pattern).

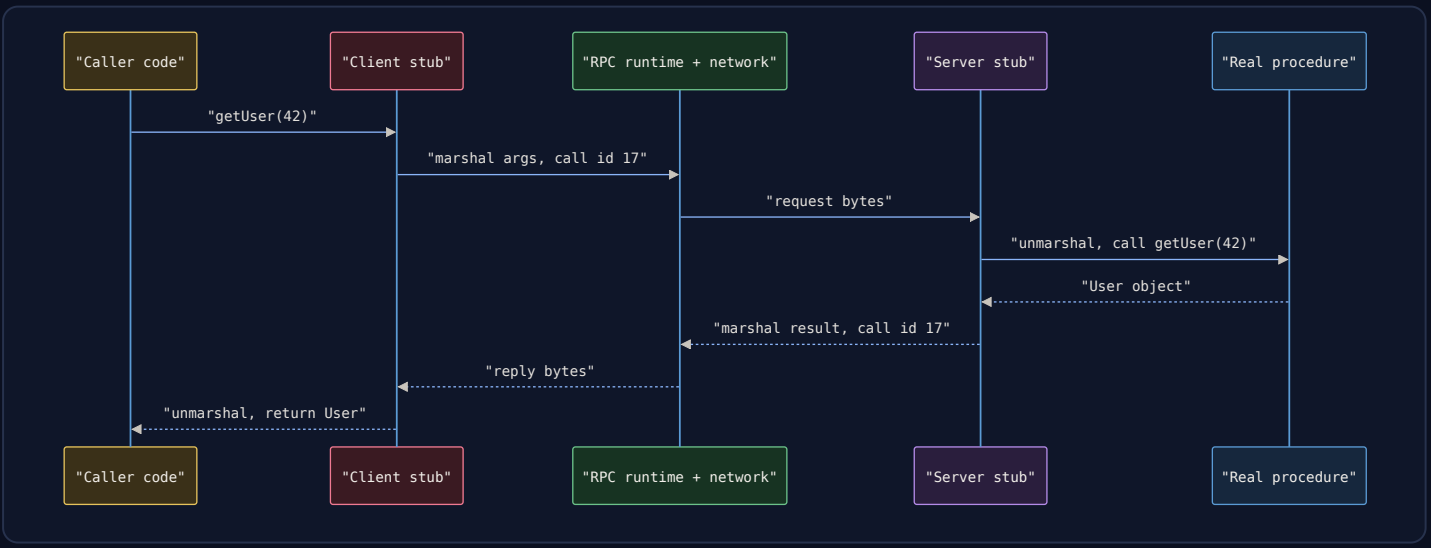
Types

	REST	gRPC	GraphQL
Model	Resources + HTTP verbs	Functions	Typed graph, client picks fields
Format	Usually JSON	Protobuf (binary)	JSON
Transport	HTTP/1.1 or 2	HTTP/2	HTTP, one endpoint
Best for	Public APIs, CRUD (create/read/update/delete), caching	Internal service calls, streaming	Many client types, complex screens
Weak spot	Over/under-fetching	Browser support, readability	Caching, query cost control

RPC internals: stubs, marshalling, binding and call semantics

RPC's promise is that `getUser(42)` on another machine looks like a local call. The idea was set out by Birrell and Nelson (Xerox PARC, 1984), and every RPC system since, from Sun's ONC RPC (used by NFS) to Java RMI, Thrift and gRPC, has the same five parts:

Part	What it does	In gRPC
Interface definition (IDL)	A language-neutral file listing the procedures and their argument and result types; both sides compile against it	The `.proto` file
Client stub	Generated code with the same signature as the remote procedure. It packs the arguments, hands them to the runtime, waits, and unpacks the result, so the caller never sees the network	The generated client class
Server stub (skeleton)	Generated code on the server that unpacks the request, calls the real procedure and packs the result	The generated service base class you implement
Marshalling	Turning in-memory arguments into a byte sequence (and back: **unmarshalling**). It must fix byte order, sizes and character encoding, and it cannot send pointers: a linked structure is copied by value	Protobuf encoding
RPC runtime	Opens and reuses connections, matches each reply to its call by a **call ID**, applies timeouts and retransmissions	HTTP/2 streams, deadlines, retry policy



Binding is how the client finds a server for the interface. It can be static (an address in config), through a directory (Birrell and Nelson used the Grapevine name database; ONC RPC uses the `rpcbind` service, which maps program numbers to ports), or, today, through DNS or a service-discovery registry feeding a client-side load balancer (gRPC's name resolver and balancing policies; see *Load balancers* and Part 5's *Service discovery*).

Why a remote call is not a local call. It is thousands of times slower (half a millisecond inside a data centre versus nanoseconds), it can't share memory, and above all it can **fail partially**: the request may be lost, the server may crash halfway through, or the reply may be lost. When no reply arrives, the caller **cannot know** whether the procedure ran. That uncertainty is why each RPC system states its **call semantics**:

Semantics	Guarantee	How it is built	Safe for
Maybe	Ran once or not at all; no retry	Send once, give up on timeout	Hints, metrics
At-least-once	Ran one or more times if a reply comes back	Retry until a reply arrives	Idempotent operations only (reads, `PUT`, "set X to 5")
At-most-once	Never runs twice; may not run at all	Server remembers recent call IDs and returns the saved reply to a duplicate instead of re-executing	Non-idempotent operations, with the caller told "unknown outcome" on failure
Exactly-once	Ran exactly once	Not achievable in general across crashes; approximated by at-least-once retries **plus** at-most-once deduplication with durable call IDs (idempotency keys)	Payments, orders

Real systems show the spectrum. Birrell and Nelson's RPC promised that a normal return meant the procedure ran exactly once, while an exception meant it ran at most once. ONC RPC (RFC 5531) leaves reliability to the transport: over UDP a reply proves only that the call ran at least once, and servers may use the call's transaction ID (`xid`) to drop duplicates. gRPC sets **no deadline by default** (always set one); it retries automatically only when it is certain the server application never saw the call, and otherwise retries only by an explicit per-method policy listing retryable status codes, which you should enable only for idempotent methods. The same ideas reappear for message queues in Part 4's *Delivery guarantees, idempotency and retries*.

Sources: A. Birrell and B. Nelson, "Implementing Remote Procedure Calls", *ACM TOCS* 2(1), 1984; RFC 5531 (ONC RPC v2) and RFC 1833 (rpcbind); gRPC documentation, "Deadlines" and "Retry" guides (grpc.io); J. Waldo et al., "A Note on Distributed Computing", Sun Labs TR-94-29, 1994.

Trade-offs

REST is universally understood and cache-friendly but may need many calls per screen. gRPC is fast, strongly typed and streams well, but binary (hard to debug with `curl`) and needs a proxy for browsers. GraphQL gives clients flexibility but moves complexity and performance risk to the server.

Common mistakes

- Verbs in REST URLs (`/getUser?id=42`) instead of methods on resources.
- GraphQL "because it is modern" for a simple public API where cached REST is cheaper.
- Changing Protobuf field numbers, which silently breaks old clients.

Real-world examples

Stripe and GitHub's v3 API are classic REST; GitHub also offers GraphQL (v4), and Shopify's storefront API is GraphQL. Google, Netflix and many others use gRPC between internal microservices; Kubernetes uses gRPC for plug-in interfaces such as kubelet ↔ container runtime.

In the interview

Q: You are designing a mobile app's backend and an internal ranking service. Which API styles?

A: REST (or GraphQL if screens need many related objects) for the public mobile API — easy to consume, cache and debug. gRPC between backend and ranking service for compact binary messages, typed contracts and streaming.

Q: What makes a REST call safe to retry?

A: An idempotent method (GET, PUT, DELETE), or an idempotency key on a POST so the server returns the original result instead of repeating the action.

Q: An RPC to `chargeCard` times out. What do you tell the caller, and how do you make a retry safe?

A: The outcome is unknown: the charge may or may not have happened. Make the call at-most-once on the server with a client-generated idempotency key stored with the result; the client can then retry (at-least-once) and the combination behaves like exactly-once.

TL;DR

REST = resources and HTTP verbs, simple and cacheable. gRPC = typed remote functions over HTTP/2 with Protobuf, great inside the data centre. GraphQL = clients query exactly the fields they need from a typed schema. Under every RPC: stubs and marshalling hide the network, binding finds the server, and the call semantics (at-least-once, at-most-once, exactly-once via idempotency keys) say what a timeout means.

Real-time communication — polling, long polling, WebSockets, SSE, webhooks

What it is

Plain HTTP only lets the client ask, but many features need the server to *tell* the client as soon as something happens: a chat message, a notification, a stock price, a driver's location. The client is usually behind NAT and cannot be dialled, so every solution keeps a channel open from the client side — or, between servers, calls back a public URL.

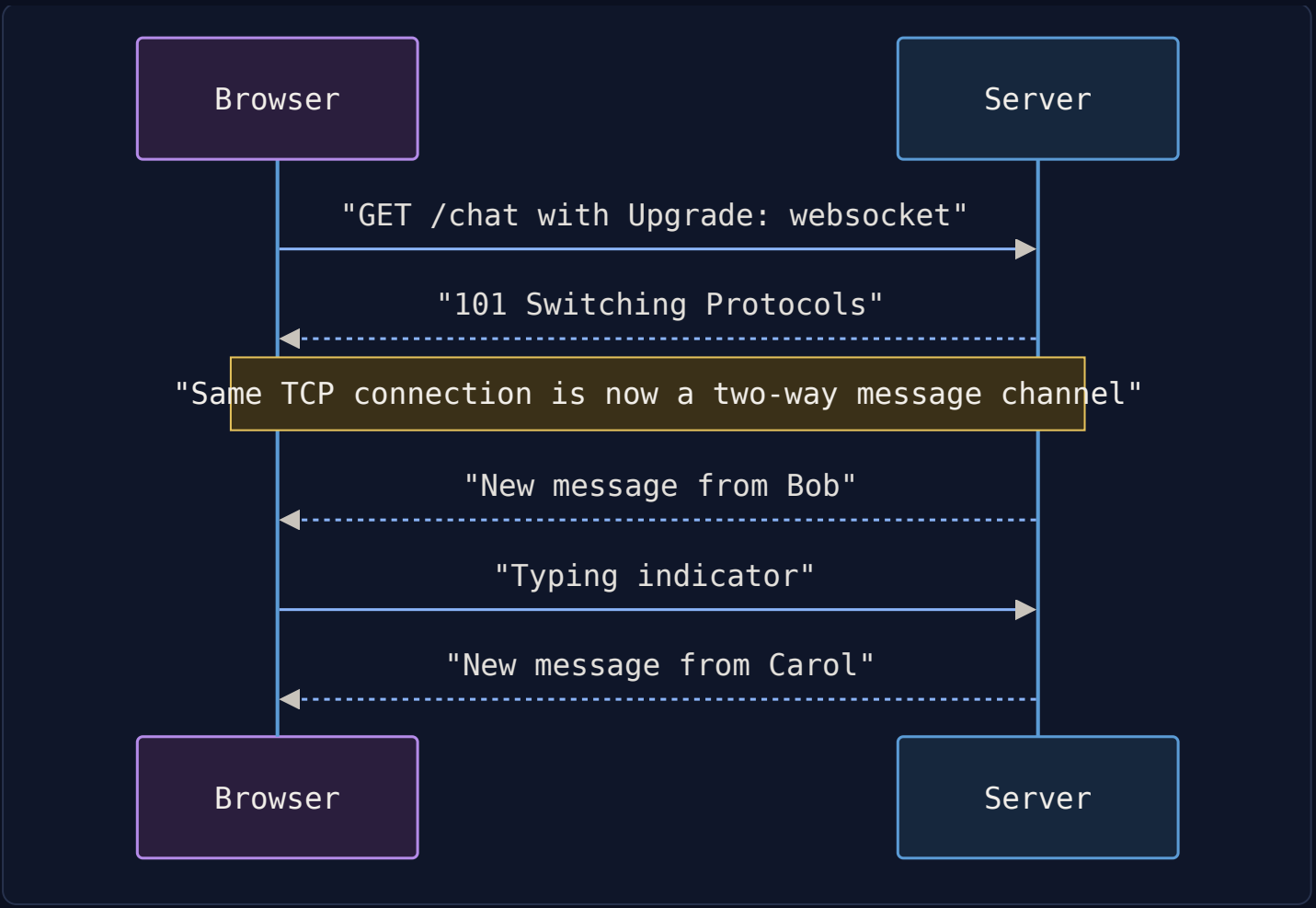
- **Short polling** — the client asks "anything new?" every few seconds ("are we there yet?").
- **Long polling** — the client asks and the server holds the request open until there is news or a timeout (say 30 s); the client then immediately asks again.
- **WebSockets** — one long-lived, two-way connection over which both sides send messages any time (an open phone line).
- **Server-Sent Events (SSE)** — one long-lived HTTP response the server keeps writing events into; one-way, server to client (a radio broadcast).
- **Webhooks** — server-to-server: when an event happens, service A sends an HTTP POST to a URL service B registered (leaving your number so the shop calls you).

Why it matters

The wrong choice wastes resources (polling a million clients every second is a million requests per second, mostly "nothing new") or adds needless complexity. Long-lived connections change capacity planning: a chat server's limit is often open connections (memory per connection, file descriptors), not requests per second.

How it works

WebSocket upgrade. Starts as an ordinary HTTP/1.1 request, so it passes through normal ports and proxies:



1. The client sends `GET` with `Upgrade: websocket` and `Connection: Upgrade` headers (scheme `ws://` or encrypted `wss://`).
2. The server answers `101 Switching Protocols`.
3. Both sides then exchange small **frames** (a few bytes of overhead each) until either closes.
4. Applications send **heartbeats** (ping/pong) every 20–60 s so NAT routers or load balancers don't silently cut idle connections; clients reconnect with backoff if the link drops.

SSE. The client opens `GET /events` with `Accept: text/event-stream`; the server never finishes the response, writing `data: {"price": 101.5}` plus a blank line per event. The browser's built-in `EventSource` reconnects automatically and sends `Last-Event-ID` so the server can resume. Plain HTTP, so it works with HTTP/2 and existing infrastructure; large language model chat interfaces commonly stream tokens this way.

Webhooks. The receiver registers `https://myapp.com/hooks/payments`; when a payment succeeds the provider POSTs a JSON event there. Good practice: sign each request (an HMAC header — a hash of the body computed with a shared secret key) so the receiver can verify the sender; retry with backoff on failure; make the receiver idempotent because an event may arrive twice.

Types

Technique	Direction	Latency	Cost / complexity	Use when
Short polling	Client pulls	Up to the interval	Many wasted requests	Rare updates, simplicity first
Long polling	Server push (emulated)	Low	Held requests, reconnect loops	Fallback where WebSockets are blocked
WebSocket	Two-way	Lowest	Stateful connections, sticky routing	Chat, games, collaboration
SSE	Server to client	Low	Simple, auto-reconnect	Feeds, notifications, live scores, token streaming
Webhook	Server to server	Low	Receiver needs public URL	Payment and Git events

Trade-offs

Persistent connections deliver instantly but make servers **stateful**: each user is attached to one server, so messages must be routed to the right server (often via a pub/sub layer such as Redis Pub/Sub or Kafka between servers), and deployments must drain connections gracefully. Polling is stateless and trivially scalable but wasteful and slower.

Common mistakes

- WebSockets for a one-way feed where SSE is simpler.
- Forgetting a load balancer's idle timeout (60 s by default on AWS ALB) closes quiet WebSockets unless you send heartbeats.
- Assuming webhooks arrive exactly once and in order. They do not.

Real-world examples

Slack and Discord use WebSockets for their real-time clients; GitHub, Stripe and Shopify deliver events by webhooks; many dashboards and AI chat apps stream with SSE.

In the interview

Q: How would you push new chat messages to 10 million online users?

A: WebSockets terminated on a fleet of connection servers (perhaps 50,000–100,000 connections each). A registry maps user → connection server, and a pub/sub layer delivers each message to the right server, which writes it to the socket. Clients reconnect with backoff and fetch missed messages by last seen ID.

Q: When would you choose SSE over WebSockets?

A: When only the server pushes (notifications, live scores, streaming text): SSE is plain HTTP, reconnects automatically, needs no protocol upgrade.

TL;DR

Polling asks repeatedly, long polling waits for news, WebSockets keep a two-way channel, SSE streams one-way from server to client, webhooks are server-to-server callbacks. Persistent connections are fast but make servers stateful.

API design basics

What it is

The practical rules that make an API pleasant and safe to evolve: versioning, paging through big lists, telling clients about limits and errors.

Why it matters

Once thousands of clients use a public API you cannot change it freely; pagination and rate limits protect servers from clients asking for too much at once.

How it works

Versioning. For a breaking change (removing or renaming a field, changing meaning), publish a new version and keep the old one running for a deprecation period.

- **URL versioning**: `/v1/orders`, `/v2/orders`. Most common, very visible.
- **Header versioning**: `Accept: application/vnd.example.v2+json` or a custom header. Cleaner URLs, harder to test in a browser.
- **Date-based versions**: Stripe pins each account to a dated version and lets clients opt in to newer ones.

Prefer **additive** changes (new optional fields) that need no new version, and tell clients to ignore unknown fields.

Pagination. Never return an unbounded list.

- **Offset pagination**: `GET /posts?limit=20&offset=40`. Simple, allows "jump to page 5", but the database still reads and skips the first 40 rows (slow at offset 1,000,000), and inserts while paging shift items, causing duplicates or misses.
- **Cursor (keyset) pagination**: `GET /posts?limit=20&after=eyJpZCI6MTIzNH0`. The cursor is an opaque token encoding the last item's sort key (`id=1234`); the server runs `WHERE id < 1234 ORDER BY id DESC LIMIT 20`, which uses an index, is equally fast on page 1 and page 50,000, and stays stable as new items arrive. No jumping to an arbitrary page. Feeds and timelines (Twitter/X, Facebook, Slack) use cursors.

	Offset	Cursor
Jump to page N	Yes	No
Speed deep into list	Gets slower	Constant
Stable under inserts	No	Yes
Best for	Admin tables, small data	Feeds, large or changing data

Rate-limit headers. Tell limited clients (say 5,000 requests per hour) where they stand with `X-RateLimit-Limit` (quota), `X-RateLimit-Remaining` and `X-RateLimit-Reset` (GitHub's headers; the IETF is standardising unprefixed `RateLimit` headers), and answer excess with **429 Too Many Requests** plus `Retry-After` (seconds to wait). Example response and algorithms: *Rate limiting and throttling* (Part 5).

Other basics. Consistent error bodies (machine-readable code plus human message), idempotency keys on POST, filtering and sorting via query parameters (`?status=paid&sort=-created_at`), and documentation as an **OpenAPI** (formerly Swagger) specification.

Common mistakes

- Breaking changes without a version bump.
- Offset pagination on an infinite-scroll feed with millions of rows.
- Returning 500 for bad input (that is a 400) or hiding rate limiting behind 503.

TL;DR

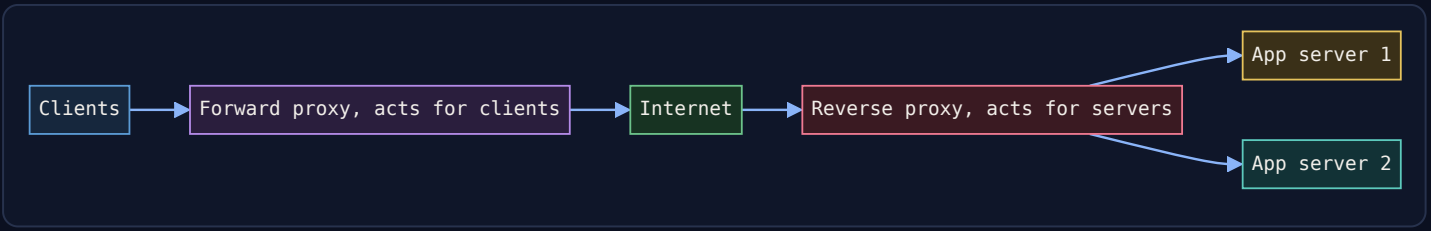
Version only for breaking changes, prefer cursor pagination for big or changing lists, and communicate limits with rate-limit headers and 429 + Retry-After.

Proxies and API gateways

What it is

A **proxy** is a middleman that receives a request and forwards it on someone else's behalf.

- A **forward proxy** acts for **clients**: they send all traffic through it and the internet sees the proxy, not the client. Used to filter and log employee browsing, by VPN-like services (virtual private networks tunnelling your traffic through another machine) to hide the user's IP, and to cache outbound downloads.
- A **reverse proxy** acts for **servers**: clients think they talk to the website, but the proxy receives the request and forwards it to one of the hidden servers behind it. Used for load balancing, TLS termination, caching, compression and hiding internal structure.



An **API gateway** is a reverse proxy specialised for APIs: the single front door to many backend services (often microservices), handling cross-cutting concerns once instead of in every service.

Why it matters

Reverse proxies and gateways hold shared concerns — authentication, rate limiting, TLS, logging, routing by path — and let you change the backend (split a monolith, move a service) without clients noticing.

How it works

An API gateway receiving `GET /orders/7` typically:

1. Terminates TLS.
2. Authenticates the caller (API key, a JWT — JSON Web Token, a signed token carrying the user's identity — or an OAuth access token from a login service).
3. Applies per-client rate limits and quotas.
4. Routes by path, host or header: `/orders/*` → order service, `/users/*` → user service.
5. Optionally transforms the request (REST to gRPC, adds a request ID header) or **aggregates** several backend calls into one response.
6. Records logs, metrics and traces, then returns the response.

A variant is **BFF (Backend for Frontend)**: a separate gateway per client type (web, mobile) that shapes responses for that client.

Types

Component	Main job	Examples
Forward proxy	Act for clients, filter, hide identity	Squid, corporate proxies
Reverse proxy	Act for servers, TLS, caching, routing	NGINX, HAProxy, Envoy
Load balancer	Spread traffic over healthy servers	AWS ALB/NLB, HAProxy
API gateway	Auth, rate limits, routing for APIs	AWS API Gateway, Kong, Apigee

The lines blur: NGINX is a reverse proxy, load balancer and basic gateway at once. A load balancer is a reverse proxy whose main job is distribution; an API gateway one whose main job is API policy.

Trade-offs

Every proxy adds a network hop (well under 1 ms inside a data centre, more with heavy logic) and is a potential bottleneck and single point of failure, so it must be redundant and scaled. A gateway that accumulates business logic becomes a hard-to-change "god service".

Common mistakes

- Confusing forward and reverse proxies. Ask: does it represent the client or the server?
- Putting business rules in the gateway. Keep it to cross-cutting policy.

- Losing the client's real IP: proxies must pass it in `X-Forwarded-For` (or the PROXY protocol for L4), and backends must trust that header only from known proxies.

In the interview

Q: What is the difference between a reverse proxy, a load balancer and an API gateway?

A: All sit in front of servers. A reverse proxy is the general idea (forwarding on behalf of servers); a load balancer focuses on spreading traffic across healthy instances; an API gateway on API concerns — auth, rate limits, routing to many services, transformation. One product often plays all three roles.

TL;DR

A forward proxy represents clients, a reverse proxy represents servers, and an API gateway is a reverse proxy enforcing API policy (auth, rate limits, routing) at the front door.

Load balancers

What it is

A **load balancer (LB)** receives incoming traffic and spreads it across a group of servers (a **pool**, **backend set** or **target group**), sending each request or connection only to healthy servers. Clients see one address; behind it may be 3 or 3,000 servers — like a restaurant host who seats each party with a free waiter and skips the one who went home sick.

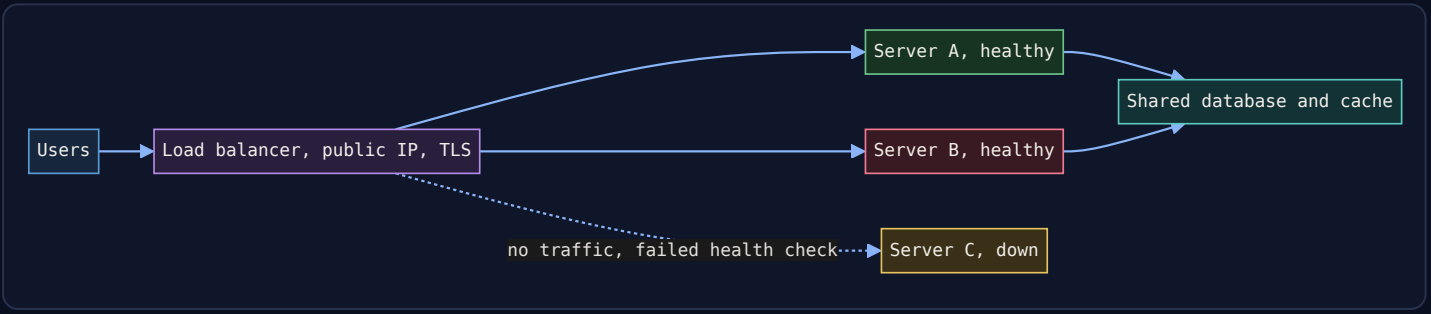
Why it matters

The load balancer makes **horizontal scaling** (Part 1) work:

- **Scalability** — add servers to the pool; capacity grows with no change for clients.
- **Availability** — health checks notice a dead server and traffic flows to the rest; a pool behind an LB is not a single point of failure.
- **Zero-downtime deployments** — take servers out one by one, update, put back.
- **Central place for shared work** — TLS termination, compression, request logging, basic security.
- **Hiding the backend** — servers keep private IPs; only the LB is public.

Almost every design has an LB in front of each tier of stateless servers, and often internal ones between services.

How it works



1. DNS resolves `api.example.com` to the LB's address (a **virtual IP**, VIP — an address belonging to the service, not one machine).
2. The client connects to the LB. An L7 LB terminates TLS and reads the HTTP request; an L4 LB sees only the TCP or UDP connection.
3. The LB picks a backend with its **algorithm** (below), among servers currently passing **health checks**.
4. It forwards the request (L7, over its own new or reused backend connection) or the connection's packets (L4).
5. The response flows back through the LB (in most designs; see "direct server return").
6. The LB records metrics: requests per second, errors, latency per backend.

Since any server may receive any request, servers behind an LB should be **stateless**, with sessions and data in a shared cache or database.

Types

By network layer: L4 vs L7

Layer 4 (transport) LB. Works on TCP/UDP connections by IP and port. It picks a backend when a connection opens, then shuttles that connection's packets (rewriting addresses NAT-style, or proxying the TCP stream) without looking inside — no URLs, headers or cookies; with TLS only encrypted bytes (unless it terminates TLS itself, which some L4 products can).

- Pros: very fast (microseconds of overhead), millions of connections, any protocol (databases, MQTT messaging for IoT devices, game servers, raw TCP), cheap.
- Cons: no content-based routing; balancing is per connection, so one long-lived connection carrying many requests (HTTP/2, gRPC) is stuck on one backend.

Layer 7 (application) LB. Terminates the client connection, parses the HTTP request, then makes a separate request to the chosen backend. It can:

- Route by path (`/api` → API servers, `/images` → image servers), host (`shop.example.com` vs `blog.example.com`), headers, cookies or query string.
- Balance **per request**, spreading many HTTP/2 or gRPC requests on one client connection across backends.
- Terminate TLS, compress, cache, retry, redirect (HTTP → HTTPS), add headers like `X-Forwarded-For` and request IDs, enforce auth and rate limits, block malicious patterns (web application firewall, WAF, features).
- Pros: smart routing and rich features. Cons: more CPU and latency per request (typically still around a millisecond), must hold TLS keys, HTTP-family protocols only.

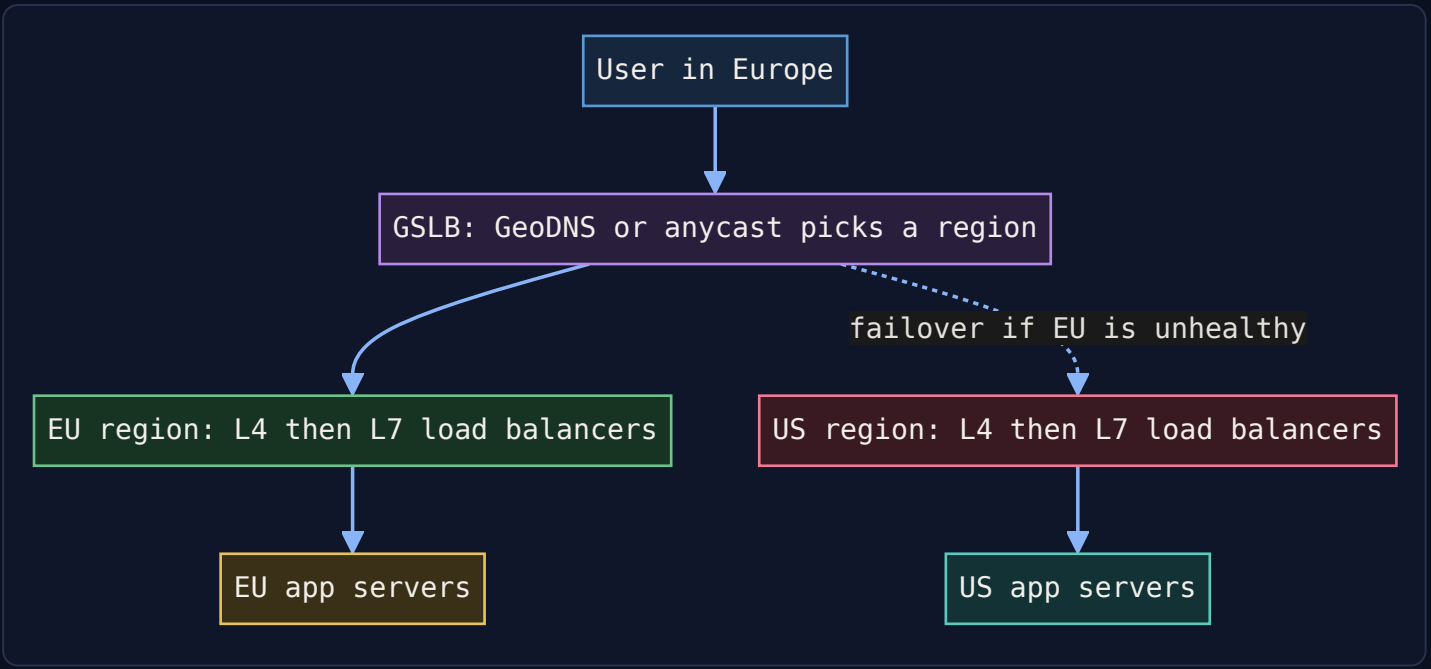
	L4 load balancer	L7 load balancer
Sees	IPs, ports, TCP/UDP	Full HTTP request
Decides per	Connection	Request
Routing by URL, header, cookie	No	Yes
Speed and cost	Faster, cheaper	More CPU per request
Protocols	Any TCP/UDP	HTTP, HTTP/2, gRPC, WebSocket
AWS example	NLB	ALB

A common layered setup: an L4 balancer in front (cheap, huge scale, static IPs) feeding a fleet of L7 proxies (smart routing), which feed the app servers.

By form: hardware, software, cloud, DNS, global

- **Hardware LBs** — dedicated appliances with specialised chips (F5 BIG-IP, Citrix NetScaler, now NetScaler ADC). Very high throughput and vendor support, but expensive (tens of thousands of dollars each), scaled by buying boxes, slow to change. Common in traditional enterprises and banks.
- **Software LBs** — programs on ordinary servers: **NGINX**, **HAProxy**, **Envoy**, Traefik. Cheap, flexible, configurable as code, scale by adding instances. Hyperscalers built software L4 balancers too: Google's **Maglev** and Meta's **Katran**, on commodity Linux servers.
- **Cloud (managed) LBs** — the provider runs and scales the fleet: AWS **ALB** (L7), **NLB** (L4) and **Gateway Load Balancer** (routing through firewalls and appliances); Google Cloud Load Balancing; Azure Load Balancer (L4) and Application Gateway (L7). Pay per hour plus traffic; automatic scaling and high availability across zones.
- **DNS load balancing** — returning different IPs (round robin, weighted, GeoDNS, latency-based; DNS section). Coarse and cached, but global with no extra hop.
- **Global server load balancing (GSLB)** — distributing users across **data centres or regions** by geography, latency, region health and capacity, failing over whole regions. Built from DNS-based steering (Route 53, Cloudflare Load Balancing, NS1, F5 DNS) and/or **anycast** (one IP announced from

many locations; CDN section), as in Google Cloud's global LB and Cloudflare. Regular L4/L7 balancers take over inside each region.



Where it runs: client-side vs server-side

Most LBs are **server-side** (a separate box or service in the middle). In **client-side load balancing** each client gets the healthy-instance list from a service registry and picks one itself (gRPC clients, Netflix's Ribbon library, service-mesh sidecars like Envoy). This removes a hop and a bottleneck but puts logic in every client. Service discovery is in Part 5.

Load-balancing algorithms

The **algorithm** picks which healthy backend gets the next request or connection: **static** algorithms follow a fixed rule; **dynamic** ones use live load signals.

Round robin. Servers in turn: A, B, C, A, B, C… Simple and fair when servers are identical and requests cost about the same; blind to load, so a server stuck with heavy requests (a 10-second report) still gets its "turn".

Weighted round robin. Weights proportional to capacity: with A=3, B=1, A gets 3 of every 4 requests. For mixed server sizes (16-core next to 8-core) and for sending a small share to a new version (**canary release**: weight 1 new, 99 old).

Least connections. The server with the fewest active connections now. Good when request durations vary a lot or connections are long-lived (WebSockets, database proxies). **Weighted least connections** divides the count by each server's weight.

Least response time (least latency). Prefer the lowest recent response time, often combined with fewest active connections; routes around a server slowed by a noisy neighbour, garbage collection or a bad disk. Needs latency measurement and can over-react — everyone rushes to the fastest server, which then slows (**herding**) — so implementations smooth measurements: Envoy's and NGINX Plus's "least time"-style policies, and the **EWMA** (exponentially weighted moving average) balancer of some proxies.

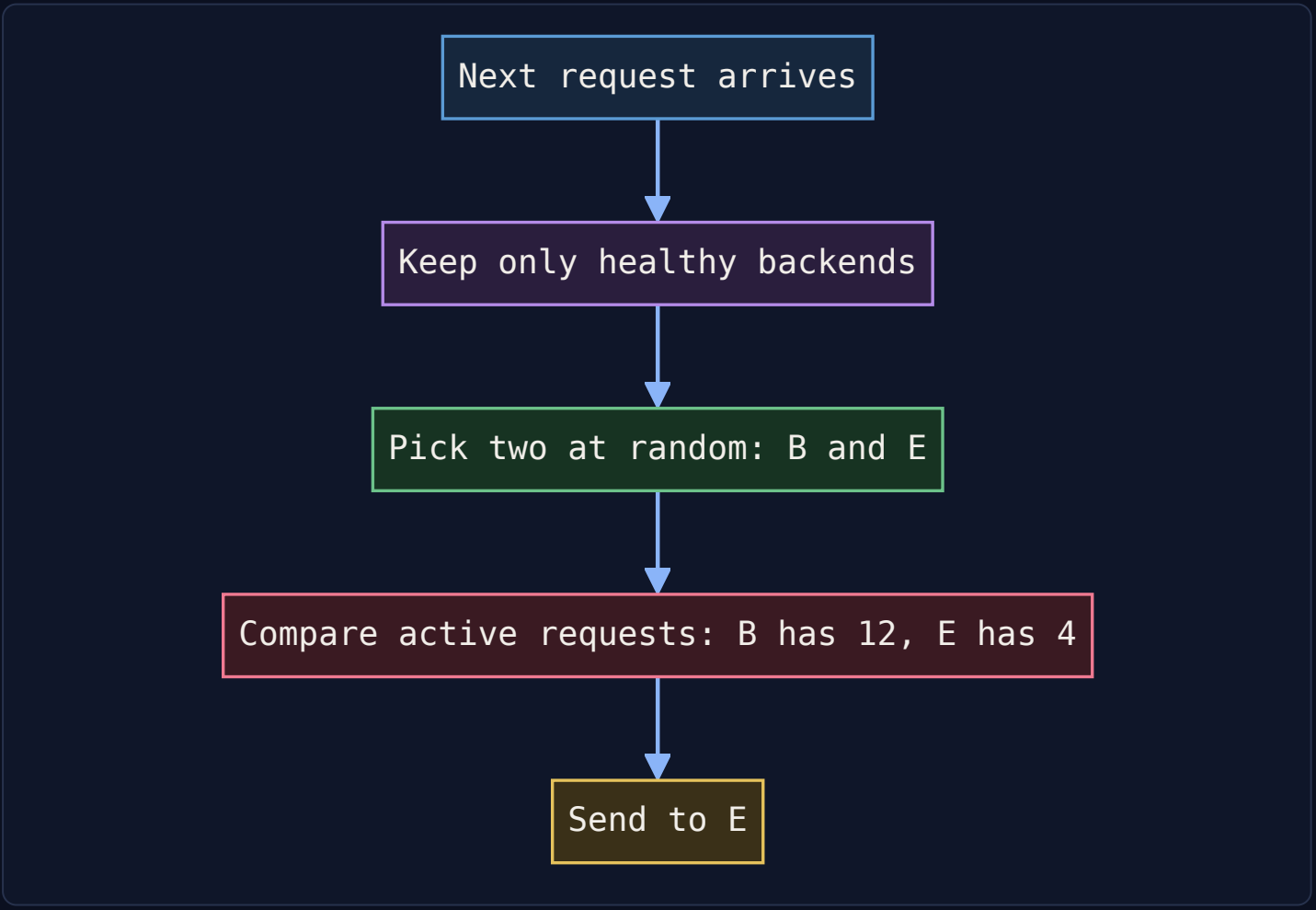
IP hash (source hash). `hash(client IP) mod N` picks the server, so the same client always lands on the same server — stickiness without cookies. Problems: many users behind one NAT hit one server, and when N changes almost every client is remapped.

Consistent hashing. Fixes the remapping: servers sit at several points (**virtual nodes**) on a hash ring; a key (client IP, user ID, URL) goes to the next server clockwise, so adding or removing a server moves only about 1/N of keys. Used when the *same key should reach the same server* so its local cache stays useful (caching proxies, sharded caches, chat servers). Google's Maglev uses its own variant; HAProxy and Envoy offer **ring hash** and **Maglev** policies. Full explanation: *Consistent hashing and virtual nodes* (Part 3).

Random. Evens out statistically over many requests, needs no state shared between LB instances, and is surprisingly good; weakness is short-term imbalance.

Power of two choices (P2C). Pick **two** servers at random and send to the less loaded (fewest active requests or lowest latency). Mitzenmacher's "two choices" analysis shows this sharply reduces worst-case load versus pure random, while avoiding herding: many independent LB instances with slightly stale data won't all pick the same "best" server. Envoy's least-request policy uses P2C by default (Envoy's overall default is round robin); NGINX supports `random two least_conn`.

Others: least bandwidth (fewest bytes per second), **resource-based / adaptive** (backends report CPU or load via an agent), **URL hash** (same URL to the same cache server), **locality-aware** (prefer same-zone backends to save cross-zone latency and cost).



Algorithm	Uses live load?	Sticky by key?	Best for
Round robin	No	No	Identical servers, similar requests
Weighted round robin	No	No	Mixed server sizes, canaries
Least connections	Yes	No	Long or uneven requests, WebSockets
Least response time	Yes	No	Avoiding slow servers
IP hash	No	Yes, by client IP	Simple stickiness, fixed pool
Consistent hashing	No	Yes, by any key	Cache affinity, changing pool
Random	No	No	Many LB instances, simplicity
Power of two choices	Yes, sampled	No	Large fleets, distributed LBs

Health checks

- **Active health checks** — the LB probes each backend on a schedule: a TCP connect (L4) or HTTP `GET /health` expecting 200 (L7). Typical: every 10 s, timeout 5 s, **unhealthy after 2–3 consecutive**

failures, **healthy again after 2–5 consecutive successes**, so one blip doesn't flap a server in and out.

- **Passive health checks (outlier detection)** — the LB watches real traffic; a backend returning several 5xx errors in a row or timing out is ejected for a while (say 30 s), then retried. Envoy calls this outlier detection; NGINX uses `max_fails` and `fail_timeout`.

Combine both, and make the health endpoint meaningful but cheap. A **liveness** check says "the process is running"; a **readiness** check says "I can serve traffic now" (caches warm, database reachable). Don't make health checks depend deeply on shared dependencies: if the database blips and every check fails, the LB removes all servers and turns a partial problem into a total outage. So many LBs **fail open** — if every backend looks unhealthy they send traffic to all of them (AWS target groups, Envoy's "panic threshold").

Sticky sessions (session affinity)

Sticky sessions send all of one user's requests to the same backend, via a **cookie** (the LB's own, such as AWS ALB's `AWSALB`, or the app's session cookie) or by **hashing the client IP** (L4 or L7).

Use it when the server holds in-memory state (an old app storing sessions locally, a WebSocket connection, an in-memory cart) or a warm per-user cache. Costs: uneven load (heavy users stuck on one server), lost sessions when a server fails or scales in, harder deployments. Better: stateless servers with session data in a shared store like Redis or a signed client token. Use stickiness only where state is unavoidable (WebSockets) — and even then design for reconnecting elsewhere.

SSL/TLS termination

TLS termination: the LB decrypts HTTPS and forwards plain HTTP to backends inside the private network.

- **Benefits**: certificates live and renew in one place; backends save handshake CPU; the LB can read requests for L7 routing, caching and WAF rules.
- **Risk**: LB-to-server traffic is unencrypted — acceptable inside a trusted private network for many companies, not for regulated data or zero-trust designs.

Variants: **TLS re-encryption (bridging)** — decrypt, inspect, open a new TLS connection to the backend (secure end to end, more CPU). **TLS passthrough** — the LB (typically L4) doesn't decrypt; the backend terminates TLS and keys stay on the servers. No content routing, but it can still route by the SNI host name visible in the ClientHello.

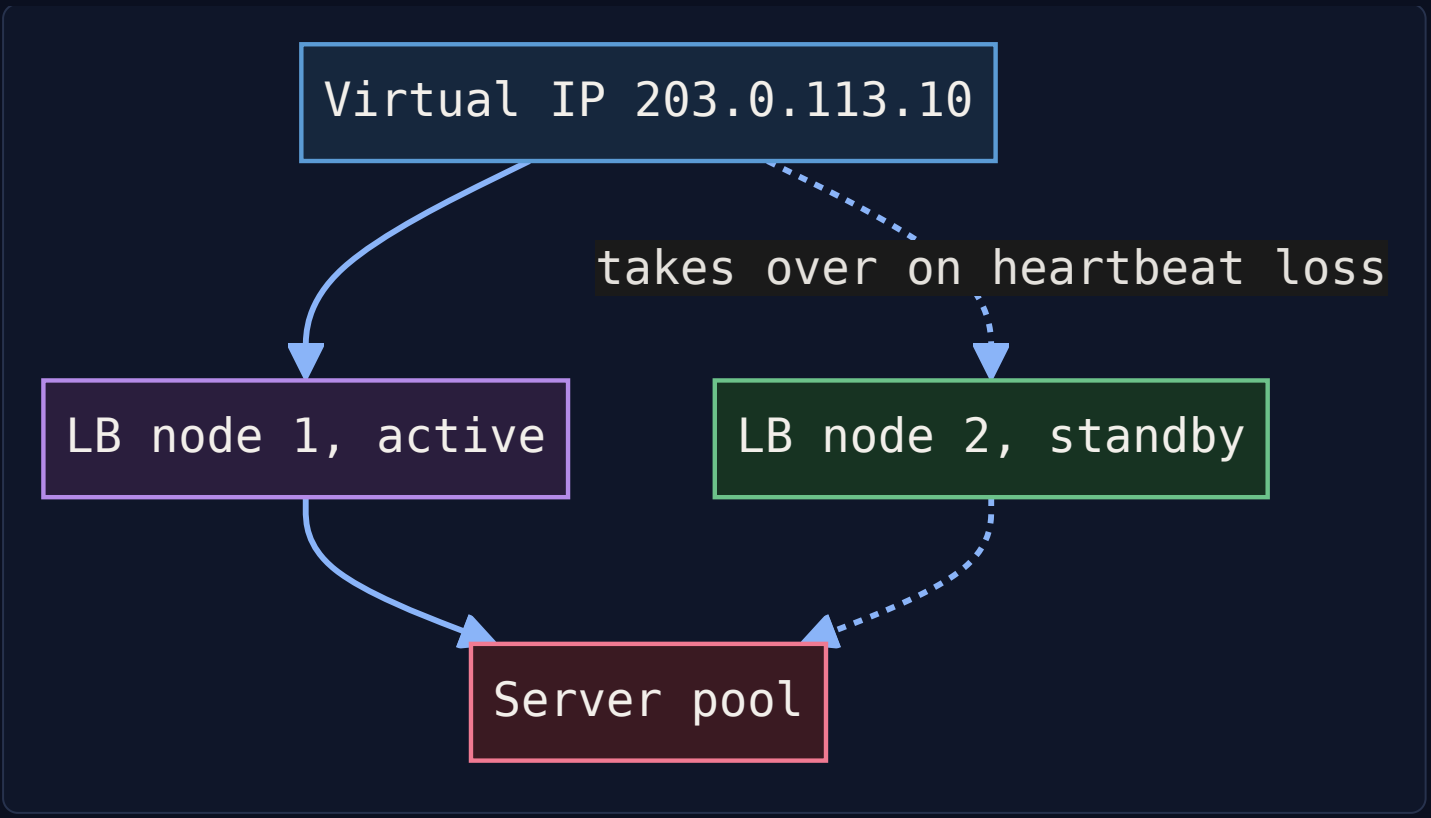
Connection draining (deregistration delay)

Removing a server (deployment, scale-in, maintenance) instantly would break in-flight requests. **Connection draining** stops **new** requests to it but lets **existing** ones finish, up to a timeout — AWS's **deregistration delay** (default 300 s, often lowered to 30–60 s for short API requests). Long-lived WebSockets may never finish, so the server should also tell clients to reconnect elsewhere. Draining plus health checks make **rolling deployments** invisible to users.

Making the load balancer itself highly available

A single LB is a single point of failure. Options:

- **Active–passive** — two LB nodes share one virtual IP; the passive one watches the active via heartbeats and takes over the VIP if it dies (commonly **VRRP**, the Virtual Router Redundancy Protocol, via `keepalived`). Simple, but half the hardware idles and failover takes a second or more.
- **Active–active** — all nodes serve at once, traffic spread by DNS (multiple IPs), anycast, or **ECMP** (equal-cost multi-path routing: routers spread flows across several LB machines announcing the same IP). No idle standby; each node must absorb the others' load if they fail. This is how Maglev, Katran and cloud LBs work.
- **Managed cloud LBs** do it for you: AWS ALB and NLB run nodes in every enabled Availability Zone, and NLB offers one static IP per zone.



Direct server return (bonus detail)

Responses are usually much larger than requests. In **direct server return (DSR)** an L4 LB forwards incoming packets to the server, which replies **straight to the client**, bypassing the LB; handling only the small inbound half, one LB fronts far more traffic. Used by Maglev, Katran and some LVS (Linux Virtual Server) setups.

Trade-offs

Decision	Option A	Option B
Layer	L4: fast, any protocol, blind to content	L7: smart routing, per-request, more CPU
Form	Managed cloud LB: no operations, pay per use	Self-run NGINX/HAProxy/Envoy: full control, you operate it
TLS	Terminate at LB: simpler, inspectable	Passthrough or re-encrypt: end-to-end encryption
Algorithm	Static (round robin): predictable, simple	Dynamic (least conn, P2C): adapts to load
Sessions	Sticky: supports local state	Stateless: even load, easy failover

Common mistakes

- Drawing a single LB box and forgetting it needs redundancy.
- Round robin for long-lived WebSocket connections; least connections is usually better.
- Sticky sessions as a substitute for a shared session store.
- Health checks that only test "process is up" (it may be unable to reach its database) — or, the opposite, deep checks that fail the whole fleet at once.
- An L4 LB in front of gRPC, leaving one backend overloaded: gRPC multiplexes many requests on one long connection, so you need per-request (L7 or client-side) balancing.
- Forgetting backends see the LB's IP, not the client's, unless `X-Forwarded-For` or the PROXY protocol is used.

Real-world examples

- **NGINX** — web server and L7 reverse proxy/LB; round robin by default, also least connections, IP hash, generic hash with consistent option, random with two choices.

- **HAProxy** — very fast L4/L7 LB; round robin, static round robin, least connections, source hash, URI hash, consistent hashing; used at GitHub, Stack Overflow and many others.
- **Envoy** — L7 proxy created at Lyft, the data plane of service meshes like Istio; round robin, least request (P2C), ring hash, Maglev, random, plus outlier detection and automatic retries.
- **AWS Application Load Balancer (ALB)** — L7; routes by path, host, headers and query; HTTP/2, gRPC and WebSockets; round robin, least outstanding requests or weighted random; cookie stickiness.
- **AWS Network Load Balancer (NLB)** — L4 TCP/UDP/TLS; millions of requests per second; static IP per zone; preserves the client source IP.
- **Google Maglev** and **Meta Katran** — software L4 balancers on commodity servers, using consistent hashing and ECMP to scale out.

In the interview

Q: What is the difference between an L4 and an L7 load balancer, and when would you pick each?

A: L4 balances TCP/UDP connections by IP and port without reading content: fast, cheap, any protocol. L7 parses HTTP: routes by path or header, balances per request, terminates TLS, applies policies. L7 in front of HTTP services and gRPC; L4 for non-HTTP protocols, extreme throughput, or as the first tier in front of L7 proxies.

Q: Which algorithm for identical stateless API servers? For a cache tier?

A: API servers: least outstanding requests or power of two choices, adapting to uneven request cost and slow servers. Cache tier: consistent hashing on the cache key, so each key hits the same node and adding a node moves only ~1/N of keys.

Q: How do you deploy a new version without dropping requests?

A: Rolling deployment: per server, deregister with connection draining, update, wait for readiness checks, re-register. For extra safety, canary the new version on a small weighted share of traffic first.

Q: Isn't the load balancer a single point of failure?

A: Only if there is one. Run LB nodes active–passive with a floating virtual IP, or active–active behind DNS, anycast or ECMP, across availability zones. Managed cloud LBs already do this.

TL;DR

A load balancer spreads traffic across healthy servers so you can scale out and survive failures. Know L4 vs L7, the main forms, the algorithms (round robin to power of two choices), health checks, stickiness, TLS termination, draining, and how to make the LB itself redundant.

CDNs and anycast

What it is

A **CDN (Content Delivery Network)** is a worldwide network of caching servers that keep copies of your content close to users: a user in Sydney fetches an image from a Sydney CDN server (a few milliseconds) instead of your server in Virginia (~200 ms round trip).

Key terms:

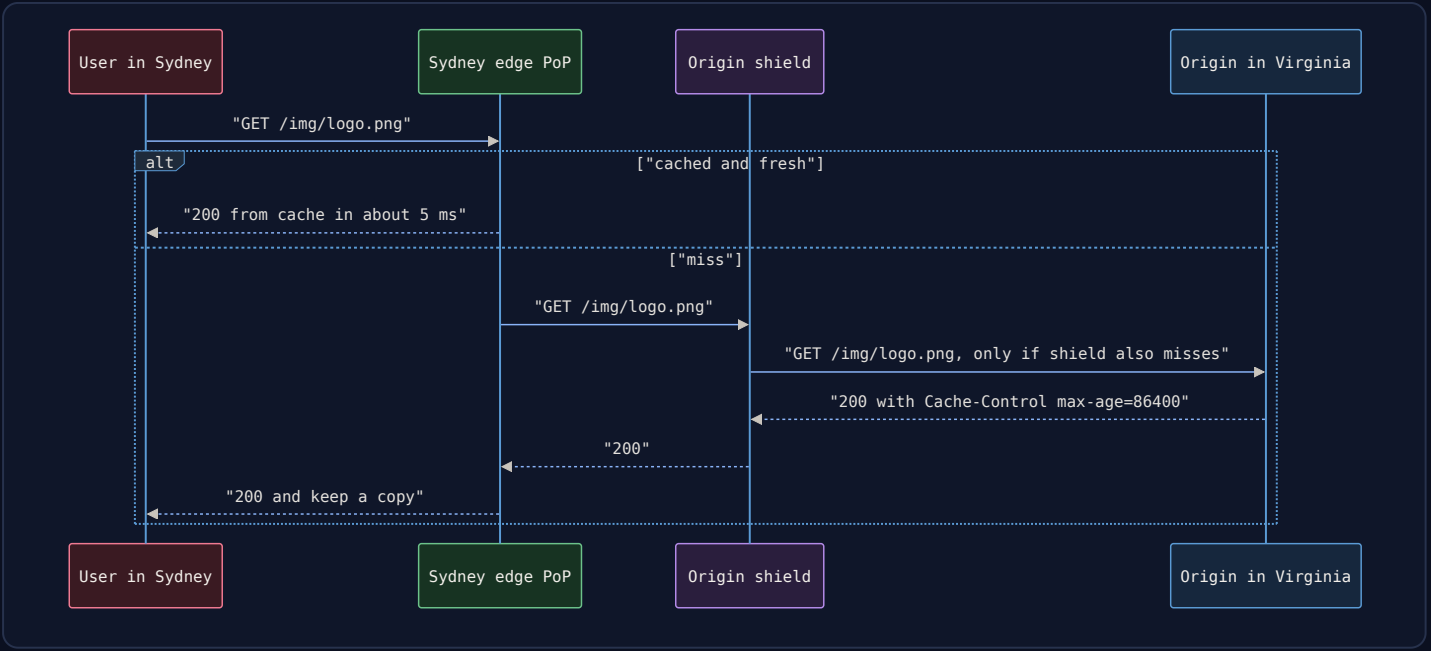
- **Origin** — your real server (or storage like Amazon S3) holding the master copy.
- **Edge server** — a CDN cache server near users.
- **PoP (Point of Presence)** — a CDN site in a city containing many edge servers; large CDNs have hundreds.
- **Cache hit** — the edge has a fresh copy and answers directly. **Cache miss** — it must fetch from the origin (or an upper tier) first.
- **Cache hit ratio** — hits ÷ total requests; 95% means only 1 in 20 requests reaches the origin.

Why it matters

- **Lower latency** — static files (images, JavaScript, CSS, video) come from nearby, and TLS handshakes finish against a nearby edge, shortening every round trip.

- **Origin offload** — at 95% hit ratio the origin handles 20× less traffic: far fewer servers and less egress bandwidth.
- **Absorbing spikes and attacks** — the CDN's huge capacity soaks up surges and DDoS (distributed denial-of-service) attacks, where many machines flood a site with junk traffic.
- **Video and large downloads** at global scale would be impossible from one origin.

How it works



1. DNS for `cdn.example.com` returns a nearby edge (via GeoDNS or anycast).
2. The edge looks up the URL in its cache (the **cache key**: usually host + path + some query parameters and headers).
3. On a fresh hit, it answers immediately.
4. On a miss, it fetches from the origin, stores the copy for as long as the **TTL** allows, and returns it.
5. The origin controls caching with HTTP headers: `Cache-Control: public, max-age=86400` (one day), `s-maxage` (TTL for shared caches like CDNs only), `no-store` (never cache), and validators (`ETag`, `Last-Modified`) so an expired copy can be re-checked cheaply with a 304 Not Modified.

Types

Pull CDN vs push CDN.

- **Pull (origin pull)** — the CDN fetches from the origin lazily on the first request; zero upload effort, only requested content cached. The first requester at each PoP waits for the origin trip. Most websites use pull.
- **Push** — you upload content to the CDN's storage ahead of time; good for large files you know will be requested (game patches, video libraries) and rarely-changing content. You manage what is stored and pay for storage.

	Pull CDN	Push CDN
How content arrives	Fetched on first miss	Uploaded in advance
Effort	Low	You manage uploads
First request	Slower (miss)	Fast
Best for	Websites, mixed traffic	Big known files, video

Invalidation (purging). When content changes before its TTL expires:

1. **Purge** — ask the CDN to delete a URL, prefix or tag from all edges (seconds to minutes).

2. **Versioned file names (cache busting)** — publish `app.3f9a1c.js` instead of overwriting `app.js`; the HTML references the new name, so nothing needs invalidating. Standard for static assets, and allows very long TTLs (a year) safely.
3. **Short TTLs** for often-changing content, accepting more origin traffic.

Origin shield (tiered caching). An extra cache layer between edges and origin. Without it, a new globally popular file misses at each of hundreds of PoPs, each hitting the origin; with it, edges miss to the shield and the origin sees roughly one request per object — higher hit ratio, protected origin.

Static vs dynamic content. CDNs mostly cache static, shared content. For personalised or dynamic responses they still help by terminating TLS near the user over optimised, warm connections back to the origin, and some run code at the edge (Cloudflare Workers, Lambda@Edge, Fastly Compute).

Anycast. Normally one IP belongs to one location (**unicast**). With **anycast** the same IP is announced from many locations via **BGP** (Border Gateway Protocol — how internet networks tell each other which IP ranges they can reach), and routers send each user to the "nearest" location in network terms. Benefits: automatic proximity routing without DNS tricks, DDoS traffic naturally spread across all sites, and fast failover (a site that stops announcing has its routes shift to the next). Cloudflare's network, Google's `8.8.8.8`, Cloudflare's `1.1.1.1` and the DNS root servers all use anycast. "Nearest" means the best BGP path by routing policy (often fewest networks crossed) — usually, not always, the lowest-latency site.

Trade-offs

Long TTLs: high hit ratio but stale content; short TTLs: freshness but more origin load. CDNs cost money per GB served, and caching anything personalised under a wrong cache key risks leaking one user's data to another.

Common mistakes

- Caching user-specific responses under a shared key (use `Cache-Control: private` or vary the key).
- Overwriting files in place with long TTLs, then being unable to update them. Use versioned file names.
- Forgetting the origin must survive a full cache flush (a "cold" CDN); the origin shield and gradual warm-up help.

Real-world examples

Akamai, Cloudflare, Fastly, Amazon CloudFront, Google Cloud CDN. Netflix runs its own CDN, **Open Connect**, placing caching appliances inside internet providers' networks so most video bytes never cross the wider internet; YouTube uses Google's global cache network similarly.

In the interview

Q: How does a CDN reduce latency and origin load?

A: It serves cached copies from PoPs near users, cutting round-trip time; at a high hit ratio most requests never reach the origin, and an origin shield collapses many edge misses into one origin request.

Q: You deploy new JavaScript and some users still get the old one. Why, and the fix?

A: Edges and browsers serve the cached file until its TTL expires. Use content-hashed file names so each version has a new URL, keep long TTLs, and purge only in emergencies.

TL;DR

A CDN caches content at edge PoPs near users to cut latency and offload the origin. Pull fetches on demand, push uploads ahead; manage freshness with TTLs, versioned names and purges; an origin shield protects the origin; anycast routes users to the nearest site using one IP.

Bandwidth vs latency, and network partitions

What it is

From Part 1: **bandwidth** is how much data a link carries per second (1 Gbps ≈ 125 MB/s); **latency** is how long one piece of data takes from A to B. A wider road does not make the trip shorter.

A **network partition** is when the network splits so some machines cannot reach others, though all are still running — like a jammed corridor door: both sides keep working but can't talk.

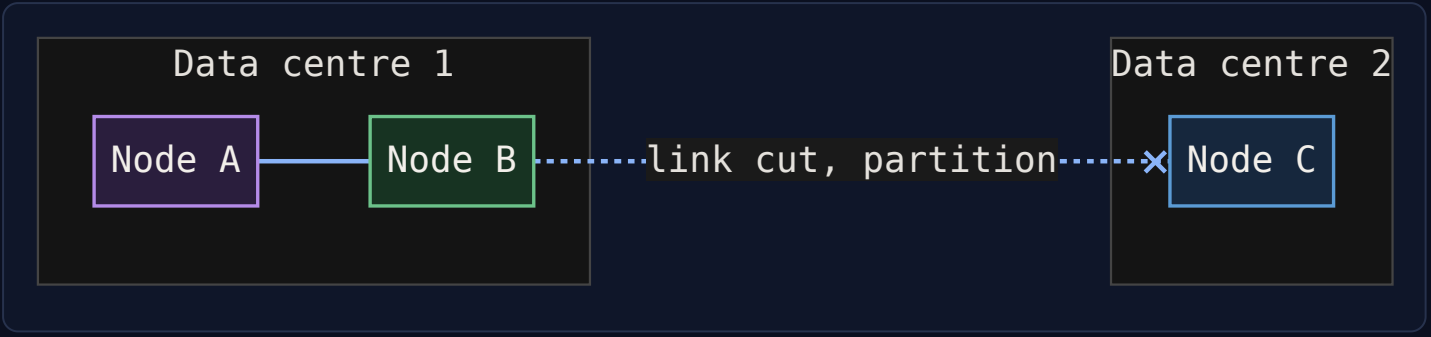
Why it matters

- Many designs are latency-bound: 10 sequential trans-Atlantic round trips take at least 700 ms however fast the connection. Fix with fewer round trips (batching, HTTP/2, connection reuse) and moving data closer (CDN, regional replicas).
- Large transfers are bandwidth-bound: 10 TB over 1 Gbps takes about 22 hours (10 × 8 × 10^12 bits ÷ 10^9 bits/s ≈ 80,000 s) — why cloud providers offer physical transfer devices (AWS Snowball).
- Latency has a physical floor, the speed of light in fibre (~200 km per millisecond, Part 1); no protocol trick removes it.
- **Bandwidth-delay product** (bandwidth × RTT) is the data that must be "in flight" to fill a link: 1 Gbps at 100 ms RTT needs 12.5 MB, which is why TCP window sizes and congestion control matter on long links.
- Partitions are not rare — a failed switch, a misconfigured firewall, a cut cable between data centres — and every distributed system must decide what to do during one (the CAP theorem, Part 4).

How it works

During a partition a node cannot tell "the other node crashed" from "it's fine but unreachable" — both are silence. Systems detect trouble with **timeouts** and **heartbeats**, then choose:

1. **Stay consistent** — refuse writes (or reads) on the side that can't reach a majority, so data never diverges; the minority side becomes unavailable.
2. **Stay available** — keep serving on both sides and reconcile after the network heals, accepting temporary inconsistency.



Nodes on both sides each believing they are leader cause **split brain** (two leaders accepting conflicting writes); quorum-based systems prevent it by requiring a majority (Parts 3 and 4).

Common mistakes

- Assuming "the network is reliable" — the first of the classic **fallacies of distributed computing** (others: latency is zero, bandwidth is infinite, the network is secure, topology does not change).
- Treating a timeout as proof the remote operation failed; it may have succeeded with only the reply lost — hence idempotent retries.
- Buying more bandwidth to fix a latency problem.

In the interview

Q: Our API is slow for users in Asia but fine in the US. More bandwidth?

A: Probably not — it's latency from distance multiplied by round trips. Cut round trips (keep-alive, HTTP/2 or HTTP/3, batching) and move things closer: CDN for static content and edge TLS termination,

then regional deployments or read replicas.

Q: What is a network partition and why can't you just avoid it?

A: A break in communication between groups of healthy nodes. Hardware and configuration failures make it unavoidable at scale, so the design must choose between consistency and availability while it lasts.

TL;DR

Bandwidth is capacity, latency is delay — round trips usually dominate. Partitions split healthy nodes, can't be avoided, and force a choice between consistency and availability.

Part 2 glossary

Every Part 2 term is defined in one line in the Master A–Z glossary, which merges the glossaries of all six parts into a single alphabetical list.

Part 3 — Data storage

Almost every system exists to remember things: users, purchases, messages, photos. Code can be thrown away and redeployed in minutes; data cannot, and losing it may end the business. This part explains, from zero, where data lives, how it is organised, how it is kept safe when machines crash, and how it is spread over many machines once one is not enough. Read it in order: each section uses words defined before it (all collected in the master A–Z glossary).

What a database is

What it is

A **database** is a program that stores data, keeps it safe, and lets other programs find it again quickly. Properly the program is a **database management system (DBMS)**; "database" is used for both the program and its data. (Like a library: books shelved in agreed places, a catalogue — an index — to find one without walking every aisle, and a librarian making sure two people can't check out the same copy and nothing is lost if the lights go out mid-shelving.)

Plain files work for a tiny program, but with many concurrent users they give you none of these:

- **Fast lookup.** Finding one customer among 50 million by reading the whole file takes seconds; an index takes well under a millisecond.
- **Safe concurrent access.** Two servers writing the same file at once can corrupt it; a database coordinates them.
- **Crash safety.** If power fails mid-write, the write either fully happened or not at all.
- **A query language.** Ask "all orders over $100 from last week" and the database works out how.

Why it matters

App servers are **stateless**: they keep nothing important between requests, so you can add or remove them freely. All the important **state** (stored facts that must survive) lives in the database — the hardest part to scale, the slowest to change and the most expensive to get wrong. Interviews' deep questions usually land on the data model and database choice.

How it works

For each request a database:

1. **Parses** it (e.g. a SQL statement) and checks it is valid.
2. **Plans** the cheapest way to answer ("use the index on `email`", not "read every row"); this is the **query planner** (optimizer).
3. **Executes** the plan, reading data pages from memory if present (the **buffer cache**) or from disk.
4. For writes, **logs** the change to a durable file first, then applies it (Durability section).
5. **Returns** the result over the network.

A **read** fetches data without changing it; a **write** inserts, updates or deletes. Most consumer apps read far more than they write: 10:1 or even 100:1 is common for social feeds and product catalogues.

TL;DR

A database stores data safely, finds it quickly, and coordinates many users at once. It holds the state the rest of the system can afford to forget.

Relational databases and SQL

What it is

A **relational database** stores data in **tables**, like spreadsheets:

- Each **row** (record, tuple) is one thing, such as one user.
- Each **column** (field, attribute) is one property, such as `email`.
- The **schema** formally describes the tables: names, columns, and each column's **data type** (integer, text, date…). Data that doesn't fit is refused.

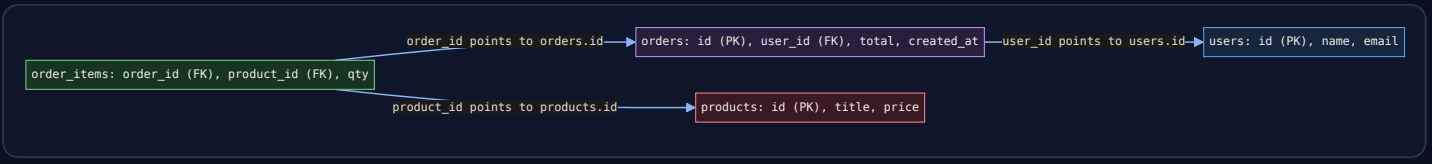
"Relational" comes from mathematics (a table is a "relation"), not from "relationships", though tables do relate through keys.

You talk to it in **SQL (Structured Query Language)**, which describes *what* data you want, not *how* to fetch it. Examples: PostgreSQL, MySQL, MariaDB, Oracle Database, Microsoft SQL Server, SQLite; in the cloud Amazon RDS and Aurora, Google Cloud SQL, Azure SQL Database.

How it works

Keys tie tables together (like an enrolment card that stores only the student's ID, not their address):

- A **primary key** is a column (or set) whose value is unique and never empty for every row — the row's identity (`users.id`).
- A **foreign key** holds the primary key of a row in another table (`orders.user_id` → `users.id`). The database can enforce that the referenced row exists — **referential integrity** — so you can't create an order for a non-existent user.
- A **unique constraint** says no two rows may share a value (no two users with the same `email`).



A **join** combines rows from two tables that match on a key. Each order with the buyer's name:

```sql
SELECT o.id, u.name, o.total
FROM orders o
JOIN users u ON u.id = o.user_id
WHERE o.created_at > '2026-01-01'
ORDER BY o.total DESC
LIMIT 10;
```

Join kinds: **inner** (only rows matching on both sides), **left** (every left row, empty values where the right has no match), **right** (the mirror) and **full outer** (every row from both sides).

`SELECT` reads; `INSERT`, `UPDATE` and `DELETE` change rows; `CREATE TABLE`, `ALTER TABLE` and `DROP TABLE` change the schema — **DDL (data definition language)**.

Why it matters

Relational databases have been the default for about 50 years because they give:

- A **flexible query language**: ask questions you didn't plan for when designing the tables.
- **Strong guarantees** (transactions and ACID, next) protecting money, inventory and anything that must add up.

- **Enforced structure**: bad data is rejected at the door.
- **Mature tools**: backups, monitoring, replication, decades of operational knowledge.

Trade-offs

- **Schema changes take care.** Adding a column to a 2-billion-row table can lock it or take hours without online migration tools (`gh-ost` , `pt-online-schema-change` for MySQL).
- **Scaling writes past one machine is hard.** Classic relational databases centre on one main server; sharding (later) loses cross-machine joins and transactions unless you use a distributed SQL database (Google Spanner, CockroachDB, YugabyteDB, TiDB).
- **Joins get expensive** on huge tables when they can't use an index.

Common mistakes

- Choosing NoSQL "for scale" when the data is a few hundred gigabytes that fit one PostgreSQL server. One modern server (64 cores, 512 GB memory, fast SSDs) handles tens of thousands of simple transactions per second.
- Storing a list as a comma-separated string in one column; it can't be indexed or joined. Use a separate table.
- A natural value such as an email as primary key. Emails change; IDs should not.

Real-world examples

Shopify, GitHub and Basecamp run large parts of their products on MySQL. Instagram began on PostgreSQL and sharded it as it grew. Stack Overflow runs one of the web's busiest sites on a handful of SQL Server machines.

TL;DR

A relational database stores data in tables with a fixed schema, links them with primary and foreign keys, and combines them with joins in SQL. It is the safe default for most systems.

Transactions and ACID

What it is

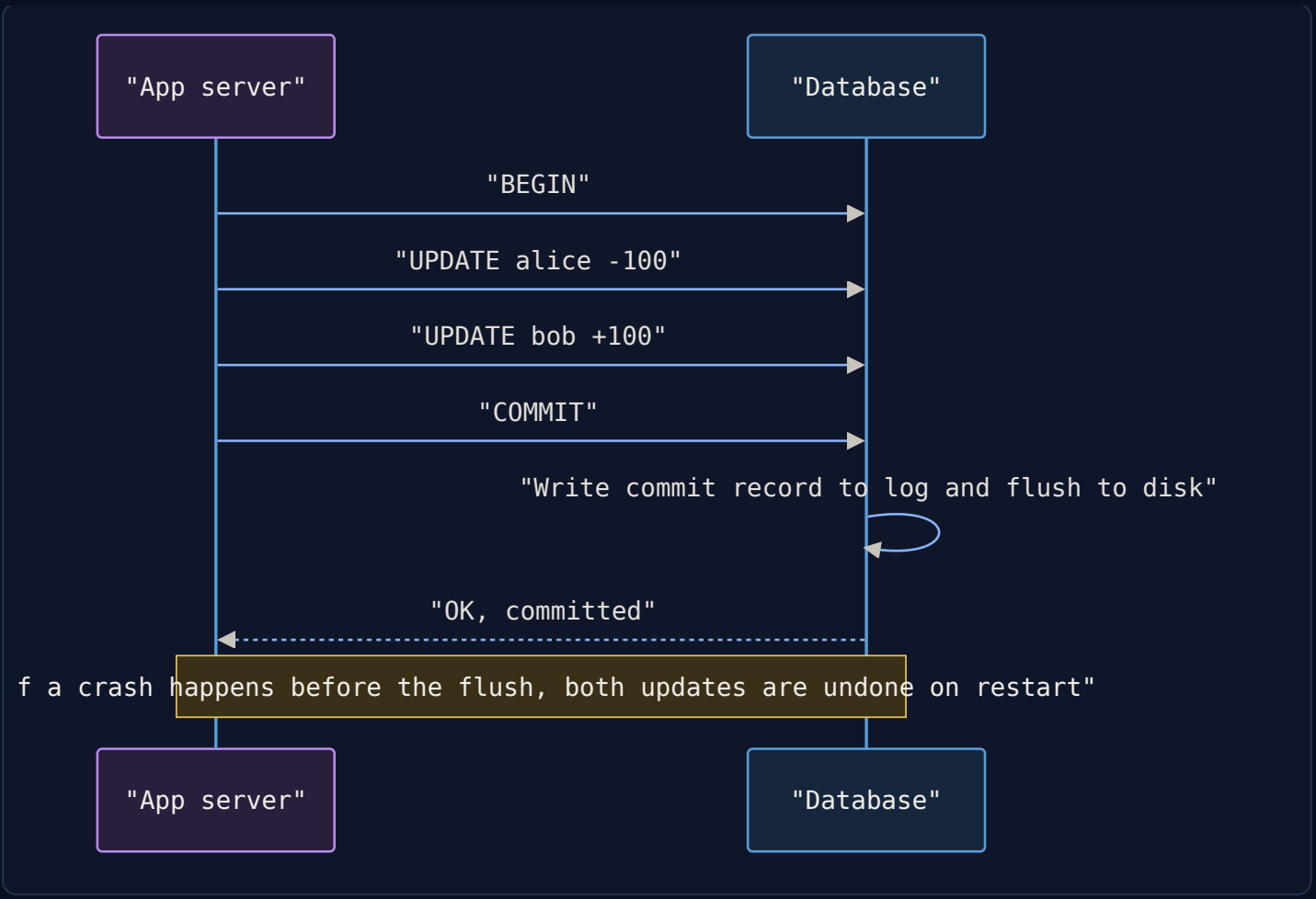
A **transaction** is a group of reads and writes treated as one unit: all take effect, or none do. Start with `BEGIN` , finish with `COMMIT` (make permanent) or `ROLLBACK` (discard everything).

Moving money is two steps — take $100 from Alice, add $100 to Bob; a crash in between makes $100 vanish. A transaction is a sealed envelope acted on only when both instructions are inside:

```
BEGIN;
UPDATE accounts SET balance = balance - 100 WHERE id = 'alice';
UPDATE accounts SET balance = balance + 100 WHERE id = 'bob';
COMMIT;
```

ACID names the four promises:

- **A — Atomicity.** All or nothing: if any step fails or the server crashes before `COMMIT` , every change is undone. "Atomic" means "cannot be split", nothing to do with speed or concurrency.
- **C — Consistency.** The transaction moves data between valid states, "valid" meaning the rules you declared: constraints, foreign keys, "balance must not go below zero" checks. Mostly the application's job (you define the rules; the database enforces only the ones you tell it). A different meaning from the "C" in CAP (Part 4).
- **I — Isolation.** Concurrent transactions don't see each other's half-finished work — ideally as if they ran one after another. In practice databases offer weaker **isolation levels** for speed (next section).
- **D — Durability.** Once "committed", data survives a crash or power cut: the change is written to a log on disk and forced to the physical device before replying (Durability section).



Why it matters

Without transactions every piece of code must handle "what if we crash halfway?" and "what if someone changed this row while I read it?" — rare, hard-to-reproduce bugs that cost real money: double charges, oversold tickets, negative stock.

Trade-offs

Stronger guarantees cost speed: isolation needs locks or version tracking, durability a disk flush per commit (often 0.1–2 ms on SSD, more on spinning or network disks). A transaction across several machines (a **distributed transaction**, Part 4) is far slower again, as the machines must agree over the network.

Common mistakes

- Holding a transaction open while calling a slow external service (a payment API): it keeps its locks and blocks others. Call outside the transaction.
- Assuming an "ACID" database gives full isolation by default. Most don't (next section).

In the interview

Q: What does ACID stand for, and which letters are the database's job versus the application's?

A: Atomicity, Consistency, Isolation, Durability. A, I and D are database mechanisms; consistency is mostly the application's rules, which the database helps enforce as declared constraints.

Q: Why not use transactions for everything across services?

A: Spanning several databases needs a coordination protocol such as two-phase commit, which is slow and blocks if the coordinator fails. Across services, use sagas and idempotent operations (Part 4).

TL;DR

A transaction groups operations so they all succeed or all fail. ACID: all-or-nothing, rules respected, no interference, nothing lost after commit.

Isolation levels and anomalies

What it is

Concurrency means transactions running at overlapping times. When they touch the same data, wrong results called **anomalies** (race conditions) can appear. An **isolation level** chooses which anomalies the database prevents; stronger levels prevent more but run slower or abort more transactions.

Types

The anomalies, simplest to subtlest:

- **Dirty read.** A reads a value B wrote but hasn't committed; if B rolls back, A acted on data that never officially existed.
- **Dirty write.** A overwrites B's uncommitted write. Almost every database prevents this at every level.
- **Non-repeatable read (read skew).** A reads a row, B updates and commits it, A re-reads and gets a different value in the same transaction.
- **Phantom read.** A queries "all bookings for room 7 on Friday" (2 rows); B inserts a matching booking and commits; A's same query now returns 3. The new row is the "phantom".
- **Lost update.** A and B both read a counter as 10, both add 1, both write 11; one increment is lost (should be 12).
- **Write skew.** A and B read the same rows, each decides something is safe, and each writes a *different* row. A hospital requires at least one doctor on call; two on-call doctors click "go off call" at once, each sees "two on call, fine for me to leave" and updates only its own row — zero on call. No row was written twice, so row locks don't catch it.

SQL-standard isolation levels, weakest to strongest:

Level	Dirty read	Non-repeatable read	Phantom read
Read uncommitted	Possible	Possible	Possible
Read committed	Prevented	Possible	Possible
Repeatable read	Prevented	Prevented	Possible (per standard)
Serializable	Prevented	Prevented	Prevented

Snapshot isolation (widely used, not in the original standard): each transaction reads from a consistent **snapshot**, a frozen picture of the database as of its start. It prevents dirty reads, non-repeatable reads and most phantoms, but still allows write skew. It is usually built with **MVCC (multi-version concurrency control)**: rows aren't overwritten; several versions are kept, each stamped with its creating transaction, so readers see the old version while a writer prepares a new one. Pay-off: a long report and a busy update stream run side by side without waiting on each other's locks.

Serializable: the result equals some one-at-a-time order of the transactions, preventing every anomaly above including write skew. Achieved by literally running one at a time (Redis, VoltDB), by **two-phase locking** (readers and writers take locks and hold them until commit), or by **serializable snapshot isolation (SSI)**, which runs optimistically and aborts a transaction when a dangerous pattern is detected (PostgreSQL).

How it works

Defaults that surprise people:

- **PostgreSQL**: *read committed*; its *repeatable read* is really snapshot isolation; *serializable* uses SSI.
- **MySQL (InnoDB)**: *repeatable read*, via MVCC snapshots plus some locking.
- **Oracle**: *read committed*; its "serializable" is really snapshot isolation.

Preventing lost updates and write skew without full serializability:

1. **Atomic operations**: `UPDATE counters SET n = n + 1 WHERE id = 7` does the read-modify-write in one step.
2. **Explicit locks**: `SELECT ... FOR UPDATE` locks the rows you read until you commit.

3. **Optimistic concurrency control**: keep a `version` column and write `UPDATE ... SET ..., version = 6 WHERE id = 7 AND version = 5`; zero rows updated means someone got there first, so retry. Also called **compare-and-set**.
4. **Constraints**: a unique index on `(room_id, date)` rejects the second double-booking.

Trade-offs

Higher isolation means more aborted transactions (to retry) or more lock waiting, so lower throughput. Most web apps run read committed or snapshot isolation and add explicit locks or constraints only where money or inventory is involved.

Common mistakes

- "Read the balance in the app, check it, write the new balance" with no lock or version check — a lost-update bug waiting to happen.
- Believing "repeatable read" means the same in every database.

In the interview

Q: Two users try to buy the last ticket at once. How do you stop overselling?

A: Make check and decrement one atomic step — `UPDATE tickets SET left = left - 1 WHERE id = 9 AND left > 0` and verify one row was affected — or use `SELECT ... FOR UPDATE`, or an optimistic version column with retry.

Q: What is write skew?

A: Two transactions read overlapping data, each decides based on it and writes a different row, so together they break a rule each alone respects. Snapshot isolation allows it; serializable prevents it.

TL;DR

Isolation levels decide which concurrency bugs the database blocks. Know the anomalies (dirty read, non-repeatable read, phantom, lost update, write skew) and that most databases default to something weaker than serializable.

NoSQL database families

What it is

NoSQL ("not only SQL") covers databases that don't use the relational table model as their main storage. The term spread around 2009, when Google (Bigtable), Amazon (Dynamo) and Facebook (Cassandra) published or open-sourced systems spreading huge data across hundreds of cheap machines.

There is no single "NoSQL design": each family is shaped around one access pattern and gives up something (often joins, flexible queries or strong transactions) to excel at it — specialist tools (pizza oven, espresso machine) versus the general-purpose kitchen.

Types

Key-value stores. A giant dictionary: `put(key, value)`, `get(key)`; the database never looks inside the value. Extremely fast (often well under 1 ms) and easy to spread across machines, since each key can live anywhere; weak at "find all items where X". Examples: Redis (in memory; also lists, sets, sorted sets), Memcached (cache only), Amazon DynamoDB (key-value plus document features), etcd (small, strongly consistent configuration data), Riak. Uses: sessions, caches, carts, feature flags, rate-limit counters.

Document stores. Each record is a **document** (usually JSON or a binary form) with nested fields and arrays; documents in one **collection** needn't share a schema, and fields inside them can be queried and indexed. Examples: MongoDB (max document 16 MB), Couchbase, Amazon DocumentDB, Google Firestore. Good when each item is a self-contained tree (a product with its variants, a post with its tags) usually read whole.

```
{ "_id": "p42", "title": "Trail shoe", "price": 89.0,
  "sizes": [40, 41, 42], "seller": { "id": "s7", "name": "RunCo" } }
```

Wide-column stores (column-family stores). Data is grouped by a **partition key**; within a partition rows are sorted by a **clustering key**, and rows can have many, differing columns. You design one table per query; built to absorb huge write volumes across many machines. Examples: Apache Cassandra, ScyllaDB, Apache HBase, Google Bigtable. Uses: message histories, activity logs, IoT readings — "write a lot, read by key and time range". "Wide-column" ≠ "columnar" (column-oriented analytics storage, later).

Graph databases. Nodes (things, such as people) and **edges** (relationships, such as "follows"), both with properties. Following edges is cheap however big the graph, so "friends of friends in Paris" is natural. Examples: Neo4j (query language Cypher), Amazon Neptune, JanusGraph. Uses: social graphs, fraud rings, recommendations, knowledge graphs.

Time-series databases. For time-stamped points arriving in time order ("CPU 73% at 10:00:05 on host 12"): heavy compression, fast "average per minute over the last 6 hours", automatic expiry of old data. Examples: InfluxDB, TimescaleDB (a PostgreSQL extension), Prometheus (monitoring), QuestDB. Uses: metrics, sensor data, stock ticks.

Search engines. Built on an **inverted index**: for each word, the list of documents containing it (like a book's index), so "red" AND "shoe" intersects two lists instead of scanning. They also do **tokenising** (splitting text into words), stemming ("running" matches "run"), typo tolerance and **relevance ranking** (e.g. the BM25 scoring formula). Examples: Elasticsearch and OpenSearch (both on the Apache Lucene library), Apache Solr. Usually a *secondary* store, copied from the source-of-truth database.

Vector databases. Store **embeddings**: vectors of numbers (often hundreds or thousands long) produced by a machine-learning model so similar items (images, sentences) are close. The key query is **nearest-neighbour search** ("the 10 vectors closest to this one"); exact search over millions is slow, so they use **approximate nearest neighbour (ANN)** indexes such as HNSW (a layered graph walked towards the target). Examples: Pinecone, Milvus, Weaviate, Qdrant, pgvector for PostgreSQL. Uses: semantic search, recommendations, retrieval-augmented generation (feeding relevant documents to a large language model).



Trade-offs

NoSQL usually gives up some of: joins, ad hoc queries, multi-record transactions, enforced schemas — gaining easy horizontal scaling, high write throughput, flexible shapes, or a very fast specialised query. The lines have blurred: MongoDB added multi-document transactions in 4.0, DynamoDB has transactions, PostgreSQL stores and indexes JSON with `jsonb`.

Real-world examples

Discord stores trillions of messages in a wide-column store (moved from Cassandra to ScyllaDB around 2022–2023). Amazon's shopping cart motivated the 2007 Dynamo paper. LinkedIn and Uber have published on their graph and time-series systems. Most large sites run Elasticsearch or OpenSearch beside their main database.

TL;DR

NoSQL is a family of specialist databases — key-value, document, wide-column, graph, time-series, search, vector — each trading general querying for speed or scale on one access pattern.

SQL vs NoSQL and BASE

What it is

SQL vs NoSQL is one of the most common interview decisions; there is no universal winner, only "which access patterns and guarantees does this data need?"

BASE, a tongue-in-cheek opposite of ACID describing many early NoSQL systems:

- **Basically Available**: keeps answering when some machines fail, even if some answers are a bit old.
- **Soft state**: copies may change without new writes, as they catch up with each other.

- **Eventually consistent**: if writes stop, all copies eventually agree; until then two readers may see different values.

(ACID is a bank teller who won't hand over cash until the ledger is checked; BASE is a coffee shop that takes your order at once and fixes the odd mix-up later.)

Trade-offs

Need	Lean towards
Money, inventory, bookings, strong correctness	SQL (or distributed SQL)
Complex queries, reporting, joins across many entities	SQL
Data shape changes often, nested records read whole	Document store
Massive write volume, simple key lookups, many regions	Wide-column or key-value
Relationship-heavy queries many hops deep	Graph
Team already knows PostgreSQL and data fits on one machine	SQL, almost always

Name the access pattern first: "Users are read by ID and email, orders need transactions with payments: PostgreSQL. The activity feed is append-heavy and read by user and time: Cassandra." Several database types in one system is normal — **polyglot persistence**.

Common mistakes

- "NoSQL is faster." Only for the pattern it was designed for; often slower or impossible for others.
- "NoSQL has no schema." It lives in the application instead (**schema-on-read**, vs SQL's **schema-on-write**).
- Eventual consistency for data users would notice, such as an account balance.

In the interview

Q: When would you choose NoSQL over SQL?

A: When access is simple (mostly by key), data or write volume exceeds one machine, the schema varies a lot, or you need multi-region writes with high availability — and you can live without joins and multi-row transactions.

TL;DR

Pick from access pattern and correctness needs, not fashion. SQL is the default; reach for NoSQL when scale, shape or availability justify the lost features.

Indexes

What it is

An **index** is an extra structure kept beside a table to find rows by a column's value without reading the whole table. Without one, `email = 'a@b.com'` needs a **full table scan** of every row — many seconds at 100 million rows; with one, a few page reads, well under a millisecond if cached. (Like a textbook's index: look up "photosynthesis", jump to page 212.)

How it works

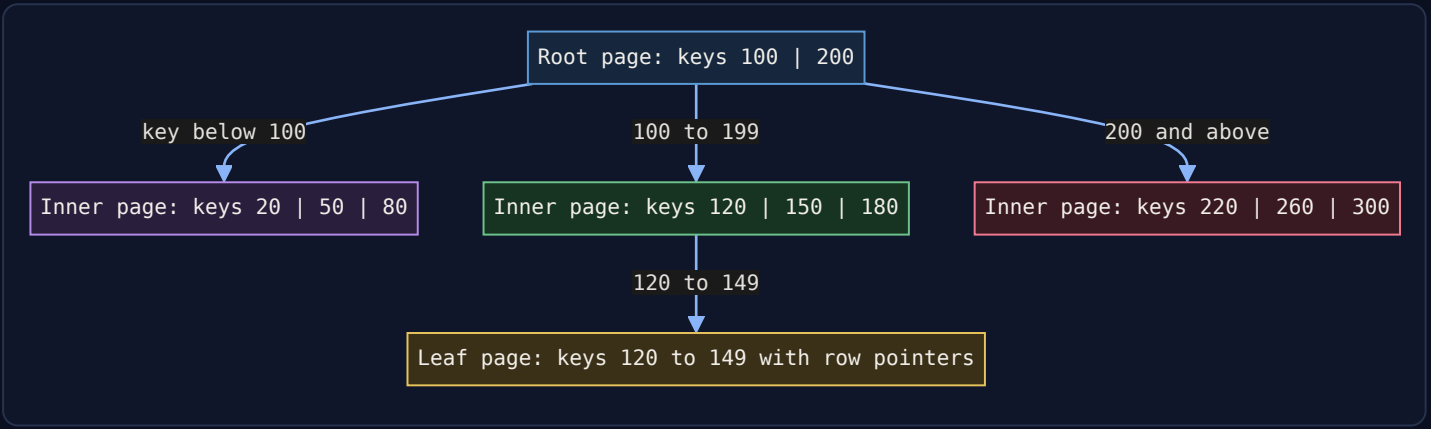
The index stores the column's values in a fast-to-search structure, each paired with a pointer to the full row (or its primary key), and is updated on every insert, update and delete.

```
CREATE INDEX idx_orders_user_created ON orders (user_id, created_at);
```

`EXPLAIN` (PostgreSQL, MySQL) shows the plan the query planner chose, including whether an index is used.

Types

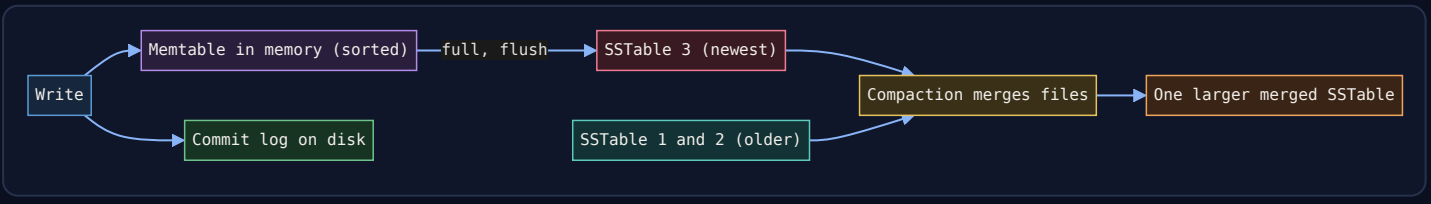
B-tree index. The default in almost every relational database: a balanced tree of fixed-size **pages** (storage blocks, typically 4–16 KB: PostgreSQL 8 KB, MySQL InnoDB 16 KB), each holding hundreds of sorted keys and child pointers. This high **branching factor** keeps the tree shallow: about 4 levels cover billions of rows. A lookup walks root to **leaf**, one page per level. Sorted keys support exact matches (`=`) and **range queries** (`BETWEEN` , `>` , `ORDER BY`). Updates are in place — random disk writes.



LSM-tree (log-structured merge tree). Used by write-heavy stores (Cassandra, ScyllaDB, HBase, RocksDB, LevelDB). Writes never modify data in place:

1. A write is appended to an on-disk log (crash safety) and inserted into an in-memory sorted **memtable**.
2. At a size limit (say 64 MB) the memtable is written out as an immutable sorted **SSTable (sorted string table)** — a fast **sequential write**.
3. Background **compaction** merges SSTables, keeping the newest version of each key and dropping deleted ones (a deletion is first recorded as a **tombstone** marker).
4. A read checks the memtable, then SSTables newest to oldest; each SSTable's **Bloom filter** ("definitely not here", Part 4) and sparse index avoid opening every file.

LSM-trees: very high write throughput, slower reads, background compaction cost. B-trees: predictable reads, slower writes. This is the **read amplification versus write amplification** trade-off: extra disk reads per lookup versus how often each byte is rewritten.



Hash index. A hash map from key to value location: O(1) (constant-time) exact lookups, but no range queries or sorting, since hashing scatters neighbouring keys. Bitcask (Riak's storage engine) keeps every key in an in-memory hash map pointing into append-only files; PostgreSQL offers a `HASH` index type.

Composite (multi-column) index. One index on `(user_id, created_at)` , sorted by the first column then the second (a phone book by last then first name). It helps queries on the **leftmost prefix** — `user_id` , or `user_id` plus `created_at` — but *not* `created_at` alone. Put the equality-filter column first, the range or sort column after.

Covering index. Contains every column a query needs, so it is answered from the index alone (an **index-only scan**). PostgreSQL and SQL Server add non-key columns with `INCLUDE` .

Clustered vs non-clustered (secondary) index. In a **clustered index** the rows themselves are stored in index order (one per table; MySQL InnoDB always clusters by primary key). A **secondary index** is any other; it points back to the row (in InnoDB, via the primary key).

Others: unique index (also enforces uniqueness), **partial index** (only rows matching a condition, such as `WHERE deleted = false`), **full-text index** (an inverted index inside a relational database), PostgreSQL **GIN/GiST** indexes for JSON, arrays and geometry, **geospatial indexes** (R-trees, geohashes).

Trade-offs

- **Every write gets slower**: an insert into a table with 6 indexes updates 7 structures.
- **Space**: sometimes as much as the table.
- **Memory**: fast only when hot pages fit in the buffer cache.
- **Low-selectivity columns** (few distinct values, like boolean `is_active`) rarely benefit — the database would read most of the table anyway.

Index the columns your frequent queries filter, join and sort on, and nothing else; read-heavy tables can afford more indexes than write-heavy ones.

Common mistakes

- Wrapping an indexed column in a function (`WHERE LOWER(email) = ...`), which stops the plain index being used (index the expression instead).
- Leading-wildcard `LIKE '%shoe'` can't use a B-tree; use full-text search.
- Building an index on a big production table in a way that locks writes (PostgreSQL offers `CREATE INDEX CONCURRENTLY`).

In the interview

Q: Why do LSM-tree databases handle writes better than B-tree databases?

A: LSM-trees buffer writes in memory and flush large sequential files, never updating in place; B-trees do random in-place page updates. The cost: slower reads and background compaction.

Q: You have an index on `(country, city)`. Does `WHERE city = 'Lyon'` use it?

A: Generally no — it is sorted by country first. It helps queries on `country`, or `country` and `city` together.

TL;DR

An index is a sorted (or hashed) lookup structure turning a full scan into a few page reads. B-trees suit reads and ranges, LSM-trees heavy writes; every index slows writes.

Normalization and denormalization

What it is

Normalization organises tables so each fact is stored in exactly one place. **Denormalization** deliberately stores the same fact in several places so reads are faster — like writing the customer's address on every invoice: quick to read, but every copy must change when they move.

How it works

Normalization is described by **normal forms**; beginners need three:

1. **First normal form (1NF)**: each cell holds one value, not a list (no `tags = "red,blue,green"`).
2. **Second normal form (2NF)**: every non-key column depends on the *whole* primary key. In `order_items(order_id, product_id, qty, product_title)`, `product_title` depends only on `product_id`, so it belongs in `products`.
3. **Third normal form (3NF)**: non-key columns depend only on the key, not other non-key columns. If `users` has `zip_code` and `city` and the city follows from the zip, the city belongs in a zip-code table.

The mnemonic usually credited to William Kent (1983): every column describes "the key, the whole key, and nothing but the key".

Normalization prevents **update anomalies**: a product title stored in 10,000 order rows must be updated 10,000 times on rename, and partial failure leaves them disagreeing.

Denormalization techniques: copying a column (`author_name` on each post), storing a precomputed total (`comment_count` instead of counting each time), nesting children in a document (a document database's natural style), and **materialized views** (a stored, periodically refreshed query result).

Trade-offs

	Normalized	Denormalized
Storage	Smaller	Larger (duplicates)
Reads	May need joins	Fewer or no joins, faster
Writes	One place to update	Many places to update
Risk	Slow joins at scale	Copies drift out of sync

Start normalized; denormalize specific hot read paths once measured, and decide how copies stay in sync (same transaction, background job, or events). At large scale and in NoSQL stores without joins denormalization is the norm: in Cassandra you often keep one table per query, each holding a copy arranged for that query.

Common mistakes

- Denormalizing before there is a performance problem, then spending months fixing drift bugs.
- Forgetting a counter like `like_count` updated by many users at once needs an atomic increment.

TL;DR

Normalize so each fact lives once; denormalize selected data for fast hot reads, accepting extra write work and the risk of stale copies.

Replication

What it is

Replication keeps copies of the same data on several machines, each a **replica** (node), kept in sync by sending every change from one machine to the others — like head office sending updated price lists to every branch, any of which could become the new master if head office burns down.

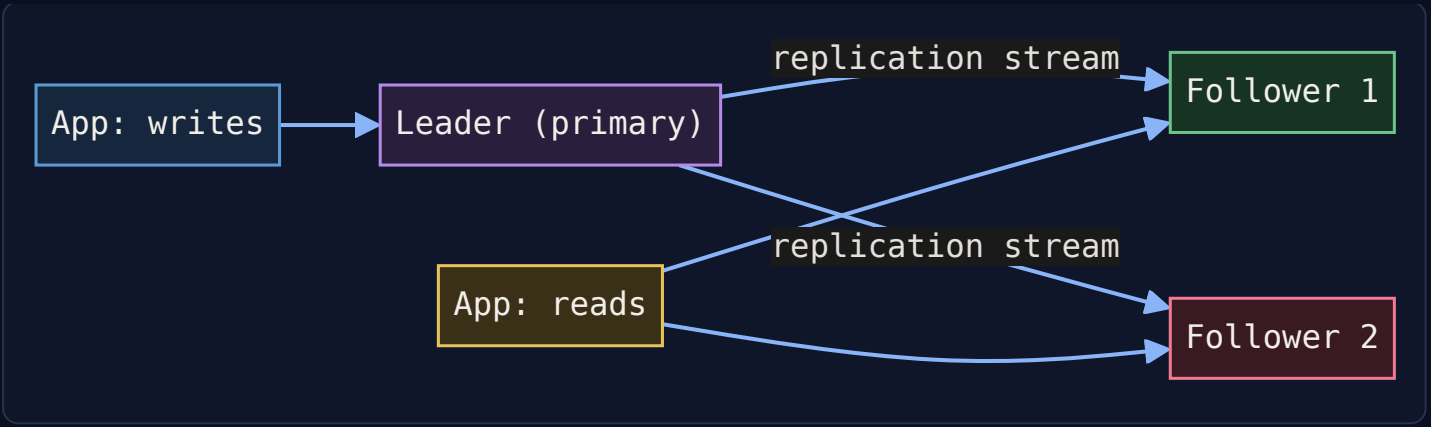
Why it matters

1. **Availability and fault tolerance.** If one machine dies, another copy serves requests.
2. **Read scaling.** Ten replicas serve roughly ten times the reads.
3. **Lower latency.** A Frankfurt replica answers European readers in ~10–20 ms instead of an ~80–100 ms round trip to Virginia.

Replication does *not* help write throughput: every copy must still apply every write. To scale writes, split the data (partitioning, next). Rule: **copy for reads, split for writes**.

Types

Leader-follower (primary-replica, master-slave, active-passive replication). One **leader** (primary) accepts all writes, records each change in its replication log and streams it to **followers** (replicas, secondaries), which apply changes in the same order. Clients read from leader or followers. Default in PostgreSQL, MySQL, MongoDB replica sets, SQL Server and Redis.



Read replicas are followers used mainly for reads (e.g. Amazon RDS read replicas) — ideal for product pages, reports, search indexing.

Multi-leader (multi-primary, active-active). Several nodes accept writes, usually one per data centre or region, replicating to each other: fast local writes, and a region keeps working if cut off. The hard part is **write conflicts** (two leaders change the same record at once). Resolutions:

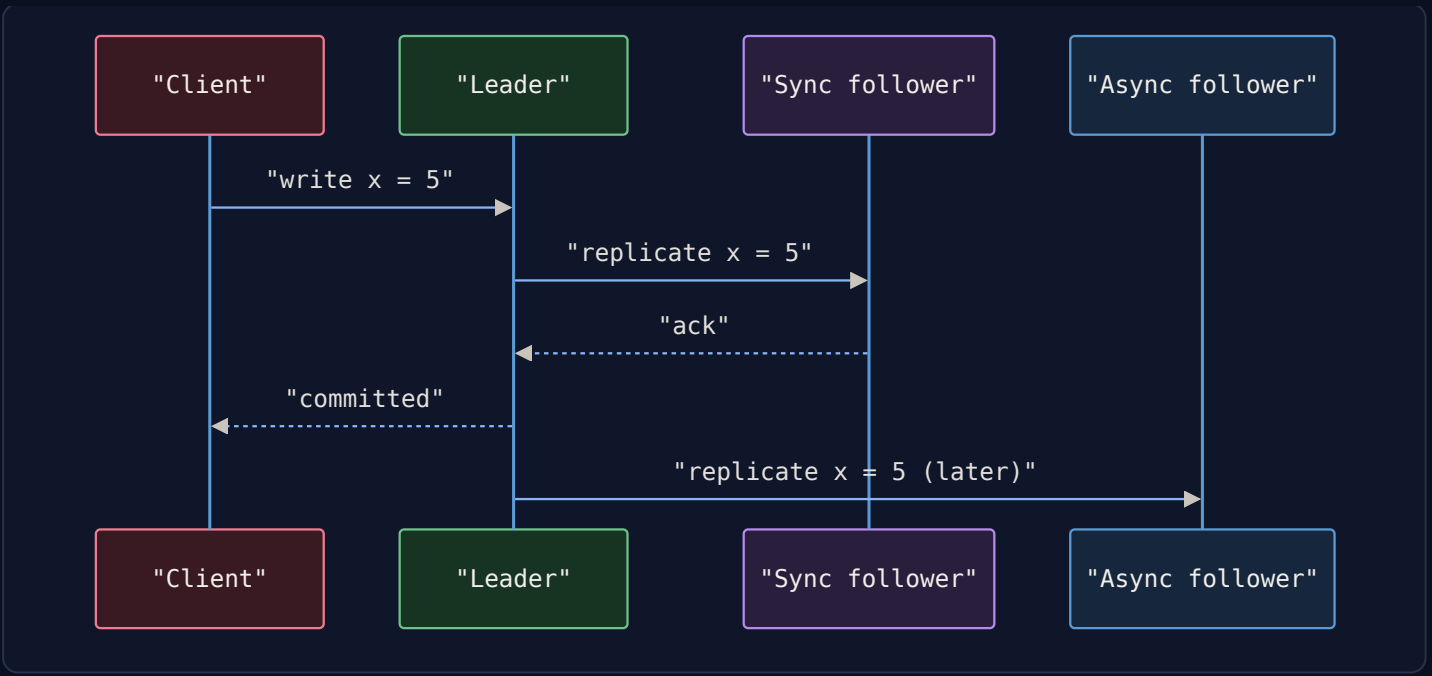
- **Last write wins (LWW)**: keep the latest-timestamped write, silently dropping the other. Simple but loses data, and clocks disagree (Part 4).
- **Merge** values with application logic, or ask the user.
- **CRDTs (conflict-free replicated data types)**: data types (counters, sets) whose concurrent changes always merge to the same result.

Examples: MySQL and PostgreSQL with multi-primary tools, CouchDB, collaborative editors and offline-first mobile apps (each device effectively a leader).

Leaderless. Any replica accepts reads and writes. A client (or coordinator node) sends each write to several replicas and succeeds when enough confirm; reads query several and take the newest value. With N copies, writing W and reading R, **R + W > N** guarantees every read overlaps an up-to-date copy (a **quorum**, Part 4); commonly N=3, W=2, R=2. Stale copies are fixed by **read repair** (the reader writes the newest value back to replicas that returned old ones), **anti-entropy** (background comparison of replicas, often with Merkle trees) and **hinted handoff** (a healthy node holds writes meant for a down node and passes them on later). Examples: Amazon's original Dynamo, Apache Cassandra, ScyllaDB, Riak.

Synchronous vs asynchronous replication — *when* the leader tells the client "done":

- **Synchronous**: wait until a follower confirms. No committed data lost if the leader then dies, but every write waits a round trip, and a down or slow follower stalls writes.
- **Asynchronous**: reply once the leader's own write is durable and ship changes in the background. Fast and unaffected by slow followers, but a dying leader loses the last changes no follower had.
- **Semi-synchronous**: wait for *one* follower (or a quorum), the rest asynchronous. MySQL has a semi-synchronous mode; PostgreSQL lets you list synchronous standbys.



How it works

Replication lag is how far a follower trails the leader — usually milliseconds, sometimes seconds or minutes when a follower is overloaded or applying a huge transaction. It causes visible bugs:

- **Read-your-own-writes violation**: a user edits their profile, the reload hits a lagging replica, and the change seems gone. Fix: read the user's own data from the leader for a short time after they write, or track the write's log position and read only from a replica that has caught up.
- **Monotonic reads violation**: two refreshes hit different replicas and the second shows *older* data, as if time went backwards. Fix: pin each user to one replica.

Failover promotes a follower when the leader fails:

1. **Detect**: the leader misses **heartbeats** (regular "I am alive" messages) for a timeout such as 10–30 seconds.
2. **Choose** a new leader, ideally the most up-to-date follower, by election among replicas or an external coordinator.
3. **Reconfigure**: point clients and other followers at it (DNS, a proxy, or service discovery).

Failover is **automatic** (Patroni for PostgreSQL, Orchestrator for MySQL, managed cloud databases failing over in under a minute or two) or **manual**.

Split brain: two nodes both believe they lead and accept writes, so data diverges — typically an old leader that was only cut off, not dead, returning still in charge. Defences: a **majority (quorum)** must agree on the leader, so an isolated minority can't elect one; **fencing**, forcibly stopping or blocking the old leader (STONITH, "shoot the other node in the head"); **fencing tokens**, increasing numbers that let storage reject writes from a stale leader (Part 4).

Trade-offs

Style	Writes go to	Strength	Weakness
Leader-follower	One leader	Simple, no write conflicts	Leader limits writes, failover needed
Multi-leader	Several leaders	Local writes per region, survives region loss	Conflicts must be resolved
Leaderless	Any W replicas	No failover, high availability	Weaker consistency, more complex reads

Common mistakes

- Treating a replica as a backup: an accidental `DELETE FROM users` replicates to every copy within milliseconds. You need real backups (Durability).
- Sending all reads to replicas without thinking about read-your-own-writes.
- A failover timeout so short that a network hiccup triggers a needless, risky failover.

Real-world examples

GitHub's October 2018 incident: a ~43-second network interruption between its US East Coast hub and data centre triggered automatic MySQL failover to the West Coast; some writes had reached only the old primary, and untangling the two data sets left the site degraded for just over 24 hours. Amazon Aurora keeps six copies of each piece of data across three **availability zones** (separate data centres in one region, two copies each) and treats a write as durable after four of six confirm; reads and repairs need three of six.

In the interview

Q: Synchronous or asynchronous replication for a payments database?

A: At least one synchronous (or quorum) copy so a committed payment survives losing the leader, plus asynchronous replicas for reads and other regions. Fully synchronous to all copies makes every write as slow as the slowest replica.

Q: A user updates their bio and immediately sees the old one. Why, and the fix?

A: The read hit a lagging replica. Route that user's reads of their own data to the leader for a short window after a write, or read from a replica only once it has applied the write's log position.

TL;DR

Replication keeps copies for availability, read scaling and locality. Know leader-follower, multi-leader and leaderless; sync vs async; replication lag; failover; split brain.

Partitioning and sharding

What it is

Partitioning splits a large dataset into smaller **partitions**; **sharding** is partitioning where the pieces (**shards**) live on *different machines*. The words are often used interchangeably, but strictly you can partition inside one server (PostgreSQL's declarative partitioning by month), whereas sharding always spreads data across servers — like a library opening branches (A–F north, G–M east) with a catalogue saying which branch to visit.

Why it matters

Replication copies everything to every node, so it can't help when the data is too big for one machine (say 50 TB) or the *writes* per second exceed one machine (say 200,000). Sharding divides both: with 10 shards each holds ~a tenth of the data and takes ~a tenth of the writes. Each shard is usually also replicated: split for writes, copy for reads and safety.

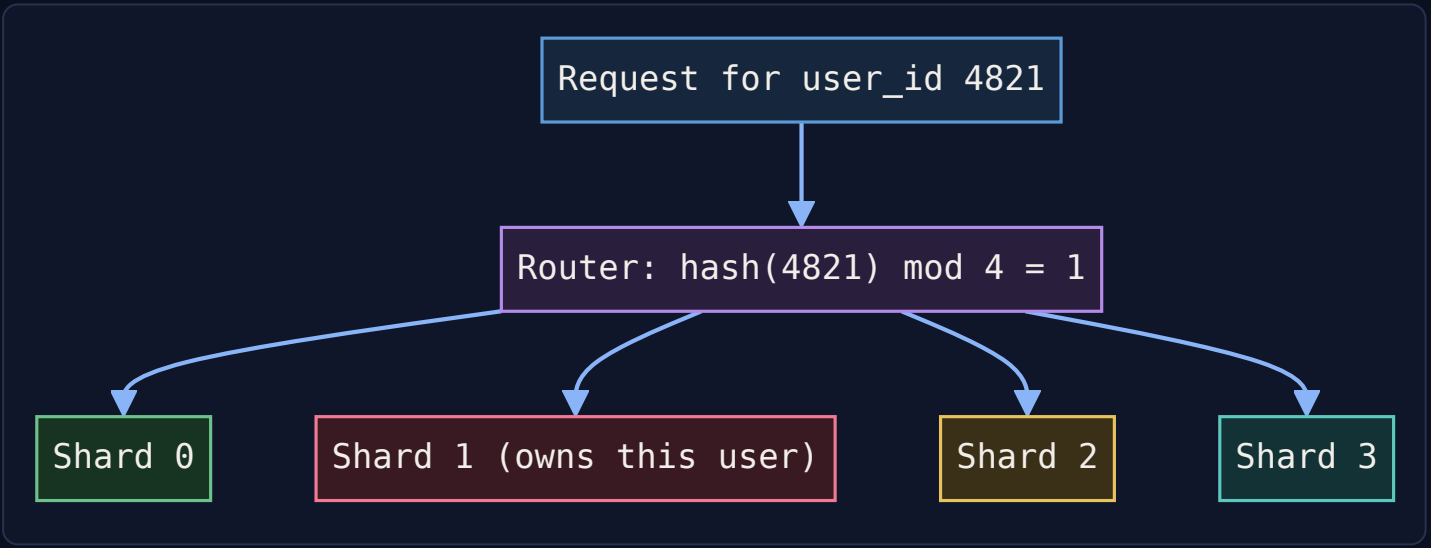
Types

Horizontal vs vertical partitioning.

- **Horizontal partitioning** splits *rows* (users 1–1,000,000 in one partition, the next million in another), every partition with the same columns. Sharding is horizontal partitioning across machines.
- **Vertical partitioning** splits *columns*: small, frequently read ones (`id` , `name` , `avatar_url`) in one table, large rarely read ones (`bio` , `settings_json`) in another, joined by ID — keeping hot data compact. Splitting whole tables or features onto different databases is federation (below).

Strategies for choosing a row's shard, by the **shard key** (partition key):

- **Range-based.** Each shard owns a continuous key range (A–F, G–M; January, February). Range queries hit one or a few shards. Danger: uneven load — with a timestamp key *all* new writes hit the newest shard. Used by HBase, Bigtable, MongoDB (ranged sharding) and CockroachDB, which split ranges automatically as they grow.
- **Hash-based.** A **hash function** (any input → random-looking number) picks the shard: `shard = hash(user_id) mod 8`. Even spread and balanced load, but neighbouring keys scatter, so range queries ask every shard; and with plain `mod N`, going from 8 to 9 shards moves almost every key — hence consistent hashing (next section).
- **Directory-based (lookup table).** A separate service maps key (or range) → shard. Fully flexible (move one big customer to its own shard), but an extra lookup per request, and the directory must be highly available and cached.
- **Geo-based (geographic).** Shard by user region (EU users on EU shards). Lower latency and helps data-residency laws such as the EU's GDPR; uneven if regions differ in size.



Where routing happens. In the application; in a proxy or **router** (Vitess for MySQL, `mongos` for MongoDB, Citus for PostgreSQL); or any node accepts and forwards (Cassandra).

Federation (functional partitioning). Split the database *by feature* — users, products, orders databases on their own servers. Each is smaller with its own traffic share, but you lose joins and transactions between them, and it scales only as far as the number of features. Often a first step before true sharding; maps naturally onto microservices each owning its database.

How it works

Choosing a shard key is the most important sharding decision; changing it later is very expensive. A good shard key:

1. **Has high cardinality** (many distinct values): `country` is poor (~200 values, very uneven); `user_id` is good.
2. **Spreads load evenly**, both data size and traffic.
3. **Matches the main query**, so most requests touch one shard: if you look up "orders for a user", shard orders by `user_id`.
4. **Doesn't grow in one direction** under range sharding (avoid auto-increment IDs and timestamps as the range key).

Queries including the shard key are **single-shard** and fast; others go to every shard and merge results (**scatter-gather**), slower and worse as shards are added.

Hot spots and hot keys. A **hot spot** is a shard with far more load than others; a **hot key** is a single key causing it (a celebrity account with 300 million followers, one viral flash-sale product). Even perfect hashing can't split one key. Fixes:

- **Key salting**: append a small random number (`celebrity_42#0` … `#9`) to spread its writes over 10 shards; read all 10 and combine.
- **Caching** the hot item's reads in front of the database.

- **Dedicated shards** for known whales (big customers), via a directory.
- **Write buffering**: collect increments in memory and apply in batches.

Rebalancing (resharding). Shards must be added as data grows, or data moved off hot shards:

- **Fixed number of logical partitions**: create far more partitions than machines from day one (1,024 on 8 machines, 128 each); adding a machine moves whole partitions and keys never change partition. Elasticsearch, Couchbase, Riak, and many hand-rolled systems.
- **Dynamic splitting**: split a range partition past a size limit (by default 512 MB per range in CockroachDB, 128 MB per chunk in recent MongoDB, 10 GB per region in HBase) and move one half.
- **Consistent hashing** (next section).

Rebalancing moves data while serving traffic: copy in the background, catch up on new writes, switch routing. Automatic rebalancing is convenient but risky; many teams keep a human in the loop.

Trade-offs

Sharding costs:

- **Cross-shard joins** become application code or scatter-gather.
- **Cross-shard transactions** need distributed transactions or sagas (Part 4).
- **Globally unique IDs** can't come from one auto-increment counter: UUIDs or Snowflake-style IDs (Part 4).
- **Secondary indexes** get harder: per-shard **local index** (queries hit all shards) or a **global index** sharded by the indexed value (fast reads, writes update another shard).
- **Operations** (backups, schema changes, monitoring) multiply by shard count.

So shard late: first optimise queries, add indexes, caching, read replicas, scale up. Shard when one primary truly can't handle the writes or data size.

Common mistakes

- Range-sharding by timestamp, so today's shard takes every write.
- A shard key absent from the main queries, making most reads scatter-gather.
- `hash mod N`, then adding one server moves nearly all data.
- Forgetting a single hot key can't be fixed by adding shards.

Real-world examples

Instagram sharded PostgreSQL into thousands of logical shards on fewer physical servers, so shards move without changing keys. Notion split its single PostgreSQL database into 480 logical shards on 32 physical databases in 2021, and in 2023 spread the same 480 over 96 machines without changing keys. YouTube created Vitess to shard MySQL (now used by Slack, GitHub and others). Pinterest puts a shard ID inside every object ID.

In the interview

Q: How would you shard a chat application's messages table?

A: By hashed `conversation_id`, since messages are read per conversation (one shard per read), ordered by time inside the shard. Watch for hot conversations such as huge group channels.

Q: What makes a good shard key?

A: High cardinality, even spread of data and traffic, present in most queries, and not monotonically increasing when range-sharded.

TL;DR

Sharding splits rows across machines so data and writes scale. The shard key decides everything: many values, even load, aligned with the main queries; plan for hot keys and rebalancing.

Consistent hashing and virtual nodes

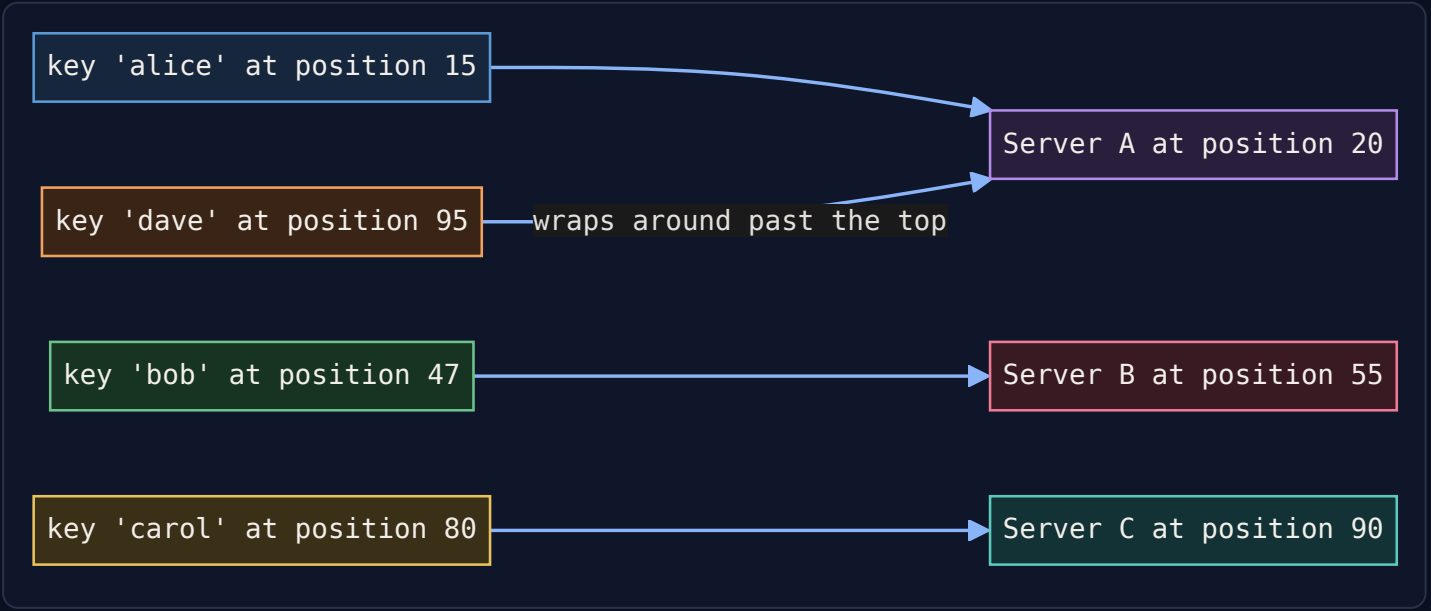
What it is

Consistent hashing assigns keys to servers so adding or removing a server moves only about 1/N of the keys (N servers), instead of almost all.

Why plain modulo fails. With `server = hash(key) mod 4`, hash 10 goes to server 2; add a fifth server and `10 mod 5 = 0`. About 80% of keys move going from 4 to 5 — a flood of cache misses, or a huge database migration.

How it works

1. Bend the hash output range (e.g. 0 to 2^32 − 1) into a circle: the **hash ring**.
2. Hash each server's name to a position on the ring.
3. Hash each key to a position and walk clockwise to the first server, which stores it.
4. **Adding a server**: it takes over only the keys between its predecessor and itself — on average 1/N of keys move.
5. **Removing a server**: its keys move to the next server clockwise; nothing else changes.
6. **Replication**: store each key on its first server plus the next 2 distinct servers clockwise for 3 copies (Cassandra, Dynamo).



Virtual nodes (vnodes). With few servers, random positions give some big arcs and some small (uneven load), and a departing server dumps *all* its load on one neighbour. Fix: give each physical server many ring positions — 100–256 **virtual nodes**, hashing "A-1", "A-2", … "A-256" — so each owns many small scattered arcs:

- Load evens out statistically.
- A departing server's arcs are picked up by many servers, spreading the extra load.
- A more powerful machine gets more virtual nodes for a bigger share.

Cassandra defaulted to 256 vnodes per node for years, lowering it to 16 in 4.0 with a smarter token-allocation algorithm.

Types

Alternatives: **rendezvous hashing** (highest random weight: score each key-server pair, pick the highest), **jump consistent hash** (Google; very fast and memory-free, but servers can only be added or removed at the end), and **Maglev hashing** (Google's load balancer; a lookup table for even spread).

Trade-offs

Minimal data movement and no central directory, but plain consistent hashing balances poorly without virtual nodes, and like any hashing it destroys key order, so range queries become scatter-gather.

Real-world examples

Amazon Dynamo (2007 paper), Apache Cassandra, ScyllaDB, Riak, Memcached client libraries (ketama), Discord's routing of users and guilds to servers, and many load balancers (sticky routing, Part 2).

In the interview

Q: Why consistent hashing instead of `hash mod N`?

A: `mod N` remaps almost every key when N changes; consistent hashing moves only ~1/N of keys when one of N servers is added or removed.

Q: What problem do virtual nodes solve?

A: Uneven arcs and load. Many positions per server spread ownership in small pieces, smooth load, and let a departing server's keys spread across many others.

TL;DR

Consistent hashing puts servers and keys on a ring so a membership change moves only ~1/N of keys; virtual nodes make the spread even.

Storage types: block, file and object

What it is

Beneath every database and uploaded photo is raw storage, sold by cloud providers in three shapes. (Block storage is an empty plot you build on; file storage a shared office building with labelled cabinets; object storage a coat check — hand over an item, get a ticket (the key), and you can only check in, retrieve or discard whole items.)

Types

Block storage. A raw disk of fixed-size, addressed **blocks**. The OS puts a **file system** (ext4, NTFS) on top, or a database manages blocks directly. Usually attached to one server at a time; low latency (well under a millisecond for local NVMe SSDs, roughly one to a few milliseconds for network block storage); you can change a few bytes mid-file. Measured in **IOPS** and throughput. Examples: a laptop SSD, Amazon EBS, Google Persistent Disk, Azure Managed Disks. Use for database data files and VM boot disks.

File storage. A shared file system of folders and files, accessed over the network by many servers at once via **NFS** (Network File System, common on Linux) or **SMB** (Windows). Examples: Amazon EFS, Azure Files, Google Filestore, a NAS (network-attached storage) box. Use for content shared between servers, home directories, legacy apps expecting paths.

Object storage. Objects (a blob of bytes plus metadata) addressed by a unique key inside a **bucket** (top-level container). No real folders (`photos/2026/cat.jpg` only looks like a path), and no partial edits — replace the whole object. Accessed over HTTP (`PUT`, `GET`, `DELETE`). Practically unlimited size, cheapest per gigabyte, extremely durable; higher latency (often tens of milliseconds to first byte). Examples: Amazon S3, Google Cloud Storage, Azure Blob Storage, Cloudflare R2, open-source MinIO and Ceph.

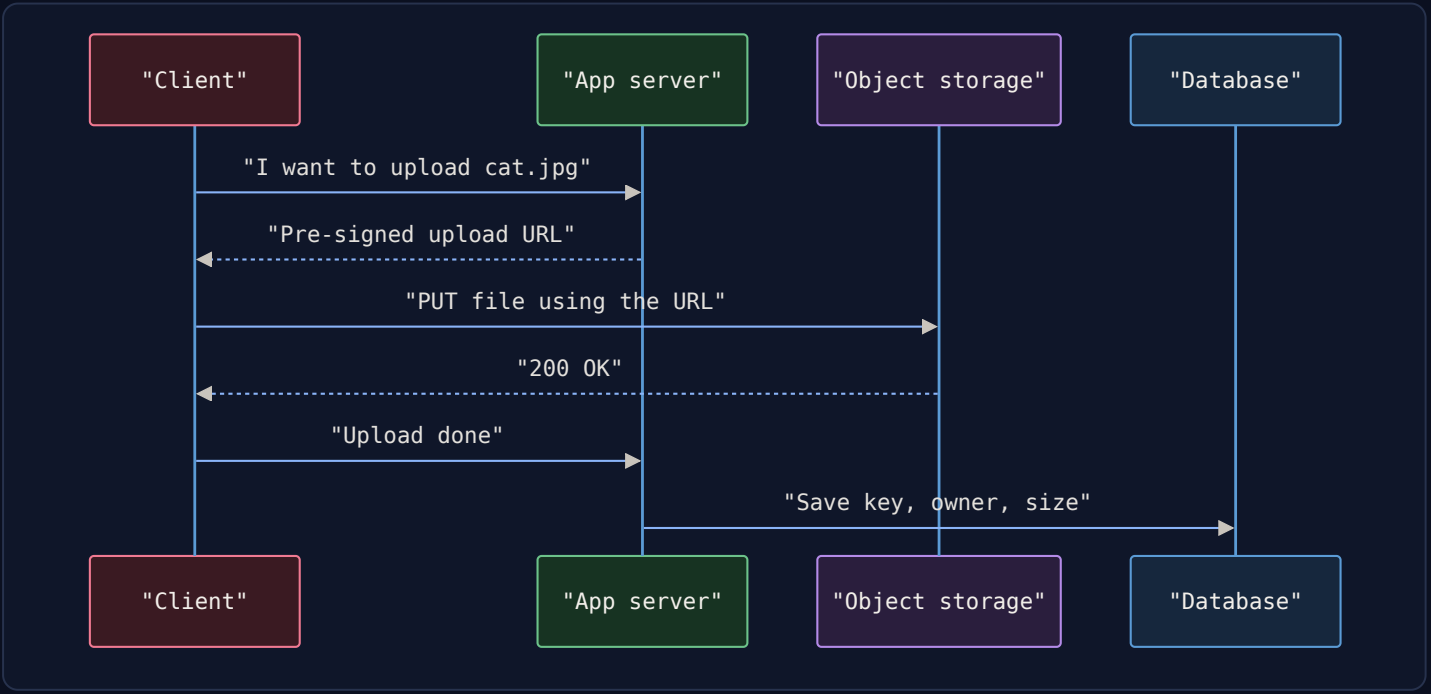
A **blob** (binary large object) is any large chunk of unstructured binary data — image, video, PDF, backup. "Blob storage" is another name for object storage (Azure's service is literally Blob Storage).

	Block	File	Object
Unit	Fixed-size block	File in folders	Whole object with key
Access	Attached disk	Network file protocol	HTTP API
Shared by many servers	Usually no	Yes	Yes
Edit part in place	Yes	Yes	No, replace whole
Typical use	Databases, VM disks	Shared files	Media, backups, data lakes

How it works

The standard upload pattern (photos, videos, documents):

1. The client asks the app server for permission to upload.
2. The app server returns a **pre-signed URL**: a time-limited link allowing one upload to one key.
3. The client uploads straight to object storage, so gigabytes never pass through app servers.
4. The app server stores only the key and metadata (owner, size, type) in the database.
5. Downloads go through a CDN (Part 2) in front of the bucket.



Trade-offs

Never store large files in the relational database: it bloats backups, wastes the buffer cache and slows replication. File in object storage, pointer in the database.

Real-world examples

Amazon S3 is designed for 99.999999999% ("eleven nines") object durability, stores each object across at least three availability zones in its standard classes, allows single objects up to 50 TB (raised from 5 TB in December 2025), and has offered strong read-after-write consistency since December 2020. Netflix, Dropbox (which later moved much of its data to its own Magic Pocket system) and almost every photo app store media in object storage.

TL;DR

Block storage is a raw fast disk for one server, file storage a shared folder tree, object storage cheap, huge, durable key-to-blob storage over HTTP. Media goes in object storage; only its key in the database.

OLTP vs OLAP, row vs column storage, lakes and warehouses

What it is

Two very different workloads:

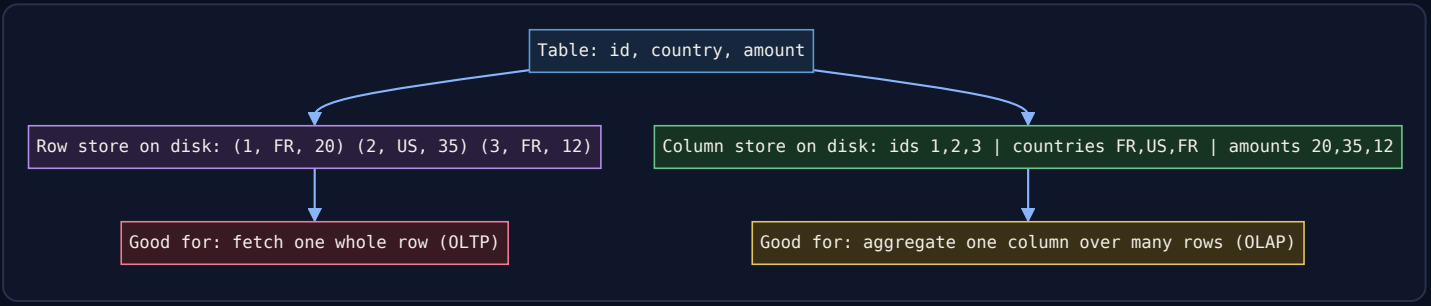
- **OLTP (online transaction processing)**: the app's everyday work — many small, fast reads and writes touching a few rows ("load user 42", "insert this order"), millisecond latency, thousands per second. PostgreSQL, MySQL, DynamoDB.
- **OLAP (online analytical processing)**: analysis and reporting — few big queries scanning millions or billions of rows but few columns ("revenue per country per month for three years"), seconds each, run by analysts and dashboards.

(The supermarket checkout vs the accountant combing a year of receipts.) Running heavy OLAP queries on the OLTP database is a classic mistake — one big report slows every checkout — so data is copied into separate analytical systems.

Types

Row-oriented vs column-oriented storage.

- A **row store** keeps all columns of a row together: "fetch user 42" reads one place. Great for OLTP.
- A **column store** (columnar storage) keeps each column's values together (all `country` values in one file, all `amount` in another): "sum of `amount` by `country`" reads two columns and skips 50 others. Similar values compress very well (often several times smaller), and the CPU processes them in batches. Great for OLAP.



Columnar databases and formats: ClickHouse, Apache Druid, Apache Pinot, Amazon Redshift, Google BigQuery, Snowflake, DuckDB; file formats **Apache Parquet** and **ORC**.

Data warehouse. A central analytical database of cleaned, structured data from many sources (orders, payments, marketing), loaded on a schedule with a schema defined in advance (schema-on-write). Often a **star schema**: a big **fact table** of events (each sale) surrounded by **dimension tables** (product, customer, date). Examples: Snowflake, BigQuery, Amazon Redshift, Azure Synapse.

Data lake. A huge, cheap store of raw data in any format (logs, JSON, CSV, images, Parquet), usually in object storage such as S3: store first, decide structure on read (schema-on-read). Flexible and cheap, but without discipline a "data swamp". Query engines such as Apache Spark, Trino/Presto and Amazon Athena read it directly.

Lakehouse. Lake files plus a table format adding transactions, schemas and versioning: Delta Lake, Apache Iceberg, Apache Hudi.

ETL vs ELT. A **pipeline** moves data from OLTP systems into analytics. **ETL (extract, transform, load)** cleans and reshapes before loading; **ELT (extract, load, transform)** loads raw data and transforms inside the warehouse, popular since cloud warehouses became cheap and powerful. **Change data capture (CDC)** streams every change from the OLTP database's log (e.g. Debezium into Kafka) so analytics lag minutes or seconds, not a day.

Trade-offs

	OLTP	OLAP
Query shape	Few rows, by key	Many rows, few columns
Users	Customers via the app	Analysts, dashboards
Storage layout	Row-oriented	Column-oriented
Latency target	Milliseconds	Seconds to minutes
Examples	PostgreSQL, MySQL	Snowflake, BigQuery, ClickHouse

In the interview

Q: Why are column stores faster for analytics?

A: Analytic queries read a few columns across many rows; a column store reads only those columns, compresses them well and processes them in tight batches, moving far less data than a row store.

Q: Data lake or data warehouse?

A: A warehouse holds curated, structured data with a fixed schema for reliable reporting; a lake holds raw data of any type cheaply for exploration and machine learning. Many companies have both, or a lakehouse.

TL;DR

OLTP is many small transactional queries on row stores; OLAP big analytical scans on column stores. Warehouses hold curated structured data, lakes raw data cheaply, and pipelines (ETL, ELT, CDC) move data between them.

Durability: WAL, fsync, backups and recovery

What it is

Durability means committed data stays saved through crashes, power cuts, disk failures and human mistakes. Different tools cover different threats:

Threat	Protection
Process crash or power cut	Write-ahead log plus fsync
One disk or machine dies	Replication, RAID
Whole data centre lost	Copies in another zone or region
Bug or human deletes data	Backups and point-in-time recovery
Ransomware or account takeover	Immutable, separate-account backups

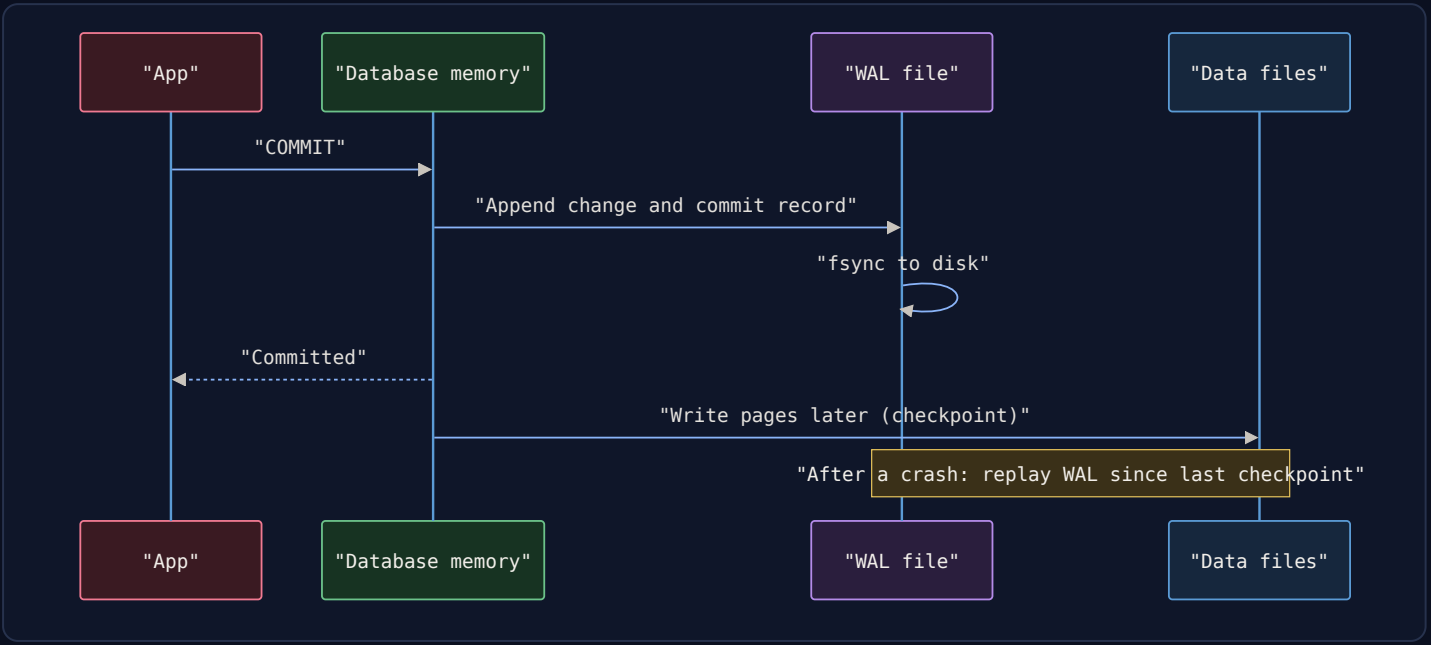
(Like an accountant who writes each transaction in an ink journal *before* updating the ledger, and photocopies the books nightly to another building.)

How it works

Write-ahead log (WAL). Updating data files in place is slow (random writes) and dangerous (a crash leaves a half-written page). So the database first appends each change to a sequential **write-ahead log** (redo log in MySQL InnoDB, commit log in Cassandra). Rule: *log first, then data*.

1. A transaction changes rows in memory.
2. On `COMMIT`, the changes and a commit record are appended to the WAL.
3. The WAL is forced to physical storage with **fsync**, the system call meaning "really write this to the device now, not just your memory buffer".
4. Only then is the client told "committed".

5. A background process later writes changed pages to the data files; a **checkpoint** records how far the files are up to date, so old WAL can be recycled.
6. After a crash, the WAL is replayed from the last checkpoint, redoing committed changes and discarding uncommitted ones.



The same log feeds replication (followers replay the leader's WAL stream) and change data capture.

Why fsync matters. The OS keeps written data in its memory **page cache** and flushes later; without fsync a power cut can lose writes the program thought were saved. fsync is slow-ish (tens of microseconds on enterprise SSDs with power-loss protection to several milliseconds on cheaper or network disks), so databases batch many commits into one fsync (**group commit**). Some systems trade safety for speed: PostgreSQL's `synchronous_commit = off` and MySQL's `innodb_flush_log_at_trx_commit = 2` can lose the last moment of commits on a crash; Redis's append-only file defaults to fsync once per second (up to ~1 second of writes lost).

RAID (redundant array of independent disks) combines disks so one failure loses no data: RAID 1 mirrors two disks, RAID 5/6 store parity, RAID 10 stripes across mirrored pairs. It protects against disk failure only.

Backups. A **backup** is a separately stored copy kept for days or months, recovering from things replication happily copies, such as an accidental `DROP TABLE`.

- **Full backup**: complete copy. Simple, slow, large.
- **Incremental backup**: only changes since the last backup. Small and fast, but restore needs the chain.
- **Logical backup**: data exported as SQL or rows (`pg_dump`, `mysqldump`). Portable, slow for big databases.
- **Physical backup**: copy of raw data files (`pg_basebackup`, Percona XtraBackup).
- **Snapshot**: point-in-time image of a disk or volume (an EBS snapshot), taken in seconds with copy-on-write — only blocks changed afterwards need extra space.

Point-in-time recovery (PITR) restores to any moment (say 14:31:07, just before a bad deploy deleted data) by restoring the last full backup and replaying archived WAL up to that moment. Managed databases such as Amazon RDS offer it with retention up to 35 days.

The **3-2-1** rule: 3 copies, on 2 kinds of media, 1 off-site (today often another region or cloud account).

Test your restores. A backup is unproven until you've rebuilt from it. Measure restore time; it feeds your recovery time objective (RTO, Part 5).

Common mistakes

- Backups in the same account or region as production: one compromised credential or regional outage takes both.
- Never testing restores. In January 2017 GitLab accidentally deleted a production database directory and found several backup and replication mechanisms hadn't been working; it restored a copy about

six hours old and lost roughly six hours of changes.

- Disabling fsync for benchmarks and forgetting to re-enable it.

In the interview

Q: How does a database guarantee durability without writing every change to the data files immediately?

A: It appends each change to a sequential write-ahead log and fsyncs the log before acknowledging; data files are updated later, and the log is replayed after a crash.

Q: You have three replicas. Do you still need backups?

A: Yes. Replicas cover hardware failure, but a bad deploy, bug or human error reaches every replica in milliseconds. Backups with point-in-time recovery go back to before the mistake.

TL;DR

Durability is layered: WAL and fsync survive crashes, replication survives machine loss, cross-region copies survive site loss, tested backups with point-in-time recovery survive mistakes.

Connection pooling, ORMs and N+1 queries

What it is

How application code talks to the database — the source of a surprising share of real-world performance problems.

Connection pooling. Opening a database connection is expensive — TCP handshake, often TLS, authentication, session memory — several milliseconds, sometimes tens. PostgreSQL starts a whole OS process per connection (several megabytes each) and defaults to `max_connections` = 100. A **connection pool** keeps open connections ready; a request borrows one and returns it (a taxi rank, rather than building a car per passenger).

How it works

A pool lives in one of two places:

1. **In the application** (client-side): a library in each app process, such as HikariCP for Java or the pools built into most drivers, with min and max size and a wait timeout when all are busy.
2. **As a separate proxy** (server-side): between all app servers and the database — **PgBouncer** or Pgpool-II for PostgreSQL, ProxySQL for MySQL, Amazon RDS Proxy. Useful with hundreds of app instances or serverless functions that would otherwise open thousands of connections. PgBouncer's **transaction pooling** lends a server connection only for one transaction, so a few dozen real connections serve thousands of clients.

Bigger is not better: a 16-core database can't run 500 queries at once; extra connections wait and compete for memory. Pools of a few times the core count often perform best. Mind the multiplication: 50 app servers × 20 connections = 1,000 at the database.

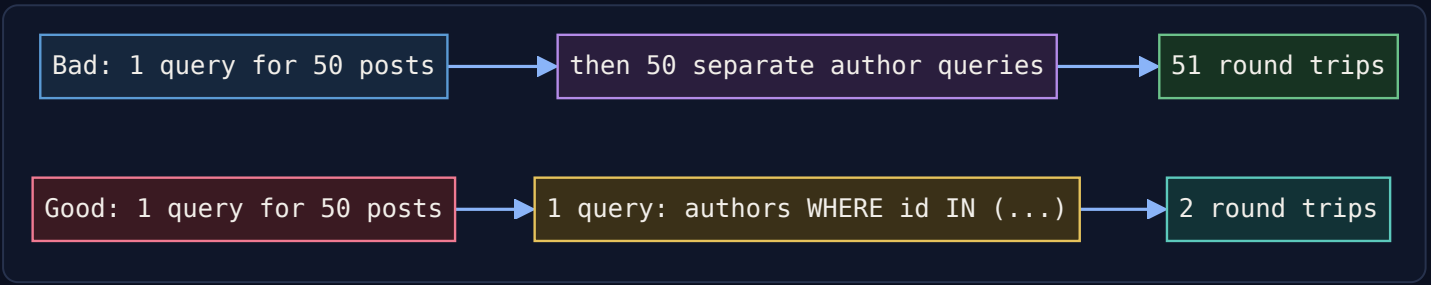
ORMs. An **ORM (object-relational mapper)** maps rows to objects so you write `user.orders` instead of SQL: Hibernate (Java), Django ORM and SQLAlchemy (Python), ActiveRecord (Rails), Entity Framework (.NET), Prisma and Sequelize (JavaScript/TypeScript). They speed development, prevent SQL injection by using parameters, and handle schema **migrations** (versioned scripts that change the schema). The cost: they hide the queries actually run, making slow code easy to write unnoticed. Use the ORM for everyday work, read the generated SQL on hot paths, and hand-write SQL where needed.

The N+1 query problem. The most famous ORM trap: 1 query loads N items, then the loop quietly runs 1 more query per item for a related object — N+1 queries.

```
posts = Post.objects.all()[:50]       # 1 query
for p in posts:
    print(p.author.name)              # 50 more queries, one per post
```

At 1 ms per round trip that's ~51 ms instead of 2 ms; with 1,000 items, a full second. Fixes:

- **Eager loading**: fetch related data up front in a join or one extra query (`select_related` / `prefetch_related` in Django, `includes` in Rails, `JOIN FETCH` in Hibernate).
- **Batching**: collect IDs and fetch them in one `WHERE id IN (...)`; the **DataLoader** pattern does this automatically in GraphQL servers.
- **Detect it**: log query counts per request in development; tools such as Bullet (Rails) flag N+1 patterns.



Common mistakes

- A new connection per request with no pool, then "too many connections" at peak.
- Pool size × instance count exceeding the database limit after autoscaling adds servers.
- Serverless functions each holding connections without a proxy such as RDS Proxy or PgBouncer.
- Trusting the ORM blindly and shipping N+1 queries on a list page.

In the interview

Q: A list page slows as the list grows, and the database shows thousands of tiny identical queries. What's happening?

A: N+1: one query for the list and one per item for related data. Eager-load or batch the lookups into one `IN` query.

Q: Why put PgBouncer in front of PostgreSQL?

A: PostgreSQL connections are heavy (one process each) and limited; PgBouncer lets thousands of client connections share a few real ones.

TL;DR

Reuse connections through a pool sized to what the database can actually run in parallel. Use ORMs for productivity but watch their SQL, especially for N+1 queries.

Choosing a database

What it is

A decision aid for this part. Justify the choice from the **access pattern** (how data is read and written), the **consistency** needed, and the **scale** (data size, requests per second) — not brand names.

How it works

Ask, in order:

1. **What are the main queries?** By key, by range, full-text, joins, aggregations, graph traversal, similarity?
2. **How much data, growing how fast?** Gigabytes fit anywhere; tens of terabytes push towards sharding or a distributed store.
3. **Read-to-write ratio and peak writes per second?** Read-heavy suggests replicas and caches; write-heavy suggests LSM-based or sharded stores.
4. **How strong must consistency be?** Money and inventory need transactions; likes and view counts can be eventual.
5. **Across regions?** Multi-region writes push towards multi-leader, leaderless or globally distributed databases.

6. **What can the team run?** Operating a database well is a major cost; a managed service often wins.

Real-world examples

Need	Good choice	Examples
General app data, transactions	Relational	PostgreSQL, MySQL, Aurora
Relational at global scale	Distributed SQL	Spanner, CockroachDB, YugabyteDB
Cache, sessions, counters	In-memory key-value	Redis, Memcached
Simple key access at huge scale	Managed key-value	DynamoDB, Bigtable
Flexible nested records	Document	MongoDB, Firestore, Couchbase
Heavy writes by key and time	Wide-column	Cassandra, ScyllaDB, HBase
Relationships many hops deep	Graph	Neo4j, Neptune
Metrics and sensor data	Time-series	Prometheus, InfluxDB, TimescaleDB
Full-text search, logs	Search engine	Elasticsearch, OpenSearch
Semantic or image similarity	Vector	pgvector, Pinecone, Milvus
Analytics and reporting	Columnar warehouse	Snowflake, BigQuery, ClickHouse
Photos, videos, backups	Object storage	S3, GCS, Azure Blob
Small strongly consistent config	Coordination store	etcd, ZooKeeper

Common mistakes

- Naming a database without saying why.
- One database type for everything, or a different exotic one for every table.
- Forgetting operational cost: backups, upgrades, monitoring, on-call.

In the interview

Q: Which database would you use for a Twitter-like timeline?

A: Users, follows and tweets as source of truth in a relational database or a wide-column store sharded by user ID; precomputed home timelines in Redis for fast reads; media in object storage behind a CDN; a search engine for tweet search. Each choice follows the access pattern.

TL;DR

Start with PostgreSQL or MySQL unless the access pattern or scale clearly needs otherwise, and add specialist stores (cache, search, object storage, analytics) beside it as needs appear.

Part 3 glossary

Every Part 3 term is defined in one line in the Master A–Z glossary, which merges the glossaries of all six parts into a single alphabetical list.

Part 4 — Caching, async processing and consistency

Parts 1 to 3 built a working system. This part covers the three ideas that keep it fast and correct once it grows past one machine:

1. **Caching**: keep a copy of popular data somewhere quicker to reach, to do less slow work.
2. **Asynchronous processing**: hand slow work to a queue and answer the user straight away.
3. **Consistency in distributed systems**: once data lives on many machines, decide what "the current value" means and how machines agree on it.

Most of the vocabulary that makes system design sound hard (CAP, quorum, Raft, idempotency, saga, vector clock, Bloom filter, Snowflake ID) is here, each explained from zero.

Caching basics

What it is

A **cache** is a small, fast store keeping copies of data that is slow or expensive to get from its real home, the **source of truth** or **origin** (usually a database or another service) — like a chef's bowl of chopped onions by the stove instead of trips to the cold room: within reach, small, and slowly going stale.

- **Cache hit**: the data was in the cache (fast path). **Cache miss**: it wasn't, so you go to the source of truth (slow path) and usually store the answer for next time.
- **Hit ratio** (hit rate): hits ÷ total lookups; 0.95 means 95 of 100 requests never touch the database.
- **TTL (time to live)**: how long an entry may stay before it is treated as expired and dropped or refreshed.
- **Stale data**: a cached copy that no longer matches the source of truth because the source changed afterwards.
- **Warm / cold cache**: a warm cache already holds the popular data; a cold one (just started or emptied) misses on almost everything until it fills.

Why it matters

Memory is many times faster than a database across the network:

Where the data is	Typical time to read a small item
CPU cache / process memory	well under 1 microsecond
Redis or Memcached on the same data-centre network	about 0.2–1 ms
Database query hitting an index on SSD	about 1–10 ms
Database query doing a big join or scan	50 ms to seconds
Call to a service in another continent	70–200 ms

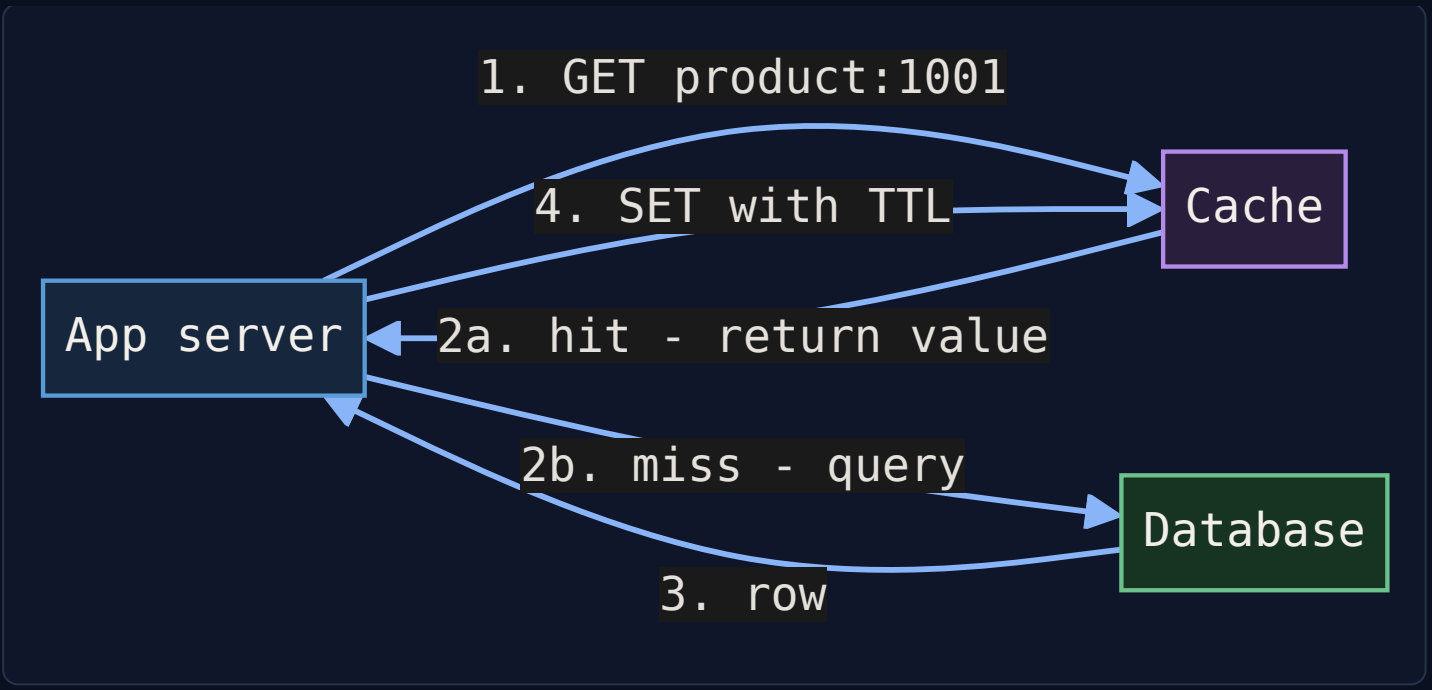
Three wins: **lower latency** (a 1 ms cache read instead of a 30 ms query); **less database load** (at 95% hits the database sees 5% of reads — the difference between one database and twenty); **lower cost** (memory reads are cheaper than recomputing or buying database capacity).

Hit-ratio maths: average read time = hit ratio × cache time + miss ratio × (cache time + database time). With a 1 ms cache, 20 ms database and 90% hits: 0.9 × 1 + 0.1 × 21 ≈ 3 ms; at 99%, ≈ 1.2 ms. Watch the miss ratio — that's what the database feels: 90% → 99% hits cuts misses from 10 per 100 reads to 1, a tenfold drop in database reads.

How it works

A cache is usually a **key-value store**: look up `user:42`, get back user 42's profile JSON. Read path:

1. Build a key from the request (`product:1001`).
2. Ask the cache.
3. On a hit, return the value.
4. On a miss, read the database, store the result with a TTL (say 5 minutes), return it.



Caches work because real traffic is **skewed**: a small fraction of items gets most requests (the top 20% of products might get 80% of views), so caching that hot fraction gives a high hit ratio.

Trade-offs

- **Speed vs freshness**: longer caching is faster and cheaper but more likely stale.
- **Memory cost**: RAM costs far more per gigabyte than disk; you can't cache everything.
- **Extra moving part**: the cache can fail, fill up, or disagree with the database — two copies of the truth to keep in line.
- **Cold-start risk**: a wiped cache sends every request to a database sized for 5% of traffic.

Common mistakes

- Caching data rarely read twice (low hit ratio, wasted memory).
- Forgetting a TTL, so a wrong value lives forever.
- Treating the cache as the source of truth; it may lose data at any time.
- Sizing the database only for miss traffic, then crashing when the cache restarts.

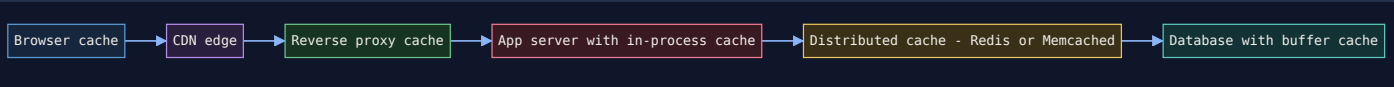
TL;DR

A cache is a fast copy of slow data: it wins on latency and database load, at the price of memory and the risk of stale data.

Where caches live

What it is

Copies of data can be kept at every hop between user and database. The closer to the user, the faster the hit but the harder to update or delete.



Types

Client or browser cache. Copies of files (images, CSS, JavaScript, sometimes API responses) on the user's device, controlled by HTTP headers: `Cache-Control: max-age=86400` ("reuse for one day without asking"); an `ETag` version fingerprint lets the browser ask "changed?" and get a tiny `304 Not Modified`. Mobile apps do the same with local storage. Fastest hit (no network), but you can't delete a stale copy

on a user's device, so rely on TTLs and changing the file name when content changes (`app.3f9a1c.js` — **cache busting**).

CDN cache. Edge servers in many cities (Part 2) cache responses near users — great for static files and public pages identical for everyone.

Reverse proxy cache. NGINX, Varnish and similar proxies in front of app servers (Part 2) store whole HTTP responses and serve repeats without touching the app.

Application or in-process cache. An in-memory map inside the app process (Java Caffeine, Python `functools.lru_cache`): nanosecond reads, no network hop. But each server has its own copy (ten servers, ten copies that can disagree), memory is limited to the process, and it's lost on every restart or deploy.

Distributed cache. A separate cluster shared by all app servers over the network:

- **Memcached**: simple, very fast, multi-threaded key-value cache of plain strings/bytes; no persistence or built-in replication. Facebook ran one of the world's largest deployments.
- **Redis**: key-value server with rich data types (strings, lists, hashes, sets, sorted sets, streams, HyperLogLog), optional persistence, replication and a cluster mode splitting keys across nodes. Also used for leaderboards, rate limiters, session stores, simple queues. (Valkey is a community fork created in 2024 when Redis changed its licence.)

Shared by all servers, survives app restarts, holds hundreds of gigabytes across nodes — at the cost of a network hop (well under a millisecond in a data centre) and another system to run.

Database buffer cache. Databases keep recently used pages in RAM — PostgreSQL's **shared buffers**, MySQL InnoDB's **buffer pool** — for free; why a query's second run is often much faster.

Others: **DNS caches** (laptop and resolver cache lookups for their TTL, Part 2), CPU caches (L1/L2/L3), and the OS **page cache** of recently read file data.

Trade-offs

Layer	Speed of hit	Shared by all users?	Easy to invalidate?
Browser	fastest (no network)	no, one user	no
CDN	very fast (nearby city)	yes, per region	yes, with purge API, takes seconds
In-process	nanoseconds	no, one server	hard across many servers
Distributed (Redis)	under 1 ms	yes	yes, delete the key
DB buffer cache	fast inside DB	yes	automatic

Real-world examples

- Facebook's 2013 paper "Scaling Memcache at Facebook" describes a Memcached layer handling billions of requests per second in front of MySQL.
- Twitter (X) and GitHub use Redis and Memcached heavily for timelines, sessions and counters.
- Netflix runs EVCache, its own Memcached-based distributed cache replicated across AWS regions.

In the interview

Q: Where would you put a cache in this design?

A: Outside in: browser and CDN for static assets and public pages, a distributed cache like Redis in front of the database for hot reads such as profiles, and the database's own buffer cache. State what you cache, the key format and the TTL.

Q: In-process cache or Redis?

A: In-process is fastest but per-server, so it suits small, rarely changing data (config, feature flags). Redis is shared, larger and survives deploys, so it suits user data and anything that must look the same from every server.

TL;DR

Caches exist at every hop: browser, CDN, reverse proxy, app memory, a shared cache like Redis, the database. Closer to the user is faster but harder to invalidate.

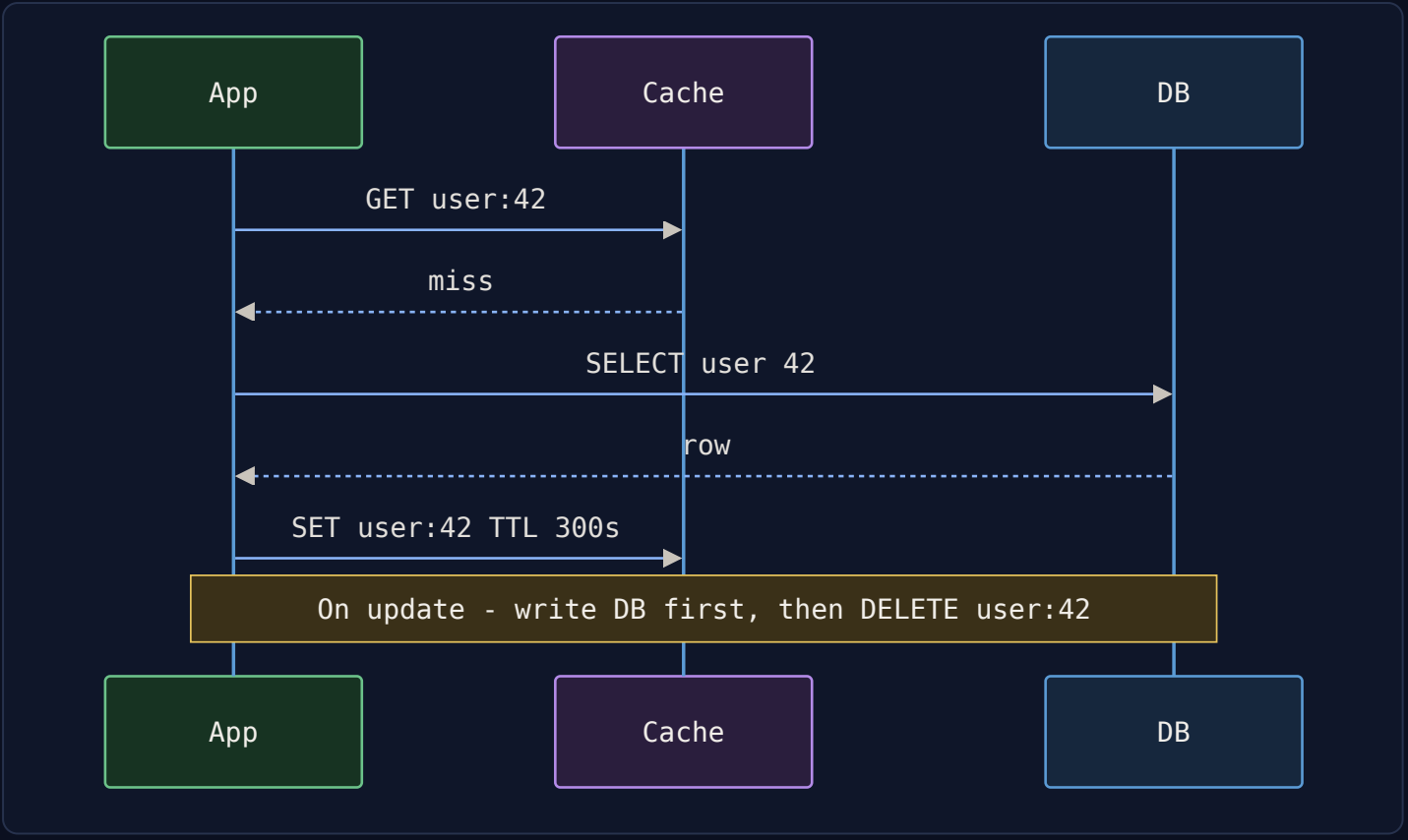
Caching strategies

What it is

A **caching strategy** (pattern) is the rule for who reads and writes the cache, and in what order relative to the database. It decides how stale data can get and what happens on a half-way failure.

Types

Cache-aside (lazy loading). The app talks to both cache and database. Read: check the cache; on a miss read the database and fill the cache. Write: update the database, then **delete** (invalidate) the key so the next read reloads it. The most common pattern: only requested data is cached, and if the cache dies the app still works (slower).



Why delete rather than update the cache on writes? Racing writers can leave the older value cached last with "update cache"; deleting makes the next reader fetch the fresh row. A small race remains (a slow reader can put an old value back just after the delete), so keep a TTL as a safety net.

Read-through. Like cache-aside, but the cache itself loads from the database on a miss; the app only talks to the cache. Simpler app code, but needs a cache library or product that can call your database (a loader function in Caffeine; Amazon DynamoDB Accelerator, DAX, in front of DynamoDB).

Write-through. Every write goes to the cache, which synchronously writes the database before confirming. Cache and database stay in step and reads after a write hit; but every write pays twice, and the cache fills with data that may never be read.

Write-behind (write-back). The write goes only to the cache, the user gets "done" immediately, and the cache flushes to the database later, often in batches (say every second). Very fast, fewer database writes (ten updates to one counter become one) — but a cache crash before flushing loses those writes. For view counters and the like, where losing a few is acceptable. CPUs and disk controllers use write-back caches internally too.

Write-around. Writes go straight to the database, skipping the cache, which is filled only by reads (usually with cache-aside). Good when written data is rarely read soon after (logs, bulk imports), so the cache isn't flooded with cold data.

Refresh-ahead. The cache reloads popular entries **before** they expire: an item with a 60-second TTL read at second 50 is returned and refreshed in the background, so users never wait on a hot-key miss. Cost: refreshing items nobody asks for again.

Strategy	Who writes the DB	Read after write	Main risk
Cache-aside	App	miss, then reload	brief stale window
Read-through	App (cache loads reads)	miss, then reload	same as cache-aside
Write-through	Cache, synchronously	hit, fresh	slower writes
Write-behind	Cache, later	hit, fresh	data loss on crash
Write-around	App, cache skipped	miss	first reads slow
Refresh-ahead	n/a (read side)	depends	wasted refreshes

Trade-offs

Read-heavy and tolerant of seconds of staleness (product pages, profiles): cache-aside with TTL. Must read your own write immediately: write-through. Extremely write-heavy and loss-tolerant (likes, view counts): write-behind. Written once, rarely read: write-around.

Common mistakes

- Updating cache and database in two steps with no TTL, so a failure between them leaves them different forever.
- Write-behind for money or orders, where a lost write is unacceptable.
- Never caching `null` results, so lookups for non-existent things always hit the database (exploitable with random IDs). Caching a short-lived "not found" marker — **negative caching** — fixes it.

In the interview

Q: Which caching pattern for user profiles?

A: Cache-aside: read Redis, on miss load from the database and set a TTL of a few minutes; on update write the database then delete the key. Simple, survives cache failure, caches only what's read.

Q: When is write-behind acceptable?

A: When speed matters more than losing a few recent writes (counters, analytics events). Never for payments or anything a user was told is saved.

TL;DR

Cache-aside is the default: read the cache, fall back to the database, delete the key on writes. Write-through trades write speed for freshness; write-behind trades safety for speed.

Eviction policies

What it is

When a full cache gets a new item it must throw one out; the **eviction policy** picks the victim. Eviction (removal because the cache is full) differs from **expiration** (removal because the TTL ran out); many caches do both.

Types

- **LRU (least recently used)**: evict the item longest without a read. A hash map plus doubly linked list moves each read item to the front in O(1). The default almost everywhere, since recent use predicts near-future use.

- **LFU (least frequently used)**: evict the item with fewest reads; better for steadily popular items. Weakness: yesterday's hit keeps a high count and squats, so real LFU decays counts over time.
- **FIFO (first in, first out)**: evict the oldest inserted item regardless of use; simple, but can evict a hot item that arrived early.
- **TTL-based**: items leave when their TTL expires — not strictly "who to evict when full", but bounds staleness and frees memory.
- **Random**: surprisingly decent and very cheap.
- Others: **MRU** (most recently used, useful for some scan patterns) and modern designs like **W-TinyLFU** (Java Caffeine), combining recency and frequency.

Redis's `maxmemory-policy` chooses, e.g. `allkeys-lru`, `allkeys-lfu`, `volatile-ttl` or `noeviction` (refuse new writes when full). Redis uses approximate LRU: it samples a handful of keys and evicts the best candidate, saving the memory of a full linked list.

Common mistakes

- `noeviction` on a cache, so it rejects writes when full.
- One huge scan (a nightly report reading every row) flushing all hot items out of an LRU cache; LFU or scan-resistant policies avoid this.

TL;DR

A full cache evicts by rule: LRU is the safe default, LFU for steady favourites, FIFO and random for simplicity, TTL to bound staleness.

Cache invalidation, stale data and stampedes

What it is

Cache invalidation is removing or updating a cached copy when the source of truth changes. Phil Karlton's joke: "there are only two hard things in computer science: cache invalidation and naming things". It is hard because cache and database are separate systems with no single moment where both change.

How it works

From simplest to most precise:

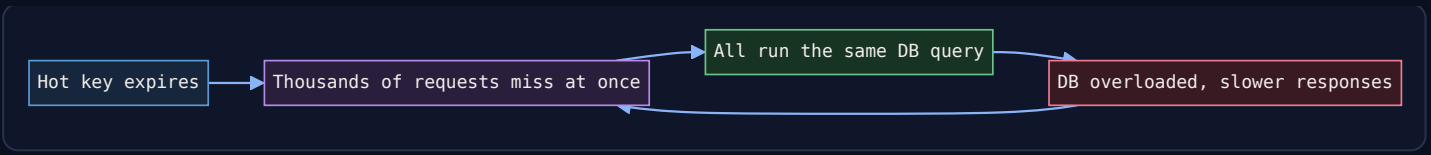
1. **TTL only.** Accept data up to TTL seconds old (a 60-second TTL on a price means a change may take a minute to show).
2. **Delete on write.** After writing the database, delete the key (cache-aside).
3. **Event-driven invalidation.** The database publishes change events (e.g. **change data capture, CDC**, reading the database's own change log) and a consumer deletes or updates affected keys — works even when many services write.
4. **Versioned keys.** Put a version in the key (`user:42:v7`); a write bumps it to `v8`, readers look for the new key and the old one ages out. No delete needed.
5. **CDN purge.** An API purges a URL or tag, usually worldwide in seconds.

Stale data

Stale reads come from TTLs, lost delete messages, reader/writer races, and replication lag (the cache filled from a lagging replica). Decide per feature how stale is acceptable: a follower count can be minutes old; a bank balance on the transfer screen cannot.

The thundering herd / cache stampede

A **cache stampede** (**thundering herd**, **dog-piling**): a very popular key expires or is deleted and thousands of requests miss at once, all querying the database for the same value; the database slows or falls over, nothing refills the cache, and more requests pile in. E.g. a home-page key read 10,000 times per second expires; in the 200 ms rebuild, 2,000 requests hit the database with the same expensive query.



Fixes for stampedes

- **Request coalescing / single flight**: only one request per key rebuilds; others wait for its result (in one process) or briefly retry the cache. CDNs such as Fastly and Cloudflare call it **request collapsing**.
- **Lock or lease**: the first miss takes a short lock (Redis `SET key:lock 1 NX PX 5000` — set only if absent, expire in 5 s); the winner rebuilds, others serve the old value or wait. Facebook's Memcache paper calls it a **lease**.
- **Serve stale while revalidating**: return the expired value while one background job refreshes it; HTTP has `Cache-Control: stale-while-revalidate=30`.
- **Early or probabilistic refresh**: as expiry approaches each reader has a small, growing chance to refresh early, so one request refreshes before the deadline instead of everyone at it.
- **TTL jitter**: randomise TTLs (300 s ± 30 s) so keys written together don't expire together. Also prevents a **cache avalanche**, where a mass expiry or cache restart sends everything to the database at once.
- **Warm-up**: pre-load hot keys after a restart or deploy before real traffic.

Related: **cache penetration** (requests for keys that exist nowhere, each hitting the database; fixed by negative caching or a Bloom filter, later) and **hot key** (one key overloading the single cache node holding it; fixed by replicating it to several nodes or caching it in-process too).

Common mistakes

- The same fixed TTL on millions of keys loaded at the same moment.
- Every miss rebuilding independently with no coalescing.
- Assuming "delete after write" is atomic: if the delete fails, the stale value lives until TTL — so always have a TTL.

In the interview

Q: A celebrity's profile key expires and the database spikes. What happened, and the fix?

A: A cache stampede: thousands of concurrent misses on one hot key all query the database. Use single-flight or a short lock so one request rebuilds, serve the stale value meanwhile, refresh hot keys early, and jitter TTLs.

Q: How do you keep a cache consistent with the database?

A: Not perfectly, cheaply. Cache-aside with delete-on-write plus a TTL safety net; if many services write, invalidate from a CDC stream; choose the TTL per data type by how much staleness users tolerate.

TL;DR

Invalidation is the hard part of caching: delete-on-write plus TTLs, and protect hot keys from stampedes with locks, stale-while-revalidate, early refresh and TTL jitter.

Message queues

What it is

A **message queue** is a middleman that holds messages (small pieces of data describing work or an event) until another program is ready to handle them. Three roles:

- **Producer** (publisher): sends a message ("send welcome email to user 42").
- **Broker**: the queue server storing messages safely until taken — RabbitMQ, Amazon SQS, Apache Kafka.
- **Consumer** (worker, subscriber): takes messages and does the work (an email worker).

Like a restaurant's order rail: the waiter clips a ticket and returns to customers; cooks take tickets when free; tickets queue up when the kitchen is busy.

Why it matters

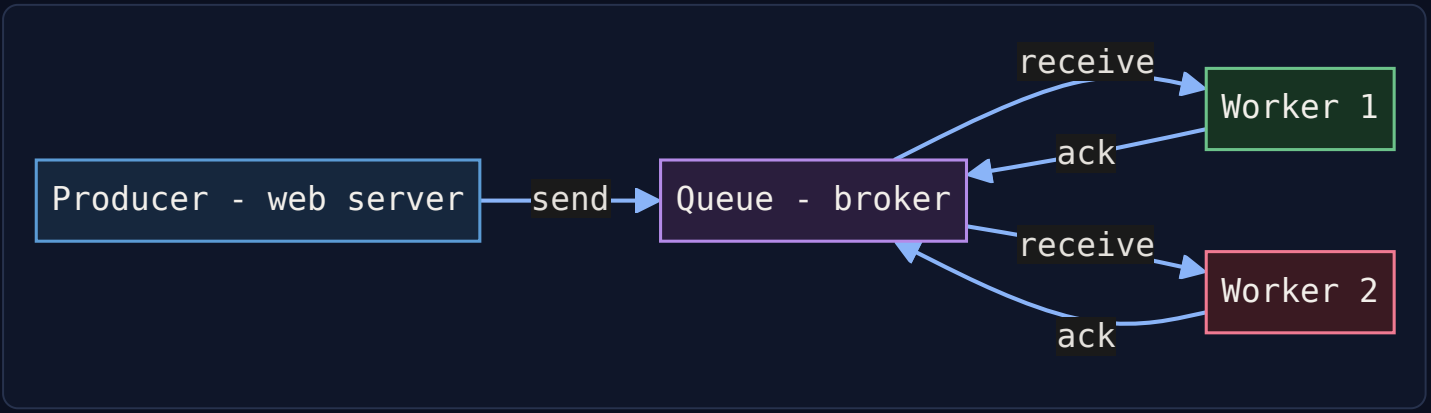
Without a queue the user waits for the email to send. Queues give:

1. **Faster responses**: "signed up!" in 50 ms while slow work (email, photo resizing, card charging, PDF generation) runs in the background — **asynchronous** processing, the caller not waiting for the result.
2. **Decoupling**: the producer needn't know who consumes, how many, or whether they're up; add a consumer (analytics) without touching the producer.
3. **Absorbing spikes (load levelling)**: 10,000 orders in a minute with capacity for 2,000 a minute — the queue holds the backlog and workers catch up in five minutes instead of falling over.
4. **Reliability**: if a worker crashes mid-task, the message is still in the broker for another worker.
5. **Independent scaling**: add workers as the queue grows, remove them as it shrinks.

How it works

A work-queue lifecycle (SQS, RabbitMQ):

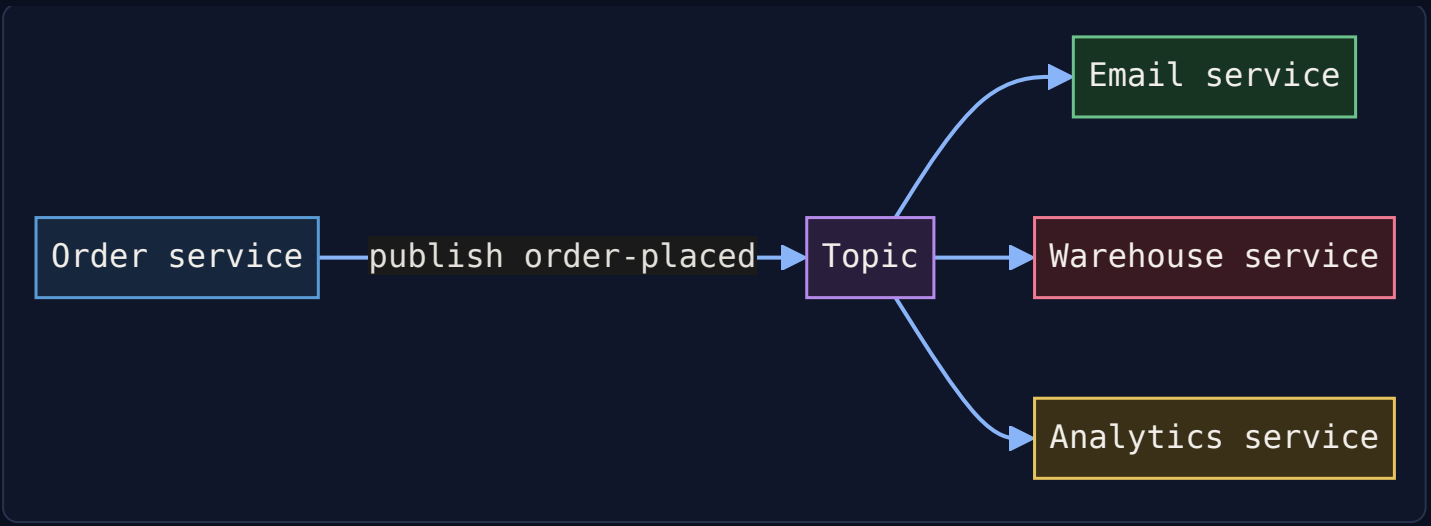
1. The producer sends; the broker writes to disk (often to replicas) and replies "accepted".
2. A consumer receives a message, which the broker hides from others for a while (SQS's **visibility timeout**, 30 s by default).
3. The consumer does the work.
4. It sends an **acknowledgement (ack)**: "done, delete it".
5. With no ack in time (crash or too slow), the message becomes visible again for another consumer — why messages can be delivered more than once (below).



Types

Point-to-point (work queue). Each message goes to exactly one of many consumers, which compete for messages (**competing consumers** pattern) — for jobs like "resize this image" that must happen once.

Publish/subscribe (pub/sub). A message published to a **topic** (a named channel, like "order-placed") gives **every** subscriber its own copy: "order 9 placed" reaches email, warehouse and analytics services. For events many parts of the system care about.



Often combined: a topic fans out to several queues, each with competing workers (an Amazon SNS topic feeding several SQS queues; a RabbitMQ fanout exchange feeding several queues).

Trade-offs

- Resilience and speed, but no immediate answer: "email will be sent", not "email sent".
- Harder debugging, as work happens later elsewhere: you need tracing and monitoring of **queue depth** (messages waiting) and **consumer lag** (how far behind consumers are).
- Messages may arrive twice or out of order (delivery guarantees), so consumers need care.
- The broker is another system to run and size.

Common mistakes

- A queue for something the user must see immediately ("is my password correct?").
- Not monitoring queue depth, so a dead worker goes unnoticed while millions of messages pile up.
- Huge payloads (videos) in messages. Store the file in object storage and send its URL — the **claim check** pattern.

TL;DR

A message queue lets producers hand work to consumers through a broker: slow tasks run in the background, spikes are absorbed, services decoupled. Point-to-point gives each message to one worker; pub/sub gives every subscriber a copy.

Event streaming: topics, partitions, consumer groups and offsets

What it is

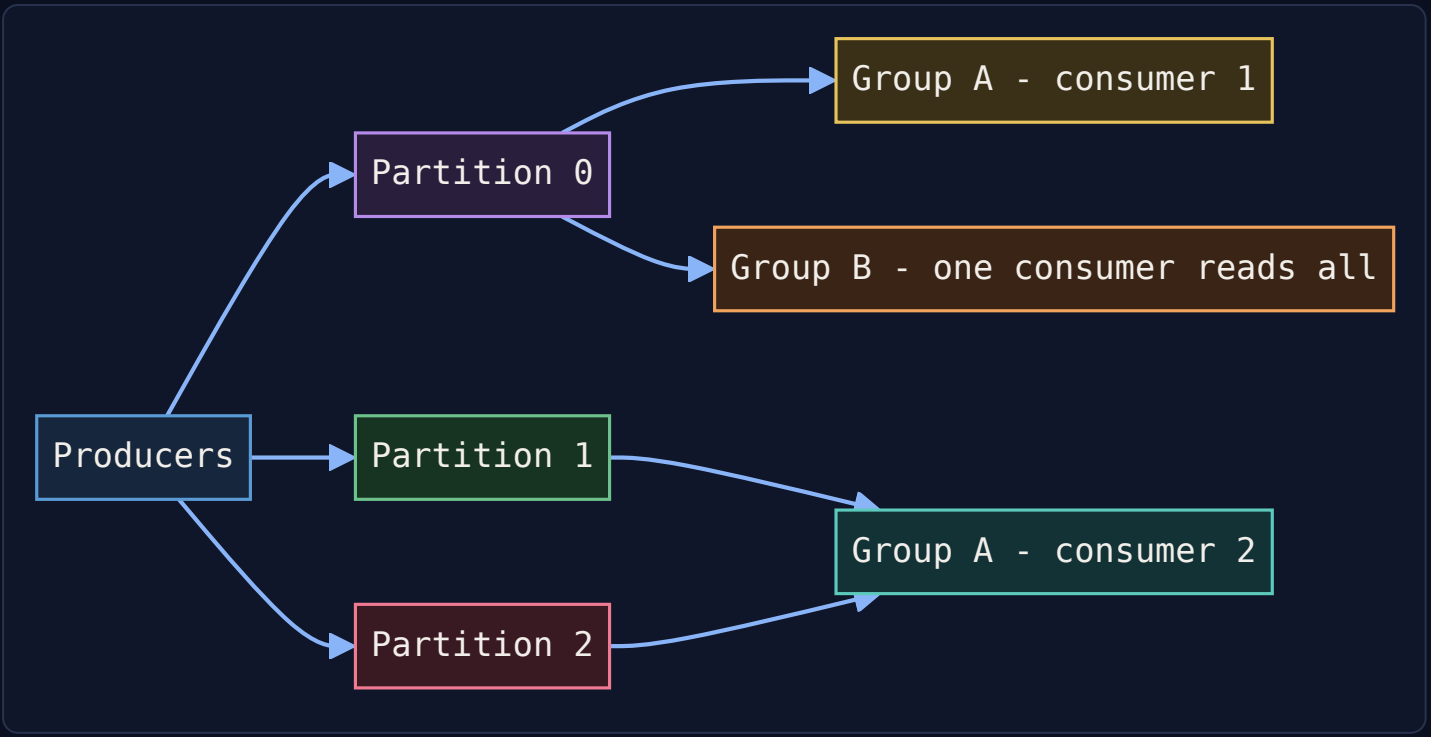
Classic queues delete a message once acknowledged. An **event streaming platform** (**log-based broker**) such as **Apache Kafka** keeps every message in an append-only **log** for a retention period (Kafka's default 7 days), read or not. Consumers just remember how far they've read, so many independent consumers can read the same log and any can rewind and re-read history — a newspaper archive with a bookmark per reader, rather than a to-do pile thrown away as you go.

How it works

Kafka vocabulary:

- **Topic**: a named stream of events, such as `payments`.
- **Partition**: a topic is split into partitions, each an ordered log on one broker (copied to others for safety). Partitions are how Kafka scales: 12 partitions can be written and read by 12 machines in parallel.
- **Key**: a message's key (such as a user ID) is hashed to choose the partition, so all of user 42's events land in one partition, in order.

- **Offset**: a message's fixed position number within a partition: 0, 1, 2… Ordering is guaranteed only **within** a partition, not across the topic.
- **Consumer group**: consumers sharing the work for one application; each partition goes to exactly one consumer in the group (12 partitions, 4 consumers → 3 each). Another group (analytics) independently gets its own full copy of the stream.
- **Committed offset**: each group periodically records "partition 3 processed up to offset 1,500"; after a crash, the replacement resumes from there.
- **Rebalance**: partitions are reassigned when a consumer joins or leaves.
- **Consumer lag**: newest offset minus committed offset — the main health metric.



More consumers than partitions doesn't help — extras sit idle — so choose the partition count with future parallelism in mind.

Types

	Classic queue (RabbitMQ, SQS)	Event stream (Kafka, Kinesis, Pulsar)
After consumption	message deleted	message kept until retention ends
Re-reading history	no	yes, reset the offset
Many independent readers	needs one queue per reader	natural, one group per reader
Ordering	weak (FIFO queues exist)	strict per partition
Best for	background jobs, task distribution	event pipelines, analytics, CDC, replay

RabbitMQ: a traditional broker with flexible routing — producers send to an **exchange**, which routes to queues by rules (direct, fanout, topic patterns). **Amazon SQS**: fully managed; standard queues give at-least-once delivery and best-effort ordering, FIFO queues add strict ordering and de-duplication. **Kafka**: created at LinkedIn, now Apache; runs its own Raft-based metadata quorum (KRaft) instead of ZooKeeper, and Kafka 4.0 (2025) removed ZooKeeper support entirely. **Amazon Kinesis**, **Apache Pulsar** and **Redpanda** are similar streaming systems.

In the interview

Q: Kafka or RabbitMQ/SQS?

A: Background jobs that run once and are forgotten: a queue like SQS or RabbitMQ is simpler. A high-volume event stream that several services read independently, replay, or need ordered per key: Kafka.

Q: How do you keep all events for one order in order?

A: Use the order ID as the message key, so they share a strictly ordered partition, with one consumer per partition.

TL;DR

Kafka-style streams keep an ordered, replayable log split into partitions; consumer groups share partitions and track position with offsets. Ordering holds only within a partition.

Delivery guarantees, idempotency and retries

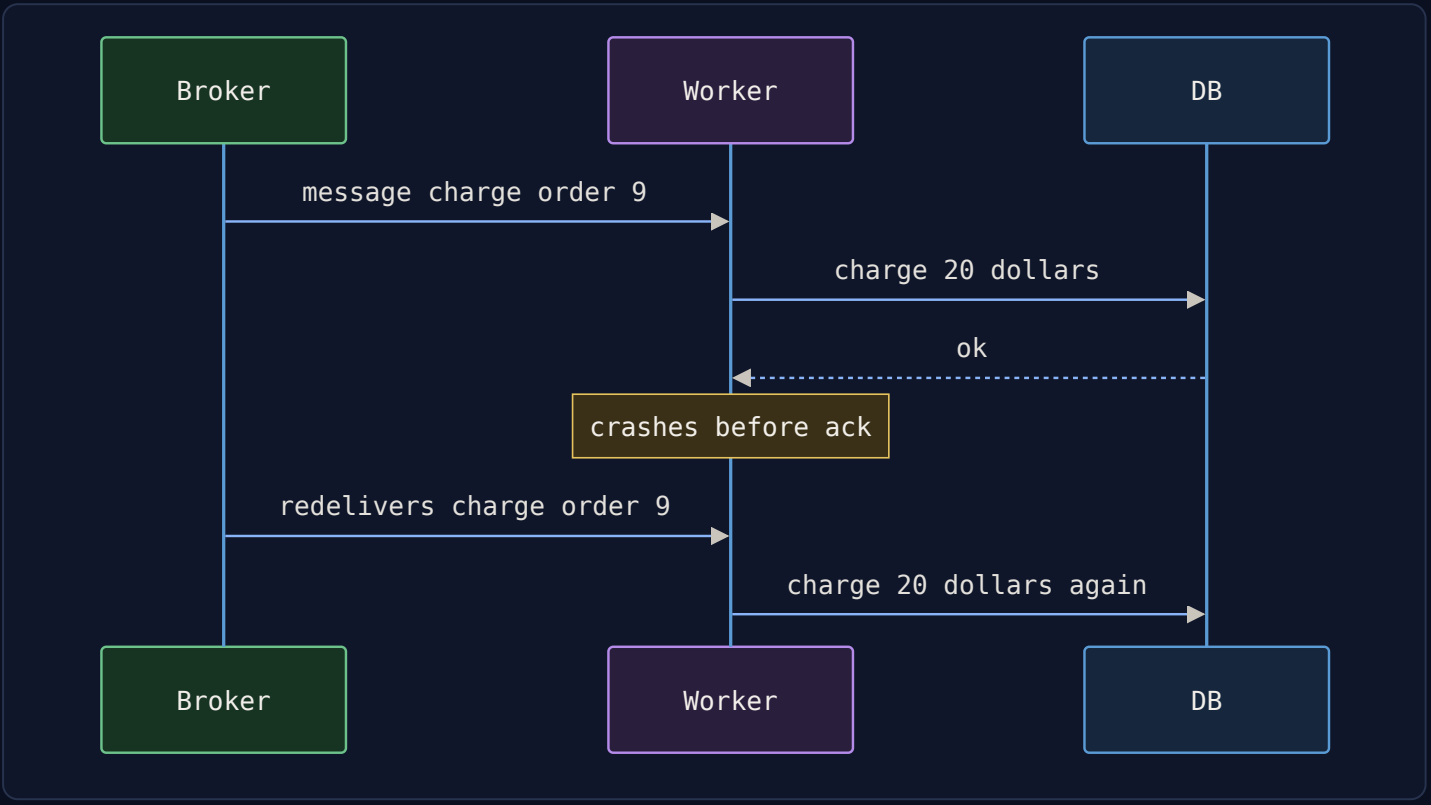
What it is

Networks drop packets and machines crash, so a broker can never be sure a message was processed exactly once. A **delivery guarantee** states the promise:

- **At-most-once**: delivered zero or one times — lost on failure, never repeated. Acknowledge (or commit the offset) **before** processing. Fine for metrics where a lost point doesn't matter.
- **At-least-once**: delivered one or more times — never lost, maybe repeated. Acknowledge **after** processing: a crash after the work but before the ack brings the message back. The default for most systems.
- **Exactly-once**: each message affects the result exactly once. True exactly-once **delivery** over an unreliable network is impossible in general; systems offer **exactly-once processing** (**effectively-once**): duplicates may be delivered but are detected and ignored.

How it works

Why at-least-once means duplicates:



The fix is an **idempotent** consumer. **Idempotency**: doing an operation twice has the same effect as once. "Set balance to 100" is idempotent, "add 20" is not; "delete order 9" is, "create a new order" is not.

Making non-idempotent work idempotent:

1. Give each operation a unique **idempotency key** (a UUID the client generates once for "pay for order 9" and reuses on every retry).
2. Before working, check a table of processed keys.

3. If new, do the work **and** record the key in the same database transaction.
4. If seen, skip the work and return the saved result.

Payment APIs such as Stripe accept an `Idempotency-Key` HTTP header so a phone that loses signal and retries "pay $20" isn't charged twice. Other techniques: **upserts** (insert or overwrite by unique ID), database unique constraints, conditional updates ("set SHIPPED only if PAID").

Kafka provides exactly-once semantics inside Kafka (since 0.11, 2017) by combining an **idempotent producer** (the broker drops duplicate sends by producer ID and sequence number) with **transactions** writing output messages and the consumer's offset atomically. Once a side effect leaves Kafka (an email, a bank call), you need idempotency in that system again. SQS FIFO queues de-duplicate messages with the same deduplication ID within a 5-minute window.

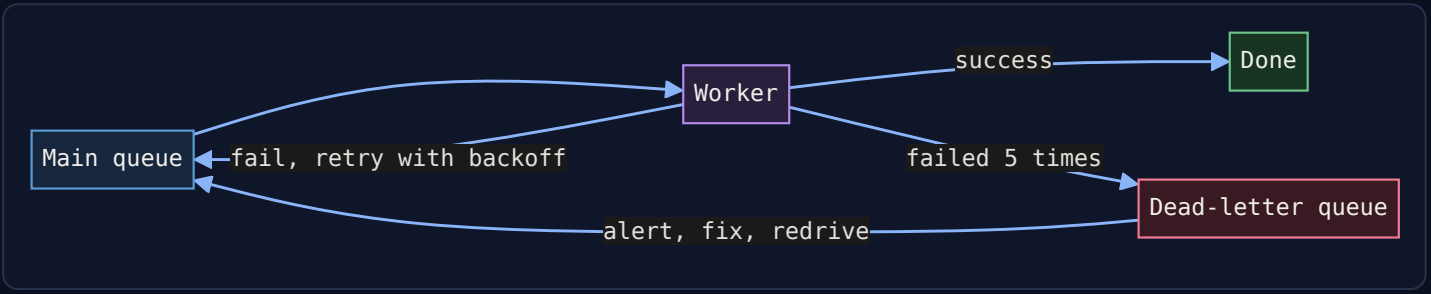
Ordering

Strict global order costs throughput (one line of processing), so practical systems order **per key** (per user, per order) by routing a key to one partition or FIFO group. Retries can break order too (message 5 retried after 6 succeeds); consumers that care carry a version number and ignore an older version arriving after a newer one.

Retries, dead-letter queues and backpressure

Retries. For temporary failures (a timeout, a database briefly down), try again with **exponential backoff** (1 s, 2 s, 4 s, 8 s) so a struggling dependency isn't hammered, plus **jitter** (random variation) so thousands of clients don't retry in sync, and a cap on attempts. Depth in Part 5.

Dead-letter queue (DLQ). A message that keeps failing — a **poison message** (malformed JSON, an order for a deleted product) — would be retried forever, blocking or wasting workers. After N attempts (say 5) the broker moves it to a separate dead-letter queue; engineers are alerted, fix the bug, and **redrive** (replay) the messages to the main queue. SQS, RabbitMQ and most Kafka frameworks support DLQs.



Backpressure. Any way for a slow downstream part to signal "slow down" upstream instead of silently drowning:

- **Buffer**: let the queue grow (fine for spikes, dangerous if it never drains).
- **Block or throttle the producer**: refuse or delay new messages (HTTP `429 Too Many Requests` or `503 Service Unavailable`).
- **Drop**: discard low-priority messages (load shedding).
- **Scale out consumers**: autoscale workers on queue depth.

TCP flow control (Part 2) and reactive-streams libraries are low-level forms of backpressure.

Common mistakes

- Assuming a vendor's "exactly-once" covers your email sends and database writes. It rarely does; design idempotent consumers anyway.
- Retrying instantly in a tight loop, worsening an outage.
- No DLQ, so one poison message blocks a partition (Kafka processes a partition in order, so a stuck message stalls everything behind it).
- Unbounded in-memory queues growing until out of memory.

Real-world examples

- Stripe's API idempotency keys, described in its public docs.

- Uber and LinkedIn run Kafka at trillions of messages per day, relying on per-key partitioning for ordering.
- AWS SQS redrive policies send messages to a DLQ after a configured `maxReceiveCount`.

In the interview

Q: How do you guarantee a payment isn't charged twice when the queue is at-least-once?

A: An idempotent consumer: each payment carries an idempotency key, recorded in the same transaction as the charge; any seen key is skipped, so duplicates are harmless.

Q: What is a dead-letter queue?

A: A side queue for messages that failed a set number of times, so a poison message doesn't block or waste workers; it is monitored and replayed after the fix.

TL;DR

Assume at-least-once delivery: idempotent consumers with idempotency keys, retries with backoff and jitter, a dead-letter queue for repeated failures, backpressure when consumers fall behind.

Batch vs stream processing

What it is

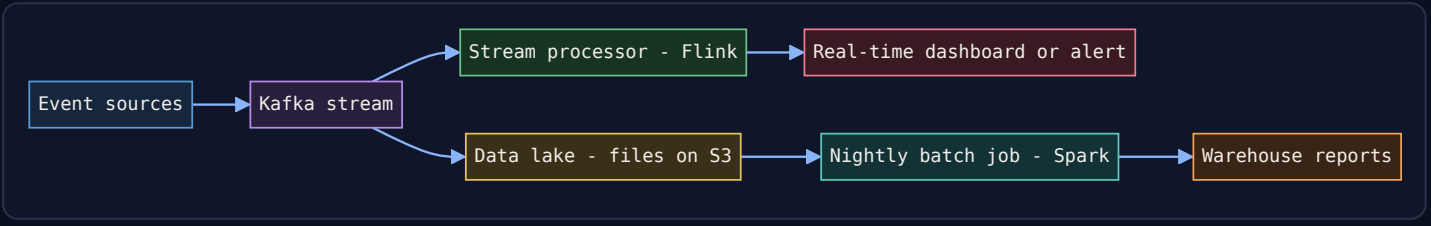
Computing things from flowing data — daily revenue, recommendations, fraud scores, search indexes:

- **Batch processing** handles a large, bounded, already-collected pile (yesterday's 2 TB of logs) in one job, usually scheduled (nightly, hourly) — baking the day's bread before opening.
- **Stream processing** handles each event (or small window) continuously as it arrives, within milliseconds or seconds — a barista making each coffee as ordered.

How it works

MapReduce (Google, 2004; copied by Apache Hadoop) is the classic batch model. Input files are split across machines. **Map**: each machine reads its chunk and emits key-value pairs (word count: `("cat", 1)` per "cat"). **Shuffle**: the framework sends all pairs with the same key to the same machine. **Reduce**: each machine combines its keys' values (`("cat", 1+1+1) = ("cat", 3)`). A dead machine's piece is simply re-run. **Apache Spark** keeps the idea but holds intermediate data in memory, far faster; SQL warehouses (BigQuery, Snowflake, Redshift) run batch-style queries over huge tables.

Stream processors — **Apache Flink**, **Kafka Streams**, **Spark Structured Streaming** — read a stream (usually Kafka), keep state (running counts per user), and group events into **windows**: "orders per minute" (a tumbling one-minute window) or "logins in the last 5 minutes, updated every 10 seconds" (a sliding window). Events can arrive late or out of order, so engines separate **event time** (when it happened) from **processing time** (when it arrived) and use **watermarks** — estimates that all events up to a time have arrived — to decide when a window is complete.



Trade-offs

	Batch	Stream
Latency of results	minutes to hours	milliseconds to seconds
Input	bounded, complete	unbounded, arriving
Complexity	simpler, easy to re-run	harder: state, late events, failures
Typical uses	reports, billing, ML training, backfills	fraud detection, live metrics, alerts

Lambda architecture runs a batch layer (accurate, slow) and a speed layer (fast, approximate) and merges them. **Kappa architecture** uses only a stream, reprocessing by replaying the log from the start; many companies now prefer it to avoid writing logic twice.

In the interview

Q: Should "trending hashtags" use batch or streaming?

A: Streaming — users expect the last few minutes. Count hashtags in sliding windows with a stream processor reading Kafka, perhaps with a count-min sketch, and recompute exact daily numbers in batch.

TL;DR

Batch processes a complete pile on a schedule (MapReduce, Spark); streaming processes events continuously (Flink, Kafka Streams). Choose by how fresh the result must be.

Event-driven architecture, event sourcing and CQRS

What it is

Event-driven architecture (EDA): services communicate by publishing **events** — facts about something already done ("OrderPlaced", "PaymentFailed") — rather than calling each other with commands ("ship this now"). The publisher doesn't know or care who listens, like a town crier.

An **event** reports something done, so a listener can ignore it but not refuse it; a **command** is a request the receiver may reject. Name events in the past tense (`OrderPlaced`), commands as orders (`ShipOrder`).

How it works

The order service saves an order and publishes `OrderPlaced`; payment, inventory, email and analytics each subscribe and react. A later loyalty-points service needs no change to the order service.

A service must both update its database and publish the event; crashing between the two leaves them disagreeing — the **dual write** problem. Standard fix, the **transactional outbox**: insert the event into an `outbox` table in the same transaction as the business change; a separate relay (or a CDC tool like Debezium) reads the outbox and publishes, retrying until it succeeds. With idempotent consumers nothing is lost.

Event sourcing. Store not just the current state (balance = 70) but the sequence of events that led there (Deposited 100, Withdrew 50, Deposited 20) as the source of truth; state is computed by replay, with periodic **snapshots** to avoid replaying millions each time. Benefits: complete audit trail, rebuild state or answer "what was the balance last Tuesday?", easy new projections. Costs: complexity, immutable events make schema evolution harder, queries need separate read models. Bank ledgers and accounting books have always worked this way.

CQRS (Command Query Responsibility Segregation). One model (often one database) for writes (commands), another for reads (queries). Writes go to a normalized database; events or CDC update read-optimised views — a denormalized table, an Elasticsearch index, a cache. Reads scale independently and each screen gets exactly the shape it needs. Cost: the read side lags slightly (eventual consistency), and more moving parts. Pairs naturally with event sourcing but doesn't require it.



Trade-offs

EDA gives loose coupling and easy extension, but flows are harder to follow (no single place shows the whole order process), with eventual consistency, duplicates and ordering issues. Use it where many services react to the same facts; keep request-response for immediate answers.

In the interview

Q: How do you reliably update the database and publish an event?

A: Transactional outbox: the event goes into an outbox table in the same transaction as the change; a relay publishes with retries; consumers are idempotent because it may publish twice.

Q: What is CQRS and when would you use it?

A: Separate write and read models, for when reads and writes differ greatly in shape or scale — e.g. writes to a relational DB, searches served from Elasticsearch kept current by events.

TL;DR

Event-driven systems react to published facts instead of direct calls. Event sourcing stores the events as the truth; CQRS splits the write model from read-optimised views. Both add flexibility at the cost of eventual consistency and complexity.

Consistency models

What it is

With replication (Part 3) a read may hit a copy that hasn't received the latest write. A **consistency model** is the system's promise about what a read can return after a write — a contract: "if you do X, you will never see Y". This is not the C in ACID (transactions keep data valid by your rules): same word, different idea.

(A shopping list copied on three family fridges: strong consistency means everyone's fridge shows "milk" the instant it's added; eventual consistency means phone calls update the copies over minutes — they differ for a while, but converge once editing stops.)

Types

Strongest (easiest to reason about, most expensive) to weakest:

Linearizability (strong consistency). The system behaves as if there is one copy and every operation happens at a single instant between its start and end: once any read returns the new value, no later read anywhere returns the old one — "it looks like one machine". Needed for username uniqueness checks, locks, balance checks before withdrawal. Cost: every write must reach a majority or a leader before confirmation (latency, especially across regions), and requests may be refused during a partition.

Sequential consistency. All clients see all operations in the same order, each client's own in issue order, but that order needn't match real time. Mostly met in theory and CPU memory models.

Causal consistency. If one operation could have caused another, everyone sees them in that order: nobody sees Bob's "Congratulations!" without Alice's "I'm engaged!". Unrelated operations (Carol's post elsewhere) may appear in different orders to different people. Research shows causal consistency (slightly strengthened) is about the strongest model that stays available during a partition, so it's attractive for global systems.

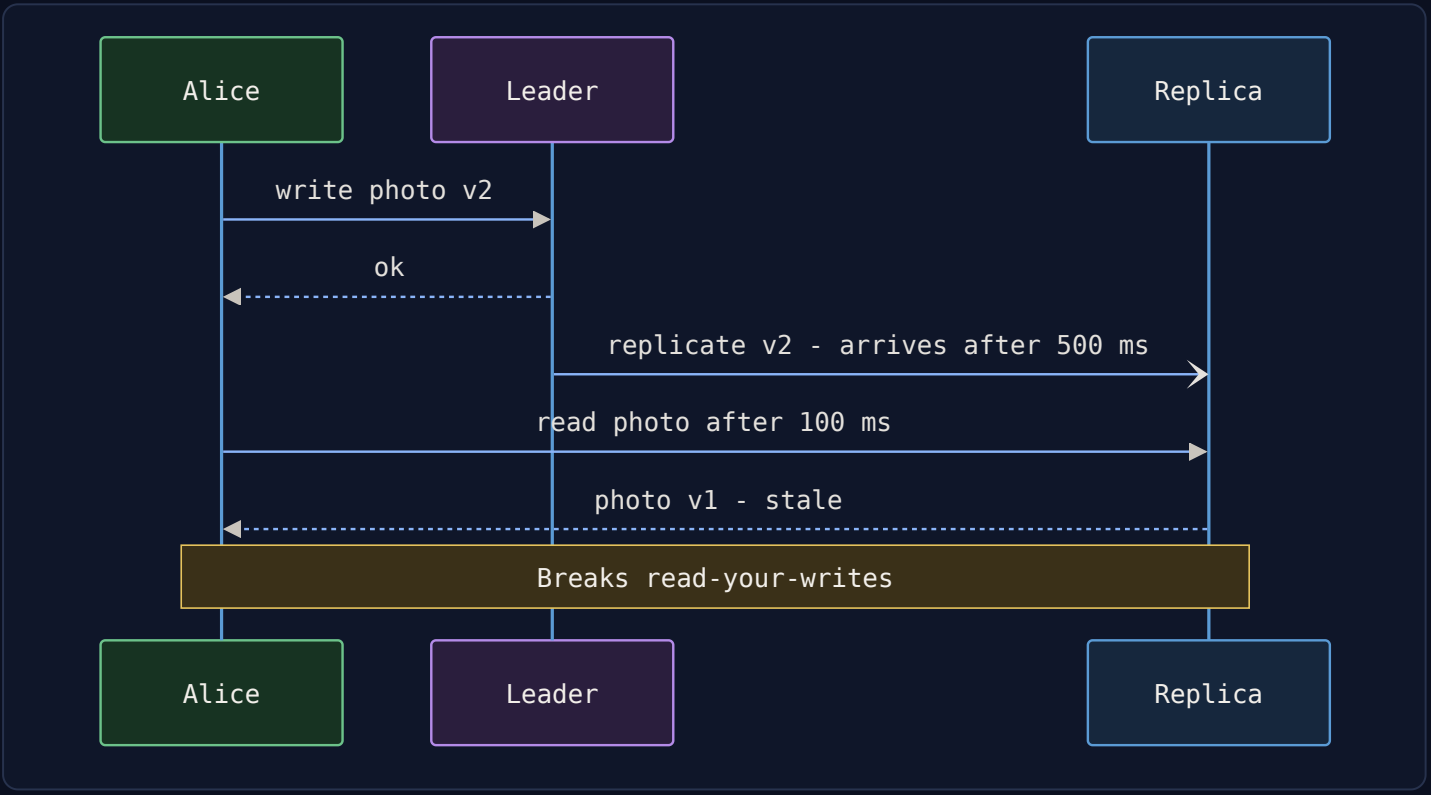
Session guarantees (client-centric consistency): cheap promises to a single user, often enough:

- **Read-your-writes** (read-your-own-writes): after changing your profile photo you always see the new one, even if others see the old for a few seconds. Read your own data from the leader briefly after

writing, or track the write's position and read only from caught-up replicas.

- **Monotonic reads**: once you've seen a value you never see an older one (no comment appearing, vanishing, reappearing across refreshes hitting different replicas). Stick each user to one replica.
- **Monotonic writes**: your writes apply in the order you made them.
- **Writes-follow-reads**: a write based on something you read is ordered after it.

Eventual consistency. If writes stop, all replicas eventually return the same value — no promise how long (usually milliseconds to seconds) or what you see meanwhile. DNS is the classic example: resolvers return old or new answers until TTLs expire. Cheap, fast, highly available.



Choosing per request. DynamoDB reads are eventually consistent by default; a strongly consistent read costs double. Cassandra sets a consistency level per query (ONE, QUORUM, ALL). Google Spanner offers externally consistent (linearizable) transactions globally.

Trade-offs

Model	Promise	Cost	Good for
Linearizable	looks like one copy	high latency, less available	locks, balances, unique names
Causal	cause before effect	moderate	comments, chat, social feeds
Read-your-writes	you see your own changes	low	profiles, settings
Eventual	converges eventually	lowest	likes, view counts, DNS

Common mistakes

- Reading "eventually consistent" as "consistent after a short fixed time". There is no bound.
- Reading from replicas in "save, then redirect to the page showing it" flows: users see their change missing. Use read-your-writes.
- Confusing CAP's consistency (linearizability) with ACID's (valid data).

In the interview

Q: Where is eventual consistency acceptable?

A: Where brief staleness is harmless: like, follower and view counts, recommendations, search indexes. Not balances, inventory at checkout, or uniqueness checks.

TL;DR

Consistency models describe what reads may return after writes: linearizable (acts like one copy), causal, session guarantees, eventual (converges sometime). Stronger means simpler reasoning but slower and less available.

CAP theorem and PACELC

What it is

The **CAP theorem** (Eric Brewer, 2000; proved by Gilbert and Lynch, 2002): a distributed data store facing a **network partition** must choose between consistency and availability.

- **C, consistency**: linearizability — every read sees the latest write, as if one copy.
- **A, availability**: every request to a working (non-crashed) node gets a non-error answer, without waiting forever.
- **P, partition tolerance**: the system keeps operating when the network between nodes drops or delays messages.

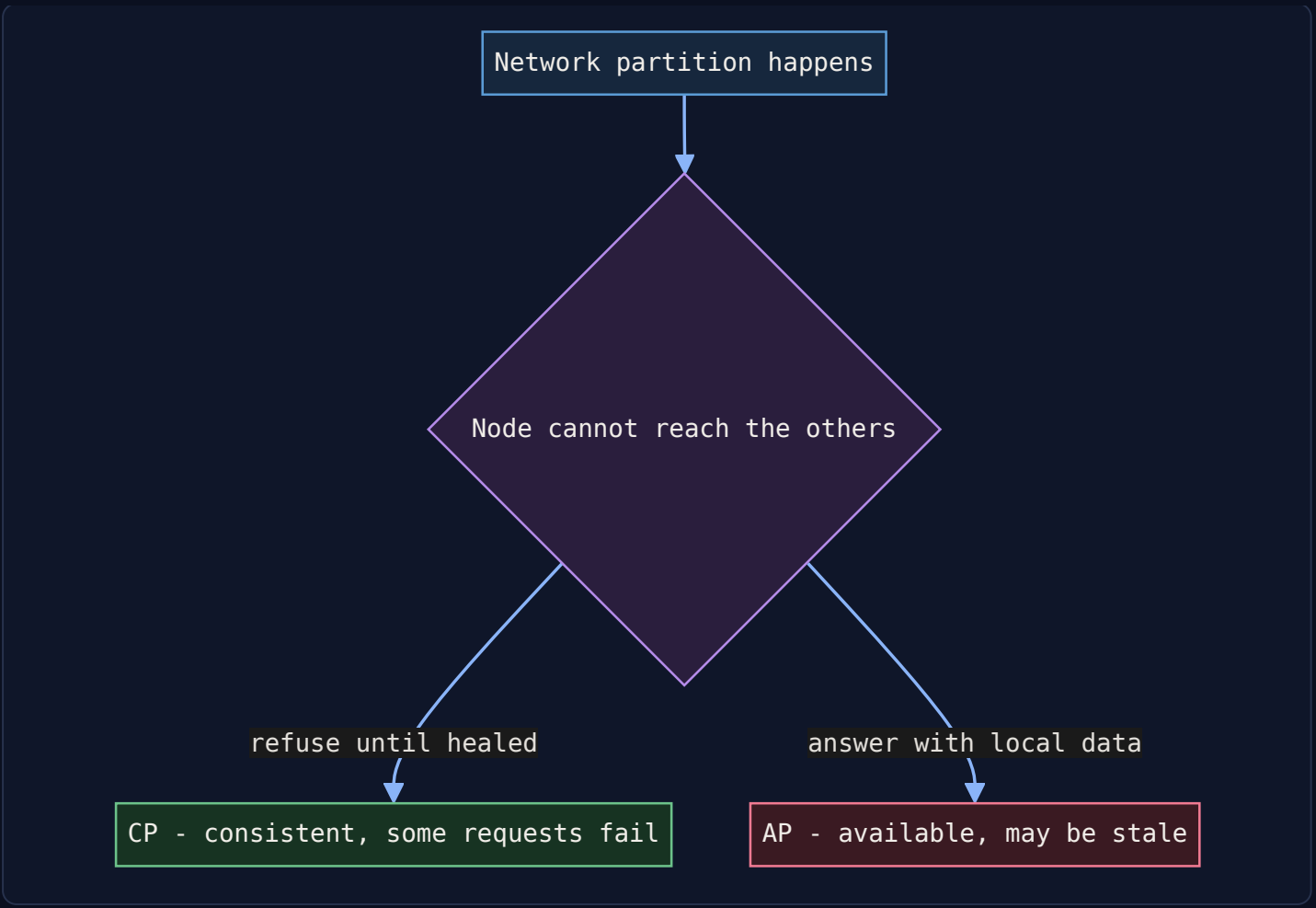
A **network partition** splits nodes into groups that can't talk, though each still runs — e.g. the US–EU link fails; both sides are healthy and serving users, but neither knows what the other is doing.

How it works

Two replicas, US and EU, link cut. An EU user writes `x = 2`; a US user reads `x`. The US replica can:

1. **Refuse or wait** (error or block) since it can't confirm the latest value: **consistent** but not **available** — **CP**.
2. **Answer with what it has** (`x = 1`, old): **available** but not **consistent** — **AP**.

There's no third option: it can't know a write it can't hear about. Since real partitions aren't optional (cut cables, failed switches, cloud zones losing connectivity), the real choice is C or A **during a partition**; with a healthy network a system can provide both.



Types

CP examples: consensus or single-leader systems — ZooKeeper, etcd, Consul, Google Spanner, HBase, MongoDB with majority reads and writes. The minority side stops accepting writes.

AP examples: Cassandra, Riak, DynamoDB-style leaderless stores at low consistency levels, CouchDB, DNS. Both sides keep serving and reconcile later (timestamps, version vectors, merge rules).

Many systems are tunable: "Cassandra is AP" really means "Cassandra at consistency level ONE behaves AP".

PACELC

CAP covers only the rare partition case. **PACELC** (Daniel Abadi, 2010) adds the everyday case: **if** **P**artition, choose **A** or **C**; **E**lse, choose **L**atency or **C**onsistency — even on a perfect network, strongly consistent replicas mean waiting for other machines.

- **PA/EL**: Cassandra, DynamoDB (defaults): available in partitions, low latency normally, consistency relaxed both times.
- **PC/EC**: Spanner, single-leader setups with synchronous replication: always consistent, paying in availability and latency.
- **PA/EC** and **PC/EL** exist too (MongoDB is often described as PA/EC under some settings; descriptions vary with configuration).

Common mistakes (misreadings of CAP)

- **"Pick any two of three."** You can't opt out of partitions, so CA isn't available to a distributed system; a single-node database is "CA" only because it isn't distributed.
- **"AP means no consistency ever."** It may return stale data during a partition; normally it's consistent within milliseconds.
- **"CP means the system goes down."** Only requests that can't be safely served fail, typically on the minority side.

- **Mixing CAP's C with ACID's C.** CAP's C is linearizability.
- **Treating CAP as a whole-system label.** Features of one product can choose differently (cart AP, checkout CP).

In the interview

Q: Explain CAP in one breath.

A: When the network splits, a replicated system must either refuse some requests to stay consistent (CP) or keep answering with possibly stale data (AP); partitions are a fact of life, so the real choice is C or A during them.

Q: Is your design CP or AP?

A: Per component: payment and inventory reservation are CP (better to fail than double-sell); catalogue, likes and feeds AP (stale is fine, downtime isn't). Then PACELC: even without partitions we accept slightly stale reads for lower latency on the AP parts.

TL;DR

CAP: during a network partition, choose consistency (refuse) or availability (answer, maybe stale). PACELC adds: otherwise, choose latency or consistency.

Quorums

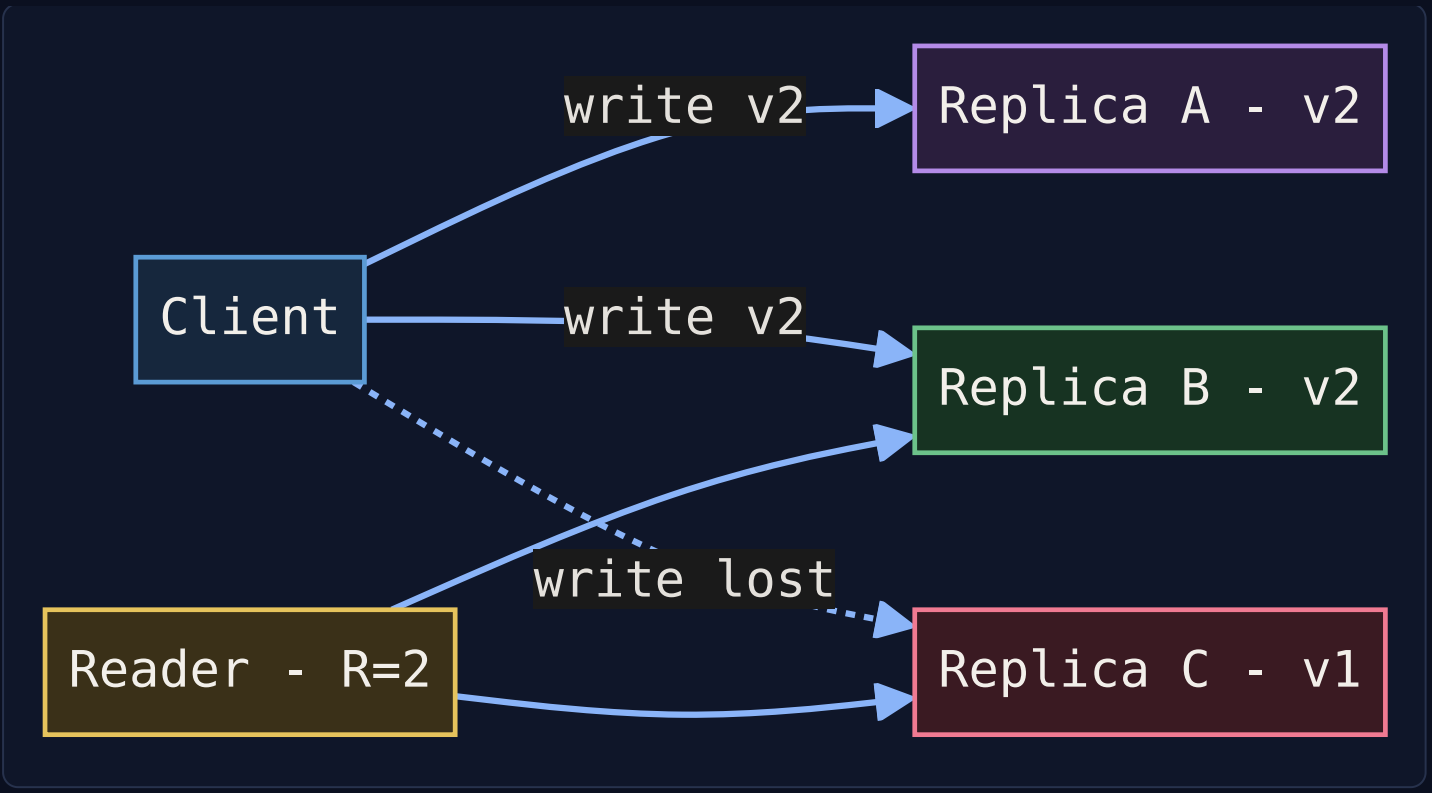
What it is

A **quorum** is the minimum number of replicas that must take part in an operation for it to count. In **leaderless replication** (Part 3: Dynamo, Cassandra, Riak), clients read and write several replicas at once, and quorums decide how many must answer:

- **N**: replicas storing each piece of data (often 3).
- **W**: replicas that must confirm a write.
- **R**: replicas that must answer a read.

How it works

If **R + W > N**, every read set overlaps every write set in at least one replica, so a read reaches at least one replica with the latest successful write; the reader keeps the newest version (by version number or timestamp). With N = 3, W = 2, R = 2: a write reaches A and B; a read asks B and C; B has it (2 + 2 = 4 > 3).



The reader gets v2 from B and v1 from C and keeps v2, possibly fixing C on the spot (**read repair**); background **anti-entropy** compares replicas (often with Merkle trees, below) to fix the rest.

Common settings for N = 3:

W	R	Effect
2	2	balanced, tolerates 1 replica down for both reads and writes
3	1	fast reads, any replica down blocks writes
1	3	fast writes, any replica down blocks reads
1	1	fastest, R + W = 2 is not > 3, so reads may be stale

A **majority quorum** is more than half (2 of 3, 3 of 5). Two majorities always overlap, which is why consensus uses them. N nodes tolerate `floor((N - 1) / 2)` failures — 1 with 3 nodes, 2 with 5 — hence odd cluster sizes: 4 nodes tolerate only 1 failure, like 3.

Sloppy quorum and hinted handoff. When home replicas are unreachable, Dynamo-style systems may write temporarily to other nodes that hand the data back later — more available, but R + W > N doesn't hold meanwhile.

Trade-offs and caveats

Quorums give tunable consistency, but R + W > N alone isn't full linearizability: concurrent writes, sloppy quorums and failed partial writes can still produce odd results. Larger W makes writes slower and less available; larger R the same for reads.

In the interview

Q: With 3 replicas, how do you make sure reads see the latest write?

A: Quorums with R + W > N, e.g. W = 2 and R = 2, so read and write sets always overlap — also tolerating one replica down.

TL;DR

Write to W and read from R of N replicas. When R + W > N, reads overlap writes and see the latest value; tune R and W to trade read speed, write speed and availability.

Consensus, leader election and distributed locks

What it is

Consensus is getting several machines to agree on one value (or sequence of values) even when some crash or messages are lost. "Who is leader?", "Was this transaction committed?", "Who holds the lock?", "What is the next log entry?" are all consensus problems — like friends choosing a restaurant over unreliable texts so nobody ends up elsewhere, even if a phone or two dies.

Why it matters

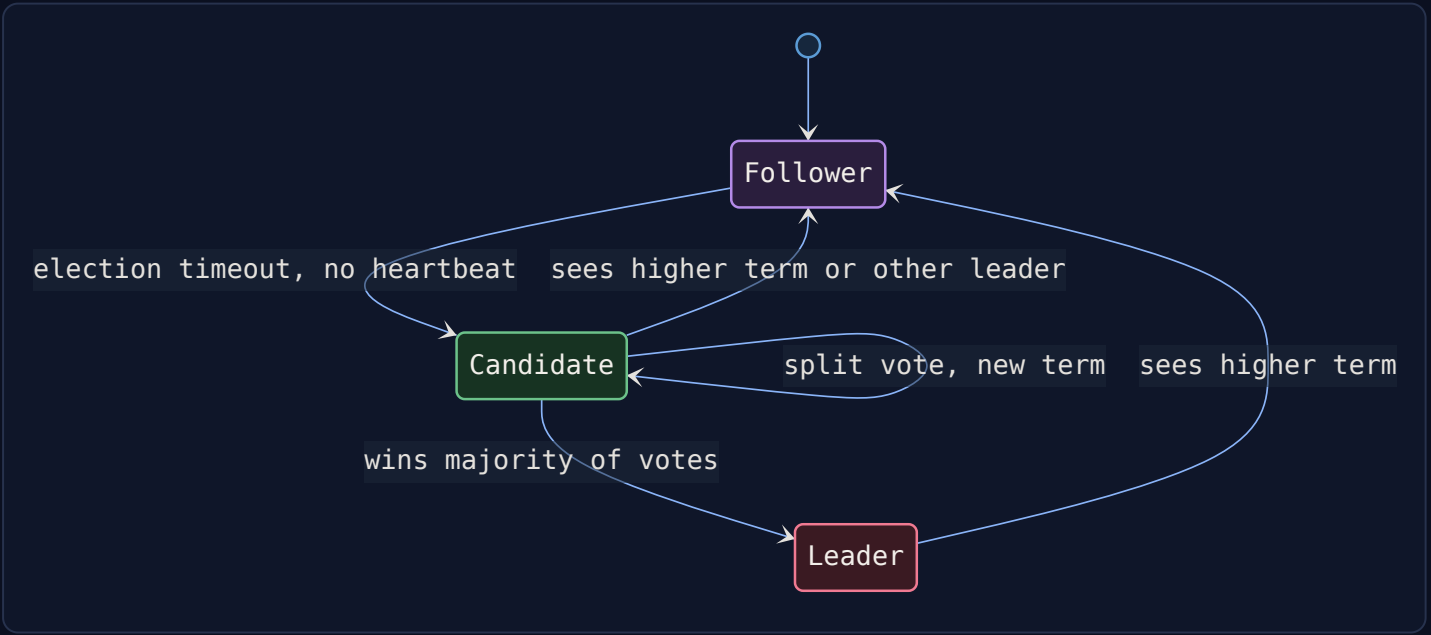
Without consensus you get **split brain** (Part 3): two self-declared leaders accept writes and data diverges. Consensus is the backbone of coordination services — **etcd** (all Kubernetes cluster state), **ZooKeeper** (Hadoop, older Kafka), **Consul** — and of CockroachDB, TiDB, YugabyteDB and Spanner.

How it works

- **Paxos**, Leslie Lamport (written up around 1989–1990, published 1998): the first widely known proven solution; correct but famously hard to understand and implement.
- **Raft**, Diego Ongaro and John Ousterhout (2014), designed to be understandable; used by most new systems (etcd, Consul, CockroachDB, Kafka's KRaft).

Raft intuition:

1. Each node is a **follower**, **candidate** or **leader**. Time is divided into numbered **terms** (**epochs** or generations elsewhere), each with at most one leader.
2. The leader sends regular **heartbeats**. A follower hearing nothing for its **election timeout** (randomised, e.g. 150–300 ms, so nodes don't all time out together) assumes the leader is dead.
3. It becomes a candidate, increments the term (4 → 5), votes for itself and requests votes.
4. Each node votes for at most one candidate per term, and only one whose log is at least as up-to-date as its own. A **majority** makes the candidate leader for term 5.
5. Clients write to the leader, which appends to its log and sends to followers; once a majority store it, the entry is **committed**, applied, and the client gets "ok".
6. A returning term-4 leader's messages are rejected by nodes that have seen term 5, and it steps down. Terms detect stale leaders.



Needing a majority, a 5-node Raft cluster works with 2 nodes down; with 3 down it stops accepting writes (it chooses consistency: CP). Clusters are usually 3 or 5 nodes: more nodes, more fault tolerance but slower writes.

Leader election is the commonest use: many replicas, but only one should run the nightly billing job or act as database primary. Use a coordination service (etcd, ZooKeeper, Consul) rather than writing your

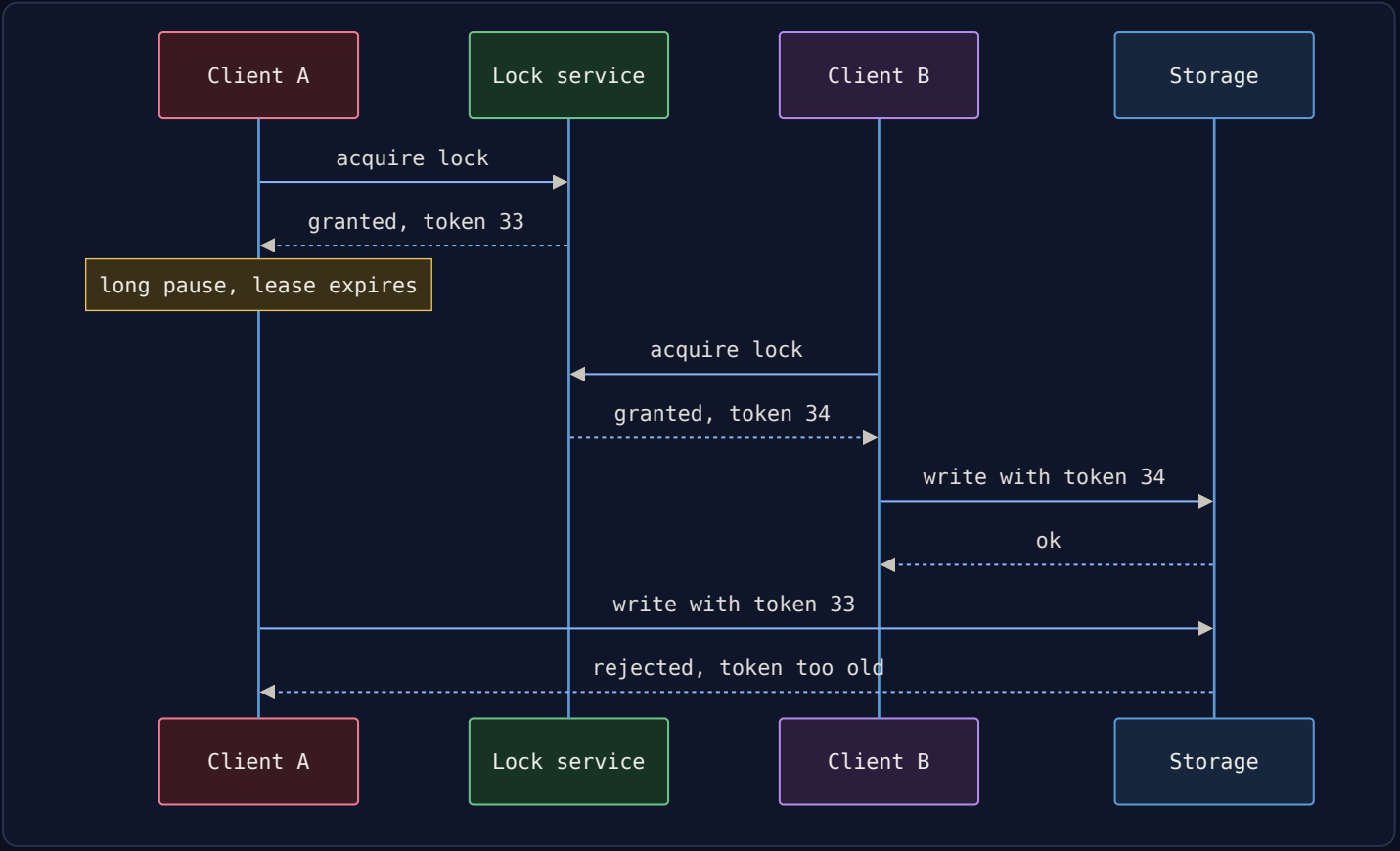
own consensus.

Distributed locks and fencing tokens

A **distributed lock** ensures only one process across many machines does something at a time (only one worker processes refund 77) — usually a key in a coordination service or Redis with a **lease** (expiry time) so a crashed holder doesn't block everyone forever.

The danger: a process takes a 10-second lease, then pauses (a long garbage-collection pause, a slow disk, its VM being migrated) for 15 seconds. The lease expires, another process takes the lock and writes; the first wakes, still believes it holds the lock, and writes too — two "holders".

The fix is a **fencing token**: each grant comes with an ever-increasing number (33, then 34). The holder sends its token with every write, and storage rejects any token lower than one already seen, so the paused holder's 33 is rejected after 34 has written.



ZooKeeper's zxid and etcd's revision numbers can serve as fencing tokens. A simple Redis `SET lock NX PX` lock is fine for efficiency (avoiding duplicate work) but not alone for correctness, since it gives no fencing token.

Common mistakes

- Home-made consensus or leader election with timestamps and hope. Use etcd, ZooKeeper or Consul.
- Consensus on 2 or 4 nodes (no extra fault tolerance over 1 or 3).
- Trusting a lease-based lock without fencing for data correctness.
- A consensus cluster across continents, so every write waits an intercontinental round trip (70+ ms).

In the interview

Q: How do you ensure only one instance runs a scheduled job?

A: Leader election via etcd or ZooKeeper, or a lock with a lease. Make the job idempotent and, if it writes shared data, pass a fencing token so a stale leader's writes are rejected.

Q: Why are consensus clusters 3 or 5 nodes?

A: A majority is needed to make progress: 3 tolerate 1 failure, 5 tolerate 2. Even numbers add cost without tolerance, and more nodes slow every write.

TL;DR

Consensus (Raft, Paxos) lets machines agree on a leader or log despite failures by requiring a majority and numbering leadership terms. Use a coordination service for leader election and locks, and protect locks with fencing tokens.

Clocks and ordering

What it is

Comparing timestamps to find "which write happened last?" is tempting, but each machine's clock differs:

- **Clock skew**: the difference between two machines' clocks at the same moment (A says 12:00:00.000, B 12:00:00.150).
- **Clock drift**: a clock running slightly fast or slow; quartz can drift tens of microseconds per second — seconds per day if uncorrected.
- **NTP (Network Time Protocol)**: machines periodically ask time servers and adjust — typically within a few to tens of milliseconds over the internet, better than a millisecond in a good data centre. Clocks can still jump backwards after a correction.

Every machine has a **wall clock** (time of day, can jump) and a **monotonic clock** (only moves forward; for durations like timeouts, meaningless across machines).

Why it matters

Last-write-wins (LWW) keeps the highest-timestamped write. If B's clock is 150 ms fast, B's write can "win" over a later write on A, which silently disappears. Ordering cross-machine events by wall-clock timestamps is unsafe.

How it works

Logical clocks order events without real time.

Lamport clock (Leslie Lamport, 1978). Each process keeps a counter:

1. Increment it before each local event.
2. Attach it to each message sent.
3. On receiving counter `c`, set yours to `max(yours, c) + 1`.

If X caused Y (directly or via a chain of messages), X's Lamport time is smaller than Y's; ties broken by process ID give a total order. But a smaller number doesn't prove causation, so Lamport clocks can't detect concurrent events.

Vector clock. Each process keeps one counter per process, e.g. `[A:2, B:0, C:1]` : increment your own entry per event; on receiving, take the element-wise maximum then increment your own. Comparing vectors gives one of three answers: X before Y (every entry of X ≤ Y's, at least one smaller), Y before X, or **concurrent** (each has some larger entry). Detecting concurrency is exactly what spotting conflicting writes needs (two people editing one cart offline). Dynamo-style stores use similar **version vectors** to keep both conflicting versions ("siblings") for the application to merge. Cost: the vector grows with the number of participants.

Clock	Detects causality?	Detects concurrency?	Size
Wall clock	no (skew)	no	one number
Lamport	if X caused Y, then L(X) < L(Y)	no	one number
Vector	yes	yes	one number per node

Hybrid logical clocks (HLC) combine wall time with a logical counter: timestamps close to real time that still respect causality (CockroachDB, YugabyteDB). **Google TrueTime** (Spanner) uses GPS receivers and atomic clocks in every data centre and returns an interval "now is between T1 and T2" with typically a few milliseconds of uncertainty; Spanner waits out the uncertainty before committing, so timestamps are globally ordered.

Common mistakes

- Using `System.currentTimeMillis()` from different servers to decide which update wins.
- Measuring timeouts with the wall clock (an NTP jump can make a duration negative). Use a monotonic clock.

In the interview

Q: Two replicas accepted conflicting writes. How do you tell which came first?

A: Not by wall clocks across machines. Vector clocks (version vectors) say whether one write happened before the other or they were concurrent — concurrent writes need a merge rule or a user choice. Lamport clocks give a consistent total order but can't detect concurrency.

TL;DR

Machine clocks disagree (skew) and drift even with NTP, so don't order cross-machine events by timestamps. Lamport clocks give a causality-consistent order; vector clocks also detect concurrent, conflicting updates.

Distributed transactions: two-phase commit and sagas

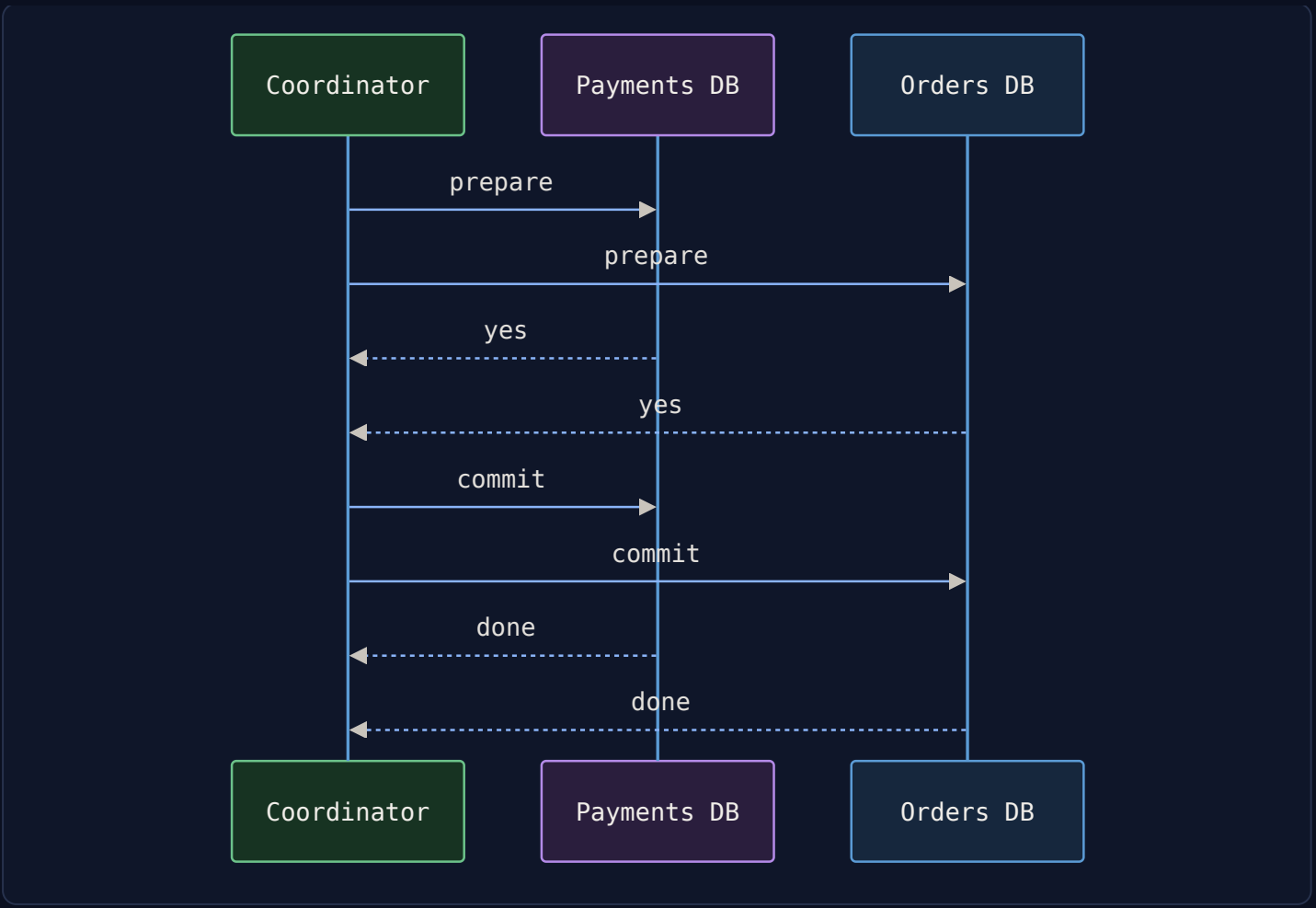
What it is

A **distributed transaction** is a change across several databases or services that must happen as one unit, all or nothing. Booking a trip charges the card (payment service), reserves a seat (flight service) and books a room (hotel service); charging without the seat costs real money. Inside one database ACID transactions (Part 3) handle this; across several there are two main approaches.

Types

Two-phase commit (2PC). A **coordinator** (transaction manager) runs the transaction across **participants** (the databases):

1. **Prepare phase**: "can you commit?" Each participant does the work, writes it durably, locks the rows, and votes **yes** (promising it can commit even after a crash) or **no**.
2. **Commit phase**: if all voted yes, the coordinator logs "commit" and tells everyone to commit; if any voted no or didn't answer, everyone aborts.



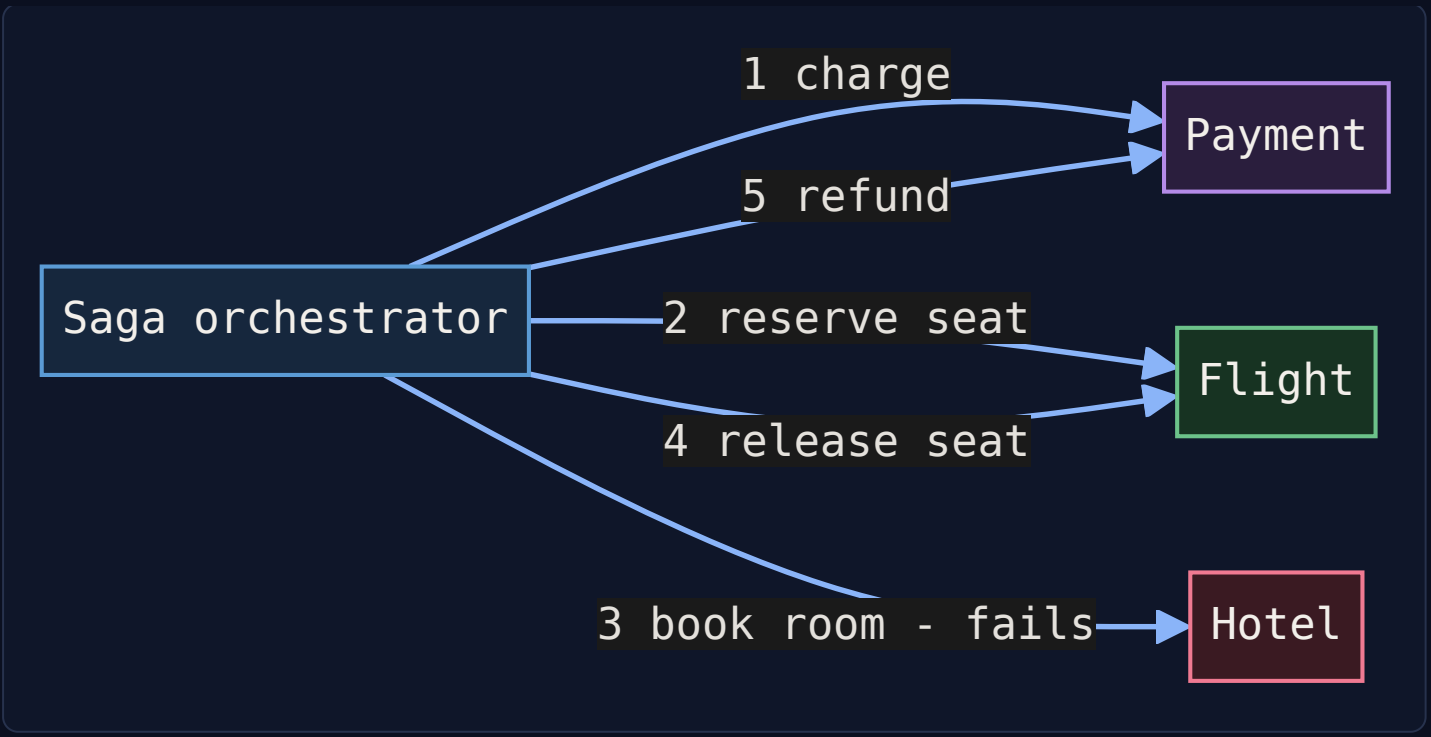
Strength: real atomicity and isolation across databases. Weakness: it's **blocking** — if the coordinator crashes after yes votes, participants keep rows locked and wait, possibly long, since they promised to obey its decision. Several round trips with locks held hurt throughput and availability. Used inside some distributed databases (Spanner runs 2PC over Paxos-replicated groups, so the coordinator is itself replicated and can't simply vanish) and via the XA standard (a common interface letting one transaction manager drive several databases), but rarely across microservices. (Three-phase commit was proposed to reduce blocking but is little used.)

Saga. Split the big transaction into a sequence of **local transactions**, one per service, each committed on its own; if a later step fails, run **compensating transactions** that semantically undo earlier steps in reverse order. Trip example:

1. Payment charges the card. (Compensation: refund.)
2. Flight reserves a seat. (Compensation: release.)
3. Hotel booking fails — no rooms.
4. Compensate: release the seat, refund the card.

Two ways to coordinate:

- **Choreography**: no central controller; services react to events ("PaymentCompleted" → flight reserves; "HotelFailed" → flight releases). Simple for 2–3 steps, hard to follow with many.
- **Orchestration**: a central **orchestrator** (a workflow engine such as Temporal, AWS Step Functions or Camunda, or your own service) tells each service what to do next and triggers compensations on failure; the whole flow is easier to see and change.



Trade-offs

	2PC	Saga
Atomicity	true all-or-nothing	eventual, via compensations
Isolation	yes, locks held	no, others can see partial results
Availability	blocks if coordinator fails	each step independent
Fits	few databases, short transactions	microservices, long business flows

Sagas give up isolation (someone may see "card charged, no seat yet"); designs use states like `PENDING`, and compensations must be idempotent and retryable since they may run more than once. Some actions can't be undone (a sent email), so put them last or compensate with another action (a "sorry" email).

In the interview

Q: How do you keep data consistent across order, payment and inventory microservices?

A: A saga: each service does a local transaction and publishes an event (via outbox), with compensations on failure, coordinated by an orchestrator for visibility. Avoid 2PC across services: it blocks and couples availability.

Q: What's the main weakness of 2PC?

A: Blocking: if the coordinator fails after yes votes, participants hold locks and wait. It also needs every participant up for every transaction.

TL;DR

2PC gives true atomic commits across databases but blocks on coordinator failure; sagas chain local transactions with compensating undo steps, trading isolation for availability — the norm in microservices.

Heartbeats, gossip and failure detection

What it is

Nodes need to know which others are alive, but over a network a dead node, a slow node and a lossy network look the same. **Failure detection** is making a good guess.

- **Heartbeat**: a small regular "I'm alive" message (say every second). Missing several (nothing for 5 seconds) marks the node **suspected dead**. Load balancer health checks (Part 2) and Raft leader heartbeats are the same idea.
- **Timeout choice**: short timeouts detect failures fast but cause **false positives** (declaring a slow-but-alive node dead, triggering needless failovers); long ones are safe but slow.

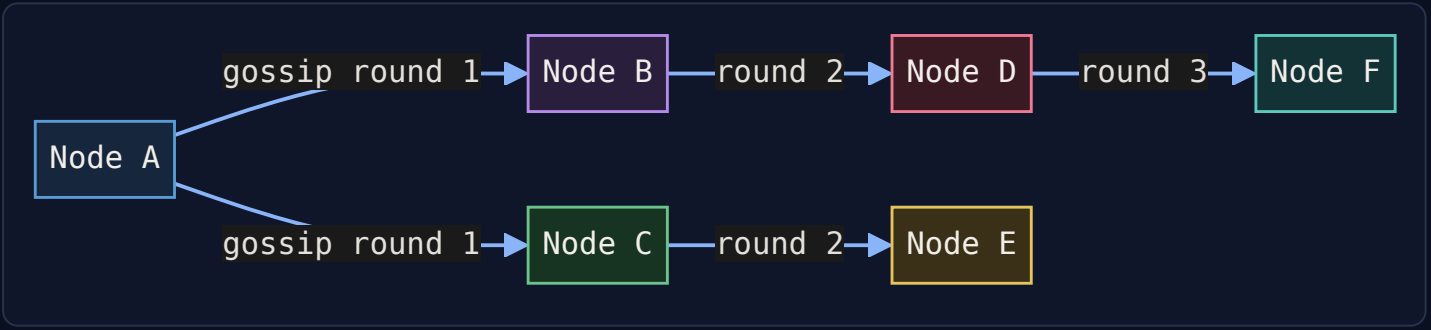
How it works

Centralised monitoring (every node heartbeats to one coordinator, e.g. ZooKeeper sessions) is simple, but the monitor must itself be highly available.

Gossip protocol. Large clusters with no central monitor spread information like office rumours:

1. Every second each node picks one or a few random peers.
2. It sends its membership list with a heartbeat counter for every node it knows.
3. The receiver merges, keeping the higher counter per node.
4. A node whose counter hasn't risen for a while is suspected dead, and the suspicion spreads too.

Information reaches all N nodes in about log(N) rounds (~10 for 1,000 nodes), with no single point of failure and constant per-node load. Cassandra, DynamoDB's original design, Consul (via memberlist, a library based on SWIM, below) and Redis Cluster use gossip for membership and failure detection.



Smarter detectors. The **phi accrual failure detector** (Cassandra, Akka) outputs a suspicion level from the history of heartbeat arrival times instead of a fixed yes/no timeout, adapting to normally slow networks. **SWIM** asks a few other nodes to ping a suspect indirectly before declaring it dead, cutting false positives from one bad link.

Common mistakes

- Very short timeouts causing **flapping**: nodes marked dead and alive repeatedly, triggering repeated failovers.
- Assuming a node declared dead has stopped; it may still be running and writing (the paused lock holder). Use fencing.

TL;DR

Missing heartbeats mean "suspected dead". Gossip spreads membership and liveness through random peer-to-peer exchanges, reaching everyone in about log N rounds with no central monitor.

Probabilistic data structures and Merkle trees

Structures that answer questions in far less memory than an exact answer needs, for a small, controlled error. Each appears often in interviews.

Bloom filter

What it is. A compact structure answering "is X in the set?" with **"definitely not"** or **"probably yes"**: never a false negative, but possibly a **false positive**.

How it works.

1. An array of m bits, all 0 (say 10 million bits, about 1.2 MB).

2. k hash functions (say 7), each mapping an item to a bit position.
3. To add "alice", set her 7 positions to 1.
4. To check "bob": any of his 7 bits 0 → **definitely never added**; all 1 → **probably** present (other items may have set them).

About 10 bits per item with 7 hash functions gives ~1% false positives. A basic Bloom filter can't delete items (a **counting Bloom filter** variant can).

Uses. LSM-tree databases (Cassandra, RocksDB, HBase) keep one per data file so a read for a missing key skips files without disk access. Caches block lookups for non-existent keys (cache penetration). Web crawlers skip visited URLs. Chrome once used one to check URLs against a malicious-site list before a slower exact check.

HyperLogLog

What it is. Estimates the **number of distinct items** (cardinality), such as unique visitors today, in a tiny fixed amount of memory.

How it works (intuition). Hash each item to a random-looking binary number. Rare patterns such as a long run of leading zeros appear only after many distinct items: a hash starting with 20 zeros suggests roughly 2^20 (~a million) distinct items. HyperLogLog splits items into many buckets, records the longest run in each, and averages to reduce luck. Duplicates hash the same, so they don't change the estimate.

Numbers. Redis's HyperLogLog uses at most 12 KB per counter with 0.81% standard error and counts billions of distinct items (an exact set of a billion IDs needs gigabytes). Sketches from different servers merge (`PFMERGE`). Used for unique visitors, distinct search queries, and approximate `COUNT(DISTINCT ...)` in BigQuery and other warehouses.

Count-min sketch

What it is. Estimates **how many times** each item appeared in a stream (frequency) in fixed memory ("how often was #worldcup posted in the last hour?").

How it works. A table of counters, d rows × w columns, one hash function per row. To add, hash the item in each row and increment that counter; to query, take the **minimum** of its d counters. Collisions only add counts, so estimates may be too high, never too low; the minimum picks the least-polluted counter.

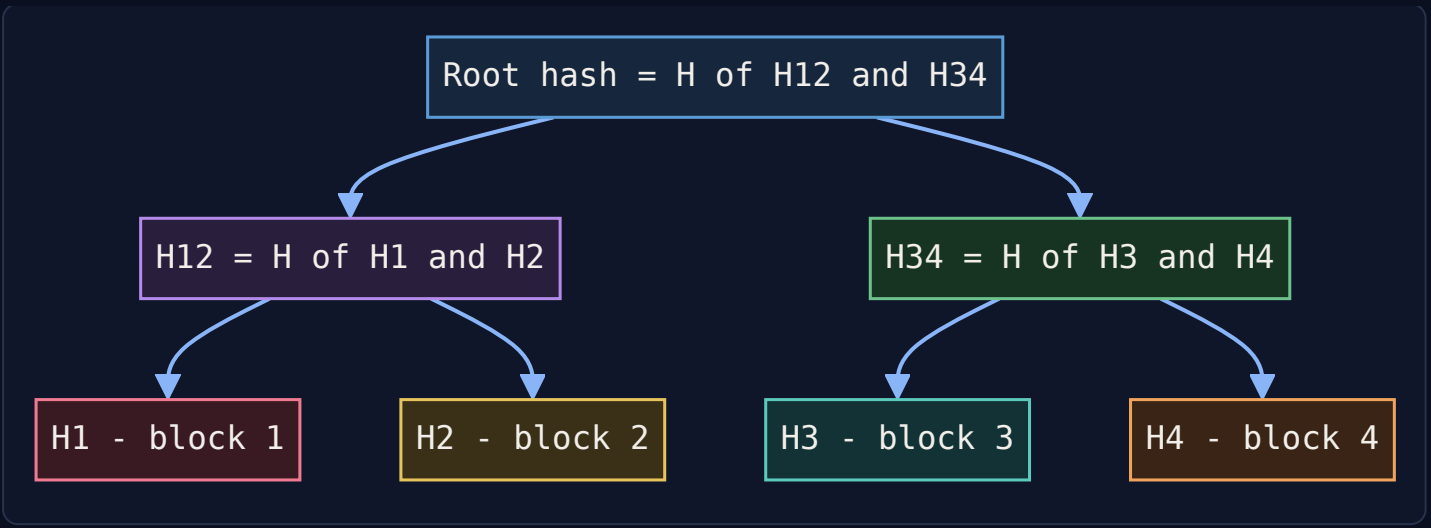
Uses. Heavy hitters and trending topics, detecting hot cache keys, network traffic monitoring, approximate top-K lists.

Merkle tree

What it is. A tree of hashes: each leaf hashes a data block, each parent hashes its children's hashes, and the top is the **root hash**. Changing any byte changes its leaf hash and every hash up to the root.

How it works for comparing replicas. Each replica builds a Merkle tree over its data ranges:

1. Compare root hashes; equal means identical data — done with one tiny message.
2. If different, compare children and descend only into halves whose hashes differ.
3. At the bottom, copy exactly the differing ranges.



A few differences among billions of records cost about log(number of blocks) comparisons instead of shipping everything. **Uses**: anti-entropy repair in Cassandra and Dynamo-style stores, Git (commit and tree objects are built from hashes), Bitcoin and other blockchains (proving a transaction is in a block), certificate transparency logs, file sync tools.

Summary table

Structure	Answers	Error	Memory
Bloom filter	is X in the set?	false positives only	~10 bits per item for 1%
HyperLogLog	how many distinct items?	~0.81% in Redis	12 KB per counter
Count-min sketch	how often has X appeared?	overestimates only	fixed table
Merkle tree	do two datasets differ, and where?	none (exact)	one hash per block

In the interview

Q: How would you count unique daily visitors for a site with 500 million users?

A: A HyperLogLog per day (Redis `PFADD`, `PFCOUNT`): ~12 KB per day, under 1% error, mergeable into weekly counts. An exact set would take gigabytes.

Q: How can a database avoid disk reads for keys that don't exist?

A: A Bloom filter per data file: "definitely not here" skips the file; only "probably yes" costs a disk read.

TL;DR

Bloom filter: set membership, false positives only. HyperLogLog: distinct counts in kilobytes. Count-min sketch: approximate frequencies. Merkle tree: find differences between large datasets by comparing hashes top-down.

Unique ID generation

What it is

Every order, user, message and photo needs a unique **ID**. Easy on one database; across many servers generating IDs without consulting each other, a design problem. Nice extras: IDs roughly sorted by creation time (good for indexes and "newest first"), short (64 bits fits a standard integer column), and not leaking business information.

Types

Auto-increment (database sequence). 1, 2, 3… — simple, short, ordered. But one database becomes a bottleneck and single point of failure; after sharding two shards would both issue ID 1000 (workarounds:

each shard steps by the shard count — shard 1 issues 1, 3, 5, shard 2 issues 2, 4, 6 — or a central **ticket server**, as Flickr used); and sequential IDs reveal volume (500 orders today) and let attackers guess other users' URLs.

UUID (universally unique identifier). 128 bits, usually 36 characters like `550e8400-e29b-41d4-a716-446655440000`, generated by any machine with no coordination.

- **UUIDv4**: 122 random bits; collisions negligible (billions of IDs per second for decades). But random order scatters inserts across a B-tree index (more page splits, cache misses), and it's twice a 64-bit integer.
- **UUIDv7** (RFC 9562, 2024): a millisecond timestamp in the first 48 bits, randomness after — unique **and** time-sorted, fixing the index problem. A good modern default. (**ULID** is a similar earlier format.)

Snowflake ID. Twitter's 2010 design: 64-bit, time-sortable IDs generated independently by many machines:

Bits	Field	Meaning
1	sign	always 0
41	timestamp	milliseconds since a custom epoch, lasts about 69 years
10	machine ID	up to 1,024 generators
12	sequence	up to 4,096 IDs per machine per millisecond

About 4 million IDs per second per machine with no coordination, sorted by time. Each machine needs a unique machine ID (config or a coordination service like ZooKeeper), and clocks must not go backwards (if they do, the generator waits or refuses). Discord, Instagram (a variant generated inside PostgreSQL shards) and many others use Snowflake-style IDs; Sonyflake and others change the bit split.



Other approaches. Segment / range allocation: a central service hands each server a block of 1,000 IDs at a time, so servers rarely contact it. **Hashing** (e.g. URL shorteners, Part 6) derives a short ID from content, with collision checks.

In the interview

Q: How would you generate unique IDs for a sharded system with high write volume?

A: Snowflake-style 64-bit IDs (timestamp, machine ID, per-millisecond sequence): no per-ID coordination, roughly time-ordered, fits a BIGINT (64-bit integer) column. If 128 bits is fine, UUIDv7 gives the same ordering with no machine IDs to manage.

Q: Why not just use UUIDv4?

A: It works without coordination, but random IDs fragment B-tree indexes and are larger; time-ordered IDs (UUIDv7, Snowflake) insert at the end of the index.

TL;DR

Auto-increment is simple but centralised and guessable; UUIDs need no coordination (v7 also sorts by time); Snowflake packs time, machine and sequence into 64 sortable bits.

Geospatial indexing: geohash and quadtree

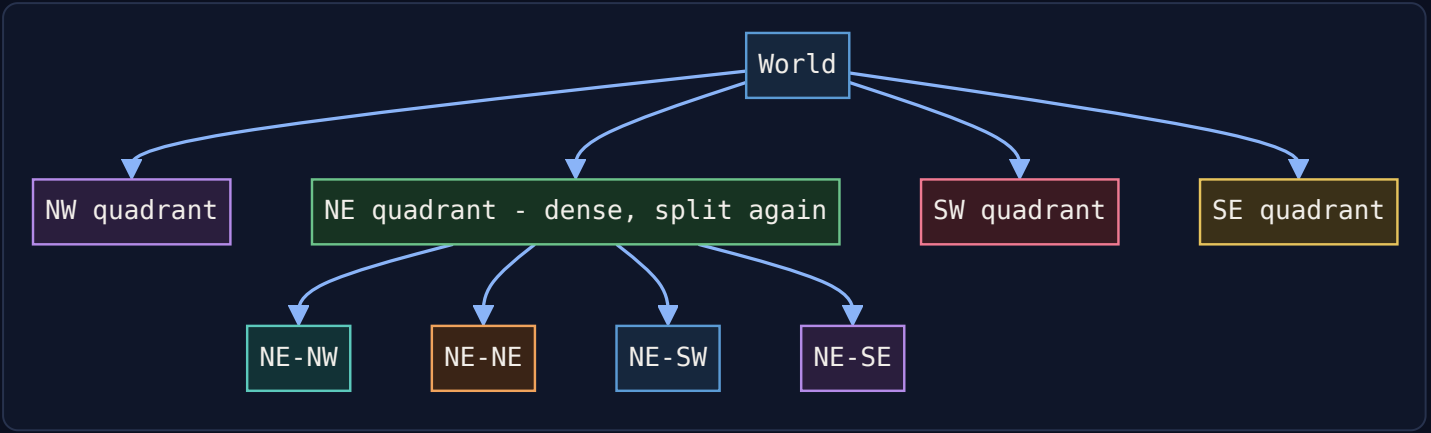
What it is

"Restaurants within 2 km" or "drivers near me" can't use a normal latitude/longitude index efficiently, because a B-tree sorts by one value at a time. **Geospatial indexes** turn 2D locations into something quickly searchable.

Types

Geohash. Halve the world map by longitude, then latitude, repeatedly, writing a bit for which half you're in each step; encode the bits as base-32 letters and digits. Longer strings are smaller boxes: 5 characters ≈ 5 km × 5 km, 6 ≈ 1.2 km × 0.6 km. A shared prefix means nearby (`gcpvj` and `gcpvm` are adjacent cells in central London), so "nearby" becomes a **prefix search** on an ordinary string index. To catch places just over a cell edge, search the cell plus its 8 neighbours. Redis's `GEOADD` / `GEOSEARCH` and Elasticsearch support geohash-style indexing.

Quadtree. Each node covers a map rectangle; a node with too many points (say over 100 businesses) splits into four children (NW, NE, SW, SE). Dense cities get small, deep cells; empty oceans stay one big cell. To find nearby places, walk down to the user's leaf and check it and its neighbours. Adapts to uneven density better than fixed-size geohash cells; typically kept in memory.



Others: **R-tree** (bounding-box tree used by PostGIS), Google's **S2** (cells on a sphere; Google Maps, MongoDB) and Uber's **H3** (hexagon grid, good for ride-demand heatmaps since every neighbour is equidistant).

In the interview

Q: How do you find drivers near a rider?

A: Drivers send locations every few seconds into an in-memory geo index (geohash cells in Redis, or a quadtree) keyed by cell. Query the rider's cell and neighbours, then compute exact distances for the few candidates.

TL;DR

Geohash turns locations into strings where a shared prefix means nearby; a quadtree recursively splits dense areas into four. Both make "what's near me?" a fast lookup.

Part 4 glossary

Every Part 4 term is defined in one line in the Master A–Z glossary, which merges the glossaries of all six parts into a single alphabetical list.

Part 5 — Architecture, reliability, operations and security

Parts 1 to 4 covered the building blocks. This part is about arranging them into a whole system, keeping it running when pieces break, shipping changes safely, seeing what it is doing, and keeping the wrong people out — what separates "it works on my laptop" from "it served a billion requests this month and nobody noticed when a data centre lost power".

Architectural styles: monolith, modular monolith, SOA, microservices and serverless

What it is

An **architectural style** answers "how is our code split into running programs, and how do they talk?" (One chef in one kitchen; one kitchen divided into stations; a food court of independent stalls; a catering agency sending a cook per order and billing per dish.)

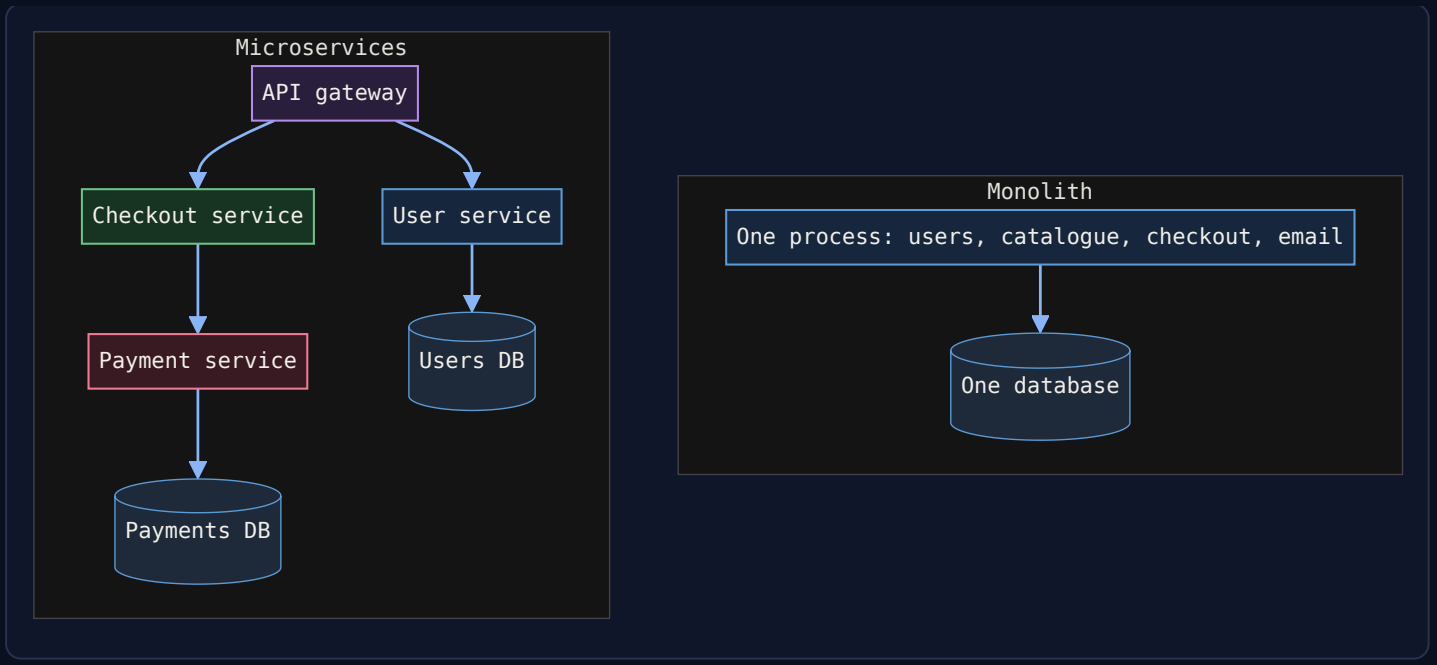
- **Monolith**: the whole application (accounts, catalogue, checkout, emails) is one codebase built into one deployable program, usually sharing one database; scaled by running identical copies behind a load balancer.
- **Modular monolith**: still one deployable program, but split into modules with strict boundaries (checkout may call the catalogue only through a defined interface, never reach into its tables) — most of the tidiness of microservices without the network.
- **Service-oriented architecture (SOA)**: a 2000s-era style splitting enterprise systems into services that talked through a central **enterprise service bus (ESB)**, middleware that routed, transformed and orchestrated messages. Services were often coarse (a whole "billing system") and shared data models.
- **Microservices**: many small services, each owning one business capability ("payments", "search") and its **own database**, deployed independently, talking over HTTP/REST, gRPC or messages. "Smart endpoints, dumb pipes": logic lives in the services, not a central bus.
- **Serverless / Function as a Service (FaaS)**: upload individual functions; the provider runs them only when an event arrives (HTTP request, file upload, queue message), scales automatically from zero to thousands of copies, and bills per invocation and millisecond of run time. You never manage a server: AWS Lambda, Google Cloud Functions, Azure Functions, Cloudflare Workers. "Serverless" also loosely covers managed services you never size, such as DynamoDB or S3.

Why it matters

The style decides how fast teams move, how failures spread, cost, and debuggability; a five-person startup and a 2,000-engineer company need different answers. **Conway's law**: a system's structure mirrors the communication structure of the organisation that builds it, so architecture is partly a decision about teams.

How it works

In a monolith a request is in-process calls — `checkout()` → `reserveStock()` → `chargeCard()` — nanoseconds to microseconds each, inside one database transaction. In microservices they become network calls (about 0.5–2 ms each inside one data centre) to inventory and payment services, with no single transaction covering both, so you need patterns like sagas (Part 4) to undo work when a later step fails.



Types

Style	Deploy unit	Data	Best for
Monolith	One app	One shared DB	Small teams, new products
Modular monolith	One app, strict modules	One DB, owned schemas	Growing teams wanting order
SOA	Coarse services + ESB	Often shared	Legacy enterprise integration
Microservices	Many small services	DB per service	Many teams, independent scaling
Serverless (FaaS)	Single functions	Managed stores	Spiky or event-driven work

Trade-offs

- **Monolith**: simple to build, test, deploy and debug; one stack trace shows everything; transactions are easy. Past a few hundred thousand lines and dozens of developers, builds slow, one bad change takes everything down, and you scale the whole thing even if only search is busy.
- **Microservices**: teams deploy independently many times a day; each service scales and fails on its own and may choose its own language or database. The price is the **distributed systems tax**: network calls fail and add latency, data spread across databases makes consistency harder, and you need service discovery, tracing, per-service CI/CD and on-call — one user action may touch 20 services.
- **Serverless**: no servers to manage, scales to zero (idle costs nothing) and up in seconds. Downsides: **cold starts** (the first call to a fresh copy waits for the runtime to load, ~100 ms to a few seconds), execution limits (AWS Lambda caps a run at 15 minutes), harder local testing, vendor lock-in, and often pricier than always-on servers at steady high load.

Common mistakes

- Starting a new product with 30 microservices: boundaries aren't known yet and you pay the tax from day one. Common advice: "monolith first", split later.
- A **distributed monolith**: services that must deploy together, share one database, or call each other synchronously in long chains — the network and operations bill of microservices while still moving in lockstep.
- Splitting by technical layer ("database service", "UI service") instead of business capability.
- Thinking microservices are faster; a network hop is thousands of times slower than a function call.

Real-world examples

- **Shopify** runs one of the largest Ruby on Rails monoliths and deliberately chose a modular monolith, enforcing component boundaries with tooling, over microservices.

- **Netflix** and **Amazon** are the classic microservices stories; Uber has written about running around 2,200 critical microservices, then grouping them into "domains" to tame complexity.
- In 2023 **Amazon Prime Video** moved an audio/video quality-monitoring tool from a distributed serverless design (Step Functions plus Lambda) into a single process running as one container task, cutting its infrastructure cost by more than 90%. Pick the style per workload.
- The **strangler fig pattern** is how most companies migrate: a proxy in front of the monolith, moving one feature at a time into new services until the old code can be deleted.

In the interview

Q: Would you start this design with microservices?

A: Usually no: a well-modularised monolith first, extracting a service when there's a clear reason — a part needing independent scaling, a team needing its own deploy schedule, a component with very different reliability needs.

Q: What does "database per service" buy and cost?

A: Each team changes its schema and scales its store without coordinating, and one service's bad query can't hurt another. It costs cross-service joins and transactions: APIs, events and sagas instead, with eventual consistency between services.

Q: When is serverless a good fit?

A: Spiky, unpredictable, event-driven work (thumbnails on upload, webhooks, nightly jobs) where scaling to zero saves money. Poor for long-running jobs, latency-critical paths that can't tolerate cold starts, or steady high traffic.

TL;DR

Monoliths are simple and fast to start; microservices trade simplicity for team independence and independent scaling; serverless trades control for zero operations. Choose by team size and workload, not fashion.

Synchronous vs asynchronous communication

What it is

When one service needs another it can **wait for the answer** (a phone call) or **leave a message and carry on** (a text).

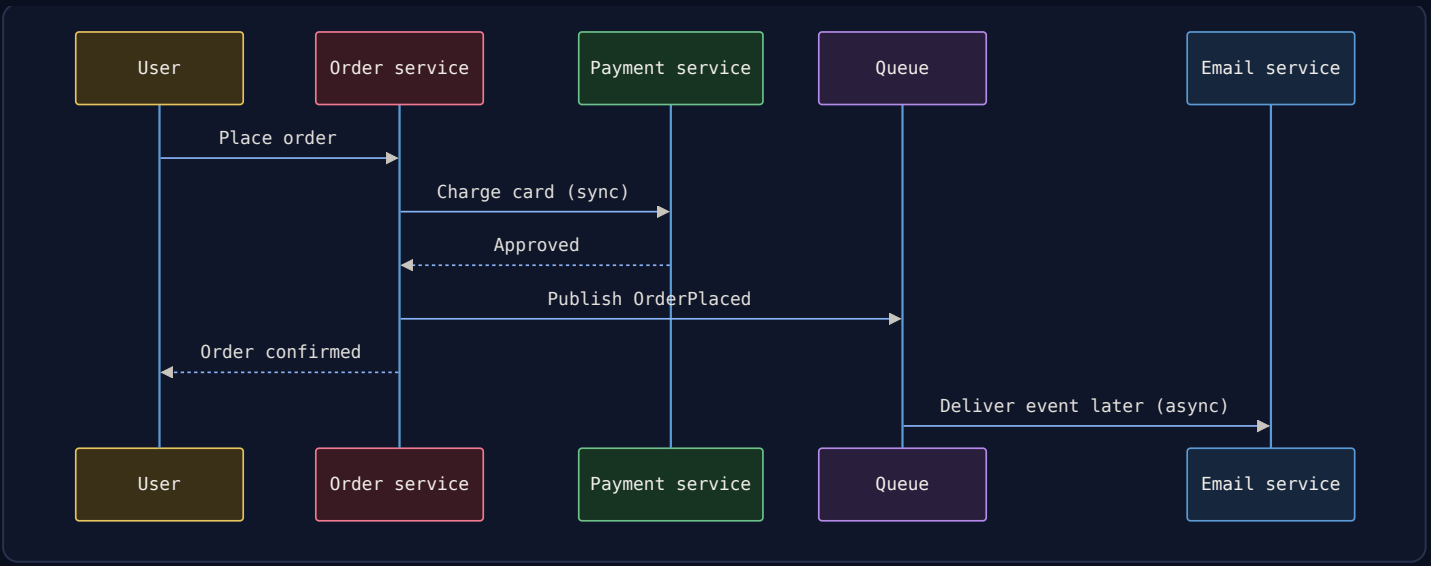
- **Synchronous**: A calls B (REST, gRPC) and blocks until B responds or a timeout fires; A needs the answer to continue.
- **Asynchronous**: A puts a message on a queue or event stream (Part 4) and returns at once; B processes it later. A may learn the result via a callback, a webhook or a status endpoint.

Why it matters

A synchronous chain ties the caller's fate to the callee: if B is slow or down, so is A. Availability multiplies too: five services each 99.9% available give at best 0.999^5 ≈ 99.5% — about 44 hours of downtime a year instead of 9. Async messaging breaks the chain: if the email service is down, orders still succeed and emails go out on recovery.

How it works

1. "Place order": the order service checks the card synchronously, because the user is waiting for "approved" or "declined".
2. Once saved, it publishes `OrderPlaced` to a queue and replies in, say, 200 ms.
3. Email, analytics and warehouse services consume the event at their own pace; if the warehouse is down for ten minutes, events wait in the queue.



Trade-offs

	Synchronous	Asynchronous
Simplicity	Easy to reason about	Harder: eventual results
Coupling	Tight in time	Loose
Latency for user	Sum of all calls	Only the critical path
Failure spread	Cascades	Absorbed by the queue
Debugging	One request, one trace	Needs correlation IDs

Common mistakes

- Making every call synchronous "because it's simpler", building deep call chains.
- Async for something the user must see immediately ("is my payment approved?"), then awkward polling.
- Forgetting async consumers must be idempotent, since queues usually deliver at least once.

In the interview

Q: How do you decide sync vs async between two services?

A: Does the caller need the result to answer the user right now? If yes, call synchronously with a timeout. If the work can finish later (notifications, analytics, search indexing), publish an event so failures and slowness don't reach the user.

Q: How do you tell the user an async job's result?

A: Return a job ID immediately (HTTP 202 Accepted), then the client polls a status endpoint, or push the result via WebSockets, server-sent events or a webhook.

TL;DR

Synchronous calls are simple but chain failures and latency together; asynchronous messages decouple services in time at the cost of eventual results and harder debugging.

Service discovery, sidecars and service mesh

What it is

With many services, each running many copies that start and stop constantly, a caller must learn "which IPs currently run the payment service?" — **service discovery**, a switchboard where you ask for "cardiology" rather than memorising who is on shift.

- **Service registry**: a database of live instances and addresses. Instances **register** on start, send **heartbeats**, and are removed when they stop or miss heartbeats. Examples: Consul, etcd, ZooKeeper, Netflix Eureka, the registry built into Kubernetes.
- **Sidecar**: a helper process deployed beside each service instance (same pod or host) handling cross-cutting jobs — networking, TLS, retries, logging — riding along everywhere the main process goes.
- **Service mesh**: a fleet of sidecar proxies, one per instance, intercepting all service-to-service traffic (the **data plane**), plus a **control plane** configuring them all. Examples: Istio, Linkerd; Envoy is the commonest sidecar proxy.

Why it matters

Hard-coded IPs break as soon as autoscaling adds a machine or a container is rescheduled. And every service needs the same plumbing (timeouts, retries, encryption between services, metrics); writing it into every codebase in every language is slow and inconsistent. A mesh moves that plumbing out of application code.

How it works

1. A new payment instance starts on `10.0.3.17:8080` and registers.
2. The registry health-checks it every few seconds.
3. Checkout asks the registry itself, or goes through a router that does.
4. The call goes to a healthy instance; a dead instance misses heartbeats and is removed, usually within seconds.

Types

- **Client-side discovery**: the caller queries the registry and picks an instance with its own load-balancing logic. Fewer hops, but every client needs a discovery library (Netflix Eureka with the Ribbon library).
- **Server-side discovery**: the caller sends to a fixed address (load balancer or router) that consults the registry and forwards. Clients stay simple (AWS load balancers with target groups, Kubernetes Services).
- **DNS-based discovery**: the registry publishes instances as DNS records (Consul, Kubernetes), so any program finds a service by name, e.g. `payments.default.svc.cluster.local`. Simple, but DNS caching can leave stale addresses.



Trade-offs

A mesh gives, without code changes, mutual TLS (both sides prove identity with certificates) between all services, uniform retries and timeouts, traffic splitting for canaries, and per-call metrics and traces. It costs an extra proxy hop on each side of every call (usually well under a millisecond, not zero), extra memory and CPU per pod, and a complex control plane to understand when things go wrong. For ten services it's often overkill.

Common mistakes

- A registry without health checks, so dead instances keep receiving traffic.
- Adopting a full mesh before having the operational skills to debug it.
- Retries both in application code and in the mesh, multiplying them (next section).

In the interview

Q: Client-side or server-side discovery?

A: Server-side keeps clients thin and language-neutral — the usual default, especially on Kubernetes. Client-side saves a hop and gives the caller load-balancing control, which matters at very high call rates.

Q: What is a service mesh in one sentence?

A: A sidecar proxy beside every service instance handling service-to-service networking (discovery, load balancing, mTLS, retries, metrics), configured centrally by a control plane.

TL;DR

Service discovery is a live phone book of instances; a sidecar is a helper process beside each instance; a service mesh is a centrally managed fleet of sidecars handling networking for you.

Timeouts, retries, exponential backoff and jitter

What it is

- A **timeout** is the maximum time a caller waits for a response before giving up.
- A **retry** repeats a failed request in case the failure was temporary (a **transient fault**, such as a dropped packet or a restarting server).
- **Exponential backoff** doubles the wait after each failed attempt: 100 ms, 200 ms, 400 ms, 800 ms, up to a cap.
- **Jitter** adds randomness to that wait so thousands of clients don't retry at exactly the same moment.

Why it matters

Without a timeout, a caller waiting on a dead service holds a thread, connection and memory forever, soon runs out and dies too — how one slow service takes down everything upstream: a **cascading failure**. Retries turn transient failures into successes, but carelessly: if a struggling service gets every request three times, its load triples exactly when it's weakest — a **retry storm**.

How it works

Setting a timeout. Base it on the callee's measured latency: with a p99 of 300 ms, around 500 ms to 1 s. Use a **deadline** across the whole request: with a 2 s user-facing budget and 1.5 s spent, the next call gets at most 0.5 s. gRPC propagates deadlines between services for this reason.

Retrying safely.

1. Retry only possibly-temporary errors: timeouts, connection resets, HTTP 502/503/504, HTTP 429 (after the `Retry-After` delay). Never 400 Bad Request or 403 Forbidden — they'll fail again.
2. Retry only **idempotent** operations, or make them idempotent with an idempotency key (Part 4); retrying "charge card $50" without one can charge twice.
3. Limit attempts, typically 2–3 retries in total.
4. Wait with exponential backoff and jitter between attempts.
5. Retry at one layer only: if browser, gateway, service A and service B each make up to 3 attempts, one click can become 3 × 3 × 3 × 3 = 81 database calls.

Backoff with jitter. With "full jitter", the wait before attempt n is random between 0 and `min(cap, base * 2^n)`:

```
base = 100 ms, cap = 10 s
attempt 1: wait random(0, 200 ms)
attempt 2: wait random(0, 400 ms)
attempt 3: wait random(0, 800 ms)
```

Without jitter, 10,000 clients failing in the same second all retry at exactly +100 ms, +200 ms… in synchronised waves; with jitter the server sees a smooth trickle.

A **retry budget** adds a guard: retries may add at most, say, 10% extra load; beyond that, stop retrying and fail fast.

Trade-offs

Short timeouts free resources quickly but may cut off requests that would have succeeded; long ones risk resource exhaustion. More retries help with rare blips but amplify load during real outages. **Hedged requests** (a second copy to another replica if the first hasn't answered by the p95 time, taking whichever answers first) cut tail latency for a few percent extra load.

Common mistakes

- No timeout at all; many HTTP client libraries default to waiting forever or for minutes.
- A caller timeout shorter than the callee's, so the callee keeps working on an answer nobody will read.
- Retrying non-idempotent writes.
- Retrying immediately in a tight loop.

Real-world examples

AWS SDKs retry throttled and transient errors with exponential backoff and jitter by default; the AWS Architecture Blog popularised the "full jitter" formula. Google's **Site Reliability Engineering (SRE)** book describes per-client retry budgets to prevent retry storms.

In the interview

Q: Why add jitter to backoff?

A: To break synchronisation: clients that fail together would otherwise retry together in waves that knock the service over again.

Q: Where should retries live in a deep call chain?

A: At one layer — ideally near the user or at each service boundary's edge — with a small attempt limit, so retries don't multiply across layers.

TL;DR

Always set timeouts; retry only transient failures on idempotent operations, a few times, with exponential backoff plus random jitter, at a single layer.

Circuit breakers, bulkheads and fallbacks

What it is

- A **circuit breaker** wraps calls to a dependency, "trips" after too many failures, then fails calls instantly for a while instead of waiting on a broken service — like the household breaker that cuts an overloaded wire to protect the rest.
- A **bulkhead** isolates resources so one failing part can't consume everything — like a ship's watertight compartments.
- A **fallback** is what you do instead when a call fails: cached data, a default, or a reduced answer.

Why it matters

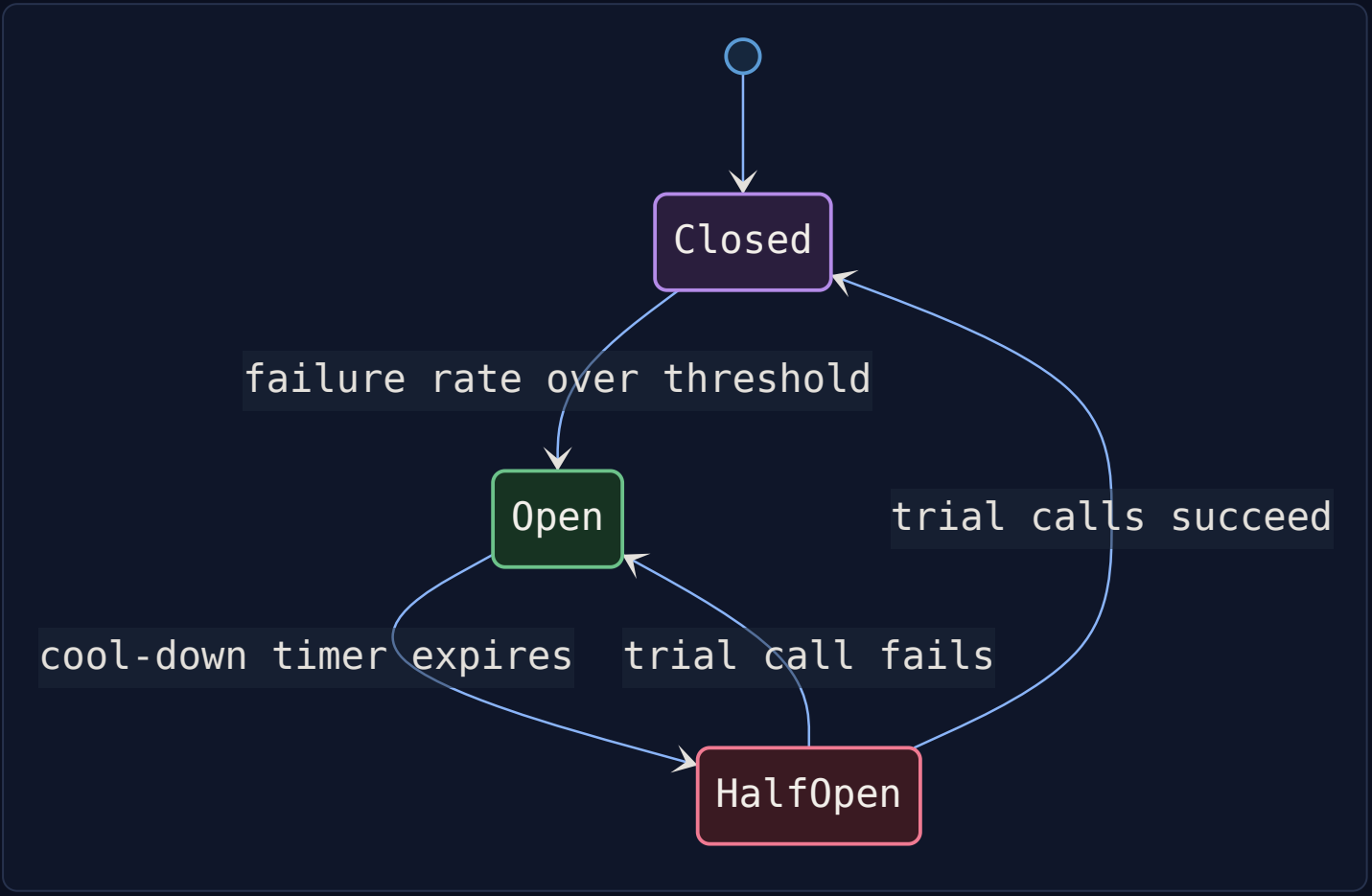
A recommendation service hangs: every product page waits 2 s for its timeout, web threads pile up, the pool fills, and product pages — which could render without recommendations — fail too. A circuit breaker stops calling the broken service; a bulkhead limits recommendation calls to, say, 20 of 200 threads; a fallback shows "popular items".

How it works

A circuit breaker has three states:

1. **Closed** (normal): calls pass; failures are counted over a rolling window (last 100 calls or 10 seconds).

2. Past a threshold (say 50% of at least 20 calls) it goes **Open**: every call fails immediately or goes straight to the fallback, giving the dependency time to recover and freeing the caller's threads.
3. After a cool-down (say 30 s) it goes **Half-open**: a few trial calls pass. Success → Closed; failure → Open for another cool-down. This is how it discovers recovery without flooding the dependency.



Bulkheads in practice: separate thread or connection pools per dependency, separate queues per customer tier, separate clusters for internal batch jobs and user traffic, or **cell-based architecture** — users split across independent copies of the whole stack ("cells"), so a bad deploy or poison request affects only one cell's users.

Fallback options: a cached last-known value, a static default ("free shipping info unavailable"), a simpler algorithm, an empty page section, or queueing the work to finish later.

Trade-offs

Breakers add state and configuration per dependency; badly tuned thresholds trip on noise or never trip. Bulkheads waste capacity (a reserved pool idles while another is overloaded). Fallbacks give stale or partial data — fine for recommendations, not a bank balance.

Common mistakes

- A fallback that calls another fragile service, which fails too.
- One global breaker for a dependency with many endpoints, so one bad endpoint blocks healthy ones.
- Never testing the open state, so the fallback path hides an unseen bug.

Real-world examples

Netflix popularised circuit breakers with Hystrix (now in maintenance mode); Resilience4j is the common Java successor, and meshes such as Istio offer breakers and outlier ejection at the proxy. Amazon describes cell-based architectures to limit the "blast radius" of failures.

In the interview

Q: Explain the half-open state.

A: After the open cool-down, the breaker lets a few test requests through: success closes it and resumes normal traffic, failure re-opens it — discovering recovery without flooding the dependency.

Q: How would you stop a slow non-critical service taking down your homepage?

A: Tight timeout, a circuit breaker, a small separate thread pool (bulkhead), and a fallback such as cached or default content, so the page renders without that section.

TL;DR

A circuit breaker stops calling a failing dependency (Closed, Open, Half-open); a bulkhead caps the resources any one dependency can consume; a fallback gives a degraded answer instead of an error.

Rate limiting and throttling

What it is

Rate limiting caps how many requests a client may make per time window (say 100 per minute per user). **Throttling** is often a synonym; strictly, throttling slows requests (delays or queues them) while rate limiting rejects the excess — a bouncer admitting 50 per hour versus one making people wait in line.

Why it matters

- **Protection**: one buggy script or abusive client can't overwhelm the service for everyone.
- **Fairness**: each customer gets a fair share.
- **Cost control**: expensive operations (sending SMS, calling a paid AI model) can't run away.
- **Security**: slows brute-force password guessing and scraping.
- **Business tiers**: free plan 1,000 calls per day, paid 100,000.

How it works

Over the limit, the server returns **HTTP 429 Too Many Requests**, ideally with `Retry-After` (seconds to wait), plus headers so clients can pace themselves:

```
HTTP/1.1 429 Too Many Requests
Retry-After: 30
X-RateLimit-Limit: 5000
X-RateLimit-Remaining: 0
X-RateLimit-Reset: 1767225600
```

Limits apply per **key**: user ID, API key, IP address, or a combination such as (user, endpoint).

Types

Algorithms.

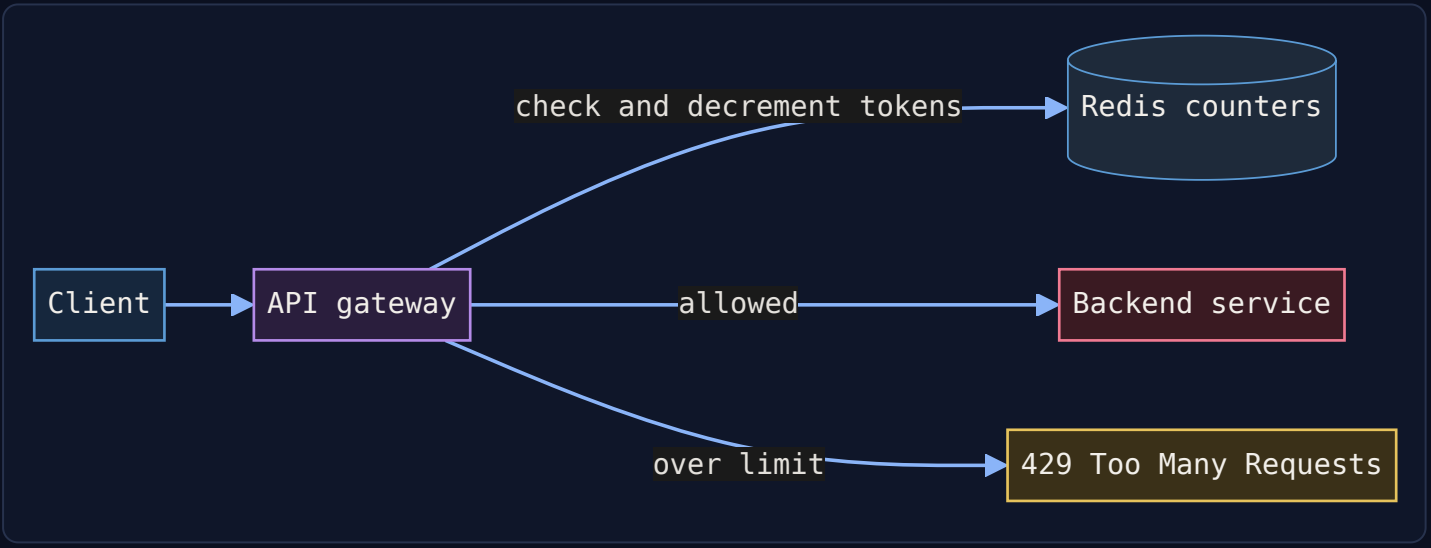
- **Token bucket**: a bucket holds up to B tokens, refilled at R per second; each request takes one, and an empty bucket rejects. Allows **bursts** up to B, then steady rate R (B = 10, R = 1/s: 10 at once, then one per second). Only two numbers per key (token count, last refill time), so cheap. The most widely used.
- **Leaky bucket**: requests enter a queue (the bucket) and leave at a fixed rate, like water dripping from a hole; a full bucket drops new ones. Smooths bursts into a perfectly steady outflow, protecting a downstream that can't handle bursts, but adds queueing delay.
- **Fixed window counter**: count requests per calendar window (12:00:00–12:00:59), reset at the boundary. One counter per key, very simple — but a client can send the full limit at 12:00:59 and again at 12:01:00, doubling the rate across the boundary.
- **Sliding window log**: store every request timestamp, drop those older than the window, count the rest. Exact, but one entry per request — memory-hungry at high limits.
- **Sliding window counter**: counters for the current and previous fixed windows, estimating `current + previous x (fraction of previous window still inside the sliding window)`. Limit 100/minute, 15 s into the minute with 30 requests, previous minute 80: 30 + 80 × 0.75 = 90 → allowed. Nearly the log's accuracy with fixed windows' memory.

Algorithm	Allows bursts	Memory per key	Accuracy
Token bucket	Yes, up to B	Tiny	Good
Leaky bucket	No, smooths	Queue size	Good
Fixed window	Yes, 2x at edges	Tiny	Rough
Sliding log	No	One entry per request	Exact
Sliding counter	Little	Two counters	Very good

Where to rate limit.

- **Client side**: polite clients self-limit, but can't be trusted.
- **Edge / CDN / WAF**: block floods by IP before they reach you (Cloudflare, AWS WAF rate rules).
- **API gateway or load balancer**: the usual place for per-API-key limits.
- **Inside each service**: protects the service itself, e.g. an expensive endpoint.
- **In front of a fragile dependency**: an outbound limiter so you never exceed a partner API's quota.

Distributed rate limiting. With 20 gateway servers, all must see the same count: a shared fast store such as Redis with an atomic increment-and-check (often a small Lua script so read and write happen in one step). To avoid a Redis call per request, some systems give each node a local share of the limit and sync periodically, accepting slight inaccuracy.



Trade-offs

Strict, globally exact limits put a central store on the hot path (latency, a dependency that can fail); local per-node limits are fast but approximate. If the limiter's store is down, choose **fail open** (allow traffic, risk overload) or **fail closed** (reject traffic, guaranteed outage); most fail open for user traffic.

Common mistakes

- Limiting only by IP, punishing many users behind one corporate NAT while attackers with many IPs dodge it.
- Returning 500 or 503 instead of 429, so clients don't know to back off.
- Not telling clients their remaining quota.

Real-world examples

GitHub's REST API allows 5,000 requests per hour authenticated and 60 unauthenticated, reporting usage in `x-ratelimit-*` headers. Stripe has written about token buckets for its API limiters. NGINX's `limit_req` module is a leaky bucket. Cloudflare has described the sliding window counter approximation at its edge.

In the interview

Q: Design a rate limiter for an API used by millions of users.

A: Token bucket per API key in Redis with an atomic script, enforced at the API gateway; 429 with `Retry-After` and rate-limit headers. Shard Redis by key, keep limits in config so they change without deploys, fail open if Redis is unavailable.

Q: Token bucket vs leaky bucket?

A: Token bucket allows bursts up to the bucket size while enforcing an average rate — suits APIs. Leaky bucket outputs a constant rate — suits protecting a downstream that can't absorb bursts.

TL;DR

Rate limiting caps requests per client per window and answers excess with 429; token bucket is the default; enforce at the edge or gateway with a shared counter store.

Load shedding and graceful degradation

What it is

- **Load shedding**: deliberately rejecting some requests when overloaded so the rest still succeed — like rolling blackouts that save the grid.
- **Graceful degradation**: turning off non-essential features under stress so the core keeps working (a shop at peak hides recommendations and reviews but keeps search and checkout).

Why it matters

Overload doesn't just slow things a bit: queues grow, every request waits longer, many time out after the server has already done the work, clients retry, and the server spends all its effort on requests nobody awaits. Throughput can collapse to near zero — **congestion collapse**. Shedding the excess early keeps **goodput** (useful completed work) high.

How it works

1. Measure an overload signal: CPU above 90%, in-flight requests above a limit, queue wait above, say, 100 ms.
2. When overloaded, reject new low-priority requests immediately and cheaply with 503 Service Unavailable (with `Retry-After`) rather than queueing them.
3. Prioritise: shed background jobs and bots first, then logged-out traffic; keep checkout and login last.
4. Drop requests that have already waited longer than the client's timeout; working on them is wasted.
5. Combine with degradation flags: a **kill switch** turns off expensive features during an incident.

Backpressure (Part 4) is the cooperative cousin: a consumer signals producers to slow down instead of silently dropping.

Trade-offs

Some users see errors on purpose, but controlled partial failure beats total failure. The hard parts: the right overload signal, and a cheap rejection path — if rejecting costs as much as serving, shedding doesn't help.

Common mistakes

- Unbounded queues hiding overload until latency is minutes long.
- Shedding at random, dropping a checkout while serving a crawler.
- No degradation switches, leaving "add servers and hope" as the only incident lever.

Real-world examples

Google's SRE book describes per-request criticality levels so overloaded servers shed the least important traffic first. Amazon has written about in-flight request limits and dropping requests whose

callers already timed out. Netflix serves a generic, non-personalised home-page row when personalisation services are degraded.

In the interview

Q: Traffic is 3x your capacity and autoscaling takes 5 minutes. What happens?

A: Unprotected, queues grow and the service collapses. Shed load at the edge by priority with fast 503s, rate limit heavy clients, switch off non-essential features, and serve cached content until new capacity arrives.

TL;DR

When overloaded, reject some work early and cheaply (load shedding) and switch off extras (graceful degradation) so the core stays up.

Autoscaling

What it is

Autoscaling automatically adds servers (containers, function copies) when load rises — **scaling out** — and removes them when it falls — **scaling in** — like opening tills as queues lengthen. It is the practical form of Part 1's **elasticity**.

Why it matters

Traffic is rarely flat: a food-delivery app may see 5× more orders at 7 pm than at 3 pm. Without autoscaling you buy for the peak (idle machines most of the day) or the average (falling over at dinner). Cloud billing by the second or hour makes matching capacity to demand real savings.

How it works

1. Choose a **metric** tracking load: average CPU, requests per second per instance, queue length, p95 latency, or a business metric.
2. Choose a **target**, e.g. "average CPU 60%"; the 40% headroom absorbs spikes while instances boot.
3. A controller checks periodically (Kubernetes' Horizontal Pod Autoscaler every 15 seconds by default) and computes `desired = current x (current metric / target)` : 10 instances at 90% CPU with a 60% target → 10 × 90/60 = 15.
4. It launches 5 more, which boot, pass health checks and join the load balancer — seconds for containers, roughly 1–5 minutes for VMs.
5. It scales in slowly after a **cool-down** (often 5–10 minutes) to avoid flapping, **draining** connections before terminating instances.
6. **Minimum and maximum** bounds prevent scaling to zero by mistake or to 10,000 machines during an attack.

Types

- **Reactive (dynamic) scaling**: responds to current metrics as above. Simple, but always slightly late — capacity arrives minutes after the spike begins.
- **Scheduled scaling**: by the clock ("weekdays at 08:30 set minimum to 40"), for known patterns: business hours, a TV ad slot, a ticket sale opening at 10:00.
- **Predictive scaling**: a model learns daily and weekly patterns and adds capacity before the forecast rise (AWS Auto Scaling offers it).
- **Horizontal vs vertical autoscaling**: horizontal adds instances (the common case); vertical gives an instance more CPU or memory (Kubernetes' Vertical Pod Autoscaler), usually needing a restart.
- **Cluster autoscaling**: Kubernetes adds whole machines (nodes) when pods can't fit.

Trade-offs

Reactive scaling is cheap but lags; keep headroom or combine with scheduled or predictive. CPU is easy but misleading for I/O-waiting services; requests per instance or queue depth is often better. The

stateless web tier autoscales easily; databases and other stateful systems scale far more slowly because data must be copied.

Common mistakes

- Scaling the web tier to 200 instances that exhaust the database's connection limit — moving the bottleneck, not fixing it.
- Targets too high (90% CPU), leaving no buffer for boot time.
- Slow startup (large images, long cache warm-up) making autoscaling useless for sudden spikes.
- No maximum, so a bug or DDoS produces a huge bill.

Real-world examples

AWS EC2 Auto Scaling groups, Google Cloud managed instance groups, Kubernetes Horizontal Pod Autoscaler, and KEDA (Kubernetes Event-driven Autoscaling, scaling on event sources such as Kafka consumer lag). Netflix built a predictive autoscaler, Scryer, to pre-provision capacity before its evening peak.

In the interview

Q: Your service gets a 10x spike every evening at 8 pm. How do you scale?

A: Scheduled or predictive scaling to raise capacity before 8 pm, reactive scaling on requests per instance for surprises, and check the database, caches and connection pools can take the higher instance count.

Q: Which metric would you autoscale a queue worker on?

A: Queue depth or consumer lag divided by one worker's processing rate; CPU doesn't say how far behind you are.

TL;DR

Autoscaling adds and removes capacity based on a metric and target; reactive is late, scheduled and predictive are early; set min and max bounds and keep headroom.

Virtual machines, containers and Docker

What it is

- A **virtual machine (VM)** is a software-emulated computer. A **hypervisor** (KVM, Xen, VMware ESXi, AWS Nitro) splits one physical server into several VMs, each with its own complete operating system and kernel — flats in a house, each with its own kitchen and plumbing.
- A **container** is an isolated process (or group) sharing the host's OS kernel but with its own view of files, network and process list, and its own CPU and memory limits — private rooms with a shared kitchen. Linux implements it with **namespaces** (what a process can see) and **cgroups** (how much it can use).
- A **container image** is a packaged, read-only snapshot of an application and everything it needs (runtime, libraries, config defaults); running it gives a container.
- **Docker** made containers popular from 2013: a simple file format (the `Dockerfile`) to build images, a way to run them, and **registries** (Docker Hub, Amazon ECR) to share them.

Why it matters

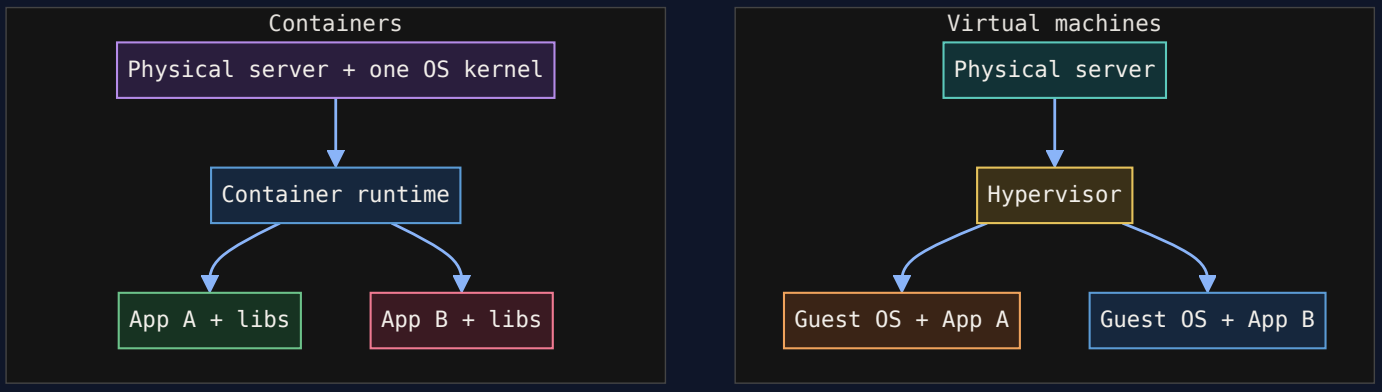
"It works on my machine" disappears: the same image runs on a laptop, in testing and in production. Containers start in under a second to a few seconds with megabytes (not gigabytes) of overhead, so dozens pack onto a server and scale quickly. They are the unit platforms like Kubernetes schedule.

How it works

A tiny Dockerfile for a web service:

```dockerfile
FROM python:3.12-slim
WORKDIR /app
COPY requirements.txt .
RUN pip install -r requirements.txt
COPY . .
CMD ["gunicorn", "-b", "0.0.0.0:8080", "app:server"]
```

1. `docker build` produces an image of stacked **layers** (base OS files, installed libraries, your code); unchanged layers are cached, so rebuilding after a code change takes seconds.
2. The image is pushed to a registry with a tag such as `shop-api:1.4.2`.
3. Any server pulls and runs it as a container with its own isolated filesystem and network interface.



Trade-offs

	Virtual machine	Container
Isolation	Strong: separate kernel	Weaker: shared kernel
Start time	Tens of seconds to minutes	Milliseconds to seconds
Size	Gigabytes	Megabytes
Density per host	Few to tens	Tens to hundreds
Typical use	Multi-tenant clouds, different OSes	Packaging and running apps

In practice they combine: clouds give you VMs and you run containers inside. "Micro-VMs" such as AWS Firecracker (used by Lambda and Fargate) blend the two, booting a minimal VM in around 125 ms.

Common mistakes

- Important data in the container's filesystem; it vanishes when the container is replaced. Use volumes or external storage.
- Huge images (several GB) slowing every deploy and scale-out.
- Running containers as root and baking secrets into images.

In the interview

Q: Container vs VM in one line?

A: A VM virtualises hardware and runs its own OS kernel; a container virtualises the OS, sharing the host kernel — lighter and faster to start but less isolated.

TL;DR

VMs are full emulated computers with their own OS; containers are isolated processes sharing one kernel; Docker packages apps as portable container images.

Kubernetes basics

What it is

Kubernetes (**K8s**) is a **container orchestrator**: it runs and manages containers across a fleet of machines. You declare *what* you want ("5 copies of `shop-api:1.4.2`, each with 1 CPU and 512 MB, reachable at `shop-api`") and it continuously works to make reality match — a harbour master assigning containers to berths and replacing any that fall in the water. Google open-sourced it in 2014, based on its internal Borg; it is now run by the Cloud Native Computing Foundation (CNCF).

Why it matters

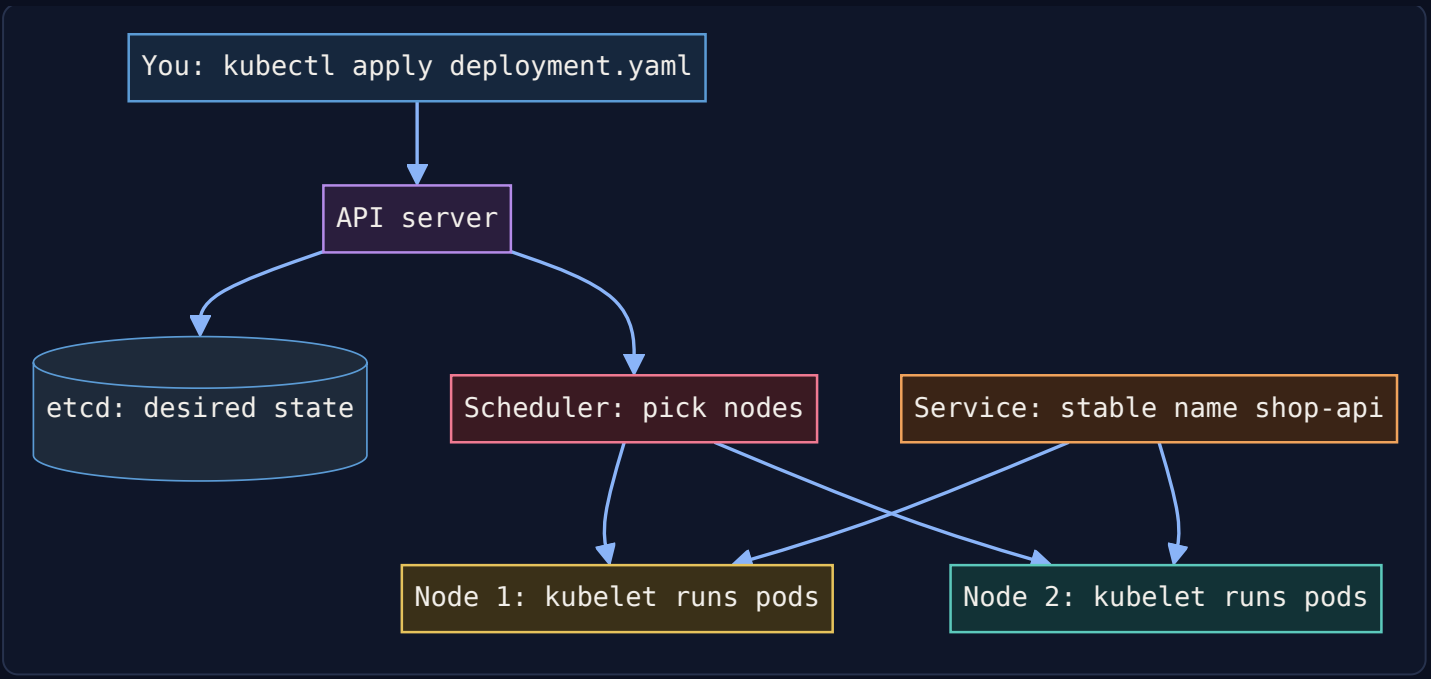
Nobody can place, restart and wire up 300 containers on 40 machines by hand. Kubernetes automates scheduling, self-healing, scaling, rolling updates, service discovery and configuration, the same way on every major cloud (Amazon EKS, Google GKE, Azure AKS) and on your own servers.

How it works

Key words:

- **Cluster**: the whole installation — machines plus the brain controlling them.
- **Node**: one machine (physical or VM) running containers, each with an agent called the **kubelet**.
- **Control plane**: the brain. The **API server** receives instructions, **etcd** stores desired and current state, the **scheduler** picks each pod's node, and **controllers** keep fixing differences.
- **Pod**: the smallest unit Kubernetes runs — one or more tightly coupled containers sharing a network address and storage (your app plus its sidecar). Pods are replaced, not repaired.
- **Deployment**: describes a stateless app ("5 replicas of this pod template"); creates and replaces pods and performs rolling updates when the image changes.
- **Service**: a stable name and virtual IP load-balancing across the current matching pods as they come and go — Kubernetes' built-in server-side service discovery.
- **Ingress** (or the newer **Gateway API**): rules routing outside HTTP traffic (`shop.com/api`) to Services.
- **ConfigMap** and **Secret**: configuration and sensitive values injected into pods.
- **Namespace**: folder-like grouping separating teams or environments.
- **StatefulSet**: a Deployment for stateful apps (databases), giving each pod a stable name and its own persistent storage.
- **Probes**: a **liveness probe** tells Kubernetes to restart a stuck container; a **readiness probe** whether to send the pod traffic yet.

At its heart is the **reconciliation loop**: controllers repeatedly compare desired with actual state and act on differences. If a node dies taking 2 of 5 pods, the Deployment controller sees "3, want 5" and creates 2 on healthy nodes.



Trade-offs

Powerful but complex: networking, storage, security policies and upgrades need expertise, and a small team can spend more time running the platform than building the product. Managed offerings (EKS, GKE, AKS) and simpler platforms (AWS ECS, Google Cloud Run, Fly.io, or a Heroku-style **platform as a service (PaaS)**, where you hand over code and the provider runs it) are often the better first choice.

Common mistakes

- No resource requests and limits, so one pod hogs a node's memory and others get killed.
- Missing readiness probes, so traffic arrives before the app has started.
- A production database on Kubernetes without understanding StatefulSets, persistent volumes and backups.

In the interview

Q: What happens if a node crashes?

A: The control plane sees it stop reporting and marks its pods lost; the Deployment controller creates replacements on healthy nodes, and the Service stops routing to the dead pods.

Q: Pod vs container?

A: A container is one isolated process image; a pod wraps one or more containers that share an IP and volumes and are scheduled together.

TL;DR

Kubernetes runs containers across many machines, keeping actual state equal to the declared desired state: pods run on nodes, Deployments manage replicas and rollouts, Services give a stable address.

Infrastructure as code

What it is

Infrastructure as code (IaC): servers, networks, load balancers, databases and permissions described in version-controlled text files applied by a tool, instead of clicks in a cloud console — a recipe anyone can reproduce, with a visible diff between versions.

Why it matters

Hand-built environments drift apart ("staging has a setting production doesn't"), can't be rebuilt quickly after a disaster and have no review trail. With IaC a new region or test environment is one

command, every change is code-reviewed, and the files document the system.

How it works

1. Write a **declarative** description of the desired resources, e.g. Terraform:

```
resource "aws_s3_bucket" "images" {
  bucket = "shop-product-images"
}
```

2. Run a **plan**: the tool compares the files with what exists and shows the diff ("will create 1 bucket").
3. A teammate reviews the plan in a pull request.
4. **Apply**: the tool calls cloud APIs to create or change resources and records what it built in a **state file**.

Declarative tools describe the end state (Terraform, AWS CloudFormation, Kubernetes YAML, Pulumi); **imperative** scripts describe steps. **Configuration management** tools (Ansible, Chef, Puppet) configure software on existing servers. **Immutable infrastructure**: never patch a running server; build a new image and replace it.

Trade-offs and common mistakes

A learning curve, and state files must be stored and locked carefully (two people applying at once can corrupt them). The classic mistake is mixing IaC with manual console edits, creating **drift** (**configuration drift**) the next apply may silently undo.

TL;DR

Infrastructure as code keeps cloud setup in reviewed, versioned files a tool applies, so environments are reproducible and changes auditable.

CI/CD

What it is

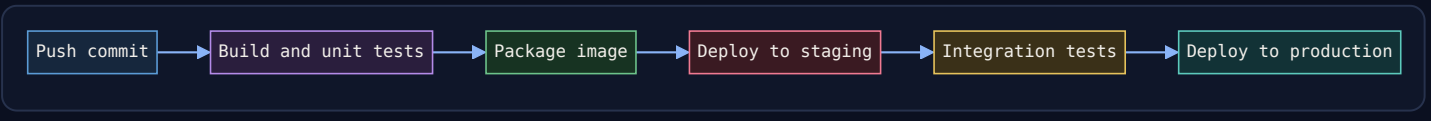
- **Continuous integration (CI)**: developers merge small changes into the main branch frequently (at least daily), each automatically triggering a build and test run, catching conflicts and bugs within minutes.
- **Continuous delivery (CD)**: every change that passes the pipeline is automatically packaged and ready to release to production at the push of a button.
- **Continuous deployment**: one step further — every passing change goes to production automatically, no button.

A **pipeline** is the automated sequence of steps, like a factory line pulling faulty cars before the showroom.

Why it matters

Small, frequent releases are less risky than big, rare ones: a release with one change that breaks something tells you exactly what to roll back. The research behind the **DORA metrics** (deployment frequency, lead time for changes, change failure rate, time to restore service) found high performers deploy more often *and* fail less.

How it works



1. A developer pushes a commit or opens a pull request.

2. The CI server (GitHub Actions, GitLab CI, Jenkins, CircleCI) compiles, runs linters, unit tests and security scans, typically in under 10 minutes.
3. It builds one versioned artifact (a container image); the **same artifact** is promoted through every environment, never rebuilt.
4. It deploys to staging and runs integration or end-to-end tests.
5. It deploys to production with a safe strategy (next section), watches health metrics and rolls back automatically if they worsen.

Common mistakes

- Flaky tests that fail randomly, teaching people to ignore red builds.
- Rebuilding separately for production, so what you tested isn't what you ship.
- Long-lived feature branches merged monthly, defeating integration.
- Schema changes that aren't backward compatible, so old and new code can't run side by side during a rollout. Use **expand and contract**: add the new column, deploy code writing both, migrate, remove the old column in a later release.

TL;DR

CI builds and tests every change automatically; CD keeps every passing build releasable (or releases it automatically); small frequent releases are safer.

Deployment strategies and rollback

What it is

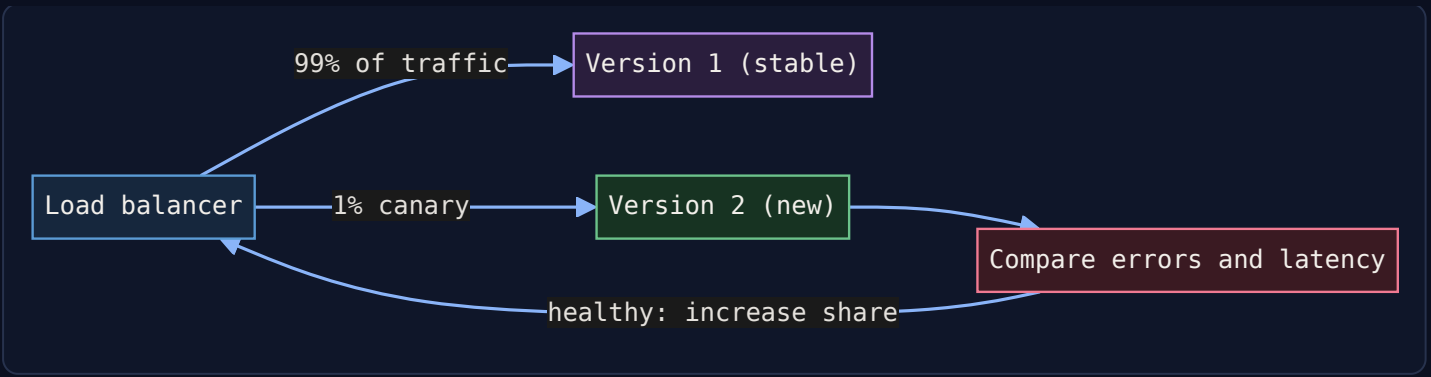
A **deployment strategy** is how you replace a running service's old version with a new one without downtime, limiting damage if the new version is bad. The **blast radius** is how many users a bad change can hurt; good strategies keep it small.

Why it matters

Most outages are caused by changes — code releases, config edits, schema migrations. How you release decides whether a bug hits 1% of users for two minutes or 100% for an hour.

Types

- **Recreate (big bang)**: stop all old instances, start new ones. Simple, with downtime; only for internal tools or maintenance windows.
- **Rolling deployment**: replace instances a few at a time (2 of 20), each batch passing health checks before the next. No extra servers or downtime, but both versions serve side by side for a while, and rollback means rolling back through batches. Kubernetes Deployments' default.
- **Blue-green deployment**: two full environments; **blue** serves live traffic on the old version, **green** gets the new one and is tested privately, then the load balancer (or DNS) switches all traffic to green at once. Instant rollback (switch back), but double capacity during the release, and the shared database means schema changes must work for both.
- **Canary release**: send a small slice of real traffic (1%) to the new version, compare its error rate and latency with the old, step up (1%, 5%, 25%, 50%, 100%) while healthy or pull back automatically. Named after miners' gas-detecting canaries. Smallest blast radius, but needs good metrics and automation.
- **Shadow (dark) launch**: copy real traffic to the new version and discard its responses — real-load testing with zero user impact, provided it writes no data and sends no emails.
- **Feature flags (feature toggles)**: ship code turned off, then enable it at runtime via configuration — internal users, then 5% of customers, then everyone. Separates **deploy** (code on servers) from **release** (users see it); a flag is also a **kill switch** (off in seconds without a deploy). Tools: LaunchDarkly, Unleash, Flagsmith, in-house systems.



Rollback returns to the last known-good version. During an incident going back beats going forward: restore, then investigate calmly. Make it boring: keep the previous artifact, automate the button, practise it. Easy for stateless code; hard when the release changed data or schema non-backward-compatibly — then **roll forward** (ship a quick fix).

Trade-offs

Strategy	Downtime	Extra capacity	Rollback speed	Blast radius
Recreate	Yes	None	Slow	Everyone
Rolling	No	Small	Medium	Grows per batch
Blue-green	No	2x briefly	Instant	Everyone at switch
Canary	No	Small	Fast	Tiny at first
Feature flag	No	None	Seconds	Chosen users

Common mistakes

- Canary without automated metric comparison, so nobody notices the 1% failing.
- Old feature flags never cleaned up, leaving confusing dead code paths.
- Treating config changes as safe and pushing them globally at once.

Real-world examples

In 2012 **Knight Capital** deployed new trading code to only 7 of 8 servers; a reused flag activated dormant old code on the eighth, which sent more than 4 million orders and lost about $440 million in roughly 45 minutes (the SEC later put the loss above $460 million). In July 2024 a faulty **CrowdStrike** content update pushed to all customers at once crashed about 8.5 million Windows machines (Microsoft's estimate); the main lesson was staged, canary-style rollout. Google and Meta release most changes through progressive rollouts with automatic rollback.

In the interview

Q: Blue-green or canary?

A: Blue-green: instant switch and rollback, but everyone exposed at once and double capacity. Canary exposes a small percentage first and grows only while metrics stay healthy, so a bad release hurts far fewer users. For high-traffic user-facing services, canary, ideally with feature flags.

Q: How do you deploy a database schema change safely?

A: Expand and contract: add new columns or tables first, deploy code that works with both shapes, backfill, switch reads, and drop the old structure in a later release once unused.

TL;DR

Release gradually (rolling, canary, feature flags) or switch whole environments (blue-green), watch metrics, and keep rollback one fast, practised step.

Observability: logs, metrics and traces

What it is

Monitoring watches known numbers to check health. **Observability** is the broader ability to understand *why* a system behaves as it does — including unpredicted problems — purely from the data it emits (a car's dashboard versus a mechanic's diagnostic computer). The data has three main kinds, the **three pillars**:

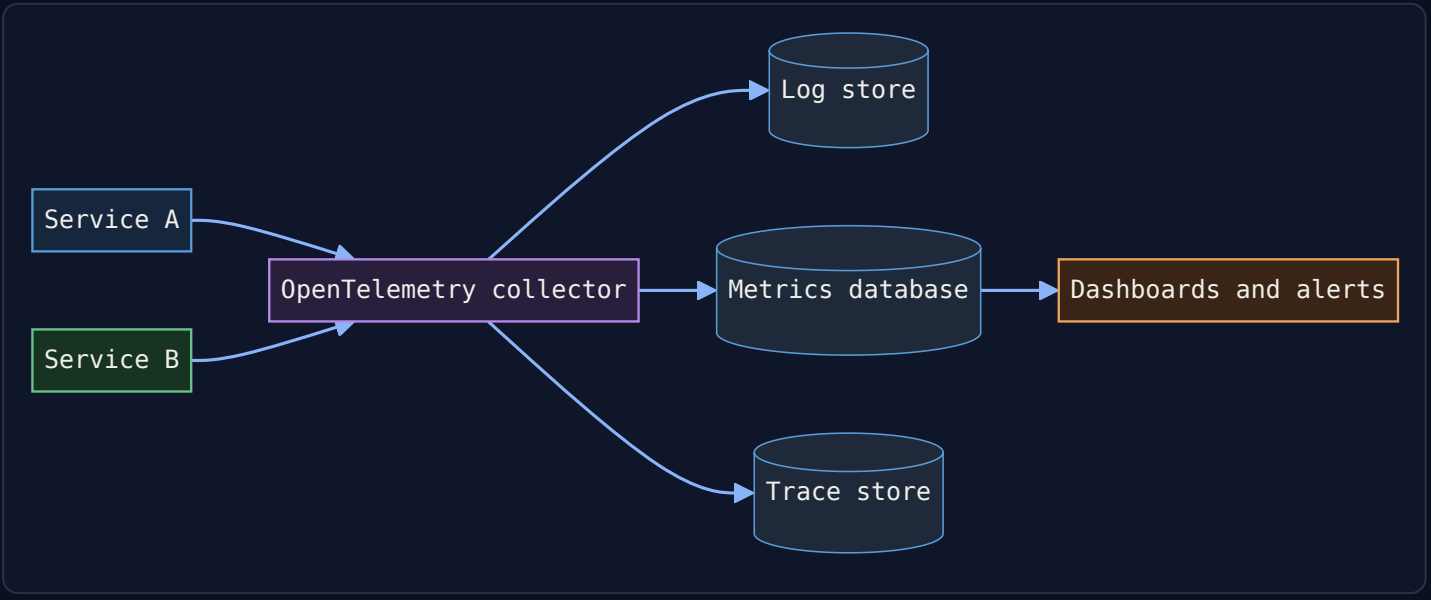
- **Logs**: timestamped records of individual events: `2026-10-07T09:14:03Z ERROR order=8812 payment declined: card expired`. Best as **structured logs** (JSON with named fields) so tools can search and filter.
- **Metrics**: numbers over time, cheap to store and fast to graph: requests per second, error count, p99 latency, CPU %, queue depth. Shapes: **counters** (only rise, e.g. total requests), **gauges** (up and down, e.g. memory in use), **histograms** (distributions, e.g. latency buckets for percentiles).
- **Traces**: one request's path through many services — a tree of **spans**, each one operation (a service handling it, a database query) with start time and duration, all sharing a **trace ID** passed in request headers, so you see that a 2.1 s checkout spent 1.8 s waiting on the tax service.

Why it matters

A monolith has one log file; in microservices one click touches 15 services on 200 machines. Without metrics you don't know something is wrong; without traces, where; without logs, why. Observability also drives capacity planning, SLOs and canary decisions.

How it works

1. **Instrument** code, usually with **OpenTelemetry**, the vendor-neutral standard for logs, metrics and traces.
2. Each incoming request gets a **correlation ID** or trace ID, logged and forwarded to every downstream call (the W3C `traceparent` header is the standard).
3. **Collect and ship**: logs to a store (Elasticsearch/OpenSearch, Loki, Splunk), metrics to a time-series database (Prometheus, Datadog, CloudWatch), traces to a tracing backend (Jaeger, Zipkin, Tempo, Honeycomb).
4. **Visualise**: dashboards (Grafana, Datadog, Kibana).
5. **Alert**: rules on metrics notify a human.



Sampling. Storing every trace at 100,000 requests per second is too expensive, so keep a sample — e.g. 1% of normal requests but 100% of errors and slow requests (**tail-based sampling**, deciding after seeing the whole trace).

Cardinality. Each unique combination of metric labels is a separate time series; a `user_id` label with 10 million values creates 10 million series and can crash a metrics system. Put high-cardinality details in logs and traces, not metric labels.

Types: what to measure

- **Four golden signals** (Google's SRE book), for any user-facing service:
  - **Latency**: request duration, successful and failed tracked separately (a fast error isn't a success).
  - **Traffic**: demand, e.g. requests per second.
  - **Errors**: rate of failed requests (explicit 5xx, wrong answers, too slow).
  - **Saturation**: how full the most constrained resource is (CPU, memory, connection pool, queue).
- **RED method** (Tom Wilkie), for request-driven services: **Rate** (requests/s), **Errors** (failed requests/s), **Duration** (latency distribution).
- **USE method** (Brendan Gregg), for resources (CPU, disks, network): **Utilisation** (% time busy), **Saturation** (extra work queued), **Errors** (error events).

RED views a service from outside (what users feel); USE the machines underneath (why it might be happening).

Trade-offs

Observability can cost a lot — large companies spend a noticeable share of their infrastructure bill on it. Logs are rich but expensive per event; metrics cheap but aggregated (they say *that* p99 rose, not *which* request); traces precise but sampled. Use all three, linked by trace ID.

Common mistakes

- Debug-level logging of everything in production, storing terabytes nobody reads.
- Logging secrets, passwords or personal data.
- Averages instead of percentiles on latency dashboards (Part 1).
- No correlation ID, so logs from different services can't be stitched together.

Real-world examples

Google's **Dapper** paper (2010) introduced large-scale distributed tracing and inspired Zipkin (Twitter) and Jaeger (Uber). Prometheus, created at SoundCloud, is the standard open-source metrics system in the Kubernetes world, usually with Grafana.

In the interview

Q: A user says checkout is slow. How do you find out why?

A: Check checkout's RED metrics to confirm and size it (which percentile, since when, which regions), correlate with recent deploys, open slow traces to find the slow span, read that service's logs for those trace IDs, then USE metrics on the slow component's machines or database.

Q: Logs vs metrics vs traces?

A: Metrics are the cheap early warning (a number moved, by how much); a trace pinpoints which hop was slow or failed; logs tell the detailed story of what one component did then.

TL;DR

Emit metrics (cheap numbers), logs (detailed events) and traces (one request across services), tied together by a trace ID; watch the golden signals, RED for services and USE for resources.

Dashboards, alerting, on-call and SLO-based alerts

What it is

- A **dashboard** is a page of graphs showing key metrics at a glance.
- An **alert** is an automated notification when a metric rule breaks ("error rate above 2% for 5 minutes").
- **On-call** is the rota of engineers responding to alerts, including at night. A **page** is an urgent alert that wakes someone (PagerDuty, Opsgenie); a **ticket** is non-urgent, handled in working hours.
- A **runbook** (playbook) is a written guide for a specific alert.

- A **postmortem** (incident review) analyses an incident afterwards — timeline, impact, root causes, follow-ups — written **blameless**: how the system allowed the mistake, not who made it.

Why it matters

Alerts tell humans something is broken before customers flood support. Too many cause **alert fatigue**: paged 30 times a night for self-healing blips, on-call starts ignoring pages and misses the real one.

How it works

Good alerting rules:

1. Page on **symptoms users feel** (high error rate, latency, checkout failures), not causes (one server at 90% CPU, which may be fine).
2. Every page must be urgent, actionable and need human judgement; otherwise make it a ticket or automate the fix.
3. Attach a runbook link and a dashboard to every alert.

SLO-based alerting. From Part 1: the **SLI** is the measured indicator (fraction of successful requests), the **SLO** the target (99.9% over 30 days), the **error budget** the allowed failure (0.1%, ~43 minutes of full downtime per 30 days). Alert not on any blip but on the **burn rate** — how fast the budget is being spent. Burn rate 1 spends exactly the whole budget in 30 days; 14.4 sustained for an hour spends 2% of the monthly budget in that hour and would exhaust it in about two days.

A common scheme from Google's SRE workbook:

Window	Burn rate	Budget used	Action
1 hour	14.4	2%	Page now
6 hours	6	5%	Page
3 days	1	10%	Ticket

Pairing each long window with a short one (1 hour and 5 minutes) makes the alert fire fast and stop quickly once fixed.

Incident response roles. An **incident commander** coordinates, others investigate, one person communicates with stakeholders and customers. The first goal is **mitigation** (stop the bleeding: roll back, fail over, shed load), not root cause.

Common mistakes

- Alerting on every metric someone thought of, never deleting noisy alerts.
- Dashboards with 80 graphs and no "is it healthy?" panel at the top.
- Debugging the root cause during an outage instead of rolling back first.
- Postmortems that blame a person, teaching people to hide mistakes.

In the interview

Q: What would you alert on for this service?

A: SLO burn rate for availability and latency of the main user journeys, plus a few saturation signals predicting imminent failure (disk nearly full, queue lag growing). Everything else on dashboards or tickets, not pages.

TL;DR

Page on user-facing symptoms and fast error-budget burn, keep every page actionable with a runbook, mitigate first, write blameless postmortems.

Disaster recovery

What it is

Disaster recovery (DR) is the plan for getting service and data back after a large failure: a data-centre fire, a whole cloud region down, ransomware, an engineer deleting the production database. **Business continuity** is the wider plan for keeping the business running. Two numbers define a DR target:

- **RPO (recovery point objective)**: how much data, measured in time, you can afford to lose ("how far back is my last save point?"). RPO 1 hour: at most the last hour of writes missing.
- **RTO (recovery time objective)**: how long the service may be down before restoration ("how long until I'm playing again?"). RTO 4 hours: running again within 4 hours.

Why it matters

Redundancy within one site (Part 1) survives a disk or server failing, not losing the whole site, nor a bug or human corruption faithfully replicated to every copy — only backups and DR plans cover those. Lower RPO and RTO cost more, so agree them with the business: a bank's payment ledger may need RPO near zero and RTO of minutes; an internal wiki can accept a day.

How it works

Backups. The kinds — full, incremental, snapshot, continuous WAL archiving for **point-in-time recovery**, and the **3-2-1 rule** (3 copies, 2 media or services, 1 off-site) — are in *Durability: WAL, fsync, backups and recovery* (Part 3). For DR add **immutable / air-gapped backups**, which can't be deleted or altered for a set period, protecting against ransomware and rogue deletes. A backup is unproven until restored: rehearse and time restores on a schedule, since restore time is part of your RTO.

Availability zones and regions. A cloud **region** is a geographic area (AWS `eu-west-1`, Ireland) containing several **availability zones (AZs)**: separate data centres with independent power, cooling and networking, a few to tens of kilometres apart, with ~1–2 ms round trips. AZs fail independently; regions are hundreds or thousands of kilometres apart.

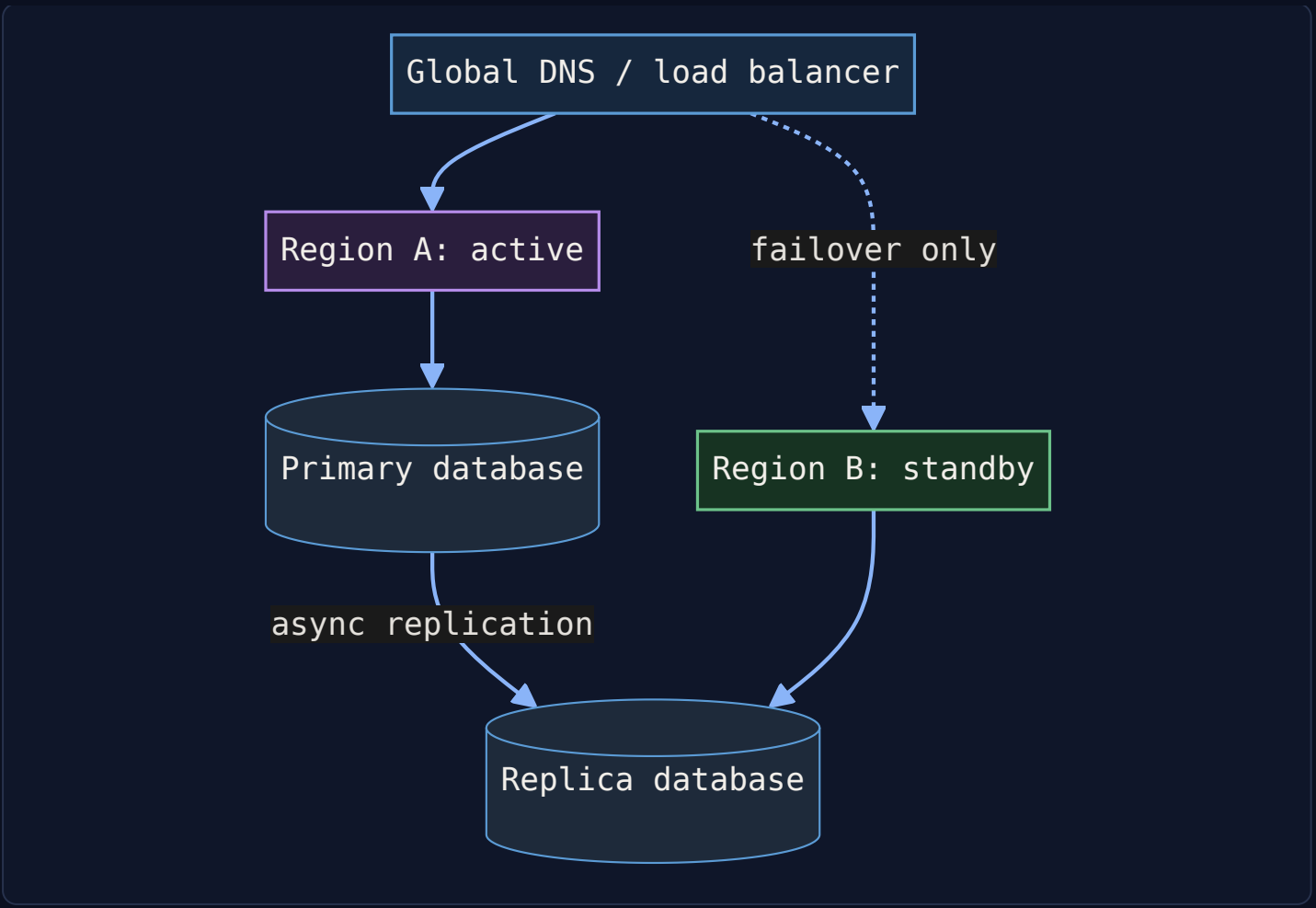
- **Multi-AZ**: copies in 2–3 AZs of one region. Survives a data-centre failure, and AZs are close enough for synchronous database replication. The normal production baseline.
- **Multi-region**: two or more regions. Survives a whole-region outage and serves users nearer home, but cross-region round trips are 50–150 ms or more, so replication is usually asynchronous (RPO > 0), and it costs much more and is far more complex.

Types: DR strategies

Cheapest and slowest to most expensive and fastest:

Strategy	What runs in the DR site	Typical RTO	Typical RPO
Backup and restore	Nothing, only backups	Hours to a day	Hours
Pilot light	Data replicated, servers off	Tens of minutes	Minutes
Warm standby	Scaled-down full copy	Minutes	Seconds to minutes
Active-active	Full copy serving traffic	Near zero	Near zero to seconds

- **Active-passive**: one site serves; the other (pilot light or warm) takes over on **failover**. Simpler, but standby capacity idles and failover is a risky, rarely exercised moment.
- **Active-active**: all sites serve live traffic and absorb a failed site's share. Fast recovery, no idle capacity, but each site needs headroom, and writes in several places raise hard consistency problems (Part 3 multi-leader replication, Part 4 conflict resolution).



Chaos engineering deliberately injects failures (killing servers, adding latency, blocking a dependency, even removing a whole AZ) in a controlled way to prove redundancy, failover and alerting work and find weaknesses before real outages do. Start small: a hypothesis ("if one cache node dies, p99 stays under 300 ms"), a limited blast radius, an abort button. **Game days** are scheduled team exercises of this kind, including full regional failover.

Common mistakes

- Backups in the same account and region as the data, so one disaster or attacker removes both.
- Never testing restores, discovering mid-disaster that backups were silently failing.
- Assuming replication is a backup; a `DELETE` replicates too.
- A DR region that depends on a service in the failed region (shared DNS, auth, deployment tooling).

Real-world examples

GitLab's 2017 database deletion (Part 3, *Durability*): none of its five backup and replication methods was working reliably, and about six hours of data were lost. Netflix created **Chaos Monkey** (open-sourced 2012) to randomly terminate production instances, later adding tools simulating whole-region outages; it runs active-active across several AWS regions. In October 2021 a mistaken maintenance command cut **Facebook**'s backbone, its DNS servers then withdrew their **BGP** routes (the announcements telling the internet how to reach them), and the ~six-hour outage also locked engineers out of internal tools needed for the fix — recovery tooling must not depend on the thing that failed.

In the interview

Q: Explain RPO and RTO with an example.

A: Backups every 6 hours and a 2-hour restore: worst-case RPO 6 hours of lost data, RTO at least 2 hours down. Continuous log shipping to another region could bring RPO to seconds; a warm standby, RTO to minutes.

Q: Multi-AZ or multi-region?

A: Multi-AZ is the baseline: survives a data-centre failure with low latency and synchronous replication. Multi-region adds regional-outage protection and lower latency for global users but costs much more and usually means asynchronous replication — only when the SLA or user geography demands it.

TL;DR

RPO is how much data you can lose, RTO how long you can be down; tested backups and multi-AZ are the baseline, multi-region active-passive or active-active buys more at a higher price, and chaos engineering proves it works.

Authentication and authorization

What it is

- **Authentication (AuthN)** answers "who are you?" — proving identity with something you know (password), have (phone, security key) or are (fingerprint). Two or more of these is **multi-factor authentication (MFA)**.
- **Authorization (AuthZ)** answers "what may you do?" — checking permissions once identity is known.

At an airport, the passport is authentication; the boarding pass deciding your plane and seat is authorization. Failed authentication returns **401 Unauthorized** (confusingly named; it means "unauthenticated"); known but not permitted returns **403 Forbidden**.

Why it matters

Nearly every system has users, and nearly every breach involves stolen identities or missing permission checks. The vocabulary is also a common interview probe.

How it works

Authorization models:

- **Access control list (ACL)**: each resource lists who may do what ("doc 42: Alice can edit, Bob can read"). Fine-grained but hard to manage at scale.
- **Role-based access control (RBAC)**: permissions attach to roles (admin, editor, viewer) given to users. Simple, auditable; most business software.
- **Attribute-based access control (ABAC)**: rules over attributes of user, resource and context ("a doctor may read a patient record if in the same department and on shift"). Very flexible, harder to reason about.
- **Relationship-based access control (ReBAC)**: permissions follow relationships in a graph ("can view if member of a group that owns the folder containing the file"). Google's Zanzibar, powering sharing in Drive and other products, is the famous example.

Check authorization on the server for every request — never only by hiding buttons in the UI.

In the interview

Q: Authentication vs authorization?

A: Authentication proves who the caller is (login, token validation; 401 on failure); authorization decides whether that identity may do this action on this resource (roles, policies; 403 on failure).

TL;DR

AuthN is who you are; AuthZ is what you may do; RBAC grants permissions through roles, ABAC through attribute rules.

Sessions, cookies, tokens and JWT

What it is

HTTP is stateless (Part 2), so after login the server needs a way to recognise you on the next request.

- **Session-based authentication**: the server creates a **session** record (user ID, expiry) in memory, a database or Redis, and gives the browser a random **session ID** in a **cookie** (small data the browser stores and automatically sends back to the same site on every request) — a cloakroom ticket the server looks up.
- **Token-based authentication**: the server gives the client a **token**, sent with each request, usually as `Authorization: Bearer <token>` (a **bearer token**: whoever holds it gets access). An **opaque token** is a random string the server must look up; a **self-contained token** carries the user's details, signed so it can't be forged.
- **JWT (JSON Web Token)**, pronounced "jot", is the commonest self-contained format: three Base64URL-encoded parts, `header.payload.signature`.

```
header:    {"alg":"RS256","typ":"JWT"}
payload:   {"sub":"user_8812","role":"editor","exp":1767225600}
signature: RSA signature over header + payload, made with the server's private key
```

How it works

JWT verification: any service with the issuer's key (a shared secret for HS256, an HMAC-based signature; the public key for RS256, an RSA signature) checks the signature and the `exp` (expiry) claim, then trusts the claims without a database call — why JWTs suit microservices, each verifying locally.

A JWT is **signed, not encrypted**: anyone can Base64-decode and read the payload, so never put secrets in it. With no lookup, a stolen JWT works until expiry and can't easily be revoked. The usual answer: short-lived **access tokens** (5–15 minutes) plus a long-lived **refresh token**, stored more carefully and checked against a database, used to get new access tokens; revoking it logs the user out within minutes.

Cookie safety flags: `HttpOnly` (JavaScript can't read it, protecting against XSS theft), `Secure` (HTTPS only), `SameSite=Lax` or `Strict` (not sent on most cross-site requests, protecting against CSRF), plus an expiry.

Trade-offs

	Server sessions	JWT access tokens
Server state	Session store needed	None for verification
Revocation	Instant: delete session	Hard until expiry
Per-request cost	Store lookup	Signature check only
Size	Small ID	Hundreds of bytes or more
Best fit	Classic web apps	APIs, microservices, mobile

Common mistakes

- Accepting JWTs with `alg: none`, or not checking signature and expiry.
- Tokens in browser `localStorage`, where any XSS script can steal them; an `HttpOnly` cookie is safer for browser apps.
- Very long-lived access tokens.

TL;DR

Sessions keep login state on the server with an ID in a cookie; tokens like JWT carry signed claims presented on each request — easy to verify anywhere, hard to revoke, so keep them short-lived.

OAuth 2.0, OpenID Connect, SSO and API keys

What it is

- **OAuth 2.0** is a standard for **delegated authorization**: letting one app access your data in another *without your password* ("Allow PhotoPrinter to read your Google Photos?") — a hotel key card for one room that expires at checkout, not the master key.

- **OpenID Connect (OIDC)** is a thin identity layer on OAuth 2.0 adding **authentication**: it returns an **ID token** (a JWT) saying who the user is. "Sign in with Google/Apple" is OIDC.
- **Single sign-on (SSO)**: log in once with one identity provider to reach many applications (sign in to Okta or Microsoft Entra ID once for email, chat, dashboards). Enterprise SSO uses OIDC or the older XML-based **SAML**.
- **API key**: a long random secret identifying a calling program (not a person), sent in a header such as `X-API-Key`. Simple for server-to-server use, metering and rate limiting, but a bearer secret with no built-in expiry or user identity.

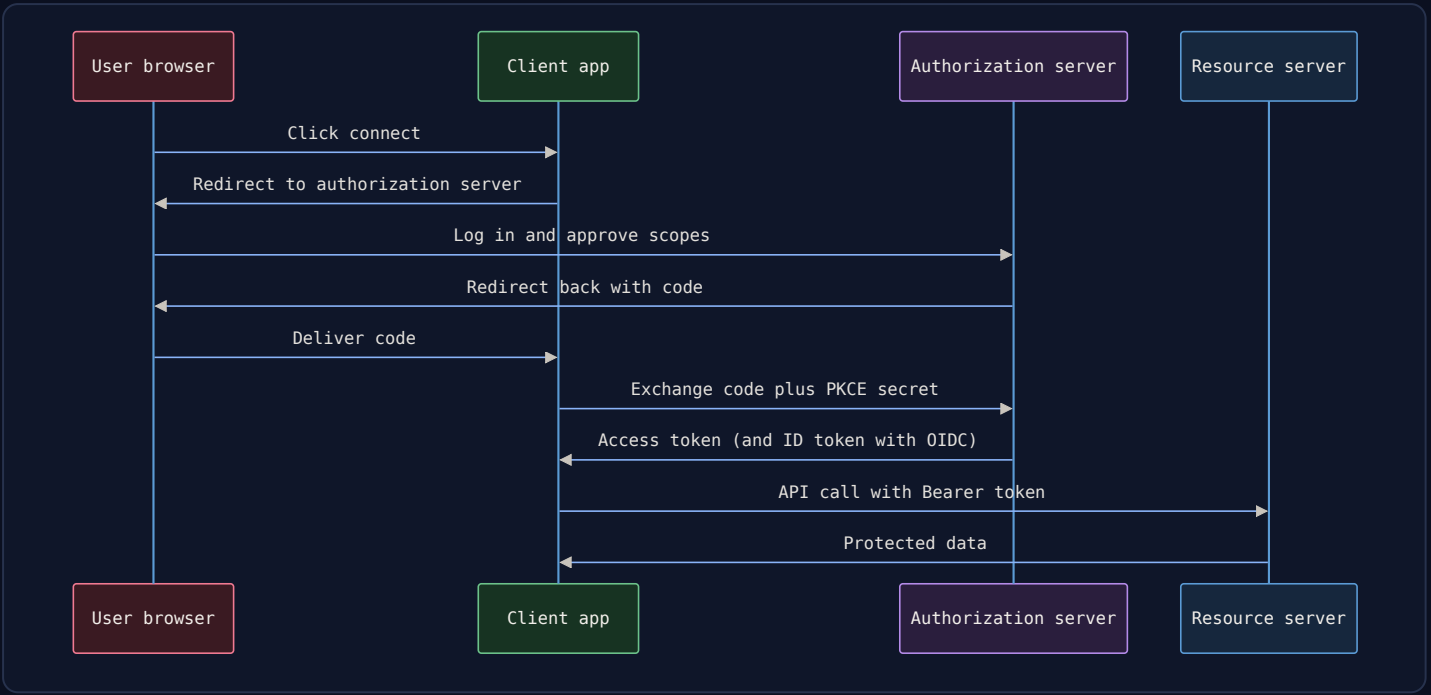
How it works

The four OAuth roles:

- **Resource owner**: the user who owns the data.
- **Client**: the app wanting access (PhotoPrinter).
- **Authorization server**: issues tokens after the user consents (Google's login and consent service).
- **Resource server**: the API holding the data (Google Photos API), accepting the access token.

Authorization code flow (with PKCE), recommended for web and mobile apps:

1. The user clicks "Connect Google Photos"; the client redirects the browser to the authorization server with its client ID, the **scopes** wanted (e.g. `photos.read`, each a named permission) and a redirect URL. With **PKCE** (Proof Key for Code Exchange) it also sends a hash of a one-time random secret.
2. The user logs in at Google (the client never sees the password) and approves the scopes.
3. The authorization server redirects back with a short-lived, single-use **authorization code**.
4. The client exchanges the code (plus the PKCE secret, and for server apps its client secret) at the token endpoint, server to server.
5. It receives an **access token** (often a refresh token; with OIDC an ID token too).
6. It calls the Photos API with `Authorization: Bearer <access token>`; the resource server validates it and checks scopes.



Other flows: **client credentials** for machine-to-machine calls with no user; **device code** for TVs and keyboardless devices. The old "implicit" and "password" flows are discouraged.

Common mistakes

- Using OAuth access tokens as proof of identity: OAuth says what a client may access; OIDC's ID token says who logged in.
- Requesting far broader scopes than needed.

- Committing API keys to Git; automated scanners find leaked keys within minutes.

In the interview

Q: OAuth vs OpenID Connect?

A: OAuth 2.0 is authorization: a scoped access token to call an API on the user's behalf. OIDC adds authentication: a signed ID token saying who the user is — what "Sign in with Google" uses.

Q: When an API key instead of OAuth?

A: Simple server-to-server access by a known customer's backend, needing mainly identification, metering and rate limiting. Acting for users, or needing scopes and expiry: OAuth.

TL;DR

OAuth 2.0 delegates scoped access without sharing passwords (owner, client, authorization server, resource server; authorization code flow with PKCE); OIDC adds an ID token for login; SSO reuses one login across apps; API keys identify programs.

Encryption, hashing, passwords and secrets

What it is

- **Encryption** scrambles data with a **key** so only the right key can **decrypt** it — two-way. **Symmetric** (same key both ways, AES; fast, for bulk data) and **asymmetric / public-key** (public key to encrypt or verify, private key to decrypt or sign; RSA, elliptic-curve; slower, for key exchange and signatures, as in TLS) are explained in *HTTPS and TLS* (Part 2).
- **Encryption in transit**: protecting data moving over a network, almost always **TLS** (the "S" in HTTPS); between services, **mutual TLS (mTLS)** with certificates on both sides.
- **Encryption at rest**: protecting stored data on disks, databases, backups and object storage so a stolen disk or backup is unreadable — in the cloud typically AES-256 with keys from a **key management service (KMS)**.
- **Hashing** turns any input into an irreversible fixed-size **fingerprint** (**hash**, **digest**); SHA-256 of any file is 256 bits. Same input, same hash; a tiny change, a completely different one. One-way, like grinding coffee beans.

How it works

Hashing vs encryption. Encrypt when you need the original back (a card number to charge later); hash when you only need to check a match (passwords, file integrity, deduplication).

Password hashing. Never store passwords in plain text or encrypted (a leaked key leaks all). Store a **slow, salted hash**:

1. Generate a random per-user **salt** (e.g. 16 bytes), so identical passwords get different hashes and precomputed "rainbow tables" are useless.
2. Hash password + salt with a deliberately slow algorithm — **bcrypt** (with a cost factor), **scrypt**, or **Argon2id** (a variant of Argon2, winner of the 2015 Password Hashing Competition; current first choice of OWASP, the Open Worldwide Application Security Project) — tuned so one hash takes tens to hundreds of milliseconds and, for scrypt and Argon2, lots of memory.
3. Store algorithm, parameters, salt and hash together.
4. At login, recompute with the stored salt and compare.

Fast hashes (SHA-256, MD5) are wrong for passwords: one modern GPU tries billions of SHA-256 guesses per second, but only thousands to tens of thousands of bcrypt guesses at typical cost.

Envelope encryption. A KMS holds a master key that never leaves it. Each piece of data is encrypted with its own **data key**, which is itself encrypted by the master key and stored beside the data; rotating or revoking the master key controls access to everything.

Secrets management. Secrets — passwords, API keys, database credentials, private keys, tokens — stay out of source code, container images and logs. Store them in a secrets manager (HashiCorp Vault, AWS Secrets

Manager, Google Secret Manager, Kubernetes Secrets backed by a KMS), give each service only its own, **rotate** them automatically, and prefer short-lived dynamically issued credentials.

Common mistakes

- Inventing your own cryptography; use well-known libraries.
- Confusing encoding (Base64, reversible by anyone) with encryption.
- Encrypting at rest but keeping the key on the same disk.
- Passwords hashed with plain SHA-256 and no salt.

In the interview

Q: How should you store user passwords?

A: Hash each with a unique random salt using a deliberately slow algorithm — Argon2id (also memory-hard) or bcrypt with tuned cost — store only hash and parameters, and compare by re-hashing at login. Never plain or reversibly encrypted.

Q: Encryption in transit vs at rest?

A: In transit protects data on networks with TLS (mTLS between services); at rest protects stored data on disks, databases and backups, usually AES with KMS-managed keys. You need both.

TL;DR

Encryption is reversible with a key (in transit via TLS, at rest via AES and a KMS); hashing is one-way; passwords get a unique salt and a slow hash like Argon2id or bcrypt; secrets live in a secrets manager and rotate.

Network security: firewalls, WAF and DDoS

What it is

- A **firewall** filters traffic by rules on IPs, ports and protocols (layers 3 and 4): "allow TCP 443 from anywhere, TCP 5432 only from the app subnet, deny everything else". Cloud **security groups** and network ACLs are firewalls.
- A **web application firewall (WAF)** inspects HTTP requests (layer 7) and blocks attack-like ones: SQL injection patterns, cross-site scripting payloads, known bad bots, too many requests from one source. AWS WAF, Cloudflare WAF, ModSecurity, often with the OWASP Core Rule Set.
- A **DDoS (distributed denial-of-service) attack** floods a service from many machines at once (often a **botnet** of hijacked devices) to make it unavailable.
- **Network segmentation**: databases and internal services in **private subnets** with no route from the internet, reachable only via load balancers or a **bastion host** (hardened jump server).

How it works

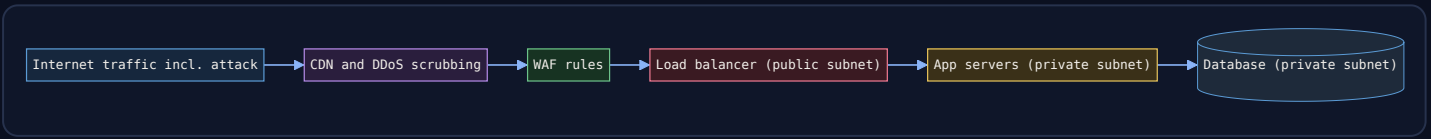
Kinds of DDoS:

- **Volumetric** (layer 3/4): saturate the pipe — UDP floods, or **amplification** attacks sending small spoofed requests to open DNS or memcached servers that reply with much larger responses to the victim.
- **Protocol**: exhaust connection state — a **SYN flood** of millions of half-open TCP handshakes.
- **Application layer** (layer 7): many legitimate-looking HTTP requests to expensive pages such as search.

Mitigation:

1. Absorb with scale: a CDN or DDoS-protection network, often using anycast (Part 2) to spread the attack over hundreds of data centres (Cloudflare, AWS Shield, Google Cloud Armor, Akamai).
2. Filter at the edge: drop malformed packets, use SYN cookies, block known bad sources.
3. Rate limit and challenge suspicious clients (CAPTCHAs, JavaScript challenges).
4. Cache aggressively so most requests never reach the origin, and hide origin IPs so attackers can't bypass the CDN.

5. Autoscale with a cost ceiling, and keep graceful degradation switches ready.



Real-world examples

In October 2023 Google disclosed that in August it had mitigated an attack peaking at about 398 million HTTP requests per second, exploiting an HTTP/2 weakness called "Rapid Reset"; Cloudflare and AWS reported record attacks from the same technique. The 2016 **Mirai** botnet of hacked cameras and routers took down DNS provider Dyn and, with it, many major websites for hours.

TL;DR

Firewalls filter by IP and port, WAFs filter malicious HTTP, private subnets hide internal systems, and DDoS is absorbed by large edge networks, filtering, rate limiting and caching.

Common web vulnerabilities: CORS, CSRF, XSS and SQL injection

Four that come up constantly — the idea plus the fix.

Same-origin policy and CORS. Browsers stop a page from one **origin** (scheme + domain + port, e.g. `https://shop.com`) reading responses from another. **CORS (Cross-Origin Resource Sharing)** is how a server opts in, replying with headers such as `Access-Control-Allow-Origin: https://shop.com`; for non-simple requests the browser first sends an `OPTIONS` **preflight** request. CORS relaxes browser safety; it doesn't protect servers (`curl` ignores it). Mistake: `Access-Control-Allow-Origin: *` on APIs using cookies or private data.

CSRF (cross-site request forgery). An evil site tricks your logged-in browser into sending a request to another site with your cookies attached (a hidden form posting "transfer $1,000" to your bank). Fixes: `SameSite` cookies, anti-CSRF tokens (a random value the real page includes and the server checks), and never changing state on GET.

XSS (cross-site scripting). An attacker's JavaScript runs inside your site in other users' browsers (a comment containing `<script>`), stealing data or acting as the user. Fixes: escape all user content when rendering (React and other modern frameworks do by default), a **Content Security Policy (CSP)** header restricting which scripts may run, and `HttpOnly` cookies scripts can't read.

SQL injection. User input glued into a SQL string lets input like `' OR '1'='1` change the query's meaning and read or delete data. Fix: **parameterised queries** (prepared statements), where input is always data, never SQL.

```
Unsafe: "SELECT * FROM users WHERE email = '" + input + "'"
Safe:   "SELECT * FROM users WHERE email = ?"   with parameter [input]
```

The **OWASP Top 10** is the widely used list of the most critical web application risks; broken access control tops its recent editions.

In the interview

Q: What is the difference between XSS and CSRF?

A: XSS runs the attacker's script inside your site, so it can do anything the user can; CSRF runs no code in your site but makes the victim's browser send a forged request with their cookies. XSS: output escaping and CSP; CSRF: SameSite cookies and CSRF tokens.

TL;DR

CORS lets servers allow cross-origin browser reads; CSRF forges requests with the victim's cookies; XSS injects scripts; SQL injection injects query text. Use SameSite cookies, CSRF tokens, escaping, CSP and parameterised queries.

Least privilege and zero trust

What it is

- **Principle of least privilege**: every user, service and process gets only the permissions its job needs, only for as long as needed (a cleaner gets office keys, not the safe). A hacked image-resizing service should read one bucket, not delete the customer database.
- **Defence in depth**: several independent layers (firewall, WAF, authentication, authorization, encryption, monitoring), so one failed layer isn't fatal.
- **Zero trust**: never trust a request for coming from "inside". The old **perimeter model** was a castle and moat: inside the office network or **VPN** (virtual private network, an encrypted tunnel into it), everything trusted you. Zero trust verifies every request wherever it comes from: authenticate user and device, check authorization, encrypt the connection (often mTLS between services), log it.

Why it matters

Attackers with one foothold (a phished laptop, a vulnerable server) move sideways — **lateral movement** — through a flat, trusting network. Least privilege and zero trust shrink what that foothold can reach.

How it works

Separate **IAM** (identity and access management) roles per service with narrowly scoped policies; no shared admin accounts; just-in-time engineer access that expires; private subnets; mTLS through a service mesh; device health checks; audit logs reviewed for unusual access. Google's **BeyondCorp** (described publicly from 2014) moved employees off a privileged corporate network, so access to internal apps depends on user and device identity, not network location.

Common mistakes

- An all-powerful cloud role for every service "to get it working", never narrowed.
- Long-lived engineer credentials on laptops.
- Treating the VPN as the only security boundary.

TL;DR

Grant the minimum access needed, layer defences, and verify every request regardless of network location.

Cost awareness

Interviewers increasingly ask "and what would this cost?":

- **Big cost drivers**: compute (servers, containers, functions), storage (per GB-month, with cheaper "cold" tiers for rarely read data), managed databases, and **data transfer**. Data in is usually free; data out to the internet (**egress**) costs roughly $0.05–0.09 per GB on AWS depending on volume, and even cross-AZ traffic costs about $0.01 per GB each way. Heavy video or download traffic is mostly a bandwidth bill — one reason CDNs matter.
- **Pricing models**: **on-demand** (per second or hour, flexible), **reserved instances or savings plans** (1–3 year commitment for large discounts), **spot or preemptible instances** (spare capacity at up to ~90% off, reclaimable at short notice; fine for batch jobs).
- **Levers**: autoscale and switch off idle environments at night; right-size oversized instances; cache to cut database and egress costs; tier old data with lifecycle rules; sample logs and traces; compare serverless vs always-on at your real traffic level.
- **Trade-off to mention**: every extra nine, extra region and lower RPO costs real money; engineering time is a cost too, so a pricier managed service may be cheaper overall.
- **FinOps**: making teams see and own their cloud spend, with per-team and per-service tags and regular reviews.

In the interview: name your design's expensive component (egress for a video site, storage for a logging system) and one way to reduce it.

Part 5 glossary

Every Part 5 term is defined in one line in the Master A–Z glossary, which merges the glossaries of all six parts into a single alphabetical list.

Part 6 — Putting it together

Parts 1 to 5 took the system apart; this part puts it back together in five self-contained sections:

1. **A request's journey** follows one click from browser to database and back, naming every concept on the way. If you can tell this story without notes, you understand the fundamentals.
2. **Worked example: a URL shortener** designs a small real system step by step with real numbers — the Part 1 interview method in action.
3. **Decision cheat sheet**: "if you need X, reach for Y".
4. **Top 60 interview questions**: the beginner questions asked most, with short model answers.
5. **Master A–Z glossary**: every term in the module, one line each.

"See *Load balancers* in Part 2" means the section of that name in Part 2; go back to it whenever a one-line reminder isn't enough.

A request's journey

What it is

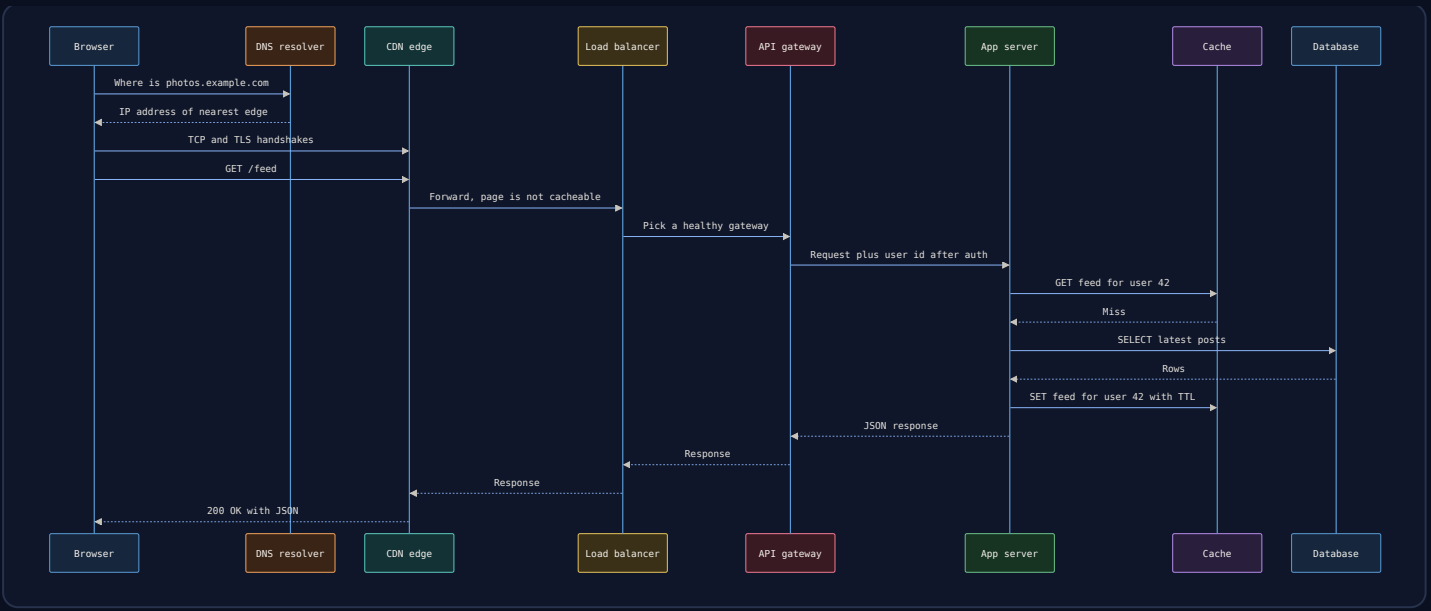
A **request's journey** is the full path one user action takes through a system. You click "Like" and a tenth of a second later the heart turns red; in that time the request may touch fifteen machines and use almost every concept in this module. Following the path is the best way to see how the pieces fit.

Why it matters

- **It is the most common warm-up question**: "What happens when you type a URL into the browser and press Enter?" — asked from internships to senior roles.
- **It shows where time goes**: every hop adds latency, so knowing the hops tells you which to remove (caches, CDNs, connection reuse) when a page is slow.
- **It shows where things break**: asking "what if this hop is down?" one hop at a time is how you find single points of failure.

The map

A user in London opens `https://photos.example.com/feed`, the latest photos from people they follow. Read top to bottom:



How it works

Times are typical for a London user and a well-run site; the order of magnitude is what matters. Each step names the section that explains it.

Step 1 — The browser checks what it already knows (0 ms). It splits the URL into **scheme** (`https` , HTTP over TLS), **host** (`photos.example.com`) and **path** (`/feed`), then checks its **browser cache** for a fresh copy of the page or a remembered IP — the cheapest cache, no network at all. (*Where caches live*, Part 4; *Clients, servers and the request/response cycle*, Part 1.)

Step 2 — DNS turns the name into an IP address (0–50 ms). The OS asks a **recursive resolver** (your ISP's, or 1.1.1.1 / 8.8.8.8); a cached answer within its **TTL** returns in milliseconds, otherwise it asks a **root server**, the `.com` **TLD server** and the **authoritative server**. Since the site uses a CDN, the answer is often a **CNAME** pointing at the CDN, whose DNS returns a nearby edge's IP (**GeoDNS** or **anycast**). (*DNS*, *CDNs and anycast*, Part 2.)

Step 3 — TCP opens a connection (about 1 round trip). A **three-way handshake** (SYN, SYN-ACK, ACK) to the edge on **port** 443: 5–10 ms to a London edge versus ~70 ms to New York — the reason CDNs exist. (*TCP, UDP and QUIC*, Part 2.)

Step 4 — TLS sets up encryption (about 1 more round trip). The edge presents a **certificate** proving it is `photos.example.com` and both sides agree keys: one extra round trip with TLS 1.3 (zero on a repeat visit with **0-RTT** resumption). **HTTP/3** over **QUIC** merges the transport and TLS handshakes into one round trip. (*HTTPS and TLS*, Part 2.)

Step 5 — The browser sends the HTTP request:

```
GET /feed HTTP/2
Host: photos.example.com
Cookie: session=7f3a9c...
Accept: application/json
```

`GET` is the **HTTP method** (read only), **headers** carry metadata, and the **cookie** carries the session token saying who you are — needed because HTTP is **stateless**. (*HTTP*, Part 2.)

Step 6 — The CDN edge serves or forwards (1–5 ms at the edge). At this **point of presence (PoP)**, static files (images, JavaScript, CSS) are almost always a **cache hit** served from London. `/feed` is personal, so it is marked uncacheable and forwarded to the **origin**, often over a fast, already-open connection. Many CDNs also absorb **DDoS** attacks and run a **WAF** here. (*CDNs and anycast*, Part 2; *Network security*, Part 5.)

Step 7 — The load balancer picks a server (under 1 ms). A **layer 7** load balancer reads path and headers, often does **TLS termination**, sends only to targets passing **health checks**, choosing by **round robin** or **least connections**. It runs as a redundant pair or managed service, so it isn't a **single point of failure**. (*Load balancers*, Part 2; *Redundancy, single points of failure and failover*, Part 1.)

Step 8 — The API gateway checks the request (1–5 ms). It **authenticates** (the session or **JWT** says you are user 42), applies **rate limiting** (**429 Too Many Requests** if exceeded) and **routes** `/feed` to the feed service. In smaller systems load balancer and gateway are one product. (*Proxies and API gateways*, Part 2; *Authentication and authorization*, *Rate limiting and throttling*, Part 5.)

Step 9 — An app server runs the business logic. The feed service runs on many **stateless** app servers, usually **containers** on **Kubernetes**: any server can handle any request, so the system **scales horizontally**, with **autoscaling** adding and removing servers. It finds other services via **service discovery**, and every outgoing call has a **timeout**. (*Stateless and stateful services*, *Vertical and horizontal scaling*, Part 1; *Service discovery*, *Autoscaling*, Part 5.)

Step 10 — Check the cache first (about 0.5 ms). Cache-aside: ask a **distributed cache** such as **Redis** for user 42's feed. A **cache hit** returns in well under a millisecond and skips steps 11–12; a well-tuned cache serves 90–99% of reads, the main reason big sites are fast and their databases survive. (*Caching strategies*, Part 4.)

Step 11 — On a miss, query the database (2–20 ms). Reads usually go to a **read replica** (kept in sync from the **primary**, leaving it free for writes), using an **index** rather than scanning every row. If the data is **sharded**, the **shard key** (user ID) picks the shard. A **connection pool** avoids paying for a new connection each time. (*Indexes*, *Replication*, *Partitioning and sharding*, Part 3.)

Step 12 — Fill the cache. Store the result with a **TTL** (say 60 s) so the next request hits. With **replication lag** and a cache up to 60 s old, the user sees **eventual consistency**: new posts within a minute — fine for a feed, not for a bank balance. (*Consistency models*, Part 4.)

Step 13 — The response travels back as **JSON** through gateway, load balancer and edge (which may **compress** it). The browser gets `200 OK`, renders, then fetches the photos — almost all CDN cache hits.

Step 14 — Asynchronous work happens later. Clicking "Like" is a **write** (`POST /photos/991/likes`) to the **primary**, answered at once. Notifying the owner, updating the counter and feeding recommendations needn't finish first, so the service publishes an event to a **message queue** or **event stream** (**Kafka**) for background **workers**. Queues usually give **at-least-once delivery**, so workers must be **idempotent** (no duplicate notifications). (*Message queues*, *Delivery guarantees, idempotency and retries*, Part 4.)

Step 15 — Everything is watched. Every hop writes **logs**, emits **metrics** (rate, errors, duration) and forwards a **trace ID** so engineers can follow this exact request later; an error rate threatening the **SLO** fires an **alert** to the **on-call** engineer. (*Observability*, Part 5.)

Where the time goes

Rough budget for the feed request, first visit, London user, origin in the same region:

Step	Typical time	How to shrink it
DNS lookup	0–50 ms	DNS caching, long TTL
TCP + TLS handshakes	10–20 ms to a nearby edge	CDN edge nearby, HTTP/3, keep-alive
Edge to origin	5–80 ms	Origin in the user's region
Gateway + app logic	5–20 ms	Fewer service hops, parallel calls
Cache hit / DB query	0.5 ms / 5–20 ms	Higher cache hit ratio, indexes
Browser rendering	50–200 ms	Smaller pages, fewer scripts

On a first visit **network round trips** often cost more than server work; repeat visits feel faster because the DNS answer is cached, the connection reused (**keep-alive**), and static files come from the browser cache.

What can go wrong at each hop

Hop	Failure	Usual protection
DNS	Provider outage, wrong record	Two DNS providers, sensible TTLs
CDN	Edge outage, stale content	Multi-CDN, cache invalidation
Load balancer	LB node dies	Redundant pair or managed LB
App server	Crash, bad deploy	Health checks, canary deploys, rollback
Cache	Cache node dies, stampede	Replicas, request coalescing
Database	Primary dies	Replica failover, backups
Queue/worker	Worker crashes mid-job	At-least-once delivery, retries, DLQ
Any call	Slow dependency	Timeouts, circuit breaker, fallback

Common mistakes

- **Forgetting DNS and TLS**, jumping from "browser" to "server"; handshakes are often the slowest part of a first visit.
- **Making everything synchronous**: if a like waits for the notification email, a slow email provider makes liking slow. Push side effects to a queue.
- **Treating the cache as the source of truth**: the database is the truth; the cache can be lost or stale any moment.
- **Forgetting every box is redundant**: one "load balancer" box means at least two machines.

In the interview

Q: What happens when you type a URL and press Enter?

A: Browser cache; DNS (resolver, root, TLD, authoritative) gives an IP, usually a nearby CDN edge; TCP connection and TLS handshake; HTTP GET. The edge serves cached static content or forwards to the origin, where a load balancer picks a healthy server, a gateway authenticates and rate-limits, and the app checks a cache before the database. The response returns the same way; the browser renders and fetches images and scripts, mostly from the CDN.

Q: Where would you put a cache in this path?

A: Wherever it pays: browser cache for static files, CDN for public content, Redis in front of the database for hot data, plus the database's buffer cache. Closer to the user saves more latency but is harder to invalidate (*Where caches live*, Part 4).

Q: Which of these steps would you make asynchronous?

A: Anything the user needn't see before the response: notifications, emails, counters, analytics, search indexing, thumbnails. Write the essential data, publish an event, return; workers do the rest with retries.

TL;DR

One click goes browser cache → DNS → TCP/TLS → CDN → load balancer → API gateway → stateless app server → cache → database, and back; side effects go through a queue to workers. Each hop costs time and can break, so good designs cache it, duplicate it or take it off the user's critical path.

Worked example: designing a URL shortener

What it is

A **URL shortener** turns `https://www.example.com/blog/2026/10/a-very-long-article-title?ref=newsletter` into `https://sho.rt/aZ3kQ9b`; opening the short link **redirects** the browser to the long one (Bitly, TinyURL). A favourite first problem: finishable in 45 minutes, yet it needs estimates, an API, a data model, ID

generation, caching, sharding and a read-heavy design. We follow the method of *How to approach a system design interview* (Part 1).

Step 1 — Requirements

Functional requirements:

- Given a long URL, create a short URL.
- Opening the short URL redirects to the long URL.
- Optional: a user-chosen **custom alias** (`sho.rt/my-party`).
- Optional: links **expire** after a date.
- Optional: **click counts** for the link's owner.

Non-functional requirements:

- **Very high availability** for redirects — if they fail, every link ever shared is broken. Target 99.99% (~52 minutes of downtime a year; *Availability and the nines*, Part 1).
- **Low latency** for redirects: p99 under 50 ms at the server.
- **Short links not guessable in order**, so nobody can walk everyone's links by counting.
- **Durability**: a created link is never lost.
- **Read-heavy**: assume 100 redirects per creation.

Out of scope: user accounts, a full analytics dashboard, link previews.

Step 2 — Estimates (back of the envelope)

Assume **100 million new links per month**; a month ≈ 2.6 million seconds (30 × 86,400 = 2,592,000).

Quantity	Calculation	Result
Writes per second (average)	100 M ÷ 2.6 M s	about 40 /s
Reads per second (average)	40 × 100	about 4,000 /s
Peak reads (3× average)	4,000 × 3	about 12,000 /s
Links over 5 years	100 M × 12 × 5	6 billion
Storage per link	URL + code + metadata	about 500 bytes
Total storage over 5 years	6 B × 500 B	about 3 TB
Read bandwidth	4,000 × 500 B	about 2 MB/s

What they tell us:

- **40 writes/s is tiny** — one ordinary database handles it.
- **12,000 peak reads/s is moderate** — one database might cope with no headroom; a cache will take most of it.
- **3 TB** exceeds one cheap machine's memory but suits a few database machines; sharding is a "later" concern.
- **Code length**: with 62 symbols per character (a–z, A–Z, 0–9: **base62**), 6 characters give $62^6 \approx 56.8$ billion codes, 7 give $62^7 \approx 3.5$ trillion. We need 6 billion, so 7 gives over 500× headroom.

Cache sizing: clicks are very uneven — a small minority of links gets most traffic (often quoted as the **80/20 rule**). A day brings ~350 million redirects (4,000 × 86,400) over far fewer distinct links, say 50 million; caching the busiest fifth, 10 million entries × ~500 bytes ≈ 5 GB, fits one cache node's memory (run a few for redundancy).

Step 3 — API

Two endpoints (*API styles*, *API design basics*, Part 2):

```
POST /api/v1/links
Content-Type: application/json
Idempotency-Key: 5d1c...

{"long_url": "https://www.example.com/blog/...", "custom_alias": null, "expires_at": null}

-> 201 Created
{"short_url": "https://sho.rt/aZ3kQ9b", "code": "aZ3kQ9b"}
```

```
GET /aZ3kQ9b

-> 302 Found
Location: https://www.example.com/blog/...
```

- `v1` in the path is **API versioning**, so the API can change without breaking old clients.
- The **idempotency key** makes a timed-out, retried create return the same link instead of two (*Delivery guarantees*, Part 4).
- Creation is **rate limited** per API key or IP (say 100 per hour) against spammers, replying `429` over the limit (*Rate limiting*, Part 5).
- A missing code returns `404 Not Found`; an expired one `410 Gone` (or 404).

301 or 302? Both are **redirect** status codes. **301 Moved Permanently** lets the browser remember the redirect and skip our server next time: less load, but repeat clicks go uncounted and we can't change or expire the link for browsers that cached it. **302 Found** (temporary) makes the browser ask every time. 302 if analytics or expiry matter, 301 if pure speed and cost do — say this trade-off out loud.

Step 4 — Data model

One main table. PostgreSQL works; so does a key-value store like DynamoDB or Cassandra, since every read is "one row by code" (*SQL vs NoSQL and BASE*, Part 3).

Column	Type	Notes
`code`	varchar(16)	Primary key, e.g. `aZ3kQ9b`
`long_url`	text	Up to about 2 KB
`owner_id`	bigint	Nullable for anonymous links
`created_at`	timestamp	
`expires_at`	timestamp	Nullable

The primary key on `code` gives an **index** for free: a redirect is one index lookup. Clicks are **not** counted by updating this row per redirect (that turns every read into a write); they go to a separate analytics pipeline (deep dives).

Step 5 — Generating the short code

The heart of the problem:

Approach	How	Pros	Cons
Hash the URL	MD5/SHA-256 of the long URL, take 7 base62 chars	Same URL gets same code	Collisions, must check and retry
Random code	7 random base62 chars	Simple, not guessable	Must check for collisions
Counter + base62	Global counter 1, 2, 3 … encoded in base62	No collisions	Counter is a bottleneck, guessable
Key generation service	Pre-make unused codes, hand out in batches	Fast, no collisions	One more service to run

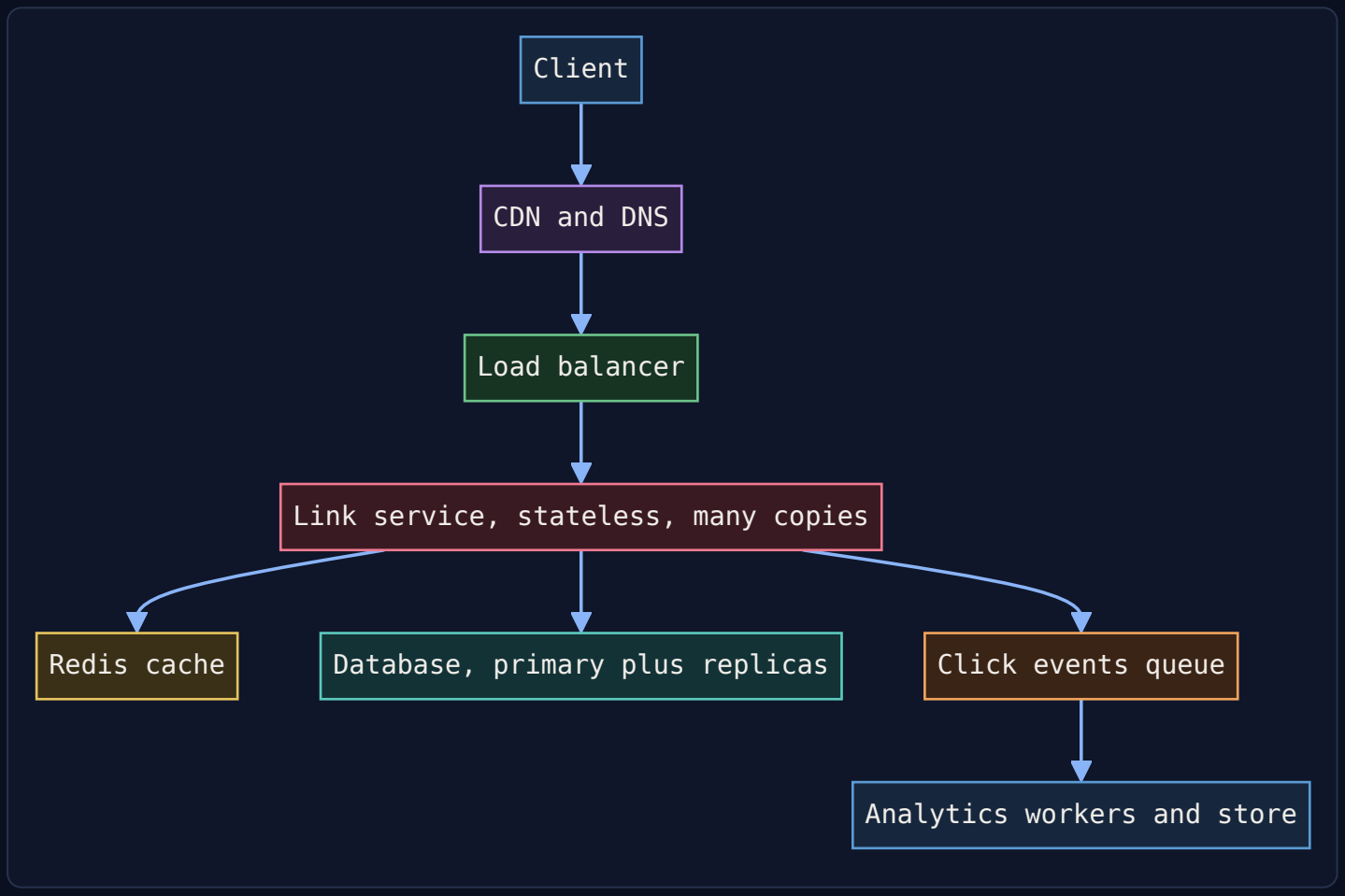
(A **hash collision** is two inputs producing the same short hash; a **key generation service (KGS)** pre-creates unused keys.) A good beginner answer is **counter ranges plus base62, with shuffling**:

1. A small coordination store (a database row, or ZooKeeper/etcd) hands each app server a **counter range** (A: 1,000,000–1,999,999; B: 2,000,000–2,999,999).
2. Each server counts through its range in memory, with no network call per link, requesting a new range when exhausted. A crashed server's leftover range is simply skipped — trillions to spare.
3. The number is **scrambled** with a reversible mixing function (or a block cipher) so consecutive links don't get consecutive codes — meeting "not guessable in order".
4. The scrambled number is written in **base62**: a 7-character code.

A **Snowflake-style ID** (64 bits from timestamp + machine ID + sequence) needs no coordination at all, but encodes to 11 base62 characters — longer than we want (*Unique ID generation*, Part 4).

Custom aliases skip the generator: insert the alias as primary key; if it exists, reply `409 Conflict`. The database's uniqueness check makes two people racing for one alias safe.

Step 6 — High-level design



Redirect path (read, 99% of traffic):

1. `GET /aZ3kQ9b` arrives via DNS, CDN and load balancer at any link-service instance.
2. It looks up `aZ3kQ9b` in **Redis** (cache-aside); a hit (most of the time) yields the long URL in under a millisecond.
3. On a miss it reads a **database replica** and writes the result to Redis with a TTL of, say, 24 hours.
4. It checks `expires_at`, publishes a small "click" event to the queue without waiting, and returns `302` with `Location`.

Create path (write, 1% of traffic):

1. `POST /api/v1/links` passes the gateway's authentication and rate limiter.
2. The service validates the URL (well-formed, not on a malware blocklist), takes the next number from its range, scrambles and encodes it.

3. It inserts into the **primary** and returns `201`. No cache write; the first redirect fills the cache.

Step 7 — Deep dives

Caching. At 90%+ hit ratio the 12,000 peak reads/s become ~1,200 database reads/s — easy for two read replicas. Links almost never change, so invalidation is rare: delete the key when a link is deleted or expires. Stop a viral link's **cache stampede** on expiry by letting one request refill it (**request coalescing**) (*Cache invalidation, stale data and stampedes*, Part 4).

Even faster: cache at the edge. For links without analytics, let the CDN cache the redirect response itself for a few minutes, so most clicks never reach our servers.

Scaling the database. 3 TB over five years fits a primary with a few replicas at first. Later, **shard by hash of** `code`: every redirect knows its code, so every query hits exactly one shard — the ideal shard key. **Consistent hashing** makes adding shards move only a small share of keys (*Partitioning and sharding*, Part 3).

Availability. Every tier runs at least two copies across **availability zones**. The stateless link service loses dead instances via load-balancer health checks; the database fails over to a replica. The redirect path needs only reads, so it keeps working from replicas and cache while the primary is replaced — **graceful degradation**: creation pauses for a minute, existing links keep working.

Analytics. A counter per click would put 12,000 writes/s on the database. Instead each click is a **Kafka** event; workers batch them and update counts once a minute in an analytics store (a column store such as ClickHouse). Counts lag slightly — **eventual consistency**, acceptable to owners. A crashed worker's events are re-read (**at-least-once**); de-duplicating by event ID keeps counts honest.

Abuse. Spammers love short links: rate-limit creation, check long URLs against malware lists, allow links to be disabled.

Trade-offs we made

- **302 over 301:** more load, but accurate analytics and expirable links.
- **Counter ranges over hashing:** no collision checks, for a small coordination store.
- **Eventual consistency for click counts:** much cheaper writes, slightly stale numbers.
- **Cache with long TTL:** fast and cheap, safe because links are effectively immutable.

Common mistakes

- **Skipping the estimates**, leaving 7 characters, cache size and sharding unjustified.
- **One auto-increment column in one database as the generator**, called scalable. Fine at 40 writes/s, but say it is a bottleneck and single point of failure, and how you'd fix it.
- **First 7 characters of an MD5 hash, ignoring collisions** — with billions of links collisions are certain; detect and retry.
- **Updating a click counter on the redirect path**, making the cheapest, commonest operation expensive.
- **Over-engineering**: no 50 shards and three regions for 40 writes/s. Start simple, then show how it grows.

In the interview

Q: How long should the short code be?

A: Depends on the link count: base62 7 characters ≈ 3.5 trillion codes for ~6 billion links in five years — plenty of headroom. 6 characters (~57 billion) would work with less margin.

Q: How would you handle one link going viral with millions of clicks a minute?

A: A hot but read-only key, so caching solves it: Redis, a small in-process cache on each app server for the very hottest keys, and briefly CDN-cached redirects. Request coalescing prevents a stampede on expiry.

Q: SQL or NoSQL?

A: Either — access is a simple key lookup at a low write rate. Start with PostgreSQL for familiarity and strong uniqueness guarantees; a key-value store like DynamoDB scales out more simply at many terabytes.

TL;DR

The service maps a code to a URL and is read ~100 times per write: generate a unique 7-character base62 code (counter ranges plus scrambling), store code → URL in a database, put Redis and optionally the CDN in front for the 100:1 reads, return a 302, and push click analytics through a queue.

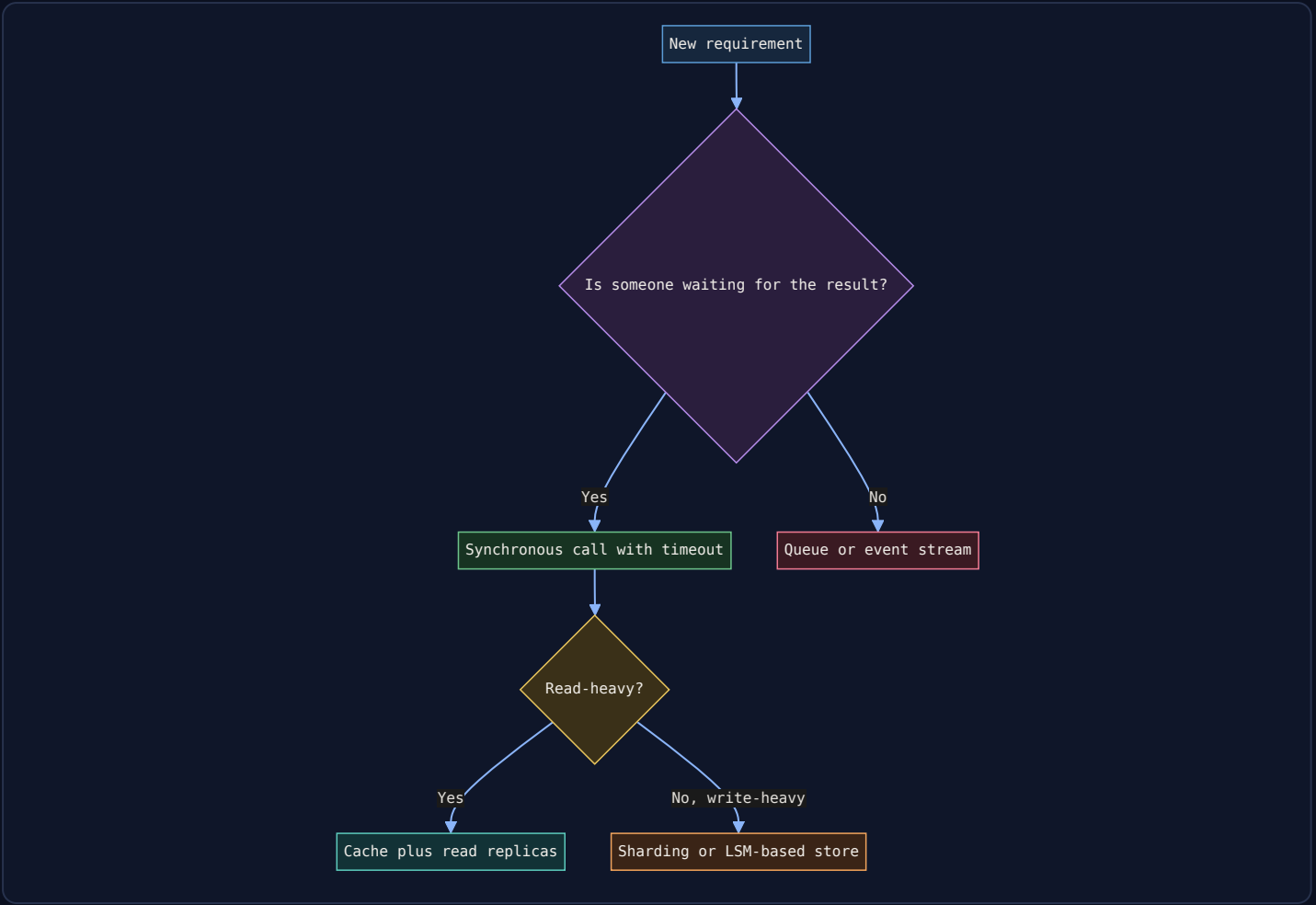
Decision cheat sheet

What it is

Common design needs and the tool or pattern usually reached for — a starting point, not a rule. Name the tool, then say **why** and **what it costs**; the "Watch out for" column earns the marks.

How to use it

Read the left column as "the interviewer just told me…" ("read-heavy" → cache and read replicas), then check the right column before committing.



The table

If you need…	Reach for…	Watch out for…
Faster reads of the same data	Cache (Redis, Memcached), cache-aside	Stale data, invalidation, stampedes
Fast static files worldwide	CDN	Invalidation delay, cost per GB
Lower latency for far-away users	CDN, edge, multi-region deployment	Data consistency across regions
More capacity for stateless servers	Horizontal scaling behind a load balancer	Must move session state out of servers
A quick capacity bump, small system	Vertical scaling (bigger machine)	Hard ceiling, still one machine
To spread traffic across servers	Load balancer (L7 for HTTP, L4 for raw TCP)	LB must be redundant itself
Traffic that grows and shrinks daily	Autoscaling	Slow start-up, scale on the right metric
One public entry point for many services	API gateway	Can become a bottleneck or a "god" layer
Strong consistency and transactions	Relational database (PostgreSQL, MySQL)	Harder to scale writes beyond one primary
Flexible, nested records	Document store (MongoDB)	Joins and multi-document transactions are weaker
Simple lookups by key at huge scale	Key-value store (DynamoDB, Redis)	Few query patterns besides by key
Very high write rate, time-ordered data	Wide-column or LSM-based store (Cassandra)	Design tables around queries, eventual consistency
Metrics and sensor readings over time	Time-series database (Prometheus, InfluxDB)	Retention and downsampling planning
Relationship queries (friends of friends)	Graph database (Neo4j)	Harder to shard
Full-text search, fuzzy matching	Search engine (Elasticsearch, OpenSearch)	It is a copy: keep it in sync, not the source of truth
Similarity search on embeddings	Vector database or vector index (pgvector)	Approximate results, memory hungry
Store large files (images, video, backups)	Object storage (Amazon S3)	Not for frequent small updates
More read capacity on the database	Read replicas	Replication lag, read-your-writes issues
More write capacity or data than one machine	Sharding with a good shard key	Hot spots, cross-shard queries, resharding
Add or remove nodes without moving most data	Consistent hashing with virtual nodes	Still some data movement, needs balancing
Fast queries on a column	Index (B-tree)	Slower writes, extra storage
Do slow work without making the user wait	Message queue + workers (SQS, RabbitMQ)	Duplicates, need idempotency and a DLQ
Many services react to one event	Pub/sub or event stream (Kafka, SNS)	Ordering, schema changes
Replay past events, high-throughput logs	Event streaming (Kafka)	Partitions limit parallelism, ops effort

If you need…	Reach for…	Watch out for…
Safe retries of a payment or order	Idempotency key	Storing keys, choosing an expiry
Absorb traffic spikes	Queue as a buffer, backpressure	Queue growth hides overload
Protect a service from abusive clients	Rate limiting (token bucket) returning 429	Fair limits per user, distributed counters
Stop a failing dependency dragging you down	Timeouts + circuit breaker + fallback	Picking thresholds, testing the open state
Retry transient failures	Retries with exponential backoff and jitter	Retry storms, only retry idempotent calls
Keep one failure from sinking everything	Bulkheads (separate pools)	Unused reserved capacity
Survive a whole data-centre outage	Multi-AZ, then multi-region, failover	Cost, replication lag, split brain
A consistent operation across services	Saga with compensating actions	Complex failure handling, interim states
Atomic commit across two databases	Two-phase commit (2PC)	Blocking if the coordinator fails, slow
Only one worker doing a job at a time	Distributed lock (with fencing tokens) or leader election	Lock expiry while still working
Nodes agreeing on a value or leader	Consensus (Raft via etcd/ZooKeeper)	Needs a majority, latency of writes
Unique IDs without coordination	UUID or Snowflake-style IDs	UUIDv4 is random (poor index locality), clock issues
"Have we seen this before?" with little memory	Bloom filter	False positives possible
Count distinct users cheaply	HyperLogLog	Approximate (about 1% error)
Find nearby places	Geohash or quadtree, geo index	Edge cases at cell borders
Real-time updates to the browser, server to client	Server-Sent Events	One direction only
Two-way real-time (chat, games)	WebSockets	Stateful connections, harder load balancing
Notify another company's system	Webhooks	Retries, signatures, duplicates
Fast internal service-to-service calls	gRPC with Protobuf	Browser support needs a proxy
Clients choosing exactly which fields they get	GraphQL	Caching, expensive queries
Large data analysis overnight	Batch processing (Spark), data warehouse	Results are hours old
Results within seconds of events	Stream processing (Flink, Kafka Streams)	Harder to get right, state management
Know what is happening in production	Metrics, logs, traces, alerts on SLOs	Alert fatigue, cost of logs
Release safely	Canary or blue-green deploys, feature flags	Database changes still need care

If you need…	Reach for…	Watch out for…
Users log in once for many apps	SSO with OpenID Connect	Central identity provider is critical
Let an app act for a user on another service	OAuth 2.0	Token storage and scopes
Stateless auth across services	Signed tokens (JWT)	Hard to revoke before expiry
Store passwords	bcrypt, scrypt or Argon2 (slow, salted hashing)	Never encrypt or plain-hash passwords
Protect secrets like API keys	Secrets manager (Vault, AWS Secrets Manager)	Rotation, access audit

TL;DR

Every "reach for" brings a "watch out for". Naming both in one sentence — "I'd add read replicas, accepting some replication lag" — is the habit that interviewers look for.

Top 60 interview questions

What it is

The beginner questions asked most in design interviews and phone screens, each with a short model answer — what you'd say before the follow-up. The bracket names the section with the full explanation; use the list as a self-test.

Foundations and performance

Q1: What is system design?

A: Deciding how the parts — servers, databases, caches, queues, networks — fit together to meet requirements for features, scale, speed and reliability; trade-offs, not one right answer. (*What system design is*, Part 1.)

Q2: Latency vs throughput?

A: Latency is how long one request takes (120 ms); throughput is how many complete per unit time (5,000/s). They can diverge: a cargo ship has huge throughput and high latency. (*Latency, throughput and bandwidth*, Part 1.)

Q3: Why measure p99 latency instead of the average?

A: The average hides the slow few; p99 is what the unluckiest 1% feel. A page making 100 backend calls almost always hits one p99-slow call, so the tail matters at scale. (*Percentiles, tail latency and why averages lie*, Part 1.)

Q4: What is Little's law and why is it useful?

A: L = λ × W: requests in the system = arrival rate × time inside. 1,000 requests/s at 0.2 s each means ~200 in flight — the threads or connections you need. (*Little's law*, Part 1.)

Q5: How do you estimate QPS?

A: DAU × actions per user (10 M × 10 = 100 M/day), ÷ ~10^5 seconds ≈ 1,000 QPS average, × 2–3 for peak. (*Back-of-the-envelope estimation*, Part 1.)

Q6: What does "four nines" mean?

A: 99.99%: about 52.6 minutes down a year, ~4.4 a month. Each extra nine is 10× less downtime and usually far more cost. (*Availability and the nines*, Part 1.)

Q7: SLA vs SLO vs SLI?

A: SLI: a measurement (% successful requests). SLO: the internal target (99.9% over 30 days). SLA: the customer contract, usually looser, with refunds or penalties. (*SLA, SLO, SLI and error budgets*, Part 1.)

Q8: What is an error budget?

A: The failure the SLO allows — 0.1% of requests (~43 minutes a month) for 99.9%. Ship fast while it lasts; focus on reliability once spent. (*SLA, SLO, SLI and error budgets*, Part 1.)

Q9: Vertical vs horizontal scaling?

A: Vertical: a bigger machine — simple, but a ceiling and still a single point of failure. Horizontal: more machines — grows further with redundancy, but needs load balancing and stateless services. (*Vertical and horizontal scaling*, Part 1.)

Q10: What is a stateless service and why does it help scaling?

A: It keeps no per-user data in memory between requests (sessions in Redis, a database, or a client token), so any instance serves any request and instances can be added or replaced freely. (*Stateless and stateful services*, Part 1.)

Q11: What is a single point of failure and how do you remove one?

A: A component whose failure takes everything down (one database, one load balancer). Add redundant copies with health checks and automatic failover. (*Redundancy, single points of failure and failover*, Part 1.)

Q12: Reliability vs availability vs durability?

A: Availability: answers now. Reliability: keeps doing the right thing over time. Durability: saved data survives (S3 is designed for eleven nines annual object durability). (*Reliability, availability and durability*, Part 1.)

Q13: How do you approach a design question?

A: Requirements (functional and non-functional), estimates, API and data model, a simple high-level design, then deep dives, trade-offs and failures — thinking aloud, simple first. (*How to approach a system design interview*, Part 1.)

Networking and communication

Q14: How does DNS work?

A: A recursive resolver walks root → TLD → authoritative server to get the IP, caching every answer for its TTL, so most lookups take milliseconds. (*DNS — the internet's phone book*, Part 2.)

Q15: TCP vs UDP?

A: TCP: handshake, complete in-order delivery, congestion control — web, APIs, databases. UDP: independent packets, no guarantees, no waiting for losses — video calls, games, DNS, and the base of QUIC. (*TCP, UDP and QUIC*, Part 2.)

Q16: HTTP/1.1 vs HTTP/2 vs HTTP/3?

A: 1.1: one request at a time per connection. 2: many multiplexed requests on one TCP connection with compressed headers, but a lost packet stalls all (head-of-line blocking). 3: over QUIC/UDP, independent streams and faster setup. (*HTTP — the language of the web*, Part 2.)

Q17: What does HTTPS add to HTTP?

A: TLS encryption so nobody on the network can read or change traffic, plus a CA-signed certificate proving you reached the real site. (*HTTPS and TLS*, Part 2.)

Q18: Which HTTP methods are idempotent and why does it matter?

A: GET, PUT, DELETE, HEAD, OPTIONS; not POST (usually not PATCH). Clients retry after network failures, and only idempotent requests are safe to repeat — others need an idempotency key. (*HTTP — the language of the web*, Part 2.)

Q19: REST vs gRPC vs GraphQL?

A: REST/JSON: the simple default for public APIs. gRPC (Protobuf over HTTP/2): fast, typed, for internal service calls. GraphQL: clients request exactly the fields they need in one call, for mobile and complex UIs. (*API styles — REST, RPC and gRPC, GraphQL*, Part 2.)

Q20: How do you push real-time updates to a browser?

A: Short polling, long polling, Server-Sent Events (one-way stream: live scores, feeds) or WebSockets (two-way: chat, games). (*Real-time communication*, Part 2.)

Q21: What is a load balancer and why?

A: It spreads requests across a server pool, enabling horizontal scaling, sending traffic only to healthy servers, and allowing invisible deploys. (*Load balancers*, Part 2.)

Q22: L4 vs L7 load balancer?

A: L4 routes by IP and port without reading content — fast, any protocol (AWS NLB). L7 understands HTTP — routes by path, header or cookie, terminates TLS, retries — at higher cost (AWS ALB, NGINX, Envoy). (*Load balancers*, Part 2.)

Q23: Name some load-balancing algorithms.

A: Round robin, weighted round robin, least connections, IP hash or consistent hashing (same key → same server), power of two choices. (*Load balancers*, Part 2.)

Q24: What are sticky sessions and why avoid them?

A: The LB sends a user to the same server (usually by cookie) so in-memory sessions work — but load gets uneven and sessions die with the server. Prefer stateless servers with a shared session store. (*Load balancers*, Part 2.)

Q25: What is a CDN and how does it help?

A: Servers in many cities caching content near users: milliseconds instead of cross-ocean round trips, origin offload, DDoS absorption. (*CDNs and anycast*, Part 2.)

Q26: Forward vs reverse proxy?

A: A forward proxy acts for clients (a company web filter); a reverse proxy acts for servers — load balancing, TLS termination, caching, protection (NGINX). (*Proxies and API gateways*, Part 2.)

Q27: What is an API gateway?

A: A reverse proxy as the single entry point to many services, handling authentication, rate limiting, routing, transformation and logging once. (*Proxies and API gateways*, Part 2.)

Q28: Offset or cursor pagination?

A: Offset (`?page=5` or `OFFSET 100`) is simple but slow on deep pages and skips or repeats items as data changes; cursor (`?after=abc`) stays fast and stable — use it for feeds and big lists. (*API design basics*, Part 2.)

Data storage

Q29: SQL or NoSQL?

A: SQL for transactions, joins and a defined schema (orders, payments); a NoSQL family for a specific shape at scale — key-value, document, wide-column, graph. Many systems use both. (*SQL vs NoSQL and BASE*, Part 3.)

Q30: What does ACID mean?

A: Atomicity (all or nothing), Consistency (valid state to valid state), Isolation (no half-finished work visible), Durability (committed data survives crashes). (*Transactions and ACID*, Part 3.)

Q31: What is an index and what does it cost?

A: A structure, usually a B-tree, that finds rows without scanning the table; faster reads, but slower writes and extra storage. (*Indexes*, Part 3.)

Q32: Replication vs sharding?

A: Replication copies all data to several machines (availability, read capacity); sharding splits data so each machine holds a part (write capacity, storage). Large systems replicate each shard. (*Replication*, *Partitioning and sharding*, Part 3.)

Q33: Synchronous vs asynchronous replication?

A: Synchronous waits for a replica's confirmation — no loss on failover, slower writes. Asynchronous confirms at once — faster, but recent writes can be lost and replicas lag. (*Replication*, Part 3.)

Q34: What is replication lag and how does it affect users?

A: The delay before a primary's write reaches a replica (milliseconds, sometimes seconds); a user may not see their own comment. Read own recent writes from the primary or track the write position. (*Replication*, Part 3; *Consistency models*, Part 4.)

Q35: How do you choose a shard key?

A: One that spreads data and traffic evenly and appears in most queries (user ID); not a timestamp, which sends all new writes to one hot shard. (*Partitioning and sharding*, Part 3.)

Q36: What is consistent hashing?

A: Servers and keys on one hash ring, each key owned by the next server clockwise, so adding or removing a server moves ~1/N of keys instead of almost all; virtual nodes even the spread. (*Consistent hashing and virtual nodes*, Part 3.)

Q37: Normalization or denormalization?

A: Normalized: each fact once, joined by keys — consistent but needs joins. Denormalized: data copied into its read shape — fast reads, more storage and sync work. Read-heavy systems denormalize hot paths. (*Normalization and denormalization*, Part 3.)

Q38: When would you use object storage?

A: Large, unstructured, write-once read-many files (images, video, backups, logs, data-lake files): cheap, extremely durable, unlimited; keep only the URL and metadata in the database. (*Storage types: block, file and object*, Part 3.)

Q39: OLTP vs OLAP?

A: OLTP: many small fast reads and writes for the live app (placing an order). OLAP: big analytical queries over history, usually in a column-oriented warehouse. (*OLTP vs OLAP*, Part 3.)

Q40: What is the N+1 query problem?

A: One query for a list, then one per item for related data — often hidden by an ORM. Fix with a join, an `IN (...)` batch or eager loading. (*Connection pooling, ORMs and N+1 queries*, Part 3.)

Caching, async and consistency

Q41: Explain cache-aside.

A: Check the cache; on a miss read the database, store with a TTL, return. On writes update the database and delete the key. The most common pattern. (*Caching strategies*, Part 4.)

Q42: Write-through vs write-back?

A: Write-through writes cache and database together — always fresh, both costs per write. Write-back (write-behind) writes the cache and flushes later in batches — fast, but data lost if the cache fails first. (*Caching strategies*, Part 4.)

Q43: What is LRU eviction?

A: Least Recently Used: when full, evict the item longest unread, since recent use predicts reuse. Alternatives: LFU, TTL expiry. (*Eviction policies*, Part 4.)

Q44: What is a cache stampede and how do you prevent it?

A: A hot entry expires and thousands of simultaneous misses hit the database. Use request coalescing, a refill lock, TTL jitter, or early refresh of hot entries. (*Cache invalidation, stale data and stampedes*, Part 4.)

Q45: Why use a message queue?

A: The sender drops a message and returns; workers process at their own pace — smoothing spikes, surviving worker downtime, scaling producers and consumers separately. (*Message queues*, Part 4.)

Q46: Queue vs publish/subscribe?

A: Queue: each message to exactly one consumer (jobs). Pub/sub: every subscriber gets a copy (events many services care about). Kafka does both via consumer groups. (*Message queues*, Part 4.)

Q47: At-most-once, at-least-once, exactly-once?

A: At-most-once may lose, never duplicates; at-least-once never loses, may duplicate (the default); exactly-once in practice means at-least-once plus idempotent processing ("effectively-once"). (*Delivery guarantees, idempotency and retries*, Part 4.)

Q48: What is idempotency and how do you implement it?

A: Doing it twice equals doing it once. Give each request a unique idempotency key, record processed keys, and return the stored result on repeats. (*Delivery guarantees, idempotency and retries*, Part 4.)

Q49: What is a dead-letter queue?

A: Where messages go after failing a set number of times, so one bad message doesn't block or retry forever and can be inspected and replayed. (*Delivery guarantees, idempotency and retries*, Part 4.)

Q50: Strong vs eventual consistency?

A: Strong: every read sees the latest write, as if one copy — simpler, but costs latency and availability. Eventual: replicas briefly disagree but converge — faster, more available. (*Consistency models*, Part 4.)

Q51: Explain the CAP theorem.

A: During a network partition a system must refuse some requests to stay consistent (CP) or answer with possibly stale data (AP); without a partition it can have both. (*CAP theorem and PACELC*, Part 4.)

Q52: What is a quorum?

A: Write to W and read from R of N replicas; R + W > N guarantees reads overlap the latest write (commonly N = 3, W = 2, R = 2). (*Quorums*, Part 4.)

Q53: Why do distributed systems need consensus like Raft?

A: Nodes must agree on one value (the leader, the log order) despite crashes and delays; Raft and Paxos do it safely while a majority is reachable, and etcd or ZooKeeper provide it ready-made. (*Consensus, leader election and distributed locks*, Part 4.)

Q54: 2PC or saga across services?

A: 2PC (prepare, then commit) gives atomicity but blocks if the coordinator fails and couples services. A saga chains local transactions with compensating actions — the usual microservices choice, accepting temporary inconsistency. (*Distributed transactions*, Part 4.)

Architecture, reliability and security

Q55: Monolith or microservices?

A: A monolith is simple to build, test and deploy — the default for small teams. Microservices let many teams deploy independently, at the cost of network calls, distributed failures and operations. Many start with a modular monolith. (*Architectural styles*, Part 5.)

Q56: What is a circuit breaker?

A: It counts failures to a dependency; past a threshold it opens and fails fast, after a pause goes half-open to test a few calls, closing if they succeed — so one slow service can't tie up all threads. (*Circuit breakers, bulkheads and fallbacks*, Part 5.)

Q57: Why add jitter to exponential backoff?

A: Backoff (1, 2, 4, 8 s) gives a struggling service room; without jitter, clients that failed together retry together in waves. Randomness spreads them. (*Timeouts, retries, exponential backoff and jitter*, Part 5.)

Q58: How does a token bucket rate limiter work?

A: Up to B tokens, refilled at R per second; each request takes one, an empty bucket returns 429 — bursts up to B, average rate R. (*Rate limiting and throttling*, Part 5.)

Q59: Authentication vs authorization?

A: Authentication: who are you (password, token check) — always first. Authorization: what may you do (may user 42 delete this post?). (*Authentication and authorization*, Part 5.)

Q60: What are RPO and RTO?

A: RPO: how much data (in time) you can lose — RPO 5 minutes means backups or replication at most 5 minutes behind. RTO: how long you can be down. Lower values cost more. (*Disaster recovery*, Part 5.)

> **TL;DR**
>
> If you can answer these 60 in your own words, each with a number or an example, you have the fundamentals an interviewer expects from a beginner.

Master A–Z glossary

What it is

Every term defined in Parts 1 to 6, in one alphabetical list with a one-line meaning (the part glossaries point here). Closely related terms share a row (for example "LRU / LFU / FIFO"); the parts themselves have the full explanations. Terms starting with a digit come first.

Term	Meaning in one line
0-RTT (resumption)	TLS 1.3 / QUIC feature letting a returning client send data in its very first message
301 Moved Permanently	Redirect the browser may cache forever
302 Found	Temporary redirect the browser re-checks each time
3-2-1 rule	Three copies, two media types, one off-site
401 / 403	Not authenticated / authenticated but not permitted
409 Conflict	Status code for a request that clashes with existing data
410 Gone	Status code for a resource that existed but was removed
429 Too Many Requests	Status code for a client that exceeded its rate limit
5-tuple	Protocol, source IP/port, destination IP/port — identifies a connection
80/20 rule	Rough rule that 20% of items get 80% of the traffic

A

Term	Meaning in one line
A / AAAA record	DNS records that map a name to an IPv4 / IPv6 address
ABAC	Attribute-based access control: permissions decided by attributes of user, resource and context
Access pattern	How an application reads and writes its data
Access token / refresh token	Short-lived API credential / long-lived credential to get new access tokens
ACID	Transaction guarantees: atomicity (all or nothing), consistency (rules kept), isolation (no interference), durability (survives crashes)
ACK	Acknowledgement that data was received
Acknowledgement (ack)	Consumer's signal that a message is done and can be removed
ACL	List of who may access a specific resource
Active-active	All redundant copies serve traffic at once and absorb a failed copy's share
Active-passive	One copy serves traffic while a standby waits to take over
Active / passive health check	Load balancer probes backends on a schedule / ejects them after errors in real traffic (outlier detection)
Alert fatigue	Ignoring alerts because there are too many noisy ones
Alt-Svc	Header advertising that a server also speaks HTTP/3
Amdahl's law	Parallel speed-up is limited by the part of the work that must run serially
Amplification attack	Spoofed small requests causing large replies to hit a victim
ANN / HNSW	Approximate nearest-neighbour search / a popular graph-based ANN index
Anomaly	An incorrect result caused by concurrent transactions
Anti-entropy	Background process comparing and syncing replicas
Anycast	Many servers announce the same IP address and the network routes each user to the nearest one
Apex domain	The bare registered domain without a subdomain, such as `example.com`
API	Application programming interface: the defined way one program asks another to do something
API gateway	Single entry point for APIs that handles auth, rate limits and routing
API key	Secret string that identifies a calling application
API versioning	Running old and new API contracts side by side for breaking changes
App server (application server)	Program that runs business logic and answers client requests
Architectural style	The overall way code is split into running programs and how they communicate
Asynchronous processing	Doing work later without the caller waiting for it
Asynchronous replication	Leader confirms before followers have the write
At-least-once / at-most-once delivery	Messages never lost but maybe duplicated / never duplicated but maybe lost
Atomicity	All-or-nothing execution of a transaction
Authentication (AuthN)	Proving who someone is

Term	Meaning in one line
Authoritative name server	DNS server that holds the real records for a domain
Authorization (AuthZ)	Deciding what an identified user may do
Authorization code flow	OAuth flow exchanging a one-time code for tokens
Auto-increment / sequence	Database-generated increasing integer IDs
Autoscaling	Automatically adding or removing capacity based on load
Availability	Fraction of time (or requests) the system is up and working
Availability zone (AZ)	One or more isolated data centres with independent power and networking inside a cloud region

B

Term	Meaning in one line
Back-of-the-envelope estimate	Quick rough calculation to find the order of magnitude of load or storage
Backpressure	A slow consumer signalling producers to slow down
Backup	A separate copy kept to recover from loss or mistakes
Backup and restore / pilot light / warm standby	DR strategies from cheapest and slowest to faster and costlier
Bandwidth	Maximum data a link or device can carry per second
Bandwidth-delay product	Bandwidth times RTT, the data in flight needed to fill a link
BASE	Basically available, soft state, eventually consistent: the NoSQL alternative to ACID
Base62	Writing numbers using 62 symbols (a–z, A–Z, 0–9) to make short codes
Bastion host	Hardened server used as the only entry point to private machines
Batching	Grouping many small operations into one to raise throughput at some latency cost
Batch processing	Processing a large set of stored data in one job, often on a schedule
bcrypt / scrypt / Argon2id	Deliberately slow, salted password-hashing algorithms
Bearer token	Token that grants access to whoever holds it
BFF	Backend for Frontend, a gateway tailored to one client type
BGP	Protocol networks use to announce which IP ranges they can reach
Binding (RPC)	How a client finds the server address for an interface: config, directory or service discovery
Bit / byte	A bit is a single 0 or 1; a byte is 8 bits
Blast radius	How much of the system or user base a failure can affect
Blob	A large chunk of unstructured binary data
Block storage	Raw fixed-size disk blocks attached to one machine, like a virtual hard drive
Bloom filter	Compact structure that says "definitely not present" or "probably present"
Blue-green deployment	Run old (blue) and new (green) versions side by side and switch traffic at once
Botnet	Network of hijacked devices controlled by an attacker
Bottleneck	The resource or component that limits overall capacity
Branching factor	Number of children per B-tree page
Browser cache	Copies of pages, files and DNS answers kept by the browser
B-tree	A shallow balanced tree of sorted pages, the default index
Bucket	A top-level container in object storage
Buffer cache / buffer pool	Database's own in-memory copy of recently used disk pages
Bulkhead	Separate resource pools so one failing part cannot exhaust everything

Term	Meaning in one line
Burn rate	How fast the error budget is being spent; 1 means exactly on budget for the SLO window

C

Term	Meaning in one line
Cache	Fast, usually in-memory store holding copies of frequently used data
Cache-aside (lazy loading)	App checks cache, on miss loads from the database and fills the cache; on write deletes the key
Cache avalanche	Mass expiry or cache loss sending all traffic to the DB
Cache busting	Changing a file name or URL when content changes so old copies aren't used
Cache-Control	Header telling caches whether and how long to store a response
Cache hit / miss	Requested data was / was not found in the cache; hits ÷ all requests is the hit ratio
Cache invalidation	Removing or updating cached copies when the source changes
Cache key	Values (host, path, some headers) identifying a cached object
Cache penetration	Requests for non-existent keys always hitting the DB
Cache stampede / thundering herd	Many requests miss the same expired key at once and overload the database
Call semantics	What a reply or timeout guarantees: maybe, at-least-once, at-most-once, exactly-once
Canary deployment (release)	Sending a small share of traffic to a new version first
CAP theorem	During a network partition a system must choose consistency or availability
Cardinality	Number of distinct label combinations in metrics
Carrier-grade NAT	NAT run by an ISP, so many customers share one public IP
Cascading failure	One component's failure spreading to the components that depend on it
Causal consistency	Causes are always seen before their effects
CDN	Content Delivery Network; servers worldwide that serve static files close to users
Cell-based architecture	Splitting users across independent copies of the stack to limit blast radius
Certificate Authority	Trusted organisation that signs certificates
Certificate (TLS)	Document signed by a trusted certificate authority (CA) proving a server owns a domain name
Chain of trust	Certificate signed by intermediate, signed by a root the browser trusts
Change data capture (CDC)	Streaming a database's changes from its log to other systems
Chaos engineering	Deliberately injecting failures to prove resilience
Checkpoint	A point up to which data files reflect the log
Choreography / orchestration	Saga steps triggered by events / by a central controller
CI/CD	Continuous integration and continuous delivery/deployment: automated build, test and release
CIDR	Notation like `10.0.0.0/8` giving an address range by its fixed leading bits
Circuit breaker	Stops calling a failing dependency for a while so it can recover
Claim check	Store a big payload elsewhere and send only its reference
Client	Program that sends requests (browser, app, or another server)

Term	Meaning in one line
Client credentials flow	OAuth flow for machine-to-machine access
Client-side load balancing	Each client picks a backend from a list of healthy instances
Client-side / server-side discovery	Caller queries the registry itself / a router or load balancer does it for the caller
Clock drift	A clock gradually running fast or slow
Clock skew	Difference between two machines' clocks
Closed / Open / Half-open	Circuit breaker states: normal, failing fast, testing recovery
Cloud load balancer	Provider-managed LB service such as AWS ALB or NLB
Cluster	The set of machines and control plane managed by Kubernetes
Cluster autoscaler	Adds or removes Kubernetes nodes as pods need room
Clustered index	An index whose order is the physical order of the rows
Clustering key	The key that sorts rows inside one partition in a wide-column store
CNAME record	DNS record that makes one name an alias of another
Cold start	Extra delay when a serverless function runs on a freshly created instance
Cold / warm / hot standby	Spares that need starting, are partly ready, or are ready instantly
Collection	A group of documents in a document store
Column-oriented storage (column store)	Stores each column's values together, great for analytics scans
Commit / rollback	Make a transaction permanent / discard all its changes
Committed offset	Saved position a consumer group has processed up to
Compaction	Merging SSTables and dropping old versions
Compare-and-set	Writing only if the current value still equals what was read
Compensating transaction (action)	Saga step that semantically undoes an earlier step
Composite index	An index over several columns in order
Concurrency	Making progress on many tasks by interleaving them
ConfigMap / Secret	Kubernetes objects for configuration and sensitive values
Configuration drift	Real infrastructure diverging from its definition
Congestion collapse	Overload where throughput falls towards zero due to wasted work
Congestion control	Sender slowing down to avoid overloading the network
Connection draining	Letting a server finish in-flight requests before removing it
Connection ID	QUIC identifier for a connection independent of IPs and ports
Connection migration	QUIC keeping a connection alive across network changes
Connection pool	A set of reusable open database connections
Consensus	Getting nodes to agree on a value despite failures
Consistency (ACID)	A transaction leaves data obeying all declared rules
Consistency (distributed)	Whether all readers see the same, latest data
Consistency model	Promise about what reads can return after writes

Term	Meaning in one line
Consistent hashing	Hash ring where adding or removing a node moves only about 1/N of keys
Consumer	Program that reads messages from a queue or stream
Consumer group	Consumers sharing a topic's partitions for one application
Consumer lag	How far a consumer is behind the newest message
Container	Lightweight packaged app with its dependencies, sharing the host OS kernel
Container image	Read-only package of an app and its dependencies
Container registry	Storage service for container images
Content Security Policy (CSP)	Header restricting which scripts and resources a page may load
Continuous delivery	Keeping every passing build releasable on demand
Continuous deployment	Automatically releasing every passing build to production
Continuous integration (CI)	Automatically building and testing every merged change
Conway's law	Systems mirror the communication structure of the teams that build them
Cookie	Small piece of data a site stores in the browser and receives with each request
Cool-down period	Wait after a scaling action before another, to avoid flapping
Coordinated omission	Load-test flaw that under-reports slow responses
Coordination service	System like etcd, ZooKeeper or Consul for consensus-backed coordination
Coordinator / participant	Runner of 2PC / database taking part in it
Correlated failure	Several "independent" parts failing together from a shared cause
CORS	Cross-origin resource sharing: headers by which a server lets pages from other origins read its responses
Counter / gauge / histogram	Metric that only rises / goes up and down / records a distribution
Counter range	Block of numbers handed to one server for ID generation
Count-min sketch	Compact structure that estimates how often items appear
Covering index	An index containing every column a query needs
CP / AP	Refuses requests to stay consistent / answers with maybe-stale data
CPU-bound	Limited by computation speed
CPU cache	Tiny, very fast memory inside the processor
CQRS	Command query responsibility segregation: separate models for writes and reads
CRDT	A data type whose concurrent updates always merge to the same result
CSRF	Cross-site request forgery: tricking a logged-in browser into sending a request
CUBIC / BBR	Common TCP congestion-control algorithms
Cursor pagination	Paging from an opaque marker of the last item seen, fast and stable
Custom alias	User-chosen short code

D

Term	Meaning in one line
Dashboard / alert	Page of graphs showing key metrics / automated notification when a metric rule breaks
Database / DBMS	A program that stores data safely, finds it quickly and coordinates concurrent users
Database per service	Each microservice owns its private data store
Datagram	Self-contained UDP message
Data lake	Cheap storage of raw data in many formats for later analysis
DataLoader	A pattern that batches per-item lookups into one query
Data plane / control plane	The proxies that carry traffic / the component that configures them
Data warehouse	Structured store optimized for analytical queries
DAU / MAU	Daily / Monthly Active Users
DDL	SQL statements that change the schema, such as CREATE TABLE
DDoS	Distributed denial of service: flooding a system from many machines
Dead-letter queue (DLQ)	Side queue for messages that failed too many times
Deadline propagation	Passing the remaining time budget along to downstream calls
Deadlock	Tasks waiting forever for locks held by each other
Decision cheat sheet	Table mapping common needs to standard tools and their costs
Decoupling	Letting components work without knowing about each other directly
Defence in depth	Multiple independent layers of security
Denormalization	Duplicating data deliberately to speed up reads
Deployment (Kubernetes)	Object managing replicas and rolling updates of stateless pods
Deployment strategy	Method for replacing a running version with a new one
Deregistration delay	AWS name for the connection-draining timeout
Diffie–Hellman	Method for two sides to agree on a secret key without sending it
Directory-based sharding	A lookup service maps keys to shards
Direct server return	Backend replies straight to the client, bypassing the LB
Dirty read	Reading another transaction's uncommitted data
Dirty write	Overwriting another transaction's uncommitted data
Disaster recovery (DR)	Plan for restoring service and data after a major failure
Distributed cache	Shared cache cluster reached over the network (Redis, Memcached)
Distributed lock	Lock that coordinates processes across machines
Distributed monolith	Microservices so coupled they must change and deploy together
Distributed SQL	Relational databases spread over many machines, such as Spanner or CockroachDB
Distributed system	Many computers working together over a network as one system
Distributed transaction	Change spanning several databases or services, all or nothing
DNS	System that translates domain names into IP addresses
Docker	Popular tool for building, sharing and running container images
Dockerfile	Text recipe for building a container image

Term	Meaning in one line
Document database (store)	NoSQL store of self-contained JSON-like documents (e.g. MongoDB)
Domain name	Human-readable hierarchical name such as `www.example.com`
DORA metrics	Deployment frequency, lead time, change failure rate, time to restore
Dual write problem	Updating DB and broker separately, risking disagreement
Durability	Saved data is not lost, even after crashes or disk failures

E

Term	Meaning in one line
Eager loading	Fetching related data up front in few queries
ECMP	Routers spreading flows across several equal-cost paths or machines
Edge computing	Running code on CDN edge servers
Edge server	CDN cache server close to users
EDNS Client Subnet	DNS extension passing part of the user's IP for better geo answers
Egress	Data leaving a cloud network, usually billed
Elasticity	Automatically adding and removing resources to match demand
Election timeout	Time without heartbeats before a follower starts an election
Embedding	A list of numbers representing an item so similar items are close
Encapsulation	Each layer wrapping data with its own header
Encryption	Reversible scrambling of data with a key
Encryption in transit / at rest	Protecting data moving over a network (TLS) / stored on disks and backups
Enterprise service bus (ESB)	Middleware that routes and transforms messages between SOA services
Envelope encryption	Encrypting data keys with a master key held in a KMS
Ephemeral port	Temporary source port the OS assigns to an outgoing connection
Error budget	Amount of failure an SLO allows (1 minus the SLO)
ETag	Version tag of a resource, used to answer 304 Not Modified
etcd	Consistent key-value store holding Kubernetes cluster state
ETL / ELT	Transform before loading / load then transform
Event-driven architecture	Services reacting to published events instead of direct calls
Event loop	Single thread handling many requests by never blocking on I/O
Event sourcing	Storing every change as an event and rebuilding state from them
Event streaming / log-based broker	Broker keeping an ordered, replayable log of events (Kafka)
Event time / processing time	When an event happened / when it was processed
Eventual consistency	Replicas may differ briefly but converge if writes stop
Event vs command	Fact that happened vs request that something happen
Eviction policy	Rule for what to remove when the cache is full
EWMA	Exponentially weighted moving average, a smoothed recent value
Exactly-once / effectively-once	Each message affects the result once; in practice at-least-once delivery plus idempotency / de-duplication
Exchange	RabbitMQ component that routes messages to queues
Expand and contract	Backward-compatible schema change done in several releases
EXPLAIN	A command showing the plan the database will use
Exponential backoff	Doubling the wait after each failed retry (100, 200, 400 ms …), up to a cap

F

Term	Meaning in one line
FaaS	Function as a Service: upload functions, provider runs them per event
Fail open / fail closed	Allowing / denying traffic when a safety component itself fails; an LB fails open by sending to all backends when all look unhealthy
Failover	Switching to a standby when the active component fails
Failure	The whole system stops providing its service
Failure detection	Deciding whether a node is probably dead
Failure domain	A group of parts that can fail together (rack, zone, region)
Failure model	Assumption about how a faulty part behaves: fail-stop, crash, crash-recovery, omission, timing, Byzantine
Fallacies of distributed computing	False assumptions such as "the network is reliable" and "latency is zero"
Fallback	Alternative response used when the normal path fails
False positive (failure)	Declaring a slow but alive node dead
Fan-out	One request triggering many parallel downstream requests
Fault	One component misbehaving (crash, slowness, wrong output)
Fault tolerance	Continuing to work despite faults in some components
Feature flag	Runtime switch that turns code paths on or off for chosen users
Federation	Splitting databases by function (users DB, orders DB)
Fencing	Forcibly stopping an old leader from acting
Fencing token	Increasing number that lets storage reject requests from a stale lock holder
File storage	Shared files and folders over the network (e.g. NFS)
FinOps	Practice of making teams see and manage their cloud costs
Firewall	Filter that allows or blocks network traffic by rules
Fixed window counter	Rate limiter counting requests per calendar time window
Flapping	Node repeatedly marked dead and alive
Flow control	Receiver limiting how fast the sender sends (receive window)
Foreign key	A column holding another table's primary key, linking the two
Forward proxy	Proxy acting for clients toward the internet
Forward secrecy	Stolen long-term keys cannot decrypt past sessions
Four golden signals	Latency, traffic, errors and saturation
fsync	A system call forcing written data onto the physical device
Full / incremental backup	Copy of everything / only changes since the last backup
Full table scan	Reading every row to answer a query
Functional requirement	A feature the system must provide

G

Term	Meaning in one line
Game day	Planned exercise practising failures and recovery
GeoDNS	DNS that answers differently depending on the user's location
Geohash	Encoding of a location as a short string where shared prefixes mean nearby
Geo sharding	Sharding by user location
Geospatial index	Index for searching by location
Goodput	Useful work completed, as opposed to raw throughput
Gossip protocol	Nodes spreading state to random peers each round
Graceful degradation	Turning off non-essential features under stress to keep the core working
Graph database	Store of nodes and edges for relationship queries (e.g. Neo4j)
GraphQL	Query language letting clients request exactly the fields they need
Group commit	Flushing many commits with one fsync
gRPC	High-performance RPC framework using Protobuf over HTTP/2
GSLB (global server load balancing)	Spreading traffic across regions or data centres

H

Term	Meaning in one line
Hardware load balancer	Dedicated appliance such as F5 BIG-IP
Hash collision	Two different inputs producing the same short hash
Hash function	A function turning any input into a fixed-size, random-looking number
Hash index	An index mapping keys to locations via a hash map
Hashing	One-way function that turns data into a fixed-size fingerprint
Hash ring	The hash output range arranged as a circle
Hash sharding	The shard is chosen from a hash of the key
HDD	Hard disk drive; spinning magnetic disk with slow seeks
Header	Key–value metadata on a request or response
Head-of-line blocking	One delayed item holding up everything queued behind it
Health check	Test that decides whether a backend should receive traffic
Heartbeat (ping/pong)	Periodic small "I am alive" message keeping a connection alive and detecting failure
Heavy hitters	The most frequent items in a stream
Hedged request	Sending a duplicate request to another replica if the first is slow
Herding	Many balancers sending traffic to the same seemingly best server at once
Hinted handoff	Holding writes for a down node and delivering them later
Histogram	Counts of measurements per value bucket, used to compute percentiles
Horizontal scaling (scale out)	Adding more machines to share the load
Horizontal / vertical partitioning	Splitting a table by rows / by columns
Host	The domain-name part of a URL, such as photos.example.com
Hot key	One key so popular it overloads the node that holds it
Hot spot	A shard receiving far more load than others
HPACK / QPACK	Header compression for HTTP/2 / HTTP/3
HTTP	Hypertext transfer protocol: request/response protocol of the web
HTTP/1.1	Text protocol, one request at a time per connection
HTTP/2	Binary protocol multiplexing many streams over one TCP connection
HTTP/3	HTTP over QUIC, removing TCP head-of-line blocking
HTTP method	Verb such as GET, POST, PUT, PATCH, DELETE
HttpOnly / Secure / SameSite	Cookie flags: no script access / HTTPS only / limit cross-site sending
HTTPS	HTTP inside a TLS-encrypted connection
Hybrid logical clock	Timestamp mixing wall time with a logical counter
HyperLogLog	Distinct-count estimator using kilobytes of memory
Hypervisor	Software that runs and isolates virtual machines on a physical host

I

Term	Meaning in one line
Idempotency (idempotent)	Doing an operation twice has the same effect as doing it once
Idempotency key	Client-supplied ID letting a server detect and ignore retries
Idempotent producer	Producer whose duplicate sends the broker discards
ID token	JWT stating who the user is
Immutable backup	Backup that cannot be altered or deleted for a set period
Immutable infrastructure	Replacing servers with new images instead of modifying them
Incident commander	Person coordinating the response to an incident
Index	An extra structure that finds rows by value without a full scan
Index-only scan	Answering a query from the index without touching the table
Infrastructure as code (IaC)	Defining servers and networks in versioned files (e.g. Terraform)
Ingress	Kubernetes rules routing external HTTP traffic to services
In-process cache	Cache inside the application's own memory
Inverted index	Map from each word to the documents containing it, used by search engines
I/O-bound	Limited by waiting for disk or network
IOPS	Input/output operations per second a storage device can perform
IP address	Numeric address of a device on a network: 32-bit IPv4 or 128-bit IPv6
IP hash	Load-balancing by hashing the client IP to pick a server
IPv4 / IPv6	32-bit addresses (~4.3 billion) / 128-bit addresses (practically unlimited)
Isolation	Concurrent transactions do not see each other's unfinished work
Isolation level	How strictly a database isolates transactions (read committed, serializable …)

J

Term	Meaning in one line
Jitter	Random variation, added to retry delays or TTLs to avoid synchronized spikes
Join	Combining rows from two tables that match on a key
JWT	Signed, self-contained JSON token: header.payload.signature

K

Term	Meaning in one line
Kafka	Distributed event streaming platform built on a partitioned log
KB / MB / GB / TB / PB	Data sizes, each about 1,000 times the previous
Keep-alive	Reusing one TCP connection for many HTTP requests
Key generation service (KGS)	Service that pre-creates unused unique keys
Key salting	Adding a suffix to spread one hot key over several shards
Key-value store	Database that stores values looked up by a key
Kill switch	Flag that instantly disables a feature
KMS	Key management service that stores and uses encryption keys
Kubelet	Agent on each node that runs its pods
Kubernetes (K8s)	Container orchestrator that keeps declared state running across machines

L

Term	Meaning in one line
L3 / L4 / L7	Network (IP), transport (TCP/UDP, ports), application (HTTP) layers
L4 load balancer	Balances TCP/UDP connections without reading content
L7 load balancer	Parses HTTP and routes or balances per request
Lakehouse	A data lake with transactional table formats on top
Lambda / Kappa architecture	Batch plus speed layer / stream-only with replay
Lamport clock	Logical counter that orders events without real time
Last-write-wins (LWW)	Conflict rule keeping the write with the highest timestamp
Latency	Time for one operation or message to complete or arrive
Latency-based routing	DNS steering to the region with the lowest measured latency
Latency budget	A latency target split into allowances for each hop of a request
Lateral movement	Attacker spreading from one compromised system to others
Leader election	Choosing one node to coordinate the others
Leader / follower / candidate	Raft node roles
Leader / follower (leader-follower replication)	One node (leader, or primary) accepts writes and copies them to followers (replicas)
Leaderless replication	Any replica accepts writes, with quorums for reads and writes
Leaky bucket	Rate limiter that processes requests at a fixed outflow rate
Lease	Lock or role that automatically expires after a time
Lease (cache)	Token granting one client the right to refill a key
Least connections	Send each request to the server with the fewest active connections
Least privilege	Give each user or service only the access it needs
Least response time	Send to the server answering fastest recently
Leftmost prefix	The rule that a composite index helps queries on its leading columns
Linearizability / strong consistency	System behaves as if there were one copy, in real-time order
Linearizability / strong consistency	Strongest single-object consistency: every read sees the latest write, as if there were one copy with instant operations in real-time order
Little's law	Items in system = arrival rate x time in system (L = lambda x W)
Liveness / readiness probe	Checks whether the process is running (else restart it) / ready to serve traffic (else send it none)
Load balancer	Component that spreads requests across servers and skips dead ones
Load levelling	Using a queue to smooth spikes into a steady processing rate
Load parameters	The numbers that describe load (QPS, data size, users)
Load shedding	Rejecting some requests on purpose to protect the system under overload
Local vs global index	Index per shard / index sharded by the indexed value
Log	Timestamped record of a single event
Logical clock	Counter-based ordering of events without real time

Term	Meaning in one line
Logical / physical backup	Exported rows or SQL / copied raw data files
Long polling	Server holding a request open until data is available
Lost update	Two read-modify-write cycles overwrite each other, losing one change
LRU / LFU / FIFO	Evict least recently used / least frequently used / oldest inserted
LSM-tree	A write-optimised structure of memtable plus merged sorted files

M

Term	Meaning in one line
MAC address	Hardware address of a network card used on the local link
Maintainability	How easily a system is operated, understood and changed (operability, simplicity, evolvability)
Majority quorum	More than half the nodes
MapReduce	Batch model: map data into key-value pairs, then reduce each key's values
Marshalling	Converting call arguments and results to bytes for the network, and back
Materialized view	A stored, refreshed result of a query
Mean (average)	Sum divided by count; hides outliers in skewed data
Median (p50)	Middle value: half are faster, half slower
Memcached	Simple, fast, multi-threaded in-memory key-value cache
Memtable	The in-memory sorted buffer of an LSM-tree
Merkle tree	Tree of hashes used to find differences between data sets quickly
Message key	Value hashed to pick a partition, keeping related events ordered
Message queue	Buffer where jobs wait to be processed later by workers
Metric	Numeric measurement tracked over time
MFA	Multi-factor authentication using two or more kinds of proof
Microservices	Many small, independently deployed services, each owning one capability and its data
Migration	A versioned script that changes the schema
Mitigation	Action that stops user impact before the root cause is fixed
Modular monolith	One deployable program with strictly separated internal modules
Monitoring	Watching known metrics to check health
Monolith	Whole application built and deployed as one program
Monotonic reads	A user never sees data go backwards in time
Monotonic writes	A user's writes apply in the order made
MTBF	Mean time between failures
mTLS	Mutual TLS: both client and server prove identity with certificates
MTTD	Mean time to detect a failure
MTTF	Mean time to failure for parts that are replaced, not repaired
MTTR	Mean time to recovery (or repair)
Multi-AZ	Running across several zones in one region
Multi-leader replication	Several nodes accept writes and replicate to each other
Multiplexing	Interleaving many independent streams over one connection
Multi-region	Running in several geographic cloud regions
Mutation / subscription	GraphQL write / live-update operation
MVCC	Keeping multiple row versions so readers and writers do not block each other

Term	Meaning in one line
MX record	Mail servers for a domain, with priorities

N

Term	Meaning in one line
N+1 query problem	One query for a list plus one per item for related data
N+1 redundancy	Running one more unit than peak load needs
Namespaces and cgroups	Linux features that isolate what a process sees and limit what it uses
NAT	Network address translation: many private IPs share one public IP
NAT gateway	Cloud service giving private servers outbound internet access
Negative caching	Caching "not found" results for a short time
Network partition	Network failure splitting healthy nodes into groups that cannot talk
NFS / SMB	Network file-sharing protocols for Linux / Windows
Nines	Shorthand for availability levels (three nines = 99.9%)
Node	One machine (or process) in a cluster or distributed system
Non-functional requirement	A quality the system must have (latency, availability, scale)
Non-repeatable read	Re-reading a row in one transaction and getting a different value
Normal forms (1NF, 2NF, 3NF)	Progressive rules removing repeated and misplaced data
Normalization	Organizing tables so each fact is stored once
NoSQL	Non-relational databases: key-value, document, wide-column, graph …
N, R, W	Replica count, read quorum, write quorum
NS record	DNS record naming a domain's authoritative servers
NTP	Protocol for synchronising clocks with time servers
NVMe	Fast interface for SSDs with very high IOPS

O

Term	Meaning in one line
OAuth 2.0	Standard for granting an app limited access to a user's resources
Object storage	Service storing large files as objects under keys (e.g. Amazon S3)
Observability	Ability to understand a system's internal state from its outputs
Offset (messaging)	Position number of a message in a partition; a consumer's position in it
Offset pagination	Paging with limit and offset, simple but slow and unstable deep in a list
OLAP	Online analytical processing: large queries for analysis
OLTP	Online transaction processing: many small live reads and writes
On-call	Engineer responsible for responding to alerts in a given period
On-demand / reserved / spot instances	Pay as you go / committed discount / cheap but reclaimable
OpenAPI	Standard format for describing HTTP APIs
OpenID Connect (OIDC)	Identity layer on OAuth that adds an ID token for login
OpenTelemetry	Vendor-neutral standard and libraries for telemetry
Optimistic concurrency control	Detecting conflicting writes with a version check and retrying
Order of magnitude	Rough size as a power of ten
Origin	Your own servers or storage holding the master copy of content behind a CDN
Origin shield	Middle cache tier that collapses edge misses before the origin
ORM	Object-relational mapper: library mapping objects to database rows
OSI model	Seven-layer reference model for networking
Over-fetching / under-fetching	Getting more data than needed / needing several calls for one screen
OWASP Top 10	Widely used list of top web application security risks

P

Term	Meaning in one line
p50 / p95 / p99 / p999	Latency below which 50 / 95 / 99 / 99.9% of requests fall
PACELC	Extends CAP: else (no partition), choose latency or consistency
Packet	A small chunk of data (about 1,500 bytes max on Ethernet) routed independently
Page	A fixed-size block of storage, typically 4 to 16 KB
Page cache	Operating system's RAM cache of recently read file data
Page / ticket	Urgent alert waking someone / non-urgent work item
Pagination	Returning large results in pages, by offset (skip N rows) or by cursor (after the last item seen)
Parallelism	Executing several tasks at the same instant on multiple cores or machines
Parameterised query	Query where inputs are passed separately as data
Parquet / ORC	Columnar file formats for analytics
Partial index	An index over only rows matching a condition
Partitioning	Splitting a dataset into smaller pieces
Partition (stream)	A subset of data stored separately; in Kafka, an ordered sub-log of a topic and the unit of parallelism
Path	The part of a URL after the host, such as /feed
Paxos	First widely known proven consensus algorithm
Peak factor	Ratio of peak traffic to average traffic
Percentile	Value below which a given percentage of measurements fall
Performance	How fast the system is for a single request at current load
Perimeter security	Trusting everything inside the network boundary
PgBouncer	A popular connection-pooling proxy for PostgreSQL
Phantom read	A repeated query returning new rows inserted by another transaction
Phi accrual detector	Failure detector giving a suspicion level, not yes/no
Pipeline	Automated sequence of build, test and deploy steps
PKCE	Proof Key for Code Exchange: one-time secret proving the app that started an OAuth flow is the one finishing it
Pod	Smallest Kubernetes unit: one or more containers sharing network and storage
Point-in-time recovery	Restoring to any moment by replaying logs over a backup
Point-to-point / competing consumers	Each message processed by one of many workers
Poison message	A message that fails every time it's processed
Polyglot persistence	Using several database types in one system
Pool / target group	The set of backend servers behind a load balancer
PoP (point of presence)	CDN or network site in a city containing many edge servers
Port	Number (0–65535) selecting which program on a machine receives data
Port exhaustion	Running out of source ports for new outgoing connections

Term	Meaning in one line
Postmortem	Blameless written review of an incident
Power of two choices	Pick two random backends and use the less loaded one
Preflight request	Browser OPTIONS request checking CORS permission first
Pre-signed URL	A time-limited link allowing one direct upload or download
Primary key	A column whose unique value identifies each row
Private IP	Address valid only inside a local network (10/8, 172.16/12, 192.168/16)
Private subnet	Network segment with no direct route from the internet
Producer / consumer / broker	Sender / processor / storage middleman of messages
Projection / read model	Read-optimised view built from events
Propagation delay	Time for a signal to travel the physical distance
Protocol	Agreed rules for how two programs exchange messages
Protocol Buffers (Protobuf)	Compact binary serialisation format with a schema
Proxy	Middleman that forwards requests on someone's behalf
PROXY protocol	Header an L4 proxy prepends to pass the original client address
Public IP	Globally unique address reachable from the internet
Public-key cryptography	Key pair where the public key verifies or encrypts for the private key
Publish/subscribe (pub/sub)	Each message delivered to every subscriber of a topic
Pull CDN	CDN fetching content from origin on first request
Purge / invalidation	Removing content from CDN caches before its TTL ends
Push CDN	You upload content to the CDN in advance

Q

Term	Meaning in one line
QPS / RPS	Queries / requests per second
Quadtree	Tree splitting map regions into four as density grows
Query planner (optimizer)	The part of a database that picks the cheapest way to run a query
Queue depth	Number of messages waiting in a queue
Queueing delay	Time spent waiting behind other work
QUIC	UDP-based transport with built-in TLS, used by HTTP/3
Quorum	Minimum number of replicas that must take part in an operation

R

Term	Meaning in one line
Race condition	Bug where the result depends on the timing of concurrent tasks
Raft	Understandable consensus algorithm with leaders and terms
RAID	Combining disks so one failure loses no data
RAM	Main working memory of a computer; fast but volatile
Random (LB)	Pick a backend at random
Range query	A query for values between two bounds
Range sharding	Each shard owns a continuous range of keys
Rate-limit headers	Headers telling clients their quota, remaining calls and reset time
Rate limiting	Capping how many requests a client may make per time window, rejecting the excess (HTTP 429)
RBAC	Role-based access control: permissions attached to roles
Reactive / scheduled / predictive scaling	Scaling on current metrics / at preset times / ahead of forecast demand
Read committed	Isolation level preventing dirty reads only
Read repair	Fixing stale replicas noticed during a read
Read replica	Copy of a database used to serve reads
Read-through	Cache loads from the DB itself on a miss
Read / write	Fetching data without changing it / inserting, updating or deleting data
Read / write amplification	Extra disk reads per lookup / extra rewrites per byte written
Read/write ratio	How many reads happen per write
Read-your-writes (read-your-own-writes)	A user always sees their own recent writes
ReBAC	Permissions derived from relationships between users and objects
Rebalance	Reassigning partitions when group membership changes
Rebalancing / resharding	Moving data between shards as the cluster changes
Reconciliation loop	Controllers continually driving actual state towards desired state
Recreate deployment	Stop old version, start new, with downtime
Recursive resolver	DNS server that finds the full answer for clients and caches it
Recursive vs iterative query	"Give me the final answer" vs "point me to who knows"
Redirect	HTTP response telling the browser to go to another URL
Redis	In-memory key-value server with rich data types, often used as a cache
RED method	Rate, errors, duration: metrics for each service
Redrive	Moving DLQ messages back to the main queue
Redundancy	Extra copies of components so one failure is survivable
Referential integrity	The guarantee that foreign keys point to rows that exist

Term	Meaning in one line
Refresh-ahead	Cache reloads hot entries before they expire
Region	Geographic cloud area containing several availability zones
Relational database	A database storing data in tables of rows and columns with a fixed schema
Relevance ranking	Ordering search results by how well they match
Reliability	Working correctly over time, even when things go wrong
Rendezvous hashing	Picking the server with the highest score for each key
Repeatable read	Isolation level also preventing non-repeatable reads
Replica	One copy of the data
Replication	Keeping copies of data on several machines
Replication lag	How far a follower is behind the leader
Request coalescing	Letting one request refill a cache entry while others wait for it
Request coalescing / single flight	Letting only one request rebuild a missing key
Request / response	A client's message asking for something and the server's reply
Request's journey	The full path one user action takes through every component and back
Resharding	Changing the number of shards and moving data
Resolver	Server function that fetches one GraphQL field
Resource	A noun the API exposes, addressed by a URL
Resource owner / client / authorization server / resource server	The four OAuth roles
Response time	Total time the client waits, including network, queueing and service
REST	API style of resources (URLs) manipulated with HTTP verbs
Retention	How long a log keeps messages
Retransmission	Resending data whose acknowledgement did not arrive
Retry	Repeating a failed request in case the fault was temporary
Retry-After	Header telling the client how long to wait before retrying
Retry budget	Cap on the share of traffic that may be retries
Retry storm	Retries multiplying load on an already struggling service
Reverse proxy	Proxy acting for servers, receiving client traffic in front of them
Rollback	Returning to the previous version after a bad deploy
Roll forward	Fixing a bad release by shipping a new fix instead of reverting
Rolling deployment	Replacing instances in small batches
Root hash	Top hash of a Merkle tree, summarising all data
Root name server	Top of the DNS tree, refers resolvers to TLD servers
Round robin	Send requests to each server in turn

Term	Meaning in one line
Round-robin DNS	Returning several IPs in rotating order to spread load
Round trip time (RTT)	Time for a message to go to a destination and back
Router	Device that forwards packets toward their destination IP
Row store	Storage keeping each row's columns together
RPC	Remote procedure call: calling a function on another machine
RPO	Recovery point objective: maximum acceptable data loss
RTO	Recovery time objective: maximum acceptable downtime
R-tree / S2 / H3	Bounding-box tree / sphere cells / hexagon grid geo indexes
Runbook	Step-by-step guide for handling a specific alert
R + W > N	Condition for read and write sets to overlap

S

Term	Meaning in one line
Safe method	Method that does not change server state
Saga	Sequence of local transactions with compensations on failure
Salt	Random per-user value mixed into a password before hashing
Same-origin policy	Browser rule blocking pages from reading other origins' responses
SAML	Older XML-based standard for enterprise SSO
Sampling	Keeping only a fraction of traces or logs to control cost
Saturation	How full a resource is (CPU, queue length)
Scalability	Handling more load by adding resources without redesign
Scale out / scale in	Adding or removing instances
Scatter-gather	Sending a query to all shards and merging results
Schema	The declared structure of tables, columns and data types
Schema (GraphQL)	Typed description of all data and operations an API offers
Schema-on-read / on-write	Structure applied when reading / enforced when writing
Scheme	First part of a URL, such as `https`, saying which protocol to use
Scope	Named permission an OAuth token grants
Search engine	A store built on an inverted index for full-text queries
Secondary index	Any index other than the clustered or primary one
Secrets management	Storing, distributing and regularly rotating (replacing) credentials securely
Security group	Cloud firewall attached to instances
SELECT FOR UPDATE	Locking rows that were read until the transaction ends
Selectivity	How few rows a filter value matches
Semi-synchronous replication	Leader waits for one follower, the rest are asynchronous
Sequence number	Counter TCP uses to order bytes and detect loss
Sequential consistency	Everyone sees the same order, not necessarily real time
Serializable	Result equals some one-at-a-time order of the transactions
Server	Program (or machine) that waits for and answers requests
Serverless	Running code or using services without managing servers, billed per use
Server-Sent Events (SSE)	One-way stream of events from server to browser over HTTP
Service discovery	Finding the current network addresses of a service's instances
Service (Kubernetes)	Stable name and virtual IP load-balancing to matching pods
Service mesh	Centrally configured sidecar proxies handling all service-to-service traffic
Service registry	Live database of service instances and their addresses
Service time	Time the server spends actually working on a request
Session	Server-side record of a logged-in user, referenced by a cookie or token
Session resumption	Reusing earlier TLS secrets to skip handshake work
Shadow launch	Copying real traffic to a new version and discarding its responses

Term	Meaning in one line
Sharding	Splitting data across machines by a shard key
Shard key	The column that decides which shard a row goes to
Shard / partition	One piece of the split data
Short code	The few characters (e.g. aZ3kQ9b) that identify a short link
Short polling	Client repeatedly asking the server for updates on a timer
Sidecar	Helper process deployed next to each service instance
Single point of failure (SPOF)	A part whose failure alone brings the whole system down
SLA	Service Level Agreement; contract with penalties for missed targets
SLI	Service Level Indicator; a measurement of service health
Sliding window counter	Rate limiter estimating a sliding count from two fixed windows
Sliding window log	Rate limiter storing each request timestamp for an exact count
SLO	Service Level Objective; internal target for an SLI
Sloppy quorum / hinted handoff	Temporarily writing to stand-in nodes that hand data back later
Slow start	TCP ramp-up that grows the sending rate each round trip
Snapshot	Point-in-time image of a disk, volume or database
Snapshot (event sourcing)	Saved state to avoid replaying every event
Snapshot isolation	Each transaction reads a frozen consistent picture of the data
SNI	Host name sent in the TLS ClientHello so one IP can serve many sites
Snowflake ID	64-bit, roughly time-ordered ID made from timestamp, machine ID and sequence
SOA	Older style of coarse services connected through a central service bus
SOA, SRV, PTR, CAA	Zone admin data, service location, reverse lookup, allowed certificate issuers
Socket	One endpoint of a connection: IP address plus port
Software load balancer	LB program on ordinary servers such as NGINX, HAProxy, Envoy
Source of truth / origin	The authoritative home of the data, usually a database
Span	One timed operation inside a trace
Split brain	Two nodes both acting as leader and accepting writes
SQL	The standard language for querying relational databases
SQL injection	Attack that smuggles SQL commands into input
SRE	Site Reliability Engineering; running production with engineering methods
SSD	Solid-state drive; flash storage with no moving parts
SSI	Serializable snapshot isolation, which aborts transactions on dangerous patterns
SSO	Single sign-on: one login for many applications
SSTable	An immutable sorted file written by an LSM-tree
Stale data	A cached or replicated copy older than the latest write
Stale-while-revalidate	Serve the expired value while refreshing in the background

Term	Meaning in one line
Star schema	A fact table surrounded by dimension tables
State	Stored facts that must survive between requests
Stateful service	Service that keeps client or data state locally
StatefulSet	Kubernetes object for stateful pods with stable identity and storage
Stateless service	Each request carries all needed context; the server keeps no per-client memory between requests
Status code	Three-digit result code (2xx success, 4xx client error, 5xx server error)
Sticky sessions	Load balancer sending a client to the same server every time
Strangler fig pattern	Migrating a monolith piece by piece behind a routing layer
Streaming RPC	gRPC call where client, server or both send a stream of messages
Stream processing	Processing events continuously as they arrive
Structured logging	Logs written as fields (such as JSON) for easy querying
Stub resolver	Minimal DNS client in the OS that asks a recursive resolver
Stub / skeleton	Generated client / server code that hides the network behind an ordinary function call
Subdomain	A name under a domain, such as `api.example.com`
SWIM	Gossip membership protocol with indirect pings
Symmetric / asymmetric encryption	One shared key encrypts and decrypts, fast / public and private key pair
Synchronous / asynchronous call	Caller waits for the response / sends a message and carries on
Synchronous replication	Leader waits for a follower before confirming a write
SYN flood	Attack exhausting servers with half-open TCP connections
System design	Choosing a system's parts and how they interact to meet its goals

T

Term	Meaning in one line
Table, row, column	A set of records, one record, one property of a record
Tail latency	Latency of the slowest requests (p99, p999, max)
TCP	Connection-based, reliable, ordered byte-stream transport
TCP/IP model	Four-layer model the internet actually uses
Term / epoch	Number that increases with each new leader
Terraform	Popular declarative infrastructure-as-code tool
Thread	Independent flow of execution inside a process
Three-way handshake	SYN, SYN-ACK, ACK exchange that opens a TCP connection
Throttling	Slowing or queuing requests that exceed a rate instead of rejecting them
Throughput	Amount of work completed per unit of time
Thundering herd	Many clients waking or retrying at the same moment
Ticket server	Central service issuing unique IDs
Timeout	Maximum time a caller waits before giving up on a call
Time-series database	A database optimised for timestamped measurements
TIME_WAIT	Short wait state after closing a TCP connection
TLD	Top-level domain such as `.com` or `.uk`
TLD name server	Refers resolvers to a domain's authoritative servers
TLS passthrough	LB forwards encrypted traffic, backend decrypts
TLS re-encryption	LB decrypts, inspects, then encrypts again to the backend
TLS / SSL	Protocol for encrypted, authenticated connections (SSL is the old name)
TLS termination	Decrypting HTTPS at the load balancer or proxy
Token	String a client presents to prove identity or access
Token bucket	Rate limiter with refilling tokens that allows bursts up to bucket size
Tokenising	Splitting text into searchable words
Tombstone	A marker recording that a key was deleted
Topic	Named stream of messages in pub/sub or Kafka
Trace	Record of one request's path across services
Trace ID / correlation ID	Identifier passed along to link all data for one request
Transaction	A group of operations that all succeed or all fail
Transactional outbox	Writing events to an outbox table in the same DB transaction
Transaction pooling	Lending a database connection only for one transaction
Transient fault	A short-lived failure likely to succeed on retry
Transmission delay	Time to push the bits of a message onto a link
TrueTime	Spanner's clock API returning a bounded time interval
TTL jitter	Randomising TTLs so keys don't expire together

Term	Meaning in one line
TTL (time to live)	How long cached data, a cache entry or a DNS answer (in seconds) stays valid before it expires
Two-phase commit (2PC)	Prepare then commit protocol for atomic distributed transactions
Two-phase locking	Taking locks and holding them until commit to get serializability
TXT record	Free-text record used for verification and email policy

U

Term	Meaning in one line
UDP	Connectionless transport without delivery or ordering guarantees
ULID	Time-sortable 128-bit ID format
Unicast	One IP address belongs to one location
Unique constraint	A rule that no two rows share a value in a column
Update anomaly	Inconsistency caused by a fact stored in many places
Upsert	Insert a row or update it if it already exists
URL shortener	Service that maps a short code to a long URL and redirects
USE method	Utilization, saturation, errors: metrics for each resource
UUID	128-bit identifier generated without coordination
UUIDv4 / UUIDv7	Random UUID / time-ordered UUID

V

Term	Meaning in one line
Vector clock / version vector	Per-node counters that also detect concurrent events
Vector database	A store for embeddings supporting nearest-neighbour search
Versioned keys	Putting a version in the cache key so new writes use new keys
Vertical scaling (scale up)	Making one machine bigger (scale up)
Virtual IP (VIP)	Address owned by a service rather than one machine
Virtual machine (VM)	Emulated computer with its own operating system
Virtual node	One of many ring positions owned by a single physical server
Visibility timeout	Time a received message is hidden before being redelivered (SQS)
VPC	Your own isolated private network inside a cloud provider
VRRP / keepalived	Protocol and tool that move a virtual IP to a standby node

W

Term	Meaning in one line
WAF	Web application firewall filtering malicious HTTP requests
Wall clock / monotonic clock	Time of day (can jump) / always-forward timer for durations
WAL (write-ahead log)	Sequential log of changes written (and fsynced) before the data files, for crash recovery
Warm / cold cache	Cache already full of hot data / empty and missing a lot
Watermark	Stream engine's estimate that all events up to a time have arrived
Webhook	HTTP callback a service sends to your URL when an event occurs
WebSocket	Persistent two-way connection between browser and server
Weighted round robin	Round robin where bigger servers get proportionally more
Wide-column store	NoSQL store that groups rows by a partition key and sorts them within it (e.g. Cassandra)
Window (streaming)	Time range grouping events, e.g. tumbling or sliding
Worker	Background program that processes jobs from a queue
Write-around	Writes skip the cache and go only to the DB
Write-behind / write-back	Writes go to cache, flushed to DB later
Write conflict	Two concurrent writes to the same record on different leaders
Write skew	Two transactions each write different rows based on a shared read and together break a rule
Write-through	Writes go to cache and DB synchronously

X, Y, Z

Term	Meaning in one line
X-Forwarded-For	Header carrying the original client IP through proxies
XSS	Cross-site scripting: injecting malicious script into pages others view
Zero trust	Security model that verifies every request, inside the network or not
Zone	Set of DNS records one authoritative server is responsible for
ZooKeeper	Coordination service for config, locks and leader election

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Part 6 glossary

Every Part 6 term is defined in one line in the Master A–Z glossary, which merges the glossaries of all six parts into a single alphabetical list.

Module 19 — Algorithms Deep Dive

Module Introduction

Earlier modules name algorithms in passing ("use consistent hashing", "a HyperLogLog counts distinct users", "the database uses MVCC"). This module opens each one up step by step, with small worked examples and numbers, so you can explain it at a whiteboard and know its limits. It assumes Module 0, where Raft, the hash ring with virtual nodes, Bloom filters, Merkle trees, gossip, token buckets and Lamport/vector clocks are explained.

Group	Algorithms
Consensus, coordination and time	Paxos and Multi-Paxos vs Raft and ZAB, 2PC vs 3PC, leases and leader election, failure detection (heartbeats, phi-accrual, SWIM), chain replication and CRAQ, hybrid logical clocks, TrueTime and commit-wait, version vectors
Partitioning and streaming sketches	Jump, rendezvous and Maglev hashing, consistent hashing with bounded loads, HyperLogLog internals, count-min sketch, heavy hitters (Misra-Gries, Space-Saving), reservoir sampling, t-digest percentiles, trending detection
Storage engines, caching and scheduling	LSM trees and compaction, B+-trees, MVCC, two-phase locking and SSI, skip lists, CLOCK / ARC / TinyLFU eviction, hierarchical timing wheels, Chandy-Lamport snapshots and Flink barriers
Search, similarity and routing	Inverted indexes, TF-IDF and BM25, SimHash / MinHash / LSH, vector search (HNSW, IVF, PQ), Dijkstra / A* / contraction hierarchies, feed ranking

Every topic follows *What it is → The problem it solves → How it works → Trade-offs → Real-world examples → In the interview → TL;DR*, with sources at the end.

Paxos and Multi-Paxos (compared with Raft and ZAB)

What it is

Paxos is a **consensus algorithm**: rules that let a group of machines agree on one value even when some crash, restart or lose messages. Leslie Lamport described it in "The Part-Time Parliament" (submitted 1990, published 1998) and plainly in "Paxos Made Simple" (2001). **Single-decree Paxos** decides exactly one value. **Multi-Paxos** runs one instance per position (**slot**) in a replicated log and adds a stable leader so most slots need one round trip.

Raft is walked through step by step in Module 0 — Consensus; here we cover the older algorithm Raft simplified, and compare both with **ZAB** (ZooKeeper Atomic Broadcast).

The problem it solves

Three replicas must agree which transaction is number 42 in the log; if two pick different ones, their data diverges forever. Messages can be delayed, duplicated or lost, and machines crash between steps. Paxos guarantees **safety** (no two machines decide different values for a slot) under all of this, and **liveness** (it eventually decides) once a majority can communicate long enough.

How it works

Three roles, often played by one process: **Proposer** (suggests a value), **Acceptor** (votes; a **majority** or **quorum**, more than half, e.g. 2 of 3 or 3 of 5, must agree), **Learner** (finds out what was chosen).

Each proposal carries a unique, ordered **proposal number** (**ballot**), e.g. `round.serverId` such as 1.1, 2.3. Each acceptor stores on disk, and persists before replying, the highest number it has **promised** and the last proposal it **accepted** (number and value). It accepts `Accept(n, v)` only if `n >= promised`.

Single-decree Paxos, step by step

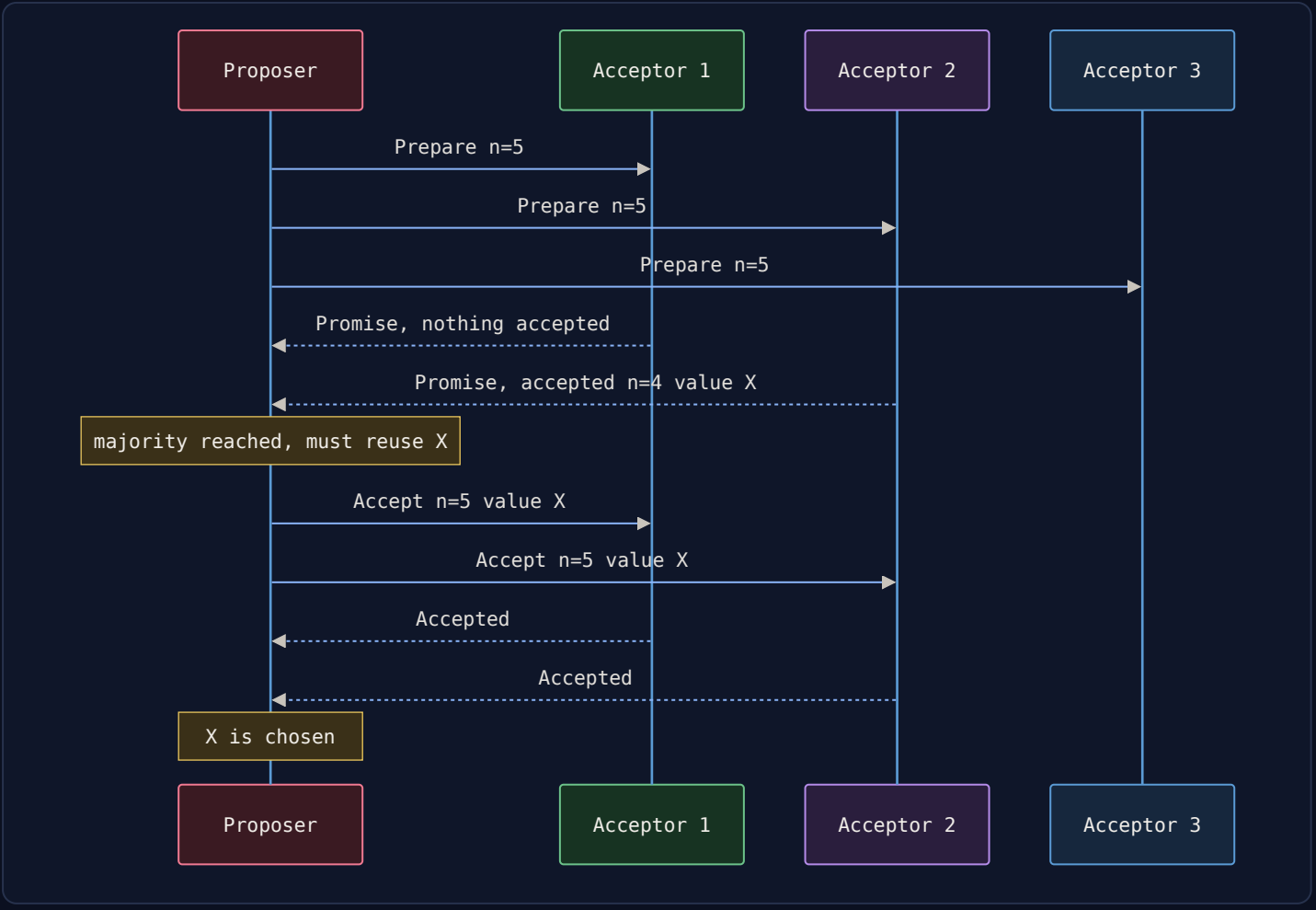
1. **Phase 1a, Prepare.** A proposer picks `n` higher than any it has used and sends `Prepare(n)`.

2. **Phase 1b, Promise.** An acceptor that has promised nothing higher replies "I will ignore proposals below `n`", including the highest-numbered proposal it already accepted, if any.
3. **Choose the value.** With promises from a majority, if any reply carries an accepted proposal the proposer **must** use the value of the highest-numbered one; only otherwise may it use its own. This rule is the heart of Paxos safety.
4. **Phase 2a, Accept.** Send `Accept(n, value)`.
5. **Phase 2b, Accepted.** An acceptor accepts unless it has since promised higher; it records `(n, value)` and tells learners.
6. **Decision.** Once a majority accepted the same `(n, value)`, it is **chosen** and never changes.

Worked example (acceptors A1, A2, A3).

- P1 sends `Prepare(1)`; all promise, none accepted anything. P1 sends `Accept(1, "X")`; A1 accepts, messages to A2/A3 are delayed, P1 crashes. Nothing chosen (1 of 3).
- P2 sends `Prepare(2)`, hears from A1 and A2. A1 reports "accepted (1, X)", so P2 must propose X, not its own "Y". A1 and A2 accept `(2, X)`: X is chosen.
- The delayed `Accept(1, "X")` reaches A3, which never saw `Prepare(2)`, so it accepts (1, X). Harmless: same value. Had A3 promised 2 first, it would reject the old message.

Key insight: any two majorities of 3 overlap in at least one acceptor, and that overlap carries forward any value that might already be chosen.



Multi-Paxos. Both phases per slot cost two round trips. Phase 1 does not depend on the value, so a proposer that wins Phase 1 for "all slots from here on" becomes the **distinguished proposer** (in practice, the leader) and sends only Phase 2 (`Accept(n, slot, value)`) per slot: one round trip. If it dies, another node runs Phase 1 with a higher number, learns values accepted in unfinished slots, re-proposes them and continues.

Cost. Steady-state Multi-Paxos: one round trip to a majority, O(N) messages for N acceptors, O(1) acceptor state per slot. Classic Paxos: two round trips. 2F+1 acceptors tolerate F crashes.

Duelling proposers. Two proposers can keep pre-empting each other (P1 prepares 3, P2 prepares 4, P1 prepares 5, ...) so nothing is chosen. Paxos stays safe but this **livelock** stops progress. Fix, as in Raft: one leader at a time plus randomised back-off.

Paxos vs Raft vs ZAB

	Multi-Paxos	Raft	ZAB
Leadership	Optional optimisation, not part of the core proof	Built in: strong leader, one per term	Built in: one primary per epoch
Era counter	Proposal number	Term	Epoch (high 32 bits of the 64-bit zxid)
Log holes	Allowed: slot 7 can be decided before slot 6	Not allowed: log is contiguous	Not allowed: strict order
Who can become leader	Anyone; new leader recovers values via Phase 1	Only a node whose log is at least as up to date (election restriction)	Node with latest epoch and highest zxid in a quorum
Leader catch-up	New leader learns from acceptors	Leader overwrites followers' conflicting entries	Synchronisation phase before new broadcasts
Spec completeness	Many details (membership, snapshots) left to implementers	Full spec, including membership change	Tailored to ZooKeeper's primary-backup model

In the normal case all three do the same thing (leader sends each entry, commits on majority ack); they differ mostly in how a new leader cleans up after the old one.

Trade-offs

- **Paxos is flexible but underspecified.** Holes and out-of-order decisions help parallelism, but every real system must invent leader election, log compaction and membership change; Google's "Paxos Made Live" (2007) describes how hard that was for Chubby.
- **Raft is easier to get right** because it fixes those choices, at the price of flexibility (no log holes).
- **Every write costs a majority round trip**: a 5-node group across three regions pays a cross-region round trip per commit.
- **More nodes, more tolerance, slower commits**: 3 nodes tolerate 1 failure, 5 tolerate 2.

Real-world examples

- **Google Chubby** (lock service) and **Spanner** (each shard replicated by its own Paxos group).
- **Apache ZooKeeper** uses ZAB; it coordinated older Kafka, HBase and Hadoop.
- **etcd**, **Consul**, **CockroachDB**, **TiKV** and Kafka's **KRaft** controller use Raft.
- **Apache Cassandra** runs single-decree Paxos per key for lightweight transactions (`IF NOT EXISTS`).

In the interview

Q: Why must a Paxos proposer reuse a previously accepted value?

A: It might already be chosen by a majority. Any majority the proposer hears from overlaps that one, so some promise reports it; reusing it means a later proposal never overwrites a decision.

Q: What does Multi-Paxos add?

A: A stable leader that runs Phase 1 once, then only Phase 2 per slot: one round trip per decision instead of two. That is the version used in practice.

Q: Raft or Paxos for a new system?

A: Use a tested library or service (etcd, ZooKeeper, a Raft library). If choosing, Raft has a complete spec and many production implementations; Paxos variants only for special needs such as out-of-order decisions.

TL;DR

Paxos agrees in two phases (promise, accept) with one golden rule: reuse the highest value already accepted in the majority you heard from. Multi-Paxos adds a stable leader so each log entry costs one round trip. Raft and ZAB are leader-centred cousins with fuller specifications.

Sources

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Two-phase commit vs three-phase commit

What it is

An **atomic commit protocol** makes a multi-machine transaction commit everywhere or abort everywhere. Data holders are **participants** (resource managers, cohorts); the **coordinator** (transaction manager) runs the protocol.

- **Two-phase commit (2PC)**: vote phase, then decision phase.
- **Three-phase commit (3PC)**: adds a "pre-commit" phase so participants can finish without the coordinator in some failures (Dale Skeen, non-blocking commit protocols, 1981).

2PC basics are in Module 7 — Two-Phase Commit; here: exactly where it blocks, how 3PC tries to fix that, and why industry instead runs 2PC on top of consensus.

The problem it solves

A transfer debits account 17 on shard A and credits 99 on shard B. If A commits and B aborts, money disappears. Each shard can make its change durable, but neither can decide alone: the other might fail its checks (constraint violation, full disk, deadlock).

How it works

2PC (coordinator C, participants A and B):

1. **Prepare.** C sends `PREPARE`.
2. **Vote.** Each participant checks it can commit, writes changes plus a "prepared" record to its durable log, keeps its locks and answers `YES`, or answers `NO` and aborts. After YES it can no longer decide alone.
3. **Decide.** All YES: C writes "commit" to its log (the actual **commit point**) and sends `COMMIT`. Any NO or timeout: `ABORT`.
4. **Finish.** Participants apply, release locks, acknowledge; C then forgets the transaction.

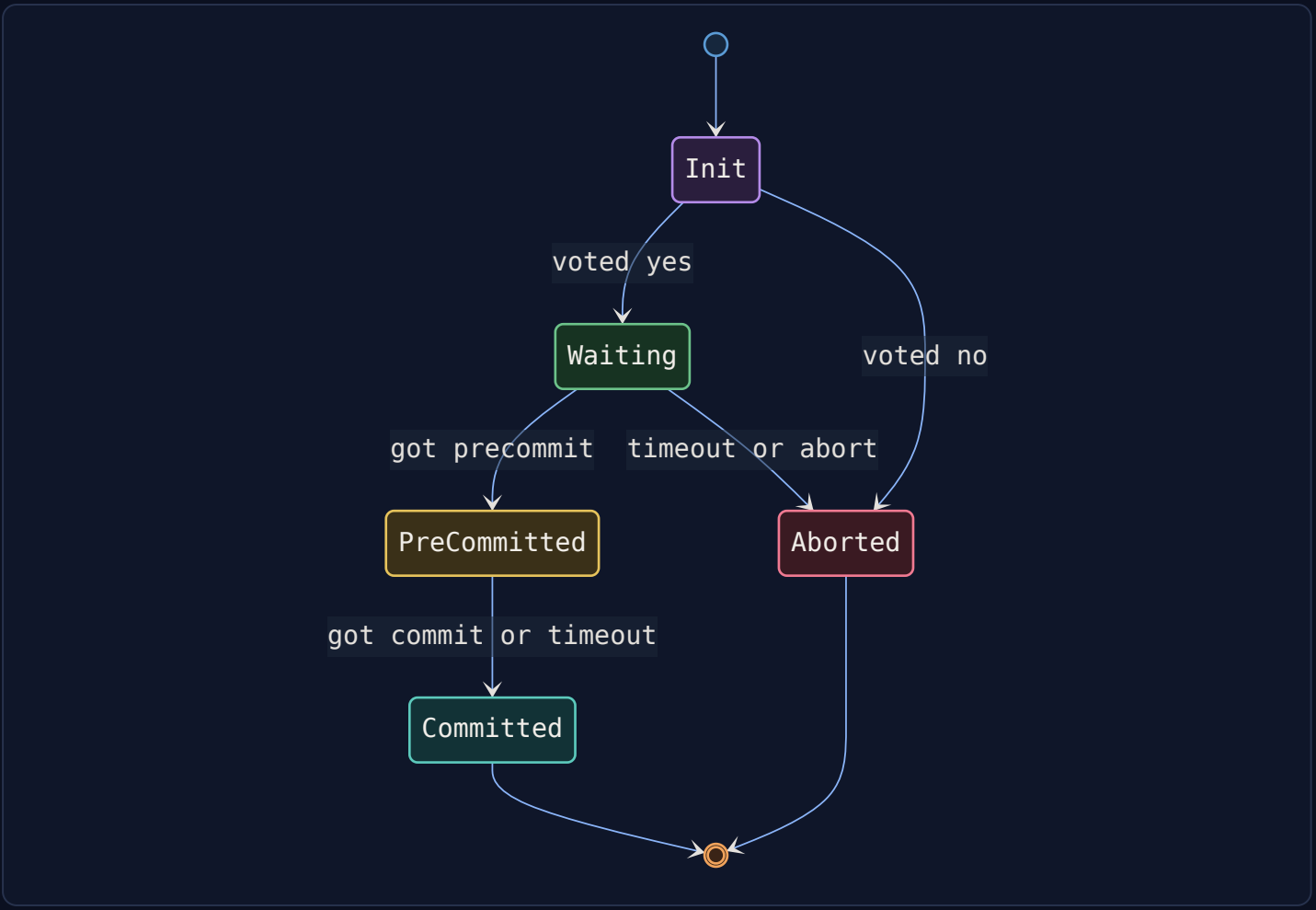
Cost: 4 messages per participant (prepare, vote, decision, ack) = 4N, plus 2 round trips and 2 forced disk writes on the critical path.

Where it blocks. A and B voted YES, then C crashes before sending the decision. They are **in doubt**: C may have told someone COMMIT, or decided ABORT. They must hold their locks until C recovers (possibly minutes), freezing rows 17 and 99. This is the **blocking problem**.

3PC:

1. **CanCommit.** Participants vote YES/NO but do not lock themselves in yet.
2. **PreCommit.** If all YES, C sends `PRECOMMIT`; participants ack. Now everyone knows everyone voted YES.
3. **DoCommit.** After all acks, C sends `COMMIT`.

Recovery rule: if C disappears, survivors confer. If any received PRECOMMIT, nobody can have aborted, so commit; if none did, nobody can have committed, so abort. Timeouts let participants proceed without C.



Worked example. Five participants; C crashes after sending PRECOMMIT to P1 only. P2 to P5 time out and elect a recovery coordinator (P2), which sees P1 pre-committed and commits all. Had P1 also crashed, survivors see nobody pre-committed and abort. With a reliable network both cases end consistently.

Why 3PC is rarely used

- **Unsafe under network partitions.** Its timeouts assume a **synchronous network** (bounded delivery, so "no reply" means "crashed"). Real networks are asynchronous: if a partition separates P1 (pre-committed) from P2 to P5, one side commits and the other aborts, which is worse than blocking.
- **An extra round trip** on every transaction, even when nothing fails.
- **A better fix exists.** Gray and Lamport's "Consensus on Transaction Commit" (2006): replicate the coordinator's decision with Paxos (**Paxos Commit**) and commit no longer blocks while a majority of coordinators lives; plain 2PC is the special case tolerating zero coordinator failures. Modern systems run ordinary 2PC where coordinator and every participant are consensus groups, so no single crash leaves anyone in doubt.

Trade-offs

	2PC	3PC	2PC over consensus groups
Round trips (normal case)	2	3	2, each also replicated
Blocks if coordinator dies	Yes	Not with reliable network	No, while a majority survives
Safe under partition	Yes (it blocks instead)	No	Yes
Typical use	XA, database-internal	Textbooks	Spanner, CockroachDB-style systems

2PC holds locks for the whole protocol, so cross-shard transactions are slower than single-shard ones. Designers keep related data on one shard, or use **sagas** (local transactions with compensations, Module 7 — Saga) when temporary inconsistency is acceptable.

Real-world examples

- **XA transactions** (X/Open standard, Java EE, many databases) and PostgreSQL's `PREPARE TRANSACTION` are plain 2PC.
- **Google Spanner** runs 2PC across Paxos groups: coordinator and participant leaders replicate prepare and commit records through Paxos.
- **Apache Kafka transactions**: a transaction coordinator writes a prepare marker to its replicated log, then commit markers to each partition (2PC on replicated logs).

In the interview

Q: What is the blocking problem in 2PC?

A: After voting YES a participant cannot decide alone; if the coordinator crashes before telling it the result, it holds its locks until the coordinator returns.

Q: Why not use 3PC?

A: It needs a missing reply to reliably mean a crash. Under partitions two groups time out in different directions, one committing and one aborting, and it adds a round trip. Replicating the coordinator with Paxos or Raft fixes blocking safely.

Q: How would you design a cross-shard transaction today?

A: 2PC over consensus-replicated shards, a durable transaction record as the commit point, idempotent retries. Better, design data so most transactions hit one shard.

TL;DR

2PC is correct but blocks if the coordinator dies after YES votes. 3PC's pre-commit avoids blocking but is unsafe under partitions and costs a round trip, so almost nobody ships it. Industry runs 2PC whose coordinator and participants are each consensus-replicated.

Sources

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Leases and lease-based leader election

What it is

A **lease** is a lock with an expiry. The holder gets a right (cache a file, be leader, serve reads) for a fixed period, say 10 s, and must **renew** before it expires. If it crashes or is cut off, renewals stop and after expiry the right can be granted elsewhere without asking it. Cary Gray and David Cheriton introduced leases for file-cache consistency (SOSP 1989). In **lease-based leader election**, whoever holds the lease is the leader.

The problem it solves

A plain lock held by a crashed node is never released, and you cannot tell crashed from slow. A lease turns "wait until the holder says done" into "wait until a known time passes", which you can always do alone. Leases also make reads cheap: a leader with a valid lease knows no other leader exists, so it serves reads locally without a majority round trip.

How it works

Lease leader election with a coordination store such as etcd:

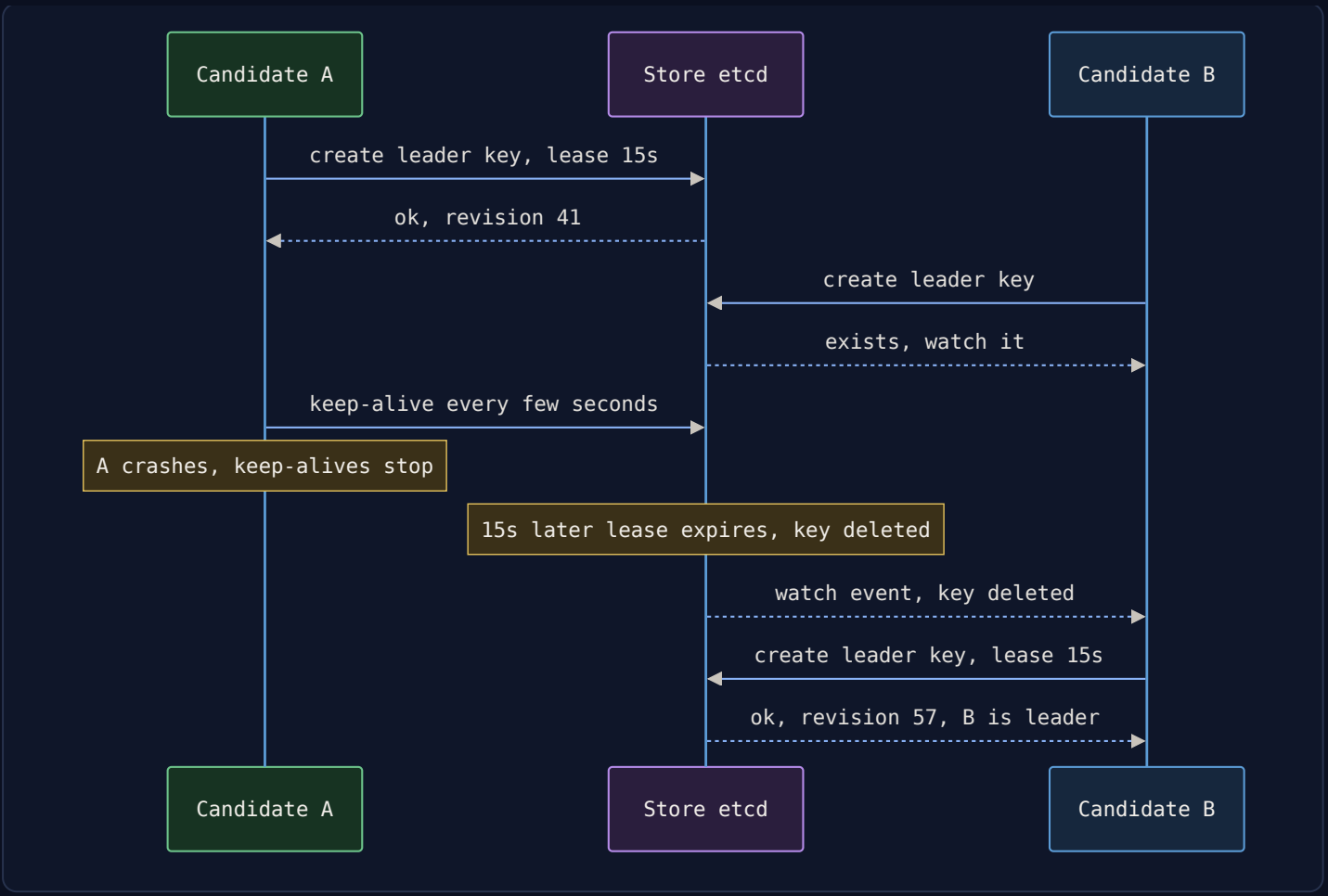
1. Each candidate tries to create `/service/leader` with its ID, attached to a lease with a **TTL** (time to live) of, say, 15 s. Only one create succeeds (compare-and-swap, "create only if absent").
2. The winner leads and renews (**keep-alives**) well before expiry, e.g. every 2 to 5 s.
3. Losers **watch** the key. If the leader dies, renewals stop, the store deletes the key at TTL, and watchers race to create it.
4. Each successful create gets a new, larger **revision number**, which the leader attaches as a **fencing token** to every write so storage rejects an old leader's writes (Module 0 explains fencing).
5. A leader that cannot renew before its **renew deadline** must stop acting as leader immediately, even if unsure it lost.

Clocks and the safety margin. Each machine measures the lease with its own clock, and clocks **drift**. Rule: the holder must believe the lease ends *earlier* than the granter does.

Worked example: L = 10 s, max drift 0.1% (1 ms/s), holder measures from when it *sent* the request.

- Request sent at local 0, grant received at 0.05 s; the lease counts from 0, so network delay only adds caution.
- Over 10 s the clock may be 10 ms off; the holder stops at 10 − 0.01 = 9.99 s minus a margin, say 9.5 s.
- The granter will not re-grant until its clock shows 10 s plus margin, say 10.5 s.
- The windows never overlap, so at most one leader exists, provided no process pauses longer than the margin without checking time. Hence fencing tokens are still needed: a 20 s garbage-collection pause makes the old leader act on stale beliefs.

Kubernetes defaults. `kube-controller-manager` and similar use a `Lease` object: lease duration 15 s, renew deadline 10 s, retry period 2 s. The leader renews every 2 s, gives up if not renewed within 10 s, and others may take over 15 s after the last renewal: worst-case failover about 15 s.



Leader leases inside consensus. A Raft or Multi-Paxos leader can hold a lease from a majority of followers, who promise not to vote for anyone else for, say, 10 s after acknowledging it. In that window the leader serves **linearizable reads** (guaranteed to see the latest committed write) locally. Spanner's Paxos leader leases default to 10 s.

Complexity. One renewal per interval, O(1) messages per period. Failover is bounded by lease length plus detection and election time.

Trade-offs

- **Short leases** (a few seconds): fast failover, more renewal traffic and more false failovers on network hiccups.
- **Long leases**: fewer renewals, longer outage after a real crash (nobody leads until expiry).
- **Leases need bounded clock drift**, not synchronised clocks. Gray and Cheriton found little extra benefit beyond about 10 s for their workload.
- **Process pauses break the assumption** that the holder checks time often: pair leases with fencing tokens when correctness matters.

Real-world examples

- **Google Chubby**: sessions are leases (default extension 12 s); after a lapse the client enters "jeopardy" and waits a 45-second grace period for a new master. The master holds a master lease of a few seconds from its replicas.
- **etcd** leases and **ZooKeeper** ephemeral nodes (deleted on session timeout) are the standard leader-election building blocks.
- **Kubernetes** control-plane components and many operators use `coordination.k8s.io` Lease objects.
- **HDFS** leases give one client at a time the right to write a file.

In the interview

Q: Lease vs lock?

A: A lease expires on its own; if the holder disappears the system waits a bounded time and moves on instead of forever.

Q: Can two leaders exist at once with leases?

A: Not if drift stays within the bound and the old leader stops before expiry. A long process pause can still make it act late, so writes carry a fencing token that storage checks.

Q: How do you choose the lease length?

A: It is the worst-case failover time: pick it from the recovery target (e.g. 10 to 15 s), then renew at a fraction of it (every 2 to 5 s) so a couple of lost renewals don't cost leadership.

TL;DR

A lease is a lock that expires; the unexpired holder leads and others take over after expiry. Holder stops early, granter waits late, fencing tokens cover pauses.

Sources

- Gray, Cheriton, "Leases: An Efficient Fault-Tolerant Mechanism for Distributed File Cache Consistency", SOSP 1989: https://courses.cs.duke.edu/spring15/cps210/readings/leases-sosp89.pdf
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- Kubernetes, `kube-controller-manager` reference (leader-elect flags): https://kubernetes.io/docs/reference/command-line-tools-reference/kube-controller-manager/
- etcd documentation, lease API: https://etcd.io/docs/latest/learning/api/
- Corbett et al., "Spanner", OSDI 2012 (Paxos leader leases, 10 s default): https://www.usenix.org/conference/osdi12/technical-sessions/presentation/corbett

Failure detection: heartbeats, phi-accrual and SWIM

What it is

A **failure detector** decides "that node is probably dead". It cannot be certain (crashed node, slow node and lost link look identical), so it is a guess tuned between **completeness** (every crashed node is eventually suspected) and **accuracy** (few **false positives** on healthy nodes). Three designs: fixed-timeout **heartbeats**, the adaptive **phi accrual** detector (Hayashibara et al., 2004), and **SWIM** (Das, Gupta, Motivala, DSN 2002), which combines probing with gossip for large clusters. Basics: Module 0 — Heartbeats, gossip and failure detection.

The problem it solves

Leader election, replica placement, load balancing and membership all need to know who is alive. Too eager a detector causes needless failovers and data movement (a "flapping" node declared dead and alive repeatedly); too slow leaves clients waiting on a dead node. At 1,000 nodes the detection traffic itself must stay small.

How it works

1. Fixed-timeout heartbeats. Each node sends "I'm alive" every interval (e.g. 1 s); the monitor records the last arrival; no heartbeat for the timeout (e.g. 5 s) means dead. One fixed number fits no network: 5 s is far too slow in a rack and too twitchy over a congested WAN. All-to-all heartbeating is $O(N^2)$: about a million messages per interval at 1,000 nodes.

2. Phi accrual detector. Outputs a suspicion level **phi (φ)** that rises the longer a heartbeat is overdue, scaled by how regular heartbeats have been.

1. Keep a sliding window of recent **inter-arrival times** (say the last 1,000).
2. Estimate mean μ and standard deviation σ.
3. With t = time since the last heartbeat, $P = 1 - F(t)$ is the probability a healthy node is *this* late (F = cumulative distribution of observed gaps).
4. $\varphi = -\log_{10}(P)$: $\varphi = 1$ is a 10% chance it is just late, $\varphi = 3$ is 0.1%, $\varphi = 8$ is 1 in 100 million.

5. The application picks a threshold; Cassandra's `phi_convict_threshold` defaults to 8.

Worked example, μ = 1,000 ms, σ = 100 ms (normal):

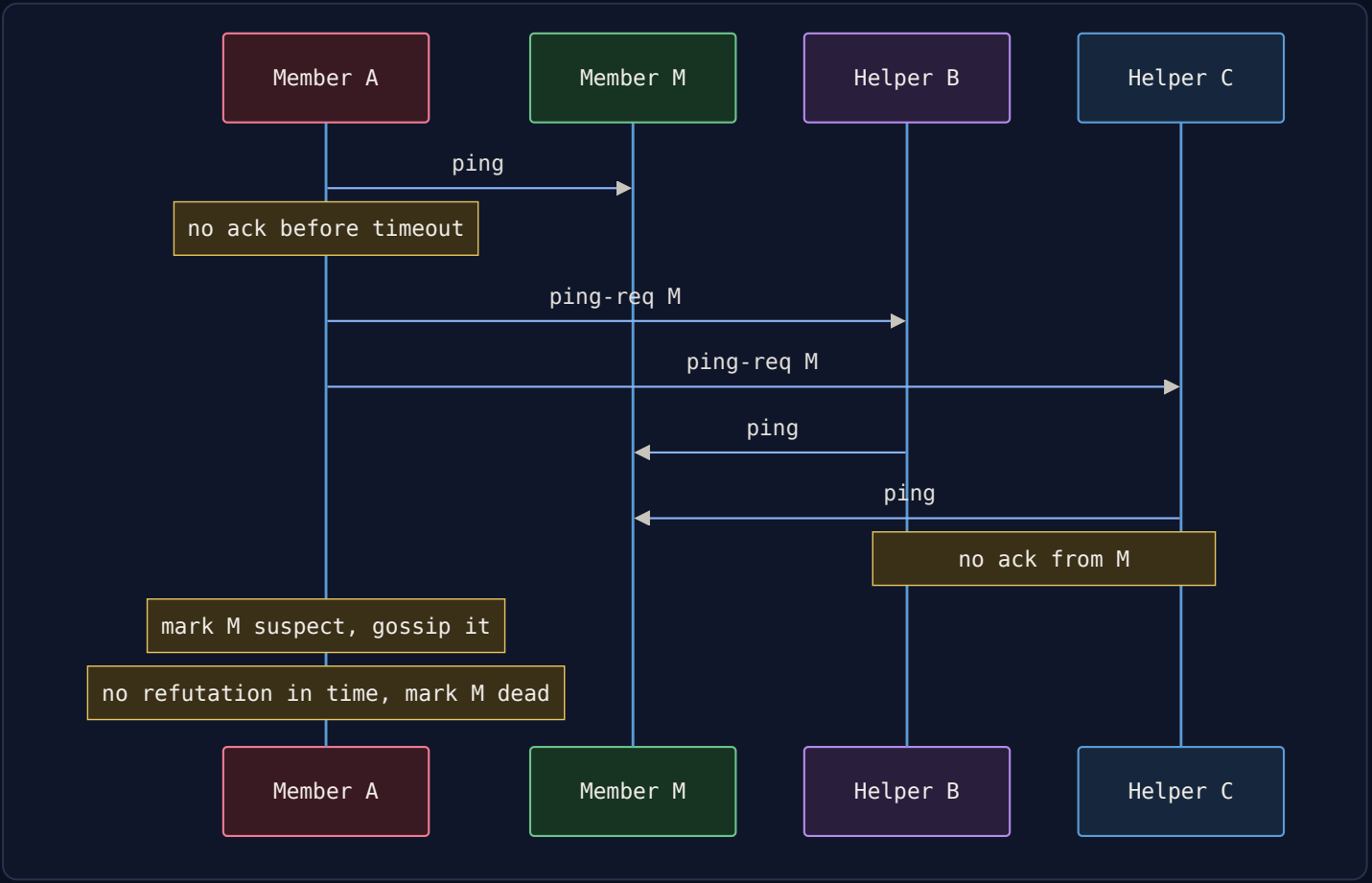
Time since last heartbeat	z = (t − μ)/σ	P(just late)	φ
1,100 ms	1	0.159	0.8
1,200 ms	2	0.023	1.6
1,300 ms	3	0.0013	2.9
1,500 ms	5	2.9e-7	6.5
1,560 ms	5.6	1.1e-8	8.0

A node is convicted about 1.56 s after its last heartbeat. If jitter grows σ to 300 ms, φ = 8 arrives only after about 2.7 s: the detector adapts itself. The paper assumed a normal distribution; some systems use a simpler exponential model.

3. SWIM (Scalable Weakly-consistent Infection-style process group Membership) separates *detecting* failures from *spreading* the news.

1. Every **protocol period** T (say 1 s), each member pings one random member M.
2. An `ack` within a short timeout ends it.
3. Otherwise it asks k random members (say 3) to `ping-req` M on its behalf; any ack is relayed back. This **indirect probe** avoids blaming M for one bad link.
4. If nobody hears from M, it is marked **suspect** (not dead) and the suspicion is gossiped.
5. M **refutes** by gossiping "alive" with a higher **incarnation number** (a counter only M may increase).
6. Unrefuted within a suspicion timeout, M is declared **dead**.
7. Membership updates (join, suspect, alive, dead) are **piggybacked** on pings and acks, spreading like an infection to all N members in about O(log N) periods with no extra messages.

Per-member cost per period is constant (one ping plus at most k indirect probes); total traffic is O(N), not $O(N^2)$, and expected first-detection time does not grow with cluster size.



Trade-offs

- **Fixed timeouts**: simple and predictable, but per-environment tuning; either slow detection or false alarms.
- **Phi accrual**: adapts and gives a tunable confidence level, but needs history (warm-up) and can be fooled when the heartbeat pattern suddenly changes.
- **SWIM**: thousands of nodes at constant per-node load, with eventual (not instant) membership agreement. An overloaded (CPU-starved) node can wrongly accuse healthy peers; HashiCorp's **Lifeguard** extensions address this, reporting about 50 times fewer false positives.
- **No detector is perfect**: pair with fencing and idempotency so a wrong guess costs a failover, not corrupt data.

Real-world examples

- **Apache Cassandra** and **Akka Cluster**: phi accrual.
- **HashiCorp memberlist** (Consul, Serf, Nomad): SWIM with Lifeguard.
- **Raft and ZooKeeper**: simple heartbeats with timeouts (randomised in Raft).
- **Kubernetes**: a node is unhealthy when its node Lease is not renewed in time (a heartbeat via a shared store).

In the interview

Q: Why not just a 5-second heartbeat timeout?

A: One number fits no network or load: too slow in a quiet data centre, false alarms during congestion. Phi accrual sets the bar from observed heartbeat timing.

Q: How does SWIM avoid $O(N^2)$ traffic?

A: Each node probes one random peer per period and piggybacks membership changes on those probes: constant per-node load, news reaches everyone in about log N rounds.

Q: Why indirect pings and a suspect state?

A: So one bad link or a short pause doesn't kill a node: indirect pings test other paths, and suspicion gives it a chance to refute.

TL;DR

Failures can only be suspected. Heartbeats with timeouts are simplest; phi accrual turns lateness into an adaptive probability-based suspicion level; SWIM probes one random peer per period with indirect probes and suspicion, gossiping results at constant per-node cost.

Sources

- Hayashibara, Défago, Yared, Katayama, "The phi accrual failure detector", SRDS 2004 (as implemented and documented in Akka): https://doc.akka.io/api/akka-core/2.9/akka/remote/PhiAccrualFailureDetector.html
- Apache Cassandra / DataStax docs, failure detection and `phi_convict_threshold`: https://docs.datastax.com/en/cassandra-oss/3.0/cassandra/architecture/archDataDistributeFailDetect.html
- Das, Gupta, Motivala, "SWIM: Scalable Weakly-consistent Infection-style Process Group Membership Protocol", DSN 2002: https://www.cs.cornell.edu/projects/Quicksilver/public_pdfs/SWIM.pdf
- HashiCorp, "Making Gossip More Robust with Lifeguard": https://www.hashicorp.com/blog/making-gossip-more-robust-with-lifeguard

Chain replication and CRAQ

What it is

Chain replication (van Renesse and Schneider, OSDI 2004) arranges a data item's replicas in a line, a **chain**, from **head** to **tail**. Writes enter at the head and flow down; the tail answers reads. A separate,

consensus-backed **configuration manager** (master) decides chain membership.

CRAQ (Chain Replication with Apportioned Queries, Terrace and Freedman, USENIX ATC 2009) keeps the write path but lets **any** node answer reads, still strongly consistent.

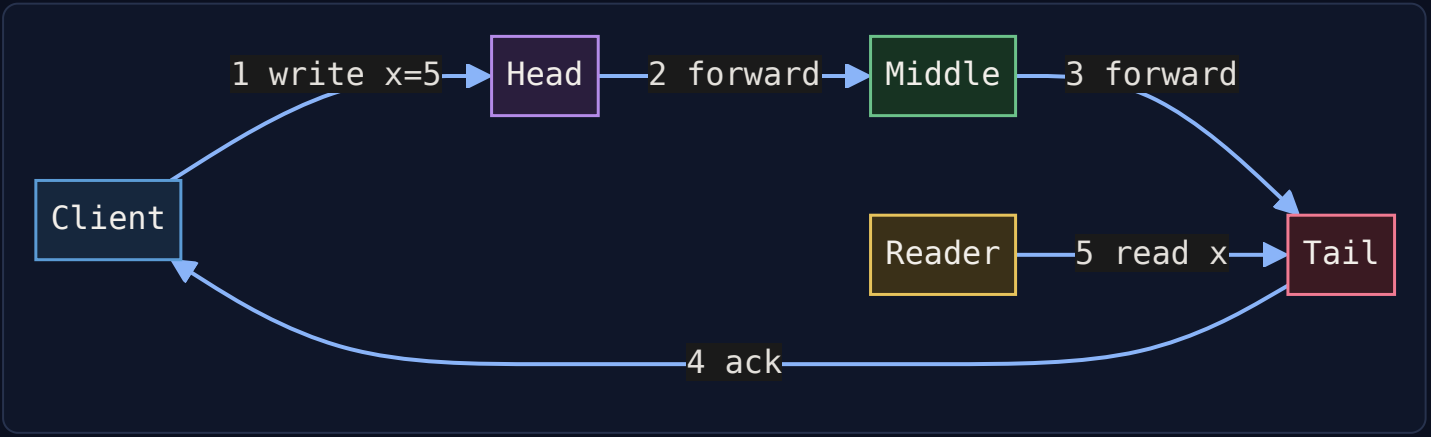
The problem it solves

In primary-backup, one primary receives each write, sends it to every backup, collects acks and serves reads, so its NIC and CPU bottleneck. A chain spreads the work: each node forwards each write to one successor, and reads (tail) are separated from writes (head), while still giving **linearizability** (every read sees the latest completed write, as if there were one copy). CRAQ fixes the chain's own weakness: the tail serving every read is a hot spot for read-heavy workloads.

How it works

Chain replication (H, M, T):

1. Client sends `put(x, 5)` to H.
2. H applies it and forwards to M; M applies and forwards to T.
3. T applies it: the write is **committed** (every replica has it). T replies to the client (or acks back up so H replies).
4. `get(x)` goes to T, which holds only committed writes; one tail means reads and writes are totally ordered.



Failure handling (by the configuration manager):

- **Head fails**: M becomes head. Writes H had not forwarded are lost, but were never acknowledged, so clients retry.
- **Tail fails**: M becomes tail; it has every write T had, maybe more, so nothing committed is lost.
- **Middle fails**: H is linked to T and re-sends every write forwarded but not yet acked downstream. Each node keeps such **pending** writes until the tail's ack flows back.
- **Adding a node**: it copies state from the tail, then is appended as the new tail.

Worked example. 0.5 ms per hop. A 3-node chain write is client→H→M→T→client: 4 hops, about 2 ms. Primary-backup (client→primary, parallel fan-out to backups, acks, reply) is also 4 hops and stays 4 however many backups, while a 5-node chain needs 6. The difference is sending work: a primary sends one copy per backup, each chain node sends 1. At 10,000 writes/s of 1 KB, the chain head sends 10 MB/s; a primary with 4 backups sends 40 MB/s.

Complexity. Write latency O(n) hops; n messages per write (one per link) plus acks; n full copies; tolerates n − 1 failures (the manager, not the chain, needs a majority).

CRAQ. Each node may store several **versions** of an object, each **clean** (known committed) or **dirty** (not yet known committed).

1. A write travels head to tail; each node stores the new version as dirty.
2. When the tail applies it, it is committed and the tail acks back up the chain.
3. As the ack passes, each node marks that version clean and deletes older ones.
4. A read can go to **any** node: if its latest version is clean, return it; if dirty, ask the tail for the latest committed version number of x (a tiny **version query**, not a full read) and return that version

from local storage.

Worked example: x is version 7 (clean) everywhere. Write 8 has reached H and M (dirty) but not T. A reader at M sees dirty, asks T, learns "7", returns 7. A millisecond later T applies 8, acks flow back, M marks 8 clean, and the next read at M returns 8 with no query. In read-mostly workloads most objects are clean, so read throughput scales with the number of nodes.

CRAQ can also relax to **eventual consistency** on request: return the newest local version, dirty or not, skipping the tail query.

Trade-offs

- **Higher write latency** than parallel fan-out: n sequential hops instead of one parallel round.
- **One slow node slows every write**, since every write passes every node.
- **Needs an external configuration manager** (Paxos, Raft or ZooKeeper) to avoid split brain; the chain itself runs no consensus.
- **Strong consistency with simple logic**: no per-write voting; the tail is the single point of truth for ordering.
- **CRAQ** removes the read hot spot at the cost of multi-version storage and extra tail queries under write-heavy load.

Real-world examples

- **Hibari** (open-source key-value store): chain replication with head, middle and tail "bricks".
- **FAWN-KV** (research store on low-power nodes) and CRAQ itself build on it.
- **HDFS** writes blocks through a **pipeline** of DataNodes, each forwarding to the next.
- The pattern recurs in object and block storage wherever per-node bandwidth matters most.

In the interview

Q: Why does the tail serve reads?

A: It is the only node guaranteed to hold only committed writes, so reading there is linearizable without coordination.

Q: How does CRAQ read from any node and stay consistent?

A: Clean versions are returned directly; for a dirty one the node asks the tail which version is committed and returns that.

Q: When prefer quorum or Raft replication to a chain?

A: When write latency or tolerance of one slow replica matters more: a majority quorum commits after the fastest majority, a chain waits for every node.

TL;DR

Chain replication pipes writes head to tail and reads from the tail: strong consistency and good throughput with simple logic, latency proportional to chain length. CRAQ lets every node serve reads by tracking clean/dirty versions and asking the tail only when unsure.

Sources

- van Renesse, Schneider, "Chain Replication for Supporting High Throughput and Availability", OSDI 2004: https://www.usenix.org/conference/osdi-04/chain-replication-supporting-high-throughput-and-availability
- Terrace, Freedman, "Object Storage on CRAQ: High-Throughput Chain Replication for Read-Mostly Workloads", USENIX ATC 2009: https://www.usenix.org/conference/usenix-09/object-storage-craq-high-throughput-chain-replication-read-mostly-workloads
- Hibari documentation, architecture: https://hibari.readthedocs.io/en/latest/admin-guide/architecture.html
- Apache Hadoop HDFS architecture guide (replication pipelining): https://hadoop.apache.org/docs/stable/hadoop-project-dist/hadoop-hdfs/HdfsDesign.html

Hybrid logical clocks

What it is

A **hybrid logical clock (HLC)** stamps each event with `l`, close to **physical (wall-clock) time**, and `c`, a small **logical counter** (Kulkarni, Demirbas et al., "Logical Physical Clocks", OPODIS 2014). It combines **physical timestamps** (human-readable, but machines disagree by milliseconds or more, **clock skew**, so a message can arrive "before" it was sent) with **Lamport clocks** (counters that guarantee **causality**, if e could influence f then e is smaller, but bear no relation to real time). Background: Module 0 — Clocks and ordering.

The problem it solves

A distributed database wants timestamps that (1) never violate cause and effect, (2) fit a 64-bit field, and (3) stay near real time for "data as of 10:00:00" or "snapshot at t". Physical clocks fail (1), Lamport clocks fail (3), vector clocks satisfy (1) but grow with node count, failing (2). HLC satisfies all three.

How it works

Node j keeps `l` (largest physical time heard of) and `c` (counter); `pt` is its physical clock.

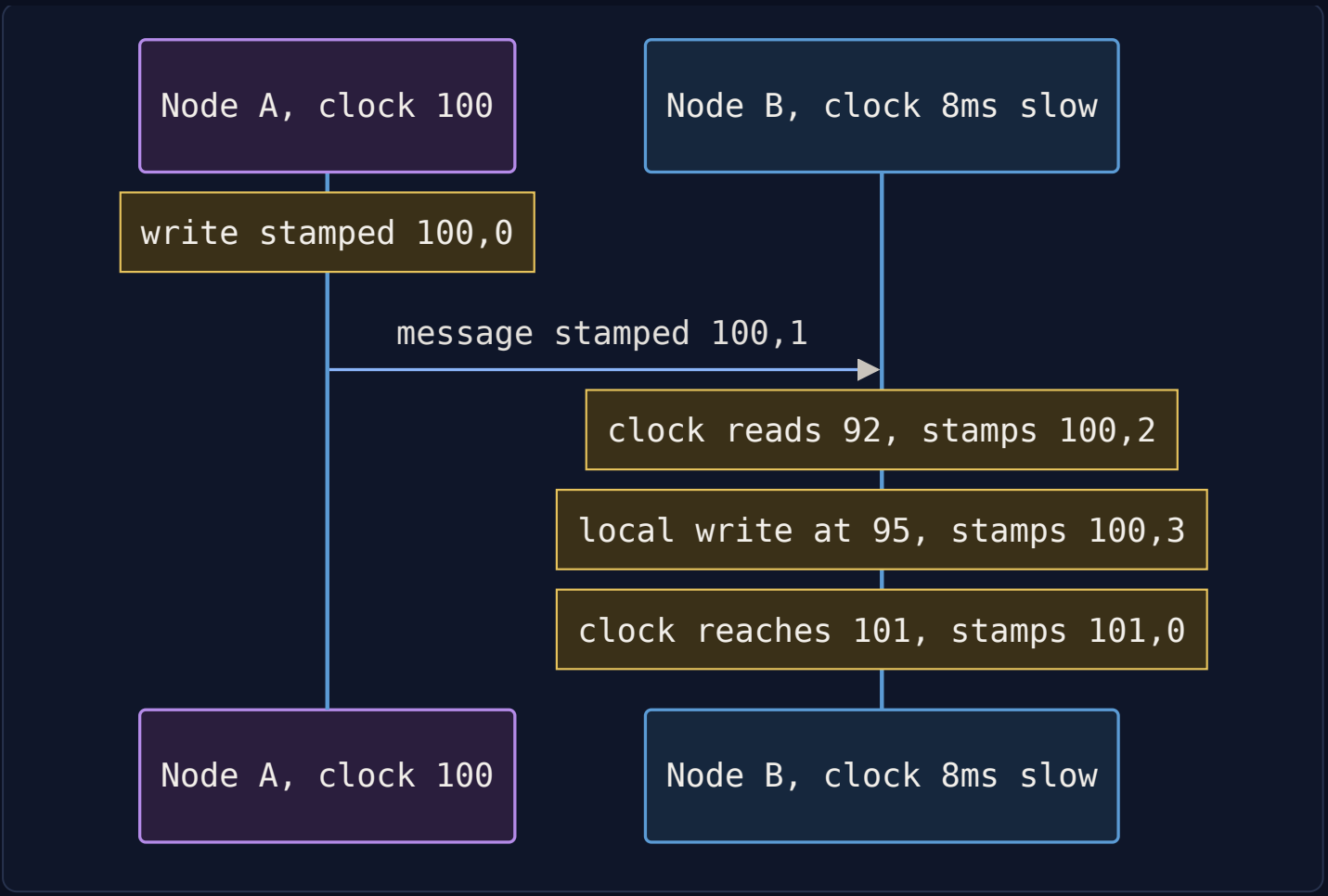
```
on local or send event:
    l_old = l
    l = max(l_old, pt)
    c = c + 1 if l == l_old else 0
    stamp event (and message) with (l, c)

on receive of message m with (l_m, c_m):
    l_old = l
    l = max(l_old, l_m, pt)
    if   l == l_old == l_m: c = max(c, c_m) + 1
    elif l == l_old:        c = c + 1
    elif l == l_m:          c = c_m + 1
    else:                   c = 0
    stamp event with (l, c)
```

Compare by `l`, then `c` (node ID breaks exact ties).

Worked example. A's clock reads 100 ms; B's is 8 ms slow and B's last stamp was (85, 0).

1. A writes at pt = 100: `l = 100`, `c = 0` → **(100, 0)**.
2. Still at 100, A sends to B: `l` unchanged, `c = 1` → **(100, 1)**.
3. B receives at its pt = 92: `l = max(85, 100, 92) = 100` = message's `l`, so `c = 2` → **(100, 2)**. Raw physical time would stamp 92, *before* the send: causality broken.
4. B writes locally at pt = 95: `l` stays 100, `c = 3` → **(100, 3)**.
5. B's clock reaches 101: `l = 101`, `c` resets → **(101, 0)**.



Properties.

- e happened before f ⇒ HLC(e) < HLC(f), as with Lamport clocks.
- `l` is never more than the clock skew ahead of the node's physical time.
- `c` stays small (bounded by node count in theory, tiny in practice). The paper packs 64 bits: 48 for `l` (NTP-format time, rounded), 16 for `c`.
- O(1) time per event and O(1) space per timestamp, vs O(N) for a vector clock.

What HLC does not do. It does not make clocks *accurate*: unrelated writes 1 ms apart on different nodes can be ordered "wrongly" if clocks disagree by more than 1 ms. External consistency ("a transaction starting after another commits sees it") needs an extra rule, such as CockroachDB's **uncertainty interval**.

Trade-offs

- **Pros**: causal, near wall time, fixed 64 bits, cheap; ideal for MVCC timestamps, snapshot reads and change-data-capture ordering.
- **Cons**: relies on a configured maximum skew. A node whose clock jumps far ahead drags everyone's `l`, so bad-clock nodes must be rejected or fenced.
- **Cannot detect concurrency**: any two HLC stamps are ordered, even for concurrent events, so it is not a conflict detector.

Real-world examples

- **CockroachDB** stamps transactions with HLC. `--max-offset` (default 500 ms) is the maximum allowed clock offset; a read treats values stamped within that offset above its own timestamp as **uncertain** and restarts higher. Nodes that drift too far shut themselves down.
- **MongoDB** uses an HLC for **cluster time**, powering causally consistent sessions across replica sets and shards.
- **YugabyteDB** uses HLC for distributed transactions.

In the interview

Q: Why not order by NTP timestamps?

A: Clocks disagree by milliseconds or more and can jump backwards, so a reply can precede its request. HLC never goes backwards and always orders a receive after its send.

Q: HLC vs vector clock?

A: HLC is a fixed-size scalar respecting causality and near real time, but cannot say two events were concurrent; vector clocks can, but grow with node count.

Q: How does CockroachDB handle skew?

A: A maximum offset: reads seeing values slightly newer than their timestamp (within the offset) restart later; nodes drifting too far stop serving.

TL;DR

HLC = max physical time heard so far plus a tie counter: causal like Lamport, within clock skew of real time, 64 bits. CockroachDB, MongoDB and YugabyteDB use it for transaction and snapshot timestamps.

Sources

- Kulkarni, Demirbas, Madappa, Avva, Leone, "Logical Physical Clocks and Consistent Snapshots in Globally Distributed Databases", OPODIS 2014 (UB tech report 2014-04): https://cse.buffalo.edu/tech-reports/2014-04.pdf
- Lamport, "Time, Clocks, and the Ordering of Events in a Distributed System", CACM 1978: https://lamport.azurewebsites.net/pubs/time-clocks.pdf
- CockroachDB docs, `cockroach start` (`--max-offset`, default 500 ms) and transaction layer: https://www.cockroachlabs.com/docs/stable/architecture/transaction-layer
- Tyulenev et al., "Implementation of Cluster-wide Logical Clock and Causal Consistency in MongoDB", SIGMOD 2019: https://doi.org/10.1145/3299869.3314049

Google TrueTime and commit-wait

What it is

TrueTime is the clock service in Google **Spanner** (Corbett et al., OSDI 2012). `TT.now()` returns an **interval** `[earliest, latest]` guaranteed to contain true absolute time; half its width is **epsilon (ε)**, the uncertainty. **Commit-wait** uses it: after choosing a commit timestamp, wait until it is definitely past before making writes visible.

The problem it solves

Spanner promises **external consistency** (the strongest form of linearizability for transactions): if T1 commits before T2 starts in real time, T2's timestamp is larger and T2 sees T1's writes. With ordinary clocks off by milliseconds, a later transaction on a slow-clock server could get a *smaller* timestamp, and a snapshot read could miss a write that had already finished. TrueTime makes clock error **explicit and bounded**; commit-wait turns the bound into a guarantee.

How it works

Building the interval.

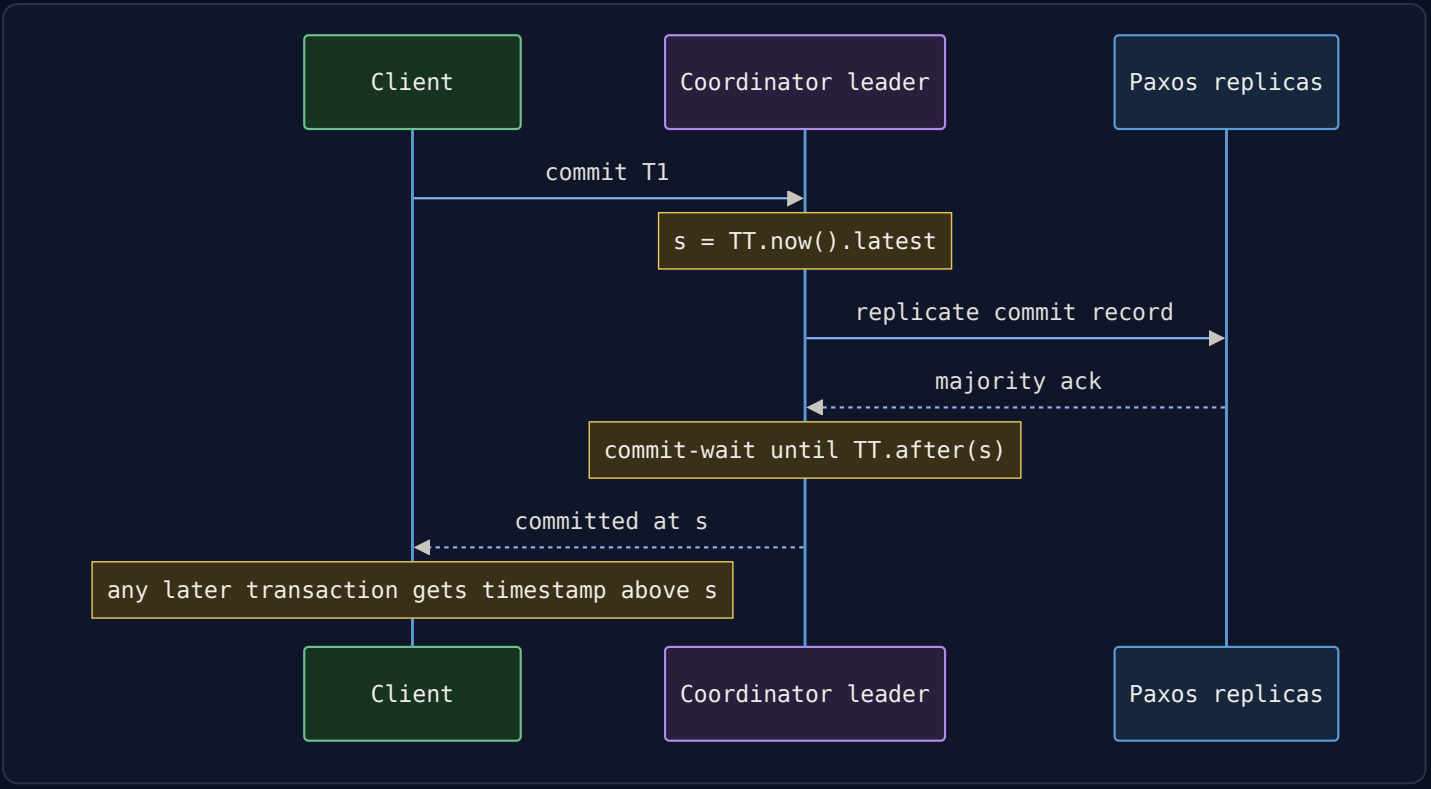
1. Each data centre has several **time masters**: most with GPS receivers, the rest (**Armageddon masters**) with atomic clocks, which fail in different ways.
2. A daemon per machine polls several masters (every 30 s in the paper), rejects outliers with a variant of Marzullo's algorithm, and syncs the local clock.
3. Between polls it assumes worst-case drift (200 µs/s in the paper): up to 6 ms after 30 s, plus about 1 ms of network delay to the masters.
4. So ε follows a **sawtooth** from about 1 ms after a poll to about 7 ms before the next; about 4 ms on average.

API: `TT.now()` → `[earliest, latest]`; `TT.after(t)` true if t has definitely passed (`t < now.earliest`); `TT.before(t)` true if t has definitely not arrived (`t > now.latest`).

Commit-wait:

1. A read-write transaction acquires locks (two-phase locking).
2. The coordinator leader picks `s = TT.now().latest`, which is not in the past.
3. It replicates the commit record through Paxos (plus 2PC across shards).
4. **Commit-wait:** wait until `TT.after(s)`.
5. Then release locks and reply "committed".

Worked example, ε = 4 ms. At true time 1000.000 s, `TT.now()` = `[999.996, 1000.004]`, so s = 1000.004. It waits until `earliest > 1000.004`, at true time about 1000.008: about 2ε = 8 ms. Paxos replication runs during the wait (often longer than 8 ms across zones), hiding much of it; the paper measured commit-wait at about 5 ms in a single-replica test. Any T2 starting after the client hears "committed" starts after 1000.008, and its timestamp ≥ its own `TT.now().latest` ≥ true time > 1000.004, so it is ordered after T1.



Why it pays off. Timestamps match real-time order, so a **read-only transaction** or **snapshot read** at t needs no locks: any replica caught up to t can answer. Spanner leaders also use TrueTime to keep their **leader leases** (10 s default) from overlapping.

Cost. About 2ε added write latency, O(1) work; smaller ε means faster commits, hence Google's GPS and atomic-clock investment.

Trade-offs

- **Write latency includes the uncertainty**: unreachable time masters grow ε and every commit.
- **Specialised infrastructure**: GPS antennas and atomic clocks per data centre plus monitoring; machines whose clocks exceed the assumed bound are evicted.
- **Reward**: lock-free consistent reads anywhere, at any past timestamp.
- **Alternatives without special hardware**: CockroachDB uses HLC with a larger maximum offset and restarts uncertain reads instead of waiting on every write. Cloud providers expose clock error bounds too: AWS ClockBound returns an `[earliest, latest]` window like TrueTime.

Real-world examples

- **Google Spanner** and Cloud Spanner (TrueTime plus commit-wait); Google notes that when uncertainty is below the Paxos round trip, commit-wait adds nothing to commit latency.

- **CockroachDB**: a similar opt-in "commit wait" for global tables, on HLC and a configured clock offset.
- **AWS ClockBound** with Amazon Time Sync Service: a bounded-uncertainty clock API for EC2.

In the interview

Q: What is commit-wait and why is it needed?

A: Having stamped the commit at the top of the uncertainty interval, the leader waits until that time is certainly past before acknowledging, so any later-starting transaction gets a larger timestamp: external consistency.

Q: How long is the wait?

A: About 2ε, a few ms in Spanner (ε was 1 to 7 ms, about 4 ms average, in the 2012 paper), overlapping Paxos replication, so the added latency is often small.

Q: Can I build this on AWS or with NTP?

A: Yes, given a trustworthy error bound; NTP's is larger, so waits are longer. That is why most open-source databases use HLC plus uncertainty restarts.

TL;DR

TrueTime returns an interval guaranteed to contain true time, a few ms wide thanks to GPS and atomic clocks. Spanner stamps commits at the top of the interval and waits until that time has passed, so timestamps follow real-time order and global snapshot reads need no locks.

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Version vectors vs vector clocks (and dotted version vectors)

What it is

All three are lists of counters, one per participant, used to decide whether one thing **happened before** another or they are **concurrent** (neither knew of the other).

- **Vector clock** (Fidge and Mattern, 1988): causality between **events**; each process keeps one counter per process, updated on every local event, send and receive.
- **Version vector** (Parker et al., 1983): causality between **versions of data**; one counter per **replica** (the server coordinating a write), changed only on writes and merges.
- **Dotted version vector (DVV)** (Preguiça, Baquero, Almeida et al., 2010): a version vector plus a **dot**, one `(replica, counter)` pair naming the exact write that created a value. It fixes false conflicts that plain version vectors create when many clients write through one server.

How a leaderless store uses these to detect conflicts is in Module 7 — Vector Clocks; here, the differences and the DVV fix.

The problem it solves

In a leaderless or multi-leader (Dynamo-style) database, two clients can update a key on different replicas at once. Is B a newer edit of A (keep B), or independent (keep both as **siblings** for the application to merge)? Timestamps cannot answer (Module 0 — Clocks and ordering); counter vectors can.

How it works

Comparing V and W (all three): V ≤ W if every entry of V ≤ W's. If V ≤ W and V ≠ W, V happened before W (W supersedes). If neither ≤ the other, they are **concurrent**: a conflict. `{A:2, B:1}` vs `{A:3, B:1}`: first is older. `{A:2, B:1}` vs `{A:1, B:2}`: concurrent.

	Vector clock	Version vector
Orders	Events (messages, computations)	Versions of a data item
One entry per	Process	Replica that coordinates writes
Updated on	Every event, send and receive	Writes and merges only
Size	Grows with processes involved	Grows with replicas (usually 3 to 5)
Typical use	Debugging, causal message delivery	Conflict detection in replicated stores

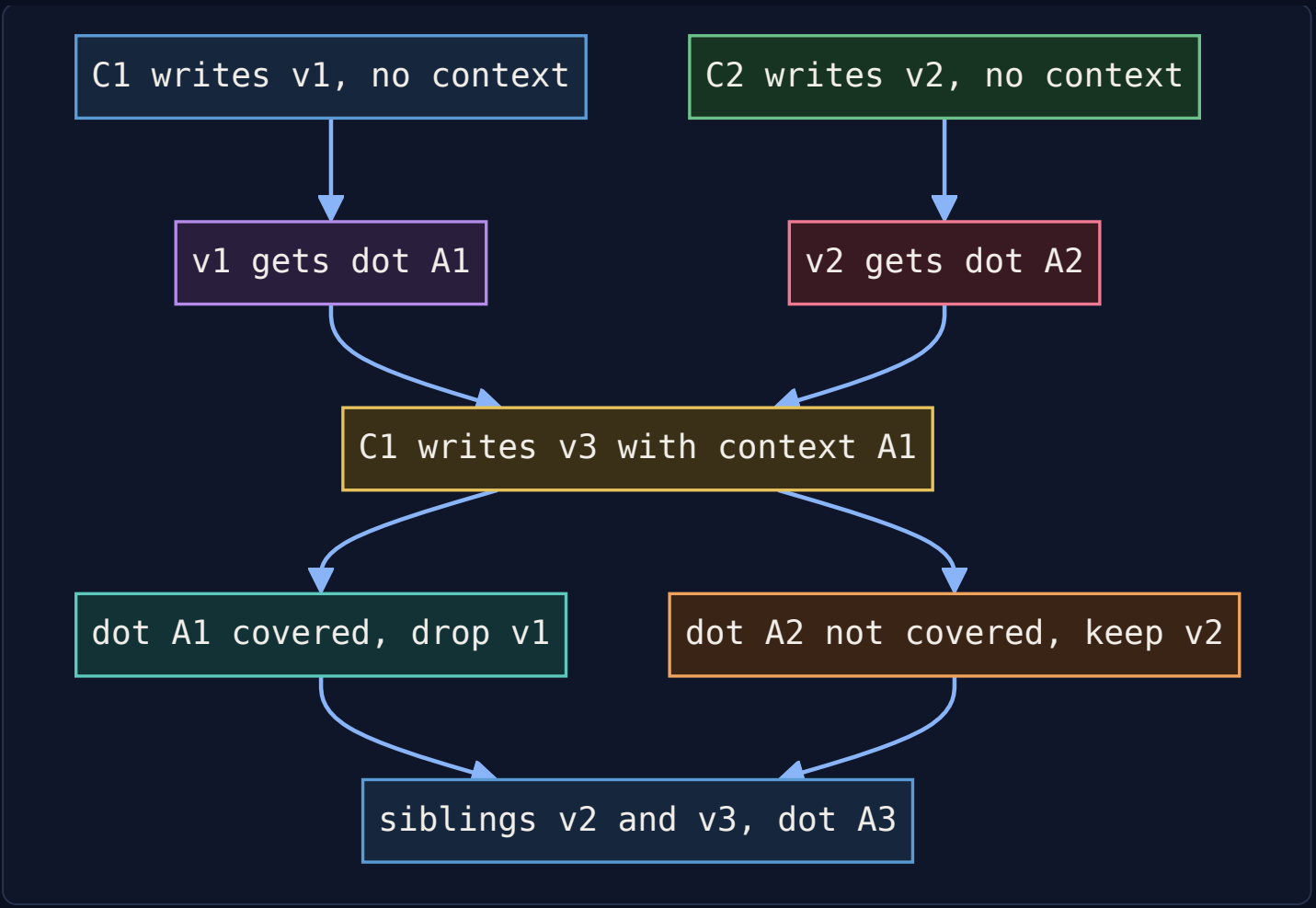
People often say "vector clock" for both, as the Dynamo paper does. It matters because **client-ID** entries grow with clients (millions), while **server-ID** entries stay small but lose information: the problem DVVs solve.

Sibling explosion, step by step. Replica A coordinates all writes to `cart`; vectors use server IDs.

1. C1 writes `v1` with no context (never read the key): A stores v1 with `{A:1}`.
2. C2, also without reading, writes `v2`. A can't tell if v2 replaces v1, keeps both as siblings and bumps to `{A:2}`; both siblings share `{A:2}`.
3. C1 writes `v3` with its context from step 1, `{A:1}`: v3 should replace v1 but stay concurrent with v2.
4. A cannot tell *which* sibling `{A:1}` covers, so it keeps v1, v2 and v3. With retries and many clients, siblings pile up without bound: **sibling explosion**, caused by **false concurrency**.

Same steps with DVVs.

1. v1 gets dot `(A,1)`; store `{v1@(A,1)}`, context returned `{A:1}`.
2. v2 (empty context) gets dot `(A,2)`; store `{v1@(A,1), v2@(A,2)}`.
3. v3 arrives with context `{A:1}`, which **covers** dot (A,1) (drop v1) but not (A,2) (keep v2); v3 gets `(A,3)`.
4. Store `{v2@(A,2), v3@(A,3)}`: exactly the two truly concurrent values.



DVV-style write at replica r:

```
on write(value, context):            # context is a version vector
    siblings = [s for s in siblings if not covered(s.dot, context)]
    vv[r] = vv[r] + 1
    siblings.append(Sibling(value, dot=(r, vv[r])))
covered(dot, ctx): return dot.counter <= ctx.get(dot.replica, 0)
```

Complexity. Comparison O(R) for R entries; version vector O(R) space per key (R = replicas, small); DVV adds one dot per sibling; vector clocks over P processes cost O(P), possibly large.

Trade-offs

- **Vector clocks** capture full event causality but grow with processes: better for debugging and messaging than storage.
- **Replica-keyed version vectors** stay small but, without dots, create false conflicts when several clients write through one replica.
- **Client-keyed vectors** are precise but unbounded; Dynamo truncated them at a threshold (the paper gives 10 pairs), risking incorrect reconciliation.
- **DVVs** keep the small server-keyed size and remove false concurrency, with slightly more complex code.
- **All only detect conflicts.** The application or a **CRDT** (conflict-free replicated data type, which merges automatically) must resolve them. Many systems use **last-write-wins** instead: simpler, but silently drops concurrent writes.

Real-world examples

- **Amazon Dynamo** (2007): vector clocks of `(node, counter)` pairs per object; siblings returned to the client to merge.
- **Riak** 2.0 added DVVs as an option and recommends them to bound siblings.
- **Project Voldemort** (LinkedIn): vector clocks in a Dynamo-style design.
- **Apache Cassandra** avoids them, using per-column last-write-wins timestamps.

In the interview

Q: Vector clock vs version vector?

A: A vector clock orders events between processes and ticks on every event and message; a version vector orders data versions across replicas and ticks only on writes. Same comparison rule, different purpose and size.

Q: Why did Riak move to dotted version vectors?

A: With one counter per server, values written by several clients through one server shared one vector, so the server couldn't tell which a new write replaced and siblings piled up. A dot names each value's write, so a new write's context discards exactly the values it saw.

Q: `{A:3, B:1}` vs `{A:2, B:2}`?

A: Neither dominates: concurrent. Keep both as siblings and merge on read (application logic, a CRDT, or last-write-wins if losing one is acceptable).

TL;DR

Vector clocks order events between processes; version vectors order data versions across replicas; both flag concurrency when neither dominates. DVVs tag each value with its creating write, avoiding false conflicts and sibling explosion with small vectors.

Sources

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Consistent hashing variants: jump, rendezvous, Maglev and bounded loads

What it is

Any method that maps a key to one of N servers so that changing N moves as few keys as possible is a **consistent hashing** scheme. The **hash ring with virtual nodes** (Module 3 — Consistent Hashing) is one; this section covers four others:

- **Jump consistent hash**: a few lines of arithmetic, no memory, for servers numbered 0 to N−1.
- **Rendezvous hashing** (**highest random weight, HRW**): every server "bids" for the key; the highest bid wins.
- **Maglev hashing**: Google's load-balancer method; a precomputed fixed-size table makes each lookup one array read.
- **Consistent hashing with bounded loads**: a ring plus a hard cap so no server exceeds, say, 25% above average.

Terms: a **key** is what you route on (user ID, URL, connection 5-tuple); a **bucket**/**backend**/**server** is the destination; **disruption** (**movement**) is the share of keys that change destination when the server set changes. The minimum going from N to N+1 servers is 1/(N+1).

The problem it solves

The ring's costs at scale: (1) **memory and lookup time** (1,000 servers × 200 vnodes = 200,000 points, binary-searched per lookup); (2) **imperfect balance** (even with 100–200 vnodes the biggest server typically carries several percent more than the smallest; more vnodes, more memory); (3) **hot keys** (it balances *keys*, not *load*, so a popular or costly key overloads a server however even the arcs are). Each variant attacks one of these.

How it works

1. Jump consistent hash (Lamping and Veach, Google, 2014). Grow the cluster one bucket at a time; when going from b to b+1 buckets each key should move to the new bucket with probability exactly 1/(b+1). Simulating that is O(N); a key-seeded pseudo-random generator instead *jumps* straight to the next bucket count where this key moves. Expected jumps ≈ ln N (about 7 for N = 1,000).

```
function jump_hash(key, num_buckets):
    b = -1            # last bucket the key moved to
    j = 0             # next bucket count at which it might move
    while j < num_buckets:
        b = j
        key = key * 2862933555777941757 + 1          # 64-bit LCG step
        j = floor((b + 1) * 2^31 / ((key >> 33) + 1))
    return b
```

Worked example (published constant): key 99999 is in bucket 0 with 1 bucket, bucket 1 for sizes 2 to 7, then jumps to bucket 7 when the 8th appears and stays for sizes 9 and 10. It moves exactly twice from 1 to 10, each time *into the newly added bucket*, never sideways. 100,000 keys going from 10 to 11 buckets moved 9,042 (9.04%) vs the ideal 1/11 = 9.09%.

Cost: O(ln N) time, O(1) memory, essentially perfect balance. Catch: buckets are numbered 0..N−1 and change only at the **end**; if bucket 3 of 10 dies you cannot delete "3". The fix is a layer mapping bucket numbers to replica groups whose members can change, which is why the paper targets sharded storage, not caches where any node can vanish.

2. Rendezvous hashing / HRW (Thaler and Ravishankar, 1996–1998). For each key compute `score = hash(key, server)` per server and pick the highest.

Server	hash("photo:7", server), scaled 0..1
S1	0.62
S2	**0.91**
S3	0.17

`photo:7` goes to S2. If S2 dies, S1 (0.62) wins; keys whose top scorer was S1 or S3 are untouched. Exactly the dead server's keys move, spread evenly over survivors (each key's runner-up is random). Adding S4 steals only keys where S4 scores highest, about 1/4.

- **Replication for free:** the top 3 scores give 3 replicas in stable order.
- **Weights:** `score = -w / ln(h)` (h = hash scaled to (0,1), w = weight); weight 2 wins about twice as often.
- **Cost:** O(N) hashes per lookup: nothing for 10 servers, too slow for 10,000, where a tree of rendezvous choices (group, then server) is used.

3. Maglev hashing (Google, NSDI 2016). Maglev, Google's software network load balancer (serving since 2008), must send every packet of a TCP connection to the same backend at line rate, so lookup is one array read into a table of prime size M (65,537 by default in the paper).

1. Per backend i: `offset = h1(name) mod M`, `skip = h2(name) mod (M-1) + 1`.
2. **Preference list** `permutation[i][j] = (offset + j * skip) mod M`; M prime means every slot is visited once.
3. Fill in rounds: each backend in turn claims its next preferred empty slot, until all M are full.
4. Lookup: `backend = table[hash(packet 5-tuple) mod M]`.

M = 7, three backends:

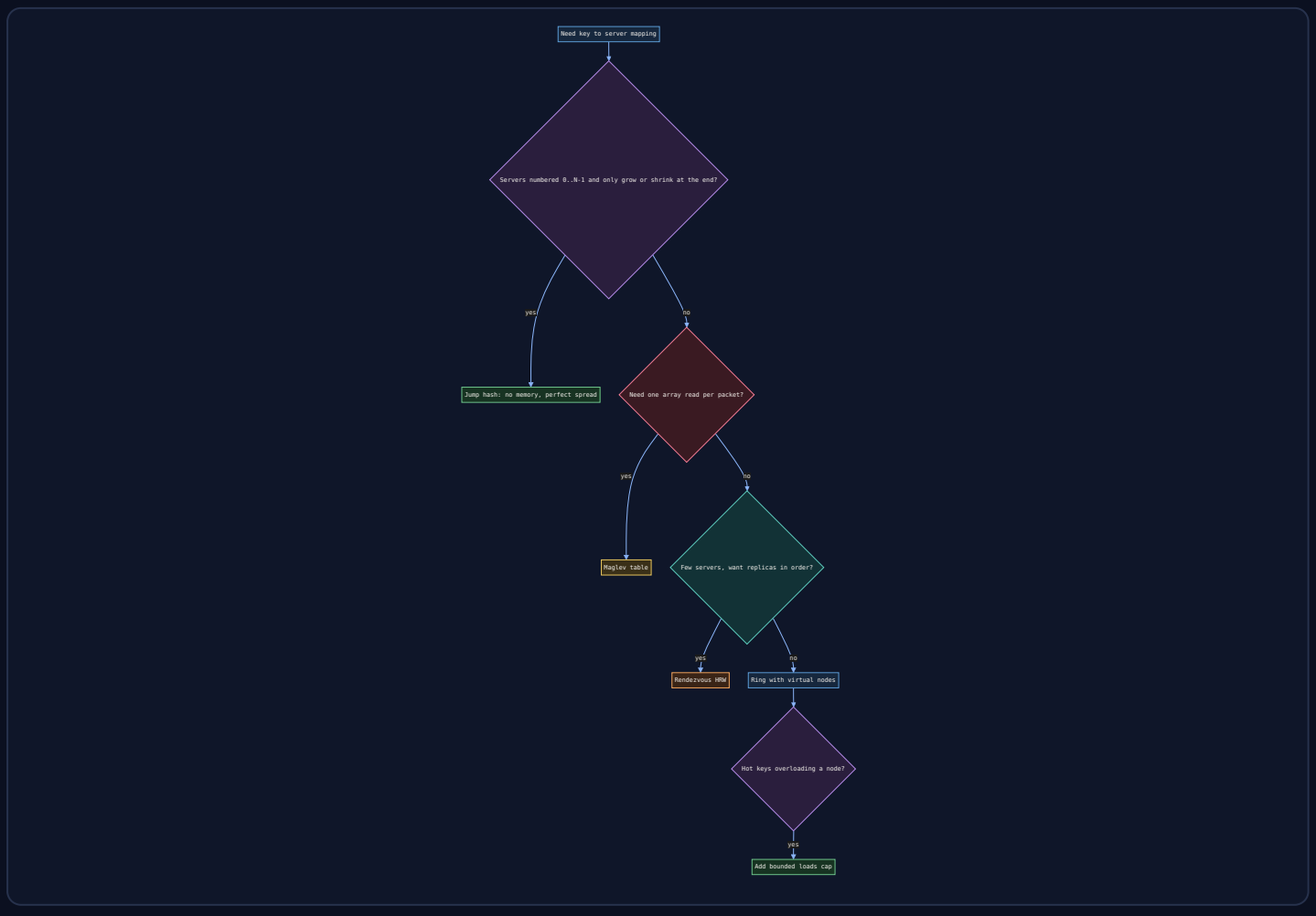
Backend	(offset, skip)	Preference list
B0	(3, 4)	3, 0, 4, 1, 5, 2, 6
B1	(0, 2)	0, 2, 4, 6, 1, 3, 5
B2	(3, 1)	3, 4, 5, 6, 0, 1, 2

Filling: B0→3, B1→0, B2→4 (3 taken), B0→1, B1→2, B2→5, B0→6. Table `[B1, B0, B1, B0, B2, B2, B0]` : 3, 2, 2 slots, as even as 7 allows. Remove B1 and refill: `[B0, B0, B0, B0, B2, B2, B2]`. B1's slots 0 and 2 had to move; slot 6 also moved B0→B2 unnecessarily. That is Maglev's deliberate trade: **near-perfect balance and O(1) lookup for a little extra disruption**, hidden by a per-machine connection-tracking table that keeps existing flows on their backend. With a 65,537-entry table the paper found Karger-style ring hashing and rendezvous would need 29.7% and 49.5% over-provisioning to absorb imbalance, while Maglev was almost perfectly even. Envoy reports the flip side: on host removal Maglev moves about twice as many keys as its ring hash.

4. Consistent hashing with bounded loads (Mirrokni, Thorup, Zadimoghaddam, Google, 2016). Balance factor c = 1 + ε (e.g. 1.25). With m active keys and n servers, no server holds more than `ceil(c * m / n)`. Lookup: hash onto the ring, take the first server clockwise if under its cap, else keep walking to the next with spare capacity.

Worked example: 12 keys, 4 servers, ε = 0.25, cap = ceil(3.75) = 4. The plain ring gives A 6, B 3, C 2, D 1; A keeps 4 and 2 walk on, giving perhaps 4, 4, 3, 1. The hottest server falls from 2× average (3) to 1.33×.

The paper proves the cap always holds and moves stay small (about $O(1/\varepsilon^2)$ times the ideal for small ε). Vimeo implemented it in HAProxy for its video cache tier and reported almost 8× less cache bandwidth; HAProxy exposes `hash-balance-factor` (a percentage, e.g. 150 = 1.5× average); Google Cloud Pub/Sub used it internally.



Comparison (N servers, V vnodes each, table size M):

Property	Ring + vnodes	Jump hash	Rendezvous (HRW)	Maglev	Bounded loads
Lookup time	O(log(N·V)) binary search	O(ln N)	O(N)	O(1) array read	Ring lookup plus a short walk
Memory	O(N·V) points	O(1)	O(N) server list	O(M), e.g. 65,537 slots	Ring plus live load counts
Keys moved on change	≈1/N	Exactly the minimum	Exactly the minimum	Minimum plus a little	More than the ring, bounded by ε
Balance	Good with many vnodes	Perfect	Good	Near perfect	Hard cap of (1+ε)× average
Remove an arbitrary server	Yes	No, only the last	Yes	Yes	Yes
Weights	More vnodes	Not natively	Weighted score	More table slots	Weighted caps
Typical home	Dynamo, Cassandra, memcached clients	Sharded storage, Guava library	Small caches, CDNs, replica placement	L4 load balancers, Envoy	Cache tiers with hot keys, HAProxy

Trade-offs

- **Jump**: fastest and smallest, but cannot drop a middle server, so needs a stable numbering layer.
- **Rendezvous**: simplest to reason about, ordered replicas for free, lookup linear in servers.
- **Maglev**: O(1) lookups and excellent balance for fixed memory per service and slightly more churn; bigger tables rebuild slower (1.8 ms at 65,537 entries vs 22.9 ms at 655,373).
- **Bounded loads**: the only one that looks at *actual load*, protecting against hot keys, but needs shared current load numbers and gives up some stickiness (an overflowing key may land elsewhere moment to moment).

Real-world examples

- **Google Maglev** and **Envoy's Maglev load balancer** (default table size 65,537) for connection-level balancing.
- **Guava's** `Hashing.consistentHash` implements jump hash.
- **Rendezvous hashing** in caching proxies and object-placement schemes where replica order matters.
- **Vimeo with HAProxy** and **Google Cloud Pub/Sub**: bounded loads.

In the interview

Q: L4 load balancer at millions of packets per second: ring hash or Maglev?

A: Maglev: one array read per packet, near-even split; the small extra movement on backend changes is covered by connection tracking that pins existing flows.

Q: Why not jump hash for a memcached fleet?

A: Any cache node can die and jump hash can only remove the highest bucket; removing node 3 of 10 remaps most keys without an indirection layer. A ring or rendezvous handles arbitrary removal.

Q: A celebrity key overloads its cache node. Do more virtual nodes help?

A: No: vnodes balance key counts, not heat. Bounded loads caps each node at (1+ε) × average and spills overflow clockwise; for one extremely hot key also replicate it or add a local cache.

Q: How does rendezvous give replicas?

A: Sort servers by score for the key and take the top R; if one dies the next steps up, and every client computes the same order without coordination.

TL;DR

All schemes keep movement near the 1/N minimum. Jump: no memory, only the last bucket changes. Rendezvous: simple, ordered replicas, O(N) lookup. Maglev: one array read, near-perfect balance, slightly more churn. Bounded loads: a hard per-server load cap that absorbs hot keys.

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HyperLogLog internals

What it is

HyperLogLog (HLL) is a **sketch** (small fixed-size summary of a big stream) estimating **cardinality**, the number of *distinct* items (Flajolet, Fusy, Gandouet, Meunier, 2007). With m small counters (**registers**) its typical relative error is 1.04/√m. Redis uses m = 16,384 six-bit registers: about 12 KB counts billions of visitors with about 0.81% error. The intuition is in Module 0 — Probabilistic data structures; here are the internals.

The problem it solves

Exact distinct counting remembers every item: 100 million 8-byte user IDs is 800 MB per counter, and you may want one per page per day. HLL uses kilobytes, and two HLLs **merge** into the union's distinct count, which plain counts cannot (users overlap).

How it works

Intuition. In a random hash, k leading zeros occur about 1 in 2^k; a longest run of 10 suggests about 2^10 ≈ 1,000 distinct hashes. Duplicates hash identically, so they don't matter. One counter is noisy (one lucky hash ruins it), so HLL keeps many and averages.

Steps (precision p, m = 2^p registers):

1. Hash the item to 64 bits.
2. First p bits = **register index**.
3. In the remaining bits, ρ ("rho") = position of the first 1-bit (leading zeros + 1).
4. `register[index] = max(register[index], ρ)`; registers only grow.
5. Estimate with a **harmonic mean**: `E = α_m · m² / Σ 2^(−register[j])`; bias constant α_m ≈ 0.7213/(1 + 1.079/m) for m ≥ 128, 0.673 for m = 16.
6. **Small-range correction:** if E ≤ 2.5·m and some registers are 0, use **linear counting** `m · ln(m / V)` (V = zero registers), far more accurate when most registers are empty.

Example 1 (update). p = 2, 4 registers. Hash `10 0001 1...`: `10` selects register 2; `0001` gives ρ = 4; register 2 goes from 2 to max(2, 4) = 4. A hash `10 1...` has ρ = 1 and changes nothing.

Example 2 (estimate). m = 16, registers `3,2,4,1,2,5,3,2,1,3,2,4,2,3,1,2`: Σ 2^(−R) = 3.656, E = 0.673 × 16² / 3.656 ≈ 47 > 40 (2.5 × 16), so about 47 items. Error at m = 16 is 1.04/4 = 26%, hence real systems use thousands of registers.

Example 3 (small range). Registers `3,0,2,0,1,0,0,2,0,1,0,0,0,1,0,0`: raw 14.2 < 40, V = 10, so 16 × ln(16/10) ≈ 7.5, about 8 items.

Why a harmonic mean? Registers `2, 2, 2, 20` (one freak hash): the arithmetic mean of 2^R is (4 + 4 + 4 + 1,048,576)/4 ≈ 262,000, ruined; the harmonic mean 4 / (0.25 + 0.25 + 0.25 + 2^(−20)) ≈ 5.3 barely notices. It is dominated by small values, right when noise is occasional huge outliers.

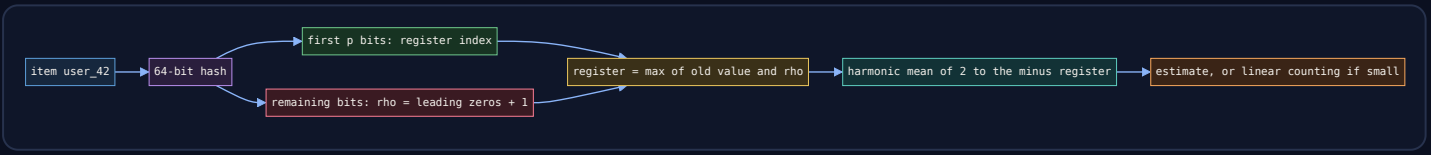
Error and size (≈ 1.04/√m):

p	m	Error	Dense size at 6 bits/register
10	1,024	3.25%	768 bytes
14	16,384	0.81%	12 KB
16	65,536	0.41%	48 KB

4× memory halves the error. Six bits suffice because ρ never exceeds 64 for a 64-bit hash.

Sparse representation. A fresh 12 KB HLL that has seen 5 items is 99.9% zeros. **HyperLogLog++** (Heule, Nunkesser, Hall, Google, EDBT 2013) adds a 64-bit hash (no large-range correction needed), an empirical bias-correction table for the awkward middle range, and a **sparse mode** storing only non-zero (index, ρ) pairs as a sorted compressed list at higher precision (p' = 25), converting to dense when the list would outgrow the array. Redis uses run-length opcodes: `XZERO:1000, VAL:2, ZERO:19, VAL:3 x2, XZERO:15362` encodes three non-zero registers in 7 bytes instead of 12 KB, switching to dense past `hll-sparse-max-bytes` (3,000 bytes default).

Merging. Same p and hash: `merged[j] = max(a[j], b[j])`, exactly the union stream's sketch. Keep one HLL per hour and merge 24 for a day, or one per shard merged at query time. Intersections only via |A ∩ B| = |A| + |B| − |A ∪ B|, where three estimates' errors add up: poor for small overlaps.



Complexity: O(1) per add (one hash, one max); O(m) per estimate (Redis caches the last estimate until a register changes); O(m) merge and memory.

Trade-offs

- Relative error (about 0.8% at 12 KB), never zero: fine for dashboards, not billing.
- No deletions (registers only grow): for "last 7 days" keep daily HLLs and merge the window.
- Counts only; cannot list items.
- Merge needs identical p and hash function across producers.
- Intersections and small overlaps are inaccurate; Theta or KMV sketches handle set operations better.

Real-world examples

- **Redis** `PFADD`, `PFCOUNT`, `PFMERGE`: up to 12 KB per key, 0.81% standard error.
- **Google** uses HLL++ internally and in BigQuery's approximate count functions.
- Druid, ClickHouse, Presto/Trino and Spark offer `approx_count_distinct`-style functions on HLL variants.

In the interview

Q: Daily unique visitors per page for 10 million pages?

A: One HLL per page per day: up to 120 GB dense, but most pages stay sparse, so far less. Weekly and monthly uniques come from merging daily sketches.

Q: Why can you merge HLLs but not add distinct counts?

A: Adding double-counts users in both sets. A register holds the max ρ seen, and the max over a union is the max of the two maxes, so register-wise max is exactly the union's sketch.

Q: What does 1.04/√m mean in practice?

A: With 16,384 registers the estimate is typically within ±0.81% (one standard error) and within about ±2.4% roughly 99% of the time (three).

TL;DR

HLL hashes items, uses p bits to pick a register and the leading-zero run of the rest as evidence of cardinality, keeps the max per register, and combines registers with a bias-corrected harmonic mean. 12 KB gives about 0.8% error; sketches merge by register-wise max; sparse encoding keeps small sketches tiny.

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Count-min sketch with conservative update

What it is

A **count-min sketch (CMS)** (Cormode and Muthukrishnan, 2005) estimates *how many times* each item appeared in a stream using a small fixed grid of counters; estimates can be too high, never too low. **Conservative update** (from Estan and Varghese's 2002 traffic-measurement work) changes the update rule to shrink the over-estimates. Basics: Module 0 — Probabilistic data structures.

The problem it solves

Per-key frequencies over a huge key space (requests per IP for rate limiting, clicks per URL, queries per term). A hash map needs memory per distinct key (maybe hundreds of millions); a CMS has a fixed size chosen from the accuracy needed, and merges across machines by adding grids.

How it works

Structure: d rows × w columns of counters, one independent hash per row.

Sizing: for estimate ≤ true + ε·N (N = total of all counts) with probability ≥ 1 − δ: `w = ceil(e / ε)`, `d = ceil(ln(1/δ))`, e ≈ 2.718. ε = 0.001, δ = 0.01 gives w = 2,719, d = 5: 13,595 counters, about 54 KB at 4 bytes, regardless of distinct keys. (Some implementations use w = 2/ε; Redis `CMS.INITBYPROB` with error 0.001 builds a 2,000-wide sketch.)

Update `add(x, c)`: for each row i, `table[i][h_i(x)] += c`. **Query** `estimate(x)`: `min over i of table[i][h_i(x)]`. Each counter holds x's count plus colliding counts; collisions only add, so the smallest is the least polluted.

Worked example. w = 5, d = 3:

Item	row 1	row 2	row 3
dog	3	4	0
cat	1	4	2
fox	1	0	3
owl (never seen)	1	4	0

Stream: dog ×5, cat ×3, fox ×2. Standard update:

```
row 1: [0, 5, 0, 5, 0]    cat 3 + fox 2 collide in column 1
row 2: [2, 0, 0, 0, 8]    dog 5 + cat 3 collide in column 4
row 3: [5, 0, 3, 2, 0]
```

cat = min(5, 8, 3) = 3, dog = min(5, 8, 5) = 5, fox = min(5, 2, 2) = 2 (all exact). owl, never seen = min(5, 8, 5) = **5**: a false over-estimate from collisions in every row.

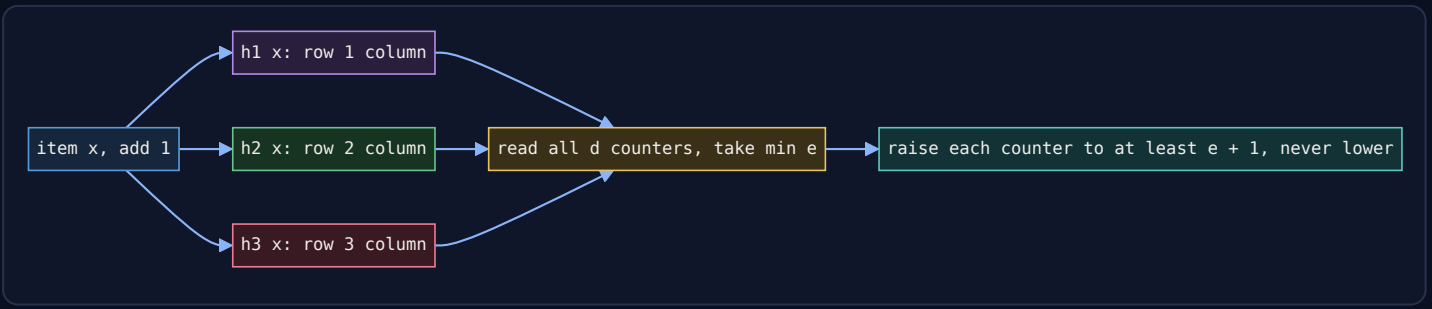
Conservative update. Before adding c to x, compute `e = min of x's counters`; the true count is now at most e + c, so raise each of x's counters to `max(counter, e + c)` and no further.

```
function add_conservative(x, c):
    e = min(table[i][h_i(x)] for i in 1..d)
    for i in 1..d:
        table[i][h_i(x)] = max(table[i][h_i(x)], e + c)
```

Same stream, one unit at a time:

- dog ×5: counters rise together to 5 (row 1 col 3, row 2 col 4, row 3 col 0).
- cat ×3: counters (0, 5, 0), estimate 0 → after three cats (3, 5, 3); row 2 col 4 **stays 5**, not 8.
- fox ×2: counters (3, 0, 0), estimate 0 → (3, 2, 2); row 1 col 1 **stays 3**, not 5.

owl = min(3, 5, 5) = **3** instead of 5. Real items are still never under-estimated; Estan and Varghese reported about an order of magnitude better accuracy on real traffic from this kind of optimisation.



Complexity: O(d) per update and query (d usually 4–8), O(w·d) memory.

Trade-offs

- **Error is relative to stream total N**, not the item: ε = 0.001, N = 10 million allows over-estimates up to 10,000. Excellent for heavy items, useless for rare ones; Redis says treat estimates below ε·N as noise.
- **Conservative update loses mergeability and deletes.** Standard sketches merge by cell-wise addition and allow negative updates; conservatively-updated sketches added together still over-estimate but lose the formal bound, and decrements are not allowed.
- **No item list**: a CMS cannot say *which* items are heavy; pair it with a heap (next section).
- **Aging**: counts grow forever; use time-bucketed sketches (one per minute) or periodically halve every counter.

Real-world examples

- **Redis** `CMS.INITBYPROB`, `CMS.INCRBY`, `CMS.QUERY`, `CMS.MERGE`.
- Router and switch traffic monitoring and DDoS detection (Estan and Varghese's motivation).

- **TinyLFU** cache admission (Caffeine, Java) uses a count-min-style sketch with periodic halving to decide whether a new key deserves a slot.
- Flink and Spark provide CMS helpers for approximate frequencies.

In the interview

Q: Can a CMS under-count?

A: Not with non-negative updates: each counter is the true count plus collisions, so the minimum is still ≥ true.

Q: Size it for "error at most 0.1% of traffic with 99% confidence".

A: ε = 0.001 → w = 2,719; δ = 0.01 → d = ceil(ln 100) = 5; about 13,600 counters, about 54 KB at 32 bits.

Q: What does conservative update buy and cost?

A: It raises only counters below the new lower-bound estimate, so collisions inflate far less, often many times smaller errors; you lose clean merging and deletions.

TL;DR

A CMS is a d × w counter grid: add increments one counter per row, estimate is the minimum, never too low, too high by at most ε·N with probability 1 − δ. Conservative update raises only counters below min + c, sharply cutting over-counting at the cost of mergeability and deletes.

Sources

- Cormode and Muthukrishnan, "An Improved Data Stream Summary: The Count-Min Sketch and its Applications": https://www.cs.tufts.edu/~nr/cs257/archive/graham-cormode/count-min.pdf (J. Algorithms 55(1), 2005)
- Estan and Varghese, "New Directions in Traffic Measurement and Accounting", SIGCOMM 2002: https://conferences2.sigcomm.org/sigcomm/2002/papers/account.html
- Redis count-min sketch docs: https://redis.io/docs/latest/develop/data-types/probabilistic/count-min-sketch/

Heavy hitters: Misra-Gries, Space-Saving and streaming top-k

What it is

Heavy hitters make up a large share of a stream (e.g. every search term above 1% of queries); **top-k** asks for the k most frequent. Two classic algorithms use k counters and keep the *items themselves*, which a CMS cannot: **Misra-Gries** (1982; when full, decrement everyone) and **Space-Saving** (Metwally, Agrawal, El Abbadi, 2005; when full, evict the smallest and let the newcomer inherit its count).

The problem it solves

"Top 10 hashtags now?", "which 100 IPs send the most traffic?" Exact answers need a counter per distinct item (maybe hundreds of millions). These keep k entries and guarantee every truly heavy item is among them, with bounded count error.

How it works

Stream (n = 10): `A B A C D A B A E A`; true A = 5, B = 2, C = D = E = 1; **k = 3**.

Misra-Gries. If x has a counter, increment; else if a slot is free, add x:1; else decrement **every** counter and drop zeros (x is not stored).

Arrival	Counters after
A, B, A, C	A:2, B:1, C:1
D (full)	decrement all: A:1 (B, C dropped)
A	A:2
B	A:2, B:1
A	A:3, B:1
E	A:3, B:1, E:1
A	A:4, B:1, E:1

Guarantee: stored counts **under-estimate** by at most n/(k+1) = 2.5 (A reported 4, true 5), and any item above n/(k+1) is in the table. Each decrement discards k+1 occurrences (k counters plus the arrival), so it happens at most n/(k+1) times.

Space-Saving. If tracked, increment; else if free, add with count 1, error 0; else replace the **minimum** item (count `min`) with x, count = min + 1, error = min.

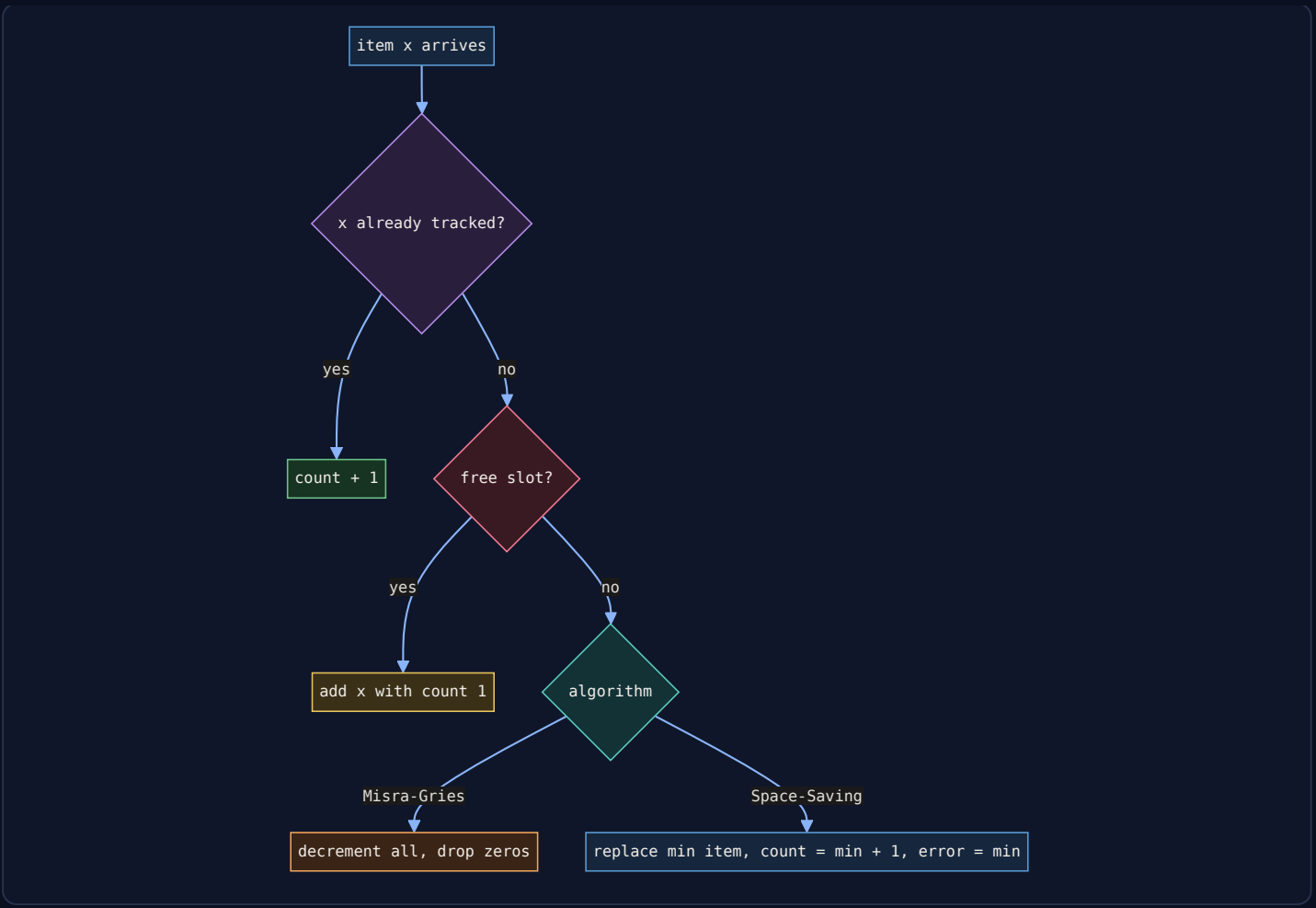
Arrival	Counters after (count, error)
A, B, A, C	A:2, B:1, C:1
D	evict B (min 1): A:2, C:1, D:2(err 1)
A	A:3, C:1, D:2
B	evict C (min 1): A:3, D:2, B:2(err 1)
A	A:4, D:2, B:2
E	evict D (min 2): A:4, B:2, E:3(err 2)
A	A:5, B:2, E:3

Counts **over-estimate** by at most n/k, and `count − error` is a guaranteed lower bound (E: true count between 1 and 3; A exact at 5). Space-Saving is usually more accurate than Misra-Gries for the same k, especially on skewed (Zipf-like) data. Updates are O(1) with a "stream summary" (linked buckets of equal counts) or O(log k) with a min-heap.

```
function space_saving(x):
    if x in counters: counters[x].count += 1
    elif len(counters) < k: counters[x] = (count=1, error=0)
    else:
        y = item with smallest count     # min-heap or stream-summary
        m = counters[y].count
        delete counters[y]
        counters[x] = (count=m+1, error=m)
```

Sketch plus heap. Common in practice: a CMS for frequencies plus a size-k min-heap; on each arrival update the CMS and, if x's estimate beats the heap minimum (or x is in the heap), update the heap. Redis `TOPK` uses **HeavyKeeper** (Gong et al., USENIX ATC 2018): a sketch-like table plus heap that *decays* counters of colliding small items so large flows keep their place.

Distributed top-k. Merging each of 50 shards' top 10 is wrong: an item 11th everywhere can be 1st globally. Options: (1) send Misra-Gries or Space-Saving summaries, designed to merge (add counts, trim to k); (2) send a larger top-k' (e.g. 10×k) and accept small error; (3) partition by item (hash the hashtag to a shard) so each item is counted in one place and local lists merge exactly.



Complexity: O(k) memory; Misra-Gries O(1) amortised per item with the right structure (O(k) naively); Space-Saving O(1) or O(log k).

Trade-offs

- **Misra-Gries**: under-counts; simple and mergeable, counts less useful as numbers.
- **Space-Saving**: over-counts but tracks per-entry error, usually more accurate; the bottom of its ranking is noisy because evicted items re-enter with inflated counts.
- **CMS + heap**: estimates for *any* item, more memory, CMS error relative to N.
- All count since the start; for "top right now" use windows (one summary per minute, merged) or decay (see trending below).

Real-world examples

- **Redis** `TOPK.ADD` / `TOPK.LIST` (HeavyKeeper).
- **Apache DataSketches** "frequent items" sketch (Druid and others), Misra-Gries-style reduction.
- Network gear and CDNs finding "elephant flows" and abusive clients.
- Trending hashtags, top queries, most-viewed products dashboards.

In the interview

Q: Top 100 search queries in the last hour at 100,000 queries/s.

A: Hash each query to one of P aggregator partitions so it is counted in one place. Each keeps a Space-Saving summary per minute (say 10,000 entries for top-100 accuracy), merges its last 60 every few seconds and publishes its top 100; a coordinator merges the P lists, exactly, since partitions are disjoint.

Q: Why not merge every server's top 10?

A: A query just below the cut everywhere can be largest overall. Partition by key, or send much larger lists or mergeable summaries.

Q: Misra-Gries says A = 4, truth is 5. Bug?

A: No: it under-counts by at most n/(k+1) and promises every item above that is present. For exact winner counts, do a second pass or keep exact counters for candidates.

TL;DR

Misra-Gries and Space-Saving find frequent items with k counters: Misra-Gries decrements all when full and under-counts by ≤ n/(k+1); Space-Saving evicts the minimum, inherits its count, records error, and over-counts by ≤ n/k. For distributed top-k, partition by item or merge proper summaries, never local top-k lists.

Sources

- Misra and Gries, "Finding Repeated Elements", Science of Computer Programming, 1982: https://users.cs.utah.edu/~pandey/courses/cs6968/spring23/papers/mg.pdf
- Metwally, Agrawal, El Abbadi, "Efficient Computation of Frequent and Top-k Elements in Data Streams" (ICDT 2005; UCSB TR 2005-23): https://old.cs.ucsb.edu/research/tech_reports/reports/2005-23.pdf
- Gong et al., "HeavyKeeper", USENIX ATC 2018: https://www.usenix.org/conference/atc18/presentation/gong
- Redis `TOPK.ADD` : https://redis.io/commands/topk.add

Reservoir sampling: Algorithm R and the weighted variant

What it is

Reservoir sampling keeps a uniform random sample of exactly k items (the **reservoir**) from a stream of unknown length, in one pass and O(k) memory. **Algorithm R** is the classic version (analysed in Vitter's 1985 "Random Sampling with a Reservoir"); **weighted reservoir sampling** (Efraimidis and Spirakis, 2006) favours heavier items.

The problem it solves

"Keep 1,000 representative log lines from today", "100 random requests for debugging", "sample 1% of traces but cap at 10,000 per minute". You can't store the stream and don't know its length; "keep each with probability p" gives a random-size sample that blows up on busy days.

How it works

Algorithm R: put the first k items in the reservoir; for item i > k (counting from 1) draw j uniformly in 1..i; if j ≤ k, replace slot j with item i, else discard it.

```
function reservoir(stream, k):
    R = []
    for i, x in enumerate(stream, start=1):
        if i <= k: R.append(x)
        else:
            j = random_int(1, i)
            if j <= k: R[j] = x
    return R
```

Worked example, k = 2, stream `a b c d e` : `[a, b]` ; c (i = 3) draws j = 1 → `[c, b]` ; d draws 4 > 2, discard; e draws 2 → `[c, e]` .

Why it is fair. Each item should end with probability k/n = 2/5. Item c enters with 2/3; survives d unless d is kept (2/4) and hits c's slot (1/2): 3/4; survives e with 1 − (2/5)(1/2) = 4/5. Total 2/3 × 3/4 × 4/5 = 2/5. The product telescopes for every position.

Complexity: O(1) per item, O(k) memory. Skip-ahead variants (Vitter's Algorithm Z, Algorithm L) compute how many items to skip before the next replacement, drawing far fewer random numbers on long streams.

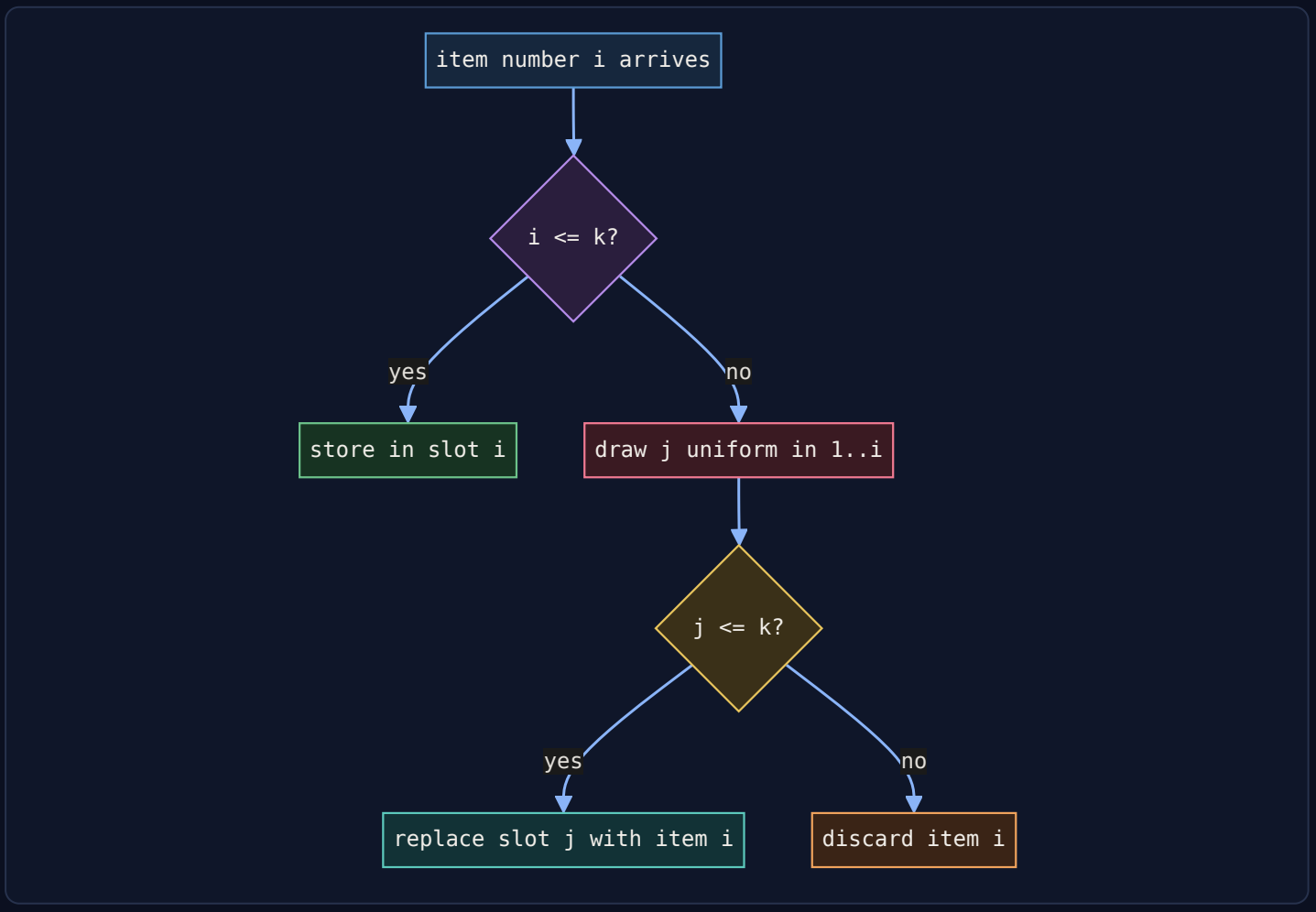
Weighted variant (A-Res). For each item with weight w > 0 draw u in (0, 1), key = `u^(1/w)` , and keep the k **largest** keys in a size-k min-heap. Higher weight pushes the key toward 1, but luck still plays a part.

k = 2:

Item	weight w	u	key = u^(1/w)
A	1	0.50	0.500
B	4	0.30	0.740
C	2	0.80	0.894
D	1	0.90	0.900

Sample {D, C}: B had the highest weight but drew a small u, which happens with the right probability. Use `log(u) / w` as the key (same order, numerically safer) for large weights.

Distributed sampling. "Random key, keep top k" is **mergeable**: each shard keeps its top-k keys, the coordinator keeps the top k of the union, exactly a sample of the whole stream. Uniform sampling is the case w = 1 (key = u).



Trade-offs

- Exactly k items regardless of volume (bounded memory and storage), but the effective rate drops as volume grows: record n to scale counts back up.
- Uniform samples miss rare important events (errors, slow requests): use weights or an always-keep rule for errors.
- Algorithm R draws one random number per item; skip-based versions are faster on huge streams.
- A sample is not a sketch: it cannot reliably answer "how many distinct" or "how often" for rare X.

Real-world examples

- **Dropwizard Metrics** (Java): reservoir-based histograms (uniform and exponentially decaying) for latency samples.
- Tail-based trace samplers and log samplers capping traces per minute.

- Database `TABLESAMPLE` and approximate query engines building statistics.

In the interview

Q: One random line from a file of unknown length, one pass, O(1) memory?

A: Reservoir with k = 1: keep line 1, replace with line i with probability 1/i; every line is chosen with probability 1/n.

Q: Uniform sample across 100 shards?

A: Random key u per item, top k per shard, top k of the union at the coordinator; for weights use u^(1/w).

Q: Why not keep each request with probability 1%?

A: Sample size grows with traffic and is unbounded in spikes; a reservoir guarantees exactly k per window.

TL;DR

Algorithm R keeps the first k items, then item i with probability k/i in a random slot, so each item is sampled with probability k/n in one pass and O(k) memory. Weighted sampling keys each item u^(1/w) and keeps the top k, which also merges across shards.

Sources

- Vitter, "Random Sampling with a Reservoir", ACM TOMS 11(1), 1985: https://doi.org/10.1145/3147.3165
- Efraimidis and Spirakis, "Weighted random sampling with a reservoir", Information Processing Letters 97(5), 2006: https://doi.org/10.1016/j.ipl.2005.11.003

t-digest and percentile estimation

What it is

A **percentile** (**quantile**) is the value q% of observations fall below; p99 latency is the value 99% of requests beat. A **t-digest** (Ted Dunning, refined with Otmar Ertl, 2019 paper) is a small distribution summary that answers any percentile approximately, is very accurate at the extremes (p99, p99.9), and **merges** across machines without losing accuracy. Why averages and percentiles matter: Module 0 — Percentiles, tail latency.

The problem it solves

Exact percentiles need every value sorted, impractical for a billion requests a day across 500 servers, so each server keeps a summary. **Averaging percentiles** is wrong:

- Server A: 9,900 requests, 9,800 at 20 ms and 100 at 50 ms; p99 (9,801st value) = 50 ms.
- Server B: 100 requests at 1,000 ms; p99 = 1,000 ms.
- Average of p99s: **525 ms**. True p99 of all 10,000: the 9,900th value, **50 ms** (the 1,000 ms requests are exactly the slowest 1% and appear in p99.5).

10× off. Percentiles can't be averaged or added: merge mergeable summaries, *then* query.

How it works

Centroids. A t-digest is a sorted list of **centroids**, (mean, count) pairs standing for clusters of nearby values. Rule: **centroids near the middle may be large, near the tails tiny**, so the tails (where p99 lives) keep nearly every point and the middle is summarised coarsely.

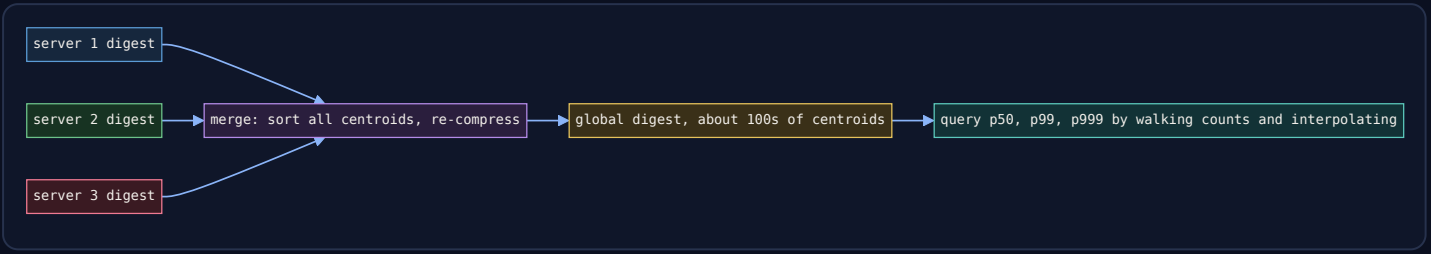
Size limit. A **scale function** k(q), steep near q = 0 and 1 and flat in the middle, sets how wide a centroid may be: one covering q_left to q_right is allowed only if `k(q_right) − k(q_left) ≤ 1`. A **compression** parameter δ (often 100) controls total centroids. The original paper's simpler bound: a centroid at q holds about `4 · N · q · (1 − q) / δ` points. N = 10,000, δ = 100:

Quantile q	Max points per centroid
0.5 (median)	4 × 10,000 × 0.25 / 100 = 100
0.1	4 × 10,000 × 0.09 / 100 = 36
0.01	4 × 10,000 × 0.0099 / 100 ≈ 4
0.001	≈ 0.4, so single points

Adding points (merging variant): buffer values; when full, sort buffer and centroids together by mean; walk left to right greedily merging each item into the current centroid if the limit at that quantile allows, else start a new one (weighted mean, summed count).

Querying p99: target rank 0.99 × N; walk centroids accumulating counts past it; **interpolate** between neighbouring centroid means. Exact min and max are kept for the ends.

Merging digests: concatenate all centroids, sort, run the same greedy merge. The size rule depends only on quantile position, so the merged digest keeps the same accuracy guarantees, which averaged percentiles lack.



Complexity: amortised O(log δ) or better per insert with buffering; O(δ) centroids (Elasticsearch caps at 20 × compression nodes, about 64 KB worst case at the default 100); merge O(total centroids × log).

Alternatives:

- **HDR Histogram**: fixed log-scale buckets with chosen significant digits; fast, mergeable by adding buckets, bounded relative error, needs a known value range. Very common for latency.
- **DDSketch** (Datadog): log-scale buckets with guaranteed relative error, fully mergeable.
- **KLL** and other rank-error sketches (Apache DataSketches): proven worst-case rank error bounds.

Trade-offs

- t-digest is small and very accurate at the tails, but its error bounds are mostly empirical and results can depend slightly on insertion order.
- Middle percentiles are less precise than tails: usually the right trade for latency.
- HDR or DDSketch are simpler with hard relative-error guarantees, but need a range (HDR) or more memory for wide ranges.
- Any sketch beats averaging per-host percentiles.

Real-world examples

- **Elasticsearch / OpenSearch** `percentiles` aggregation: t-digest by default (compression 100), HDR optional.
- Many metrics libraries and stream processors offer t-digest. Prometheus **histograms** (fixed buckets, mergeable) are the bucket alternative; Prometheus **summaries** compute per-process quantiles that cannot be aggregated across instances, for the reason above.
- Datadog uses DDSketch for distribution metrics.

In the interview

Q: Your dashboard averages p99 across 200 hosts. What's wrong?

A: Percentiles don't average: a host with few slow requests drags the average far off, or a busy host's tail is diluted. Ship mergeable sketches (t-digest, HDR, DDSketch) or bucketed histograms, merge, then compute p99 once.

Q: Why is t-digest more accurate at p99.9 than p50?

A: The scale function forces tail centroids to hold very few points, often single values, while median centroids hold many: precision is spent where tail questions are asked.

Q: Weekly p99 from daily p99s?

A: No. Merge seven daily digests, then query.

TL;DR

Never average percentiles. A t-digest keeps sorted (mean, count) centroids, tiny at the tails and larger in the middle, answers percentiles by walking counts and interpolating, and merges across hosts and windows without losing accuracy. HDR Histogram and DDSketch are bucket-based alternatives with hard relative-error bounds.

Sources

- Dunning and Ertl, "Computing Extremely Accurate Quantiles Using t-Digests" (2019): https://arxiv.org/abs/1902.04023
- t-digest reference implementation: https://github.com/tdunning/t-digest
- Elasticsearch percentiles aggregation: https://www.elastic.co/docs/reference/aggregations/search-aggregations-metrics-percentile-aggregation
- Prometheus histograms and summaries: https://prometheus.io/docs/practices/histograms/

Trending detection with decayed counts and z-scores

What it is

Trending detection finds items whose activity is *unusually high for them*, not simply the most popular. "#WorldCup" at a steady 50,000 mentions an hour is popular; a small band jumping from 20 to 2,000 an hour is trending. Building blocks: **exponentially decayed counts** (recent events weigh more, old ones fade smoothly) and **z-scores** (how many standard deviations the current rate is above the usual).

The problem it solves

Raw counts always rank the same giants. Fixed windows ("last hour") are expensive (old events must expire) and jumpy (a burst falls off a cliff 60 minutes later). Wanted: (1) react within minutes to a real surge; (2) ignore big-but-steady items; (3) O(1) state and work per item per event; (4) no firing on tiny-count noise (2 to 6 mentions is "3×" but meaningless).

How it works

1. Exponentially decayed count. Per item keep c and t_last; on an event at t:

```
c = c * 2^(-(t - t_last) / H) + 1     # H = half-life
t_last = t
```

With H = 10 min, 1,000 mentions around 12:00 give c ≈ 1,000; with no new mentions c = 500 at 12:10, 250 at 12:20, 125 at 12:30. No window bookkeeping. Keep two half-lives, e.g. 5 minutes (short-term rate) and 24 hours (baseline): a high short/long ratio means "much hotter than usual".

2. EWMA baseline. Count per item in fixed buckets (per minute, x_t) and track an **exponentially weighted moving average (EWMA)** mean μ and variance σ² with smoothing λ (NIST suggests 0.2–0.3 for control charts; trending systems often go smaller for longer memory):

```
diff  = x - mu
z     = diff / sqrt(var)            # score BEFORE updating the baseline
mu    = mu + lam * diff
var   = (1 - lam) * (var + lam * diff * diff)
```

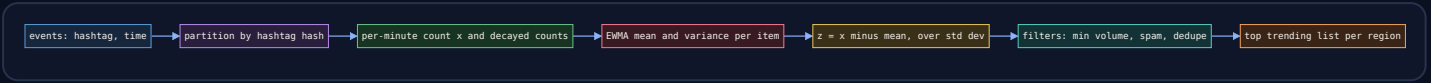
3. z-score threshold. z = (x − μ)/σ says how unusual this minute is for *this item*; flag when z exceeds a threshold (e.g. 3) and volume passes a minimum.

Worked example, λ = 0.2:

- Small topic: μ = 100/min, σ² = 400 (σ = 20), x = 180 → z = **4.0**, trending. Update: μ = 116, σ² = 0.8 × (400 + 0.2 × 80²) = 1,344, σ ≈ 36.7. Next minute x = 260 → z ≈ **3.9**, still trending. The baseline catches up, so a surge lasting hours eventually stops trending, as intended.
- Giant topic: μ = 10,000, σ = 1,000, x = 10,500: higher raw count, z = 0.5, **not trending**.

4. Small-number guards. μ = 2, σ = 1, jump to 6 gives z = 4. Require a minimum count (x ≥ 50), add a variance prior (`z = diff / sqrt(var + c)`), or score a smoothed ratio `(x + β) / (μ + β)` with β around 10–50.

5. At scale. Find candidates with a heavy-hitter front end (Space-Saving or CMS + heap on decayed counts), keep μ and σ² only for candidates plus a tail of recently seen items, and partition by item hash across workers; each emits top trending scores every few seconds to a merging coordinator. Then product filters: de-duplicate near-identical terms, remove spam and bot bursts, localise by region.



Complexity: O(1) time and state per tracked item per event; total memory bounded by the heavy-hitter front end.

Trade-offs

- **Half-life / λ**: short reacts fast but flags noise; long is stable but late. Many systems run two or three timescales.
- **z-scores assume roughly normal noise**, but counts are bursty with daily cycles: compare with the same hour last week or model seasonality, so "every morning" isn't a trend.
- **Small-count bias** needs explicit guards.
- **Manipulation**: coordinated accounts manufacture spikes; count distinct users (HLL per item), weight by account trust.
- Front-end sketches trade a little accuracy for large memory savings.

Real-world examples

- **Twitter / X trends** compare a topic's volume with its own past rather than raw popularity; the company rebuilt the trends back end on stream processing (Storm) to do this in real time.
- **EWMA control charts**: decades-old industrial quality control (NIST handbook); the same math drives monitoring anomaly alerts.
- News, e-commerce "trending now" and music charts use decayed scores so yesterday's hits fade smoothly.

In the interview

Q: Design trending hashtags.

A: Stream posts to workers partitioned by hashtag. Per tracked hashtag keep per-minute counts, a short-half-life decayed count and EWMA mean and variance, with a Space-Saving front end bounding what is tracked. Score by z with a minimum-volume floor, counting distinct authors to resist spam. Every 30 s publish each worker's top N per region and merge at a coordinator into the cache the app reads.

Q: Why not rank by last-hour count?

A: That shows always-popular topics. Trending means unusual for this item, so normalise against its own baseline.

Q: Why exponential decay instead of a sliding window?

A: One number and timestamp, O(1) update, no per-event storage or expiry, and smooth fading instead of a cliff when a burst leaves the window.

TL;DR

Trending = much higher than the item's own normal. Keep decayed counts (one number per item with a half-life) and EWMA mean and variance, score each interval with z = (x − μ)/σ, guard against small counts and spam, and bound tracked items with a heavy-hitter sketch partitioned by item.

Sources

- NIST/SEMATECH e-Handbook, EWMA control charts: https://www.itl.nist.gov/div898/handbook/pmc/section3/pmc324.htm
- Finch, "Incremental calculation of weighted mean and variance" (2009): https://fanf2.user.srcf.net/hermes/doc/antiforgery/stats.pdf
- Coverage of Twitter's trends back-end rebuild (Storm): https://www.smartcompany.com.au/technology/twitter-overhauls-trending-topics-adds-more-contextual-information/
- Nikolov and Shah, trend detection research with Twitter data (MIT, 2012), summary: https://flowingdata.com/2013/06/19/twitter-trend-detection-algorithm/

LSM trees in depth

What it is

An **LSM tree** (log-structured merge tree; O'Neil et al., 1996) stores sorted key-value data by turning every write into a fast sequential append and sorting later in the background. It powers LevelDB, RocksDB, Cassandra, ScyllaDB and HBase. Parts:

- **Memtable:** in-memory sorted structure (RocksDB default: a skip list) receiving every write.
- **SSTable** (sorted string table): an immutable on-disk file of key-sorted pairs plus a small index and filter; never edited, changes go to newer files.
- **Compaction:** background merging of SSTables into new ones, dropping overwritten values and deleted keys.
- **Write-ahead log (WAL):** append-only file recording each write first so the memtable can be rebuilt after a crash (Module 3 — Write-Ahead Log).

The problem it solves

A B-tree updates in place: changing a 100-byte row may rewrite a 16 KB page at a random disk location. Random writes are slow on spinning disks and wear SSDs. Write-heavy workloads (logging, metrics, messaging, time series) are better served by batching writes in memory and flushing large sequential files, absorbing much higher write rates on the same hardware.

How it works

Write path

1. Append (`put user:42 = "Ana"`) to the WAL and `fsync` it (force to physical disk), or batch several writes per `fsync`.
2. Insert into the memtable and acknowledge: one sequential append plus an O(log n) in-memory insert.
3. At its size limit (RocksDB default 64 MB) the memtable becomes **immutable** and a fresh one takes writes.
4. A background **flush** writes it out in key order as a new SSTable; the matching WAL segment can then be deleted.
5. A **delete** is a write of a **tombstone** marker ("gone as of now"); the old value disappears only when compaction merges the tombstone with it.

Read path (point lookup)

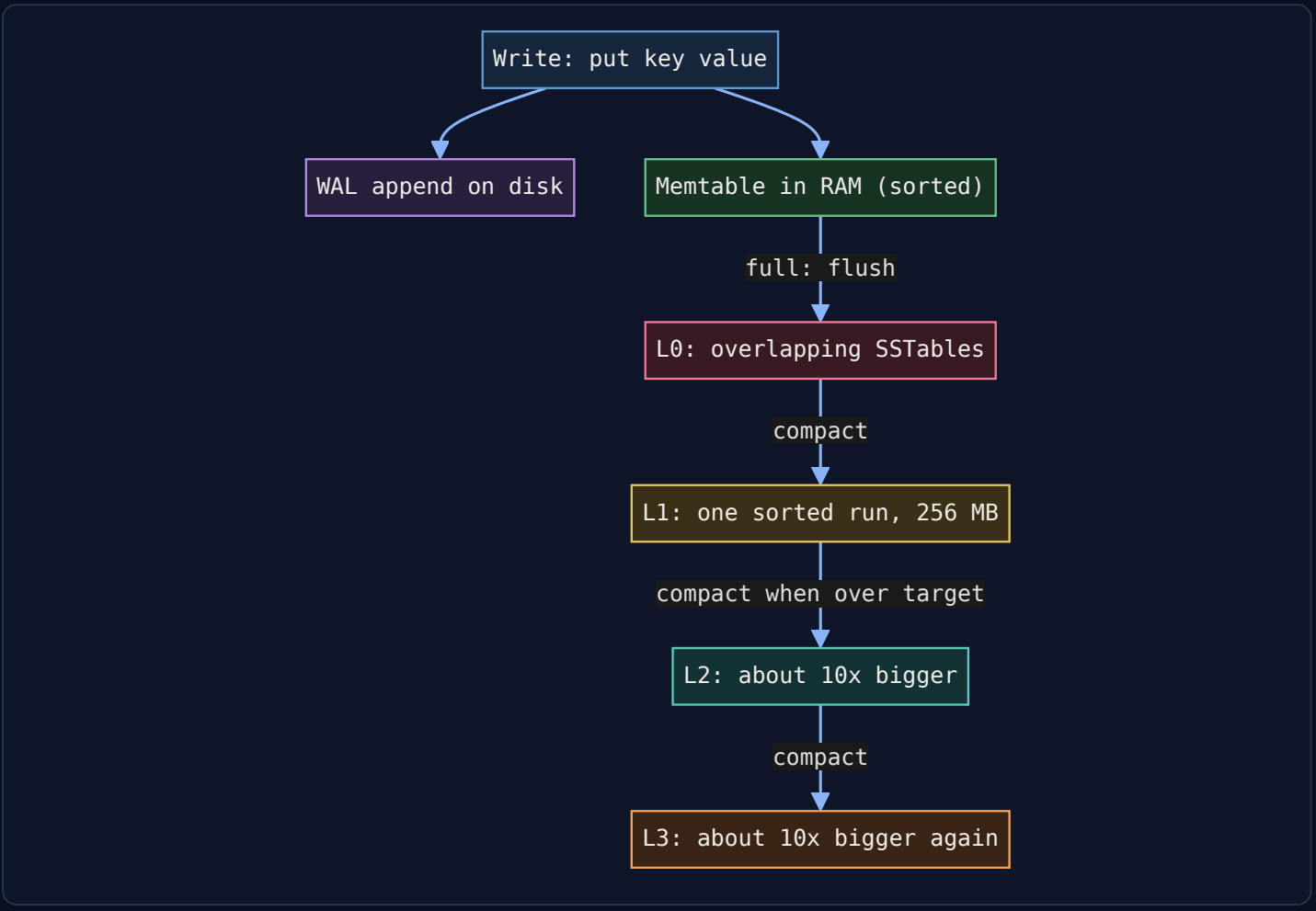
1. Check the active memtable, then immutable memtables; newest wins.
2. Check SSTables newest to oldest; first ask each file's **Bloom filter** ("definitely not here" / "maybe here") and skip on "definitely not".
3. On "maybe", use the index block to find the ~4 KB data block, read it, binary search.
4. Stop at the first file holding the key (or its tombstone).

Bloom filter per SSTable, built when the file is written: about 10 bits per key gives about 1% false positives (RocksDB measures about 0.95% for its cache-friendly variant). A missing-key lookup across 8 SSTables would read 8 blocks; with 1% filters it reads 8 × 0.01 = 0.08 on average, which keeps LSM point reads fast despite many files.

Compaction strategies (where LSM designs differ most):

- **Size-tiered** (Cassandra's STCS, RocksDB's "universal"): when enough similar-sized SSTables pile up (Cassandra `min_threshold` default 4), merge them into one bigger file: four 64 MB flushes → one ~256 MB file; four of those → ~1 GB.
- **Leveled** (LevelDB, RocksDB default): L0 holds raw flushes with overlapping key ranges; from L1 down each level is one sorted run with non-overlapping files, so a key is in at most one file per level. Each level's target is about 10× the previous (`max_bytes_for_level_multiplier`). When a level exceeds target, pick a file, merge it with the overlapping files in the next level, and write the result there.

Leveled sizing with L1 = 256 MB, multiplier 10: L2 = 2.56 GB, L3 = 25.6 GB, L4 = 256 GB. The last level holds about 90% of data, so upper-level copies add only about 1/9 = 11% overhead.



The three amplifications (vocabulary for comparing engines):

- **Write amplification** = disk bytes written / user bytes written. Leveled: moving one file down can rewrite ~10 overlapping files, so up to about the fanout (10) per level, tens across 4 to 5 levels. Size-tiered rewrites each byte roughly once per tier: much lower.
- **Read amplification** = disk reads per query. Leveled: at most one file per level plus L0 files. Size-tiered: possibly many overlapping files per tier; range scans suffer most.
- **Space amplification** = disk bytes / live data bytes. Leveled: about 1.1. Size-tiered: old and new copies coexist across tiers, and a big merge needs room for inputs and outputs at once; RocksDB universal compaction allows up to 200% extra by default (100 bytes of data may need 300 bytes of disk).

Complexity. Writes O(log m) in memory (m = memtable entries) plus amortised compaction I/O; point reads O(levels) filter checks, usually one disk read; space as above.

Trade-offs

- Excellent write throughput and compression (sorted immutable files compress well), paid for by background compaction stealing disk bandwidth and CPU, with latency spikes when it falls behind.
- Leveled favours reads and space; size-tiered favours writes; hybrids exist (Cassandra's time-window compaction for time series).
- Deletes are not instant: tombstones linger until compaction, and scans over many tombstones are slow.
- Range scans must merge the memtable and every level, so they cost more than in a B-tree.

Real-world examples

- **RocksDB** (Meta) underpins MyRocks, Kafka Streams state stores, CockroachDB's earlier storage and more; defaults: skip-list memtable, leveled compaction, per-file Bloom filters.
- **Cassandra** defaults to size-tiered, with leveled and time-window per table.
- **Bigtable/HBase** use the same memtable-plus-immutable-files idea.

In the interview

Q: Why are LSM trees faster for writes than B-trees?

A: Writes hit a sequential log and an in-memory table and are flushed in big sorted batches; a B-tree rewrites a page at a random location per update.

Q: How does an LSM tree avoid checking every SSTable on a read?

A: Per-SSTable Bloom filters (10 bits per key skip about 99% of non-holding files without disk reads), and leveled compaction guarantees at most one candidate file per level below L0.

Q: Leveled or size-tiered for a write-heavy metrics store?

A: Size-tiered (or time-window): lower write amplification for append-mostly data read in recent ranges. Leveled suits read-heavy or space-constrained workloads.

TL;DR

An LSM tree writes to a log and an in-memory sorted table, flushes immutable sorted files, and merges them in the background; Bloom filters keep reads cheap. Leveled compaction: low read and space amplification, higher write cost; size-tiered: the opposite.

Sources

- O'Neil, Cheng, Gawlick, O'Neil, "The Log-Structured Merge-Tree (LSM-Tree)", Acta Informatica 33, 1996: https://www.cs.umb.edu/~poneil/lsmtree.pdf
- RocksDB wiki, MemTable: https://github.com/facebook/rocksdb/wiki/MemTable
- RocksDB wiki, Leveled Compaction: https://github.com/facebook/rocksdb/wiki/Leveled-Compaction
- RocksDB wiki, Compaction (amplification, universal compaction): https://github.com/facebook/rocksdb/wiki/Compaction
- RocksDB wiki, Bloom Filter: https://github.com/facebook/rocksdb/wiki/RocksDB-Bloom-Filter
- Apache Cassandra docs, Size Tiered Compaction Strategy: https://cassandra.apache.org/doc/latest/cassandra/managing/operating/compaction/stcs.html

B+-trees

What it is

A **B+-tree** is a balanced search tree for storage read in fixed-size **pages** (InnoDB default 16 KB, PostgreSQL 8 KB). Each node is one page holding many keys, so the tree is wide and shallow. All records (or pointers) live in the **leaf** nodes, linked left to right; **internal** nodes hold only separator keys that steer the search. Balanced: every leaf is at the same depth.

The problem it solves

Storage reads whole pages, each expensive compared with memory work. A one-key-per-node binary tree needs about 30 page reads to find one row among a billion; a B+-tree with hundreds of keys per node needs 3 or 4, with the top levels almost always cached. Linked leaves make range queries (`WHERE created_at BETWEEN ...`) and `ORDER BY` cheap: find the start, walk sideways.

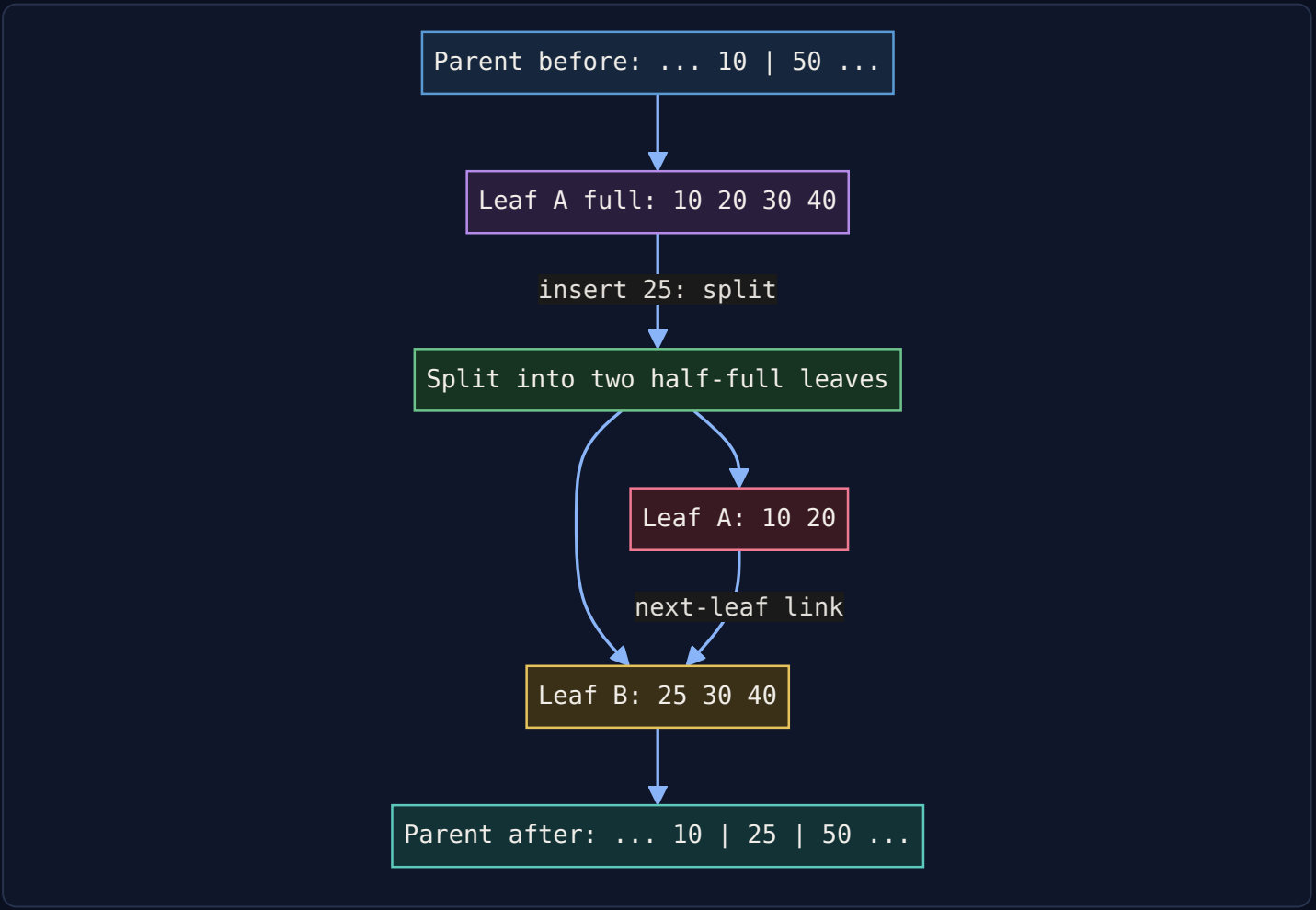
How it works

Fan-out arithmetic. An internal entry of an 8-byte key plus 6-byte child page number (~14 bytes) gives a 16 KB page about 16,384 / 14 ≈ 1,170 children. With leaves of about 15 rows of 1 KB: height 2 ≈ 17,500 rows; height 3 ≈ $1{,}170^2 \times 15 \approx 20$ million; height 4 ≈ 24 billion. Any row is 3 to 4 page reads away, usually with only the leaf missing the cache.

Search: from the root, binary search each page's keys for the child whose range holds the target, down to a leaf, then binary search the leaf. Cost O(log_B n) page reads (B = fan-out), O(log n) comparisons.

Insert with page split (max 4 keys per leaf):

1. Find the leaf: A = [10, 20, 30, 40]; insert 25.
2. With room, insert in order and finish; here A is full.
3. **Split:** new leaf B; A keeps [10, 20], B gets [25, 30, 40].
4. Fix sibling links: A → B, B → A's old right neighbour.
5. Insert separator 25 (B's first key) and a pointer to B into the parent.
6. A full parent splits too, pushing a separator up; a root split creates a new root, the only way the tree grows taller, so all leaves stay at the same depth.



Delete mirrors it: remove the key; a page below threshold borrows from or merges with a sibling. InnoDB tries to merge below 50% full (`MERGE_THRESHOLD` default); many systems lazily leave half-empty pages.

Fill factor = how full pages are packed on index build or append; free space lets future mid-range inserts land without splits.

- PostgreSQL B-tree indexes default to fill factor 90 (leaf pages packed to 90% on build and when appending new largest keys).
- InnoDB leaves 1/16 of each clustered-index page free: sequential inserts give pages about 15/16 full; random inserts give 1/2 to 15/16, because splits leave two half-full pages.

So a random UUID primary key in InnoDB causes more splits, partly-empty pages and a bigger index than an auto-increment key that always appends at the right edge.

Complexity. Search, insert, delete O(log_B n) page I/Os; range scan of k results O(log_B n + k/B); space O(n) at roughly 50% to 100% page utilisation.

Trade-offs

- Great reads and range scans; predictable latency (no background compaction).
- Every update rewrites a whole page in place, after first writing the WAL, so write amplification is high for small random updates.
- Random-key inserts cause splits and fragmentation; periodic rebuilds (`REINDEX`, `OPTIMIZE TABLE`) may be needed.
- In-tree concurrency control (latching pages during splits) is subtle; engines use latch crabbing or B-link trees.

Real-world examples

- **InnoDB** (MySQL): the table is a B+-tree keyed by primary key (a **clustered index**, rows in the leaves); secondary indexes are separate B+-trees whose leaves hold the primary key.
- **PostgreSQL**: tables are unordered heaps; B-tree indexes point into the heap.
- **SQLite**, Oracle, SQL Server, and file systems NTFS, XFS and Btrfs use B-tree variants.

In the interview

Q: Why B+-trees rather than binary trees or hash tables?

A: Pages are the I/O unit and a B+-tree packs hundreds of keys per page: 3 to 4 page reads for billions of rows. Unlike a hash table it keeps keys sorted, so ranges and ordering are cheap.

Q: Why is a random UUID a poor InnoDB primary key?

A: Inserts hit random leaves: page splits, half-full pages, more cache misses, larger index. A monotonic key (auto-increment or time-ordered UUIDv7) appends at the right edge.

Q: B+-tree or LSM tree?

A: B+-tree for read-heavy, range-heavy, latency-sensitive OLTP (online transaction processing); LSM for write-heavy workloads tolerating some read and compaction cost.

TL;DR

A B+-tree is a short, wide, balanced tree of pages with all data in linked leaves: lookups are a handful of page reads, ranges a sideways walk, inserts split full pages and push separators up; fill factor postpones splits.

Sources

- Bayer and McCreight, "Organization and Maintenance of Large Ordered Indexes", Acta Informatica, 1972, and Comer, "The Ubiquitous B-Tree", ACM Computing Surveys, 1979 (background).
- MySQL 8.0 Reference Manual, The Physical Structure of an InnoDB Index: https://dev.mysql.com/doc/refman/8.0/en/innodb-physical-structure.html
- PostgreSQL docs, CREATE INDEX (fillfactor): https://www.postgresql.org/docs/current/sql-createindex.html

MVCC step by step

What it is

MVCC (multi-version concurrency control) keeps several versions of a row so readers see a consistent **snapshot** (frozen view of committed data as of a moment) while writers create new versions: readers never block writers and vice versa. Used by PostgreSQL, MySQL InnoDB, Oracle, SQL Server (with snapshot isolation), CockroachDB and many others.

The problem it solves

With plain locking, a long report reading a million rows holds shared locks that stall every update to them (or updates block the report). MVCC lets the report read old versions while updates write new ones. It underlies **snapshot isolation** (every read in a transaction sees the same snapshot; see Module 0 — Isolation levels).

How it works

Core ideas: each transaction gets an increasing **transaction ID** (XID); each row version records its creator and, if replaced or deleted, its deleter; a snapshot records which transactions had committed when taken; a **visibility rule** decides which versions it sees; versions no snapshot can see are garbage-collected (PostgreSQL **vacuum**, InnoDB **purge**).

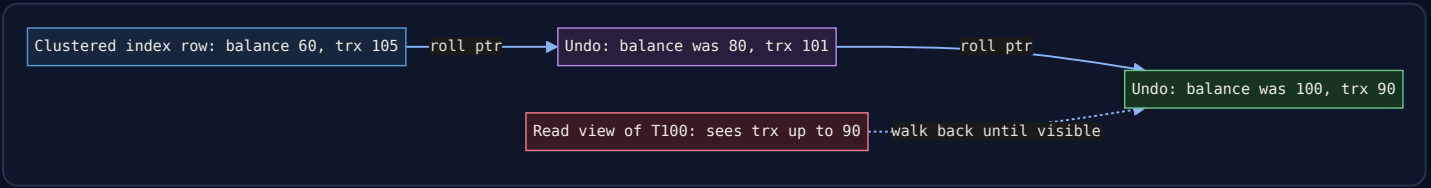
PostgreSQL: new versions in the table. Each version (**tuple**) has hidden `xmin` (inserting XID) and `xmax` (deleting/updating XID, or empty). `UPDATE` sets `xmax` on the old tuple and inserts a new one with `xmin` = updater's XID; the version chain lives in the heap.

Worked example. `account 7, balance 100` inserted by XID 90, long committed.

1. T100 starts; snapshot: "committed XIDs below 100 visible, T100 in progress".
2. T101 sets balance = 80: old tuple xmin 90, xmax 101; new tuple xmin 101. T101 commits.
3. T100 reads: old tuple created by 90 (visible), deleted by 101 (after T100's snapshot, so invisible) → visible, balance 100; new tuple invisible. T100 consistently sees 100.
4. T102, starting after T101 committed, sees 80.

Visibility rule (simplified): visible to snapshot S if the creator committed before S (or is S's own transaction) and the deleter is empty, aborted, or not committed as of S.

InnoDB: newest version in place, older ones rebuilt from undo. Rows are updated in place in the clustered index. Each carries a hidden 6-byte `DB_TRX_ID` (last modifier) and 7-byte `DB_ROLL_PTR` (roll pointer) to an **undo log** record describing the previous version; following roll pointers gives the **version chain**. A consistent read creates a **read view** (transactions active at that moment plus ID bounds); if a row's `DB_TRX_ID` is not visible, InnoDB applies undo records back to a visible version. REPEATABLE READ (InnoDB default) creates the read view at the transaction's first consistent read; READ COMMITTED takes a fresh one per statement.



Garbage collection.

- **PostgreSQL VACUUM:** finds tuples whose `xmax` committed before the oldest running snapshot, marks space reusable, removes index entries pointing to them. It also **freezes** old tuples: XIDs are 32-bit and compared in a circle, so after about 2 billion transactions an old XID would look "in the future"; freezing marks a tuple visible to everyone forever. Autovacuum forces an anti-wraparound vacuum when a table's oldest unfrozen XID is 200 million transactions old (`autovacuum_freeze_max_age` default).
- **HOT updates** (heap-only tuples): if no indexed column changes and the page has room, the new version goes on the same page with no new index entries; a lower table fill factor makes this likelier.
- **InnoDB purge:** insert-undo is dropped at commit; update-undo stays until no read view needs it, then a purge thread removes it and physically removes delete-marked rows.

Complexity. Recent version O(1) extra; under a long-running snapshot O(version-chain length); space proportional to live old versions.

Trade-offs

- **PostgreSQL style** (append versions): instant rollback (new tuple ignored), but tables and indexes bloat until vacuum, and every non-HOT update writes new index entries.
- **InnoDB style** (in place plus undo): compact table and fast current reads, but old snapshots walk undo, and long transactions grow the undo log (the "history list length" problem).
- In both, one forgotten long transaction blocks cleanup for everyone.
- Snapshot isolation still allows **write skew** (next section), so it is not full serializability.

Real-world examples

- PostgreSQL: heap versions, `xmin` / `xmax`, VACUUM and autovacuum.
- MySQL InnoDB, Oracle: undo-based versions.
- CockroachDB and Spanner: versions keyed by timestamp in a key-value store, garbage-collected after a configurable time.

In the interview

Q: How do PostgreSQL readers avoid blocking writers?

A: An update writes a new tuple and stamps the old one with its XID; readers pick the visible version from their snapshot and never wait.

Q: Why can a long-running transaction hurt an otherwise idle database?

A: Its snapshot pins every old version created since it started: PostgreSQL can't vacuum them, InnoDB can't purge undo, so tables and undo logs grow and reads slow.

Q: What is transaction ID wraparound?

A: PostgreSQL XIDs are 32-bit, compared modulo 2^32; without freezing, after about 2 billion transactions old rows would appear created in the future and vanish. Vacuum freezing prevents it.

TL;DR

MVCC keeps row versions tagged with transaction IDs; each transaction reads through a snapshot via visibility rules. PostgreSQL stores versions in the table and needs VACUUM; InnoDB updates in place, keeps old versions in undo logs and purges later.

Sources

- PostgreSQL docs, Routine Vacuuming: https://www.postgresql.org/docs/current/routine-vacuuming.html
- PostgreSQL docs, Heap-Only Tuples: https://www.postgresql.org/docs/current/storage-hot.html
- MySQL 8.0 Reference Manual, InnoDB Multi-Versioning: https://dev.mysql.com/doc/refman/8.0/en/innodb-multi-versioning.html
- MySQL 8.0 Reference Manual, Consistent Nonlocking Reads: https://dev.mysql.com/doc/refman/8.0/en/innodb-consistent-read.html

Two-phase locking and Serializable Snapshot Isolation

What it is

Both give **serializability**: concurrent transactions produce the same result as some one-at-a-time order.

- **Two-phase locking (2PL)**: classic, pessimistic; transactions lock data and wait for each other.
- **Serializable Snapshot Isolation (SSI)**: optimistic, on MVCC snapshots; transactions never block, and the database aborts one when it detects a pattern that could break serializability (Cahill, Röhm, Fekete, SIGMOD 2008; PostgreSQL's SERIALIZABLE since 9.1).

The problem it solves

Snapshot isolation permits **write skew**: two transactions read overlapping data, decide based on it, and each write a *different* row, so no write-write conflict is detected. Example: at least one doctor must be on call; Alice and Bob both are. Each simultaneously sees 2 on call and takes themselves off. Both commit under snapshot isolation; nobody is on call, a result no serial order could produce.

How it works

2PL.

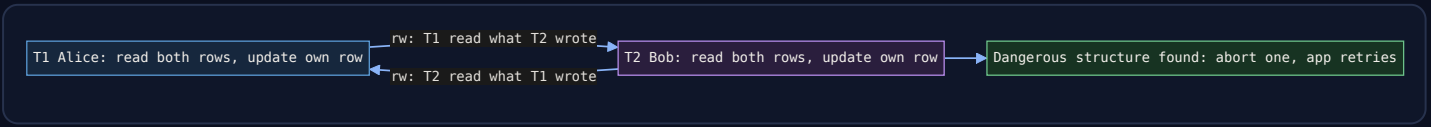
1. **Shared lock** before reading (many holders), **exclusive lock** before writing (one holder, no readers).
2. **Growing phase:** acquire, never release. **Shrinking phase:** after the first release, acquire nothing more.
3. **Strict 2PL** (what databases do): hold all locks until commit or abort, so nobody reads uncommitted data.
4. Stop **phantoms** (new rows matching a query's condition) by locking ranges: **predicate locks** or, in InnoDB, **next-key/gap locks** on index ranges.
5. **Deadlock**: T1 waits for T2 and vice versa; the database finds the cycle in its wait-for graph (or times out) and aborts one.

Doctors under 2PL: both take shared locks on the rows they read. Alice's update needs an exclusive lock on her row, which Bob holds shared: wait. Bob's update waits on Alice. Deadlock detected, one aborted, the other proceeds: correct, but blocking.

SSI.

1. Every transaction reads from an MVCC snapshot as in snapshot isolation; no read blocks.
2. Reads leave non-blocking **SIREAD locks** (`SIReadLock` in PostgreSQL's `pg_locks`) on rows, pages or relations read.
3. A write to something a concurrent transaction read records a **rw-antidependency** (edge reader → writer: the reader didn't see the change, so must come first serially).
4. Cahill's theorem: every snapshot-isolation anomaly contains a **dangerous structure**, two consecutive rw edges T1 → T2 → T3 among concurrent transactions (T1 and T3 may be the same); T2 is the **pivot**.
5. On such a structure, abort one transaction with a serialization failure (SQLSTATE `40001`); the application retries.

Doctors under SSI: Alice reads Bob's row and Bob writes it (Alice → Bob); Bob reads Alice's row and Alice writes it (Bob → Alice). Two rw edges: one is aborted at commit, and neither ever waited.



False positives. SSI tracks a sufficient condition, not the exact cycle, and may coarsen tuple locks to page or table locks to save memory, so it sometimes aborts safe transactions. PostgreSQL warns that sequential scans take relation-level predicate locks and raise the failure rate.

Complexity. 2PL: O(1) per lock, waits depend on contention. SSI: O(1) bookkeeping per read/write, memory proportional to concurrent read sets; a retry costs a whole transaction.

Trade-offs

- **2PL:** simple guarantee and no anomaly retries, but readers block writers, throughput collapses under contention, deadlocks need handling.
- **SSI:** read-only and low-conflict workloads run near snapshot-isolation speed, but applications *must* retry on `40001`, aborts waste work under contention, and some are false positives.
- Neither replaces good design: short transactions, rows touched in a consistent order.

Real-world examples

- **PostgreSQL** SERIALIZABLE = SSI (since 9.1); REPEATABLE READ = plain snapshot isolation.

- **MySQL InnoDB** SERIALIZABLE is lock-based: plain `SELECT`s become locking reads with next-key locks against phantoms.
- **SQL Server** offers lock-based SERIALIZABLE and a separate SNAPSHOT level.

In the interview

Q: What is write skew and which level prevents it?

A: Two transactions read the same data, then each update a different row based on it, breaking an invariant. Snapshot isolation allows it; serializable (2PL or SSI) prevents it. Under weaker levels, `SELECT ... FOR UPDATE` the rows the decision depends on.

Q: Why is SSI "optimistic"?

A: It runs transactions without blocking and only aborts at the end on a dangerous pattern; 2PL blocks up front in case of conflict.

Q: What must the application do differently under SSI?

A: Treat SQLSTATE 40001 as normal: retry the whole transaction, discarding anything computed in the failed attempt.

TL;DR

2PL gets serializability by locking (grow, hold until commit, break deadlocks); SSI runs on MVCC snapshots without blocking, tracks rw dependencies, and aborts when two consecutive rw edges form a dangerous structure. 2PL pays in waiting, SSI in retries.

Sources

- Cahill, Röhm, Fekete, "Serializable Isolation for Snapshot Databases", SIGMOD 2008: https://courses.cs.washington.edu/courses/csep544/09sp/resources/fekete-sigmod2008.pdf
- PostgreSQL docs, Transaction Isolation: https://www.postgresql.org/docs/current/transaction-iso.html
- PostgreSQL wiki, SSI: https://wiki.postgresql.org/wiki/SSI

Skip lists

What it is

A **skip list** (William Pugh, 1990, a simpler alternative to balanced trees) is a sorted linked list with randomly built "express lanes": the bottom level links every element; each element is randomly promoted upward, so with promotion probability 1/4 level 1 links about every 4th element, level 2 every 16th. Search starts in the top lane and drops down before overshooting.

The problem it solves

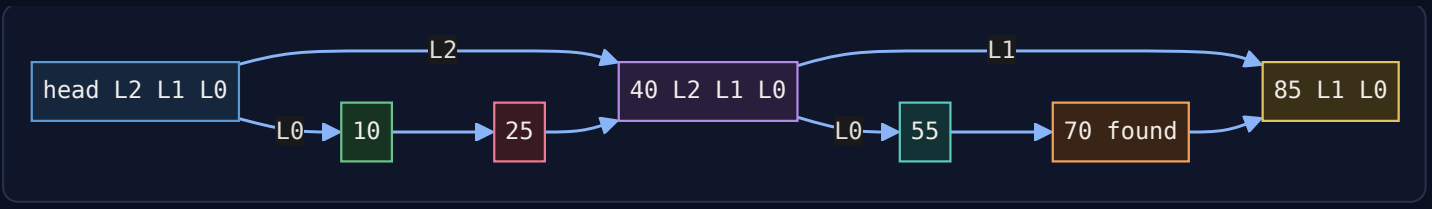
A sorted linked list searches in O(n). Balanced trees (red-black, AVL) give O(log n) but need rotations, fiddly to code and hard to make concurrent. A skip list gets expected O(log n) search, insert and delete from coin flips and pointer updates, and its bottom level is already ordered, so range scans are trivial.

How it works

Structure. Each node has a key (and value) and one forward pointer per level it belongs to; a head node has every level. Height is chosen at insert: start at 1, add a level with probability p (Redis: 1/4) up to a maximum (Redis: 32).

Search for 70. Keys 10, 25, 40, 55, 70, 85; 40 and 85 reached level 1, and 40 reached level 2.

1. Head, level 2: next is 40 < 70, move to 40.
2. 40, level 2: nothing next, drop to level 1.
3. Level 1: next is 85 > 70, drop to level 0.
4. Level 0: 55 < 70, move; next is 70: found. Touched 40, 55, 70 instead of walking from 10.



Insert: search, remembering at each level the last node before dropping (the `update` array); pick a random height h; splice the node in after `update[level]` for levels 0 to h−1. **Delete** uses the same `update` array to unsplice at each level.

```
insert(key):
  x = head
  for lvl from top down to 0:
    while x.next[lvl] and x.next[lvl].key < key:
      x = x.next[lvl]
    update[lvl] = x
  h = 1
  while random() < p and h < MAX: h += 1
  node = new Node(key, h)
  for lvl in 0..h-1:
    node.next[lvl] = update[lvl].next[lvl]
    update[lvl].next[lvl] = node
```

Complexity. Expected O(log n) search, insert, delete; expected levels log base 1/p of n (p = 1/4, one million items: log4(10^6) ≈ 10). Space: on average 1/(1 − p) pointers per node, 1.33 at p = 1/4 (vs 2 at p = 1/2, and 2 for a binary tree). Worst case O(n), astronomically unlikely and independent of input order (it depends only on the random generator).

Rank queries. Store on each forward pointer a **span** (bottom-level nodes jumped); summing spans along the search path gives rank in O(log n). Redis does exactly this.

Trade-offs

- Simple code, no rotations, ordered iteration for free.
- Expected, not guaranteed, O(log n); fine in practice.
- Pointer-chasing across random memory is less cache-friendly than a B-tree's packed arrays; on disk, B-trees win.
- Lock-free concurrent versions are comparatively easy (Java's `ConcurrentSkipListMap`), since inserts touch few local pointers.

Real-world examples

- **Redis sorted sets** (`ZADD` , `ZRANGE` , `ZRANK` , leaderboards). Beyond 128 entries or any member over 64 bytes (`zset-max-listpack-entries` / `zset-max-listpack-value` defaults), a sorted set becomes a skip list plus a hash table: the hash maps member to score for O(1) `ZSCORE` , the skip list orders by score for ranges and ranks, and each node has a `backward` pointer for reverse iteration. Small sets use a compact flat listpack.
- **Why Redis chose a skip list** (per its author): comparable memory (tunable via p), range iteration at least as cache-friendly as trees, and much simpler code, which made O(log n) `ZRANK` via spans a small change.
- **LevelDB and RocksDB memtables** default to skip lists.
- Java's `ConcurrentSkipListMap` and `ConcurrentSkipListSet` .

In the interview

Q: Real-time leaderboard with "top 10" and "my rank"?

A: A Redis sorted set: `ZADD` to update, `ZREVRANGE 0 9` for top 10, `ZREVRANK` for rank, all O(log n) via the skip list with spans. Shard by game or region if a set grows too large.

Q: Why not a balanced binary tree?

A: Same asymptotic cost, but simpler to implement and extend (rank via spans), no rebalancing, and ordered range scans along the bottom level.

Q: Expected height?

A: About log base 1/p of n: around 10 levels for p = 1/4 and a million entries.

TL;DR

A skip list is a sorted linked list with random express lanes: expected O(log n) operations without rebalancing. Redis sorted sets pair one (with spans, for ranges and ranks) with a hash table (for score lookup).

Sources

- Pugh, "Skip Lists: A Probabilistic Alternative to Balanced Trees", CACM 33(6), 1990: https://homepage.cs.uiowa.edu/~ghosh/skip.pdf
- Redis source, `server.h` (ZSKIPLIST_MAXLEVEL, ZSKIPLIST_P, zskiplistNode): https://github.com/redis/redis/blob/unstable/src/server.h
- Redis default configuration, `redis.conf` (zset-max-listpack-*): https://github.com/redis/redis/blob/unstable/redis.conf
- Redis docs, Sorted sets: https://redis.io/docs/latest/develop/data-types/sorted-sets/
- antirez on why Redis uses skip lists (Hacker News): https://news.ycombinator.com/item?id=1171423

Cache eviction beyond LRU

What it is

An **eviction policy** picks what to throw out when a cache is full; **LRU** (least recently used) evicts the item untouched longest (basics: Module 0 — Eviction policies). Three better-behaved families: **CLOCK** (a cheap LRU approximation), **ARC** (Adaptive Replacement Cache: balances recency and frequency, self-tuning) and **TinyLFU / W-TinyLFU** (an admission filter on approximate frequency counts, used by the Java library Caffeine). **Recency** = how recently used, **frequency** = how often, **hit ratio** = fraction of requests served from cache.

The problem it solves

LRU's weaknesses: (1) **cost**: every hit moves the item to the list front, needing a lock and touching shared memory on every read; (2) **scan pollution**: one big sequential scan (backup, report) reads many items once and pushes out the popular ones, because LRU knows recency, not frequency.

How it works

CLOCK. Slots in a circle, each with a **reference bit**, and a "hand" pointer.

1. On a hit, set the bit to 1; nothing moves, so hits are nearly free.
2. To evict, look under the hand: bit 1 → clear to 0 and advance (a "second chance"); bit 0 → evict.

Slots A(1) B(0) C(1) D(1), hand at A: clear A, move; B is 0, evict B. A cleared bit means "unused since the hand last passed". PostgreSQL's shared buffer cache uses a "clock sweep" with small usage counters instead of one bit.

ARC (Megiddo and Modha, IBM, FAST 2003). For a cache of c pages it keeps **T1** (pages seen once recently: recency), **T2** (seen at least twice: frequency), **ghost lists B1, B2** (keys only, no data, of pages recently evicted from T1 and T2), and an adaptive target size **p** for T1.

1. Hit in T1 or T2: move to the front of T2.
2. Miss found in B1: a recency page was evicted too early, so increase p, then load into T2.
3. Miss found in B2: a frequency page was evicted too early, so decrease p.
4. Complete miss: put in T1.
5. Evict from T1 if it exceeds p, else from T2, recording the key in the matching ghost list.

A one-off scan only touches T1, so it can't flush T2. In the paper's SPC1-like benchmark with a 4 GB cache, LRU hit 9.19% and ARC 20%. ARC was patented by IBM; PostgreSQL used it briefly in 8.0 and replaced it (2Q in 8.0.2, clock sweep in 8.1) partly to avoid the patent.

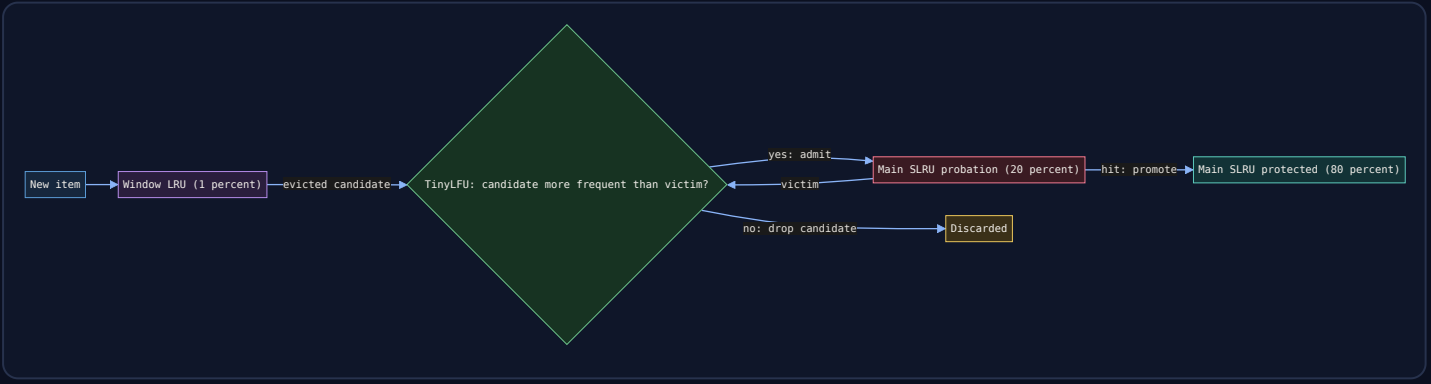
TinyLFU and W-TinyLFU (Einziger, Friedman and Manes). TinyLFU is an **admission policy**: when the cache is full, does the newcomer deserve to replace the eviction victim at all?

1. Every access increments the item in a **count-min sketch** (see Count-min sketch; overcounts slightly, never undercounts). Caffeine uses 4-bit counters.
2. **Aging (reset):** after W accesses (Caffeine: W = 10 × cache size) halve every counter so old popularity fades.
3. A **doorkeeper** (small Bloom filter) absorbs first-time items so one-hit wonders don't consume sketch counters.
4. **Admission:** admit the candidate only if its estimated frequency beats the victim's.

Plain TinyLFU rejects items about to become popular with no history (sudden bursts). **W-TinyLFU** adds a **window**:

1. New items enter a small LRU window (1% of capacity in the paper).
2. Items evicted from the window are candidates for the **main cache** (99%), a **segmented LRU** (SLRU): 20% probation, 80% protected; a hit in probation promotes to protected.
3. When main is full, TinyLFU compares the window's candidate with probation's victim; the more frequent stays.
4. Caffeine resizes the window by **hill climbing**: nudge it bigger or smaller and keep the change if hit ratio improves.

Example: 100-item cache (window 1, probation 20, protected 79). A backup reads 10,000 cold keys; each passes through the window as a candidate with estimated frequency about 1 against a probation victim of, say, 6. All are rejected; the hot set survives.



Complexity. CLOCK: O(1) hit, amortised O(1) eviction, 1 bit per entry. ARC: O(1) per request plus ghost metadata for up to c extra keys. W-TinyLFU: O(1) per request plus a sketch of a few bytes per entry (Caffeine cites about 8 bytes per entry for the 4-bit sketch).

Trade-offs

- CLOCK: very cheap and concurrency-friendly, but only approximates LRU and is not scan-resistant on its own.
- ARC: self-tuning and scan-resistant, but ghost lists add complexity; patent history limited adoption.
- W-TinyLFU: near-optimal hit ratios across many traces with tiny metadata, but approximate frequencies and more knobs (window size, sample size).
- Fancy policies pay off only with skew; under uniform random access all perform about the same.

Real-world examples

- **PostgreSQL** shared buffers: clock sweep with usage counts.
- **ZFS** uses an ARC-derived cache (also called the ARC).
- **Caffeine** (W-TinyLFU): Cassandra, Kafka, HBase, Solr, Neo4j, Druid, Spring Cache, Quarkus.

In the interview

Q: Why is LRU vulnerable to scans, and the fix?

A: It treats one access like many, so a one-time scan evicts popular items. Use frequency: ARC's T1/T2 lists, SLRU segments, or a TinyLFU admission filter refusing cold items.

Q: What does TinyLFU use instead of exact counters?

A: A count-min sketch of small, periodically halved counters: frequency for far more keys than the cache holds, in a few bytes per entry.

Q: Why does CLOCK scale better than LRU under concurrency?

A: A hit just sets a bit; no list reordering, so readers don't contend on a shared lock.

TL;DR

CLOCK makes LRU cheap with a reference bit and a sweeping hand. ARC keeps recency and frequency lists plus ghosts and adapts the split. W-TinyLFU puts a small LRU window before an SLRU main cache and admits only items a count-min sketch rates more popular than the victim, beating LRU on skewed and scan-heavy workloads.

Sources

- Megiddo and Modha, "ARC: A Self-Tuning, Low Overhead Replacement Cache", USENIX FAST 2003: https://www.usenix.org/conference/fast-03/arc-self-tuning-low-overhead-replacement-cache
- Einziger, Friedman, Manes, "TinyLFU: A Highly Efficient Cache Admission Policy" (arXiv 1512.00727): https://arxiv.org/abs/1512.00727
- Caffeine wiki, Efficiency: https://github.com/ben-manes/caffeine/wiki/Efficiency
- Caffeine README (users and integrations): https://github.com/ben-manes/caffeine
- Ben Manes, "Design of a Modern Cache", High Scalability: https://highscalability.com/design-of-a-modern-cache/
- PostgreSQL 8.0.2 release notes (2Q replaces ARC): https://www.postgresql.org/docs/release/8.0.2/
- PostgreSQL 8.1 release notes (clock sweep): https://www.postgresql.org/docs/release/8.1.0/

Hierarchical timing wheels

What it is

A **timing wheel** manages huge numbers of timers (timeouts, retries, scheduled tasks) cheaply: a clock face of slots, each holding a list of timers due in that interval, with a pointer advancing one slot per **tick** (fixed time unit, e.g. 1 ms). A **hierarchical** timing wheel stacks wheels with coarser ticks, like a clock's seconds, minutes and hours hands (Varghese and Lauck, 1987).

The problem it solves

Servers track enormous numbers of timers (idle timeout per connection, deadline per request, back-off per retry), most cancelled before firing. A **priority queue** (heap by deadline) costs O(log n) per add and cancel; a timing wheel makes both O(1) and fires timers in batches.

How it works

Single wheel. n slots, tick u; a timer due in d < n ticks goes in slot `(current + d) mod n`. Each tick, advance and fire the new slot. Cancel removes it from the slot's doubly linked list in O(1). It covers only n × u of time.

Hashed wheel. For longer delays, use slot `(current + d) mod n` and store the number of full rotations to wait; each pass decrements it and fires at zero. Simple, but long timers are visited every lap.

Hierarchical wheels (modelled on Kafka: 1 ms tick, 20 slots per wheel): level 0 = 20 × 1 ms (20 ms); level 1 = 20 × 20 ms (400 ms); level 2 = 20 × 400 ms (8 s). Kafka creates higher "overflow" wheels on demand. A timer due in 350 ms (at time 0):

1. Too far for level 0, fits level 1: slot 350 / 20 = 17 (340 to 359 ms).
2. At t = 340 the level-1 bucket comes due; its timers are **re-inserted** with remaining time, so ours (10 ms left) lands in level 0 slot 10 (relative).

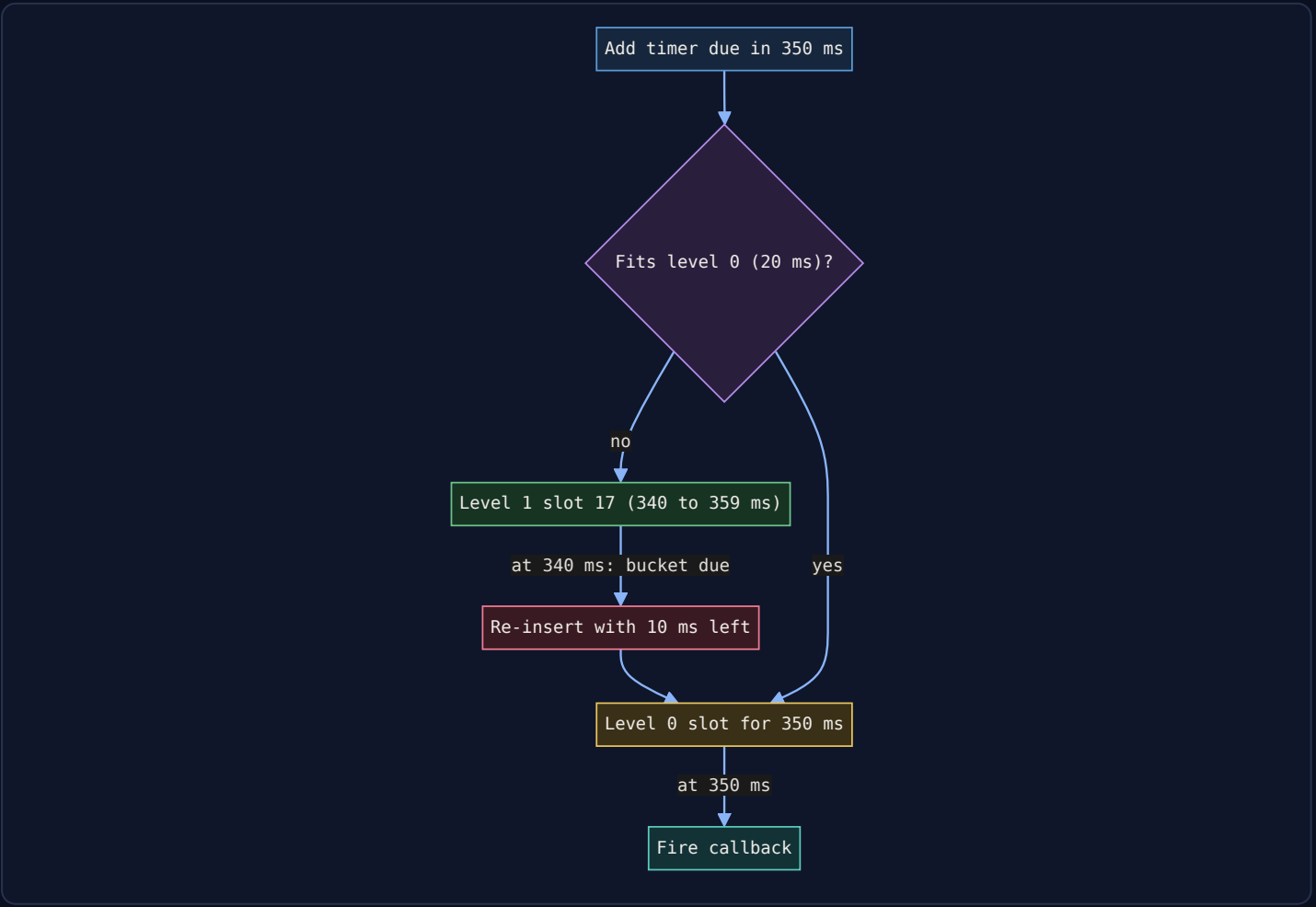
3. At t = 350 it fires.

Each timer moves down at most once per level: add O(m) for m levels (tiny), cancel O(1), per-tick work proportional only to timers coming due.

Avoiding wasted ticks. Ticking every millisecond when nothing is due wastes CPU. Kafka keeps the *buckets* (not timers) in a Java `DelayQueue` ordered by expiry, so the timer thread sleeps until the next non-empty bucket.

```
add(timer):
  if timer.deadline < currentTime + tick: run(timer) and return
  if timer.deadline < currentTime + tick * slots:
    bucket = buckets[(timer.deadline / tick) mod slots]
    bucket.append(timer)
    if bucket expiry changed: delayQueue.offer(bucket)
  else:
    overflowWheel().add(timer)   # coarser wheel, created lazily

advanceTo(t):  # called when delayQueue hands back a due bucket
  currentTime = t rounded down to tick
  for each timer in bucket: add(timer)   # re-insert lower or run
```



Complexity. Add O(m) (O(1) single wheel); cancel O(1); per tick O(1) plus expiring or cascading timers; space O(slots + timers).

Trade-offs

- Precision limited to the tick (coarser on upper levels if they fire in place): good for timeouts, wrong for high-precision scheduling.
- Cascading timers down from upper wheels costs bursts of work; the Linux kernel's 2015 rework lets coarse-level timers fire in place, slightly late, to avoid cascading.
- A heap is simpler and exact; it wins with few timers or precise firing.
- Slot-array memory is fixed even when idle.

Real-world examples

- **Apache Kafka** "purgatory" (requests awaiting acks or data) moved to hierarchical timing wheels; in Confluent's high-timeout benchmark the old design saturated at about 25,000 requests/s, the new one about 105,000.
- **Netty** `HashedWheelTimer` : default 100 ms tick, 512-slot wheel, for I/O timeouts that need not be exact.
- **Linux kernel** timer wheel for low-resolution timeouts (network, I/O).
- Job schedulers and delay queues on Redis or in-process.

In the interview

Q: Ten million connection idle timeouts, mostly reset. Data structure?

A: A hashed or hierarchical timing wheel: O(1) add and cancel, batch firing per tick; a heap pays O(log n) per reset and cancel.

Q: What do you give up versus a priority queue?

A: Precision (tick granularity) and some cascade work moving timers from coarse to fine wheels.

Q: How do you avoid ticking every millisecond?

A: Wake only when a non-empty bucket is due, e.g. buckets in a delay queue as Kafka does.

TL;DR

A timing wheel is a circular array of time slots with a moving pointer; timers drop into their deadline's slot. Stacked coarser wheels cover long delays with O(1) cancel and near-O(1) add, which is why Kafka, Netty and Linux use them for huge numbers of mostly-cancelled timeouts.

Sources

- Varghese and Lauck, "Hashed and Hierarchical Timing Wheels", SOSP 1987: http://www.cs.columbia.edu/~nahum/w6998/papers/sosp87-timing-wheels.pdf
- Confluent blog, "Apache Kafka, Purgatory, and Hierarchical Timing Wheels": https://www.confluent.io/blog/apache-kafka-purgatory-hierarchical-timing-wheels/
- Netty API, HashedWheelTimer: https://netty.io/4.1/api/io/netty/util/HashedWheelTimer.html
- LWN, "Reinventing the timer wheel" (2015): https://lwn.net/Articles/646950/

Distributed snapshots: Chandy-Lamport and Flink checkpoint barriers

What it is

A **distributed snapshot** records every process's state plus messages in flight so that together they form a **consistent cut**: a moment that *could* have happened, where no message is recorded as received but not sent. The **Chandy-Lamport algorithm** (1985) takes one without stopping the system; Apache Flink's **checkpoint barriers** are a variant for streaming dataflows.

The problem it solves

With no global clock you can't say "everyone record at 12:00:00.000", and independent recording can be inconsistent: branches P and Q each hold $100; P sends $10 to Q; if P records after sending ($90) and Q before receiving ($100), $10 vanishes ($190). Consistent snapshots are needed for checkpointing (restart after a crash), deadlock or termination detection, and debugging.

How it works

Assumptions. Reliable FIFO channels (messages arrive in the order sent), and a special **marker** message separating "before" from "after".

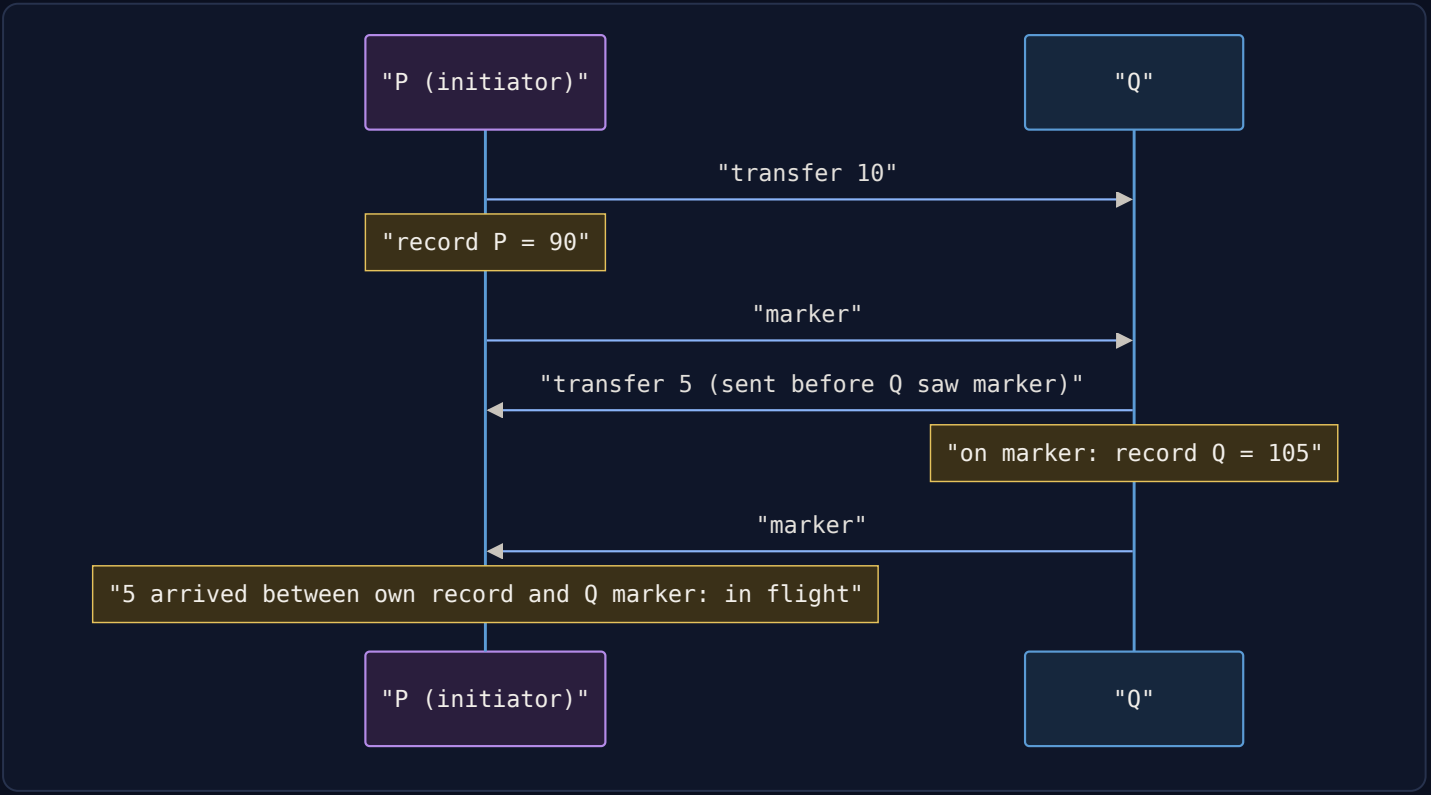
Chandy-Lamport.

1. The **initiator** records its state, then sends a marker on every outgoing channel before anything else.
2. On a process's first marker (on channel C): record own state, record C as empty, send markers on all outgoing channels, start recording messages on each *other* incoming channel.

3. On a later marker on channel C: stop recording C; messages recorded on C between the process's snapshot and this marker are C's in-flight state.
4. Done when every process has received a marker on every incoming channel.

Bank example, P initiates:

1. P sends $10 to Q (P = 90), records "P = 90", sends a marker. Meanwhile Q sends $5 to P (Q = 95).
2. Q receives the $10 (Q = 105), then the marker (FIFO guarantees order); records "Q = 105", channel P→Q empty, sends a marker to P.
3. P receives Q's $5 after its own snapshot but before Q's marker: in flight on Q→P.
4. P receives Q's marker: P = 90, Q = 105, in flight Q→P = 5. Total 200: consistent.



Flink checkpoint barriers (inspired by Chandy-Lamport; Carbone et al., "Lightweight Asynchronous Snapshots for Distributed Dataflows", 2015). A Flink job is a parallel graph of sources, operators and sinks.

1. The **checkpoint coordinator** (in the JobManager) tells every source to start checkpoint n.
2. Each source records its position (e.g. Kafka partition offsets) and injects **barrier n** into its output; barriers flow in line with records and never overtake them.
3. A single-input operator, on barrier n, snapshots its state (asynchronously to durable storage such as S3) and forwards the barrier.
4. A multi-input operator performs **alignment**: after barrier n arrives on one input, it buffers that input's further records until barrier n arrives on all inputs, then snapshots and forwards. This keeps the cut consistent: exactly-once state.
5. Sinks acknowledge when they have barrier n on all inputs; all acks complete checkpoint n.
6. On failure, every operator restores from the latest complete checkpoint and sources rewind to the recorded offsets.

No channel state is needed because the graph is acyclic and sources replayable: aligned barriers leave no record "between" operators at the cut (cyclic graphs do need recorded channel state, as in the original).

Unaligned checkpoints (Flink 1.11, 2020): under backpressure alignment can stall for a long time, so the operator forwards the barrier on its first arrival and stores the overtaken in-flight buffers in the checkpoint, closer to classic Chandy-Lamport: fast checkpoints regardless of throughput, more I/O.

At-least-once mode skips alignment: records after the barrier may be processed before the snapshot, so after recovery they replay and may appear twice.

Complexity. One marker per channel, O(E) messages for E channels; storage = process state plus in-flight messages (zero in aligned Flink); time ≈ markers crossing the longest path plus persisting state.

Trade-offs

- No global pause, but snapshots cost bandwidth and storage; frequent checkpoints shorten replay but add overhead.
- Aligned: no channel state, but alignment adds latency and stalls under backpressure.
- Unaligned: fast under backpressure, larger checkpoints and more I/O.
- End-to-end exactly-once also needs a replayable source and a transactional or idempotent sink (e.g. Flink's two-phase-commit Kafka sink); the snapshot covers only internal state (Module 10 — Exactly-Once Semantics).

Real-world examples

- **Apache Flink**: barriers, aligned and unaligned checkpoints, savepoints (always aligned).
- Distributed debuggers and deadlock or termination detectors (the 1985 paper's motivation).
- Marker-based ideas in other stream processors and consistent backups of sharded systems.

In the interview

Q: Why can't each node snapshot its own state at 12:00?

A: Clocks drift and messages are in flight, so a message can be recorded as received but not sent, or lost. Markers (or barriers) give a consistent cut.

Q: How does Flink achieve exactly-once state?

A: Sources inject barriers and record offsets, operators align barriers across inputs and snapshot, and on failure everything rolls back to the last complete checkpoint with sources replaying from stored offsets.

Q: What do unaligned checkpoints solve?

A: Under backpressure aligned barriers queue behind slow data and checkpoints take very long or time out; unaligned ones let the barrier overtake buffered data and save that data in the snapshot.

TL;DR

Chandy-Lamport snapshots without stopping: record local state, flood markers, record messages arriving on a channel before its marker. Flink injects barriers at sources and aligns them at multi-input operators, with replayable sources, for exactly-once state; unaligned checkpoints trade larger snapshots for speed under backpressure.

Sources

- Chandy and Lamport, "Distributed Snapshots: Determining Global States of Distributed Systems", ACM TOCS 3(1), 1985: https://www.microsoft.com/en-us/research/publication/distributed-snapshots-determining-global-states-distributed-system/
- Carbone et al., "Lightweight Asynchronous Snapshots for Distributed Dataflows" (arXiv 1506.08603): https://arxiv.org/abs/1506.08603
- Apache Flink docs, Stateful Stream Processing: https://nightlies.apache.org/flink/flink-docs-stable/docs/concepts/stateful-stream-processing/
- Apache Flink 1.11.0 release announcement (unaligned checkpoints): https://flink.apache.org/2020/07/06/apache-flink-1.11.0-release-announcement/

Inverted index

What it is

An **inverted index**, behind almost every search box, maps each word to the documents containing it (a forward index maps document to words), like a textbook's back-of-book index.

- **Term**: one searchable unit, usually a lower-cased cleaned word.
- **Dictionary** (term dictionary): sorted list of distinct terms.
- **Posting**: "this term appears in document N" (optionally with frequency and positions).
- **Posting list**: all postings for a term, sorted by document ID.

The problem it solves

Scanning 1 billion 10 KB documents (10 TB) at 1 GB/s takes almost 3 hours per query. An inverted index touches only the posting lists of the query's terms, and turns "apple AND pie" into intersecting two sorted integer lists.

How it works

1. Analyse. An **analyser** tokenises, lower-cases, optionally drops "stop words" (the, a, of) and applies **stemming** ("apples" → "apple").

2. Build postings. Emit (term, docID) pairs, sort by term then docID, group:

Doc ID	Text
1	Red apple pie
2	Green apple
3	Red car

Term	Posting list
apple	1, 2
car	3
green	2
pie	1
red	1, 3

"red AND apple" intersects [1, 3] and [1, 2] with two pointers: [1]. Sorted lists make intersection O(|A| + |B|); **skip pointers** (stored jumps such as "the 128th entry is doc 9,412") leap over long runs.

3. Immutable segments. Lucene (under Elasticsearch, OpenSearch and Solr) buffers new documents in memory; on fill or refresh timer the buffer becomes a small complete **immutable segment** (a mini inverted index never changed after writing). Searches query every segment and merge. A delete marks the document dead in a small "live documents" bitmap; an update is delete plus add.

4. Merge segments. Thousands of tiny segments slow every query (one lookup each) and keep dead documents forever, so a background **merge policy** combines them, dropping deletions. Lucene's default `TieredMergePolicy` groups similar sizes into tiers; Solr documents about 10 segments per tier, merging 10 at once, capped at about 5 GB per merged segment. With 10 MB flushes: ten merge into 100 MB, ten of those into 1 GB, stopping near 5 GB. Each document is rewritten about log10(5,000/10) = log10(500) ≈ 2.7, so about 3 times: the **write amplification** price of append-only segments.

5. Compress posting lists (a popular term may be in hundreds of millions of documents):

- **Delta (gap) encoding**: store the difference from the previous doc ID; sorted IDs give small positive gaps.
- **Varint** (variable-length integer, **VByte**): 7 bits of the number per byte, top bit 1 = "more bytes follow", 0 = last. Below 128: 1 byte; below 16,384: 2 bytes.

Posting list [3, 7, 150, 152, 1000] → gaps 3, 4, 143, 2, 848. 3, 4, 2 take 1 byte each. 143 = 1 × 128 + 15 → bytes 15 + 128 = 143 (0x8F), then 1 (0x01). 848 = 6 × 128 + 80 → 208 (0xD0), then 6 (0x06). Total 7 bytes vs 20 as 32-bit integers, almost 3× smaller; dense terms compress far better.

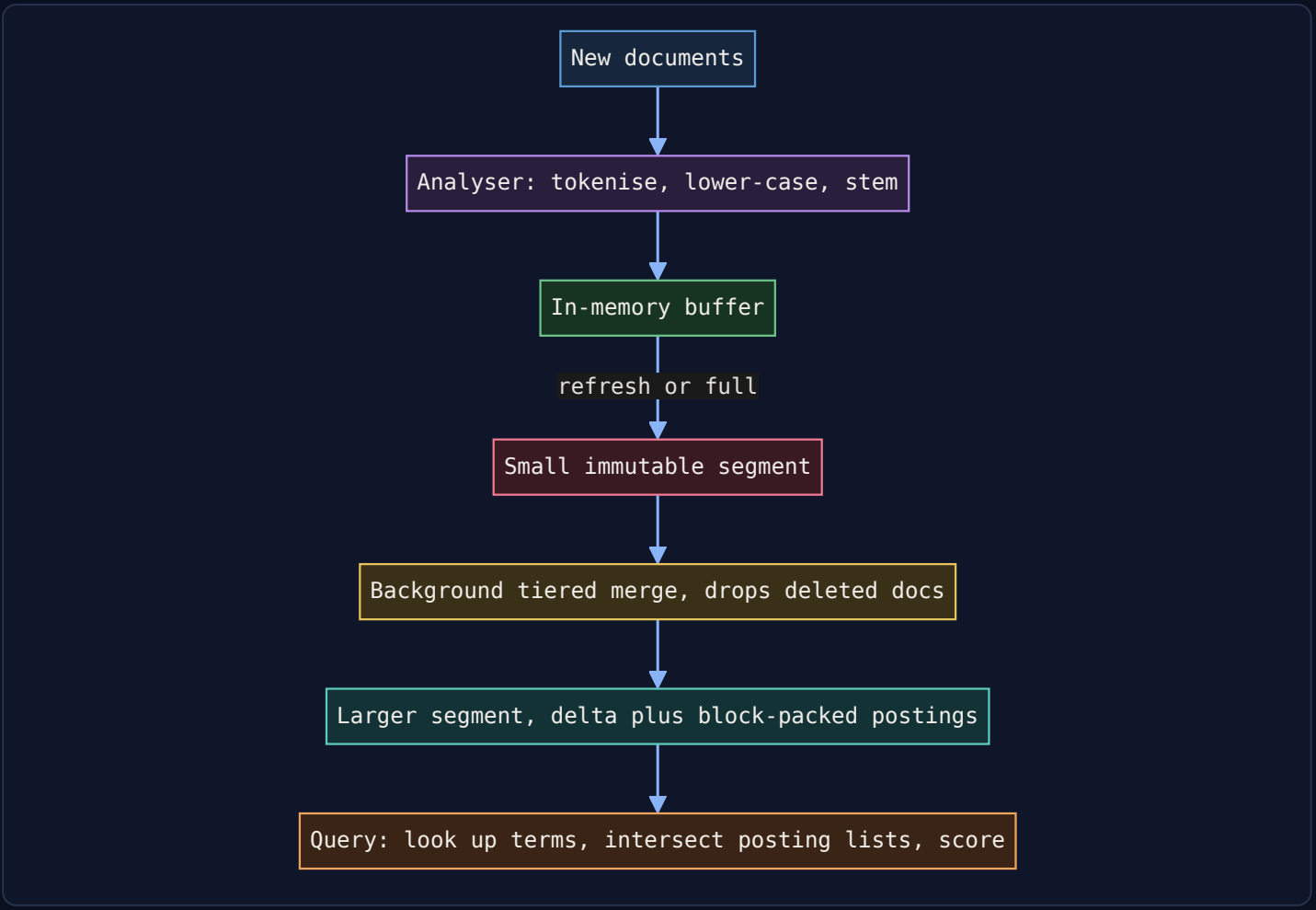
```
encode_varint(n):
    out = []
    while n >= 128:
        out.append((n % 128) + 128)   # low 7 bits, continuation bit set
        n = n // 128
    out.append(n)                     # last byte, top bit clear
    return out
```

Modern Lucene packs gaps in blocks of 128 integers at the smallest bit width fitting the block's largest gap, with varints only for the leftover tail; block packing decodes much faster on modern CPUs than byte-at-a-time varints.

6. Partition across machines.

- **Document-partitioned (local index):** each shard has a full inverted index for a subset of documents; a query goes to every shard (**scatter-gather**), each returns its local top 10, a coordinator merges.
- **Term-partitioned (global index):** each shard owns whole posting lists for some terms ("apple" on shard 7); a query visits only its terms' shards.

1 billion documents on 100 shards: document-partitioned, a two-word query fans out to all 100, each scanning lists for its 10 million documents and returning 10 hits, the coordinator merging 1,000 into a top 10; load is even, but every query touches every shard and the slowest sets response time. Term-partitioned, it touches 2 shards, but if "apple" is in 50 million documents that shard ships (or intersects) a 50-million-entry list; popular terms make **hot spots**, and indexing one document updates many shards.



Complexity: build is dominated by sorting (term, doc) pairs, O(P log P) for P postings; a query costs roughly the posting-list lengths read (less with skip pointers); space roughly proportional to P after compression, often a fraction of the raw text.

Trade-offs

- **Freshness vs throughput:** refresh every second for near-real-time search but many small segments and more merging; every 30 s is cheaper but staler.
- **Compression vs CPU:** tighter encodings save disk and memory but cost decode time per query; block packing is the usual sweet spot.
- **Document vs term partitioning:** document partitioning is simpler, balances load and keeps indexing local, at the cost of fan-out; term partitioning cuts fan-out but suffers hot terms and expensive multi-term queries. Nearly all production systems partition by document.
- **Merging vs I/O (input/output):** aggressive merging keeps queries fast but burns disk bandwidth and can starve indexing.

Real-world examples

- **Lucene, Elasticsearch, OpenSearch, Solr:** immutable segments, tiered merging, delta plus block-packed postings, document-partitioned shards.
- **Google (Barroso, Dean, Holzle, 2003):** the index was sharded by document so one query ran in parallel across many machines, each shard replicated for throughput.
- **Database full-text indexes** such as PostgreSQL's GIN (generalised inverted index) use the same term-to-rows idea.

In the interview

Q: Why are Lucene segments immutable?

A: No reader locks, aggressive OS file caching, compression tuned once at write time. Cost: deletes become tombstones and background merges reclaim space.

Q: How do you split a 10 TB index?

A: By document, into shards that fit comfortably on a node (tens of GB), each replicated for read throughput and failover; queries scatter to all shards and a coordinator merges top results. Mention term partitioning only to explain why hot terms make it a poor default.

Q: Why delta-encode before varint?

A: Varint saves space only for small numbers; raw doc IDs reach millions, but gaps between sorted IDs are small (especially for common terms), so most fit in one byte.

TL;DR

An inverted index maps each term to a sorted doc-ID list. Engines write immutable segments, merge them in the background, compress postings with delta plus varint or block packing, and shard by document so queries scatter to all shards and gather a merged top-k.

Sources

- Lucene 9.12 postings format, block size and VInt tail: https://lucene.apache.org/core/9_12_2/core/org/apache/lucene/codecs/lucene912/Lucene912PostingsFormat.html
- Solr guide, index segments and merging (TieredMergePolicy defaults): https://solr.apache.org/guide/solr/9_9/configuration-guide/index-segments-merging.html
- Barroso, Dean, Holzle, "Web Search for a Planet: The Google Cluster Architecture", IEEE Micro 2003: https://research.google/pubs/web-search-for-a-planet-the-google-cluster-architecture/
- Manning, Raghavan, Schutze, *Introduction to Information Retrieval*, chapters 1, 4 and 5 (index construction and compression): https://nlp.stanford.edu/IR-book/

Ranking with TF-IDF and BM25

What it is

After the inverted index finds matches, **ranking** orders them. Two classic formulas dominate keyword search:

- **TF-IDF** (term frequency × inverse document frequency): higher when the query word appears often in the document (TF) and is rare across the collection (IDF).
- **BM25** ("Best Matching 25", from the Okapi system): a refined TF-IDF that stops rewarding repetition after a while and adjusts for document length; the default in Lucene, Elasticsearch and OpenSearch.

Notation: **N** = documents in the collection; **df** (document frequency) = documents containing a term; **tf** (term frequency) = occurrences in one document; **dl** = document length in words; **avgdl** = average length.

The problem it solves

"apple pie" may match millions of pages and users look at the first few, so order is the product; plain matching gives no order. TF-IDF and BM25 are cheap, need no training data, and give a sensible order out of the box; modern systems use them as the first stage, with a learned model re-ranking the top few hundred.

How it works

Three ideas: (1) five mentions of "apple" suggest more about apples than one (term frequency); (2) a match on a rare word tells more than on "the" (inverse document frequency); (3) long documents mention everything more often just by being long, so discount them (length normalisation, BM25 only).

Collection (N = 4), query "apple pie":

Doc	Text	Length dl
D1	apple pie recipe apple	4
D2	apple phone review	3
D3	cherry pie	2
D4	best pie shop in town today	6

df(apple) = 2, df(pie) = 3, avgdl = 15 / 4 = 3.75.

TF-IDF with idf = log10(N / df), score = Σ tf × idf: idf(apple) = 0.301, idf(pie) = 0.125. D1 = 2 × 0.301 + 0.125 = **0.727**; D2 = **0.301**; D3 = **0.125**; D4 = **0.125**. Problems: D3 (two words, squarely about pie) ties D4 (six words, passing mention), and if D1 said "apple" 50 times its score would grow 25-fold, rewarding spam.

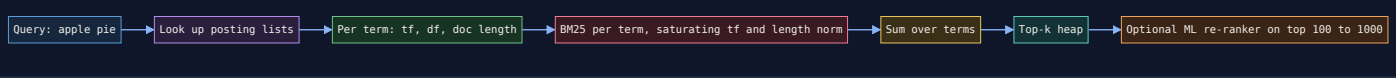
BM25, per query term in a document:

```
idf(t)  = ln(1 + (N - df + 0.5) / (df + 0.5))          # Lucene's variant, never negative
norm(d) = 1 - b + b * dl / avgdl                        # 1.0 for an average-length doc
score(t, d) = idf(t) * tf * (k1 + 1) / (tf + k1 * norm(d))
score(q, d) = sum of score(t, d) over query terms t
```

k1 controls how fast repeated terms saturate (default 1.2); **b** how strongly length matters (default 0.75; 0 ignores length, 1 fully normalises).

- IDFs: apple = ln(1 + 2.5/2.5) = ln 2 = 0.693; pie = ln(1 + 1.5/3.5) = ln 1.4286 = 0.357.
- k1 × norm: D1 norm = 0.25 + 0.75 × 4/3.75 = 1.05 → 1.26; D2 0.85 → 1.02; D3 0.65 → 0.78; D4 1.45 → 1.74.
- Scores (k1 + 1 = 2.2): D1 apple = 0.693 × (2 × 2.2)/(2 + 1.26) = 0.936, pie = 0.357 × 2.2/(1 + 1.26) = 0.347, total **1.28**; D2 = 0.693 × 2.2/2.02 = **0.755**; D3 = 0.357 × 2.2/1.78 = **0.441**; D4 = 0.357 × 2.2/2.74 = **0.286**.

Ranking D1 > D2 > D3 > D4: the tie breaks for the short, focused document. **Saturation:** with D1's length factor, tf × 2.2 / (tf + 1.26) gives 0.97 at tf = 1, 1.35 at 2, 1.95 at 10, and never exceeds k1 + 1 = 2.2: keyword stuffing stops paying.



Complexity: O(total postings read) with a size-k heap (O(log k) per insert). **WAND** and **block-max WAND** skip documents whose best possible score cannot enter the current top k.

Trade-offs

- **No semantics:** exact (analysed) words only; "car" won't match "automobile" without synonyms or vector search ("hybrid search").
- **Tuning:** k1 and b are rarely tuned, but short fields (titles) and long ones (bodies) often want different b; Elastic says defaults are a reasonable start and any change must be measured on real queries.
- **Per-shard statistics:** each shard computes df and avgdl locally, so a document can score slightly differently per shard; Elasticsearch's "DFS" mode gathers global statistics first at the cost of an extra round trip.
- **Cheap and explainable:** no training data, scores break down term by term for debugging.

Real-world examples

- Lucene switched its default similarity from classic TF-IDF to BM25 in version 6 (2016); Elasticsearch and OpenSearch inherit it.
- Retrieval-augmented generation (RAG) systems commonly run BM25 alongside embedding search and fuse the lists, since BM25 handles rare identifiers, product codes and names well.
- Web search engines historically used BM25-style features as one input among hundreds to a learned ranker.

In the interview

Q: Why does BM25 beat raw TF-IDF?

A: Term frequency saturates (the tenth mention adds much less than the second) and length normalisation stops long documents winning by size; k1 and b control both.

Q: Where does BM25 sit in a modern stack?

A: The cheap first stage: top few hundred candidates in milliseconds, then a learned re-ranker (gradient-boosted trees or a cross-encoder) reorders them with many more features.

Q: What does b = 0 do?

A: Ignores length, favouring long documents; right for roughly fixed-length fields such as titles or tags.

TL;DR

TF-IDF scores by how often a document contains query terms, weighted by their rarity. BM25 adds saturation (k1, default 1.2) and length normalisation (b, default 0.75), robust enough to be Lucene's default first-stage ranker.

Sources

- Elastic, "Practical BM25, Part 2: The BM25 algorithm and its variables": https://www.elastic.co/blog/practical-bm25-part-2-the-bm25-algorithm-and-its-variables
- Elastic, "Practical BM25, Part 3: Considerations for picking b and k1": https://www.elastic.co/blog/practical-bm25-part-3-considerations-for-picking-b-and-k1-in-elasticsearch
- Robertson and Zaragoza, "The Probabilistic Relevance Framework: BM25 and Beyond", Foundations and Trends in IR, 2009: https://www.staff.city.ac.uk/~sbrp622/papers/foundations_bm25_review.pdf
- Manning, Raghavan, Schutze, *Introduction to Information Retrieval*, chapter 6 (tf-idf) and 11 (BM25): https://nlp.stanford.edu/IR-book/

Near-duplicate detection with SimHash, MinHash and LSH

What it is

Near-duplicate detection finds items that are almost the same (a news article with different adverts, a listing reposted with one changed word, one page under two URLs). Exact duplicates just need a

cryptographic hash such as SHA-256; near-duplicates need hashes where similar inputs give similar outputs:

- **SimHash**: a short bit string (often 64 bits) where similar documents differ in few positions, measured by **Hamming distance** (number of differing positions).
- **MinHash**: a short list of numbers (**signature**) whose fraction of matches estimates **Jaccard similarity** (overlap size / union size of two sets).
- **LSH (locality-sensitive hashing)**: buckets signatures so similar items share a bucket, so you compare only within buckets instead of every pair.

The problem it solves

All pairs of 1 billion pages is about 5 × 10^17 comparisons. Crawlers must decide "seen this page?" in milliseconds, and pipelines (e.g. de-duplicating large-language-model (LLM) training data) cluster billions of documents. Both need a compact fingerprint plus an index that finds close ones without scanning everything.

How it works

1. Features. Usually **shingles**, overlapping k-word or k-character sequences: "the cat sat on" with 2-word shingles = {the cat, cat sat, sat on}. Similar documents share most shingles.

2a. SimHash (Charikar 2002; used at Google by Manku, Jain and Das Sarma). Toy 8-bit version:

Feature	Weight	8-bit hash
apple	2	10110010
pie	1	01110100
recipe	1	10011001

Keep one counter per bit; for each feature add its weight where its bit is 1, subtract where 0. Sums: +2, −2, +2, +4, −2, −2, 0, −2. Output 1 where positive: **10110000**. Document B swaps "recipe" for "easy" (00011101, weight 1): sums 0, −2, +2, +4, −2, 0, 0, −2 → **00110000**, Hamming distance 1. Unrelated documents differ in about half the bits (4 of 8, 32 of 64).

Finding close SimHashes. Manku et al. found 64-bit SimHashes with threshold k = 3 differing bits worked well for about 8 billion pages. Pigeonhole lookup: split 64 bits into 4 blocks of 16; fingerprints differing in ≤ 3 bits match exactly in at least one block. Keep 4 tables keyed by one block each, look up the new fingerprint's 4 blocks, and check full Hamming distance only on the hits.

2b. MinHash (Broder 1997, for AltaVista). Shingle sets A = {a, b, c, d}, B = {b, c, d, e}: true Jaccard = 3/5 = **0.6**. Per hash function, a document's MinHash is its smallest shingle hash; the probability two sets share the minimum equals their Jaccard similarity.

Hash	a	b	c	d	e	min of A	min of B	Match?
h1	3	7	1	9	5	1 (c)	1 (c)	yes
h2	2	8	6	4	1	2 (a)	1 (e)	no
h3	9	2	5	7	3	2 (b)	2 (b)	yes
h4	4	6	8	3	9	3 (d)	3 (d)	yes

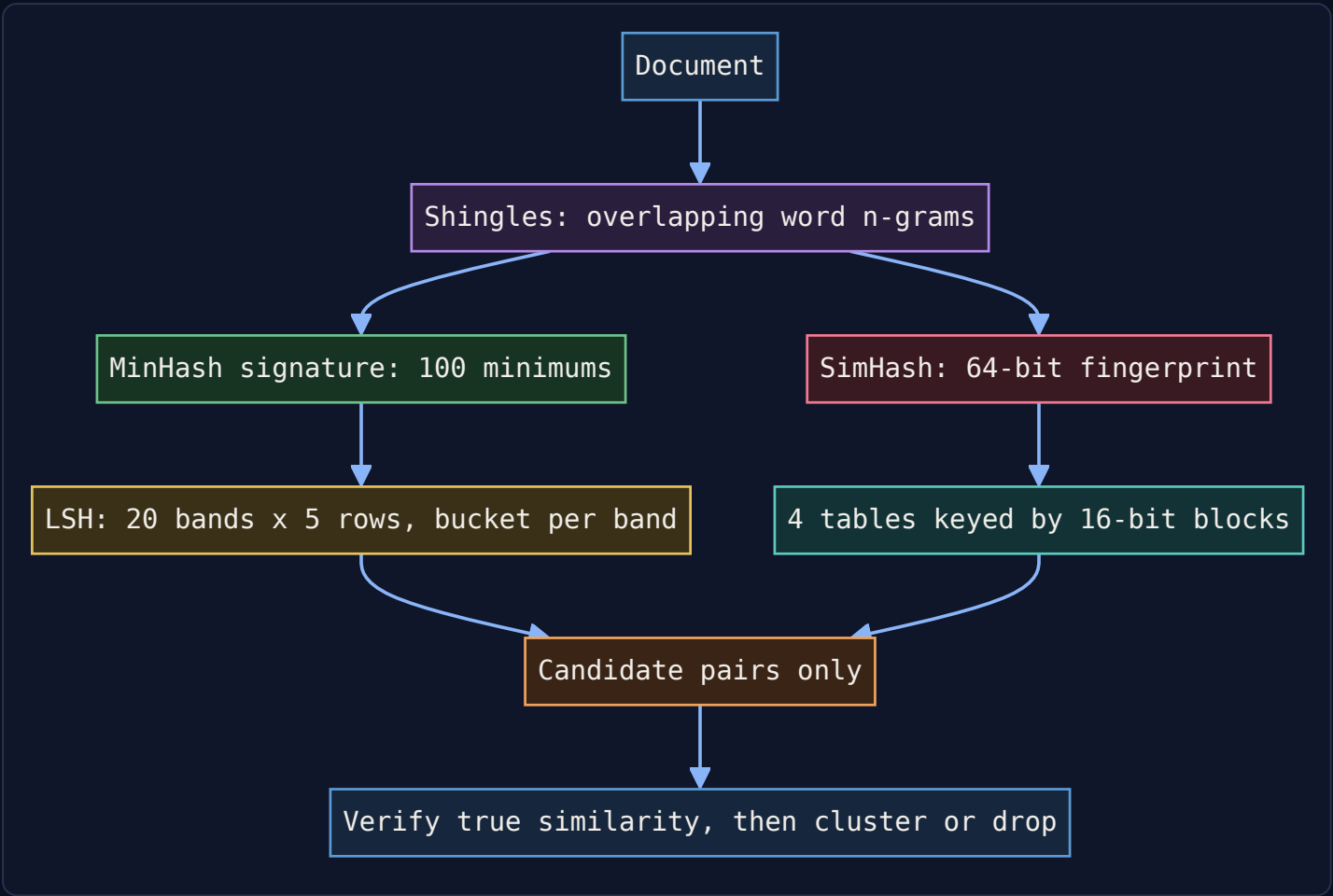
Estimate 3/4 = 0.75 vs true 0.6: noisy with 4 hashes. Standard error ≈ sqrt(J(1 − J)/n); 100 hashes give sqrt(0.6 × 0.4 / 100) ≈ 0.05. A 100-number signature (400 bytes) summarises a document of any size.

3. LSH banding. Split each 100-number signature into **b = 20 bands of r = 5 rows**, hash each band to a bucket; documents sharing any bucket are a **candidate pair**. For Jaccard s: P(band matches) = s^r; P(candidate) = 1 − (1 − s^r)^b.

Similarity s	s^5	P(candidate) with b = 20
0.3	0.0024	0.047
0.5	0.031	0.47
0.8	0.328	0.9996

An S-curve with its steep part near (1/b)^(1/r) = (1/20)^(1/5) ≈ 0.55: pairs above 0.8 are almost always caught, below 0.3 rarely compared. More rows per band raises the threshold (fewer false positives); more bands lowers it (fewer misses).

```
for doc in docs:
    sig = [min(h_i(s) for s in shingles(doc)) for i in 1..100]
    for band in 0..19:
        key = hash(band, sig[band*5 : band*5+5])
        for other in buckets[key]: candidates.add((doc, other))
        buckets[key].append(doc)
verify each candidate pair with exact Jaccard (or signature agreement)
```



Complexity: MinHash O(n × shingles) per document for n hashes, O(n) space; LSH O(b) bucket operations per document, so the job is near-linear instead of quadratic; SimHash O(features × 64) per document, 8 bytes stored.

Trade-offs

- **SimHash vs MinHash:** SimHash is tiny (8 bytes) and good for "almost identical" pages with weighted features; MinHash estimates Jaccard directly, behaves predictably at moderate similarity (0.5 to 0.8) and tunes easily with banding.
- **Banding:** b and r trade false positives (wasted verification) against false negatives (missed duplicates); choose from the target threshold.
- **Shingle size:** too small makes unrelated documents look similar; too large lets one changed word break many shingles. Word 5-grams or character 5- to 9-grams are common starts.

- **Memory:** LSH tables for billions of documents are large, so it often runs as a batch job (Spark, MapReduce) rather than an online index.

Real-world examples

- **AltaVista:** Broder's shingling and MinHash clustered the whole crawl to remove near-duplicates.
- **Google crawling:** 64-bit SimHash with k = 3 over roughly 8 billion pages (Manku paper).
- **LLM data pipelines** de-duplicate web text with MinHash plus LSH before training, since repeats waste compute and encourage memorisation.
- **Plagiarism checkers and news aggregators** group syndicated copies of a story.

In the interview

Q: How does a crawler avoid storing the same page under many URLs?

A: 64-bit SimHash of each page's text, stored in a few tables keyed by fixed bit blocks; look up each block of a new page and check full Hamming distance on hits. Within 3 bits: duplicate, skip or canonicalise.

Q: Why not compare MinHash signatures pairwise?

A: Still quadratic in documents; LSH banding puts likely-similar documents in shared buckets, making the job near-linear.

Q: How do you pick b and r?

A: Fix n = b × r, set the threshold (1/b)^(1/r) just below the similarity you care about, and check false-positive and false-negative rates with 1 − (1 − s^r)^b.

TL;DR

Hash so similar documents collide. SimHash: a 64-bit fingerprint compared by Hamming distance. MinHash: a signature whose agreement rate estimates Jaccard. LSH banding finds candidate pairs without all-pairs comparison, turning quadratic into near-linear.

Sources

- Manku, Jain, Das Sarma, "Detecting Near-Duplicates for Web Crawling", WWW 2007: https://www.www2007.cpsc.ucalgary.ca/papers/paper215.pdf
- Broder, "On the Resemblance and Containment of Documents", 1997: https://www.cs.princeton.edu/courses/archive/spring13/cos598C/broder97resemblance.pdf
- Broder et al., "Syntactic Clustering of the Web", WWW6 1997: https://archives.iw3c2.org/www6/proceedings/Technical/Paper205/Paper205.html
- Leskovec, Rajaraman, Ullman, *Mining of Massive Datasets*, chapter 3 (MinHash and LSH banding): http://www.mmds.org/

Vector search with HNSW, IVF and product quantization

What it is

Vector search finds items whose **embeddings** (a **vector** of numbers, e.g. 768 long, produced by an ML model so similar meaning gives nearby vectors) are closest to the query's, by Euclidean (L2) distance, inner product or cosine similarity. **Approximate nearest neighbour (ANN)** search accepts missing a few true neighbours for huge speed-ups; quality is **recall@k** (fraction of the true k nearest returned; recall@10 = 0.95 means 9.5 of the true top 10 on average). Building blocks: **HNSW** (Hierarchical Navigable Small World, a layered graph walked greedily), **IVF** (inverted file index: search only the clusters nearest the query) and **PQ** (product quantization: compress each vector to a few bytes). Sharding vector stores: Module 11 — Vector Database Sharding.

The problem it solves

100 million × 768 dimensions × 4 bytes = **307 GB** of floats, and a brute-force query is about 77 billion multiply-adds. Semantic search, RAG, recommendations and image search need tens of milliseconds over up

to billions of vectors; ANN indexes trade a little recall for orders of magnitude less latency and memory.

How it works

1. Brute force baseline: distance to every vector, top k in a heap, O(N × d) per query. Use it below roughly 100,000 vectors and as ground truth for recall.

2. IVF.

1. **Train:** k-means (finds nlist cluster centres) on a sample; with N = 1,000,000, nlist ≈ sqrt(N) = 1,000 is a common start.
2. **Assign:** each vector to its nearest centre's list (an inverted index whose "terms" are clusters), about 1,000 vectors per list.
3. **Query:** compare with the 1,000 centres, scan the nearest **nprobe** lists (say 10): 10,000 vectors plus 1,000 centres, about **1% of brute-force work**.
4. **Tune:** recall and latency rise roughly linearly with nprobe; a neighbour just across a cluster boundary is missed if its cluster isn't probed, which is why nprobe = 1 has poor recall.

3. PQ. Split each vector into m sub-vectors, run k-means per sub-space with 256 centres, and store each sub-vector's centre number (1 byte). Toy example, 4 dimensions, m = 2, 4 centres (2 bits) each: x = [0.1, 0.9, 0.4, 0.3] → [0.1, 0.9] and [0.4, 0.3]. Codebook 1 (c0 = [0,0], c1 = [0,1], c2 = [1,0], c3 = [1,1]): nearest c1 (squared distance 0.02). Codebook 2 (c0 = [0.5,0.5], c1 = [0,0], c2 = [1,0], c3 = [0.5,0]): nearest c0 (0.05 vs 0.10 for c3). Code (1, 0): 16 bytes became 4 bits.

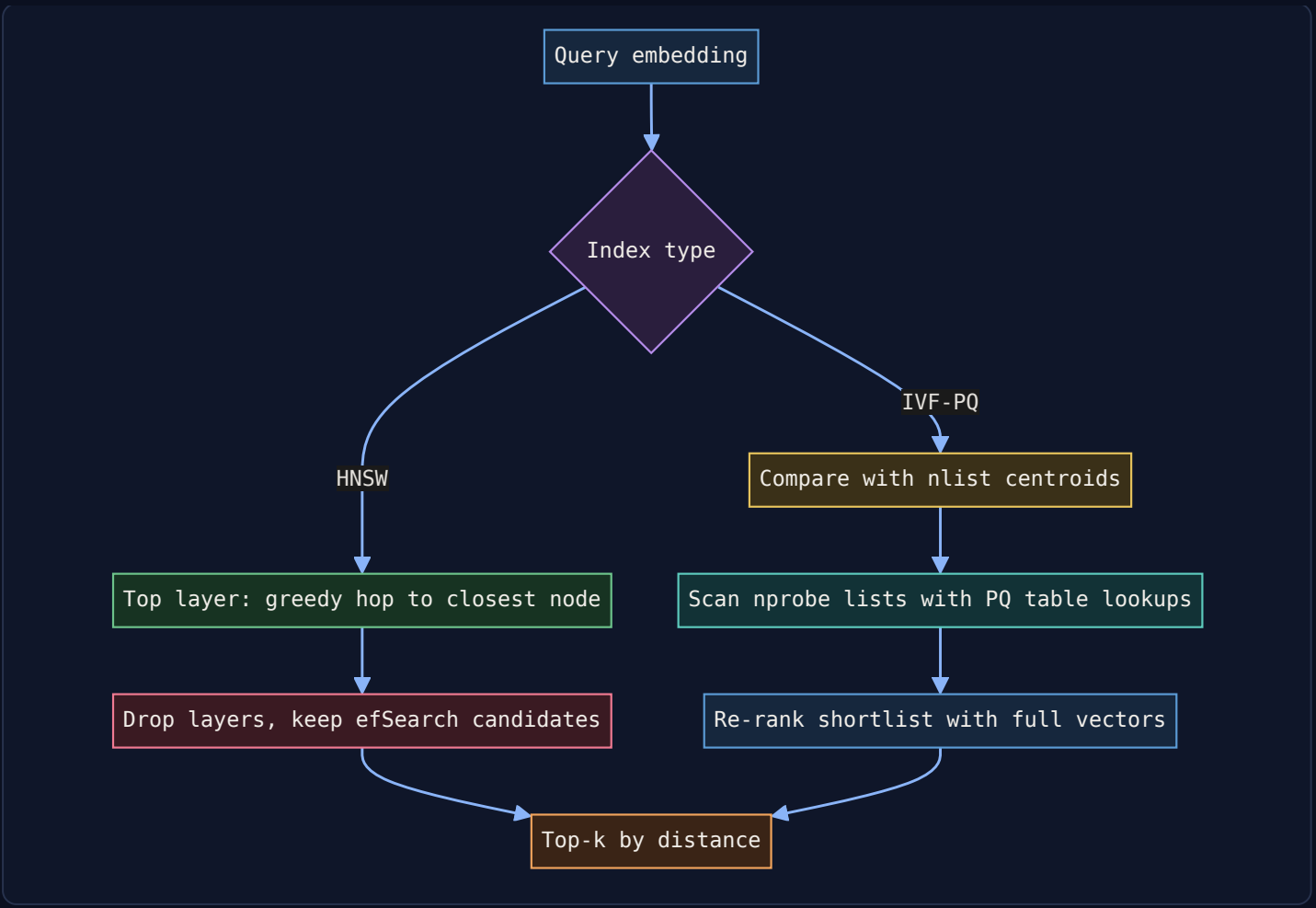
Asymmetric distance computation (ADC): at query time compute each query sub-vector's distance to every centre once (an m × 256 table); distance to any stored vector is then m table lookups summed, with no floating-point vector maths.

Real numbers: 768 dimensions, m = 96 sub-vectors of 8 dimensions at 8 bits = **96 bytes per vector** instead of 3,072 (32×); 100 million vectors shrink from 307 GB to about **9.6 GB**, fitting one server's memory. **IVF-PQ** clusters with IVF, stores residuals with PQ, and optionally **re-ranks** the top few hundred with full vectors kept on SSD.

4. HNSW.

1. Each inserted vector becomes a node linked to about **M** of its nearest existing nodes (commonly 16; the bottom layer allows 2M).
2. Each node gets a random top layer, exponentially fewer nodes higher up, like a skip list's express lanes.
3. **Search:** from the top-layer entry point, greedily hop to whichever neighbour is closer to the query until none is, drop a layer, repeat. On the bottom layer keep a candidate list of size **efSearch** (e.g. 64), explore until it stops improving, return the best k.
4. **efConstruction** (candidate list size while inserting) sets build quality: higher means better graphs, slower indexing.

Search cost grows roughly with log N, and HNSW typically gives the best recall for a given latency in RAM. With M = 16 the bottom layer stores 32 neighbour IDs × 4 bytes = 128 bytes per vector: about 13 GB of links plus 307 GB of vectors for 100 million 768-dimension vectors.



Complexity (N vectors, d dimensions):

Method	Query time	Memory per vector	Notes
Brute force	O(N d)	4d bytes	100% recall
IVF (flat lists)	O(nlist d + nprobe (N/nlist) d)	4d bytes	recall set by nprobe
IVF-PQ	same scan, but m lookups per vector	m bytes plus IDs	lowest memory, lower recall
HNSW	roughly O(log N) hops x M x d	4d + about 8M bytes	best recall/latency in RAM, slow build

Trade-offs

- **Recall vs latency:** a dial per method (nprobe, efSearch); plot recall@10 against p99 latency and choose a point, e.g. 0.95 at 10 ms.
- **Memory vs recall:** PQ cuts memory 10 to 60× but loses precision; re-ranking a shortlist with full vectors recovers much of it.
- **Updates and deletes:** HNSW inserts are cheap-ish but deletes awkward (tombstones, periodic rebuild); IVF needs retraining when the distribution drifts.
- **Filtering** ("nearest products in stock under 20 dollars") fits ANN badly: post-filtering can return too few results; filtering during graph traversal can disconnect the graph.
- **Build time:** HNSW builds are CPU-heavy; IVF-PQ needs a training pass but builds quickly.

Real-world examples

- **FAISS** (Meta): flat, IVF, PQ, IVF-PQ and HNSW indexes, plus GPU versions.
- **pgvector, Elasticsearch, OpenSearch, Redis, Qdrant, Weaviate and Milvus** offer HNSW; several also IVF or quantised variants.

- **RAG** stores chunk embeddings and retrieves the top 5 to 50 chunks per question.
- **Recommendation candidate generation** retrieves items near the user's embedding (see Feed ranking).

In the interview

Q: 1 billion 768-dimension embeddings: which index?

A: Raw floats are about 3 TB, so not HNSW on full vectors in RAM. IVF-PQ: tens of thousands of clusters, 64 to 96 byte codes (64 to 96 GB, sharded across a few machines), probe a few dozen lists, re-rank the top few hundred with full vectors from SSD.

Q: What do efSearch and nprobe control?

A: Query-time recall vs latency: candidates kept on HNSW's bottom layer, and IVF clusters scanned. Neither needs a rebuild.

Q: How do you know the ANN index is good enough?

A: Brute-force exact top-k for a sample of real queries, then recall@k and latency percentiles of the ANN index on them, plus end-to-end quality, since 0.9 recall may be fine with a strong re-ranker.

TL;DR

HNSW walks a layered graph (best recall for latency in memory); IVF searches only the closest clusters; PQ compresses vectors to tens of bytes with per-subspace codebooks. Real systems combine them and tune nprobe or efSearch for recall vs latency vs memory.

Sources

- Malkov and Yashunin, "Efficient and robust approximate nearest neighbor search using Hierarchical Navigable Small World graphs", IEEE TPAMI 2018: https://arxiv.org/abs/1603.09320
- Jegou, Douze, Schmid, "Product Quantization for Nearest Neighbor Search", IEEE TPAMI 2011: https://doi.org/10.1109/TPAMI.2010.57
- FAISS wiki, "Faster search" and index guidelines: https://github.com/facebookresearch/faiss/wiki/Faster-search
- Johnson, Douze, Jegou, "Billion-scale similarity search with GPUs" (FAISS paper), 2017: https://arxiv.org/abs/1702.08734

Shortest paths for maps: Dijkstra, A* and contraction hierarchies

What it is

A road map is a **graph**: **nodes** are intersections, **edges** road segments with a **weight** (metres or, more usefully, seconds). The **shortest path** problem asks for the route with the smallest total weight.

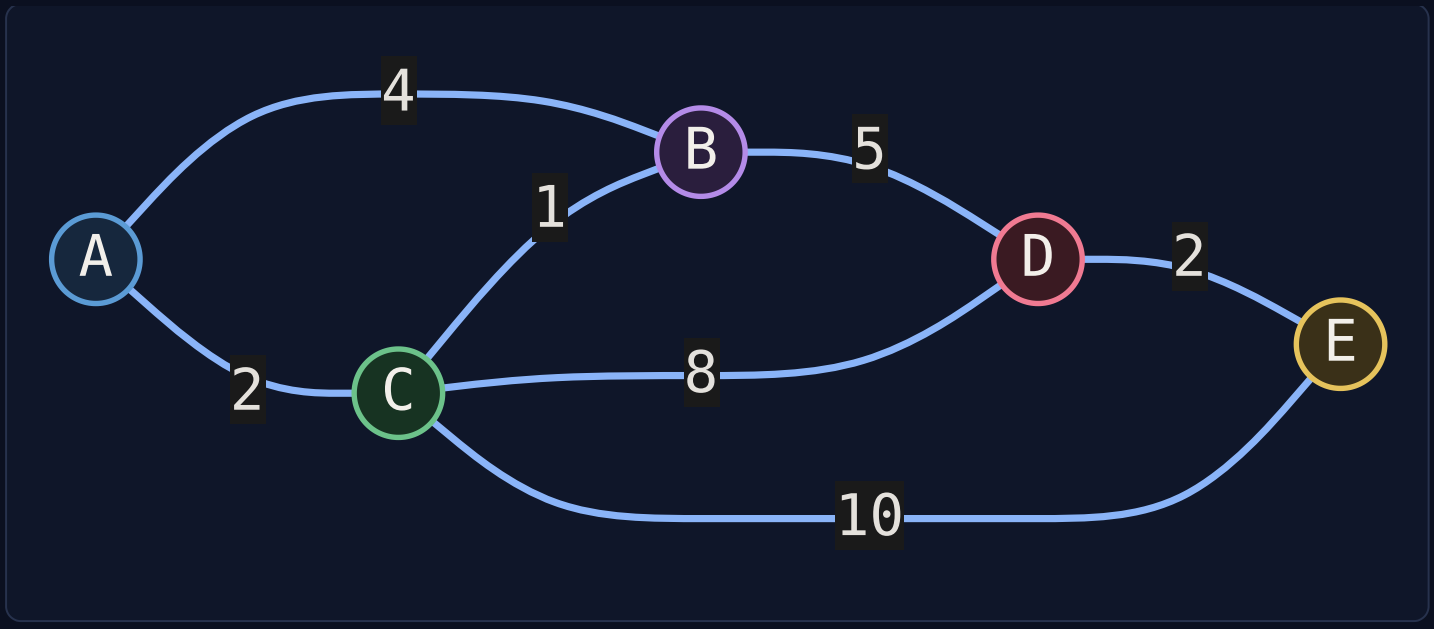
- **Dijkstra's algorithm** grows outward from the start in distance order, settling one node at a time; exact for non-negative weights.
- **A*** (A-star) is Dijkstra plus a **heuristic** (estimate of remaining distance) steering the search towards the target.
- **Contraction hierarchies (CH)** precompute **shortcut** edges once so each query explores only a few hundred nodes even on a continental map.

The problem it solves

A continental network has tens of millions of nodes; plain Dijkstra London to Rome settles millions (seconds). Navigation at thousands of requests per second, plus distance matrices (ETAs from 100 drivers to 100 riders = 10,000 routes), needs milliseconds. Road hierarchy (motorways matter more than cul-de-sacs) can be exploited by preprocessing.

How it works

Worked graph (undirected, minutes), A to E:



1. Dijkstra. Tentative distances (infinity at first) and a **priority queue** (min-heap). Repeatedly pop the closest unsettled node and **relax** its edges (update a neighbour if going through it is shorter).

```
dist[s] = 0, all others infinity, push (0, s)
while queue not empty:
    (d, u) = pop_min(queue)
    if u settled: continue
    mark u settled, if u == target: stop
    for (v, w) in edges(u):
        if d + w < dist[v]:
            dist[v] = d + w, prev[v] = u, push (dist[v], v)
```

Pop	Settled distance	Updates
A	0	B = 4, C = 2
C	2	B = 3 (via C beats 4), D = 10, E = 12
B	3	D = 8 (3 + 5 beats 10)
D	8	E = 10 (8 + 2 beats 12)
E	10	done

Answer **A-C-B-D-E, 10 minutes** (follow `prev` back from E); the direct A-B road (4) lost to the detour via C (3). Binary heap: O((V + E) log V) time, O(V) space.

2. A*. Pop by f = g + h (g = distance so far, h = heuristic to goal); on maps h is usually straight-line distance / maximum road speed. **Admissible** = never overestimates the true remaining distance, so A* stays optimal. **Consistent** (monotone) = h(u) ≤ w(u, v) + h(v) for every edge, so each node is settled once, like Dijkstra.

h(A) = 7, h(B) = 6, h(C) = 7, h(D) = 2, h(E) = 0; true remaining distances A 10, B 7, C 8, D 2, so admissible, and every edge checks consistent (e.g. C to B: 7 ≤ 1 + 6).

Pop (f)	g	Pushes (g, f)
A (7)	0	B (4, 10), C (2, 9)
C (9)	2	B (3, 9), D (10, 12), E (12, 12)
B (9)	3	D (8, 10)
D (10)	8	E (10, 10)
E (10)	10	done

Same answer; no saving on a tiny graph, but on a real map Dijkstra explores a disc around the start while A* explores an ellipse stretched towards the goal. h = 0 turns A* back into Dijkstra; an overestimating h is faster but can return a worse route.

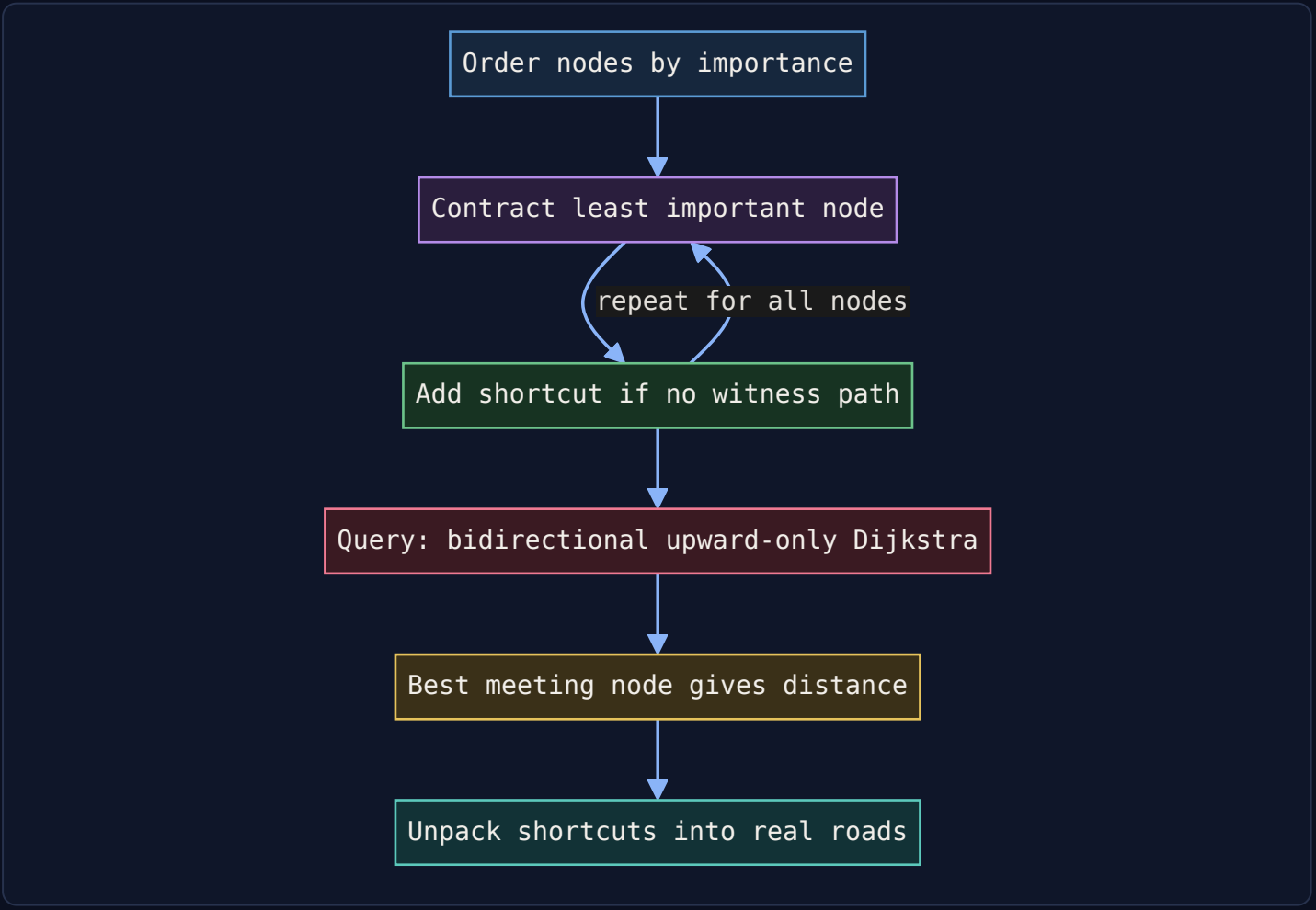
3. Contraction hierarchies.

Preprocessing (once; minutes to hours for a continent): order nodes by **importance** (heuristics such as how many shortcuts removing it would add; motorway junctions important, dead ends not), then **contract** from least to most important: removing v, for each pair of remaining neighbours u, w whose shortest path goes through v, add shortcut u-w with that weight, unless a **witness path** (equally short, avoiding v) exists.

Order C (1), A (2), B (3), D (4), E (5). Contract C (neighbours A, B, D, E): A-B via C = 3 < direct 4, no witness → **shortcut A-B = 3 (via C)**. A-D via C = 10, witness A-B-D = 9, skip. A-E via C = 12, witness A-B-D-E = 11, skip. B-D via C = 9 > 5, skip. B-E via C = 11, witness B-D-E = 7, skip. D-E via C = 18, skip. Contracting A, B, D leaves single higher neighbours: no pairs, no shortcuts.

Query: bidirectional Dijkstra where both searches follow only edges **upward** to higher-ranked nodes. Forward from A: shortcut A-B (3), B-D (8), D-E (10). Backward from E: E is top, nothing higher. Best meeting node E: **10**. Unpack A-B into A-C-B: A-C-B-D-E.

Every shortest path goes "up then down" in rank, so the restricted search stays exact. On continental networks the CH authors report queries visiting a few hundred nodes in well under a millisecond, vs millions settled by plain Dijkstra.



Trade-offs

- **Dijkstra:** no preprocessing, handles any change instantly, slow on big graphs.
- **A*:** easy speed-up with good geometry; benefit shrinks for travel-time weights on fast motorways (straight-line over top speed is a weak estimate).
- **CH:** fastest queries, excellent for distance matrices, but rebuilt when weights change. Live traffic pushes systems towards **customisable** techniques such as multi-level Dijkstra (MLD) or customisable CH, separating the slow metric-independent step from fast reweighting.

- **Memory:** shortcuts roughly double the edge count; acceptable in practice.

Real-world examples

- **OSRM** (Open Source Routing Machine, OpenStreetMap data) ships CH and MLD pipelines; its wiki recommends MLD for most uses (frequent traffic updates) and CH where raw speed for very large distance matrices matters most.
- **GraphHopper** offers CH, A* and landmark-based A*.
- Commercial map providers combine hierarchical preprocessing with live traffic layers; ride-hailing ETA systems run millions of shortest-path queries per minute for dispatch.

In the interview

Q: Why not run Dijkstra per route request?

A: It settles millions of nodes per continental query (seconds of CPU); CH lets each query touch a few hundred nodes, under a millisecond.

Q: What makes an A* heuristic admissible, and why does it matter?

A: It never overestimates remaining cost, guaranteeing the goal's distance is optimal when first popped. Straight-line distance / maximum speed is the standard choice for travel-time maps.

Q: Live traffic with CH?

A: Full re-contraction every minute is too slow. Use a customisable variant (precompute node order or partition once, recompute only weights: OSRM's MLD or customisable CH), and fall back to A* on small regions for very fresh incidents.

TL;DR

Dijkstra settles nodes in distance order: exact but slow on huge graphs. A* aims the search with an admissible heuristic. CH precomputes shortcuts by contracting unimportant nodes so a bidirectional upward-only search answers continental queries in under a millisecond, at the cost of rebuilding (or customising) when weights change.

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Feed ranking

What it is

Feed ranking chooses which eligible posts appear in a user's home feed and in what order. A **chronological feed** shows newest first; a **ranked feed** scores each post by how likely the user finds it valuable.

- **Candidate generation (retrieval):** quickly picking a few hundred to a few thousand plausible posts from millions.
- **Ranking (scoring):** a heavier model predicting engagement such as P(like), P(comment), P(share).
- **Re-ranking:** adjusting for goals the model misses: diversity, freshness, policy, ads.
- **In-network** posts come from followed accounts; **out-of-network** from others.

The problem it solves

Following 500 accounts can mean thousands of unseen posts, and out-of-network recommendation means hundreds of millions, while a user scrolls maybe 20 to 50 items. Chronological order buries a close friend's post under prolific accounts. A big neural model can't score every post in a 200 ms budget, so work is split into stages, each cheaper per item and handling more items than the next.

How it works

1. EdgeRank, the classic intuition. Facebook's early News Feed (around 2010) was publicly described as summing, over a story's interactions ("edges"), **affinity** (u: viewer-author closeness from past interactions, one-directional), **weight** (w: value of the edge type; a comment beats a like, photos beat text) and **time decay** (d: older edges count less). Score = Σ u × w × d. With d = 1 / (1 + hours old):

- Post P, close friend (u = 0.9), one comment (w = 4), 1 hour old (d = 0.5): **1.8**.
- Post Q, acquaintance (u = 0.2), three likes (w = 1), 1 hour old: 3 × 0.1 = **0.3**.

P wins despite less raw engagement. Learned models replaced the hand-tuned formula, but relationship, content value and recency remain features.

2. Candidate generation. Several cheap **sources** return a few hundred items each: recent posts from followed accounts (a precomputed timeline cache or fan-out store); posts liked by similar people (collaborative filtering); **embedding retrieval** via a **two-tower model** that learns user and post embeddings whose dot product predicts engagement, with post embeddings in an ANN index (Vector search); trending or topic posts. Twitter's open-sourced 2023 pipeline gathered around 1,500 candidates per request from hundreds of millions, roughly half in-network.

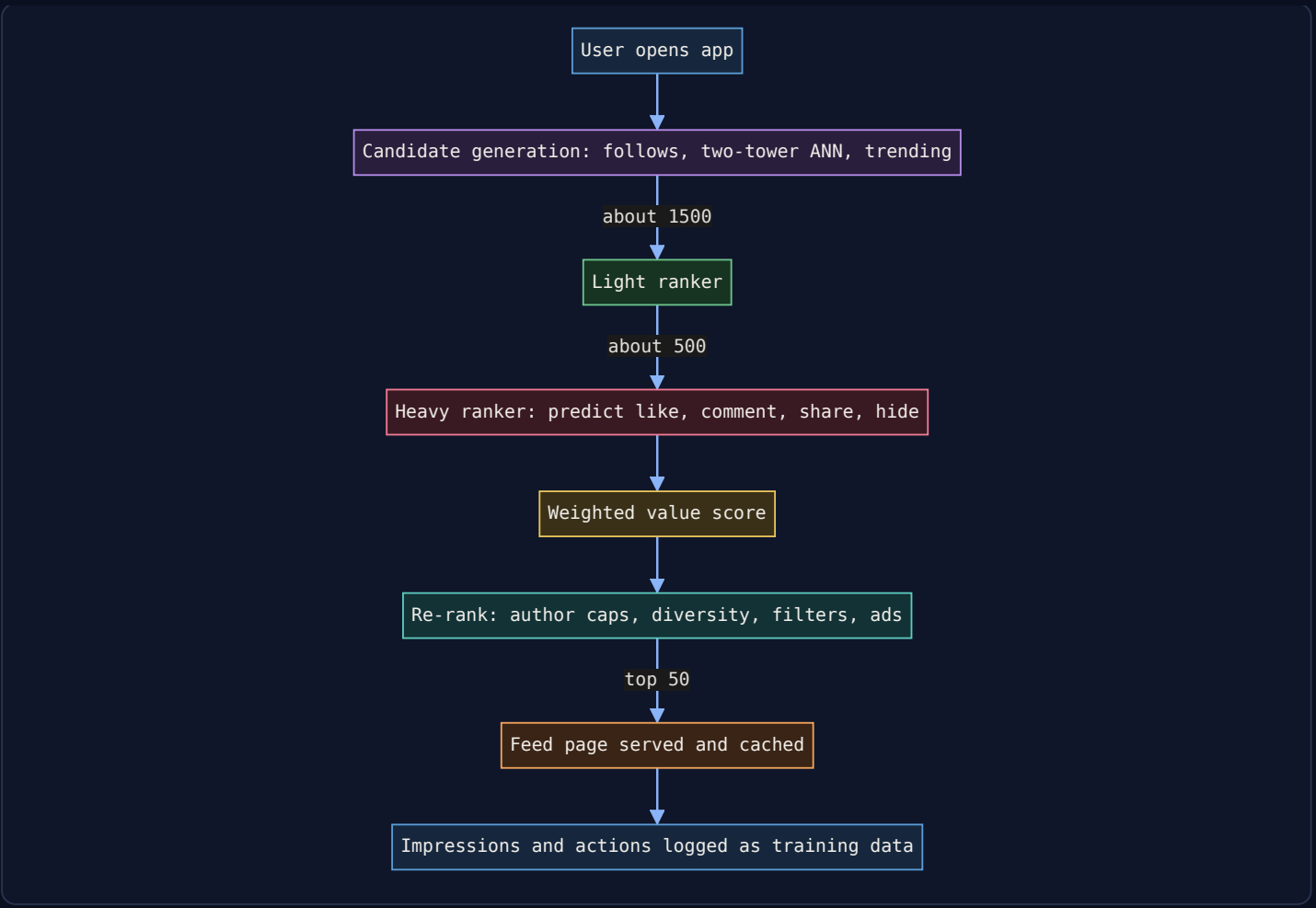
3. Light ranking (optional). A small model (logistic regression or a tiny network) trims 1,500 to about 500 on cheap features.

4. Heavy ranking. A large model predicts several probabilities per candidate from hundreds of **features** (affinity, post age, media type, viewer history, early engagement), combined with business-chosen weights. Illustrative: score = 1 × P(like) + 5 × P(comment) + 10 × P(share) − 50 × P(hide).

Post	P(like)	P(comment)	P(share)	P(hide)	Score
Friend's photo	0.30	0.05	0.01	0.001	0.30 + 0.25 + 0.10 - 0.05 = **0.60**
Viral meme	0.40	0.02	0.03	0.005	0.40 + 0.10 + 0.30 - 0.25 = **0.55**
Brand ad-like post	0.10	0.01	0.00	0.002	0.10 + 0.05 + 0 - 0.10 = **0.05**

The meme has the highest P(like) but the photo wins (comments weigh more, the meme is likelier hidden). The weights decide what the platform optimises: a product decision, not just engineering.

5. Re-ranking. Score order can show five posts from one prolific author in a row. Fixes: **author caps** (e.g. no more than 2 consecutive: A1 (0.9), A2 (0.85), A3 (0.8), B1 (0.7), C1 (0.6) becomes A1, A2, B1, A3, C1); **maximal marginal relevance (MMR)**, picking the next item maximising λ × score − (1 − λ) × similarity to items already chosen; freshness boosts, **filters** (blocked users, already seen, policy violations) and **ad insertion** at fixed slots.




```
def build_feed(user):
    cands = union(follow_source(user, 500), ann_source(user_emb(user), 800), trending(200))
    cands = dedupe_and_filter(cands, user)                # blocked, seen, policy
    cands = top_n(cands, light_model.score, 500)
    for c in cands: c.p = heavy_model.predict(user, c)     # several probabilities
    for c in cands: c.value = dot(WEIGHTS, c.p)
    return rerank_with_diversity(sort_desc(cands, "value"))[:50]
```

Budget: each stage handles fewer items with a costlier model, e.g. 20 to 40 ms retrieval, 50 to 100 ms heavy ranking on GPU or CPU batches, a few ms re-ranking, inside a 200 to 300 ms request.

Trade-offs

- **Engagement vs well-being:** optimising clicks can promote outrage or clickbait; negative signals (hide, report) and "was this worth your time?" survey labels counteract it.
- **Freshness vs quality:** new posts have no history (cold start); ranking must explore or they never get a chance.
- **Precompute vs on demand:** request-time ranking uses the freshest signals but costs compute per open; precomputed feeds are cheaper but stale.
- **Feedback loops:** trained on what it chose to show, the model reinforces its biases; random exploration and position-bias correction help.
- **Explainability:** a weighted sum of predicted actions is easier to reason about than one opaque score.

Real-world examples

- **YouTube** (Covington, Adams, Sargin, 2016): the canonical two-stage design, deep candidate generation (millions to hundreds) then a deep ranking network.
- **Twitter/X** open-sourced "For You" in 2023: candidate sources, light ranker, heavy neural ranker, heuristics and filters.
- **Facebook News Feed** evolved from EdgeRank to multi-task neural models whose predicted actions combine into a value score.

- **Instagram, TikTok and LinkedIn** describe multi-stage retrieve-then-rank systems.

In the interview

Q: Why split feed ranking into stages?

A: Cost: the heavy model may take a millisecond or more per item, so millions per request is impossible. Cheap retrieval narrows to about a thousand, a light model to a few hundred, and only those get the expensive model, keeping the request within a couple of hundred milliseconds.

Q: Stop one popular creator dominating?

A: A re-ranking pass: per-author caps in a window or MMR penalising similarity to chosen items, with a diversity metric tracked in A/B tests.

Q: What would you log, and why?

A: Every impression with position, candidate source, model scores and subsequent actions (or none): training data for the next model, position-bias correction, and per-source contribution.

TL;DR

Feeds are a funnel: candidate generation pulls about a thousand posts from follows, embedding retrieval and trends; a light ranker trims; a heavy multi-task model predicts actions combined into a weighted value score; re-ranking adds diversity, filters and ads. EdgeRank's affinity × weight × time decay lives on as features.

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